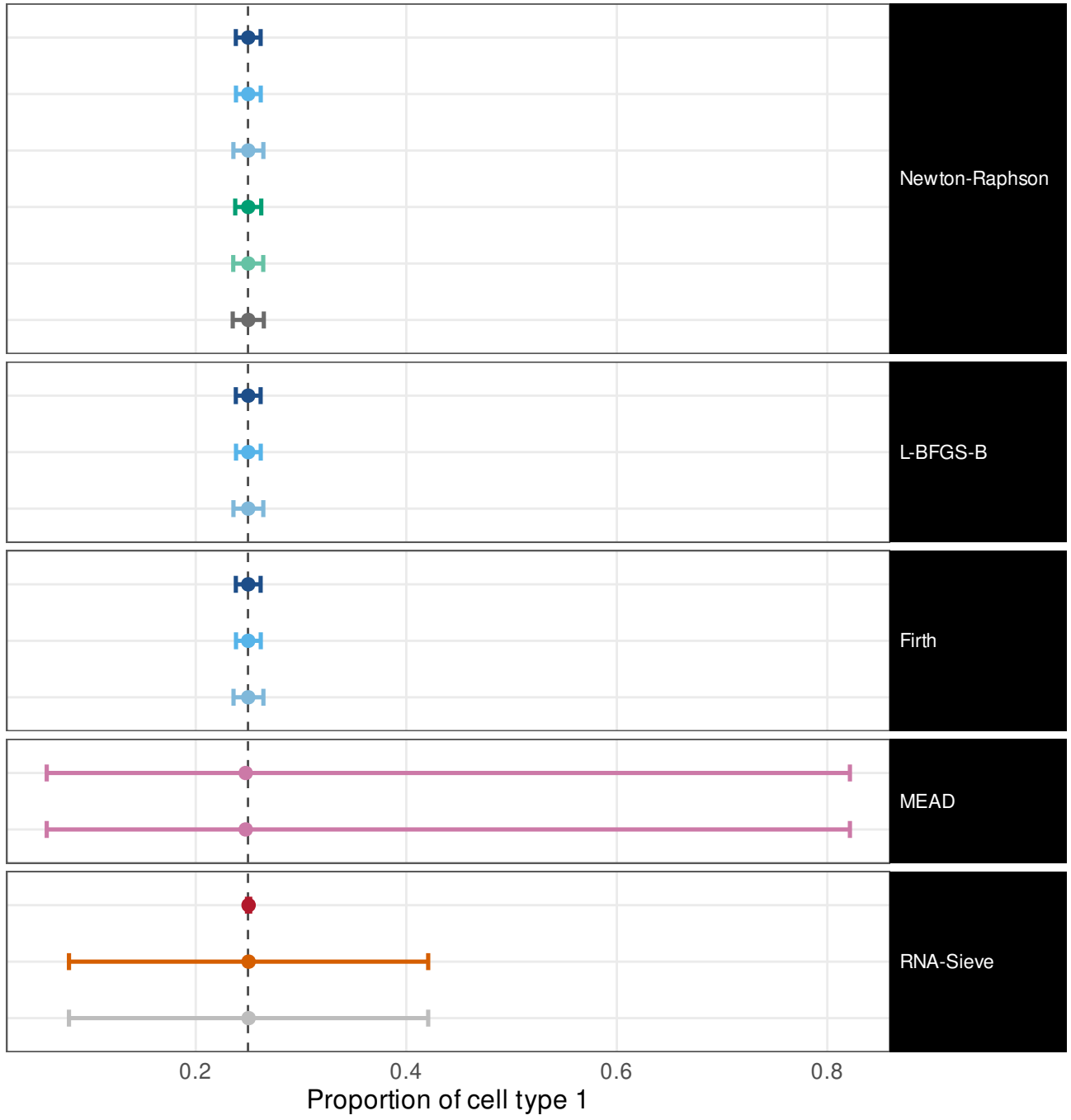
Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / balanced / low sibling overlap / a = 1 / equal sizes

Wald (delta)	0.94 [0.9, 0.97]	w = 0.0236
ILR, mapped back	0.94 [0.89, 0.96]	w = 0.0235
ILR, REML	0.96 [0.92, 0.98]	w = 0.0285
Profile likelihood	0.96 [0.92, 0.98]	w = 0.0247
Profile, REML	0.96 [0.92, 0.98]	w = 0.0285
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0295
Wald (delta)	0.94 [0.9, 0.97]	w = 0.0236
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.0235
ILR, REML	0.96 [0.92, 0.98]	w = 0.0284
Wald (delta)	0.94 [0.9, 0.97]	w = 0.0236
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.0235
ILR, REML	0.96 [0.92, 0.98]	w = 0.0284
Method's own interval	0.88 [0.83, 0.92]	w = 0.763
Own interval, transcript fractions	0.88 [0.83, 0.92]	w = 0.763
Fisher information (Wald)	0.13 [0.09, 0.18]	w = 0.00301
Godambe sandwich	1 [0.98, 1]	w = 0.341
Released code, as published	1 [0.98, 1]	w = 0.341

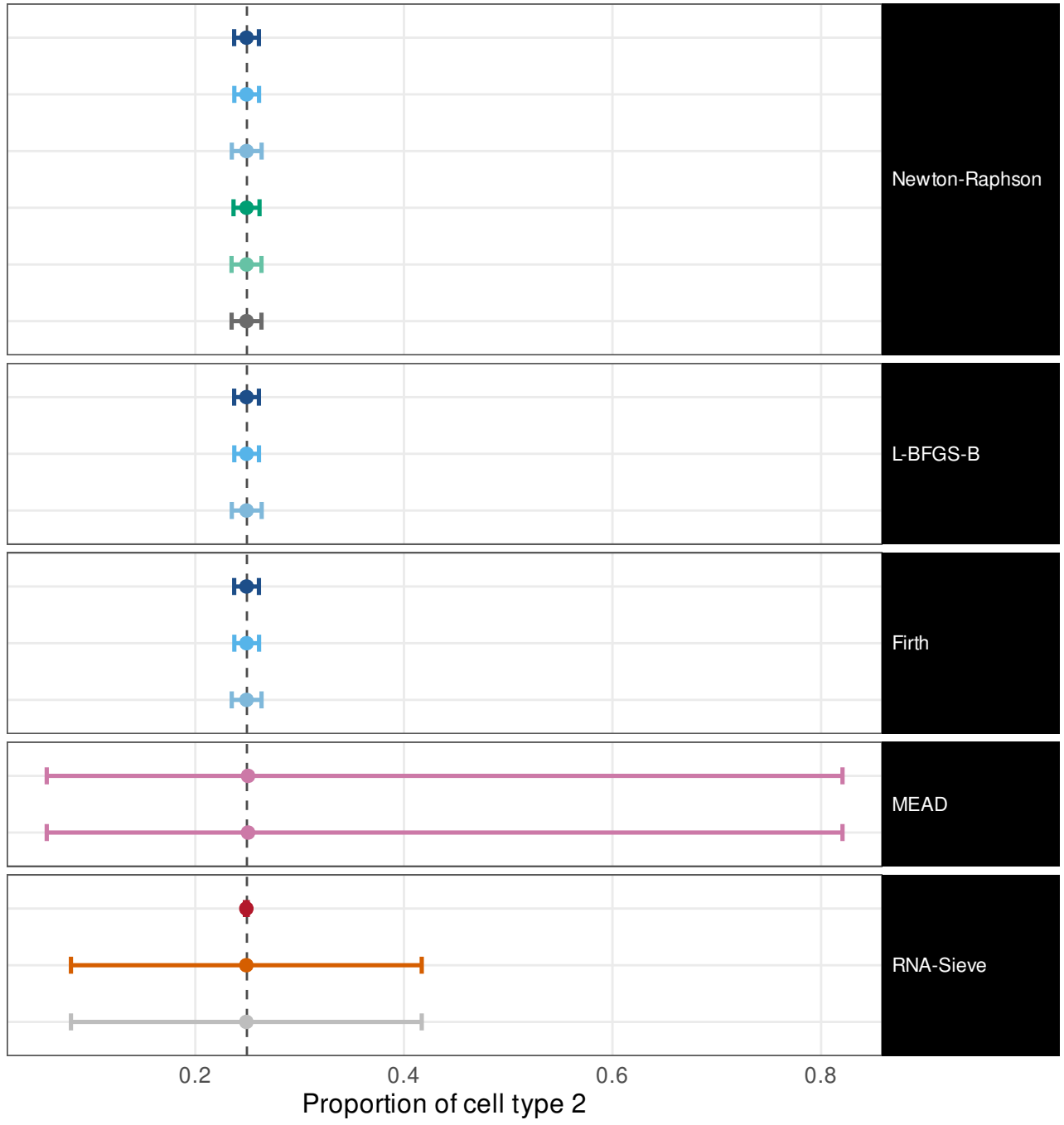


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / balanced / low sibling overlap / a = 1 / equal sizes

Wald (delta)	0.94 [0.89, 0.96]	w = 0.0237
ILR, mapped back	0.94 [0.89, 0.96]	w = 0.0237
ILR, REML	0.96 [0.93, 0.98]	w = 0.0286
Profile likelihood	0.94 [0.9, 0.97]	w = 0.0249
Profile, REML	0.96 [0.93, 0.98]	w = 0.0287
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0286
Wald (delta)	0.94 [0.89, 0.96]	w = 0.0237
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.0237
ILR, REML	0.96 [0.93, 0.98]	w = 0.0287
Wald (delta)	0.94 [0.89, 0.96]	w = 0.0237
ILR, mapped back	0.94 [0.89, 0.96]	w = 0.0237
ILR, REML	0.96 [0.92, 0.98]	w = 0.0286
Method's own interval	0.9 [0.84, 0.93]	w = 0.763
Own interval, transcript fractions	0.9 [0.84, 0.93]	w = 0.763
Fisher information (Wald)	0.14 [0.094, 0.19]	w = 0.00272
Godambe sandwich	1 [0.98, 1]	w = 0.336
Released code, as published	1 [0.98, 1]	w = 0.336

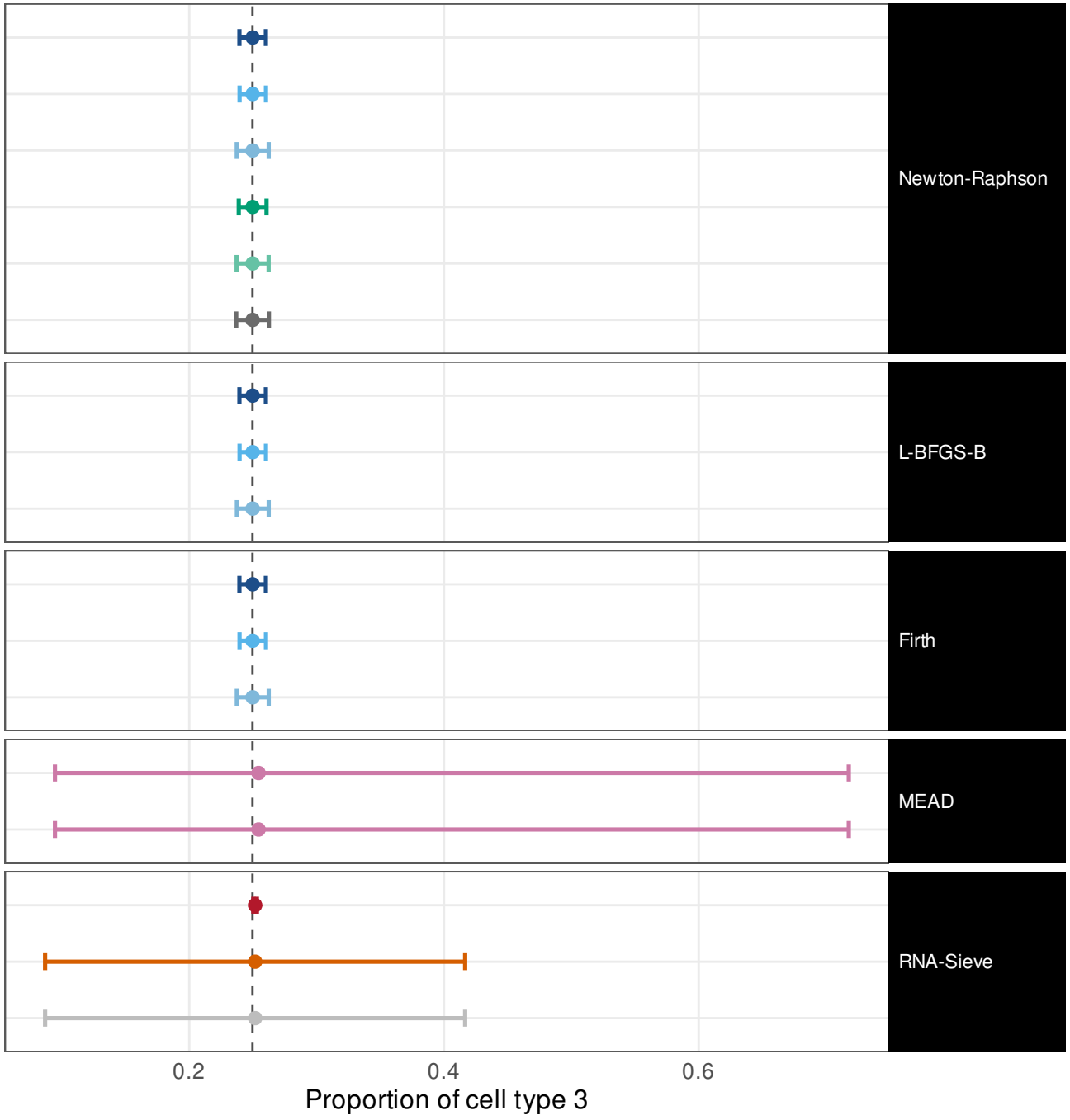


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / balanced / low sibling overlap / a = 1 / equal sizes

Wald (delta)	0.92 [0.88, 0.95]	w = 0.0208
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.0207
ILR, REML	0.94 [0.9, 0.97]	w = 0.0251
Profile likelihood	0.94 [0.89, 0.96]	w = 0.0218
Profile, REML	0.94 [0.9, 0.97]	w = 0.0251
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0257
Wald (delta)	0.92 [0.88, 0.95]	w = 0.0208
ILR, mapped back	0.92 [0.88, 0.95]	w = 0.0207
ILR, REML	0.94 [0.9, 0.97]	w = 0.025
Wald (delta)	0.92 [0.88, 0.95]	w = 0.0208
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.0207
ILR, REML	0.94 [0.9, 0.97]	w = 0.025
Method's own interval	0.8 [0.74, 0.85]	w = 0.623
Own interval, transcript fractions	0.8 [0.74, 0.85]	w = 0.623
Fisher information (Wald)	0.18 [0.13, 0.23]	w = 0.00275
Godambe sandwich	1 [0.98, 1]	w = 0.33
Released code, as published	1 [0.98, 1]	w = 0.33

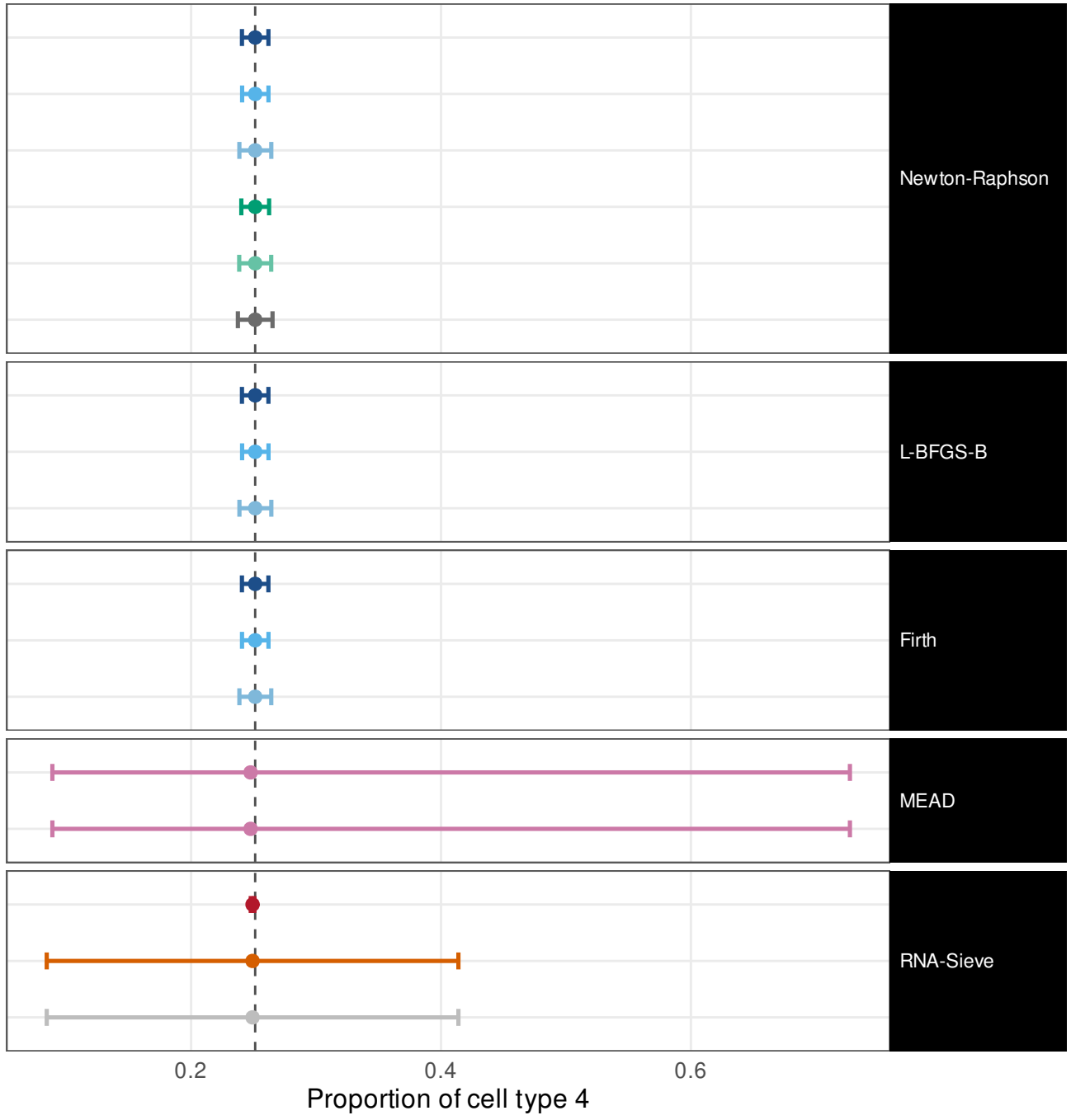


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / balanced / low sibling overlap / a = 1 / equal sizes

Wald (delta)	0.92 [0.87, 0.95]	w = 0.0212
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.0211
ILR, REML	0.93 [0.89, 0.96]	w = 0.0256
Profile likelihood	0.92 [0.87, 0.95]	w = 0.0222
Profile, REML	0.93 [0.89, 0.96]	w = 0.0256
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0277
Wald (delta)	0.92 [0.87, 0.95]	w = 0.0212
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.0211
ILR, REML	0.92 [0.88, 0.95]	w = 0.0255
Wald (delta)	0.92 [0.87, 0.95]	w = 0.0212
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.0212
ILR, REML	0.92 [0.88, 0.95]	w = 0.0255
Method's own interval	0.79 [0.73, 0.84]	w = 0.638
Own interval, transcript fractions	0.79 [0.73, 0.84]	w = 0.638
Fisher information (Wald)	0.2 [0.15, 0.27]	w = 0.00285
Godambe sandwich	1 [0.98, 1]	w = 0.329
Released code, as published	1 [0.98, 1]	w = 0.329



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / balanced / low sibling overlap / a = 1 / sizes 1:1:5:5

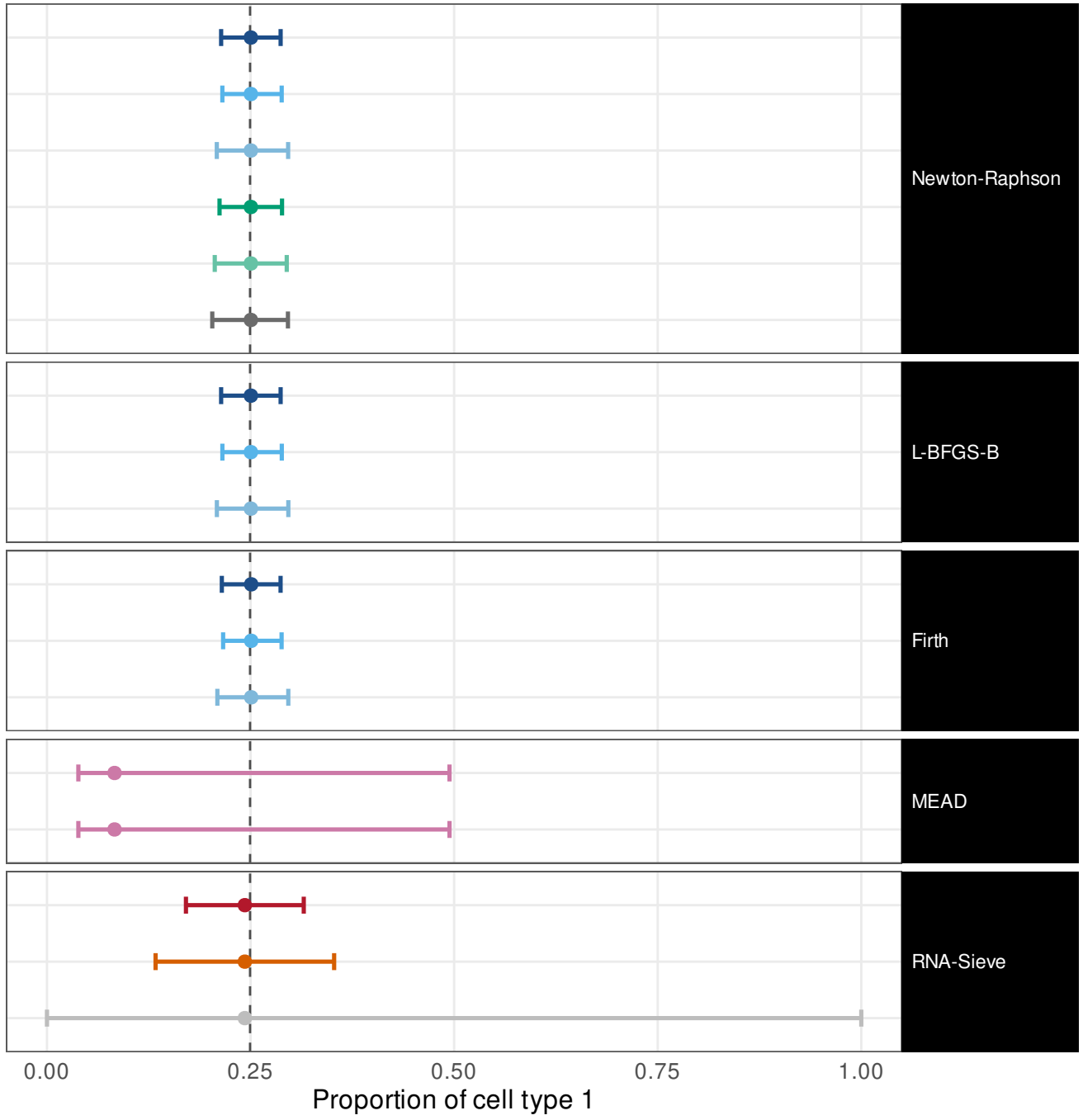
```
Wald (delta)                          0.88 [0.82, 0.91]   w = 0.0731
ILR, mapped back                      0.88 [0.83, 0.92]   w = 0.0729
ILR, REML                             0.94 [ 0.9, 0.97]   w = 0.0876
Profile likelihood                    0.88 [0.83, 0.92]   w = 0.0767
Profile, REML                         0.94 [ 0.9, 0.97]   w = 0.0884
Oracle (Fisher at p*)                    1 [0.98,    1]   w = 0.0929

Wald (delta)                          0.88 [0.82, 0.91]   w = 0.0731
ILR, mapped back                      0.88 [0.83, 0.92]   w = 0.0729
ILR, REML                             0.94 [ 0.9, 0.97]   w = 0.0877

Wald (delta)                          0.88 [0.82, 0.91]   w = 0.0721
ILR, mapped back                      0.88 [0.83, 0.92]   w = 0.0718
ILR, REML                             0.94 [ 0.9, 0.97]   w = 0.0871

Method's own interval                 0.44 [0.37, 0.51]   w = 0.456
Own interval, transcript fractions    0.91 [0.86, 0.94]   w = 0.456

Fisher information (Wald)                1 [0.98,    1]   w = 0.145
Godambe sandwich                         1 [0.98,    1]   w = 0.219
Released code, as published              1 [0.98,    1]   w =     1
```

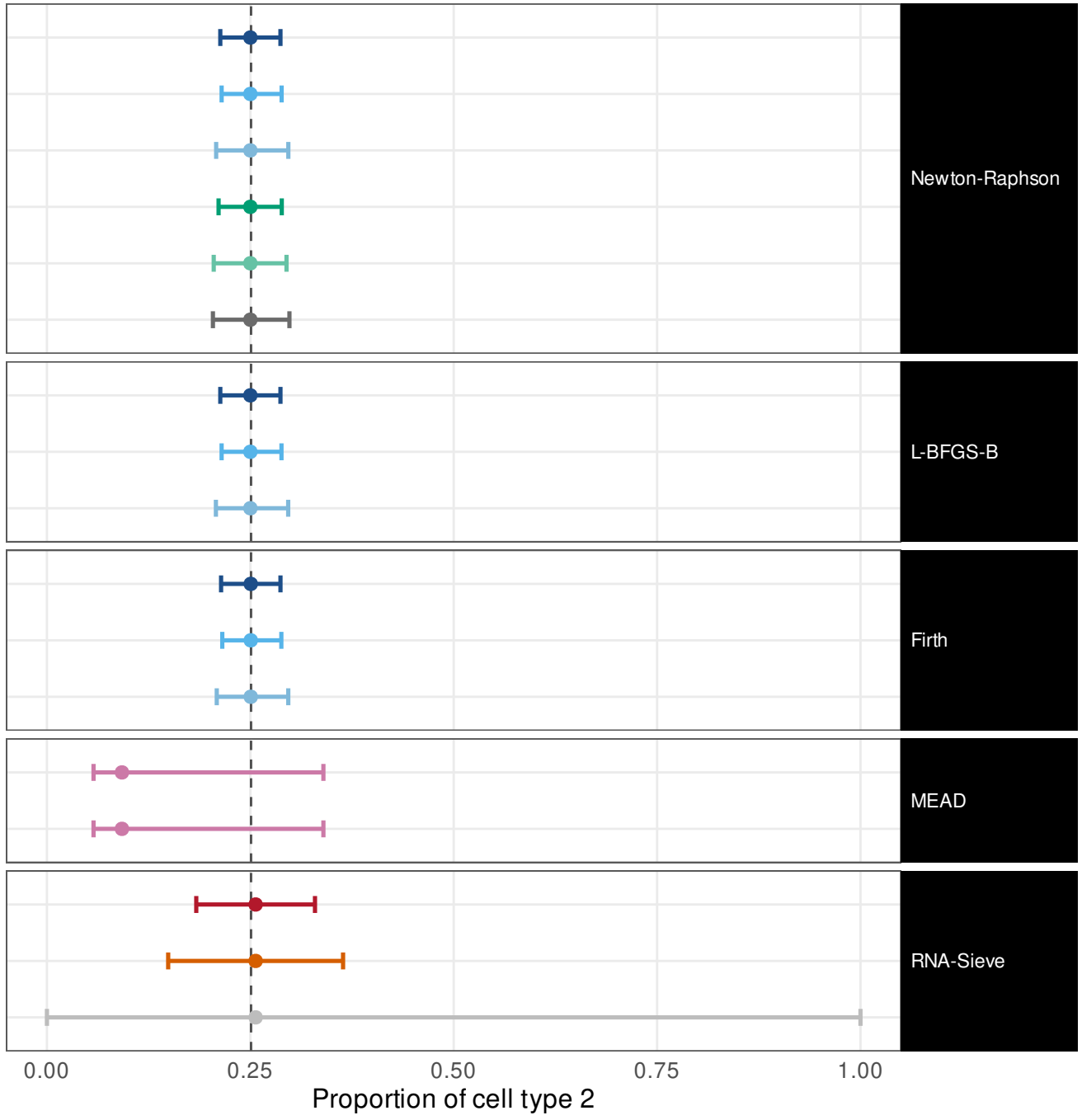


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / balanced / low sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.88 [0.83, 0.92]	w = 0.0739
ILR, mapped back	0.88 [0.83, 0.92]	w = 0.0737
ILR, REML	0.94 [0.9, 0.97]	w = 0.0887
Profile likelihood	0.9 [0.85, 0.93]	w = 0.0777
Profile, REML	0.94 [0.9, 0.97]	w = 0.0895
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.094
Wald (delta)	0.88 [0.83, 0.92]	w = 0.0739
ILR, mapped back	0.88 [0.83, 0.92]	w = 0.0736
ILR, REML	0.94 [0.9, 0.97]	w = 0.0889
Wald (delta)	0.88 [0.83, 0.92]	w = 0.073
ILR, mapped back	0.88 [0.83, 0.92]	w = 0.0726
ILR, REML	0.94 [0.9, 0.97]	w = 0.0879
Method's own interval	0.26 [0.2, 0.32]	w = 0.283
Own interval, transcript fractions	0.8 [0.73, 0.85]	w = 0.283
Fisher information (Wald)	1 [0.98, 1]	w = 0.146
Godambe sandwich	1 [0.98, 1]	w = 0.215
Released code, as published	1 [0.98, 1]	w = 1

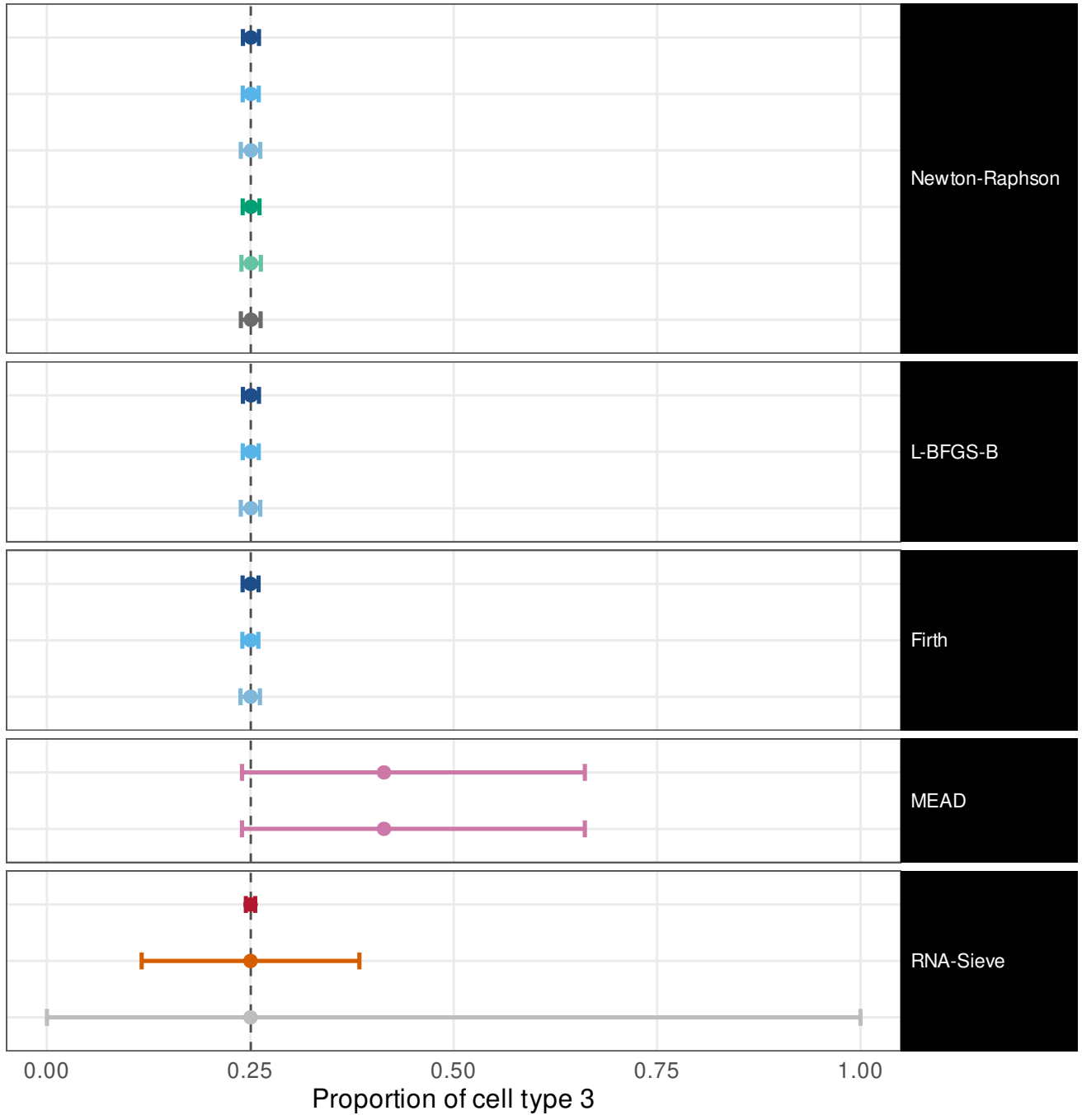


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / balanced / low sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.94 [0.89, 0.96]	w = 0.0197
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.0197
ILR, REML	0.98 [0.95, 0.99]	w = 0.0239
Profile likelihood	0.95 [0.91, 0.97]	w = 0.0207
Profile, REML	0.98 [0.95, 0.99]	w = 0.0238
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0242
Wald (delta)	0.94 [0.89, 0.96]	w = 0.0197
ILR, mapped back	0.92 [0.88, 0.95]	w = 0.0197
ILR, REML	0.98 [0.95, 0.99]	w = 0.0239
Wald (delta)	0.92 [0.88, 0.95]	w = 0.0196
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.0197
ILR, REML	0.98 [0.94, 0.99]	w = 0.0239
Method's own interval	0.41 [0.34, 0.48]	w = 0.421
Own interval, transcript fractions	0.82 [0.77, 0.87]	w = 0.421
Fisher information (Wald)	0.65 [0.58, 0.71]	w = 0.0114
Godambe sandwich	1 [0.98, 1]	w = 0.268
Released code, as published	1 [0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / balanced / low sibling overlap / a = 1 / sizes 1:1:5:5

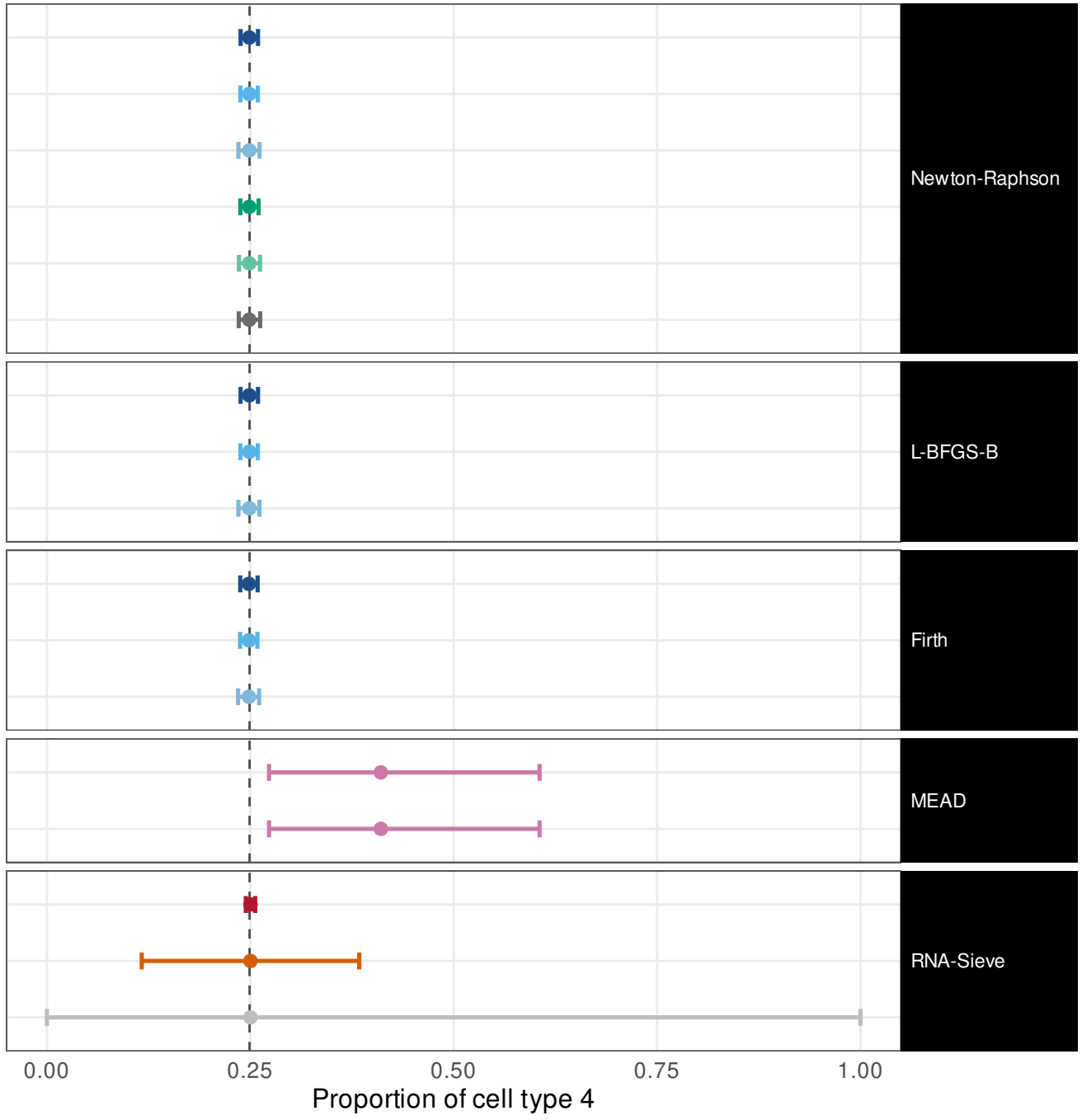
```
Wald (delta)                          0.9  [0.85, 0.93]   w = 0.0214
ILR, mapped back                      0.9  [0.85, 0.93]   w = 0.0213
ILR, REML                             0.95 [0.91, 0.97]   w = 0.026
Profile likelihood                    0.92 [0.87, 0.95]   w = 0.0224
Profile, REML                         0.94 [ 0.9, 0.97]   w = 0.0259
Oracle (Fisher at p*)                   1  [0.98,    1]   w = 0.0263

Wald (delta)                          0.9  [0.85, 0.93]   w = 0.0214
ILR, mapped back                      0.9  [0.86, 0.94]   w = 0.0214
ILR, REML                             0.94 [ 0.9, 0.97]   w = 0.0259

Wald (delta)                          0.9  [0.86, 0.94]   w = 0.0213
ILR, mapped back                      0.9  [0.84, 0.93]   w = 0.0214
ILR, REML                             0.94 [ 0.9, 0.97]   w = 0.0259

Method's own interval                 0.32 [0.26, 0.39]   w = 0.333
Own interval, transcript fractions    0.72 [0.65, 0.77]   w = 0.333

Fisher information (Wald)             0.58 [0.51, 0.65]   w = 0.0114
Godambe sandwich                        1  [0.98,    1]   w = 0.267
Released code, as published             1  [0.98,    1]   w =     1
```

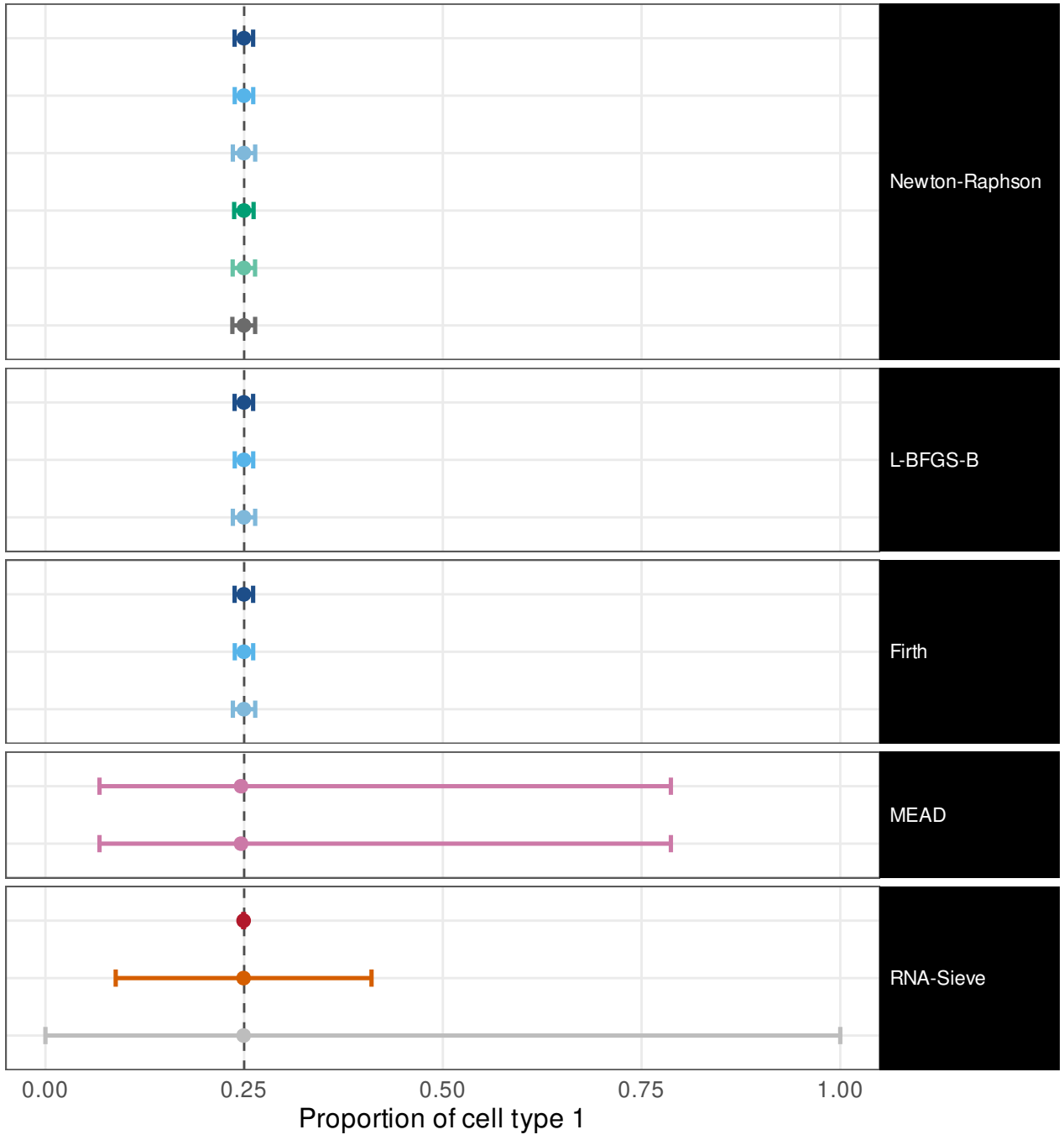


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / balanced / low sibling overlap / a = 5 / equal sizes

Wald (delta)	0.88 [0.83, 0.92]	w = 0.0233
ILR, mapped back	0.87 [0.82, 0.91]	w = 0.0232
ILR, REML	0.92 [0.87, 0.95]	w = 0.0281
Profile likelihood	0.89 [0.84, 0.93]	w = 0.0244
Profile, REML	0.92 [0.87, 0.95]	w = 0.0282
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0286
Wald (delta)	0.88 [0.83, 0.92]	w = 0.0233
ILR, mapped back	0.88 [0.83, 0.92]	w = 0.0232
ILR, REML	0.92 [0.88, 0.95]	w = 0.0281
Wald (delta)	0.88 [0.83, 0.92]	w = 0.0233
ILR, mapped back	0.87 [0.82, 0.91]	w = 0.0232
ILR, REML	0.9 [0.86, 0.94]	w = 0.0281
Method's own interval	0.88 [0.83, 0.92]	w = 0.719
Own interval, transcript fractions	0.88 [0.83, 0.92]	w = 0.719
Fisher information (Wald)	0.05 [0.027, 0.09]	w = 0.000958
Godambe sandwich	1 [0.98, 1]	w = 0.322
Released code, as published	1 [0.98, 1]	w = 1

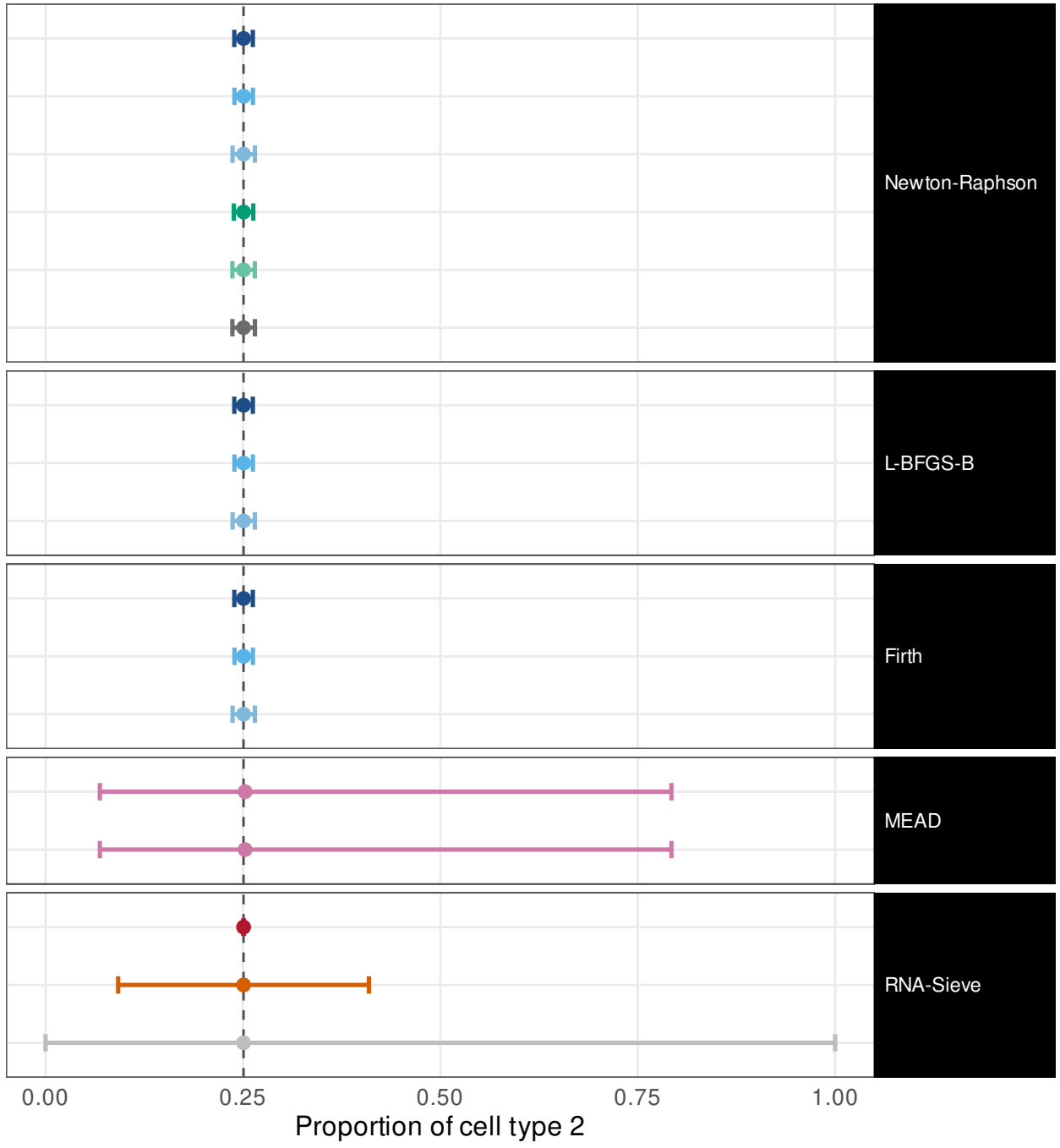


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / balanced / low sibling overlap / a = 5 / equal sizes

Wald (delta)	0.87 [0.82, 0.91]	w = 0.0235
ILR, mapped back	0.87 [0.82, 0.91]	w = 0.0234
ILR, REML	0.94 [0.9, 0.97]	w = 0.0283
Profile likelihood	0.9 [0.86, 0.94]	w = 0.0246
Profile, REML	0.93 [0.89, 0.96]	w = 0.0284
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0286
Wald (delta)	0.87 [0.82, 0.91]	w = 0.0235
ILR, mapped back	0.87 [0.82, 0.91]	w = 0.0233
ILR, REML	0.94 [0.9, 0.97]	w = 0.0283
Wald (delta)	0.87 [0.82, 0.91]	w = 0.0235
ILR, mapped back	0.86 [0.81, 0.91]	w = 0.0233
ILR, REML	0.93 [0.89, 0.96]	w = 0.0283
Method's own interval	0.9 [0.84, 0.93]	w = 0.724
Own interval, transcript fractions	0.9 [0.84, 0.93]	w = 0.724
Fisher information (Wald)	0.02 [0.0078, 0.05]	w = 0.000859
Godambe sandwich	1 [0.98, 1]	w = 0.318
Released code, as published	1 [0.98, 1]	w = 1

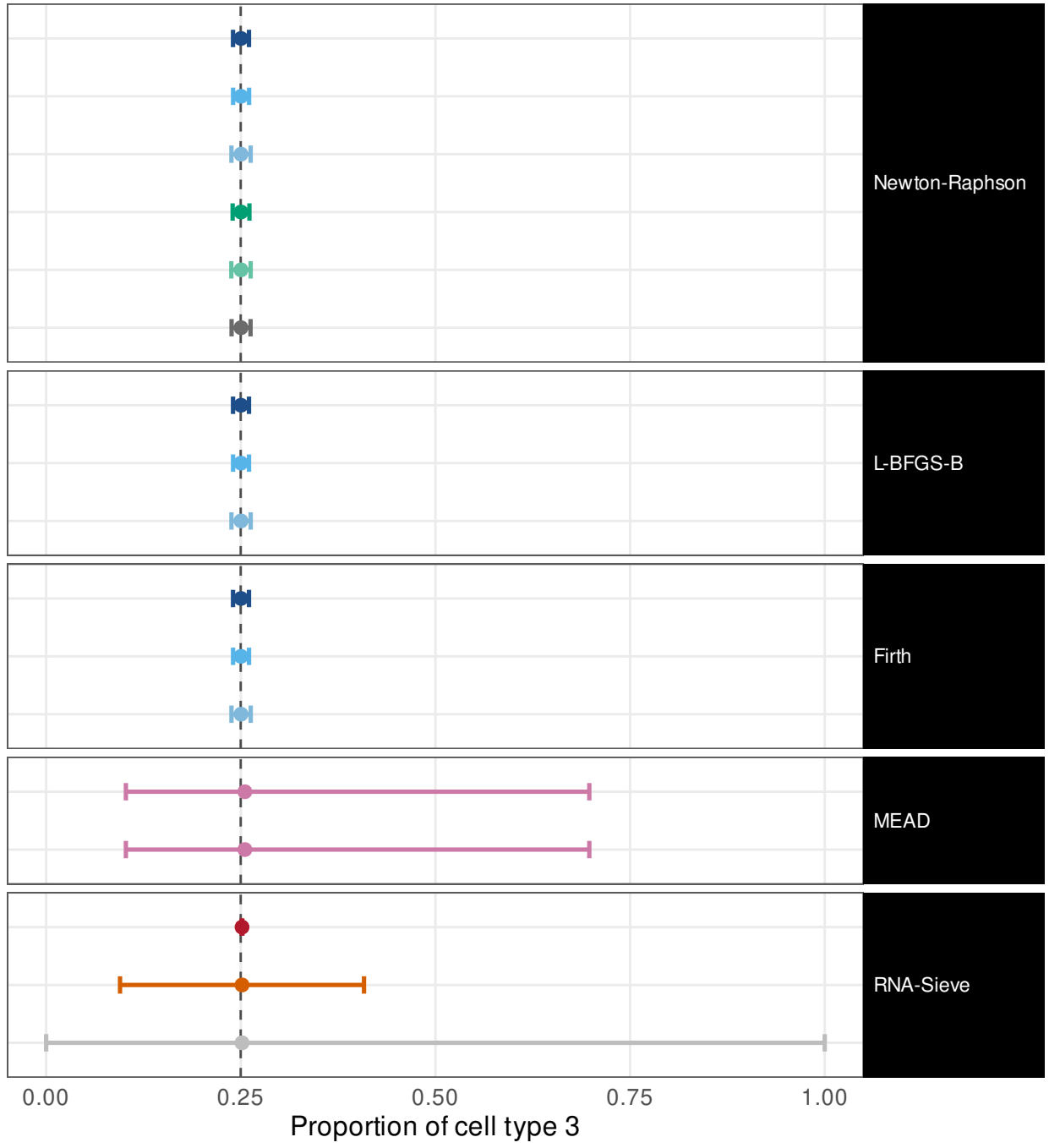


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / balanced / low sibling overlap / a = 5 / equal sizes

Wald (delta)	0.93 [0.89, 0.96]	w = 0.0205
ILR, mapped back	0.94 [0.89, 0.96]	w = 0.0205
ILR, REML	0.96 [0.93, 0.98]	w = 0.0247
Profile likelihood	0.94 [0.89, 0.96]	w = 0.0215
Profile, REML	0.97 [0.94, 0.99]	w = 0.0248
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0244
Wald (delta)	0.93 [0.89, 0.96]	w = 0.0205
ILR, mapped back	0.92 [0.88, 0.95]	w = 0.0205
ILR, REML	0.96 [0.93, 0.98]	w = 0.0247
Wald (delta)	0.93 [0.89, 0.96]	w = 0.0205
ILR, mapped back	0.94 [0.89, 0.96]	w = 0.0205
ILR, REML	0.97 [0.94, 0.99]	w = 0.0247
Method's own interval	0.78 [0.72, 0.84]	w = 0.595
Own interval, transcript fractions	0.78 [0.72, 0.84]	w = 0.595
Fisher information (Wald)	0.055 [0.031, 0.096]	w = 0.000875
Godambe sandwich	1 [0.98, 1]	w = 0.313
Released code, as published	1 [0.98, 1]	w = 1

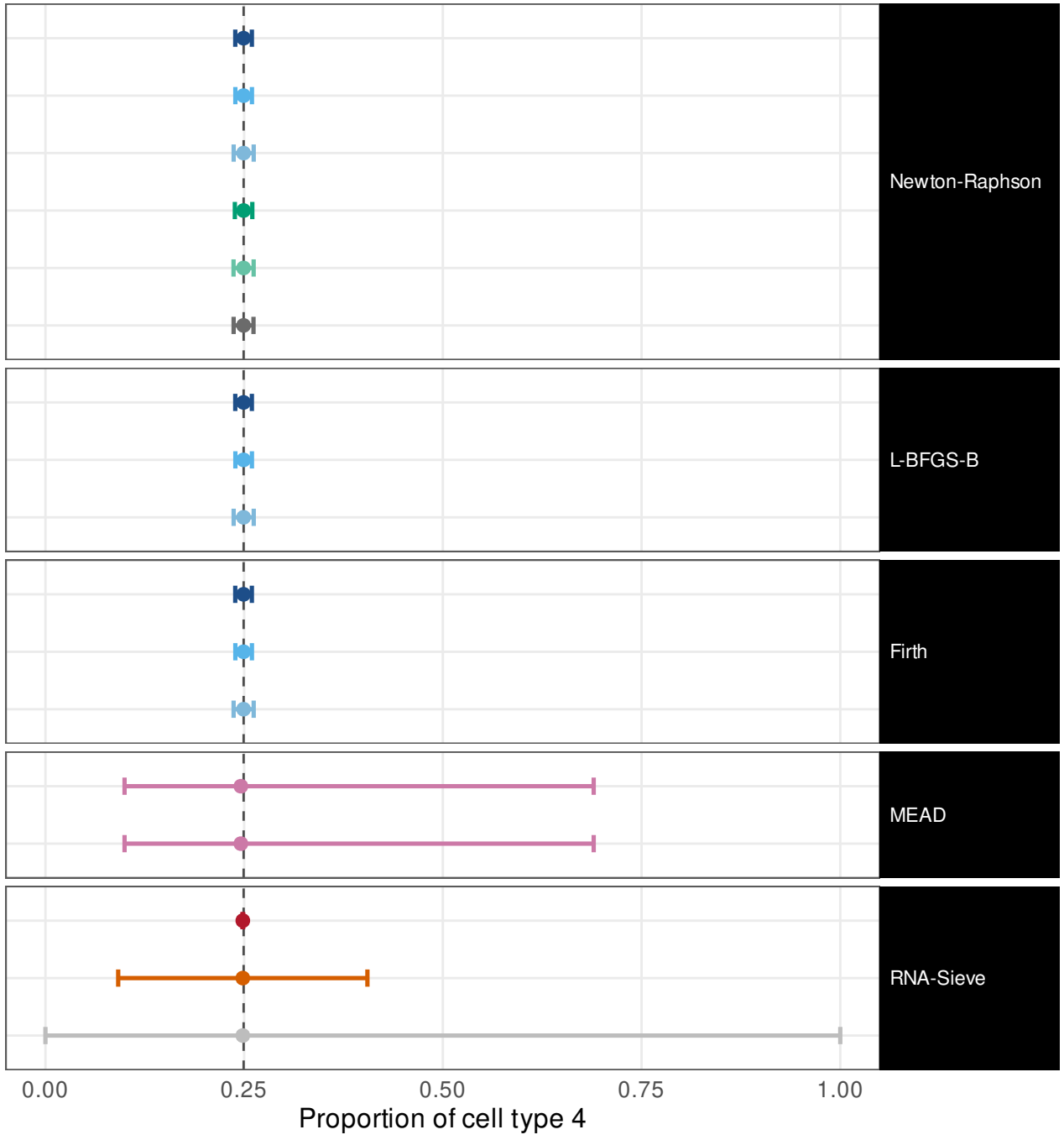


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / balanced / low sibling overlap / a = 5 / equal sizes

Wald (delta)	0.93 [0.89, 0.96]	w = 0.0209
ILR, mapped back	0.92 [0.88, 0.95]	w = 0.0209
ILR, REML	0.96 [0.92, 0.98]	w = 0.0252
Profile likelihood	0.93 [0.89, 0.96]	w = 0.022
Profile, REML	0.96 [0.93, 0.98]	w = 0.0253
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.025
Wald (delta)	0.93 [0.89, 0.96]	w = 0.0209
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.0209
ILR, REML	0.96 [0.92, 0.98]	w = 0.0253
Wald (delta)	0.93 [0.89, 0.96]	w = 0.0209
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.0209
ILR, REML	0.97 [0.94, 0.99]	w = 0.0253
Method's own interval	0.83 [0.77, 0.88]	w = 0.59
Own interval, transcript fractions	0.83 [0.77, 0.88]	w = 0.59
Fisher information (Wald)	0.06 [0.035, 0.1]	w = 0.000901
Godambe sandwich	1 [0.98, 1]	w = 0.314
Released code, as published	1 [0.98, 1]	w = 1

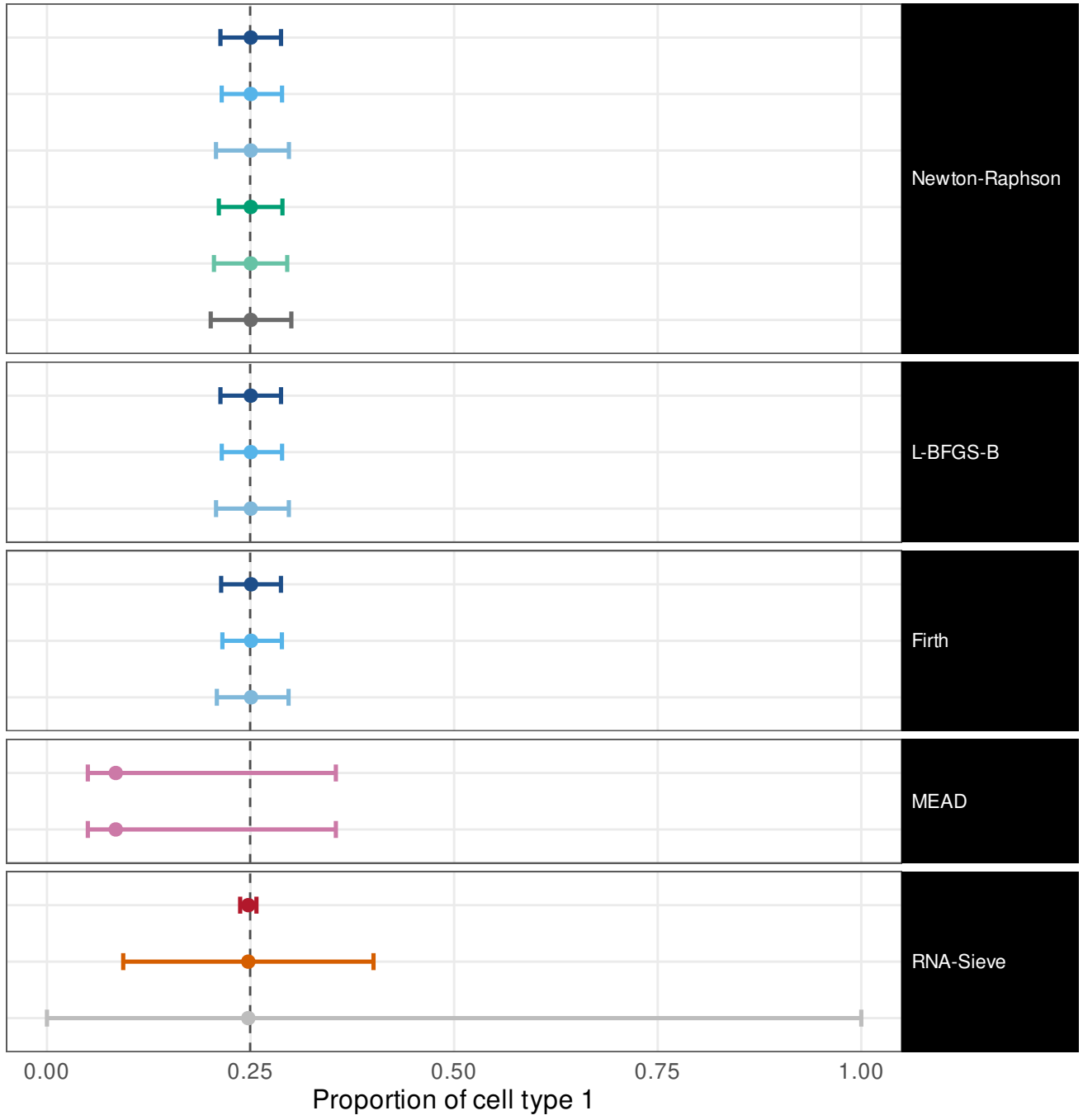


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / balanced / low sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.92 [0.88, 0.95]	w = 0.0744
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.0741
ILR, REML	0.99 [0.96, 1]	w = 0.0895
Profile likelihood	0.94 [0.9, 0.97]	w = 0.0781
Profile, REML	0.98 [0.94, 0.99]	w = 0.09
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.099
Wald (delta)	0.92 [0.88, 0.95]	w = 0.0744
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.074
ILR, REML	0.98 [0.95, 0.99]	w = 0.0894
Wald (delta)	0.92 [0.88, 0.95]	w = 0.0734
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.0731
ILR, REML	0.98 [0.96, 0.99]	w = 0.0881
Method's own interval	0.28 [0.23, 0.35]	w = 0.305
Own interval, transcript fractions	0.9 [0.84, 0.93]	w = 0.305
Fisher information (Wald)	0.31 [0.25, 0.38]	w = 0.02
Godambe sandwich	1 [0.98, 1]	w = 0.307
Released code, as published	1 [0.98, 1]	w = 1

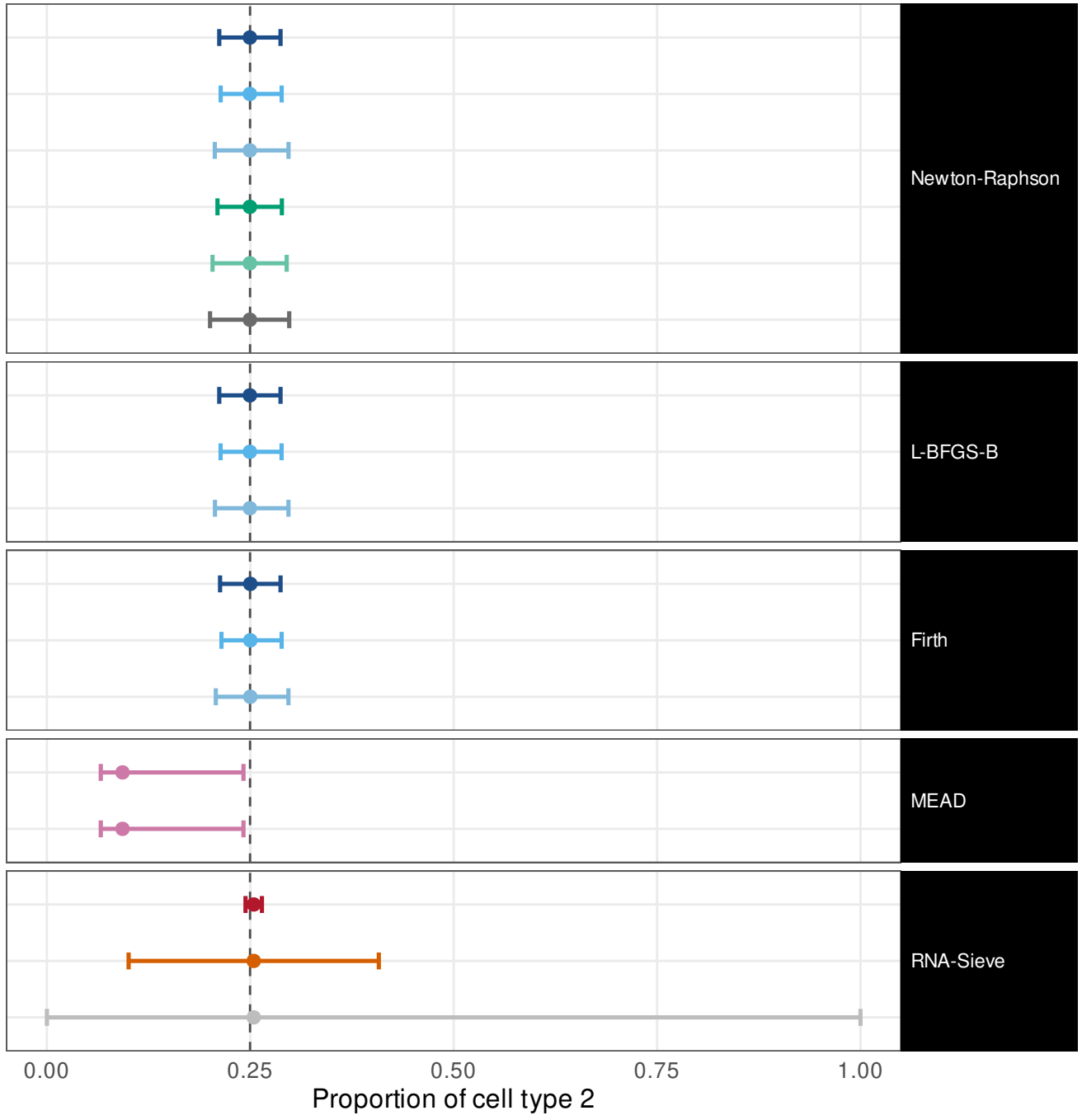


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / balanced / low sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.9	[0.85, 0.93]	w = 0.0753
ILR, mapped back	0.9	[0.86, 0.94]	w = 0.0749
ILR, REML	0.96	[0.92, 0.98]	w = 0.0906
Profile likelihood	0.92	[0.88, 0.95]	w = 0.0791
Profile, REML	0.96	[0.92, 0.98]	w = 0.0911
Oracle (Fisher at p*)	1	[0.98, 1]	w = 0.0973
Wald (delta)	0.9	[0.85, 0.93]	w = 0.0753
ILR, mapped back	0.9	[0.86, 0.94]	w = 0.075
ILR, REML	0.96	[0.92, 0.98]	w = 0.0903
Wald (delta)	0.91	[0.86, 0.94]	w = 0.0743
ILR, mapped back	0.91	[0.86, 0.94]	w = 0.074
ILR, REML	0.96	[0.92, 0.98]	w = 0.0893
Method's own interval	0.15	[0.11, 0.21]	w = 0.175
Own interval, transcript fractions	0.74	[0.67, 0.79]	w = 0.175
Fisher information (Wald)	0.3	[0.25, 0.37]	w = 0.0202
Godambe sandwich	1	[0.98, 1]	w = 0.308
Released code, as published	1	[0.98, 1]	w = 1

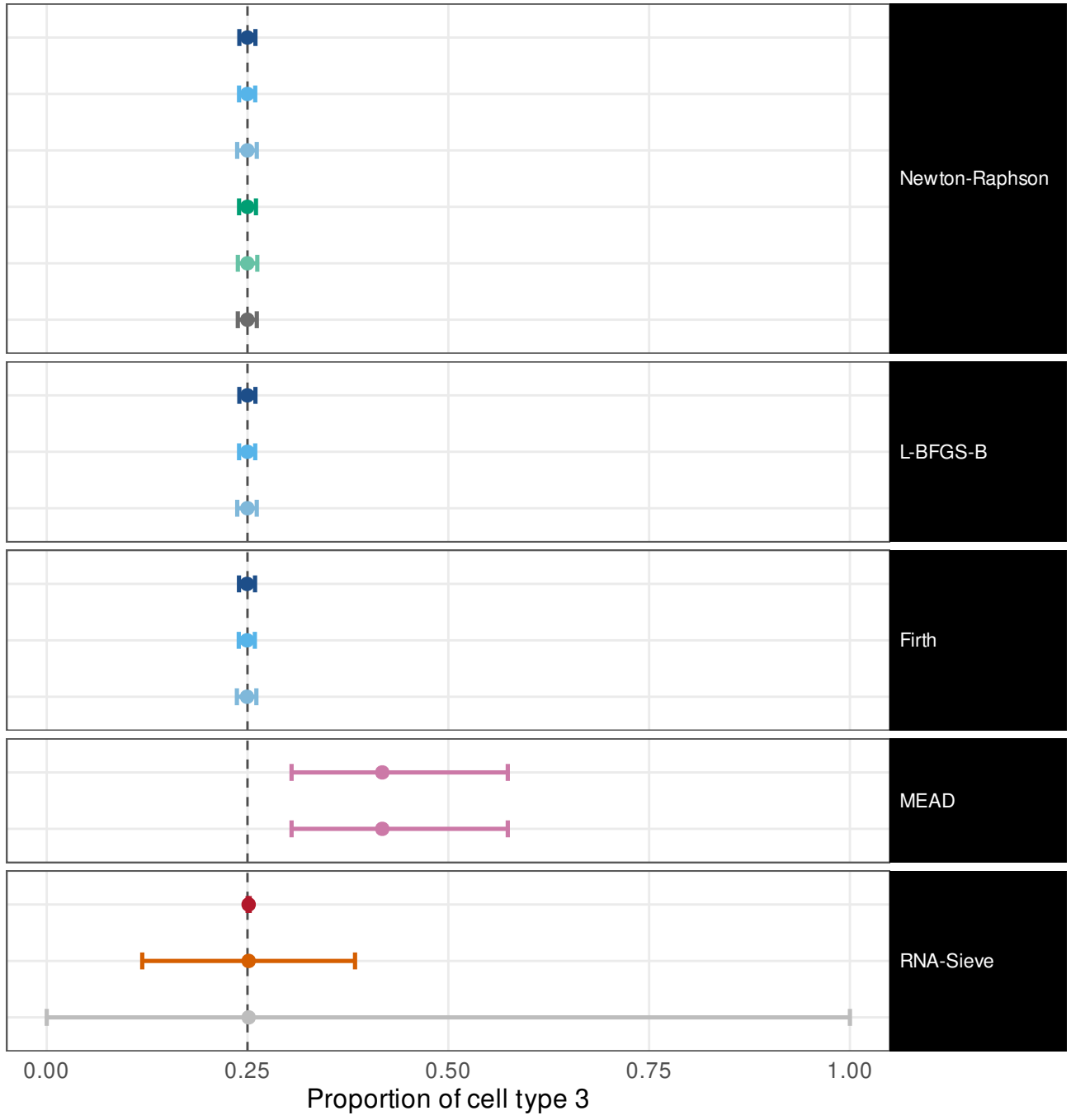


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / balanced / low sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.9 [0.86, 0.94]	w = 0.02
ILR, mapped back	0.91 [0.86, 0.94]	w = 0.0201
ILR, REML	0.94 [0.9, 0.97]	w = 0.0244
Profile likelihood	0.92 [0.87, 0.95]	w = 0.021
Profile, REML	0.94 [0.89, 0.96]	w = 0.0242
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0237
Wald (delta)	0.9 [0.86, 0.94]	w = 0.02
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.0201
ILR, REML	0.94 [0.9, 0.97]	w = 0.0243
Wald (delta)	0.92 [0.87, 0.95]	w = 0.02
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.02
ILR, REML	0.94 [0.9, 0.97]	w = 0.0243
Method's own interval	0.26 [0.2, 0.32]	w = 0.269
Own interval, transcript fractions	0.8 [0.73, 0.85]	w = 0.269
Fisher information (Wald)	0.16 [0.11, 0.21]	w = 0.00257
Godambe sandwich	1 [0.98, 1]	w = 0.265
Released code, as published	1 [0.98, 1]	w = 1

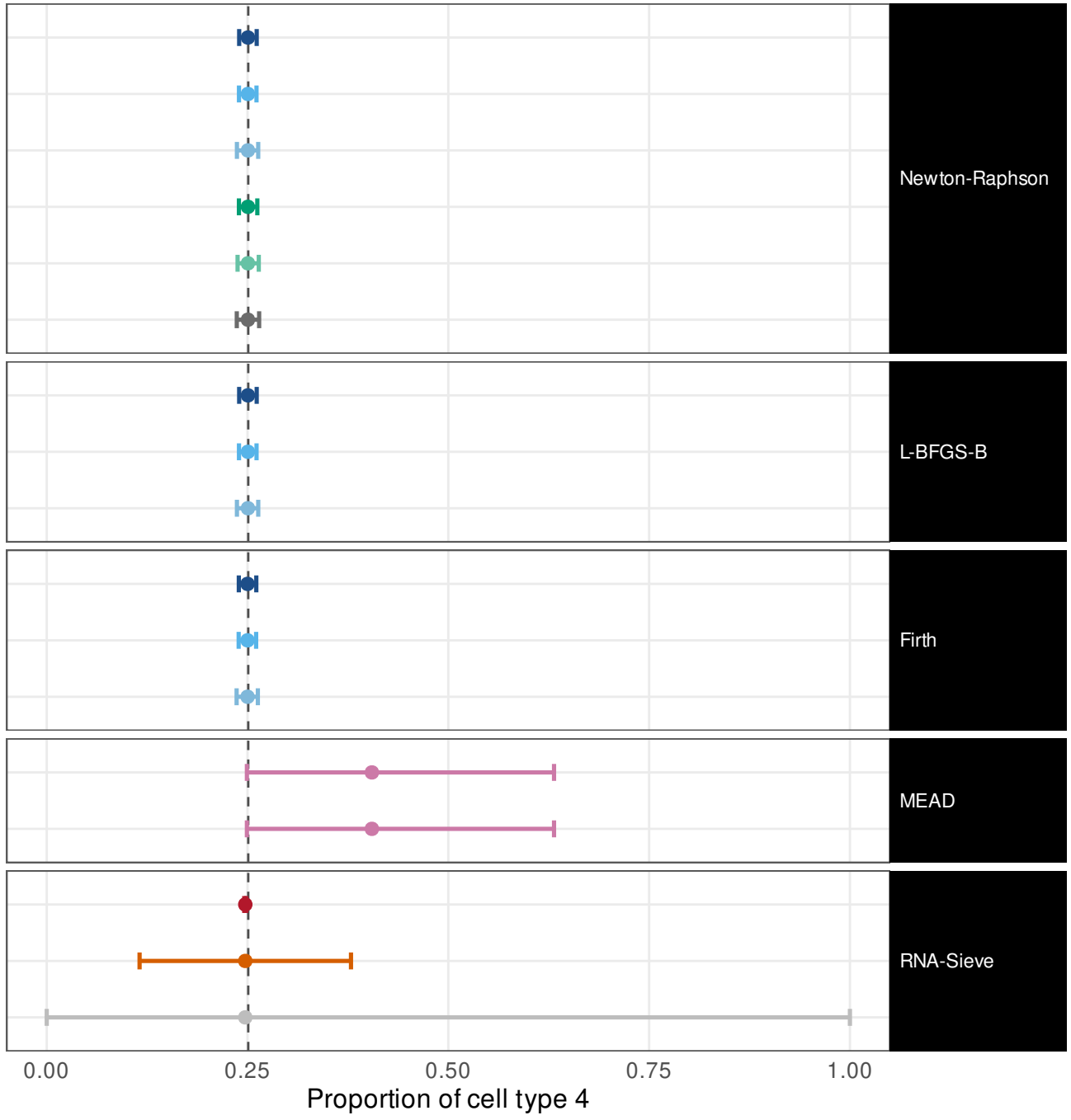


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / balanced / low sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.93 [0.89, 0.96]	w = 0.0218
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.0218
ILR, REML	0.98 [0.94, 0.99]	w = 0.0265
Profile likelihood	0.94 [0.9, 0.97]	w = 0.0229
Profile, REML	0.96 [0.93, 0.98]	w = 0.0264
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0278
Wald (delta)	0.93 [0.89, 0.96]	w = 0.0218
ILR, mapped back	0.94 [0.89, 0.96]	w = 0.0219
ILR, REML	0.98 [0.94, 0.99]	w = 0.0265
Wald (delta)	0.93 [0.89, 0.96]	w = 0.0218
ILR, mapped back	0.92 [0.88, 0.95]	w = 0.0217
ILR, REML	0.97 [0.94, 0.99]	w = 0.0264
Method's own interval	0.37 [0.31, 0.44]	w = 0.382
Own interval, transcript fractions	0.64 [0.57, 0.7]	w = 0.382
Fisher information (Wald)	0.14 [0.1, 0.2]	w = 0.00253
Godambe sandwich	1 [0.98, 1]	w = 0.263
Released code, as published	1 [0.98, 1]	w = 1

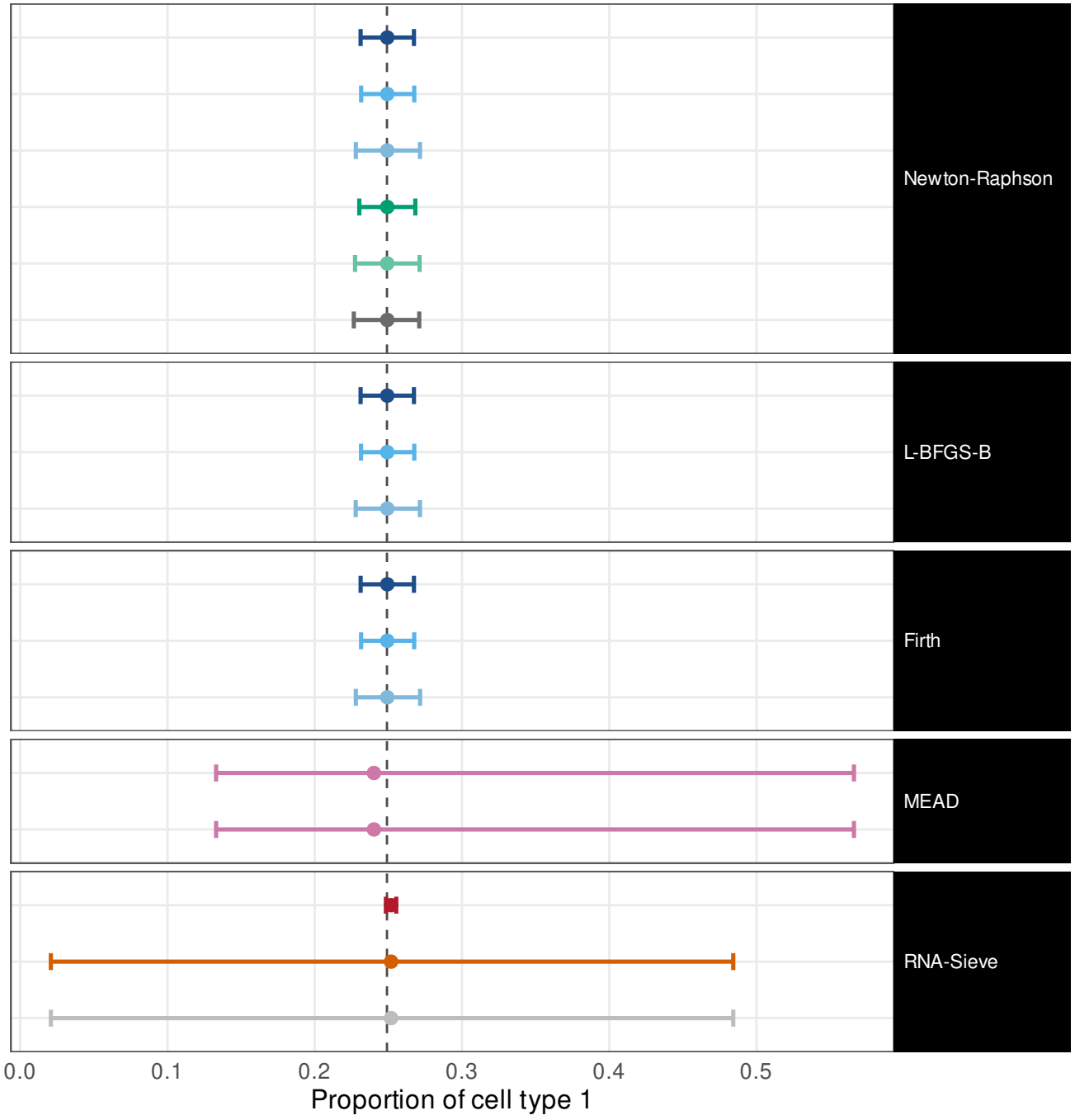


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / balanced / high sibling overlap / a = 1 / equal sizes

Wald (delta)	0.89 [0.84, 0.93]	w = 0.0362
ILR, mapped back	0.88 [0.83, 0.92]	w = 0.0361
ILR, REML	0.92 [0.88, 0.95]	w = 0.0435
Profile likelihood	0.9 [0.85, 0.93]	w = 0.038
Profile, REML	0.93 [0.89, 0.96]	w = 0.0438
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0444
Wald (delta)	0.89 [0.84, 0.93]	w = 0.0362
ILR, mapped back	0.89 [0.84, 0.93]	w = 0.0362
ILR, REML	0.92 [0.87, 0.95]	w = 0.0436
Wald (delta)	0.89 [0.84, 0.93]	w = 0.0362
ILR, mapped back	0.89 [0.84, 0.93]	w = 0.0361
ILR, REML	0.93 [0.89, 0.96]	w = 0.0436
Method's own interval	0.78 [0.72, 0.83]	w = 0.433
Own interval, transcript fractions	0.78 [0.72, 0.83]	w = 0.433
Fisher information (Wald)	0.26 [0.21, 0.33]	w = 0.00699
Godambe sandwich	1 [0.98, 1]	w = 0.463
Released code, as published	1 [0.98, 1]	w = 0.463

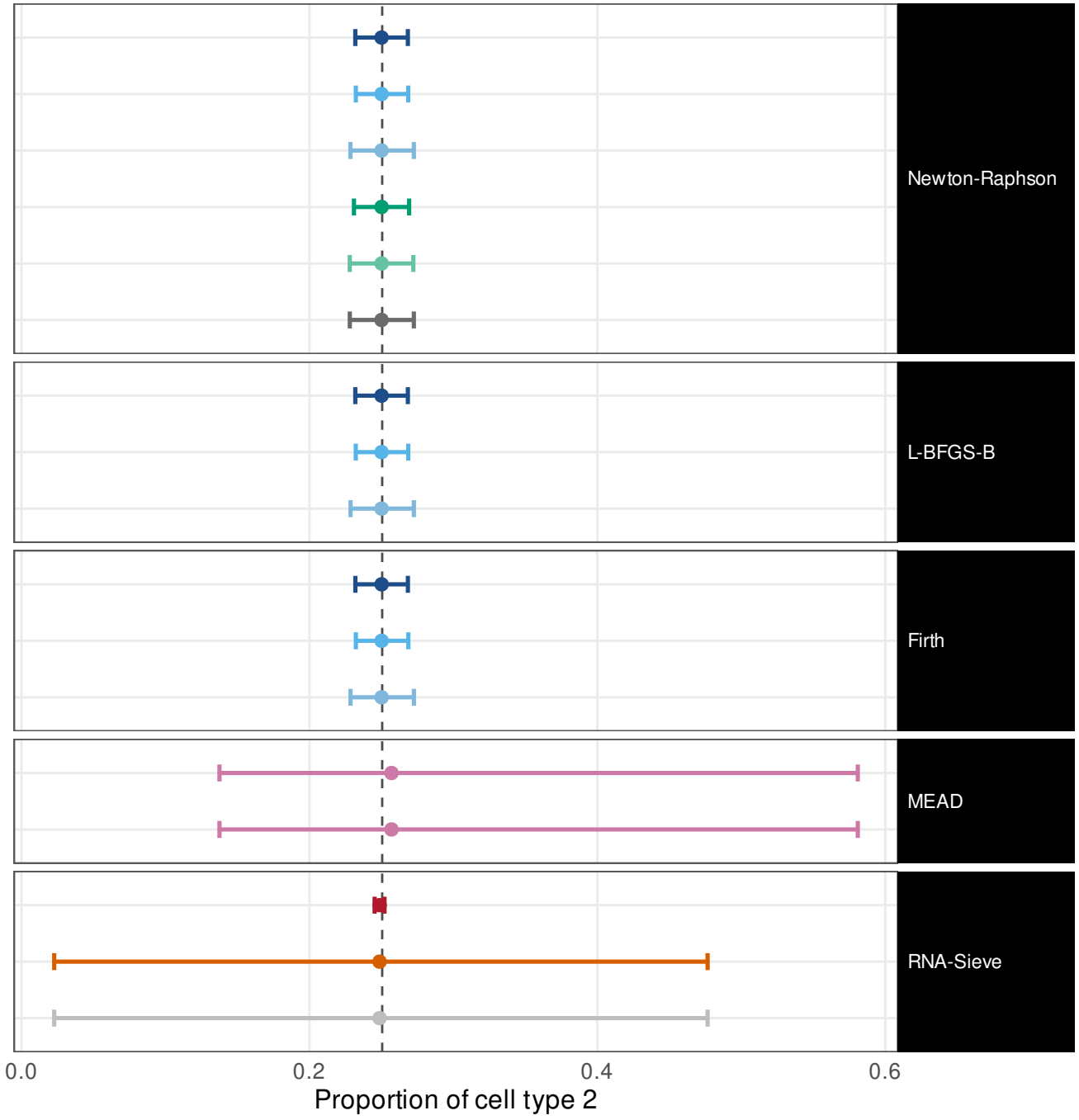


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / balanced / high sibling overlap / a = 1 / equal sizes

Wald (delta)	0.89 [0.84, 0.93]	w = 0.0365
ILR, mapped back	0.9 [0.85, 0.93]	w = 0.0364
ILR, REML	0.94 [0.89, 0.96]	w = 0.0441
Profile likelihood	0.9 [0.84, 0.93]	w = 0.0383
Profile, REML	0.94 [0.9, 0.97]	w = 0.0442
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0445
Wald (delta)	0.89 [0.84, 0.93]	w = 0.0365
ILR, mapped back	0.9 [0.84, 0.93]	w = 0.0365
ILR, REML	0.95 [0.91, 0.97]	w = 0.044
Wald (delta)	0.89 [0.84, 0.93]	w = 0.0365
ILR, mapped back	0.9 [0.85, 0.93]	w = 0.0364
ILR, REML	0.94 [0.9, 0.97]	w = 0.044
Method's own interval	0.86 [0.81, 0.9]	w = 0.443
Own interval, transcript fractions	0.86 [0.81, 0.9]	w = 0.443
Fisher information (Wald)	0.21 [0.16, 0.27]	w = 0.00684
Godambe sandwich	1 [0.98, 1]	w = 0.454
Released code, as published	1 [0.98, 1]	w = 0.454

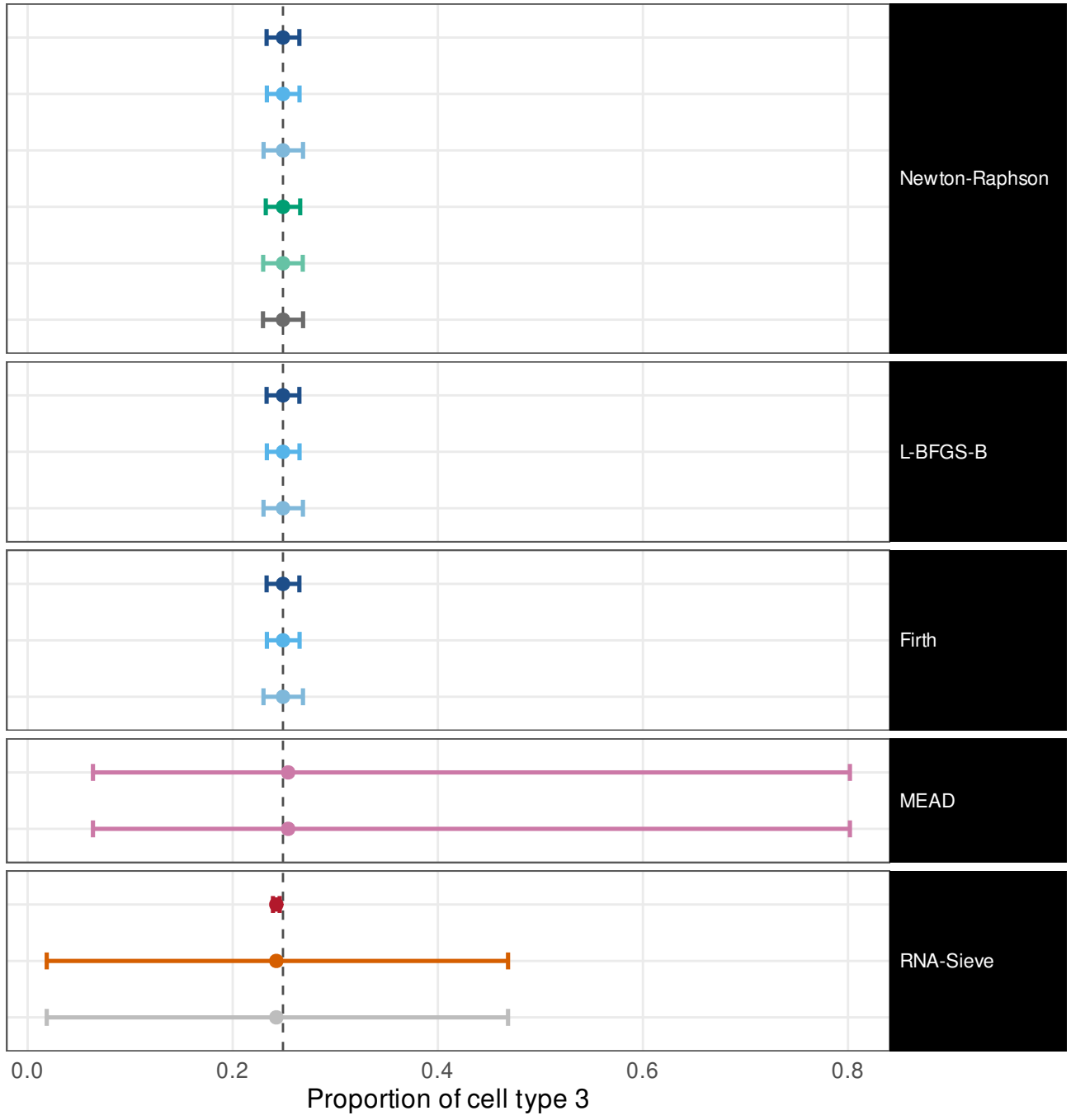


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / balanced / high sibling overlap / a = 1 / equal sizes

Wald (delta)	0.9	[0.84, 0.93]	w = 0.0319
ILR, mapped back	0.9	[0.84, 0.93]	w = 0.0318
ILR, REML	0.94	[0.89, 0.96]	w = 0.0385
Profile likelihood	0.9	[0.86, 0.94]	w = 0.0336
Profile, REML	0.94	[0.9, 0.97]	w = 0.0386
Oracle (Fisher at p*)	1	[0.98, 1]	w = 0.0391
Wald (delta)	0.9	[0.84, 0.93]	w = 0.0319
ILR, mapped back	0.9	[0.84, 0.93]	w = 0.0319
ILR, REML	0.94	[0.89, 0.96]	w = 0.0385
Wald (delta)	0.9	[0.85, 0.93]	w = 0.032
ILR, mapped back	0.9	[0.85, 0.93]	w = 0.0319
ILR, REML	0.94	[0.9, 0.97]	w = 0.0385
Method's own interval	0.93	[0.89, 0.96]	w = 0.738
Own interval, transcript fractions	0.93	[0.89, 0.96]	w = 0.738
Fisher information (Wald)	0.19	[0.14, 0.25]	w = 0.00653
Godambe sandwich	1	[0.98, 1]	w = 0.45
Released code, as published	1	[0.98, 1]	w = 0.45

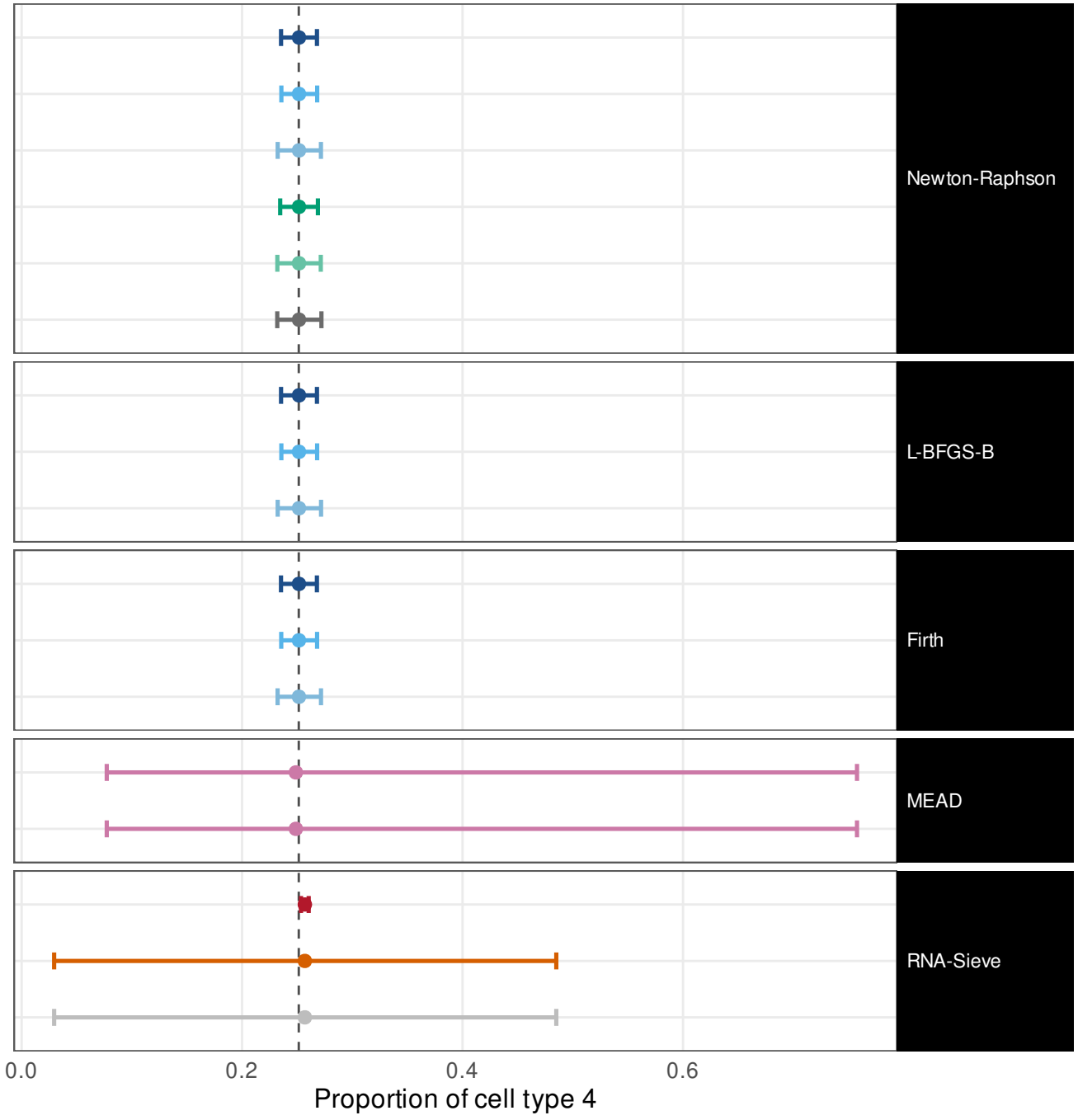


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / balanced / high sibling overlap / a = 1 / equal sizes

Wald (delta)	0.88 [0.83, 0.92]	w = 0.0326
ILR, mapped back	0.88 [0.82, 0.91]	w = 0.0325
ILR, REML	0.94 [0.9, 0.97]	w = 0.0392
Profile likelihood	0.9 [0.84, 0.93]	w = 0.0342
Profile, REML	0.95 [0.91, 0.97]	w = 0.0394
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0401
Wald (delta)	0.88 [0.83, 0.92]	w = 0.0326
ILR, mapped back	0.88 [0.83, 0.92]	w = 0.0325
ILR, REML	0.95 [0.91, 0.97]	w = 0.0393
Wald (delta)	0.88 [0.83, 0.92]	w = 0.0326
ILR, mapped back	0.88 [0.83, 0.92]	w = 0.0325
ILR, REML	0.95 [0.91, 0.97]	w = 0.0394
Method's own interval	0.94 [0.9, 0.97]	w = 0.681
Own interval, transcript fractions	0.94 [0.9, 0.97]	w = 0.681
Fisher information (Wald)	0.2 [0.15, 0.27]	w = 0.00697
Godambe sandwich	1 [0.98, 1]	w = 0.456
Released code, as published	1 [0.98, 1]	w = 0.456

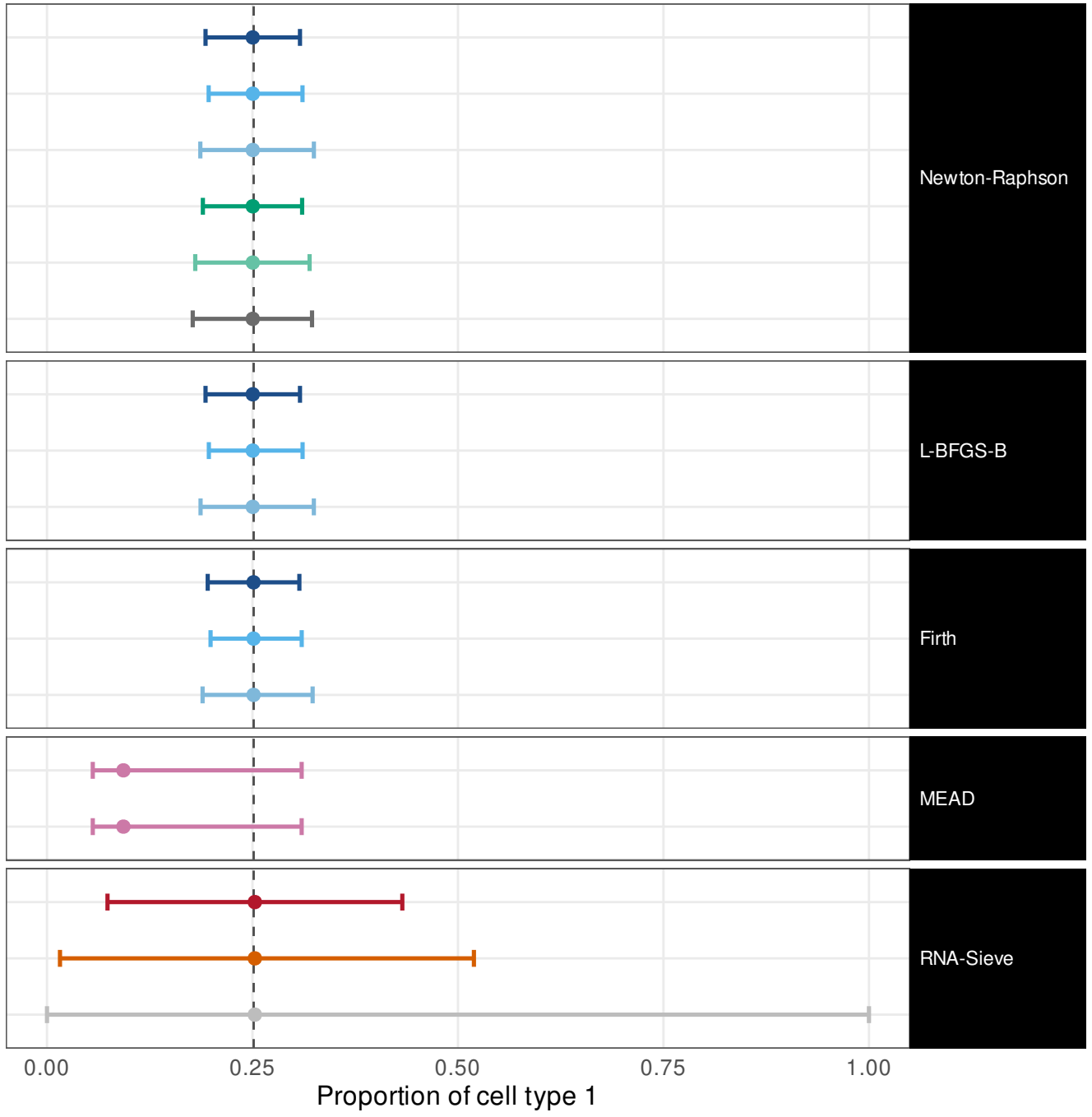


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / balanced / high sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.94 [0.89, 0.96]	w = 0.115
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.114
ILR, REML	0.98 [0.95, 0.99]	w = 0.138
Profile likelihood	0.94 [0.9, 0.97]	w = 0.121
Profile, REML	0.96 [0.93, 0.98]	w = 0.139
Oracle (Fisher at p*)	0.99 [0.96, 1]	w = 0.145
Wald (delta)	0.94 [0.89, 0.96]	w = 0.115
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.114
ILR, REML	0.98 [0.94, 0.99]	w = 0.138
Wald (delta)	0.93 [0.89, 0.96]	w = 0.112
ILR, mapped back	0.92 [0.88, 0.95]	w = 0.111
ILR, REML	0.97 [0.94, 0.99]	w = 0.134
Method's own interval	0.22 [0.17, 0.28]	w = 0.254
Own interval, transcript fractions	0.88 [0.83, 0.92]	w = 0.254
Fisher information (Wald)	1 [0.98, 1]	w = 0.359
Godambe sandwich	1 [0.98, 1]	w = 0.504
Released code, as published	1 [0.98, 1]	w = 1

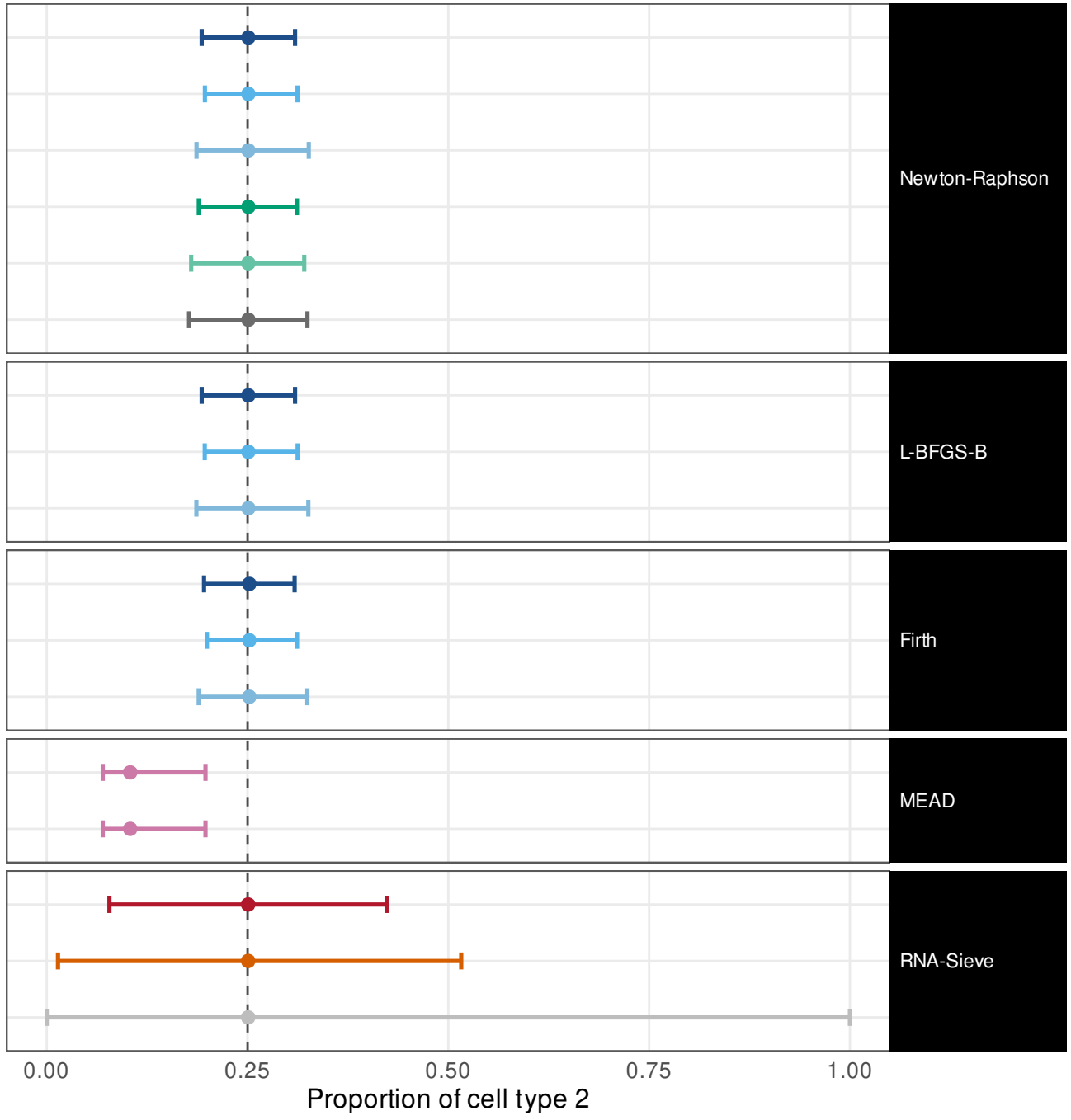


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / balanced / high sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.94 [0.9, 0.97]	w = 0.116
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.115
ILR, REML	0.97 [0.94, 0.99]	w = 0.14
Profile likelihood	0.94 [0.9, 0.97]	w = 0.122
Profile, REML	0.97 [0.94, 0.99]	w = 0.141
Oracle (Fisher at p*)	0.99 [0.97, 1]	w = 0.147
Wald (delta)	0.94 [0.9, 0.97]	w = 0.116
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.116
ILR, REML	0.97 [0.94, 0.99]	w = 0.139
Wald (delta)	0.94 [0.89, 0.96]	w = 0.113
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.112
ILR, REML	0.97 [0.94, 0.99]	w = 0.135
Method's own interval	0.075 [0.046, 0.12]	w = 0.128
Own interval, transcript fractions	0.71 [0.64, 0.77]	w = 0.128
Fisher information (Wald)	1 [0.98, 1]	w = 0.346
Godambe sandwich	1 [0.98, 1]	w = 0.502
Released code, as published	1 [0.98, 1]	w = 1

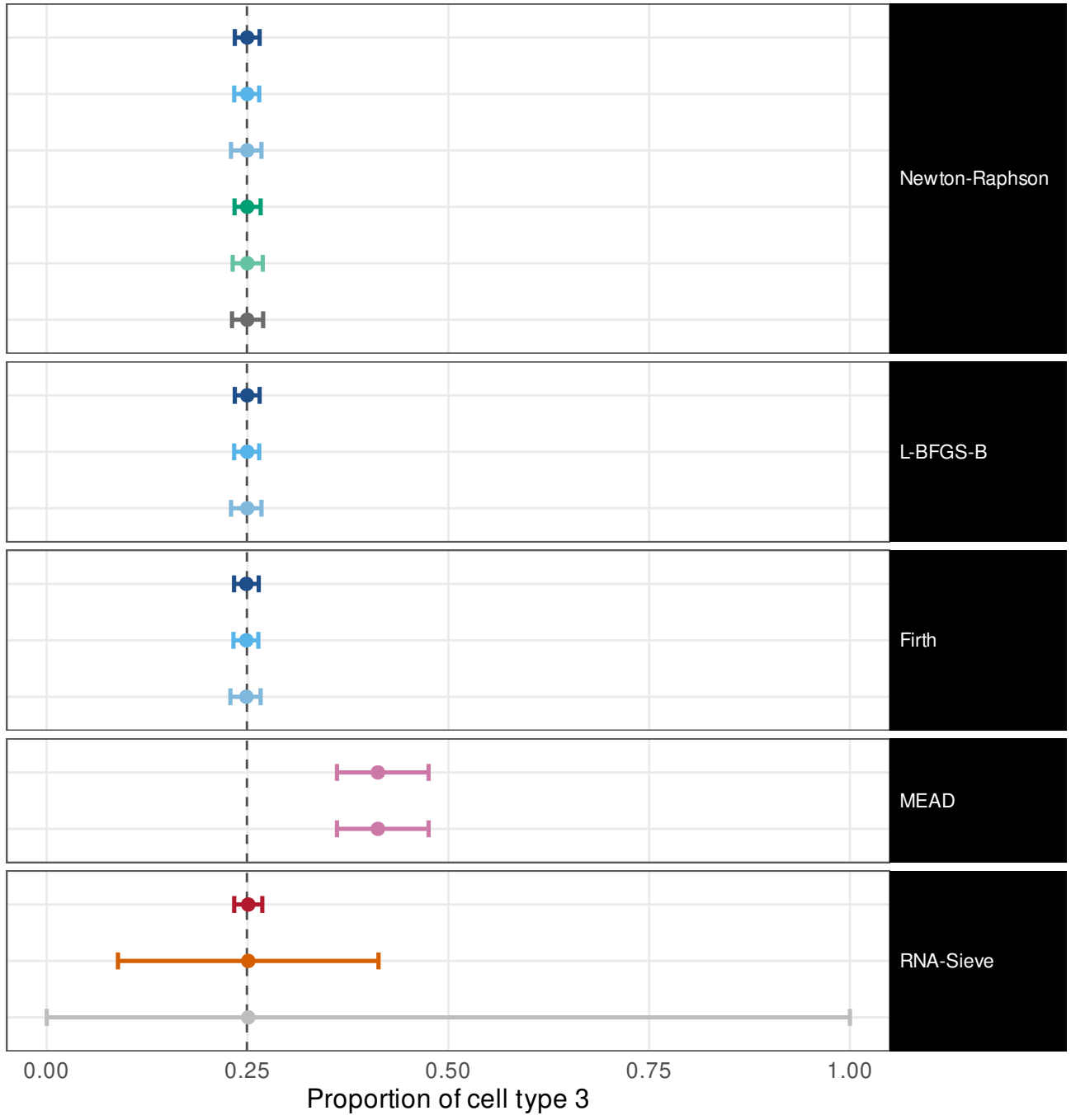


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / balanced / high sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.88 [0.83, 0.92]	w = 0.0309
ILR, mapped back	0.88 [0.83, 0.92]	w = 0.0311
ILR, REML	0.94 [0.89, 0.96]	w = 0.0379
Profile likelihood	0.9 [0.86, 0.94]	w = 0.0325
Profile, REML	0.93 [0.89, 0.96]	w = 0.0374
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0387
Wald (delta)	0.88 [0.83, 0.92]	w = 0.0309
ILR, mapped back	0.89 [0.84, 0.93]	w = 0.0312
ILR, REML	0.94 [0.89, 0.96]	w = 0.0379
Wald (delta)	0.88 [0.83, 0.92]	w = 0.0307
ILR, mapped back	0.88 [0.82, 0.91]	w = 0.031
ILR, REML	0.94 [0.89, 0.96]	w = 0.0376
Method's own interval	0.065 [0.038, 0.11]	w = 0.114
Own interval, transcript fractions	0.9 [0.85, 0.93]	w = 0.114
Fisher information (Wald)	0.91 [0.86, 0.94]	w = 0.035
Godambe sandwich	1 [0.98, 1]	w = 0.324
Released code, as published	1 [0.98, 1]	w = 1

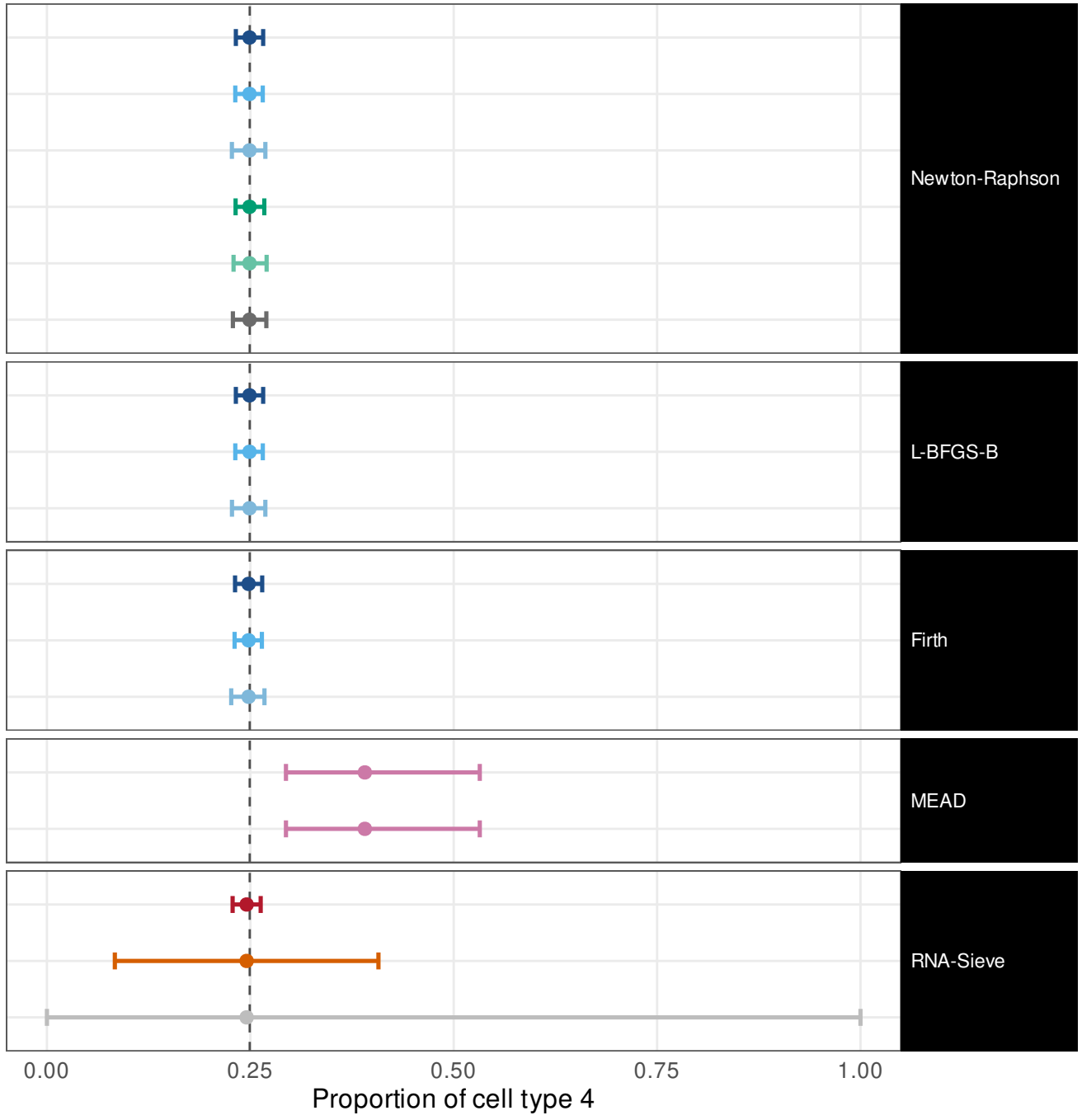


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / balanced / high sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.93 [0.89, 0.96]	w = 0.0336
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.0338
ILR, REML	0.96 [0.92, 0.98]	w = 0.0411
Profile likelihood	0.94 [0.9, 0.97]	w = 0.0353
Profile, REML	0.96 [0.92, 0.98]	w = 0.0407
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0411
Wald (delta)	0.93 [0.89, 0.96]	w = 0.0336
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.0338
ILR, REML	0.96 [0.92, 0.98]	w = 0.041
Wald (delta)	0.92 [0.87, 0.95]	w = 0.0334
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.0335
ILR, REML	0.96 [0.92, 0.98]	w = 0.0409
Method's own interval	0.2 [0.15, 0.26]	w = 0.238
Own interval, transcript fractions	0.6 [0.54, 0.67]	w = 0.238
Fisher information (Wald)	0.88 [0.82, 0.91]	w = 0.0345
Godambe sandwich	1 [0.98, 1]	w = 0.324
Released code, as published	1 [0.98, 1]	w = 1

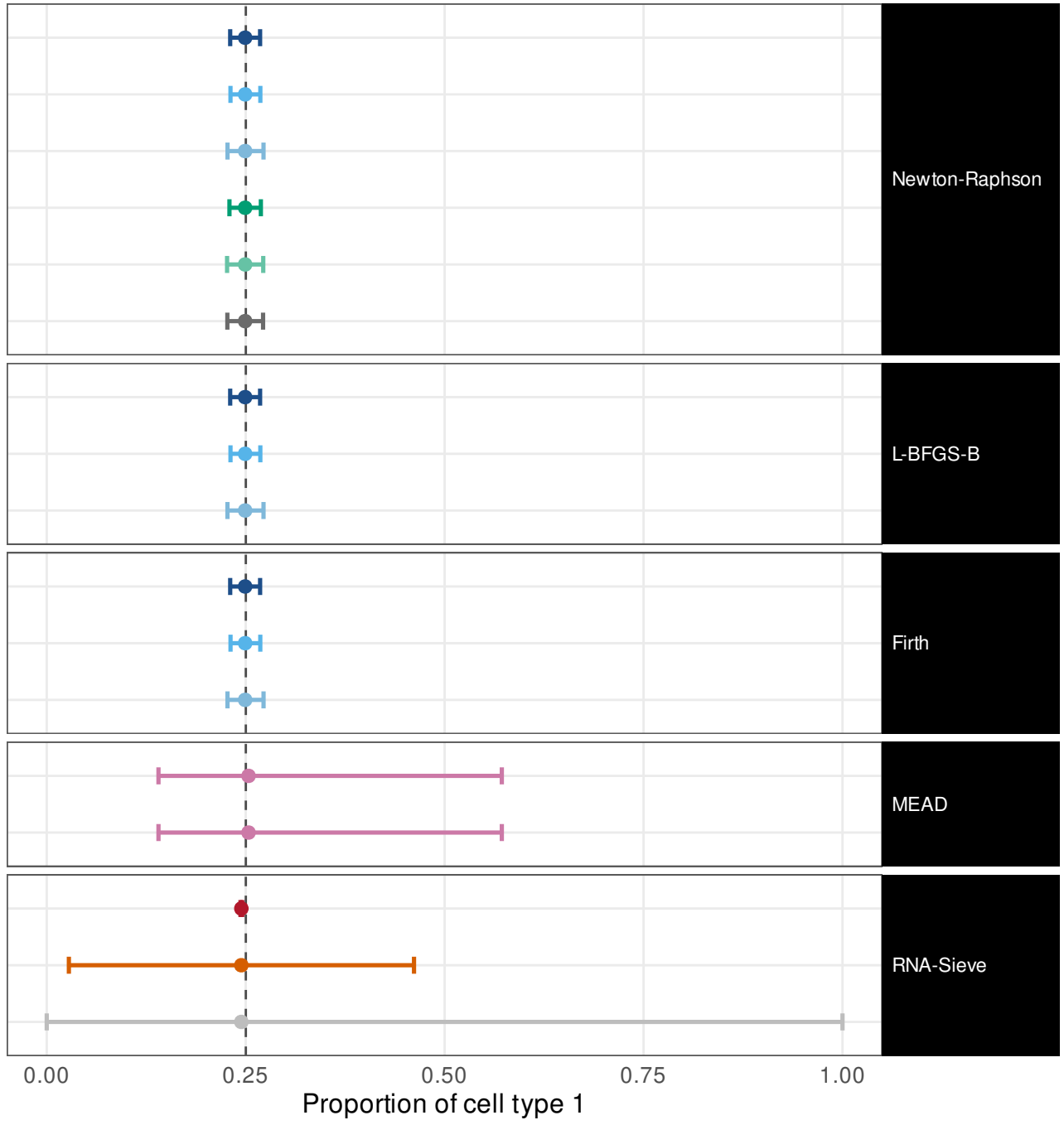


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / balanced / high sibling overlap / a = 5 / equal sizes

Wald (delta)	0.94 [0.89, 0.96]	w = 0.0376
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.0375
ILR, REML	0.97 [0.94, 0.99]	w = 0.0452
Profile likelihood	0.94 [0.9, 0.97]	w = 0.0395
Profile, REML	0.97 [0.94, 0.99]	w = 0.0455
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0449
Wald (delta)	0.94 [0.89, 0.96]	w = 0.0376
ILR, mapped back	0.94 [0.89, 0.96]	w = 0.0375
ILR, REML	0.96 [0.93, 0.98]	w = 0.0453
Wald (delta)	0.94 [0.89, 0.96]	w = 0.0376
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.0374
ILR, REML	0.98 [0.94, 0.99]	w = 0.0453
Method's own interval	0.8 [0.74, 0.85]	w = 0.431
Own interval, transcript fractions	0.8 [0.74, 0.85]	w = 0.431
Fisher information (Wald)	0.04 [0.02, 0.077]	w = 0.00213
Godambe sandwich	1 [0.98, 1]	w = 0.434
Released code, as published	1 [0.98, 1]	w = 1

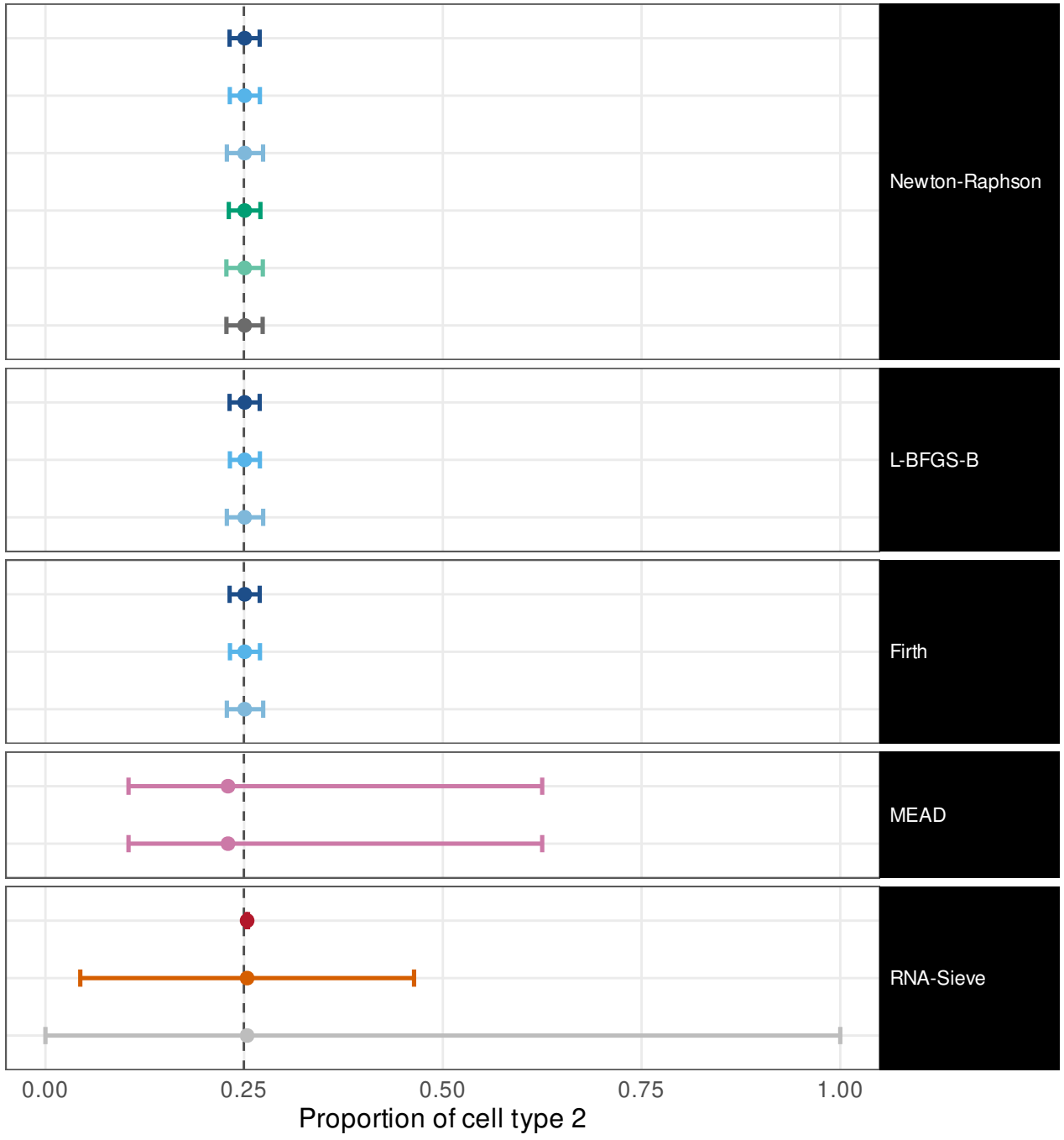


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / balanced / high sibling overlap / a = 5 / equal sizes

Wald (delta)	0.91 [0.86, 0.94]	w = 0.0379
ILR, mapped back	0.91 [0.86, 0.94]	w = 0.0378
ILR, REML	0.94 [0.9, 0.97]	w = 0.0455
Profile likelihood	0.92 [0.87, 0.95]	w = 0.0398
Profile, REML	0.95 [0.91, 0.97]	w = 0.0458
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0456
Wald (delta)	0.91 [0.86, 0.94]	w = 0.0379
ILR, mapped back	0.91 [0.86, 0.94]	w = 0.0377
ILR, REML	0.96 [0.92, 0.98]	w = 0.0457
Wald (delta)	0.91 [0.86, 0.94]	w = 0.0379
ILR, mapped back	0.9 [0.86, 0.94]	w = 0.0377
ILR, REML	0.96 [0.92, 0.98]	w = 0.0457
Method's own interval	0.64 [0.57, 0.7]	w = 0.52
Own interval, transcript fractions	0.64 [0.57, 0.7]	w = 0.52
Fisher information (Wald)	0.065 [0.038, 0.11]	w = 0.00219
Godambe sandwich	1 [0.98, 1]	w = 0.42
Released code, as published	1 [0.98, 1]	w = 1

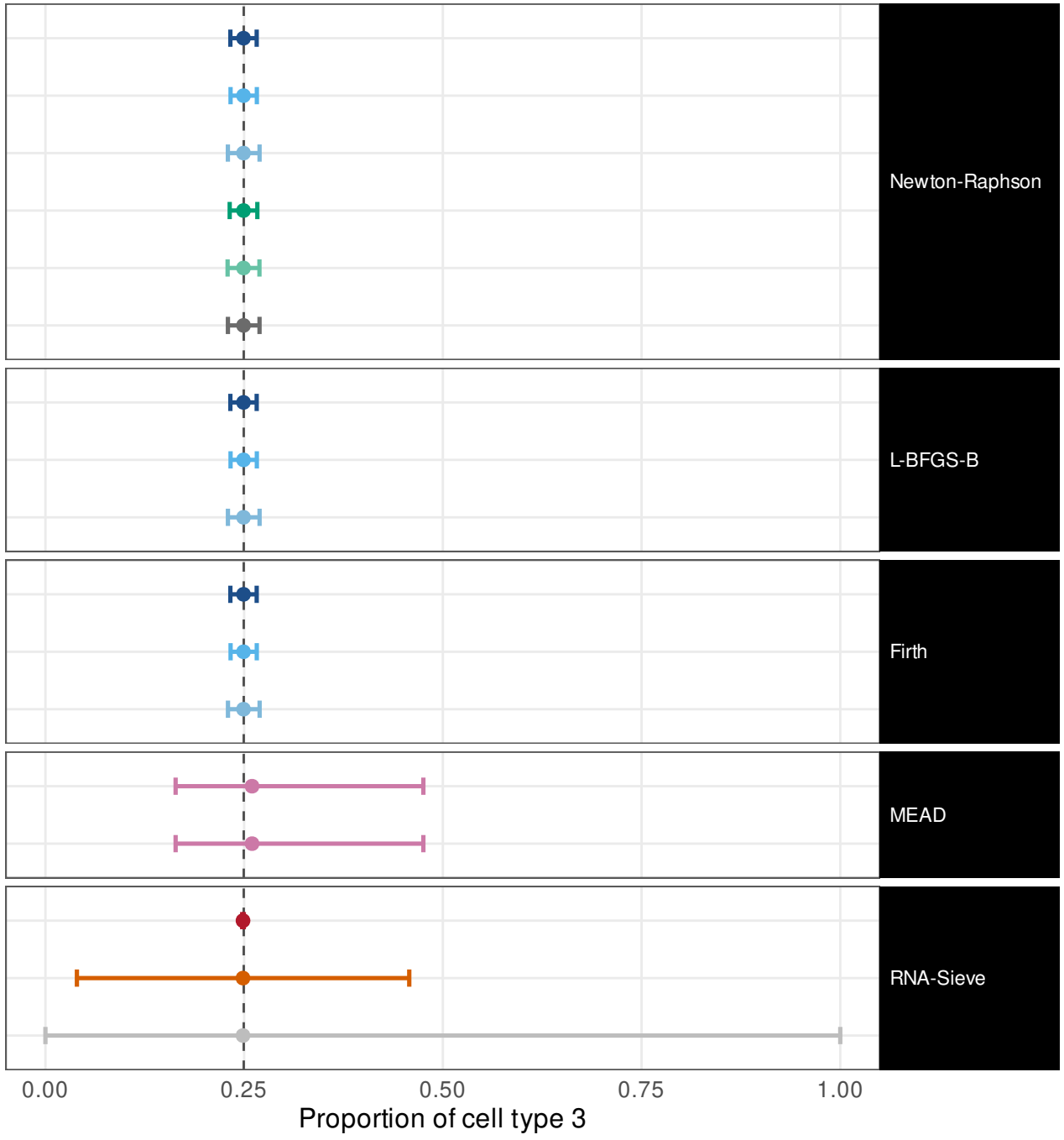


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / balanced / high sibling overlap / a = 5 / equal sizes

Wald (delta)	0.91 [0.86, 0.94]	w = 0.0331
ILR, mapped back	0.91 [0.86, 0.94]	w = 0.0331
ILR, REML	0.96 [0.92, 0.98]	w = 0.0399
Profile likelihood	0.92 [0.87, 0.95]	w = 0.0348
Profile, REML	0.96 [0.92, 0.98]	w = 0.0401
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0398
Wald (delta)	0.91 [0.86, 0.94]	w = 0.0331
ILR, mapped back	0.9 [0.86, 0.94]	w = 0.0331
ILR, REML	0.96 [0.92, 0.98]	w = 0.0398
Wald (delta)	0.91 [0.86, 0.94]	w = 0.0331
ILR, mapped back	0.91 [0.86, 0.94]	w = 0.0329
ILR, REML	0.96 [0.92, 0.98]	w = 0.0398
Method's own interval	0.82 [0.76, 0.87]	w = 0.312
Own interval, transcript fractions	0.82 [0.76, 0.87]	w = 0.312
Fisher information (Wald)	0.085 [0.054, 0.13]	w = 0.00199
Godambe sandwich	1 [0.98, 1]	w = 0.418
Released code, as published	1 [0.98, 1]	w = 1

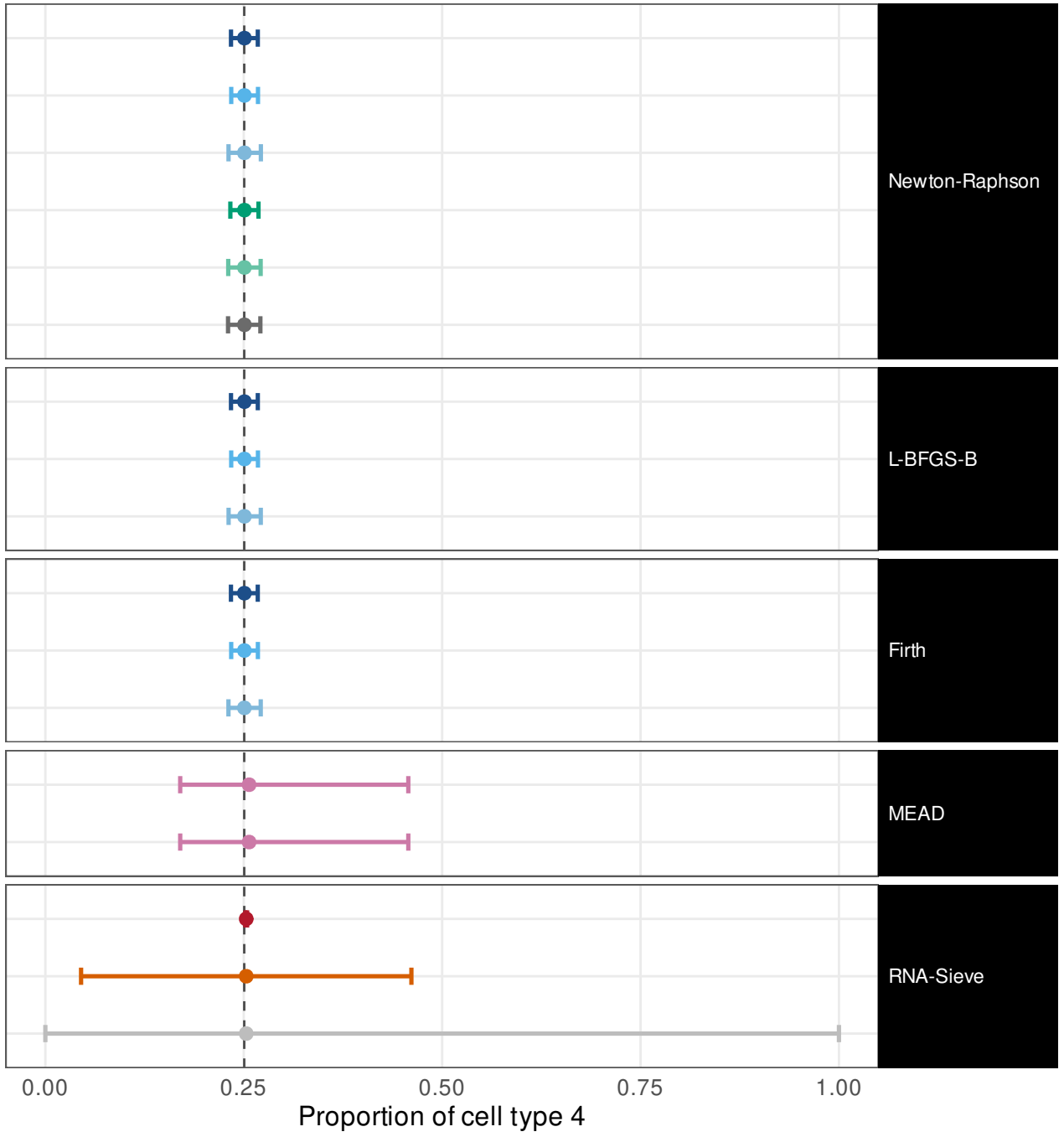


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / balanced / high sibling overlap / a = 5 / equal sizes

Wald (delta)	0.9	[0.84, 0.93]	w = 0.0338
ILR, mapped back	0.9	[0.85, 0.93]	w = 0.0337
ILR, REML	0.96	[0.92, 0.98]	w = 0.0407
Profile likelihood	0.92	[0.87, 0.95]	w = 0.0355
Profile, REML	0.95	[0.91, 0.97]	w = 0.0408
Oracle (Fisher at p*)	1	[0.98, 1]	w = 0.0407
Wald (delta)	0.9	[0.84, 0.93]	w = 0.0338
ILR, mapped back	0.88	[0.83, 0.92]	w = 0.0337
ILR, REML	0.95	[0.91, 0.97]	w = 0.0405
Wald (delta)	0.9	[0.84, 0.93]	w = 0.0338
ILR, mapped back	0.89	[0.84, 0.93]	w = 0.0336
ILR, REML	0.96	[0.92, 0.98]	w = 0.0407
Method's own interval	0.9	[0.86, 0.94]	w = 0.287
Own interval, transcript fractions	0.9	[0.86, 0.94]	w = 0.287
Fisher information (Wald)	0.035	[0.017, 0.07]	w = 0.00223
Godambe sandwich	1	[0.98, 1]	w = 0.416
Released code, as published	1	[0.98, 1]	w = 1

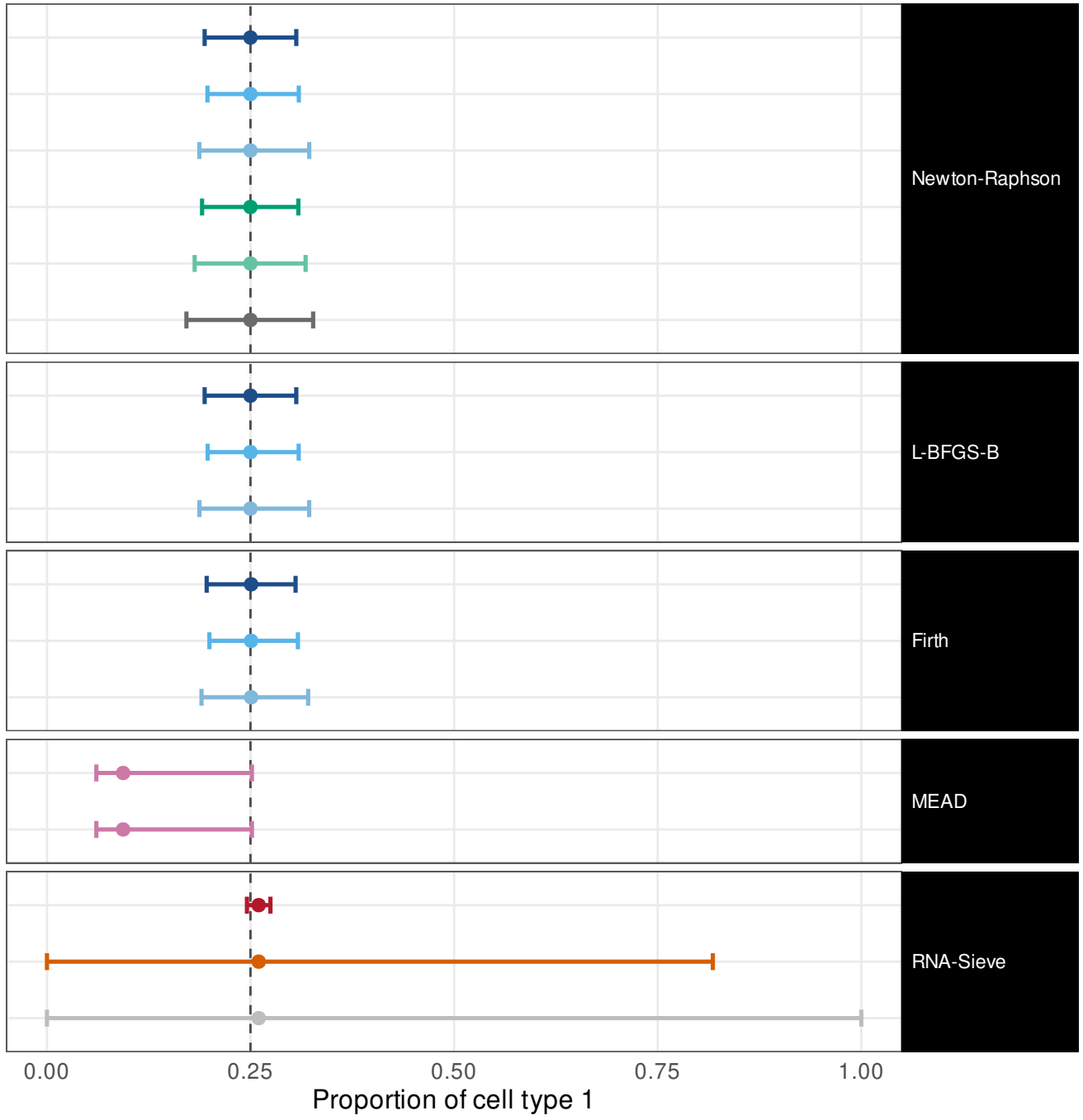


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / balanced / high sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.86 [0.81, 0.91]	w = 0.113
ILR, mapped back	0.88 [0.83, 0.92]	w = 0.112
ILR, REML	0.94 [0.89, 0.96]	w = 0.135
Profile likelihood	0.88 [0.83, 0.92]	w = 0.118
Profile, REML	0.92 [0.88, 0.95]	w = 0.136
Oracle (Fisher at p*)	0.98 [0.95, 0.99]	w = 0.156
Wald (delta)	0.86 [0.81, 0.91]	w = 0.113
ILR, mapped back	0.88 [0.82, 0.91]	w = 0.112
ILR, REML	0.93 [0.89, 0.96]	w = 0.135
Wald (delta)	0.87 [0.82, 0.91]	w = 0.109
ILR, mapped back	0.88 [0.83, 0.92]	w = 0.109
ILR, REML	0.92 [0.87, 0.95]	w = 0.131
Method's own interval	0.16 [0.11, 0.21]	w = 0.191
Own interval, transcript fractions	0.78 [0.72, 0.83]	w = 0.191
Fisher information (Wald)	0.28 [0.23, 0.35]	w = 0.0288
Godambe sandwich	1 [0.98, 1]	w = 0.818
Released code, as published	1 [0.98, 1]	w = 1

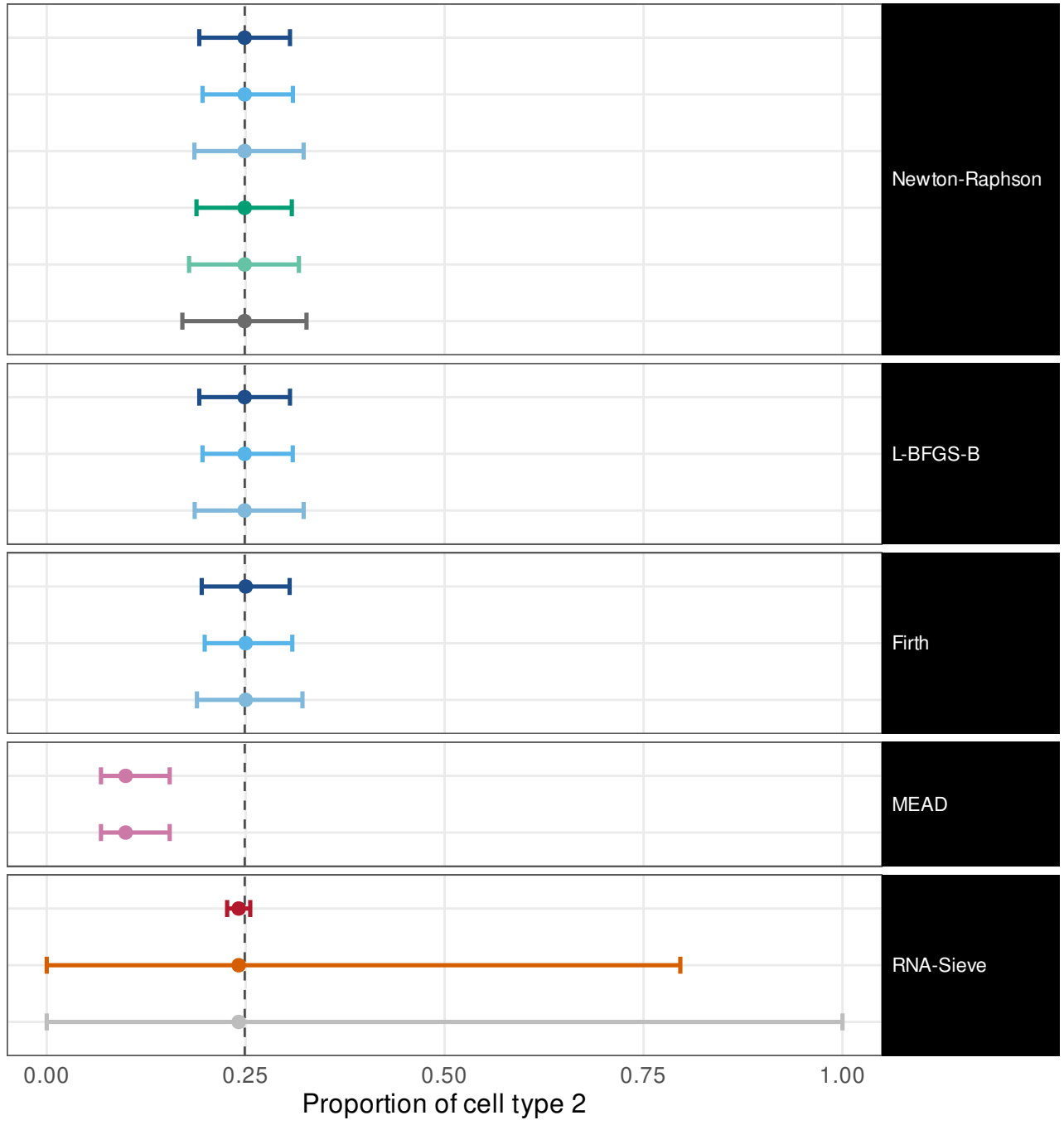


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / balanced / high sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.88 [0.82, 0.91]	w = 0.114
ILR, mapped back	0.89 [0.84, 0.93]	w = 0.113
ILR, REML	0.94 [0.89, 0.96]	w = 0.137
Profile likelihood	0.88 [0.83, 0.92]	w = 0.12
Profile, REML	0.94 [0.9, 0.97]	w = 0.138
Oracle (Fisher at p*)	0.98 [0.95, 0.99]	w = 0.156
Wald (delta)	0.88 [0.82, 0.91]	w = 0.114
ILR, mapped back	0.88 [0.83, 0.92]	w = 0.113
ILR, REML	0.94 [0.9, 0.97]	w = 0.137
Wald (delta)	0.88 [0.83, 0.92]	w = 0.11
ILR, mapped back	0.89 [0.84, 0.93]	w = 0.11
ILR, REML	0.94 [0.9, 0.97]	w = 0.133
Method's own interval	0.03 [0.014, 0.064]	w = 0.0864
Own interval, transcript fractions	0.78 [0.72, 0.84]	w = 0.0864
Fisher information (Wald)	0.3 [0.24, 0.36]	w = 0.029
Godambe sandwich	1 [0.98, 1]	w = 0.796
Released code, as published	1 [0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / balanced / high sibling overlap / a = 5 / sizes 1:1:5:5

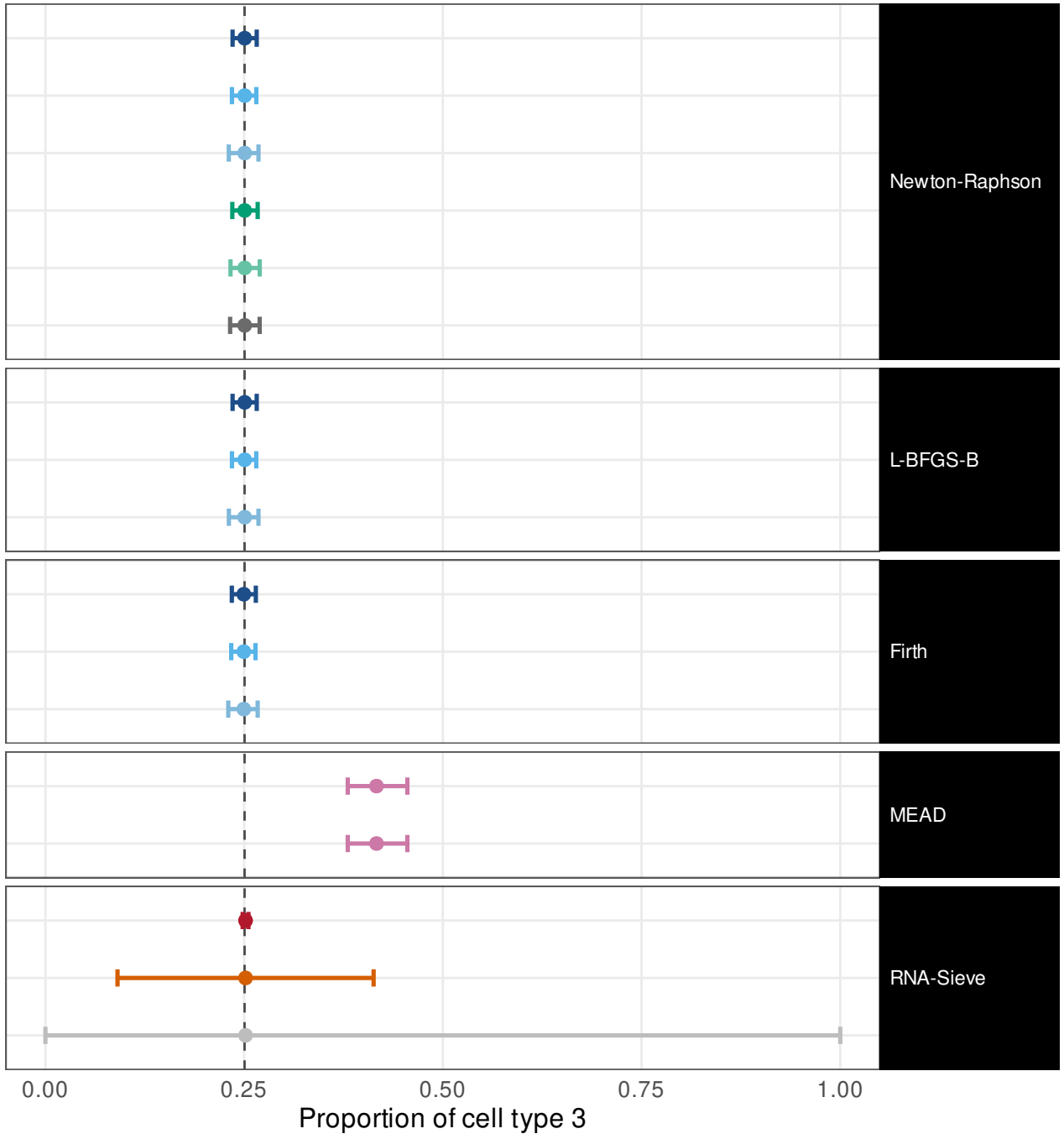
```
Wald (delta)                          0.88 [0.82, 0.91]   w = 0.0303
ILR, mapped back                      0.88 [0.83, 0.92]   w = 0.0307
ILR, REML                             0.94 [ 0.9, 0.97]   w = 0.0374
Profile likelihood                     0.9 [0.85, 0.93]   w = 0.0319
Profile, REML                         0.94 [ 0.9, 0.97]   w = 0.0368
Oracle (Fisher at p*)                    1 [0.98,    1]   w = 0.0372

Wald (delta)                          0.88 [0.82, 0.91]   w = 0.0303
ILR, mapped back                      0.88 [0.83, 0.92]   w = 0.0306
ILR, REML                             0.94 [0.89, 0.96]   w = 0.0372

Wald (delta)                          0.88 [0.83, 0.92]   w = 0.0302
ILR, mapped back                      0.88 [0.83, 0.92]   w = 0.0305
ILR, REML                             0.94 [ 0.9, 0.97]   w = 0.037

Method's own interval                0.015 [0.0051, 0.043]   w = 0.075
Own interval, transcript fractions    0.95 [0.91, 0.97]   w = 0.075

Fisher information (Wald)             0.25 [ 0.2, 0.31]   w = 0.00724
Godambe sandwich                         1 [0.98,    1]   w = 0.322
Released code, as published              1 [0.98,    1]   w =     1
```

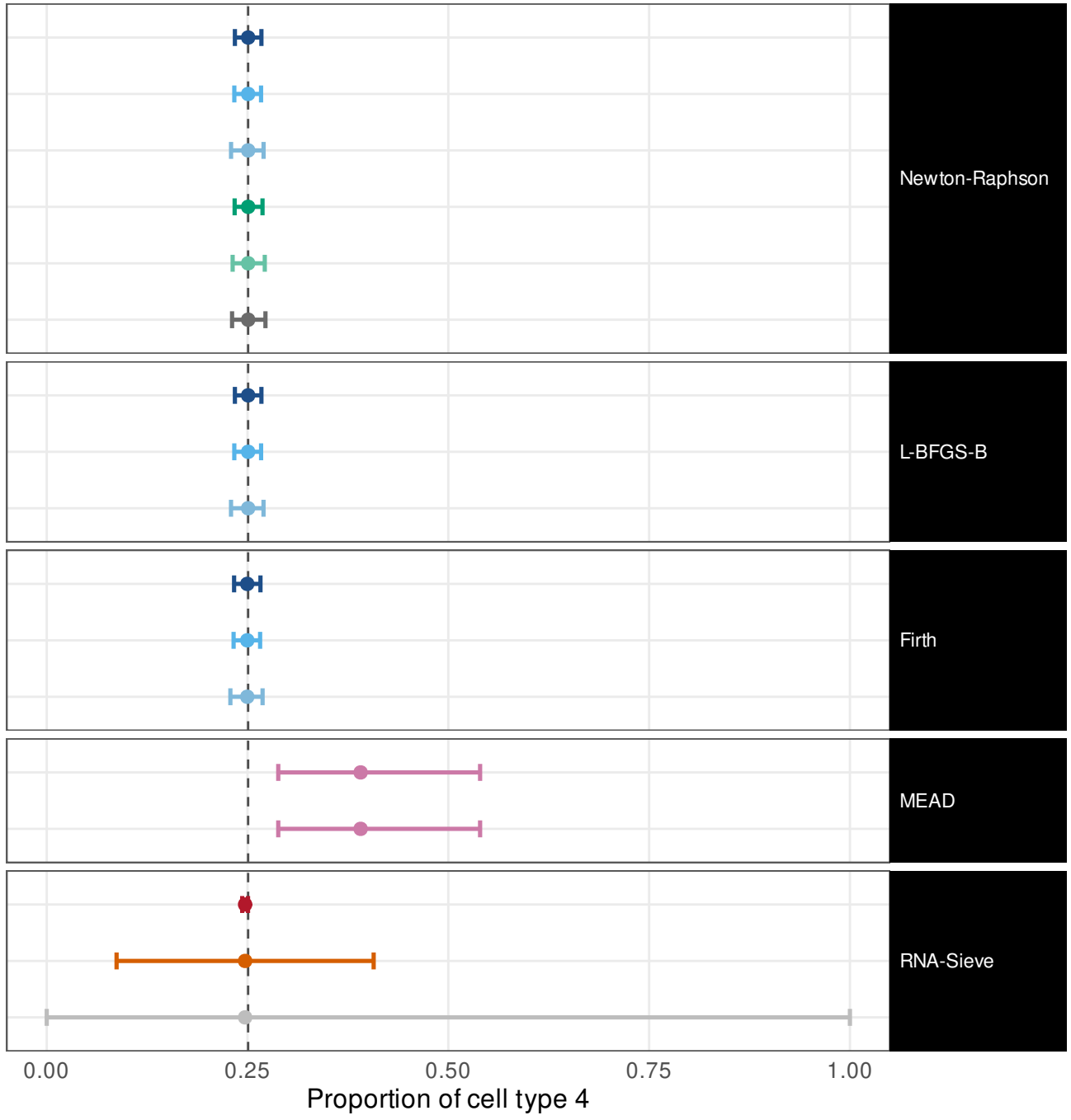


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / balanced / high sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.9 [0.86, 0.94]	w = 0.033
ILR, mapped back	0.91 [0.86, 0.94]	w = 0.0333
ILR, REML	0.94 [0.89, 0.96]	w = 0.0405
Profile likelihood	0.92 [0.87, 0.95]	w = 0.0347
Profile, REML	0.94 [0.89, 0.96]	w = 0.04
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0416
Wald (delta)	0.9 [0.86, 0.94]	w = 0.033
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.0334
ILR, REML	0.94 [0.9, 0.97]	w = 0.0405
Wald (delta)	0.91 [0.86, 0.94]	w = 0.0328
ILR, mapped back	0.91 [0.86, 0.94]	w = 0.033
ILR, REML	0.92 [0.88, 0.95]	w = 0.0401
Method's own interval	0.21 [0.16, 0.27]	w = 0.251
Own interval, transcript fractions	0.6 [0.54, 0.67]	w = 0.251
Fisher information (Wald)	0.26 [0.21, 0.33]	w = 0.00723
Godambe sandwich	1 [0.98, 1]	w = 0.32
Released code, as published	1 [0.98, 1]	w = 1

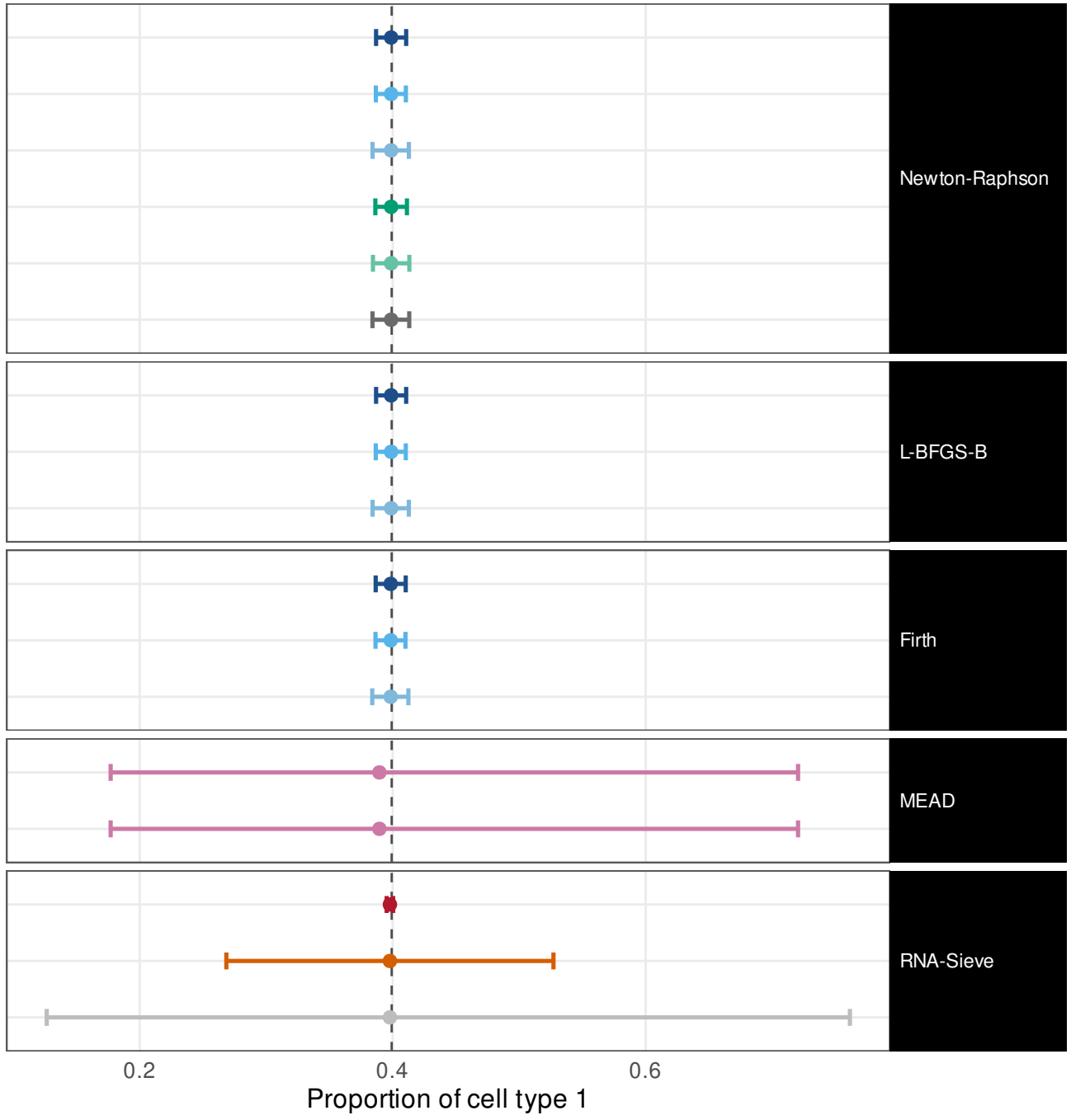


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / unbalanced / low sibling overlap / a = 1 / equal sizes

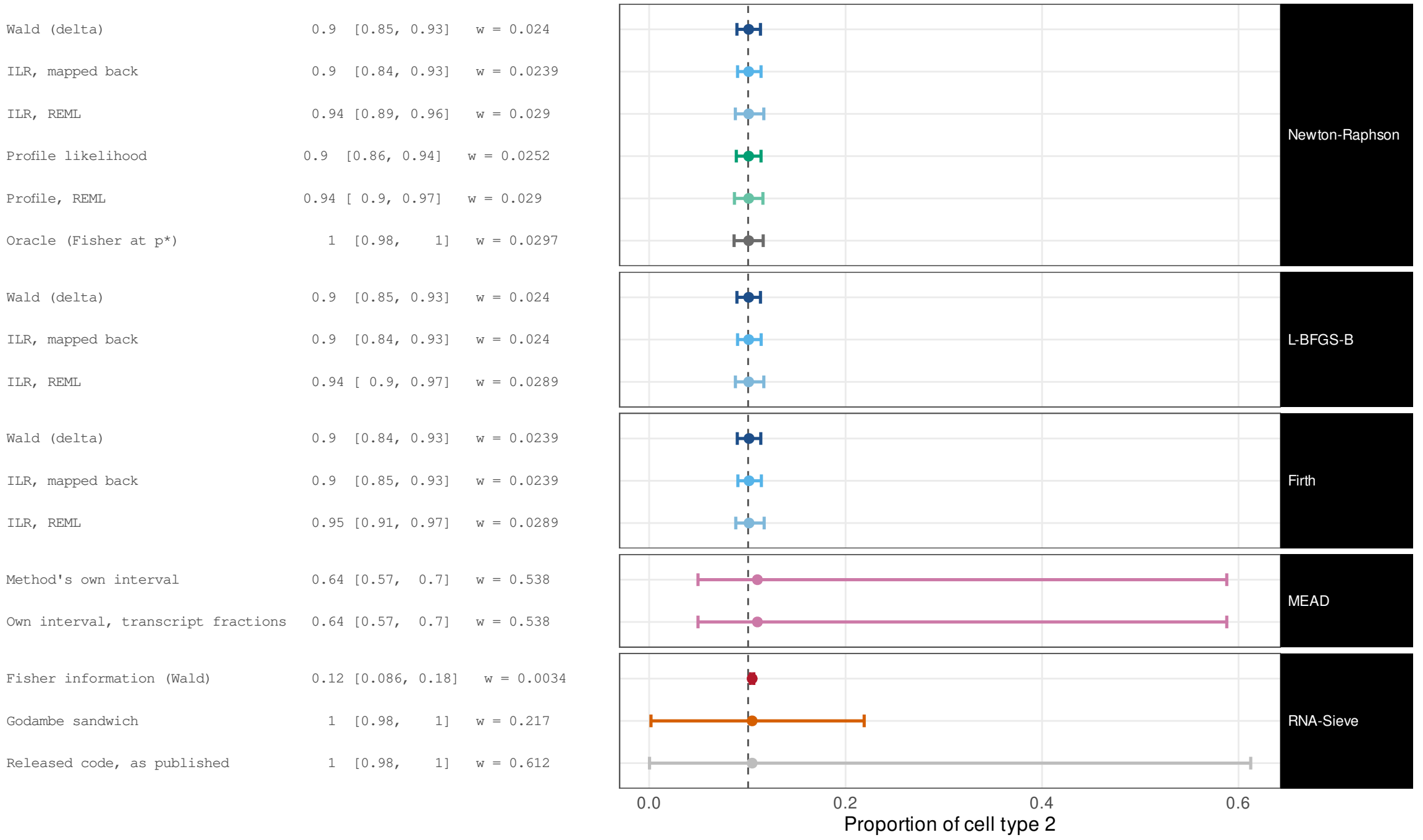
Wald (delta)	0.88 [0.83, 0.92]	w = 0.0238
ILR, mapped back	0.88 [0.83, 0.92]	w = 0.0237
ILR, REML	0.96 [0.92, 0.98]	w = 0.0287
Profile likelihood	0.92 [0.87, 0.95]	w = 0.025
Profile, REML	0.96 [0.93, 0.98]	w = 0.0288
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0291
Wald (delta)	0.88 [0.83, 0.92]	w = 0.0238
ILR, mapped back	0.89 [0.84, 0.93]	w = 0.0237
ILR, REML	0.96 [0.92, 0.98]	w = 0.0287
Wald (delta)	0.89 [0.84, 0.93]	w = 0.0237
ILR, mapped back	0.88 [0.83, 0.92]	w = 0.0237
ILR, REML	0.96 [0.93, 0.98]	w = 0.0286
Method's own interval	0.64 [0.58, 0.71]	w = 0.543
Own interval, transcript fractions	0.64 [0.58, 0.71]	w = 0.543
Fisher information (Wald)	0.22 [0.17, 0.29]	w = 0.00511
Godambe sandwich	1 [0.98, 1]	w = 0.259
Released code, as published	1 [0.98, 1]	w = 0.635



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / unbalanced / low sibling overlap / a = 1 / equal sizes

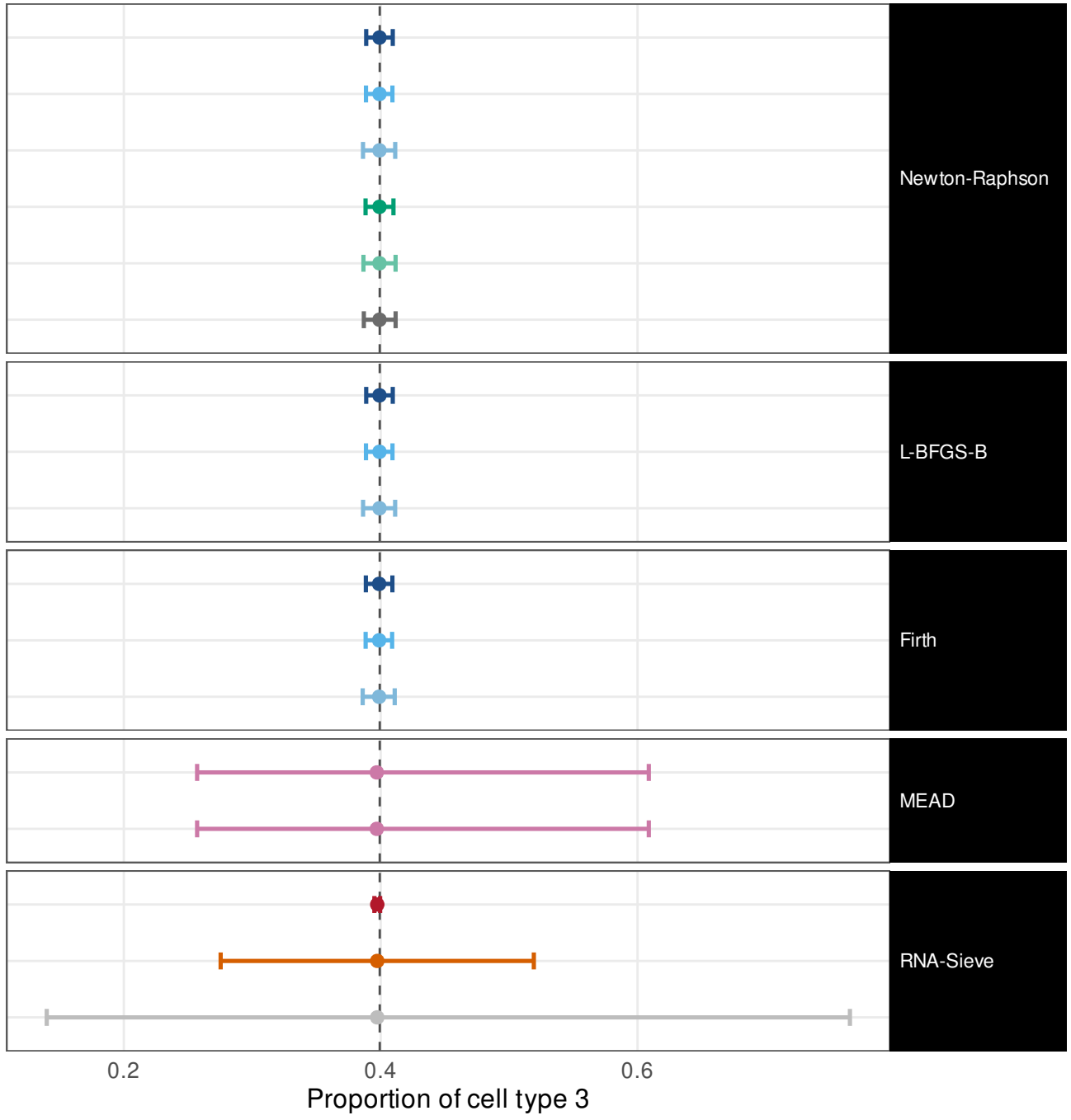


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / unbalanced / low sibling overlap / a = 1 / equal sizes

Wald (delta)	0.92 [0.87, 0.95]	w = 0.0207
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.0206
ILR, REML	0.96 [0.92, 0.98]	w = 0.0251
Profile likelihood	0.94 [0.89, 0.96]	w = 0.0217
Profile, REML	0.96 [0.92, 0.98]	w = 0.025
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0247
Wald (delta)	0.92 [0.87, 0.95]	w = 0.0207
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.0207
ILR, REML	0.96 [0.92, 0.98]	w = 0.025
Wald (delta)	0.92 [0.87, 0.95]	w = 0.0207
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.0206
ILR, REML	0.95 [0.91, 0.97]	w = 0.0249
Method's own interval	0.44 [0.37, 0.51]	w = 0.352
Own interval, transcript fractions	0.44 [0.37, 0.51]	w = 0.352
Fisher information (Wald)	0.3 [0.24, 0.37]	w = 0.00445
Godambe sandwich	1 [0.98, 1]	w = 0.244
Released code, as published	1 [0.98, 1]	w = 0.625

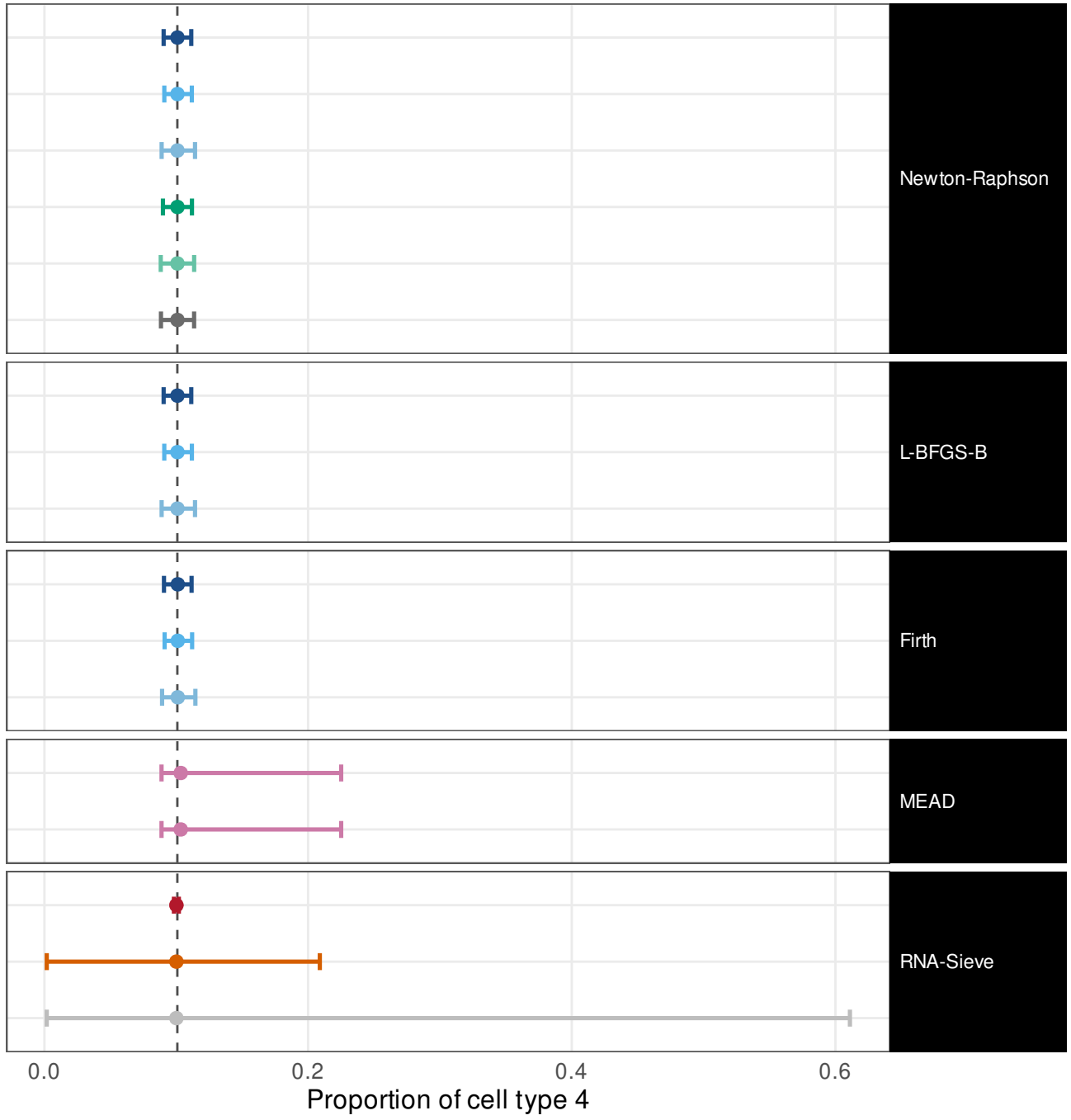


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / unbalanced / low sibling overlap / a = 1 / equal sizes

Wald (delta)	0.92 [0.87, 0.95]	w = 0.021
ILR, mapped back	0.92 [0.88, 0.95]	w = 0.0209
ILR, REML	0.96 [0.92, 0.98]	w = 0.0253
Profile likelihood	0.92 [0.88, 0.95]	w = 0.022
Profile, REML	0.95 [0.91, 0.97]	w = 0.0253
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0251
Wald (delta)	0.92 [0.87, 0.95]	w = 0.021
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.0209
ILR, REML	0.96 [0.92, 0.98]	w = 0.0253
Wald (delta)	0.92 [0.87, 0.95]	w = 0.0209
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.0209
ILR, REML	0.96 [0.92, 0.98]	w = 0.0253
Method's own interval	0.21 [0.16, 0.27]	w = 0.136
Own interval, transcript fractions	0.21 [0.16, 0.27]	w = 0.136
Fisher information (Wald)	0.2 [0.15, 0.26]	w = 0.00372
Godambe sandwich	1 [0.98, 1]	w = 0.207
Released code, as published	1 [0.98, 1]	w = 0.609

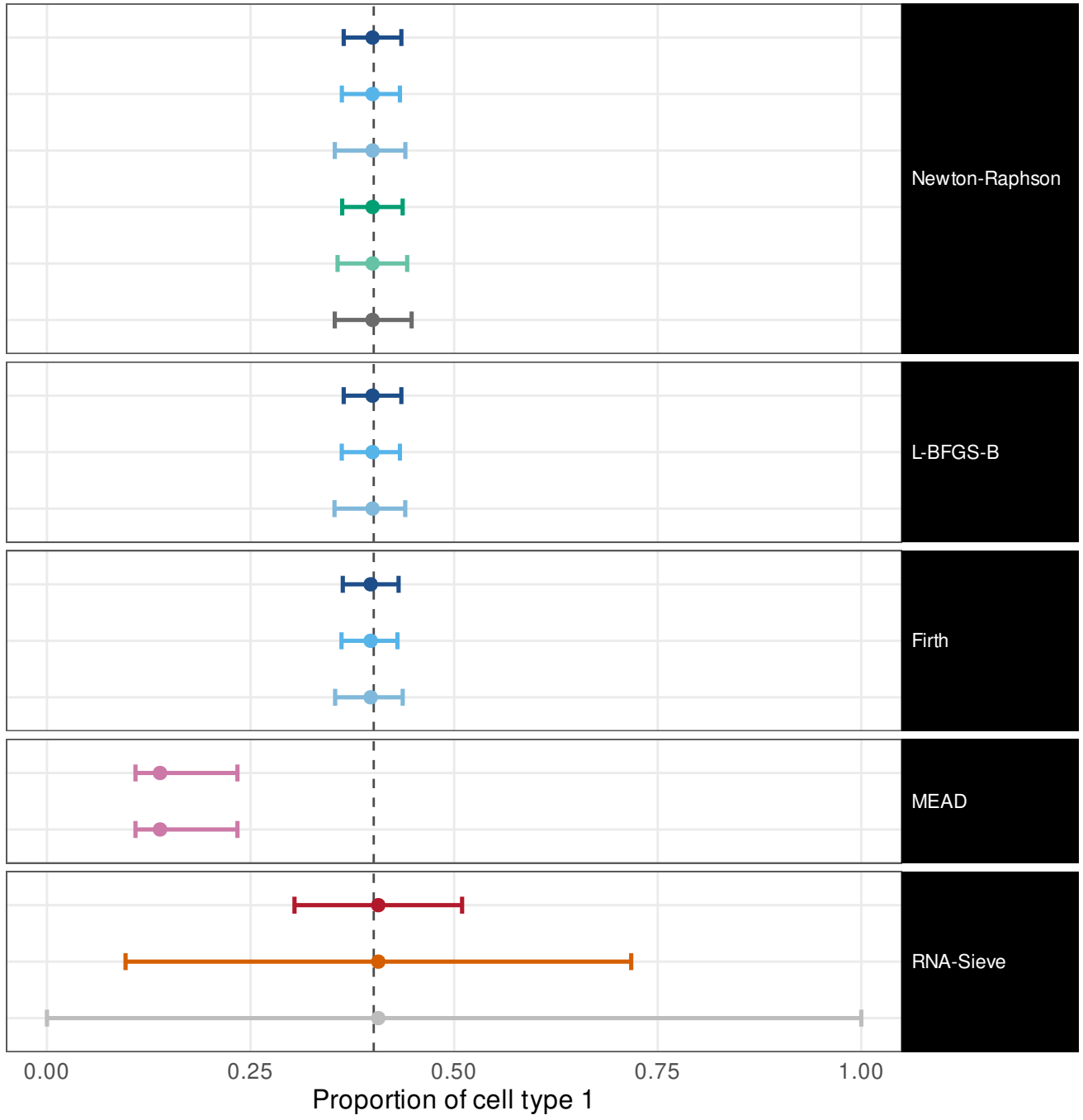


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / unbalanced / low sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.92 [0.87, 0.95]	w = 0.0708
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.0713
ILR, REML	0.98 [0.95, 0.99]	w = 0.0866
Profile likelihood	0.92 [0.88, 0.95]	w = 0.0742
Profile, REML	0.96 [0.92, 0.98]	w = 0.0855
Oracle (Fisher at p*)	0.97 [0.94, 0.99]	w = 0.0944
Wald (delta)	0.92 [0.87, 0.95]	w = 0.0708
ILR, mapped back	0.94 [0.89, 0.96]	w = 0.0714
ILR, REML	0.98 [0.95, 0.99]	w = 0.0869
Wald (delta)	0.9 [0.86, 0.94]	w = 0.0685
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.0687
ILR, REML	0.98 [0.94, 0.99]	w = 0.0829
Method's own interval	0.09 [0.058, 0.14]	w = 0.125
Own interval, transcript fractions	0.9 [0.85, 0.93]	w = 0.125
Fisher information (Wald)	1 [0.98, 1]	w = 0.206
Godambe sandwich	1 [0.98, 1]	w = 0.621
Released code, as published	1 [0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / unbalanced / low sibling overlap / a = 1 / sizes 1:1:5:5

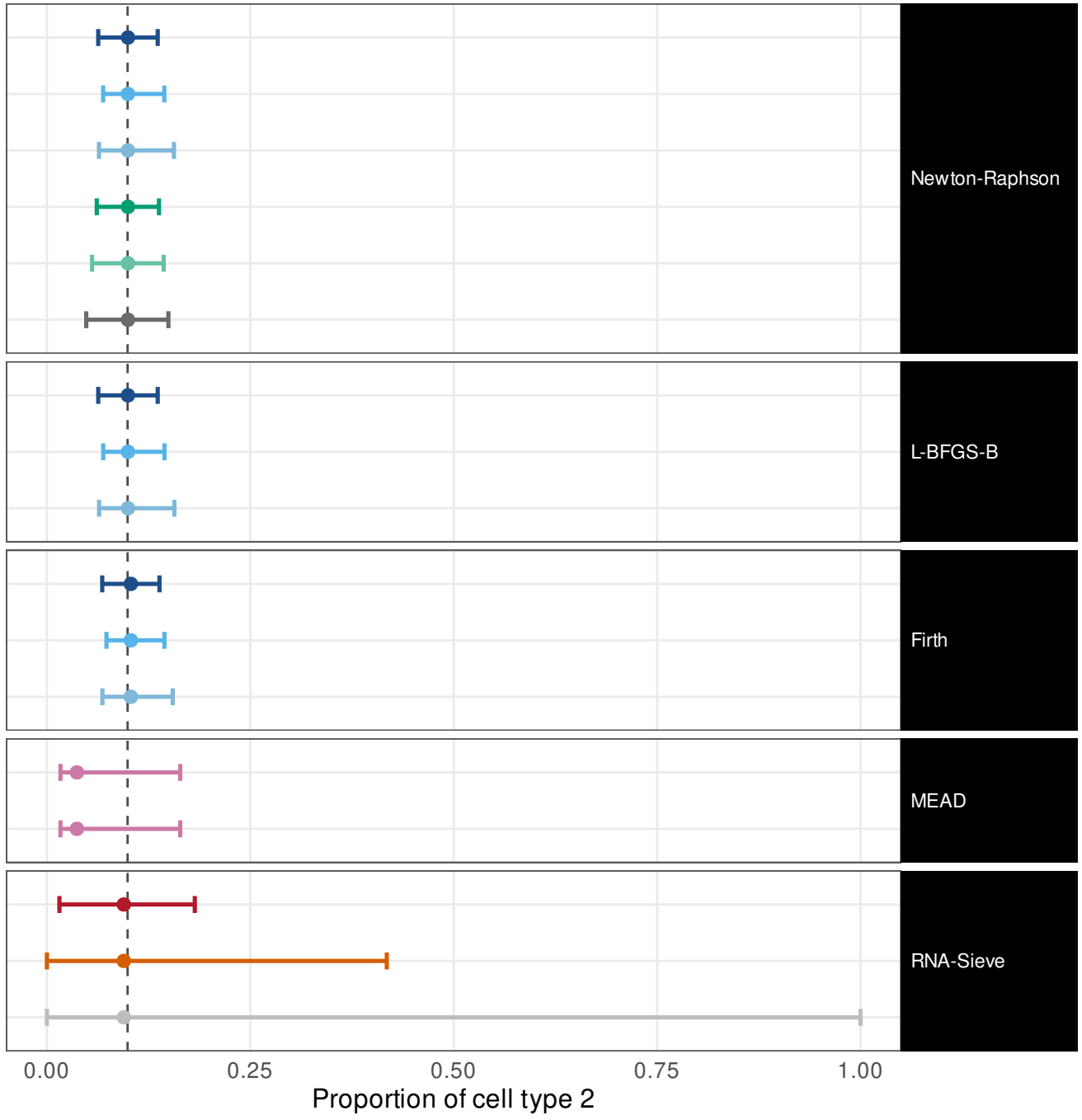
```
Wald (delta)                          0.88 [0.83, 0.92]   w = 0.0731
ILR, mapped back                      0.92 [0.87, 0.95]   w = 0.0752
ILR, REML                             0.96 [0.92, 0.98]   w = 0.0922
Profile likelihood                     0.9 [0.86, 0.94]   w = 0.0765
Profile, REML                         0.94 [ 0.9, 0.97]   w = 0.088
Oracle (Fisher at p*)                 0.96 [0.93, 0.98]   w = 0.101

Wald (delta)                          0.88 [0.83, 0.92]   w = 0.0731
ILR, mapped back                      0.91 [0.86, 0.94]   w = 0.0753
ILR, REML                             0.96 [0.92, 0.98]   w = 0.0925

Wald (delta)                          0.88 [0.83, 0.92]   w = 0.0704
ILR, mapped back                       0.9 [0.85, 0.93]   w = 0.0712
ILR, REML                             0.95 [0.91, 0.97]   w = 0.0865

Method's own interval                 0.12 [0.078, 0.17]   w = 0.147
Own interval, transcript fractions    0.87 [0.82, 0.91]   w = 0.147

Fisher information (Wald)                1 [0.97,    1]   w = 0.166
Godambe sandwich                         1 [0.98,    1]   w = 0.418
Released code, as published              1 [0.98,    1]   w =     1
```

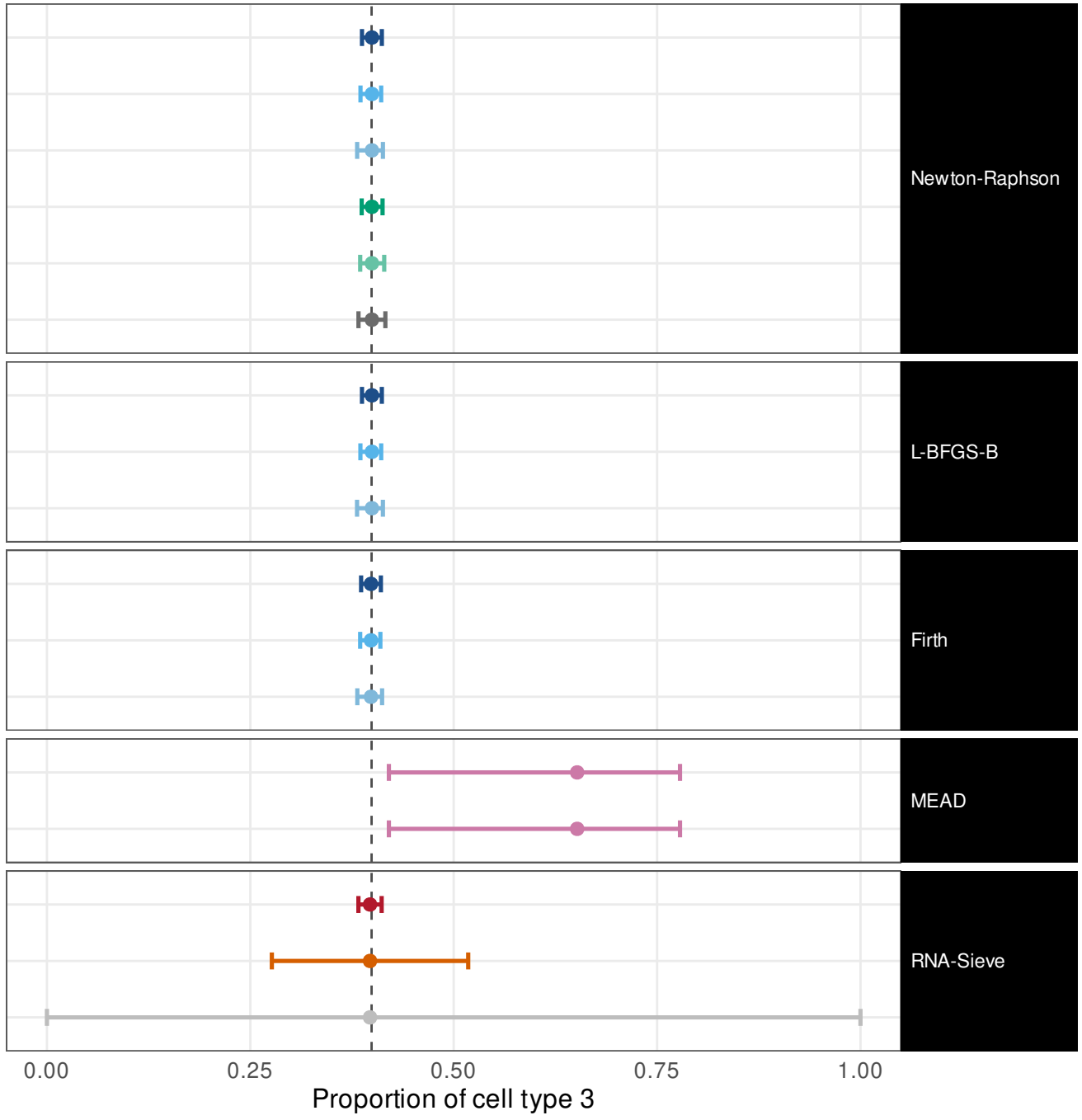


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / unbalanced / low sibling overlap / a = 1 / sizes 1:1:5:5

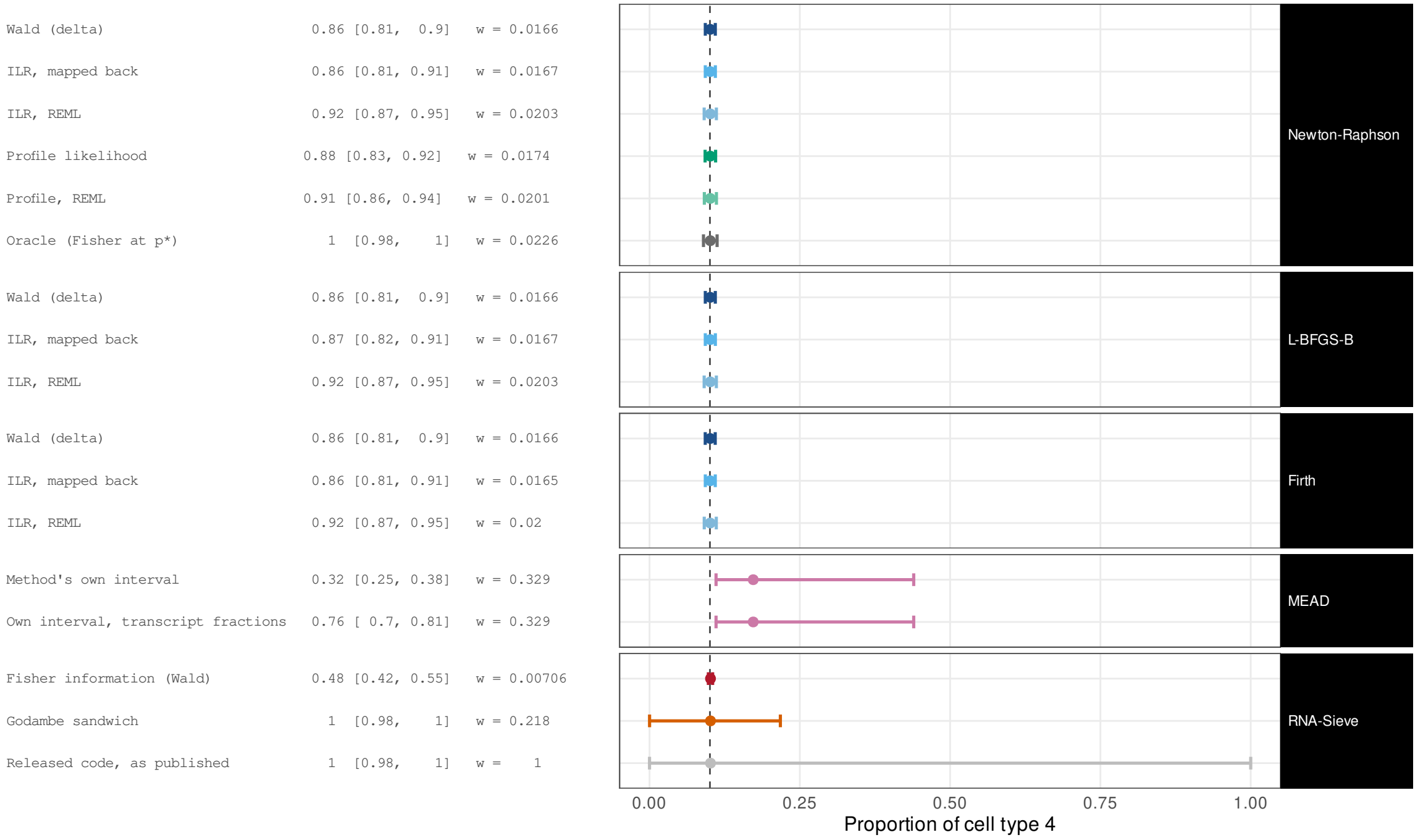
Wald (delta)	0.91 [0.86, 0.94]	w = 0.0244
ILR, mapped back	0.91 [0.86, 0.94]	w = 0.0254
ILR, REML	0.96 [0.92, 0.98]	w = 0.0317
Profile likelihood	0.93 [0.89, 0.96]	w = 0.0256
Profile, REML	0.96 [0.92, 0.98]	w = 0.0295
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.033
Wald (delta)	0.91 [0.86, 0.94]	w = 0.0244
ILR, mapped back	0.9 [0.86, 0.94]	w = 0.0256
ILR, REML	0.96 [0.92, 0.98]	w = 0.0318
Wald (delta)	0.9 [0.84, 0.93]	w = 0.0243
ILR, mapped back	0.9 [0.85, 0.93]	w = 0.0248
ILR, REML	0.94 [0.9, 0.97]	w = 0.0304
Method's own interval	0.34 [0.28, 0.41]	w = 0.358
Own interval, transcript fractions	0.65 [0.58, 0.71]	w = 0.358
Fisher information (Wald)	0.95 [0.91, 0.97]	w = 0.0285
Godambe sandwich	1 [0.98, 1]	w = 0.241
Released code, as published	1 [0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / unbalanced / low sibling overlap / a = 1 / sizes 1:1:5:5

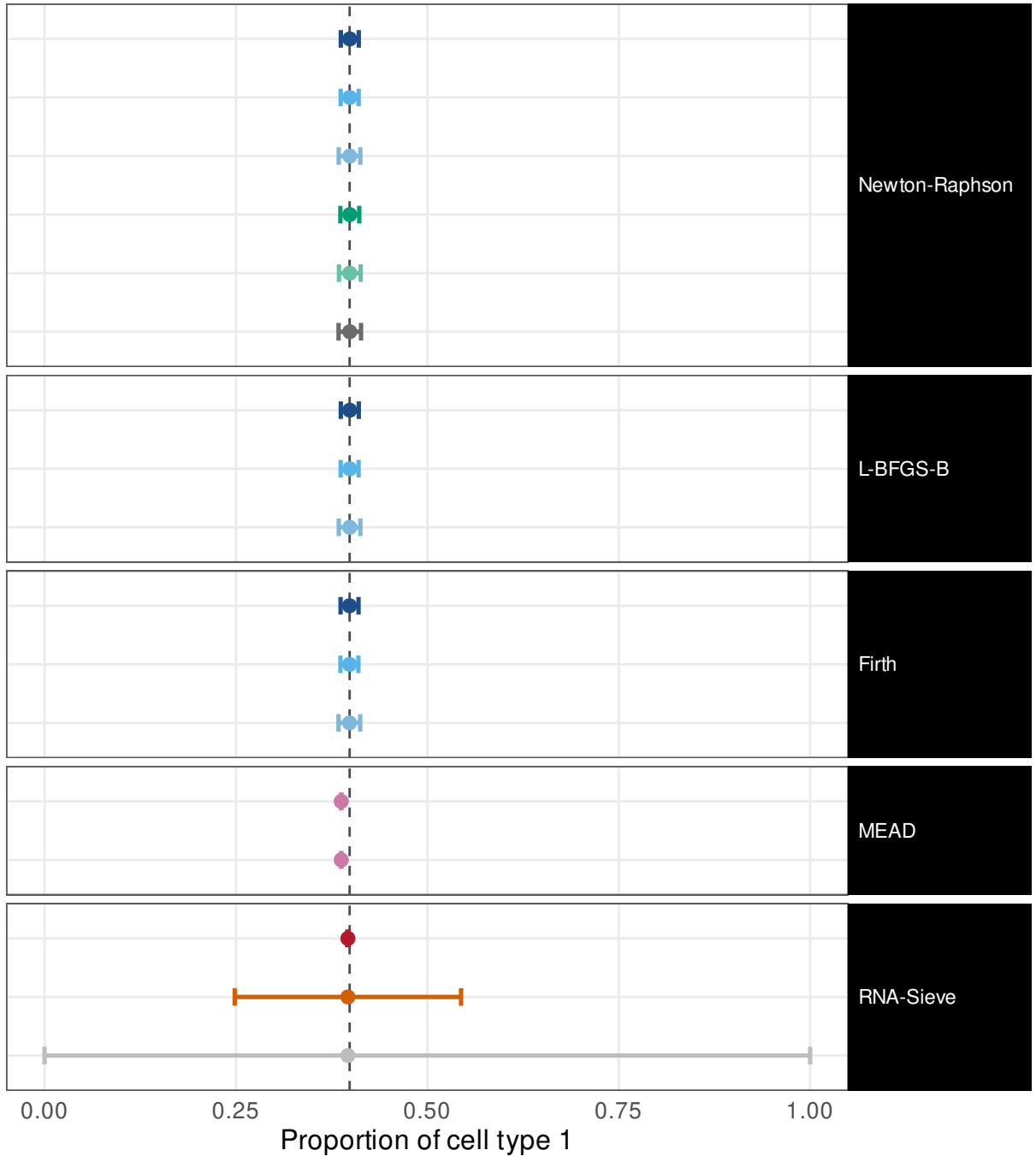


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / unbalanced / low sibling overlap / a = 5 / equal sizes

Wald (delta)	0.88 [0.82, 0.91]	w = 0.0235
ILR, mapped back	0.88 [0.83, 0.92]	w = 0.0235
ILR, REML	0.92 [0.87, 0.95]	w = 0.0283
Profile likelihood	0.88 [0.83, 0.92]	w = 0.0246
Profile, REML	0.92 [0.87, 0.95]	w = 0.0284
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0293
Wald (delta)	0.88 [0.82, 0.91]	w = 0.0235
ILR, mapped back	0.88 [0.82, 0.91]	w = 0.0234
ILR, REML	0.92 [0.88, 0.95]	w = 0.0283
Wald (delta)	0.88 [0.82, 0.91]	w = 0.0234
ILR, mapped back	0.87 [0.82, 0.91]	w = 0.0234
ILR, REML	0.92 [0.88, 0.95]	w = 0.0283
Method's own interval	0.005 [0.00088, 0.028]	w = 0.000501
Own interval, transcript fractions	0.005 [0.00088, 0.028]	w = 0.000501
Fisher information (Wald)	0.11 [0.074, 0.16]	w = 0.00177
Godambe sandwich	1 [0.98, 1]	w = 0.296
Released code, as published	1 [0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / unbalanced / low sibling overlap / a = 5 / equal sizes

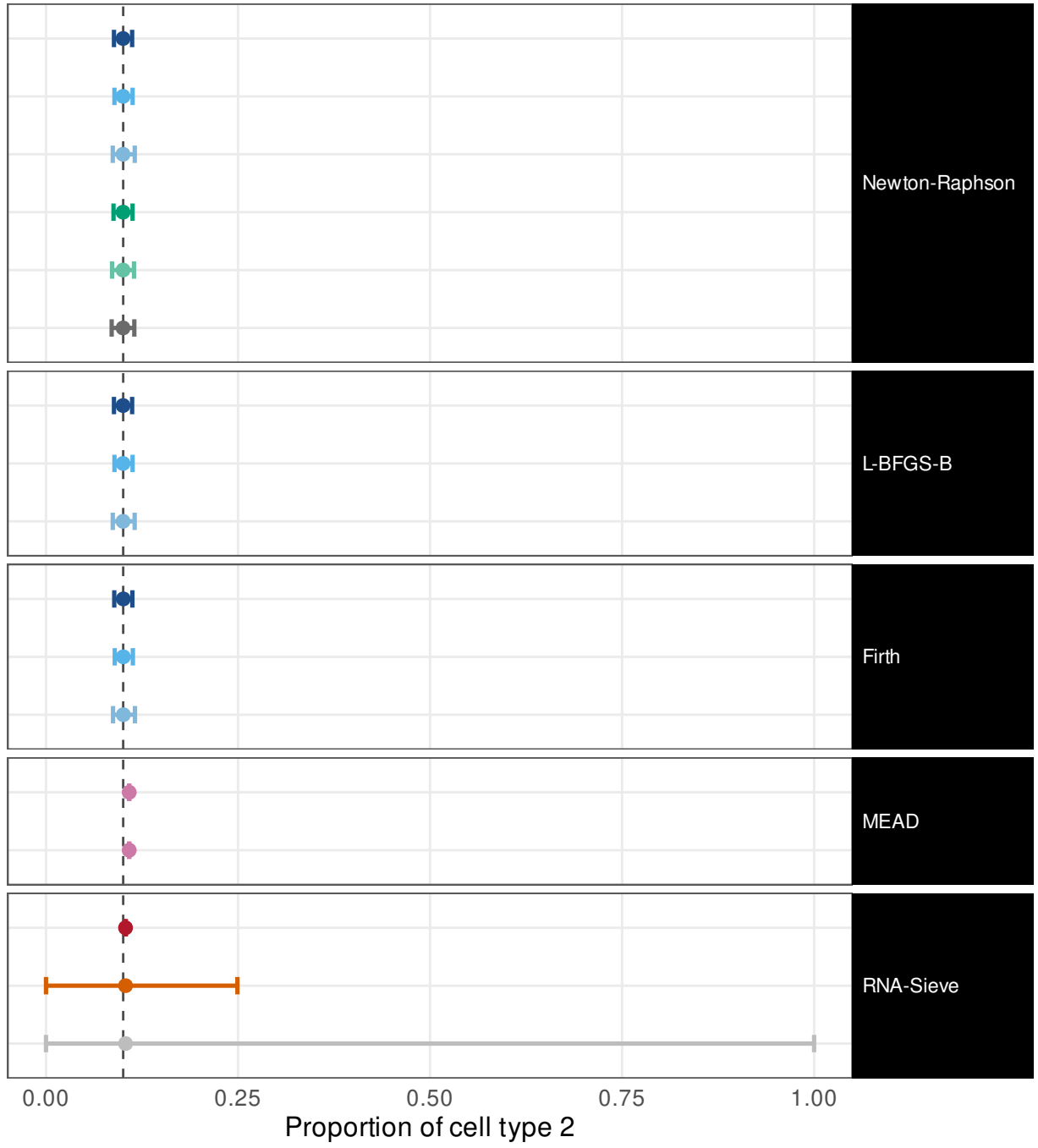
```
Wald (delta)                          0.88 [0.83, 0.92]   w = 0.0236
ILR, mapped back                      0.88 [0.82, 0.91]   w = 0.0237
ILR, REML                             0.94 [0.89, 0.96]   w = 0.0286
Profile likelihood                    0.89 [0.84, 0.93]   w = 0.0248
Profile, REML                         0.93 [0.89, 0.96]   w = 0.0286
Oracle (Fisher at p*)                    1 [0.98,    1]   w = 0.0295

Wald (delta)                          0.88 [0.83, 0.92]   w = 0.0236
ILR, mapped back                      0.86 [0.81, 0.91]   w = 0.0236
ILR, REML                             0.94 [ 0.9, 0.97]   w = 0.0285

Wald (delta)                          0.88 [0.82, 0.91]   w = 0.0236
ILR, mapped back                      0.88 [0.82, 0.91]   w = 0.0236
ILR, REML                             0.94 [ 0.9, 0.97]   w = 0.0285

Method's own interval                 0.015 [0.0051, 0.043]   w = 0.000279
Own interval, transcript fractions    0.015 [0.0051, 0.043]   w = 0.000279

Fisher information (Wald)             0.03 [0.014, 0.064]   w = 0.00088
Godambe sandwich                         1 [0.98,    1]   w = 0.249
Released code, as published              1 [0.98,    1]   w =     1
```

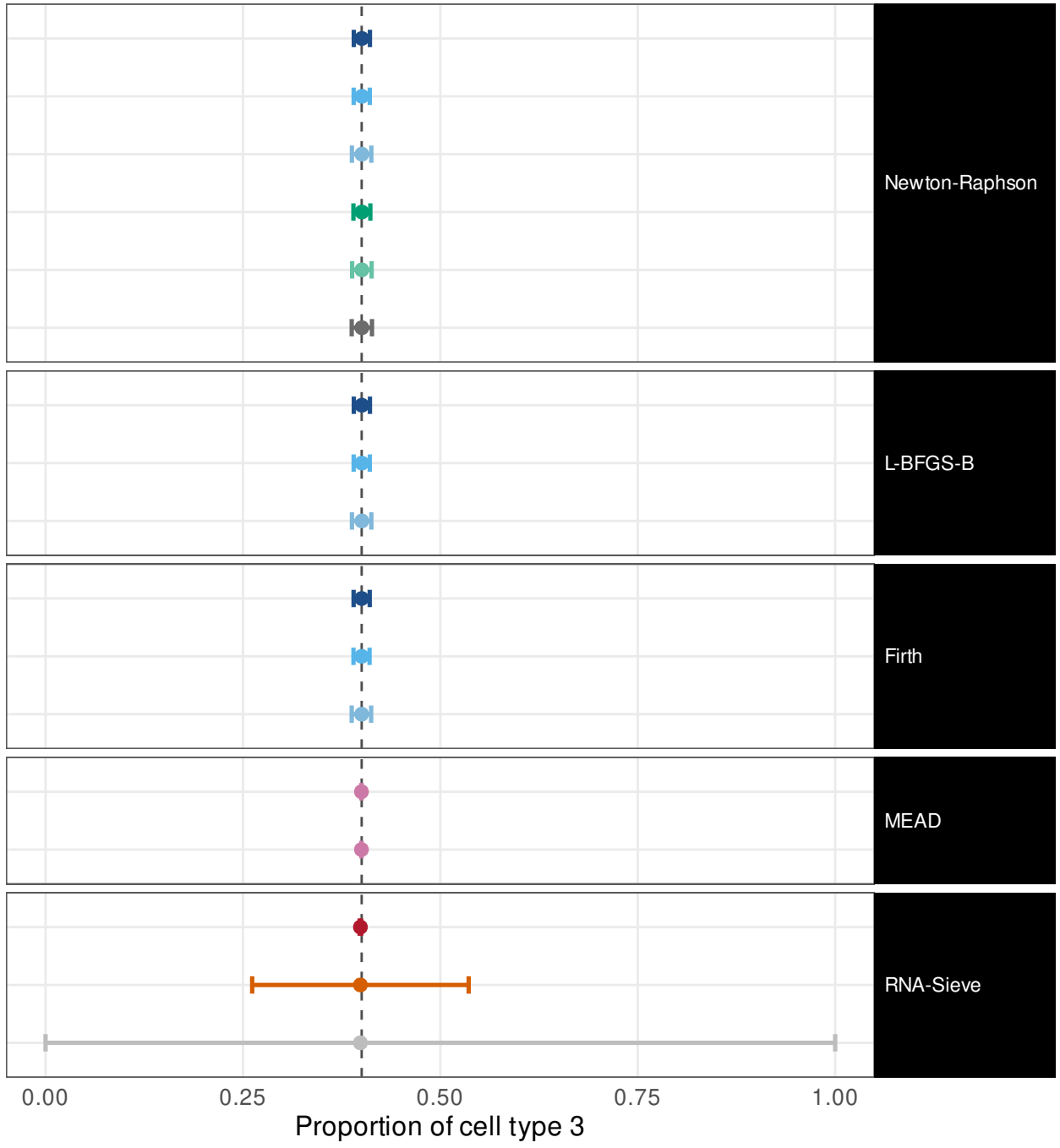


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / unbalanced / low sibling overlap / a = 5 / equal sizes

Wald (delta)	0.86 [0.81, 0.91]	w = 0.0204
ILR, mapped back	0.86 [0.81, 0.91]	w = 0.0204
ILR, REML	0.92 [0.88, 0.95]	w = 0.0246
Profile likelihood	0.88 [0.83, 0.92]	w = 0.0214
Profile, REML	0.92 [0.88, 0.95]	w = 0.0247
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0257
Wald (delta)	0.86 [0.81, 0.91]	w = 0.0204
ILR, mapped back	0.86 [0.81, 0.91]	w = 0.0204
ILR, REML	0.92 [0.87, 0.95]	w = 0.0246
Wald (delta)	0.87 [0.82, 0.91]	w = 0.0204
ILR, mapped back	0.86 [0.81, 0.91]	w = 0.0204
ILR, REML	0.93 [0.89, 0.96]	w = 0.0246
Method's own interval	0.02 [0.0078, 0.05]	w = 0.000214
Own interval, transcript fractions	0.02 [0.0078, 0.05]	w = 0.000214
Fisher information (Wald)	0.075 [0.046, 0.12]	w = 0.00187
Godambe sandwich	1 [0.98, 1]	w = 0.274
Released code, as published	1 [0.98, 1]	w = 1

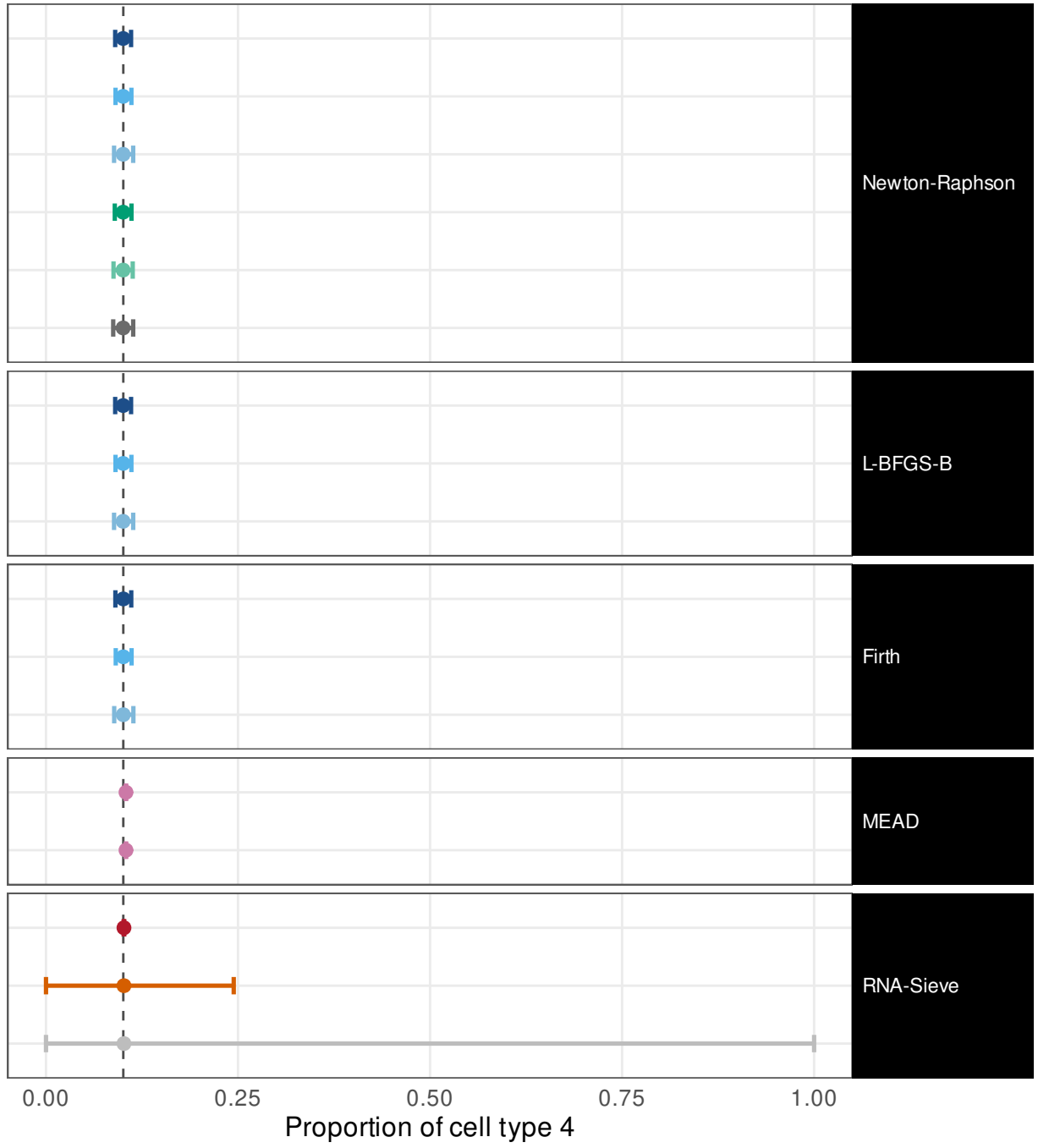


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / unbalanced / low sibling overlap / a = 5 / equal sizes

Wald (delta)	0.86 [0.81, 0.91]	w = 0.0207
ILR, mapped back	0.88 [0.83, 0.92]	w = 0.0206
ILR, REML	0.93 [0.89, 0.96]	w = 0.0249
Profile likelihood	0.88 [0.83, 0.92]	w = 0.0217
Profile, REML	0.93 [0.89, 0.96]	w = 0.025
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.026
Wald (delta)	0.86 [0.81, 0.91]	w = 0.0207
ILR, mapped back	0.86 [0.81, 0.91]	w = 0.0207
ILR, REML	0.94 [0.9, 0.97]	w = 0.0249
Wald (delta)	0.87 [0.82, 0.91]	w = 0.0207
ILR, mapped back	0.87 [0.82, 0.91]	w = 0.0206
ILR, REML	0.94 [0.89, 0.96]	w = 0.025
Method's own interval	0.03 [0.014, 0.064]	w = 0.000412
Own interval, transcript fractions	0.03 [0.014, 0.064]	w = 0.000412
Fisher information (Wald)	0.015 [0.0051, 0.043]	w = 0.000801
Godambe sandwich	1 [0.98, 1]	w = 0.244
Released code, as published	1 [0.98, 1]	w = 1

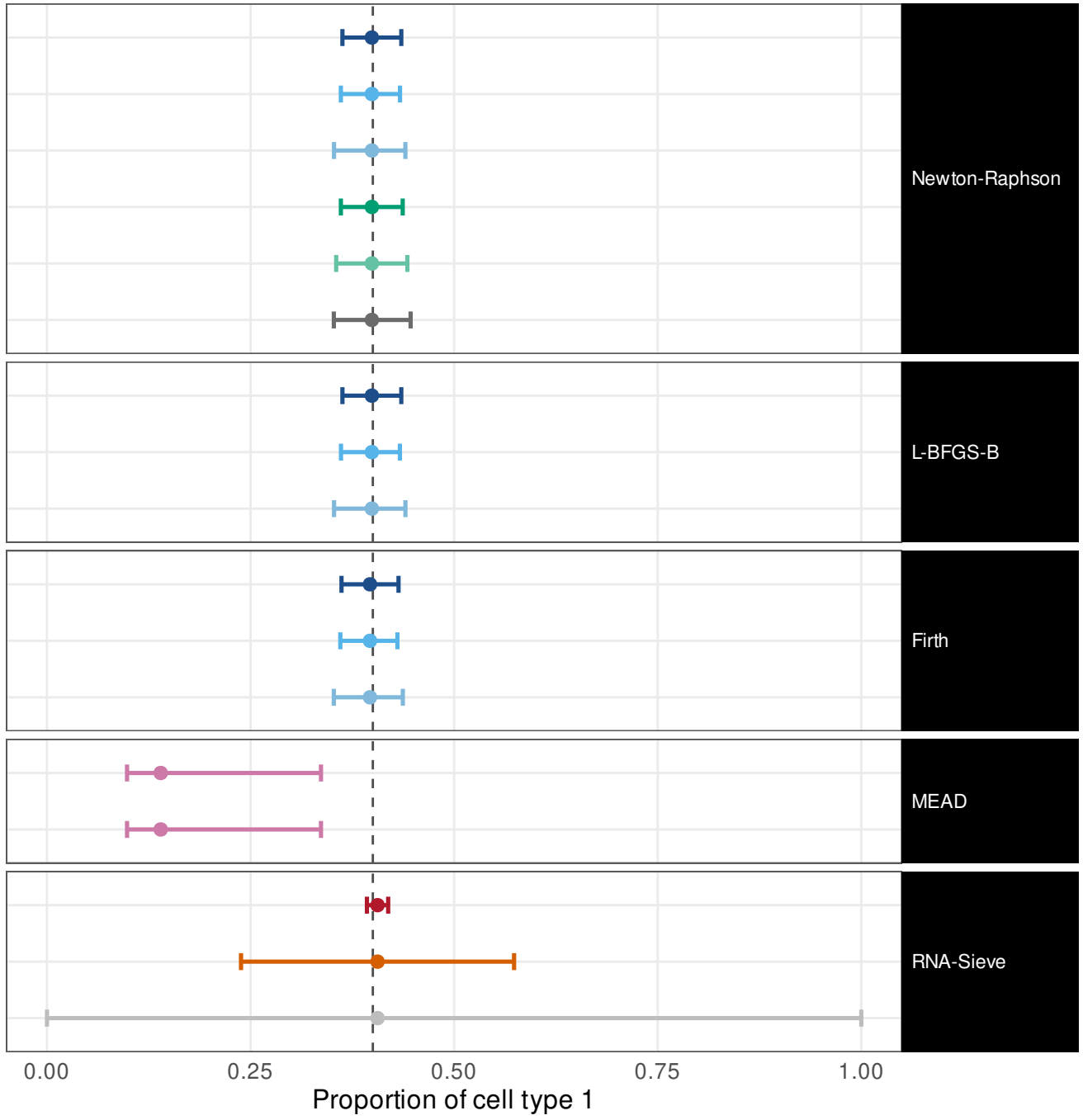


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / unbalanced / low sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.89 [0.84, 0.93]	w = 0.0723
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.0726
ILR, REML	0.96 [0.93, 0.98]	w = 0.0876
Profile likelihood	0.92 [0.87, 0.95]	w = 0.0759
Profile, REML	0.96 [0.93, 0.98]	w = 0.0874
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0943
Wald (delta)	0.89 [0.84, 0.93]	w = 0.0723
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.0723
ILR, REML	0.97 [0.94, 0.99]	w = 0.0876
Wald (delta)	0.91 [0.86, 0.94]	w = 0.07
ILR, mapped back	0.91 [0.86, 0.94]	w = 0.0701
ILR, REML	0.96 [0.92, 0.98]	w = 0.0848
Method's own interval	0.22 [0.16, 0.28]	w = 0.238
Own interval, transcript fractions	0.82 [0.77, 0.87]	w = 0.238
Fisher information (Wald)	0.38 [0.31, 0.44]	w = 0.0261
Godambe sandwich	1 [0.98, 1]	w = 0.335
Released code, as published	1 [0.98, 1]	w = 1

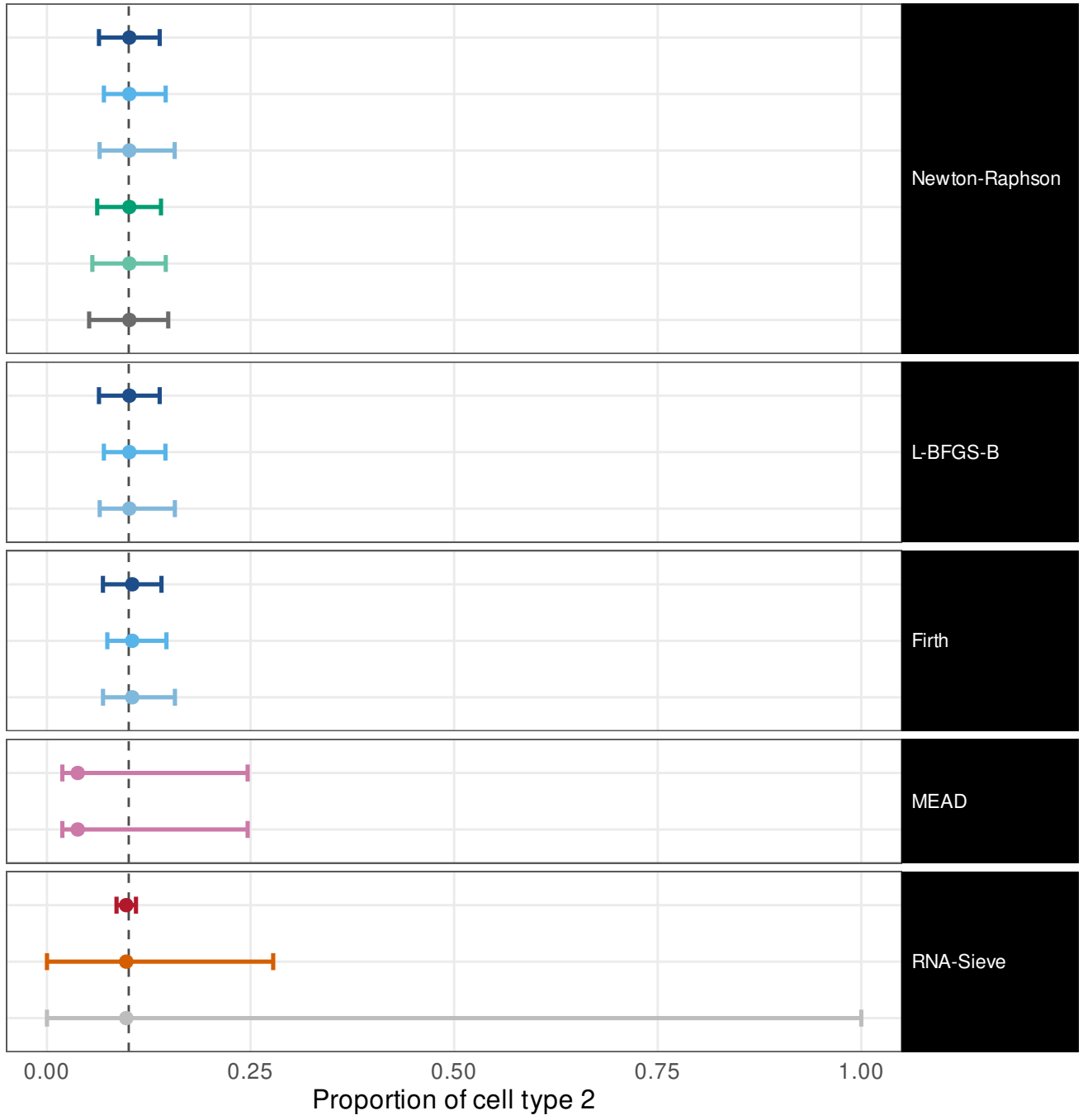


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / unbalanced / low sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.89 [0.84, 0.93]	w = 0.0746
ILR, mapped back	0.89 [0.84, 0.93]	w = 0.0759
ILR, REML	0.94 [0.9, 0.97]	w = 0.0922
Profile likelihood	0.91 [0.86, 0.94]	w = 0.0783
Profile, REML	0.96 [0.93, 0.98]	w = 0.0902
Oracle (Fisher at p*)	0.98 [0.96, 0.99]	w = 0.0971
Wald (delta)	0.89 [0.84, 0.93]	w = 0.0746
ILR, mapped back	0.9 [0.84, 0.93]	w = 0.0756
ILR, REML	0.94 [0.9, 0.97]	w = 0.0923
Wald (delta)	0.88 [0.83, 0.92]	w = 0.0719
ILR, mapped back	0.88 [0.83, 0.92]	w = 0.0726
ILR, REML	0.92 [0.88, 0.95]	w = 0.0883
Method's own interval	0.2 [0.15, 0.27]	w = 0.228
Own interval, transcript fractions	0.85 [0.79, 0.89]	w = 0.228
Fisher information (Wald)	0.4 [0.33, 0.47]	w = 0.0239
Godambe sandwich	1 [0.98, 1]	w = 0.278
Released code, as published	1 [0.98, 1]	w = 1

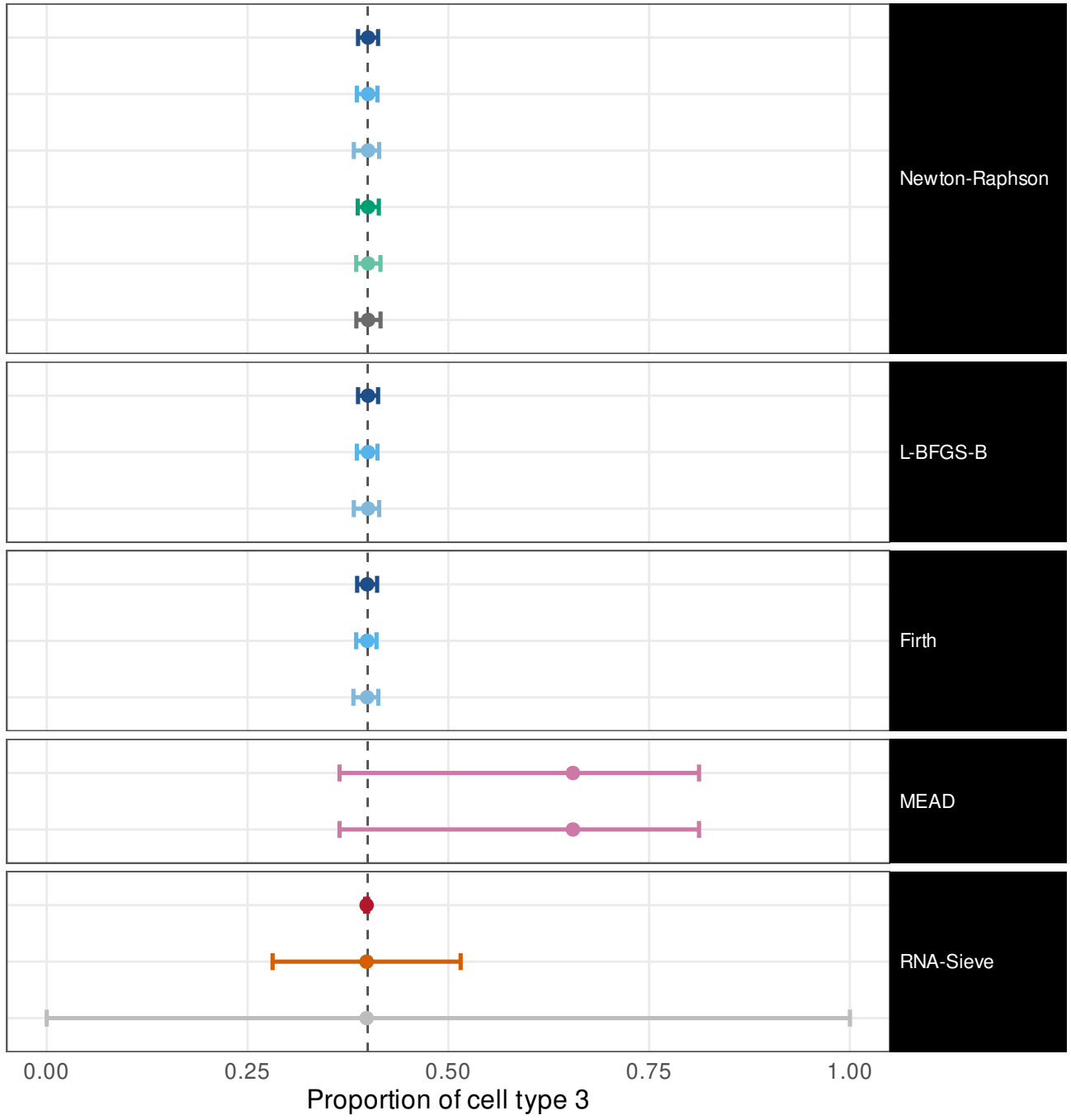


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / unbalanced / low sibling overlap / a = 5 / sizes 1:1:5:5

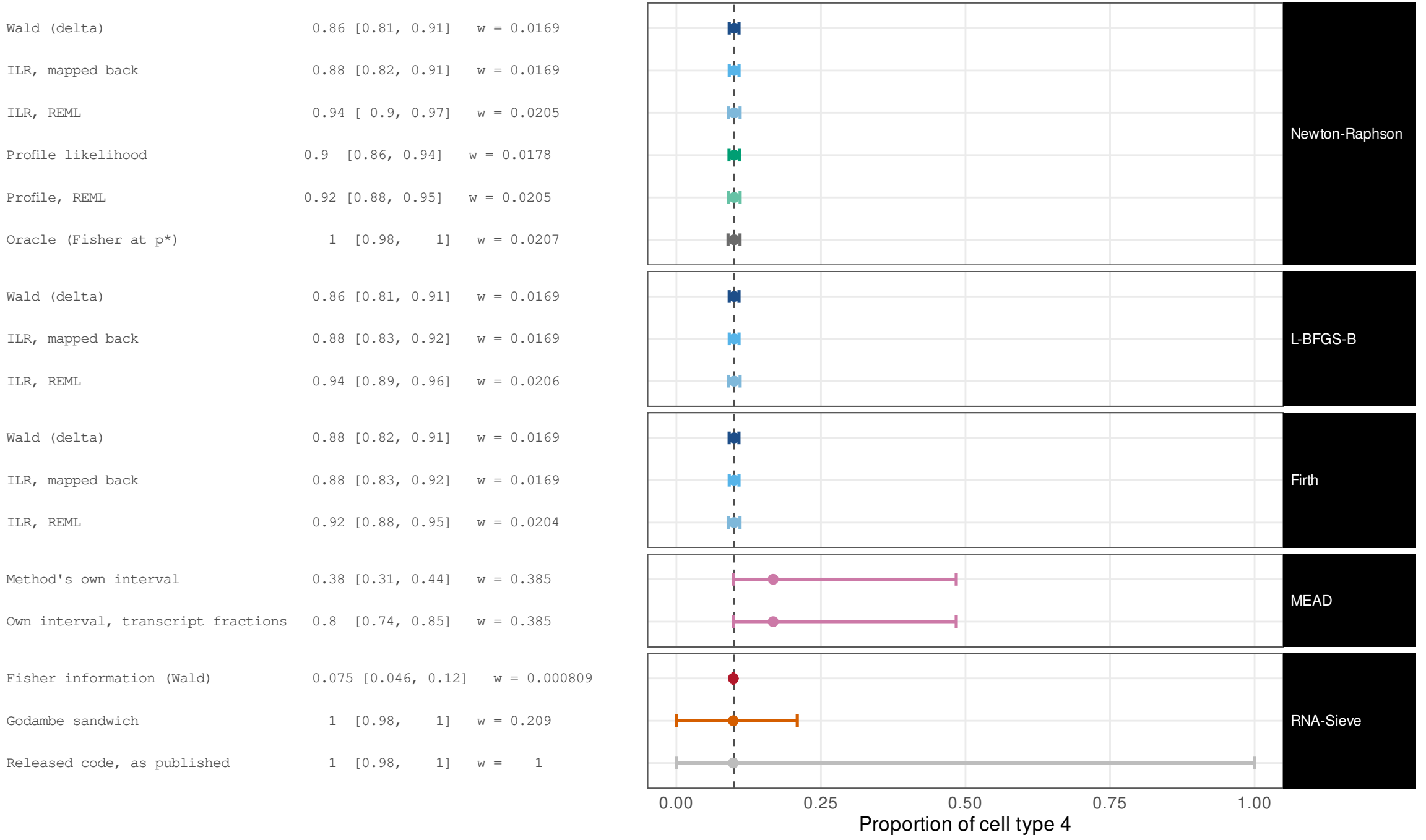
Wald (delta)	0.94 [0.9, 0.97]	w = 0.0249
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.0256
ILR, REML	0.98 [0.94, 0.99]	w = 0.0316
Profile likelihood	0.96 [0.92, 0.98]	w = 0.0262
Profile, REML	0.96 [0.93, 0.98]	w = 0.0302
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0301
Wald (delta)	0.94 [0.9, 0.97]	w = 0.0249
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.0257
ILR, REML	0.97 [0.94, 0.99]	w = 0.0315
Wald (delta)	0.93 [0.89, 0.96]	w = 0.0248
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.0253
ILR, REML	0.96 [0.93, 0.98]	w = 0.031
Method's own interval	0.44 [0.37, 0.5]	w = 0.448
Own interval, transcript fractions	0.71 [0.64, 0.77]	w = 0.448
Fisher information (Wald)	0.21 [0.16, 0.27]	w = 0.00355
Godambe sandwich	1 [0.98, 1]	w = 0.234
Released code, as published	1 [0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / unbalanced / low sibling overlap / a = 5 / sizes 1:1:5:5

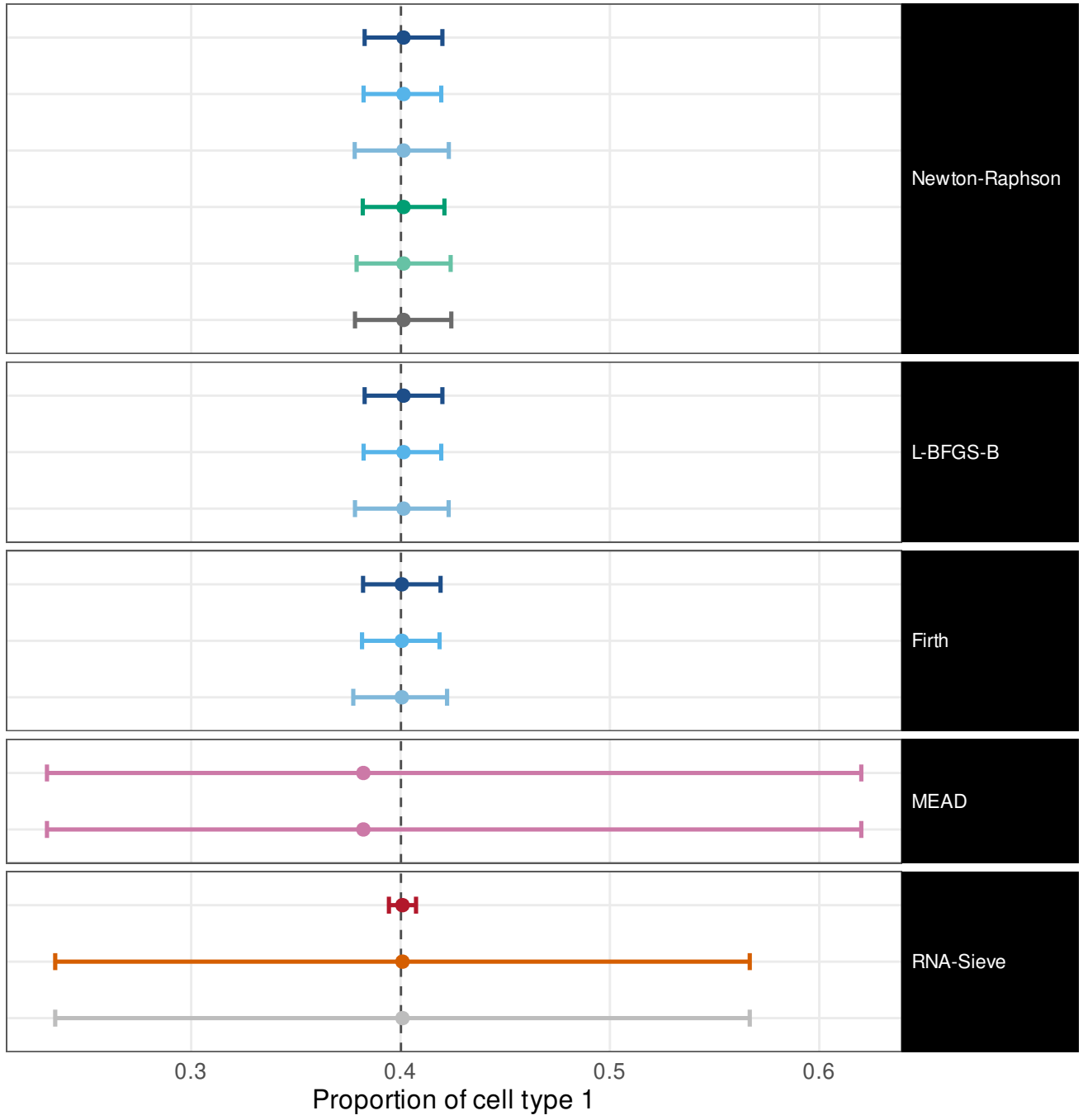


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / unbalanced / high sibling overlap / a = 1 / equal sizes

Wald (delta)	0.88 [0.83, 0.92]	w = 0.0371
ILR, mapped back	0.88 [0.83, 0.92]	w = 0.0371
ILR, REML	0.96 [0.92, 0.98]	w = 0.0449
Profile likelihood	0.9 [0.86, 0.94]	w = 0.039
Profile, REML	0.95 [0.91, 0.97]	w = 0.0449
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.046
Wald (delta)	0.88 [0.83, 0.92]	w = 0.0371
ILR, mapped back	0.88 [0.83, 0.92]	w = 0.0371
ILR, REML	0.96 [0.92, 0.98]	w = 0.0447
Wald (delta)	0.88 [0.82, 0.91]	w = 0.037
ILR, mapped back	0.88 [0.83, 0.92]	w = 0.037
ILR, REML	0.95 [0.91, 0.97]	w = 0.0448
Method's own interval	0.69 [0.62, 0.75]	w = 0.389
Own interval, transcript fractions	0.69 [0.62, 0.75]	w = 0.389
Fisher information (Wald)	0.39 [0.33, 0.46]	w = 0.0129
Godambe sandwich	1 [0.98, 1]	w = 0.332
Released code, as published	1 [0.98, 1]	w = 0.332

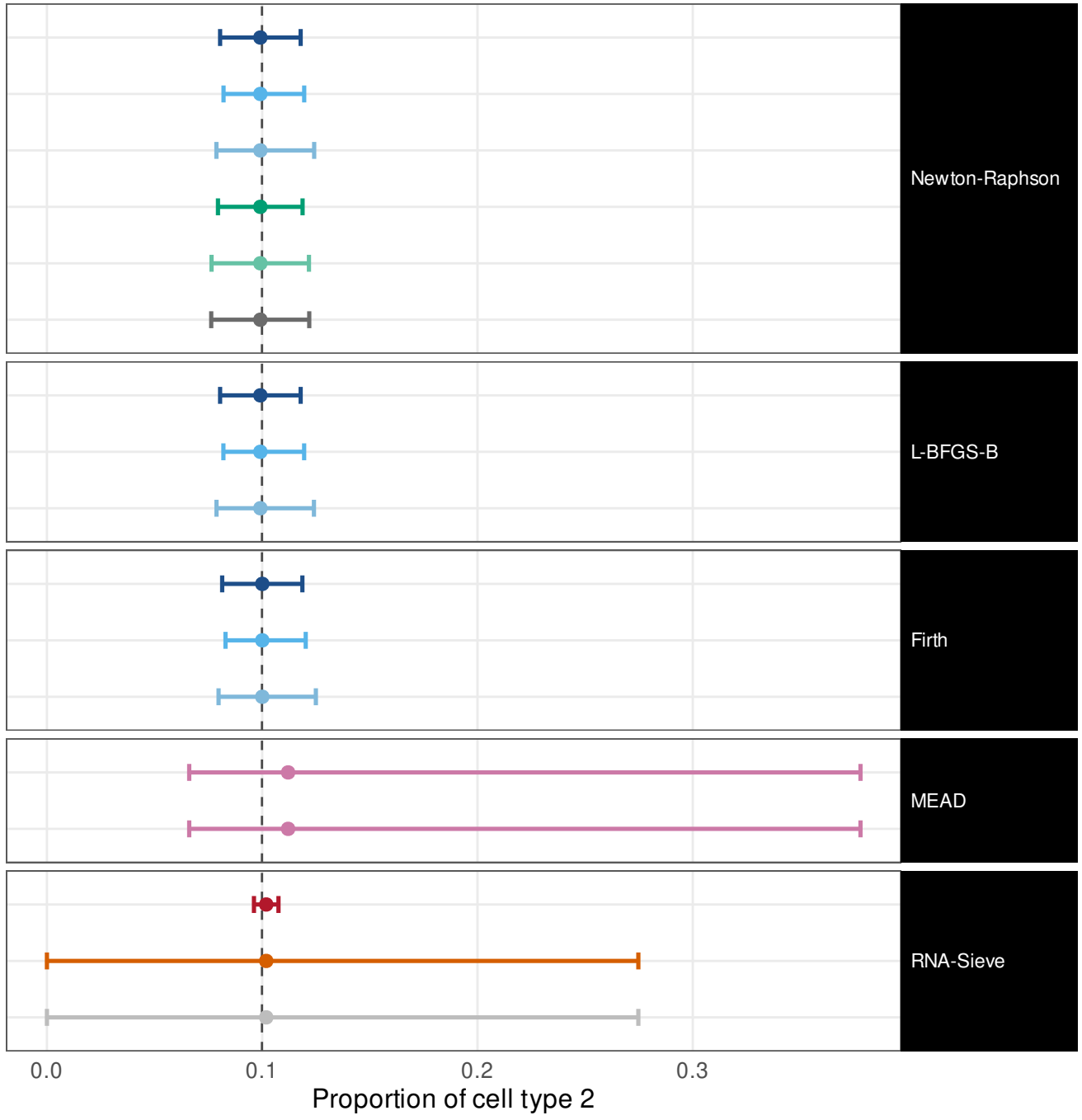


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / unbalanced / high sibling overlap / a = 1 / equal sizes

Wald (delta)	0.91 [0.86, 0.94]	w = 0.0374
ILR, mapped back	0.9 [0.84, 0.93]	w = 0.0375
ILR, REML	0.94 [0.9, 0.97]	w = 0.0454
Profile likelihood	0.92 [0.87, 0.95]	w = 0.0393
Profile, REML	0.95 [0.91, 0.97]	w = 0.0453
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0455
Wald (delta)	0.91 [0.86, 0.94]	w = 0.0374
ILR, mapped back	0.9 [0.84, 0.93]	w = 0.0375
ILR, REML	0.96 [0.92, 0.98]	w = 0.0453
Wald (delta)	0.9 [0.86, 0.94]	w = 0.0373
ILR, mapped back	0.89 [0.84, 0.93]	w = 0.0373
ILR, REML	0.95 [0.91, 0.97]	w = 0.0452
Method's own interval	0.78 [0.72, 0.84]	w = 0.312
Own interval, transcript fractions	0.78 [0.72, 0.84]	w = 0.312
Fisher information (Wald)	0.36 [0.3, 0.43]	w = 0.0114
Godambe sandwich	1 [0.98, 1]	w = 0.275
Released code, as published	1 [0.98, 1]	w = 0.275

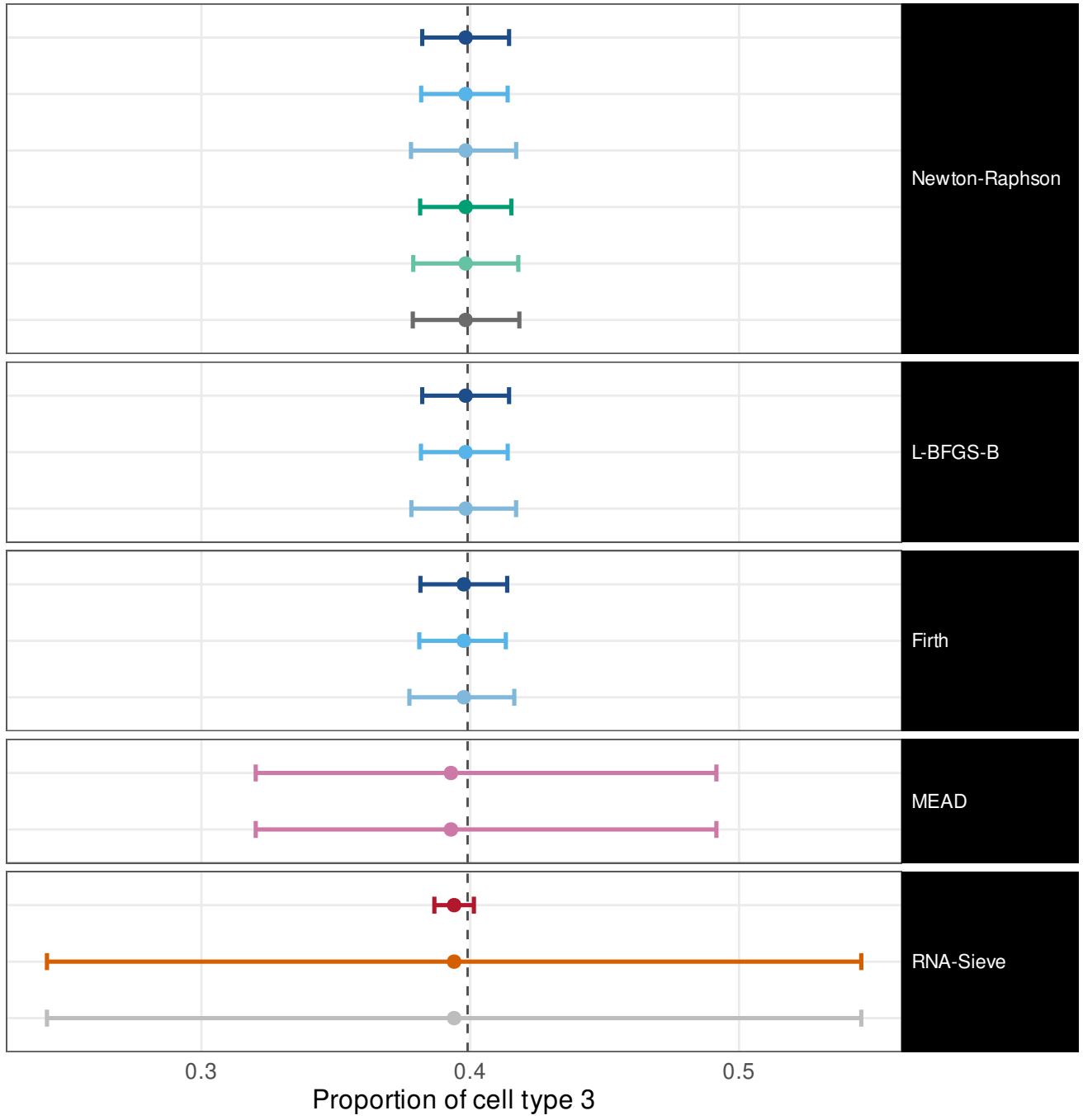


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / unbalanced / high sibling overlap / a = 1 / equal sizes

Wald (delta)	0.88 [0.83, 0.92]	w = 0.0323
ILR, mapped back	0.88 [0.83, 0.92]	w = 0.0322
ILR, REML	0.93 [0.89, 0.96]	w = 0.0391
Profile likelihood	0.9 [0.84, 0.93]	w = 0.0339
Profile, REML	0.94 [0.9, 0.97]	w = 0.0391
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0396
Wald (delta)	0.88 [0.83, 0.92]	w = 0.0323
ILR, mapped back	0.88 [0.83, 0.92]	w = 0.0323
ILR, REML	0.94 [0.9, 0.97]	w = 0.039
Wald (delta)	0.88 [0.83, 0.92]	w = 0.0323
ILR, mapped back	0.88 [0.83, 0.92]	w = 0.0322
ILR, REML	0.94 [0.89, 0.96]	w = 0.039
Method's own interval	0.94 [0.9, 0.97]	w = 0.171
Own interval, transcript fractions	0.94 [0.9, 0.97]	w = 0.171
Fisher information (Wald)	0.46 [0.4, 0.53]	w = 0.0147
Godambe sandwich	1 [0.98, 1]	w = 0.303
Released code, as published	1 [0.98, 1]	w = 0.303

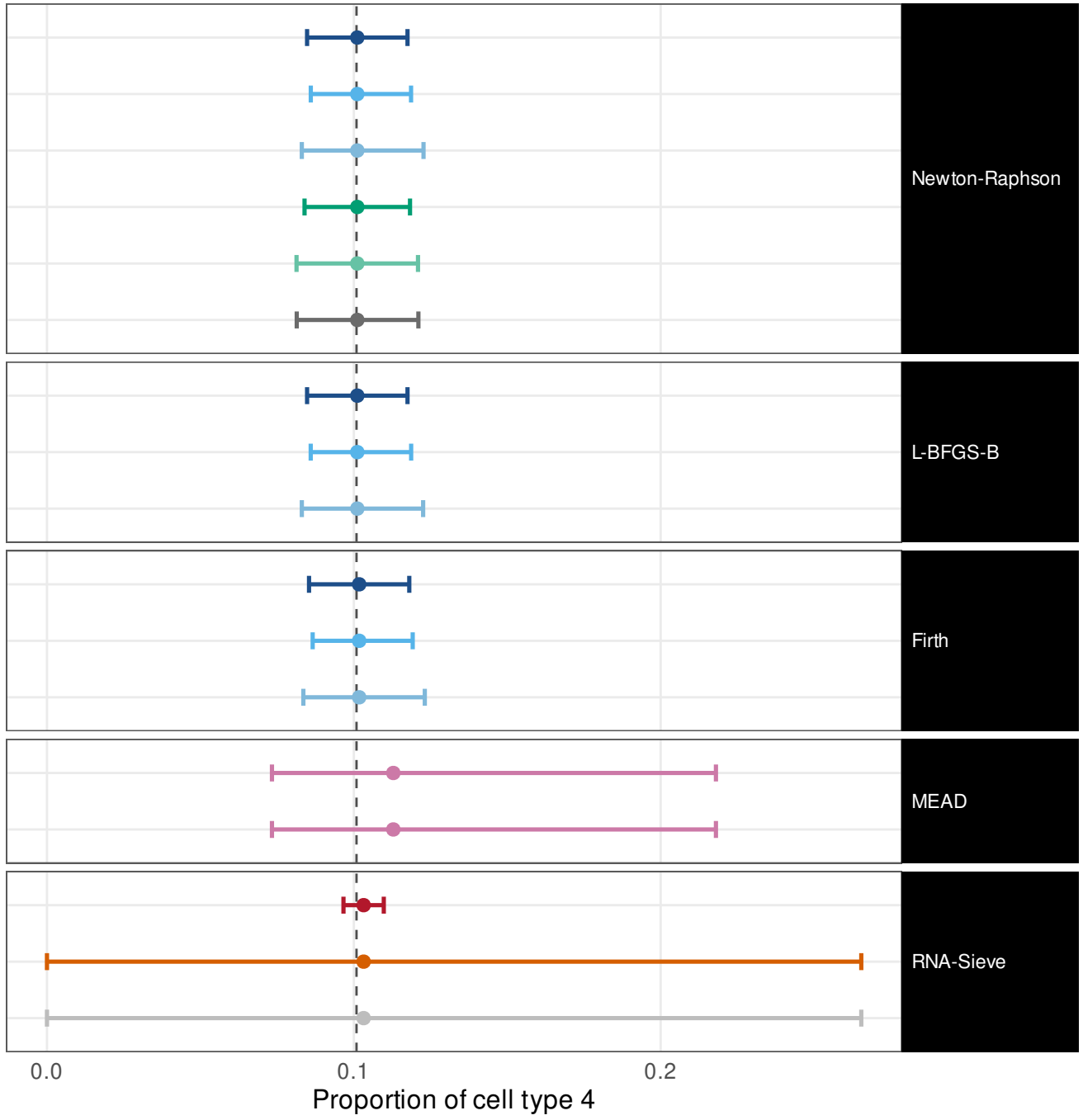


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / unbalanced / high sibling overlap / a = 1 / equal sizes

Wald (delta)	0.9	[0.85, 0.93]	w = 0.0327
ILR, mapped back	0.92	[0.87, 0.95]	w = 0.0327
ILR, REML	0.97	[0.94, 0.99]	w = 0.0397
Profile likelihood	0.92	[0.87, 0.95]	w = 0.0344
Profile, REML	0.96	[0.92, 0.98]	w = 0.0396
Oracle (Fisher at p*)	1	[0.98, 1]	w = 0.0396
Wald (delta)	0.9	[0.85, 0.93]	w = 0.0327
ILR, mapped back	0.92	[0.87, 0.95]	w = 0.0327
ILR, REML	0.96	[0.93, 0.98]	w = 0.0395
Wald (delta)	0.9	[0.86, 0.94]	w = 0.0327
ILR, mapped back	0.92	[0.87, 0.95]	w = 0.0326
ILR, REML	0.96	[0.93, 0.98]	w = 0.0396
Method's own interval	0.9	[0.86, 0.94]	w = 0.145
Own interval, transcript fractions	0.9	[0.86, 0.94]	w = 0.145
Fisher information (Wald)	0.52	[0.45, 0.58]	w = 0.0131
Godambe sandwich	1	[0.98, 1]	w = 0.265
Released code, as published	1	[0.98, 1]	w = 0.265

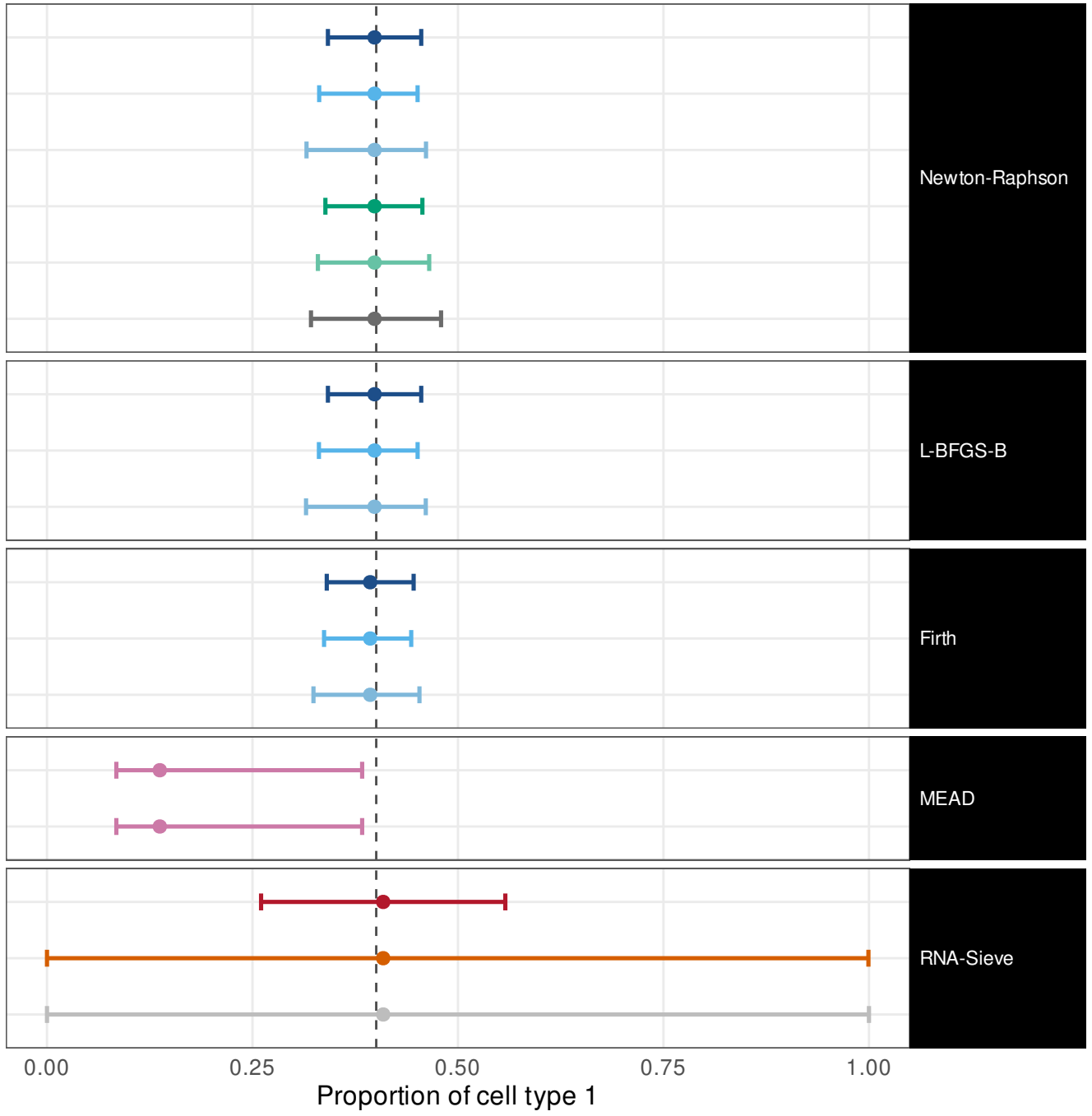


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / unbalanced / high sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.89 [0.84, 0.93]	w = 0.114
ILR, mapped back	0.91 [0.86, 0.94]	w = 0.12
ILR, REML	0.96 [0.92, 0.98]	w = 0.145
Profile likelihood	0.91 [0.86, 0.94]	w = 0.118
Profile, REML	0.92 [0.88, 0.95]	w = 0.135
Oracle (Fisher at p*)	0.97 [0.93, 0.99]	w = 0.158
Wald (delta)	0.89 [0.84, 0.93]	w = 0.114
ILR, mapped back	0.9 [0.85, 0.93]	w = 0.12
ILR, REML	0.96 [0.93, 0.98]	w = 0.146
Wald (delta)	0.88 [0.83, 0.92]	w = 0.106
ILR, mapped back	0.88 [0.83, 0.92]	w = 0.106
ILR, REML	0.94 [0.9, 0.97]	w = 0.129
Method's own interval	0.26 [0.21, 0.33]	w = 0.299
Own interval, transcript fractions	0.86 [0.81, 0.9]	w = 0.299
Fisher information (Wald)	1 [0.98, 1]	w = 0.297
Godambe sandwich	1 [0.98, 1]	w = 0.999
Released code, as published	1 [0.98, 1]	w = 1

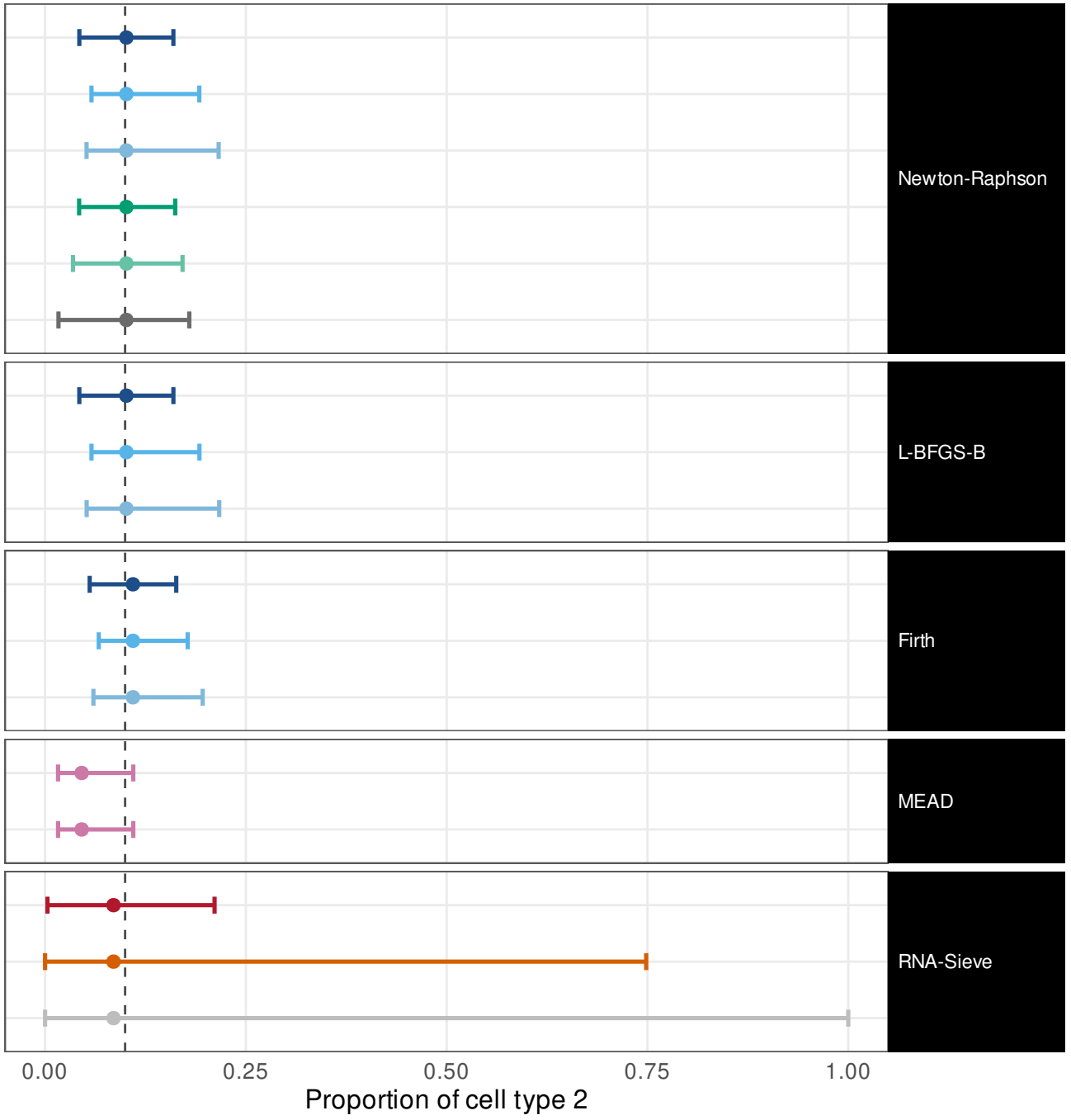


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / unbalanced / high sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.89 [0.84, 0.93]	w = 0.117
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.134
ILR, REML	0.94 [0.89, 0.96]	w = 0.164
Profile likelihood	0.89 [0.84, 0.93]	w = 0.12
Profile, REML	0.94 [0.9, 0.97]	w = 0.136
Oracle (Fisher at p*)	0.97 [0.94, 0.99]	w = 0.163
Wald (delta)	0.89 [0.84, 0.93]	w = 0.117
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.134
ILR, REML	0.92 [0.88, 0.95]	w = 0.165
Wald (delta)	0.88 [0.83, 0.92]	w = 0.108
ILR, mapped back	0.87 [0.82, 0.91]	w = 0.111
ILR, REML	0.92 [0.87, 0.95]	w = 0.136
Method's own interval	0.14 [0.099, 0.19]	w = 0.0937
Own interval, transcript fractions	0.84 [0.78, 0.88]	w = 0.0937
Fisher information (Wald)	1 [0.98, 1]	w = 0.208
Godambe sandwich	1 [0.98, 1]	w = 0.748
Released code, as published	1 [0.98, 1]	w = 1

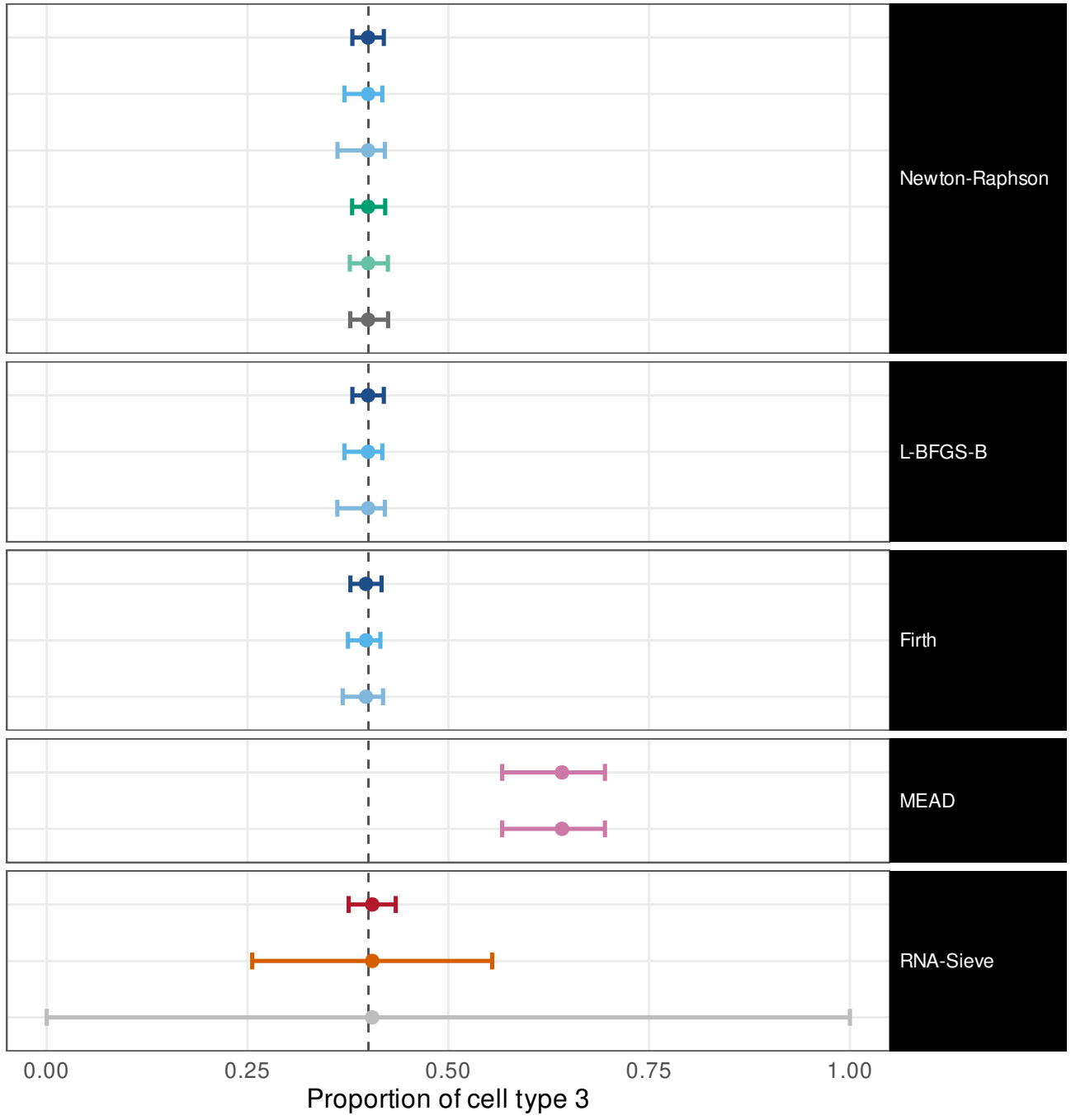


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / unbalanced / high sibling overlap / a = 1 / sizes 1:1:5:5

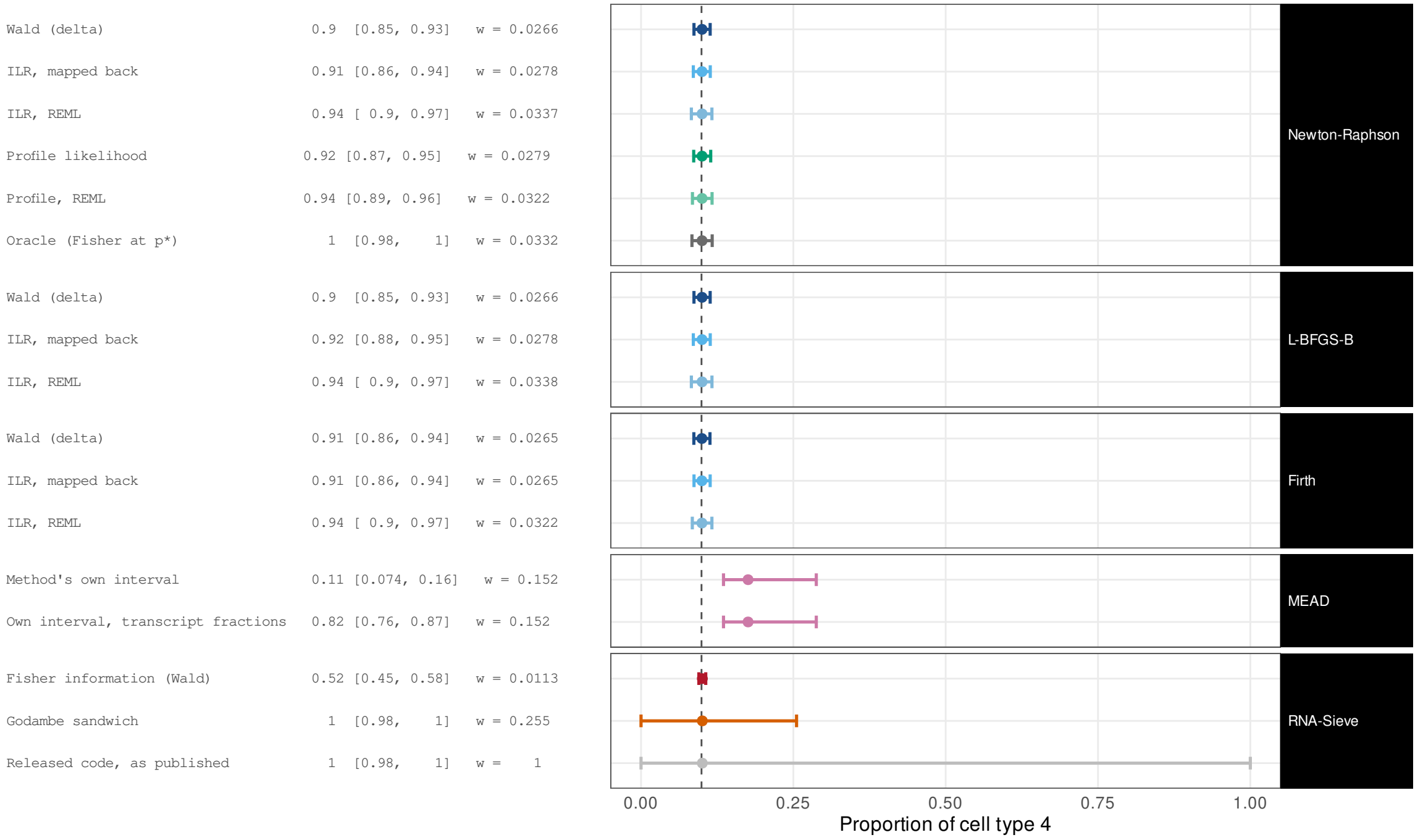
Wald (delta)	0.92 [0.88, 0.95]	w = 0.0391
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.0472
ILR, REML	0.96 [0.93, 0.98]	w = 0.059
Profile likelihood	0.93 [0.89, 0.96]	w = 0.0412
Profile, REML	0.94 [0.9, 0.97]	w = 0.0474
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0472
Wald (delta)	0.92 [0.88, 0.95]	w = 0.0391
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.0472
ILR, REML	0.96 [0.93, 0.98]	w = 0.0592
Wald (delta)	0.92 [0.87, 0.95]	w = 0.0388
ILR, mapped back	0.9 [0.86, 0.94]	w = 0.0404
ILR, REML	0.96 [0.93, 0.98]	w = 0.0502
Method's own interval	0.075 [0.046, 0.12]	w = 0.128
Own interval, transcript fractions	0.6 [0.53, 0.66]	w = 0.128
Fisher information (Wald)	0.96 [0.92, 0.98]	w = 0.0586
Godambe sandwich	1 [0.98, 1]	w = 0.299
Released code, as published	1 [0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / unbalanced / high sibling overlap / a = 1 / sizes 1:1:5:5

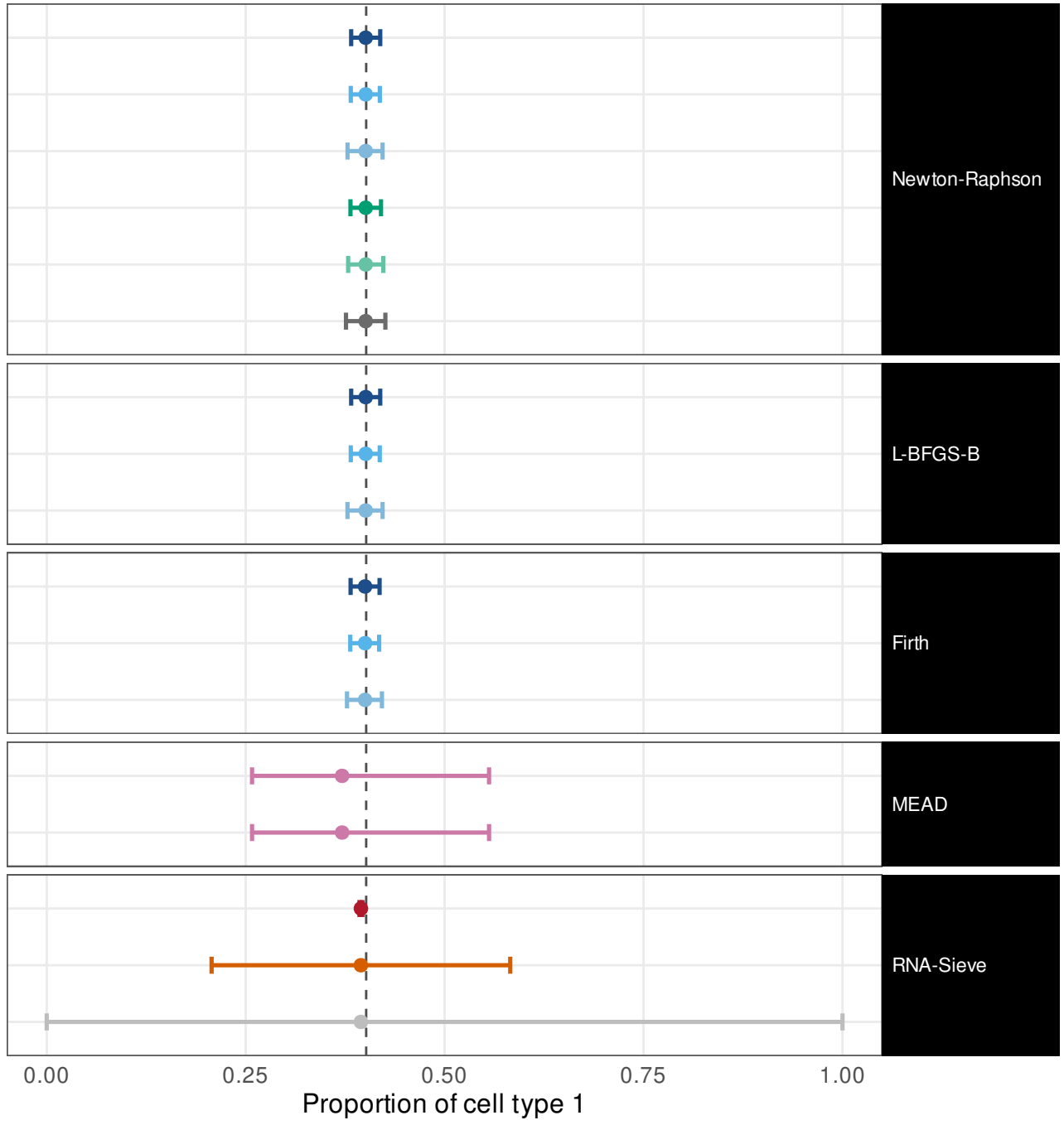


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / unbalanced / high sibling overlap / a = 5 / equal sizes

Wald (delta)	0.92 [0.87, 0.95]	w = 0.0365
ILR, mapped back	0.91 [0.86, 0.94]	w = 0.0365
ILR, REML	0.95 [0.91, 0.97]	w = 0.0441
Profile likelihood	0.92 [0.88, 0.95]	w = 0.0384
Profile, REML	0.96 [0.92, 0.98]	w = 0.0442
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0495
Wald (delta)	0.92 [0.87, 0.95]	w = 0.0365
ILR, mapped back	0.9 [0.86, 0.94]	w = 0.0364
ILR, REML	0.96 [0.92, 0.98]	w = 0.0441
Wald (delta)	0.91 [0.86, 0.94]	w = 0.0364
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.0363
ILR, REML	0.95 [0.91, 0.97]	w = 0.0439
Method's own interval	0.42 [0.35, 0.48]	w = 0.298
Own interval, transcript fractions	0.42 [0.35, 0.48]	w = 0.298
Fisher information (Wald)	0.06 [0.035, 0.1]	w = 0.00356
Godambe sandwich	1 [0.98, 1]	w = 0.375
Released code, as published	1 [0.98, 1]	w = 1

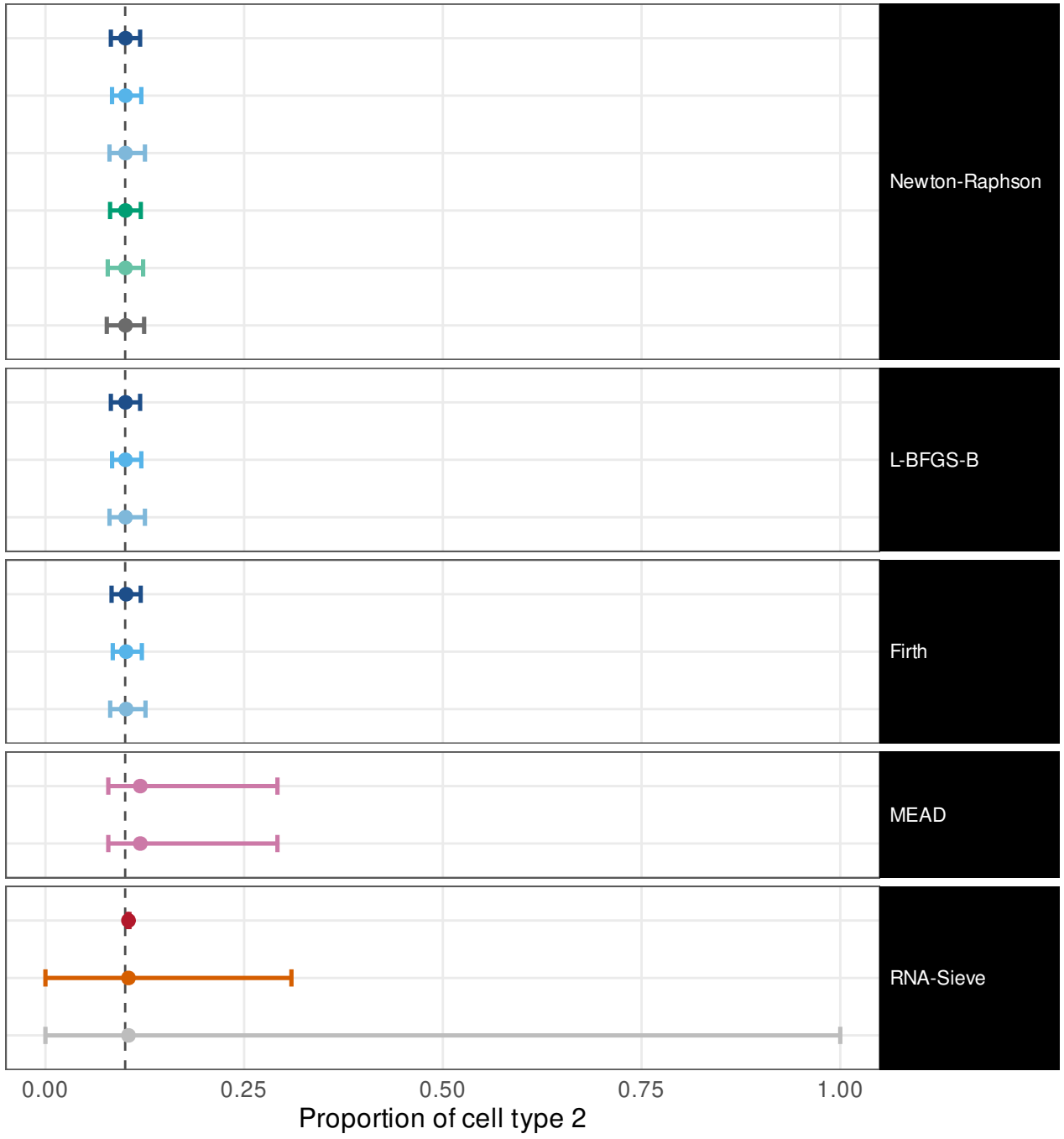


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / unbalanced / high sibling overlap / a = 5 / equal sizes

Wald (delta)	0.92 [0.87, 0.95]	w = 0.0368
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.0368
ILR, REML	0.95 [0.91, 0.97]	w = 0.0446
Profile likelihood	0.93 [0.89, 0.96]	w = 0.0386
Profile, REML	0.95 [0.91, 0.97]	w = 0.0445
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.047
Wald (delta)	0.92 [0.87, 0.95]	w = 0.0368
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.0368
ILR, REML	0.96 [0.92, 0.98]	w = 0.0446
Wald (delta)	0.92 [0.87, 0.95]	w = 0.0367
ILR, mapped back	0.91 [0.86, 0.94]	w = 0.0366
ILR, REML	0.96 [0.92, 0.98]	w = 0.0444
Method's own interval	0.67 [0.6, 0.73]	w = 0.213
Own interval, transcript fractions	0.67 [0.6, 0.73]	w = 0.213
Fisher information (Wald)	0.065 [0.038, 0.11]	w = 0.00239
Godambe sandwich	1 [0.98, 1]	w = 0.309
Released code, as published	1 [0.98, 1]	w = 1

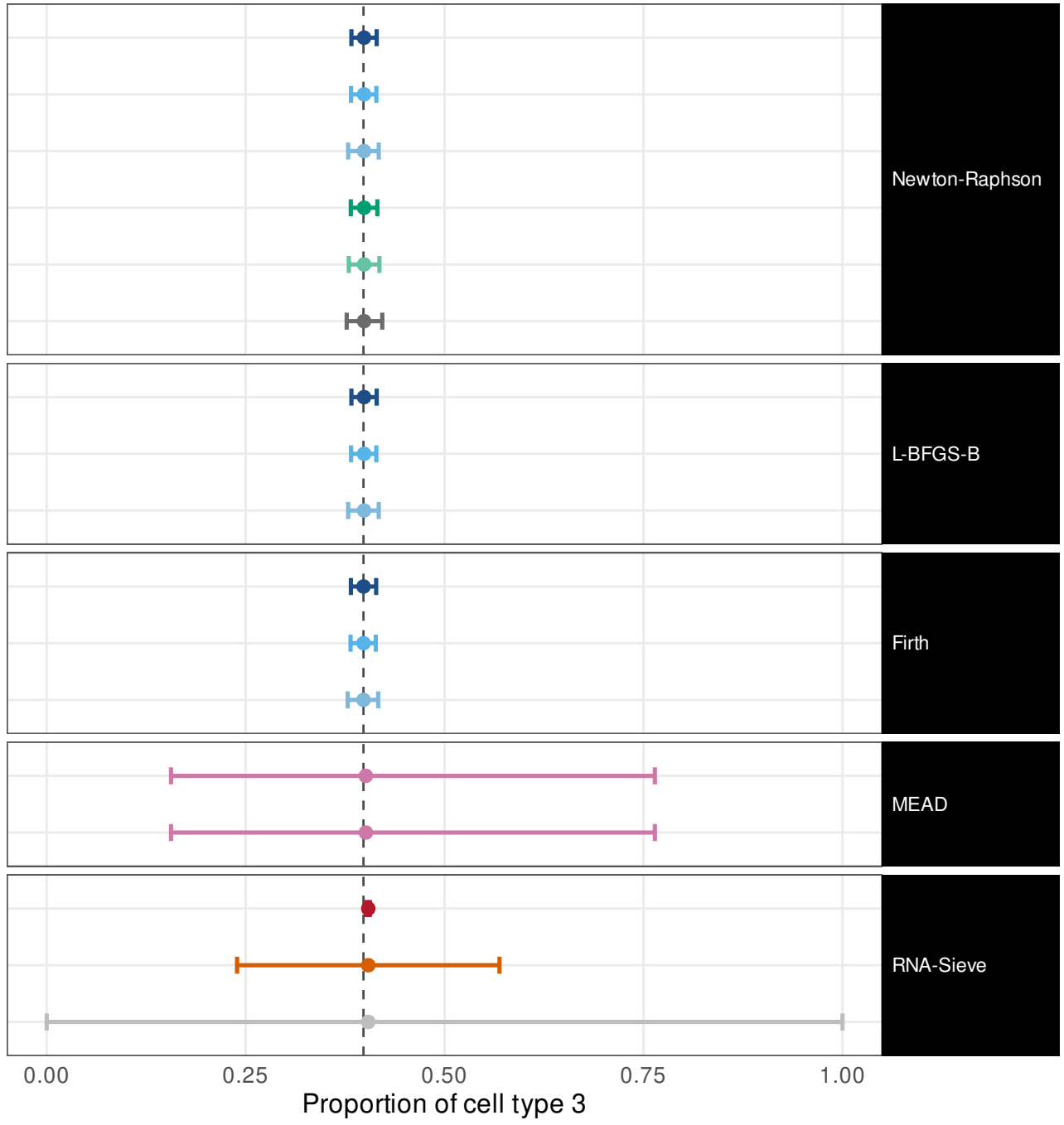


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / unbalanced / high sibling overlap / a = 5 / equal sizes

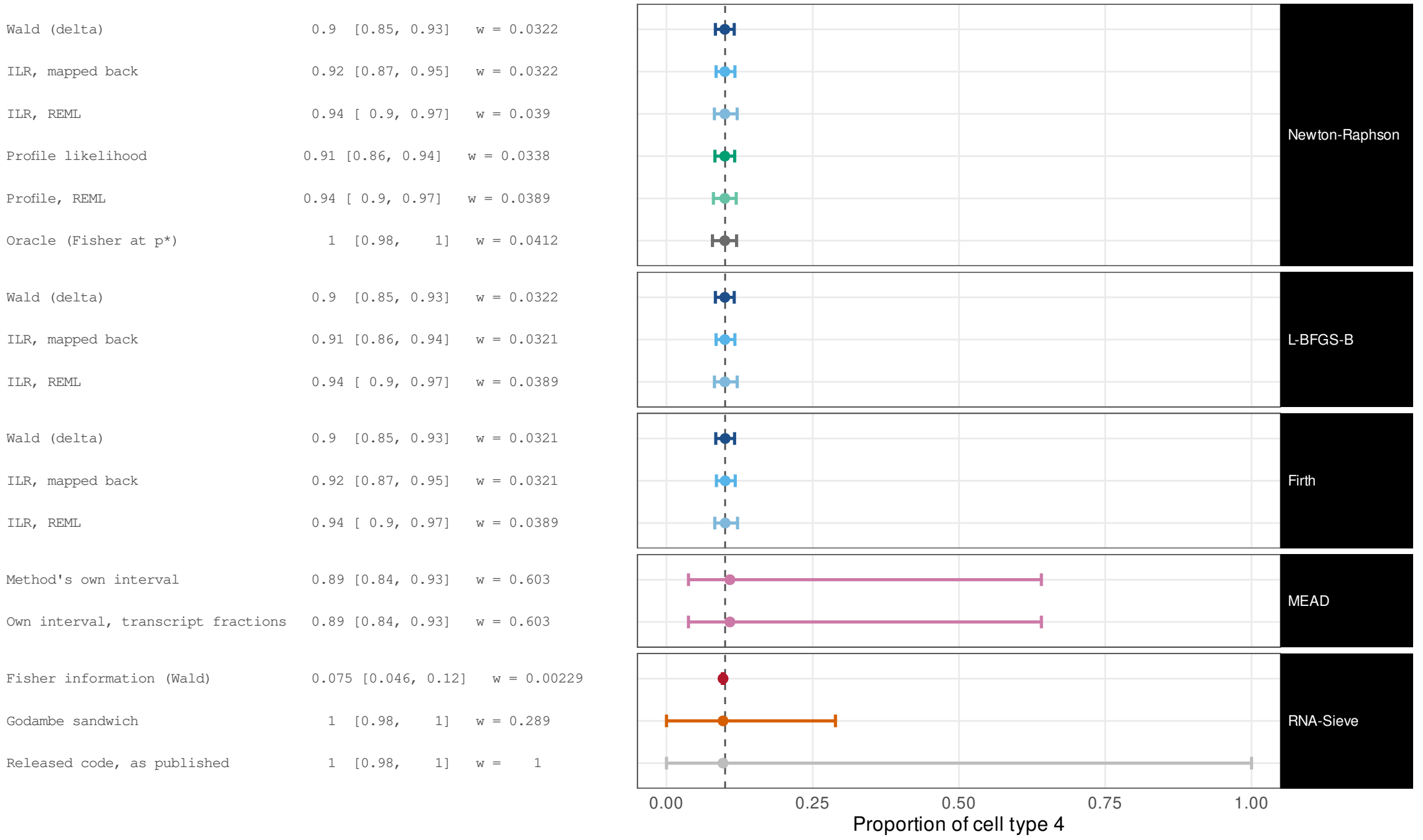
Wald (delta)	0.86 [0.81, 0.9]	w = 0.0318
ILR, mapped back	0.88 [0.82, 0.91]	w = 0.0318
ILR, REML	0.93 [0.89, 0.96]	w = 0.0384
Profile likelihood	0.89 [0.84, 0.93]	w = 0.0334
Profile, REML	0.94 [0.89, 0.96]	w = 0.0384
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0447
Wald (delta)	0.86 [0.81, 0.9]	w = 0.0318
ILR, mapped back	0.87 [0.82, 0.91]	w = 0.0317
ILR, REML	0.94 [0.89, 0.96]	w = 0.0384
Wald (delta)	0.86 [0.81, 0.91]	w = 0.0317
ILR, mapped back	0.88 [0.82, 0.91]	w = 0.0317
ILR, REML	0.92 [0.88, 0.95]	w = 0.0383
Method's own interval	0.94 [0.89, 0.96]	w = 0.608
Own interval, transcript fractions	0.94 [0.89, 0.96]	w = 0.608
Fisher information (Wald)	0.12 [0.078, 0.17]	w = 0.00388
Godambe sandwich	1 [0.98, 1]	w = 0.33
Released code, as published	1 [0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / unbalanced / high sibling overlap / a = 5 / equal sizes

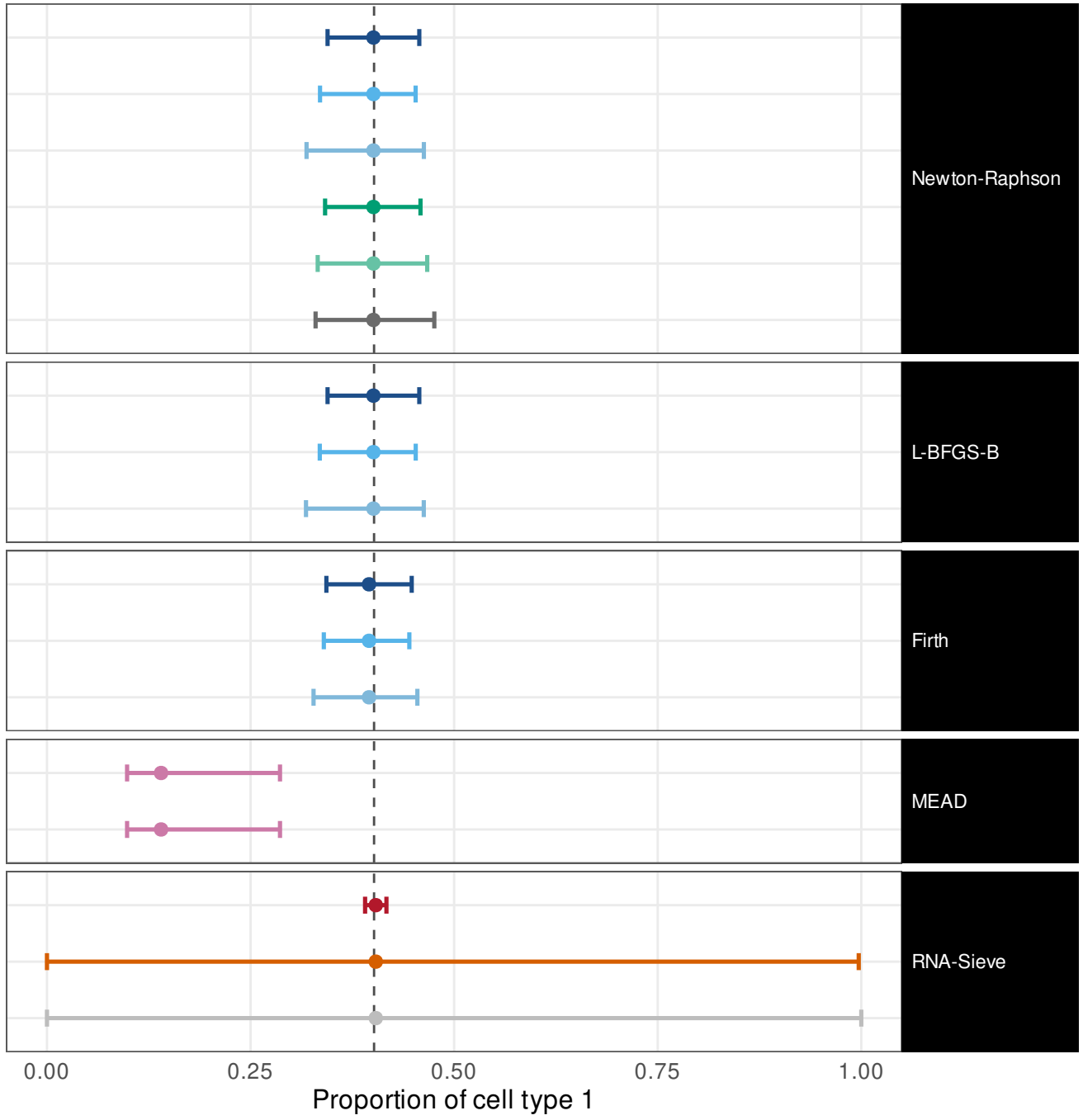


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / unbalanced / high sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.91 [0.86, 0.94]	w = 0.113
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.117
ILR, REML	0.97 [0.94, 0.99]	w = 0.144
Profile likelihood	0.92 [0.87, 0.95]	w = 0.117
Profile, REML	0.95 [0.91, 0.97]	w = 0.134
Oracle (Fisher at p*)	0.97 [0.93, 0.98]	w = 0.146
Wald (delta)	0.91 [0.86, 0.94]	w = 0.113
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.118
ILR, REML	0.97 [0.94, 0.99]	w = 0.145
Wald (delta)	0.92 [0.87, 0.95]	w = 0.105
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.105
ILR, REML	0.97 [0.94, 0.99]	w = 0.128
Method's own interval	0.14 [0.1, 0.2]	w = 0.188
Own interval, transcript fractions	0.9 [0.85, 0.93]	w = 0.188
Fisher information (Wald)	0.3 [0.25, 0.37]	w = 0.0262
Godambe sandwich	1 [0.98, 1]	w = 0.997
Released code, as published	1 [0.98, 1]	w = 1

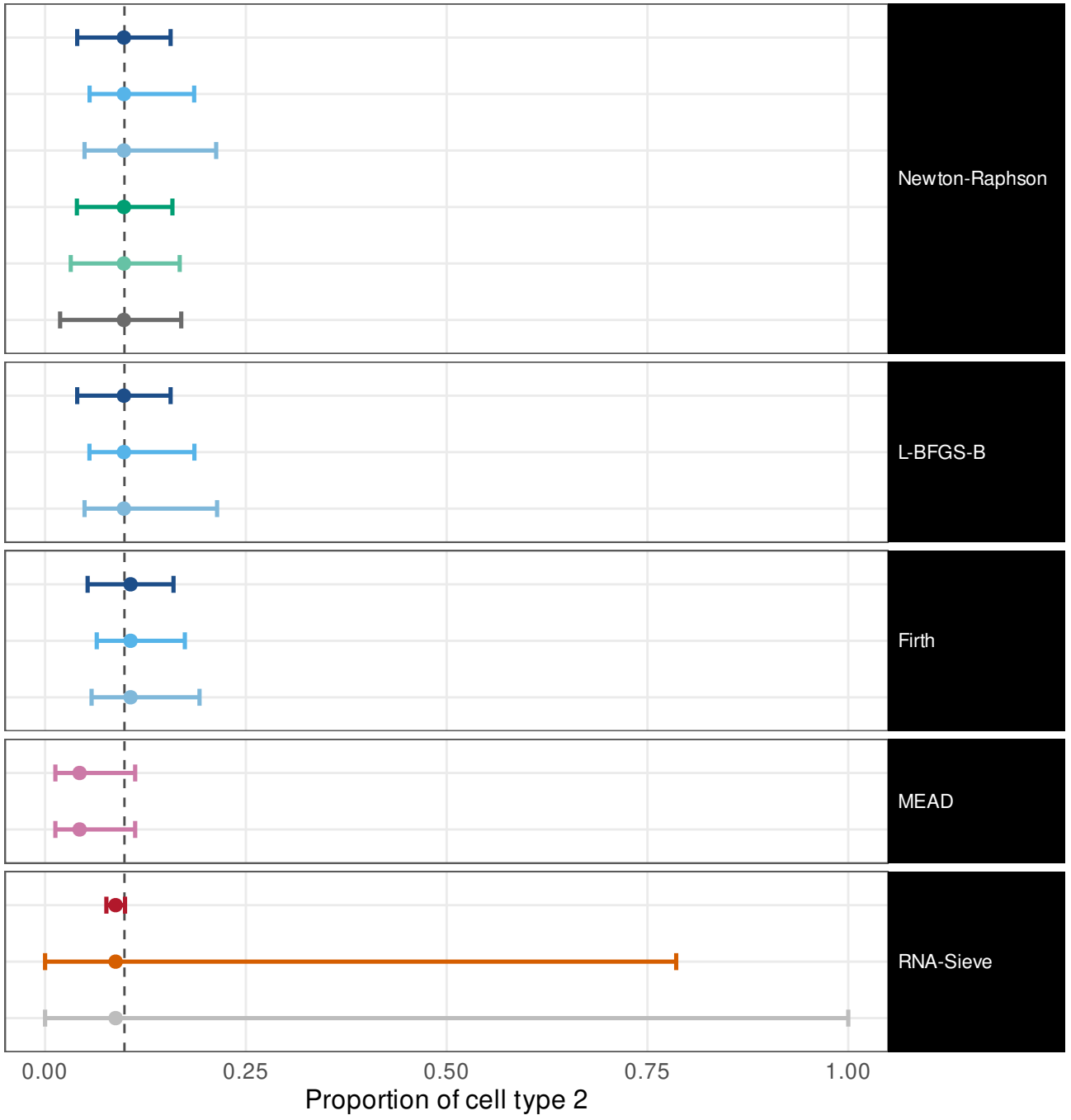


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / unbalanced / high sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.9 [0.85, 0.93]	w = 0.116
ILR, mapped back	0.92 [0.88, 0.95]	w = 0.13
ILR, REML	0.96 [0.92, 0.98]	w = 0.164
Profile likelihood	0.92 [0.87, 0.95]	w = 0.119
Profile, REML	0.95 [0.91, 0.97]	w = 0.136
Oracle (Fisher at p*)	0.96 [0.92, 0.98]	w = 0.151
Wald (delta)	0.9 [0.85, 0.93]	w = 0.116
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.131
ILR, REML	0.96 [0.92, 0.98]	w = 0.165
Wald (delta)	0.88 [0.83, 0.92]	w = 0.107
ILR, mapped back	0.9 [0.85, 0.93]	w = 0.11
ILR, REML	0.93 [0.89, 0.96]	w = 0.134
Method's own interval	0.09 [0.058, 0.14]	w = 0.0993
Own interval, transcript fractions	0.9 [0.86, 0.94]	w = 0.0993
Fisher information (Wald)	0.28 [0.23, 0.35]	w = 0.0232
Godambe sandwich	1 [0.98, 1]	w = 0.786
Released code, as published	1 [0.98, 1]	w = 1

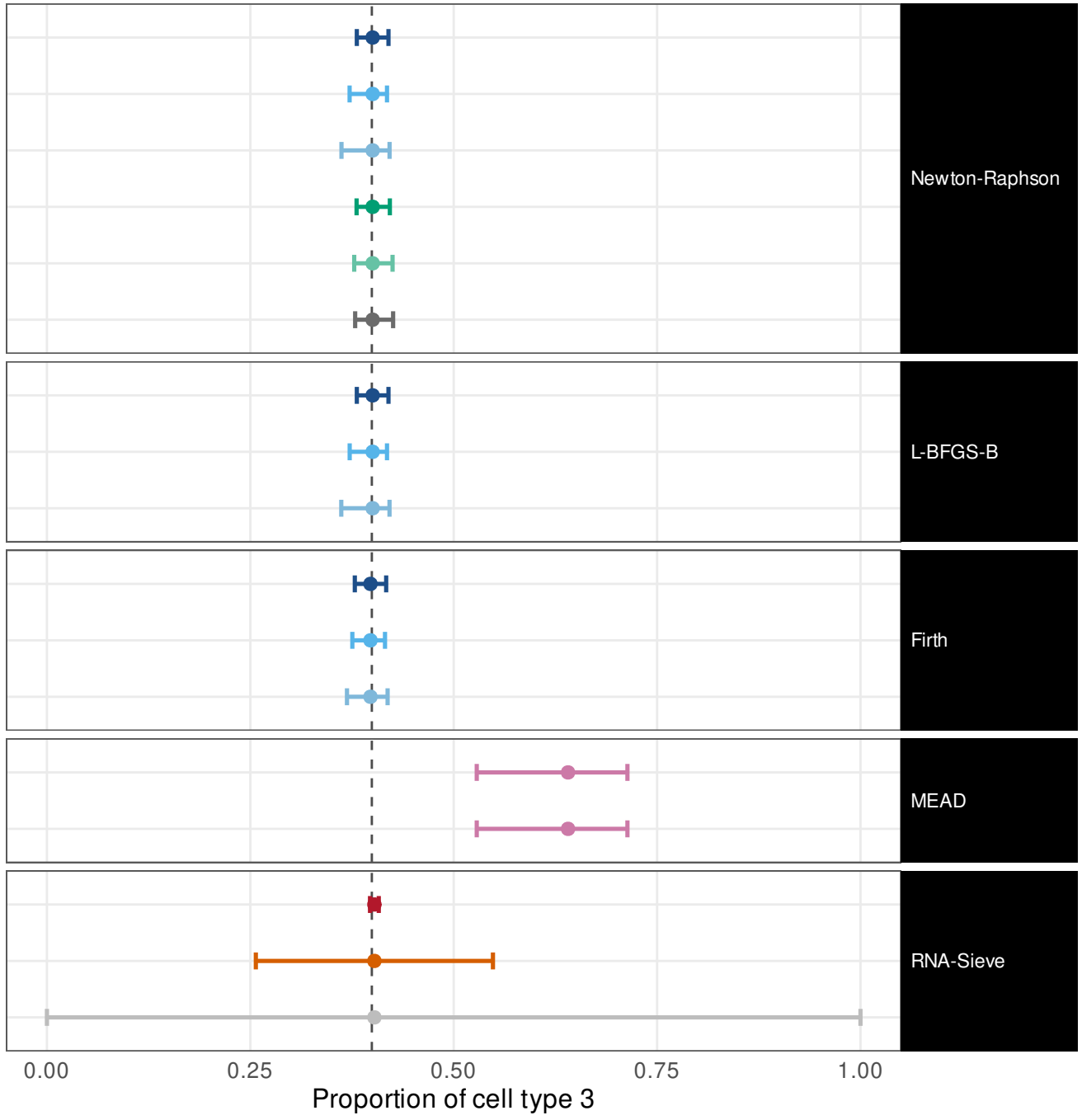


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / unbalanced / high sibling overlap / a = 5 / sizes 1:1:5:5

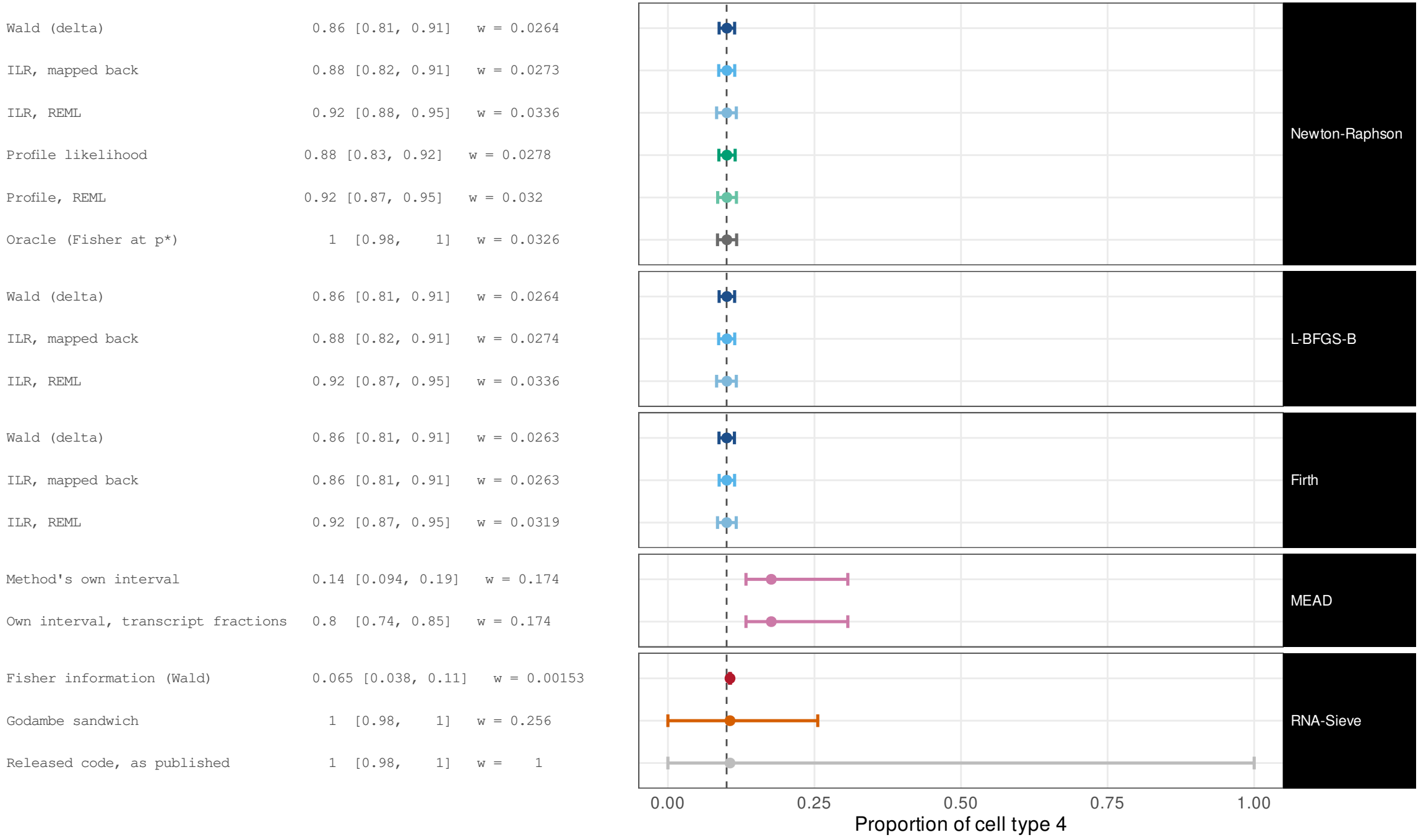
Wald (delta)	0.9 [0.84, 0.93]	w = 0.0389
ILR, mapped back	0.9 [0.86, 0.94]	w = 0.046
ILR, REML	0.96 [0.92, 0.98]	w = 0.0591
Profile likelihood	0.93 [0.89, 0.96]	w = 0.0409
Profile, REML	0.96 [0.92, 0.98]	w = 0.0471
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0466
Wald (delta)	0.9 [0.84, 0.93]	w = 0.0389
ILR, mapped back	0.91 [0.86, 0.94]	w = 0.0459
ILR, REML	0.94 [0.9, 0.97]	w = 0.0593
Wald (delta)	0.89 [0.84, 0.93]	w = 0.0385
ILR, mapped back	0.89 [0.84, 0.93]	w = 0.0401
ILR, REML	0.91 [0.86, 0.94]	w = 0.0499
Method's own interval	0.14 [0.099, 0.19]	w = 0.185
Own interval, transcript fractions	0.6 [0.54, 0.67]	w = 0.185
Fisher information (Wald)	0.3 [0.24, 0.36]	w = 0.0107
Godambe sandwich	1 [0.98, 1]	w = 0.291
Released code, as published	1 [0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1000 cells per bulk / unbalanced / high sibling overlap / a = 5 / sizes 1:1:5:5

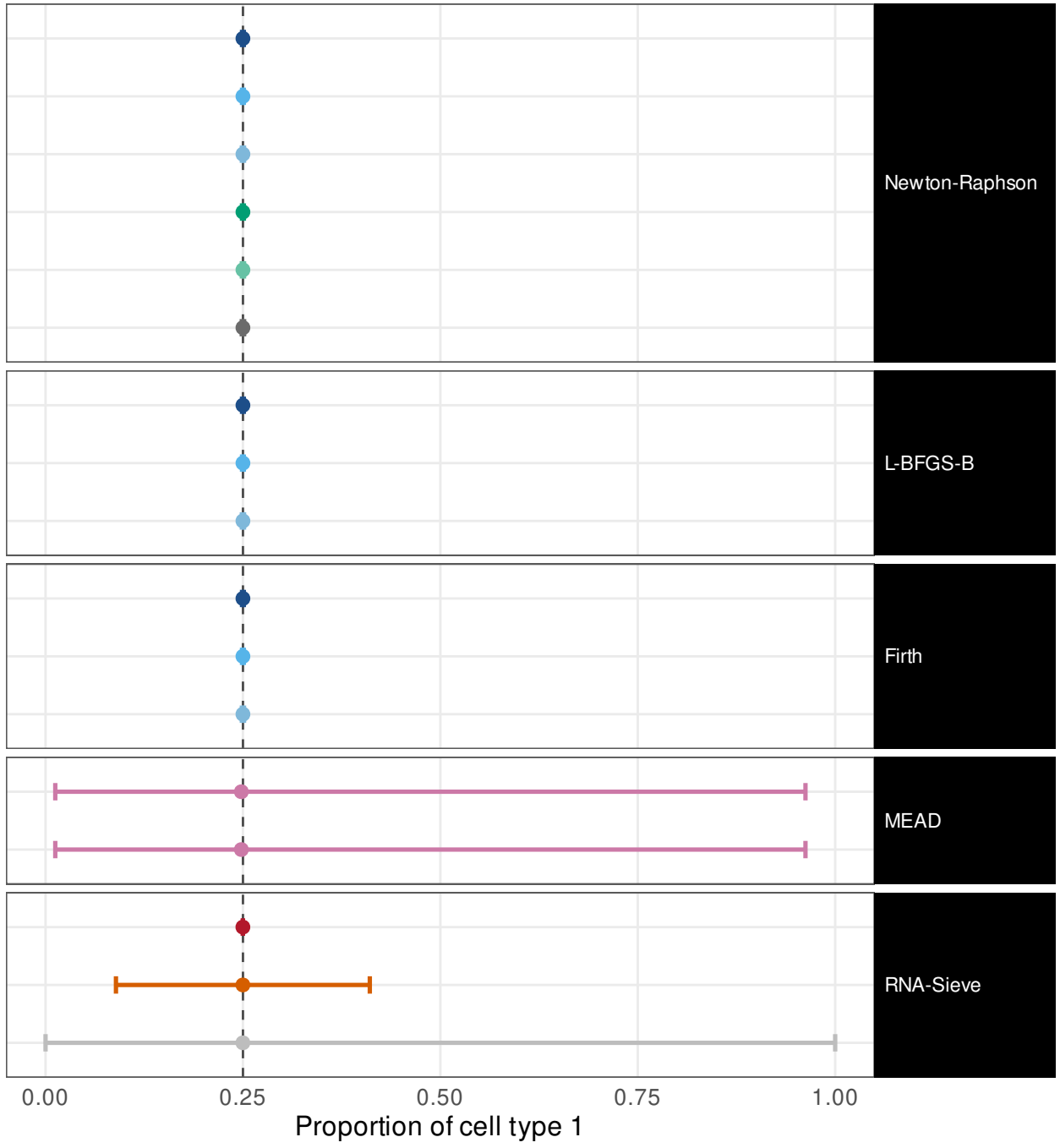


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / balanced / low sibling overlap / a = 1 / equal sizes

Wald (delta)	0.92 [0.87, 0.95]	w = 0.00228
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.00227
ILR, REML	0.96 [0.92, 0.98]	w = 0.00274
Profile likelihood	0.94 [0.9, 0.97]	w = 0.00239
Profile, REML	0.95 [0.91, 0.97]	w = 0.00275
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.00298
Wald (delta)	0.92 [0.87, 0.95]	w = 0.00228
ILR, mapped back	0.92 [0.88, 0.95]	w = 0.00227
ILR, REML	0.95 [0.91, 0.97]	w = 0.00275
Wald (delta)	0.92 [0.87, 0.95]	w = 0.00228
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.00228
ILR, REML	0.96 [0.92, 0.98]	w = 0.00274
Method's own interval	0.95 [0.91, 0.97]	w = 0.95
Own interval, transcript fractions	0.95 [0.91, 0.97]	w = 0.95
Fisher information (Wald)	0.02 [0.0078, 0.05]	w = 6.43e-05
Godambe sandwich	1 [0.98, 1]	w = 0.321
Released code, as published	1 [0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / balanced / low sibling overlap / a = 1 / equal sizes

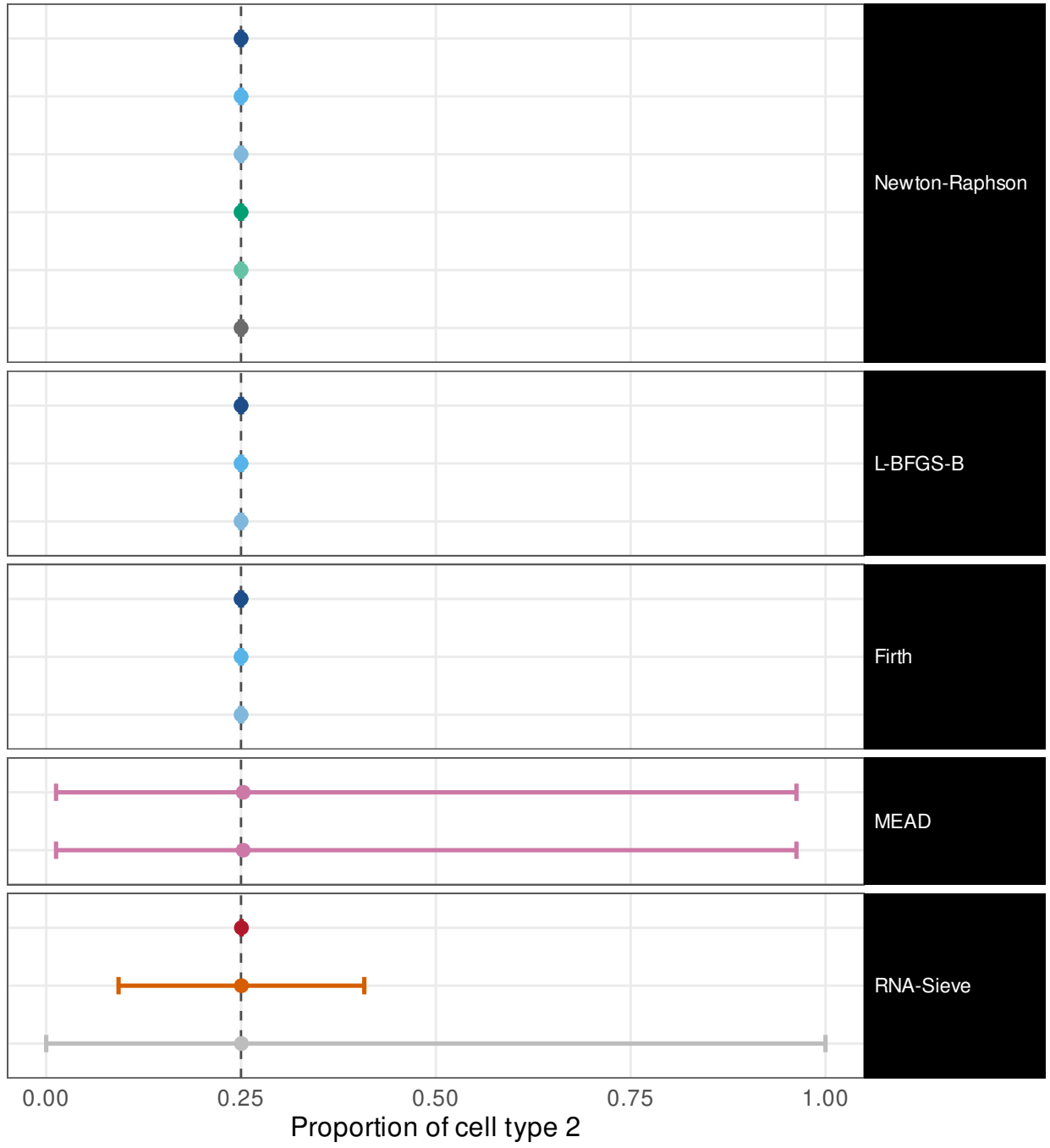
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Wald (delta)                          0.88 [0.83, 0.92]   w = 0.00229
ILR, mapped back                      0.88 [0.82, 0.91]   w = 0.00229
ILR, REML                             0.94 [ 0.9, 0.97]   w = 0.00276
Profile likelihood                    0.92 [0.87, 0.95]   w = 0.00241
Profile, REML                         0.94 [ 0.9, 0.97]   w = 0.00278
Oracle (Fisher at p*)                    1 [0.98,    1]   w = 0.00294

Wald (delta)                          0.88 [0.83, 0.92]   w = 0.00229
ILR, mapped back                      0.88 [0.83, 0.92]   w = 0.00229
ILR, REML                             0.96 [0.92, 0.98]   w = 0.00277

Wald (delta)                          0.88 [0.83, 0.92]   w = 0.00229
ILR, mapped back                      0.87 [0.82, 0.91]   w = 0.00229
ILR, REML                             0.94 [ 0.9, 0.97]   w = 0.00276

Method's own interval                 0.95 [0.91, 0.97]   w = 0.95
Own interval, transcript fractions    0.95 [0.91, 0.97]   w = 0.95

Fisher information (Wald)             0.01 [0.0027, 0.036]   w = 5.53e-05
Godambe sandwich                         1 [0.98,    1]   w = 0.315
Released code, as published              1 [0.98,    1]   w =    1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / balanced / low sibling overlap / a = 1 / equal sizes

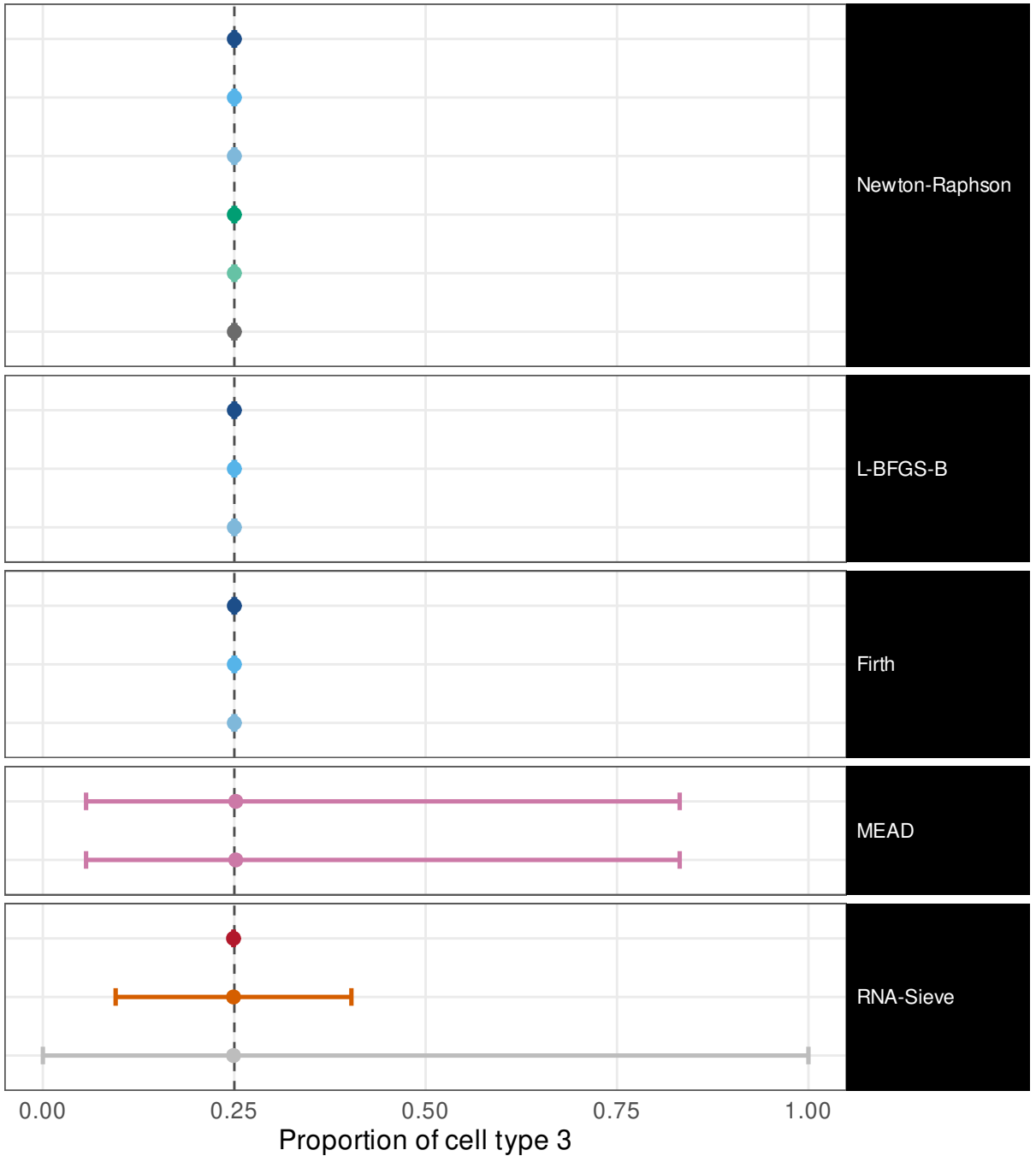
```
Wald (delta)                          0.92 [0.88, 0.95]   w = 0.00201
ILR, mapped back                      0.92 [0.87, 0.95]   w = 0.002
ILR, REML                             0.96 [0.92, 0.98]   w = 0.00242
Profile likelihood                    0.92 [0.88, 0.95]   w = 0.00211
Profile, REML                         0.96 [0.92, 0.98]   w = 0.00243
Oracle (Fisher at p*)                    1 [0.98,    1]   w = 0.00267

Wald (delta)                          0.92 [0.88, 0.95]   w = 0.00201
ILR, mapped back                      0.92 [0.87, 0.95]   w = 0.002
ILR, REML                             0.94 [ 0.9, 0.97]   w = 0.00242

Wald (delta)                          0.92 [0.88, 0.95]   w = 0.00201
ILR, mapped back                      0.92 [0.87, 0.95]   w = 0.002
ILR, REML                             0.96 [0.92, 0.98]   w = 0.00241

Method's own interval                 0.8  [0.74, 0.85]   w = 0.775
Own interval, transcript fractions    0.8  [0.74, 0.85]   w = 0.775

Fisher information (Wald)             0.005 [0.00088, 0.028]   w = 5.74e-05
Godambe sandwich                         1 [0.98,    1]   w = 0.308
Released code, as published              1 [0.98,    1]   w =    1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / balanced / low sibling overlap / a = 1 / equal sizes

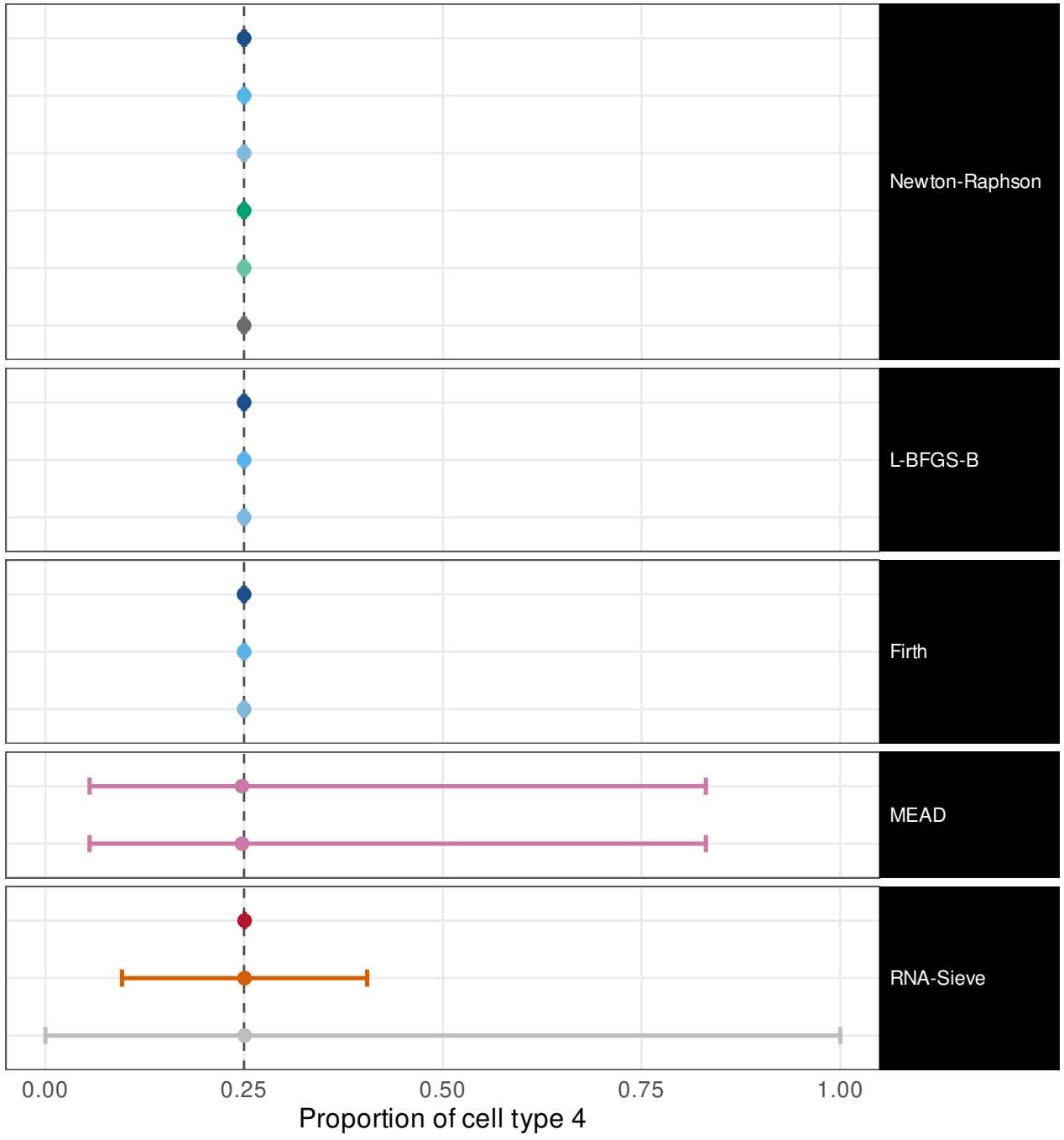
```
Wald (delta)                          0.91 [0.86, 0.94]   w = 0.00205
ILR, mapped back                      0.9  [0.86, 0.94]   w = 0.00204
ILR, REML                             0.94 [ 0.9, 0.97]   w = 0.00247
Profile likelihood                    0.92 [0.88, 0.95]   w = 0.00215
Profile, REML                         0.96 [0.92, 0.98]   w = 0.00247
Oracle (Fisher at p*)                    1 [0.98,    1]   w = 0.00265

Wald (delta)                          0.91 [0.86, 0.94]   w = 0.00205
ILR, mapped back                      0.92 [0.87, 0.95]   w = 0.00204
ILR, REML                             0.94 [ 0.9, 0.97]   w = 0.00247

Wald (delta)                          0.91 [0.86, 0.94]   w = 0.00205
ILR, mapped back                      0.91 [0.86, 0.94]   w = 0.00204
ILR, REML                             0.96 [0.92, 0.98]   w = 0.00247

Method's own interval                 0.78 [0.71, 0.83]   w = 0.775
Own interval, transcript fractions    0.78 [0.71, 0.83]   w = 0.775

Fisher information (Wald)             0.02 [0.0078, 0.05]   w = 5.9e-05
Godambe sandwich                         1 [0.98,    1]   w = 0.308
Released code, as published              1 [0.98,    1]   w =     1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / balanced / low sibling overlap / a = 1 / sizes 1:1:5:5

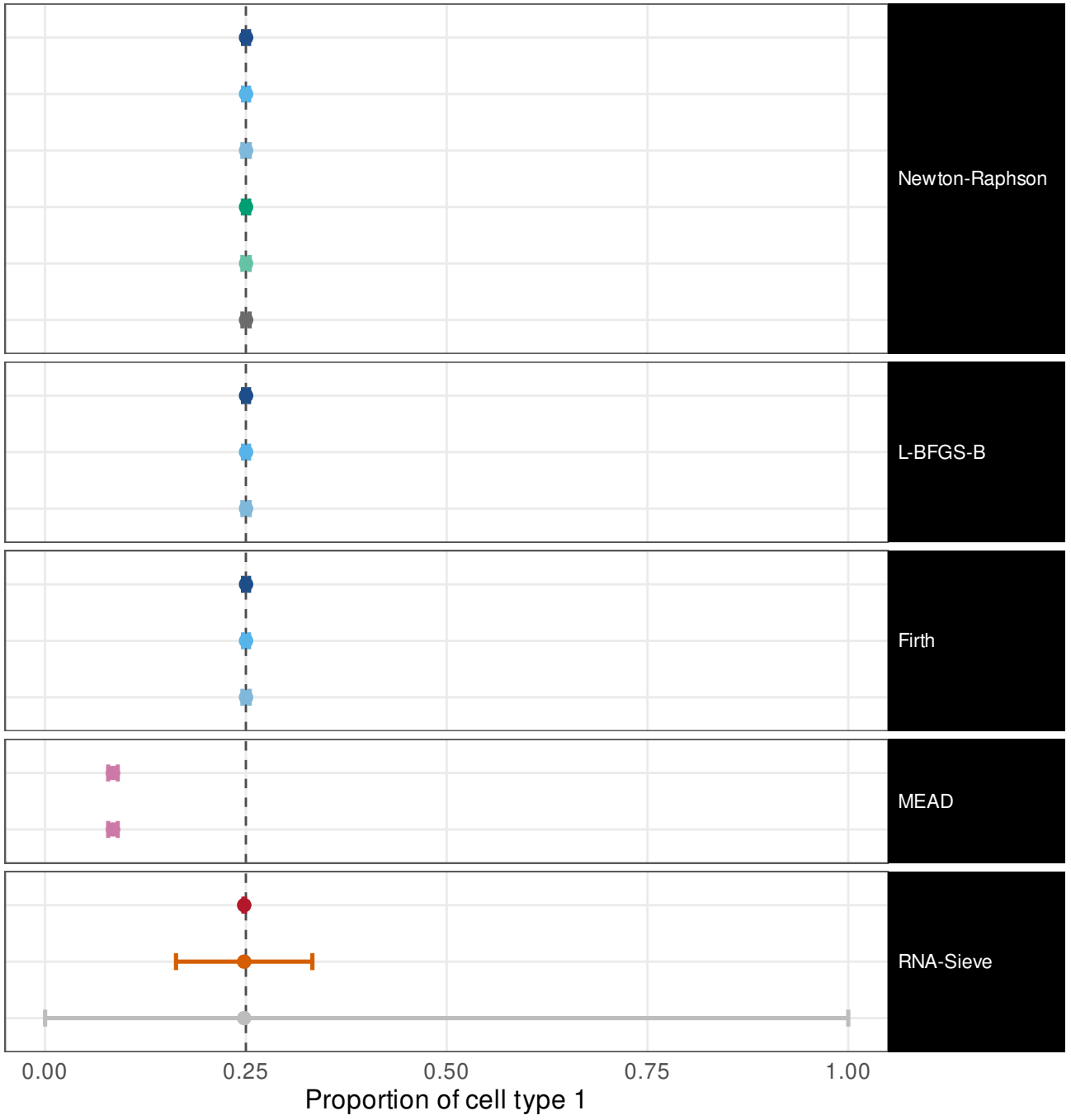
```
Wald (delta)                        0.9  [0.84, 0.93]   w = 0.00747
ILR, mapped back                    0.89 [0.84, 0.93]   w = 0.00746
ILR, REML                           0.94 [ 0.9, 0.97]   w = 0.00903
Profile likelihood                  0.9  [0.86, 0.94]   w = 0.00784
Profile, REML                       0.96 [0.92, 0.98]   w = 0.00903
Oracle (Fisher at p*)                 1  [0.98,    1]   w = 0.00889

Wald (delta)                        0.9  [0.84, 0.93]   w = 0.00747
ILR, mapped back                    0.9  [0.84, 0.93]   w = 0.00745
ILR, REML                           0.96 [0.92, 0.98]   w = 0.00899

Wald (delta)                        0.9  [0.84, 0.93]   w = 0.00746
ILR, mapped back                    0.89 [0.84, 0.93]   w = 0.00746
ILR, REML                           0.95 [0.91, 0.97]   w = 0.00899

Method's own interval                 0  [   0, 0.019]   w = 0.0118
Own interval, transcript fractions    1  [0.98,    1]   w = 0.0118

Fisher information (Wald)           0.2  [0.15, 0.26]   w = 0.00188
Godambe sandwich                      1  [0.98,    1]   w = 0.17
Released code, as published           1  [0.98,    1]   w =    1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / balanced / low sibling overlap / a = 1 / sizes 1:1:5:5

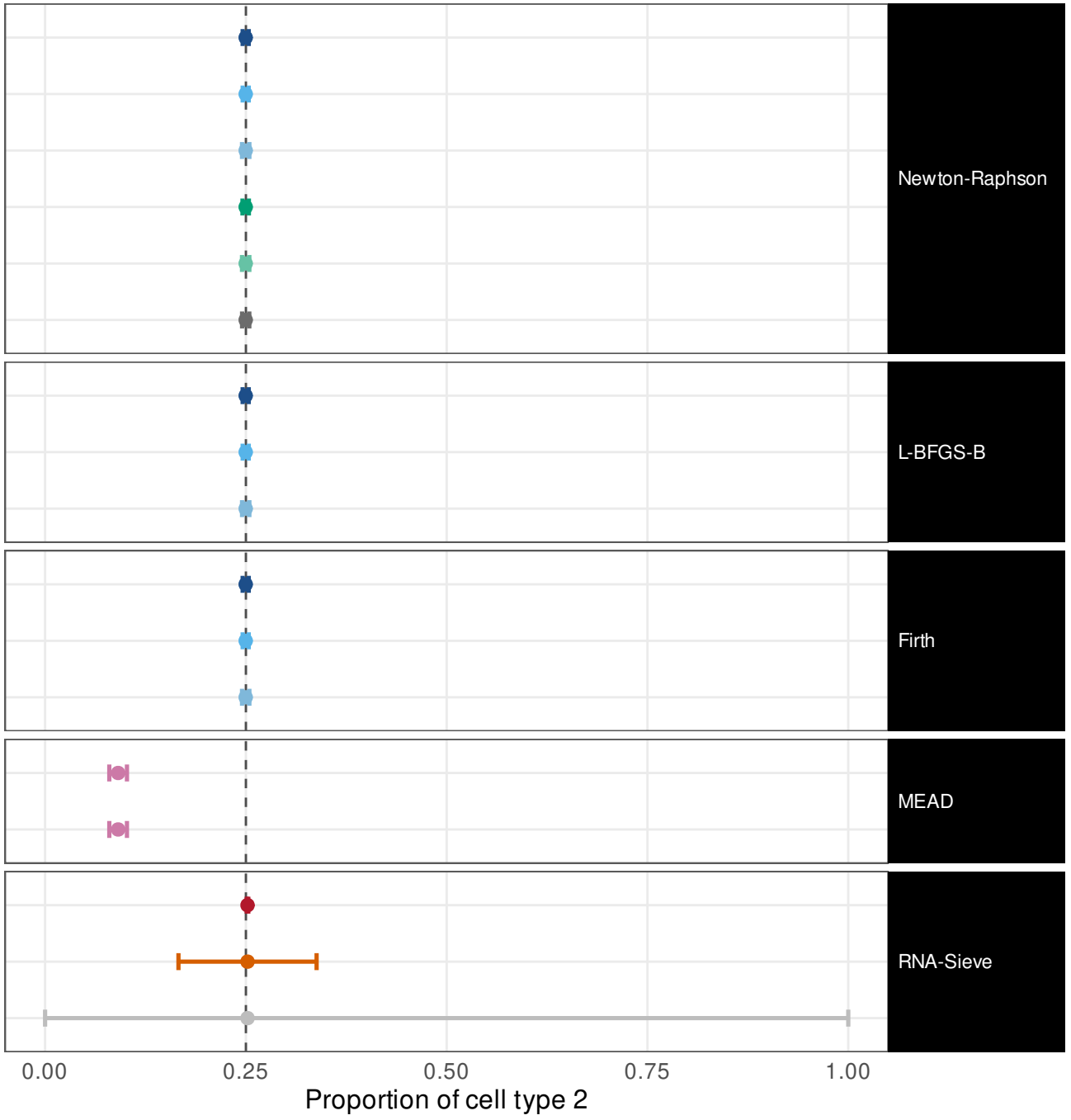
```
Wald (delta)                        0.91 [0.86, 0.94]   w = 0.00756
ILR, mapped back                    0.9  [0.86, 0.94]   w = 0.00755
ILR, REML                           0.96 [0.92, 0.98]   w = 0.00916
Profile likelihood                  0.92 [0.87, 0.95]   w = 0.00794
Profile, REML                       0.96 [0.92, 0.98]   w = 0.00914
Oracle (Fisher at p*)                  1 [0.98,    1]   w = 0.00899

Wald (delta)                        0.91 [0.86, 0.94]   w = 0.00756
ILR, mapped back                    0.9  [0.86, 0.94]   w = 0.00755
ILR, REML                           0.96 [0.92, 0.98]   w = 0.00911

Wald (delta)                        0.91 [0.86, 0.94]   w = 0.00756
ILR, mapped back                    0.91 [0.86, 0.94]   w = 0.00755
ILR, REML                           0.96 [0.92, 0.98]   w = 0.00912

Method's own interval                  0 [   0, 0.019]   w = 0.022
Own interval, transcript fractions     1 [0.97,    1]   w = 0.022

Fisher information (Wald)           0.24 [0.19, 0.31]   w = 0.00194
Godambe sandwich                       1 [0.98,    1]   w = 0.172
Released code, as published            1 [0.98,    1]   w =     1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / balanced / low sibling overlap / a = 1 / sizes 1:1:5:5

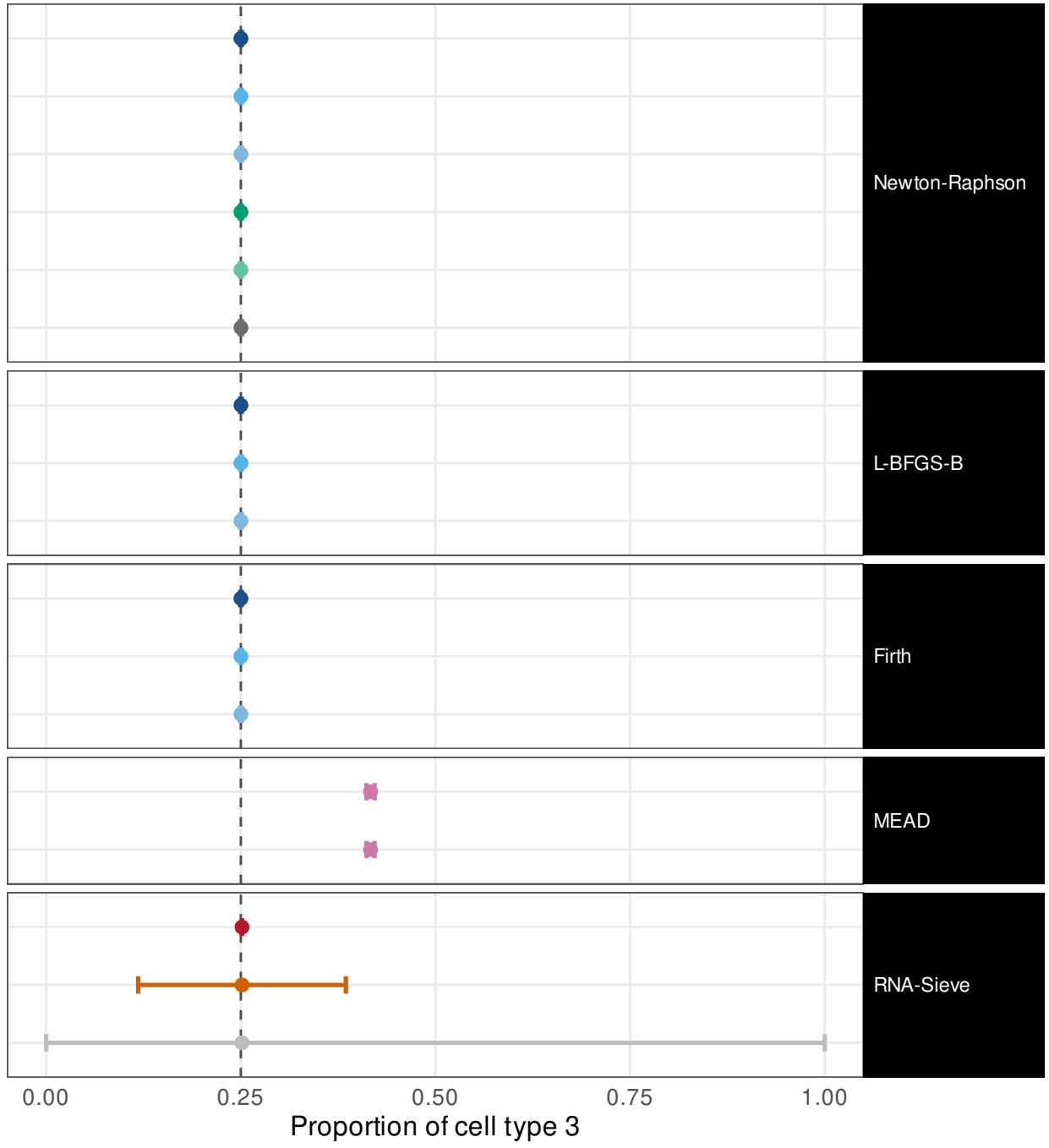
```
Wald (delta)                        0.9  [0.84, 0.93]   w = 0.00201
ILR, mapped back                    0.89 [0.84, 0.93]   w = 0.002
ILR, REML                           0.92 [0.88, 0.95]   w = 0.00243
Profile likelihood                  0.9  [0.85, 0.93]   w = 0.00211
Profile, REML                       0.92 [0.88, 0.95]   w = 0.00243
Oracle (Fisher at p*)                 1  [0.98,    1]   w = 0.00243

Wald (delta)                        0.9  [0.84, 0.93]   w = 0.00201
ILR, mapped back                    0.9  [0.84, 0.93]   w = 0.002
ILR, REML                           0.93 [0.89, 0.96]   w = 0.00242

Wald (delta)                        0.9  [0.84, 0.93]   w = 0.00201
ILR, mapped back                    0.89 [0.84, 0.93]   w = 0.00201
ILR, REML                           0.93 [0.89, 0.96]   w = 0.00242

Method's own interval                 0  [   0, 0.019]   w = 0.0105
Own interval, transcript fractions    1  [0.97,    1]   w = 0.0105

Fisher information (Wald)           0.01 [0.0027, 0.036]   w = 0.000157
Godambe sandwich                      1  [0.98,    1]   w = 0.267
Released code, as published           1  [0.98,    1]   w =     1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / balanced / low sibling overlap / a = 1 / sizes 1:1:5:5

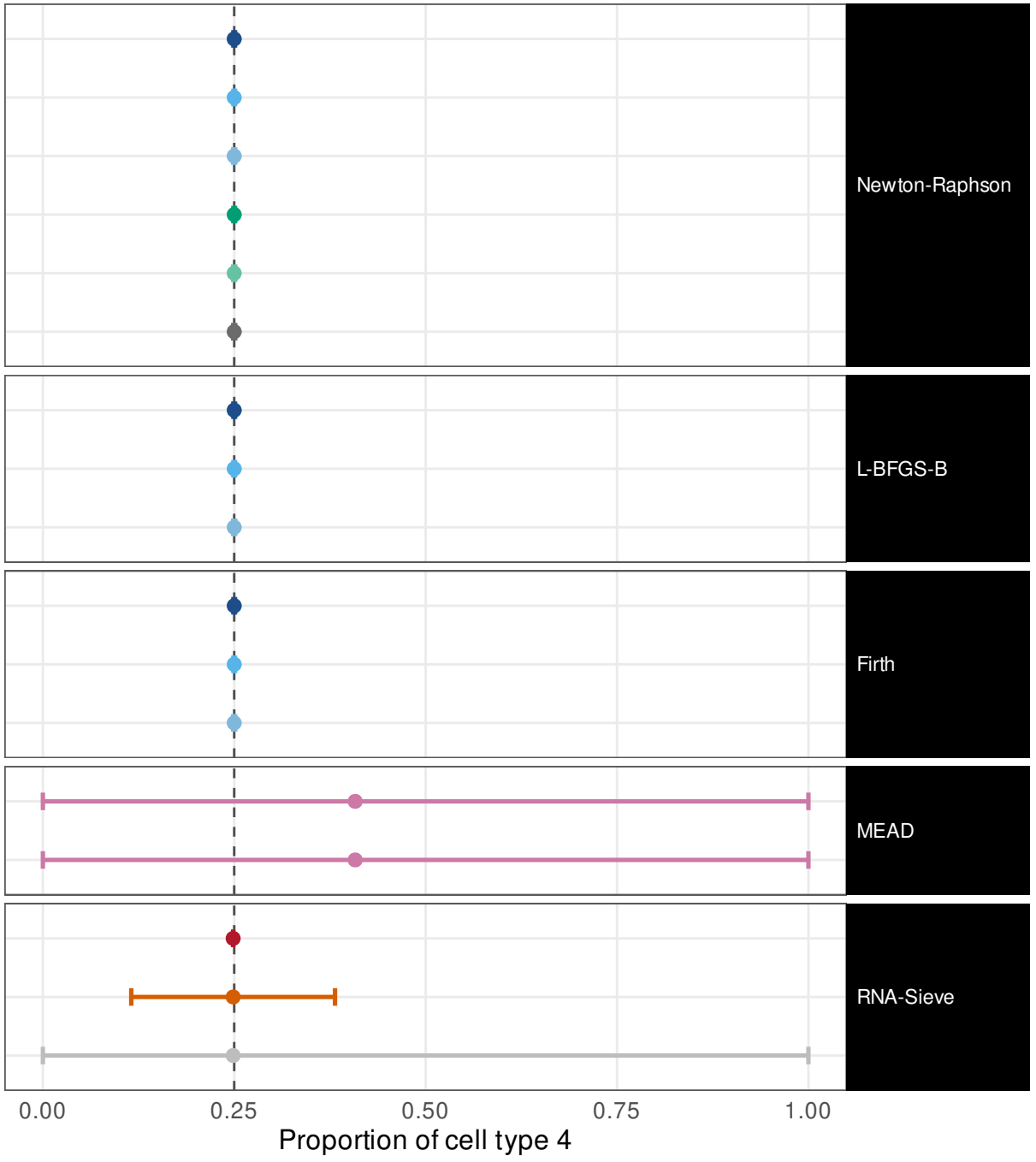
```
Wald (delta)                          0.9  [0.86, 0.94]   w = 0.00219
ILR, mapped back                      0.9  [0.86, 0.94]   w = 0.00218
ILR, REML                             0.95 [0.91, 0.97]   w = 0.00263
Profile likelihood                   0.92 [0.87, 0.95]   w = 0.00229
Profile, REML                        0.96 [0.92, 0.98]   w = 0.00264
Oracle (Fisher at p*)                   1  [0.98,    1]   w = 0.00261

Wald (delta)                          0.9  [0.86, 0.94]   w = 0.00219
ILR, mapped back                      0.9  [0.86, 0.94]   w = 0.00217
ILR, REML                             0.95 [0.91, 0.97]   w = 0.00264

Wald (delta)                          0.9  [0.86, 0.94]   w = 0.00219
ILR, mapped back                      0.91 [0.86, 0.94]   w = 0.00218
ILR, REML                             0.96 [0.92, 0.98]   w = 0.00263

Method's own interval                   1  [0.98,    1]   w =    1
Own interval, transcript fractions      1  [0.98,    1]   w =    1

Fisher information (Wald)           0.005 [0.00088, 0.028]   w = 0.000156
Godambe sandwich                        1  [0.98,    1]   w = 0.266
Released code, as published             1  [0.98,    1]   w =    1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / balanced / low sibling overlap / a = 5 / equal sizes

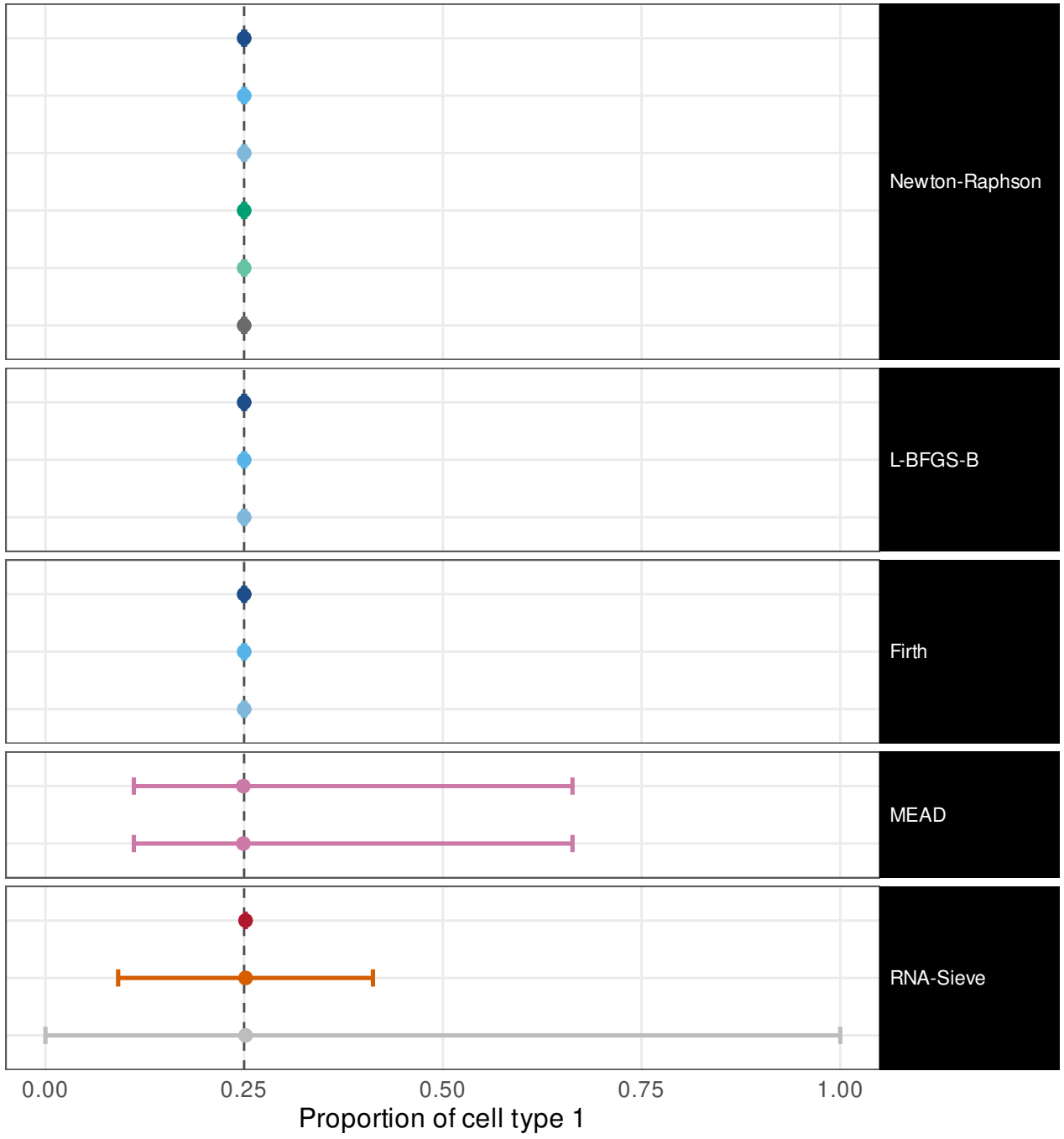
```
Wald (delta)                          0.9  [0.85, 0.93]   w = 0.00235
ILR, mapped back                      0.9  [0.85, 0.93]   w = 0.00235
ILR, REML                             0.95 [0.91, 0.97]   w = 0.00283
Profile likelihood                    0.92 [0.87, 0.95]   w = 0.00247
Profile, REML                         0.95 [0.91, 0.97]   w = 0.00284
Oracle (Fisher at p*)                   1  [0.98,    1]   w = 0.00286

Wald (delta)                          0.9  [0.85, 0.93]   w = 0.00235
ILR, mapped back                      0.9  [0.85, 0.93]   w = 0.00235
ILR, REML                             0.94 [ 0.9, 0.97]   w = 0.00283

Wald (delta)                          0.9  [0.85, 0.93]   w = 0.00235
ILR, mapped back                      0.9  [0.86, 0.94]   w = 0.00235
ILR, REML                             0.95 [0.91, 0.97]   w = 0.00283

Method's own interval                 0.96 [0.92, 0.98]   w = 0.552
Own interval, transcript fractions    0.96 [0.92, 0.98]   w = 0.552

Fisher information (Wald)               0  [   0, 0.019]   w = 1.28e-05
Godambe sandwich                        1  [0.98,    1]   w = 0.321
Released code, as published             1  [0.98,    1]   w =    1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / balanced / low sibling overlap / a = 5 / equal sizes

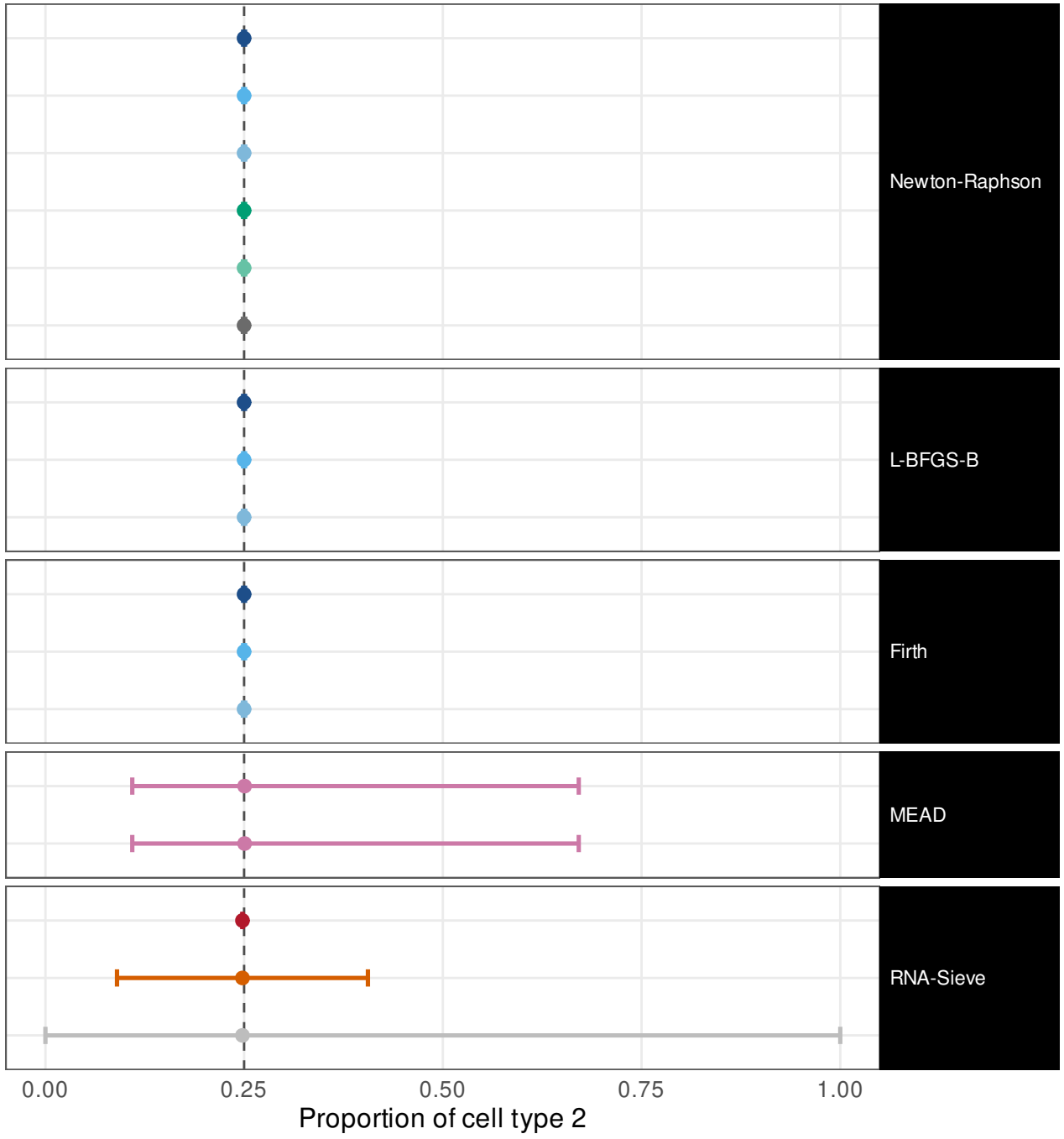
```
Wald (delta)                          0.89 [0.84, 0.93]   w = 0.00237
ILR, mapped back                      0.89 [0.84, 0.93]   w = 0.00237
ILR, REML                             0.94 [0.89, 0.96]   w = 0.00285
Profile likelihood                    0.91 [0.86, 0.94]   w = 0.00249
Profile, REML                         0.94 [0.89, 0.96]   w = 0.00286
Oracle (Fisher at p*)                    1 [0.98,    1]   w = 0.00293

Wald (delta)                          0.89 [0.84, 0.93]   w = 0.00237
ILR, mapped back                      0.88 [0.83, 0.92]   w = 0.00236
ILR, REML                             0.92 [0.88, 0.95]   w = 0.00286

Wald (delta)                          0.89 [0.84, 0.93]   w = 0.00237
ILR, mapped back                      0.88 [0.83, 0.92]   w = 0.00236
ILR, REML                             0.94 [0.89, 0.96]   w = 0.00285

Method's own interval                 0.95 [0.91, 0.97]   w = 0.562
Own interval, transcript fractions    0.95 [0.91, 0.97]   w = 0.562

Fisher information (Wald)                0 [   0, 0.019]   w = 1.12e-05
Godambe sandwich                         1 [0.98,    1]   w = 0.316
Released code, as published              1 [0.98,    1]   w =     1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / balanced / low sibling overlap / a = 5 / equal sizes

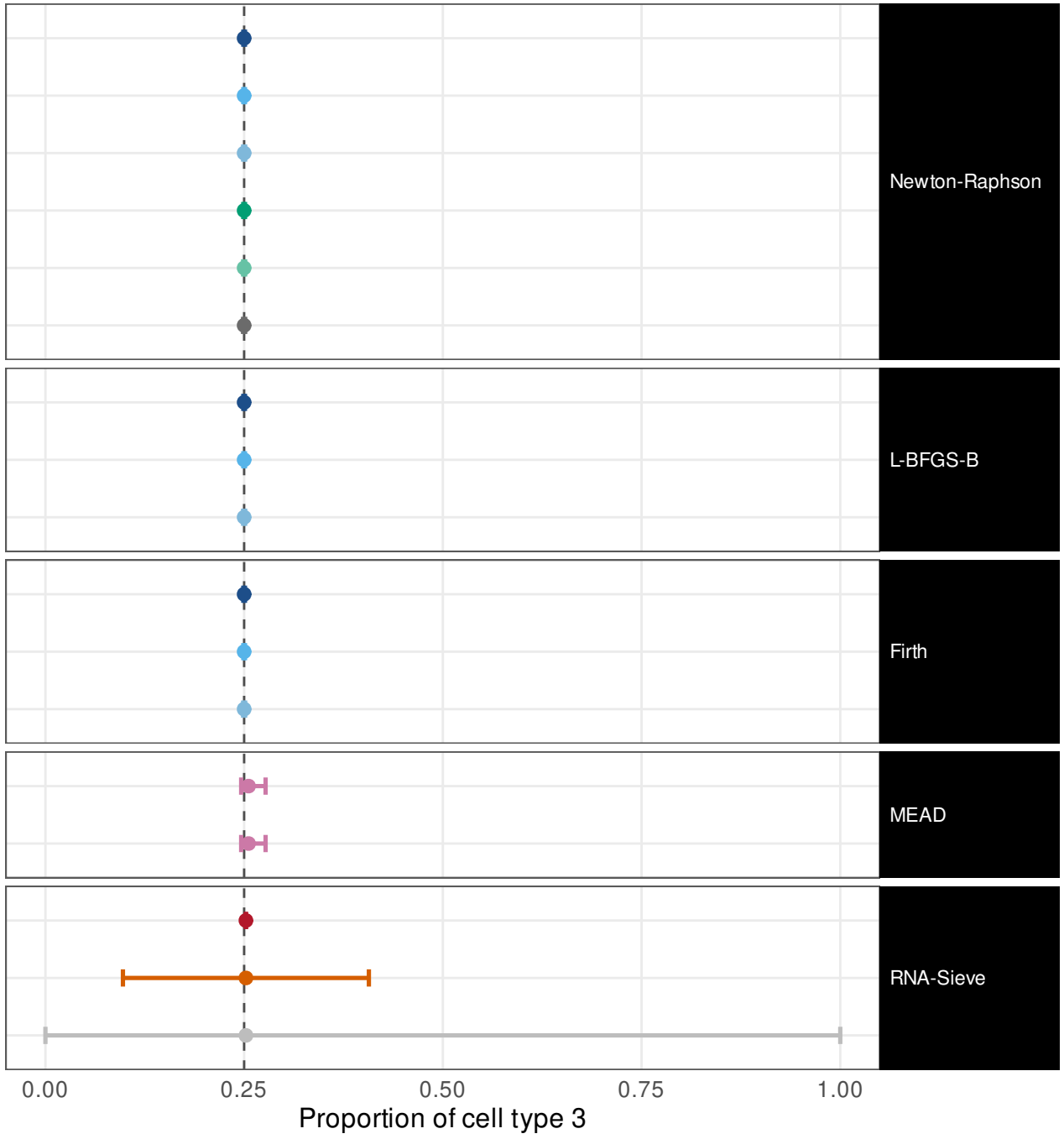
```
Wald (delta)                          0.91 [0.86, 0.94]   w = 0.00207
ILR, mapped back                      0.9  [0.86, 0.94]   w = 0.00207
ILR, REML                             0.94 [ 0.9, 0.97]   w = 0.0025
Profile likelihood                    0.93 [0.89, 0.96]   w = 0.00217
Profile, REML                         0.94 [ 0.9, 0.97]   w = 0.0025
Oracle (Fisher at p*)                    1 [0.98,    1]   w = 0.0026

Wald (delta)                          0.91 [0.86, 0.94]   w = 0.00207
ILR, mapped back                      0.9  [0.86, 0.94]   w = 0.00207
ILR, REML                             0.94 [ 0.9, 0.97]   w = 0.0025

Wald (delta)                          0.91 [0.86, 0.94]   w = 0.00207
ILR, mapped back                      0.92 [0.87, 0.95]   w = 0.00207
ILR, REML                             0.94 [ 0.9, 0.97]   w = 0.0025

Method's own interval                 0.025 [0.011, 0.057]   w = 0.0307
Own interval, transcript fractions    0.025 [0.011, 0.057]   w = 0.0307

Fisher information (Wald)                0 [   0, 0.019]   w = 1.15e-05
Godambe sandwich                         1 [0.98,    1]   w = 0.309
Released code, as published              1 [0.98,    1]   w =    1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / balanced / low sibling overlap / a = 5 / equal sizes

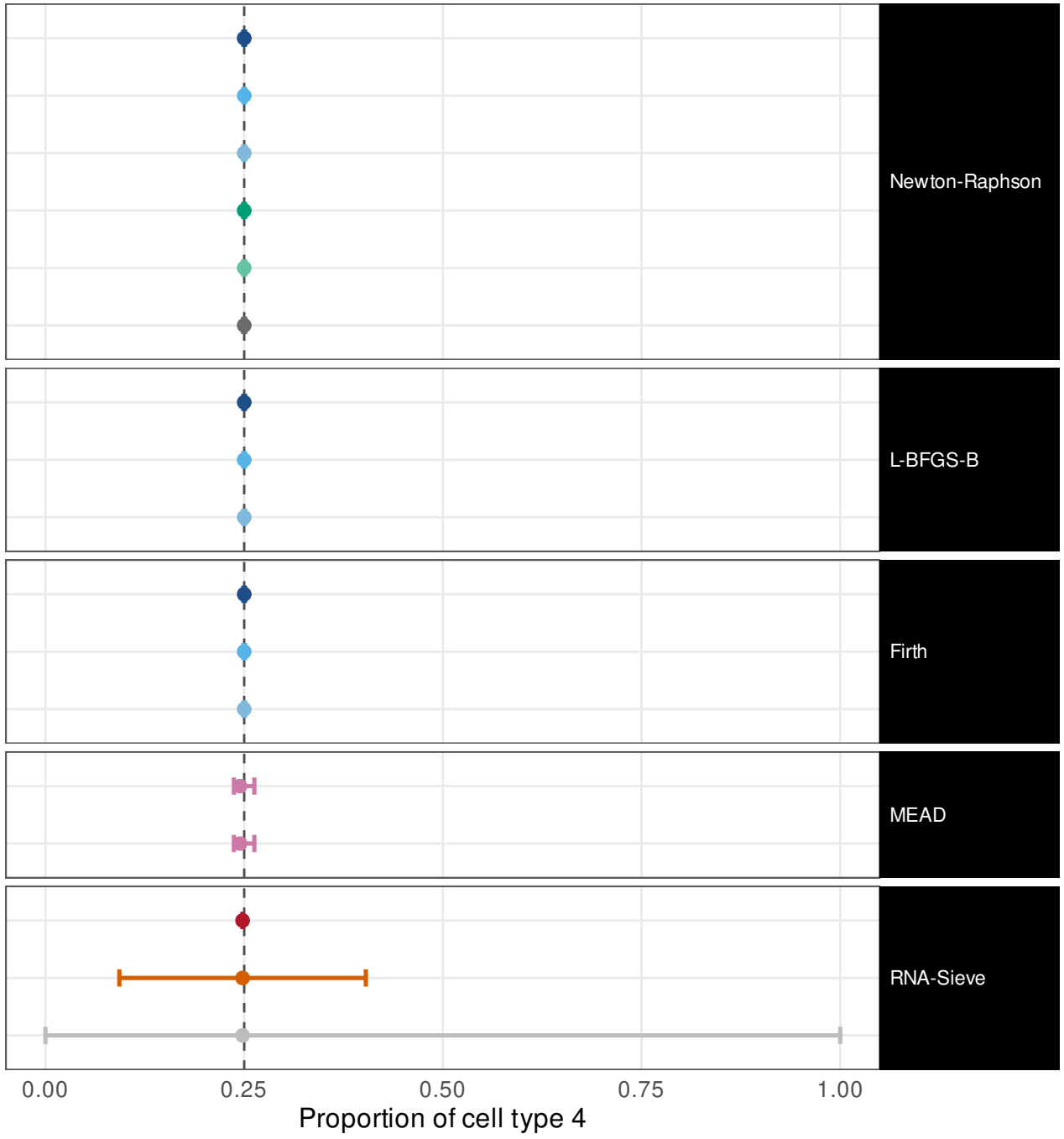
```
Wald (delta)                          0.9  [0.85, 0.93]   w = 0.00211
ILR, mapped back                      0.9  [0.86, 0.94]   w = 0.0021
ILR, REML                             0.95 [0.91, 0.97]   w = 0.00255
Profile likelihood                    0.93 [0.89, 0.96]   w = 0.00222
Profile, REML                         0.95 [0.91, 0.97]   w = 0.00255
Oracle (Fisher at p*)                   1  [0.98,    1]   w = 0.00261

Wald (delta)                          0.9  [0.85, 0.93]   w = 0.00211
ILR, mapped back                      0.91 [0.86, 0.94]   w = 0.0021
ILR, REML                             0.95 [0.91, 0.97]   w = 0.00255

Wald (delta)                          0.9  [0.85, 0.93]   w = 0.00211
ILR, mapped back                      0.9  [0.85, 0.93]   w = 0.00211
ILR, REML                             0.96 [0.92, 0.98]   w = 0.00255

Method's own interval                 0.02 [0.0078, 0.05]   w = 0.0257
Own interval, transcript fractions    0.02 [0.0078, 0.05]   w = 0.0257

Fisher information (Wald)               0  [   0, 0.019]   w = 1.19e-05
Godambe sandwich                        1  [0.98,    1]   w = 0.31
Released code, as published             1  [0.98,    1]   w =    1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / balanced / low sibling overlap / a = 5 / sizes 1:1:5:5

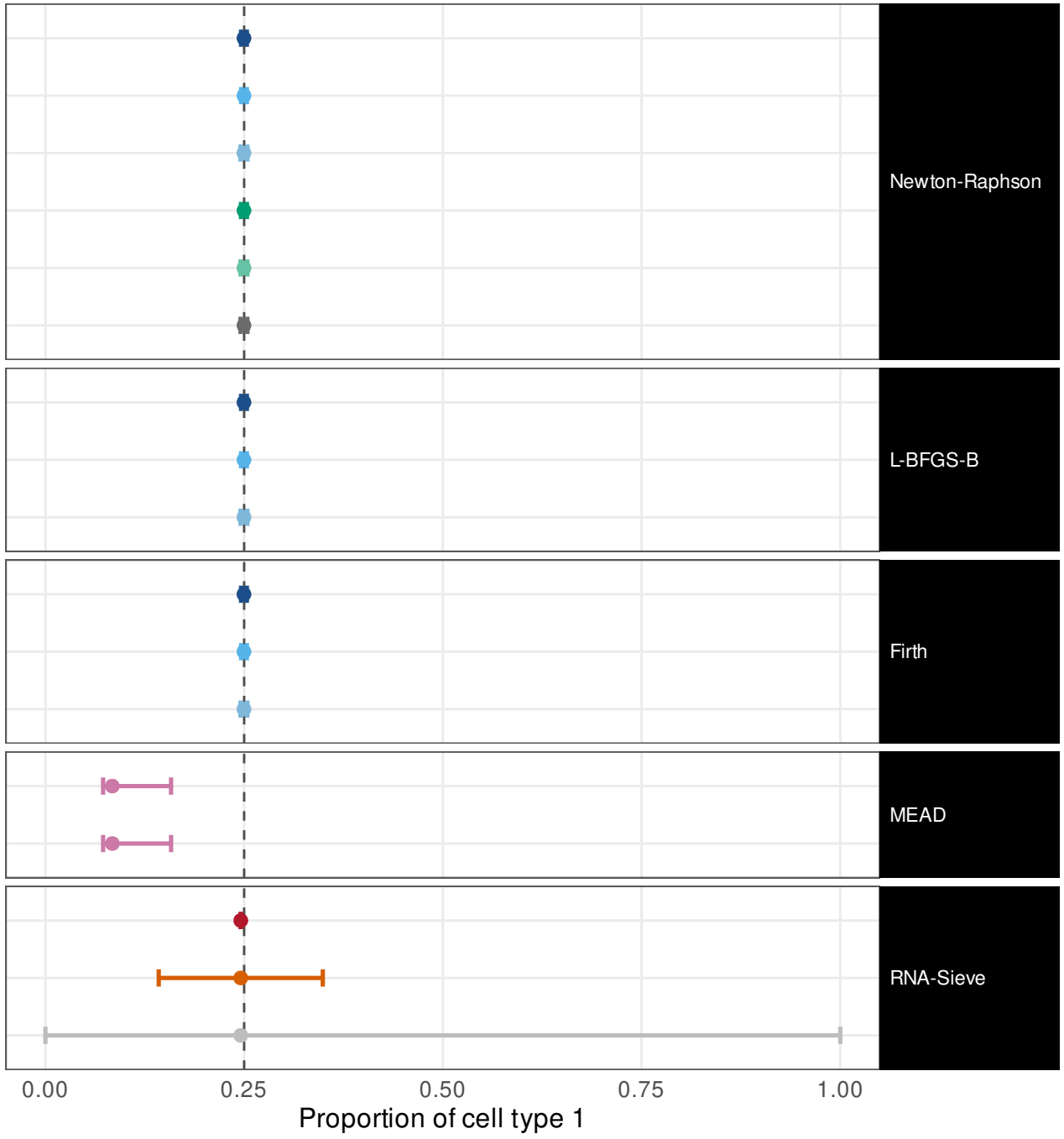
```
Wald (delta)                          0.93 [0.89, 0.96]   w = 0.00725
ILR, mapped back                      0.94 [0.89, 0.96]   w = 0.00723
ILR, REML                             0.97 [0.94, 0.99]   w = 0.00874
Profile likelihood                    0.94 [ 0.9, 0.97]   w = 0.00761
Profile, REML                         0.97 [0.94, 0.99]   w = 0.00877
Oracle (Fisher at p*)                    1 [0.98,    1]   w = 0.00862

Wald (delta)                          0.93 [0.89, 0.96]   w = 0.00725
ILR, mapped back                      0.93 [0.89, 0.96]   w = 0.00724
ILR, REML                             0.97 [0.94, 0.99]   w = 0.00872

Wald (delta)                          0.93 [0.89, 0.96]   w = 0.00725
ILR, mapped back                      0.93 [0.89, 0.96]   w = 0.00722
ILR, REML                             0.97 [0.94, 0.99]   w = 0.00875

Method's own interval                0.075 [0.046, 0.12]   w = 0.0855
Own interval, transcript fractions    0.99 [0.96,    1]   w = 0.0855

Fisher information (Wald)                0 [   0, 0.019]   w = 0.000309
Godambe sandwich                         1 [0.98,    1]   w = 0.206
Released code, as published              1 [0.98,    1]   w =     1
```

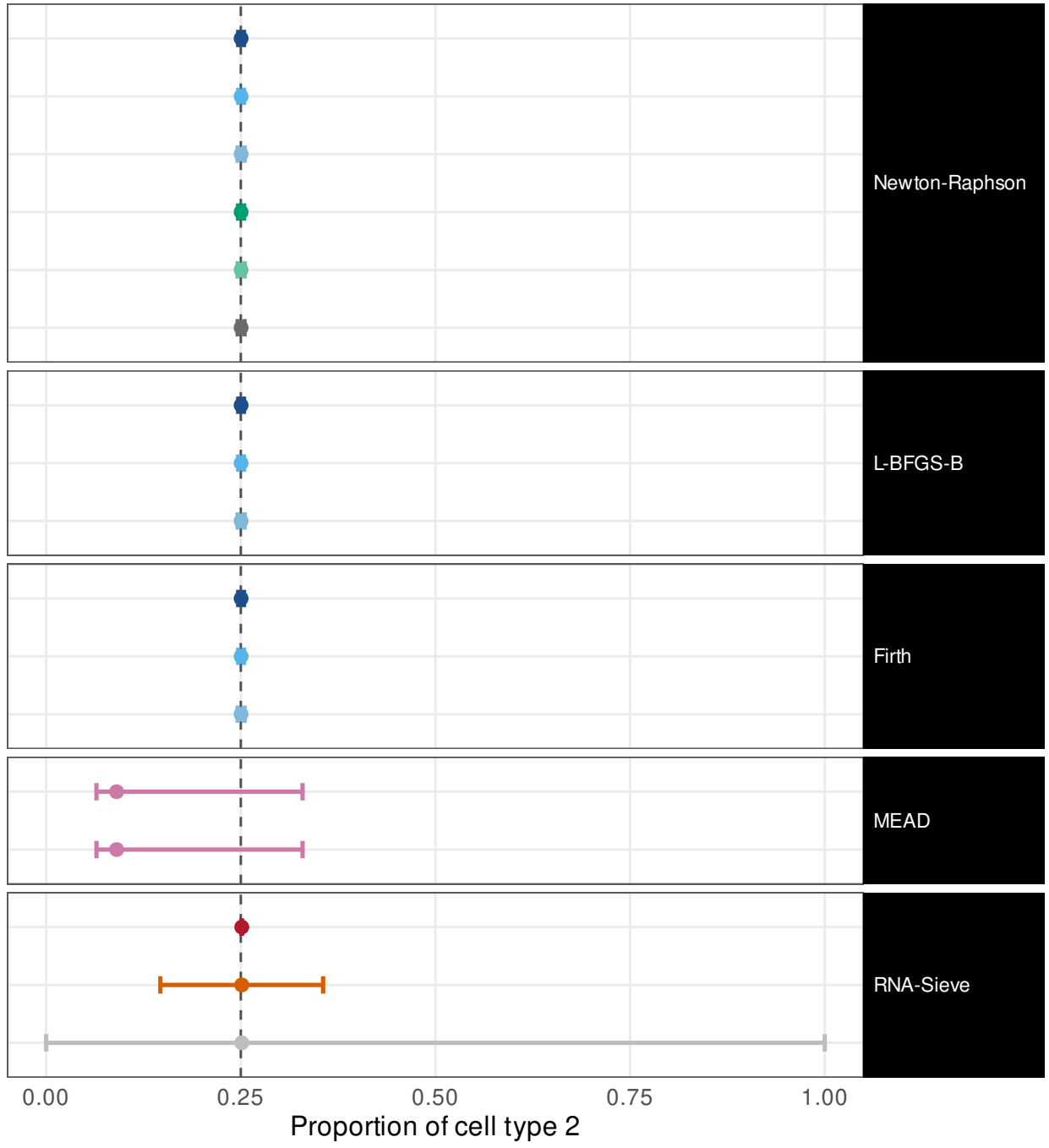


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / balanced / low sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.92 [0.87, 0.95]	w = 0.00734
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.00731
ILR, REML	0.94 [0.9, 0.97]	w = 0.00886
Profile likelihood	0.94 [0.89, 0.96]	w = 0.00771
Profile, REML	0.95 [0.91, 0.97]	w = 0.00888
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.00872
Wald (delta)	0.92 [0.87, 0.95]	w = 0.00734
ILR, mapped back	0.92 [0.88, 0.95]	w = 0.00732
ILR, REML	0.95 [0.91, 0.97]	w = 0.00884
Wald (delta)	0.92 [0.87, 0.95]	w = 0.00734
ILR, mapped back	0.92 [0.88, 0.95]	w = 0.00731
ILR, REML	0.94 [0.9, 0.97]	w = 0.00885
Method's own interval	0.26 [0.2, 0.32]	w = 0.265
Own interval, transcript fractions	0.26 [0.2, 0.32]	w = 0.265
Fisher information (Wald)	0.055 [0.031, 0.096]	w = 0.000312
Godambe sandwich	1 [0.98, 1]	w = 0.209
Released code, as published	1 [0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / balanced / low sibling overlap / a = 5 / sizes 1:1:5:5

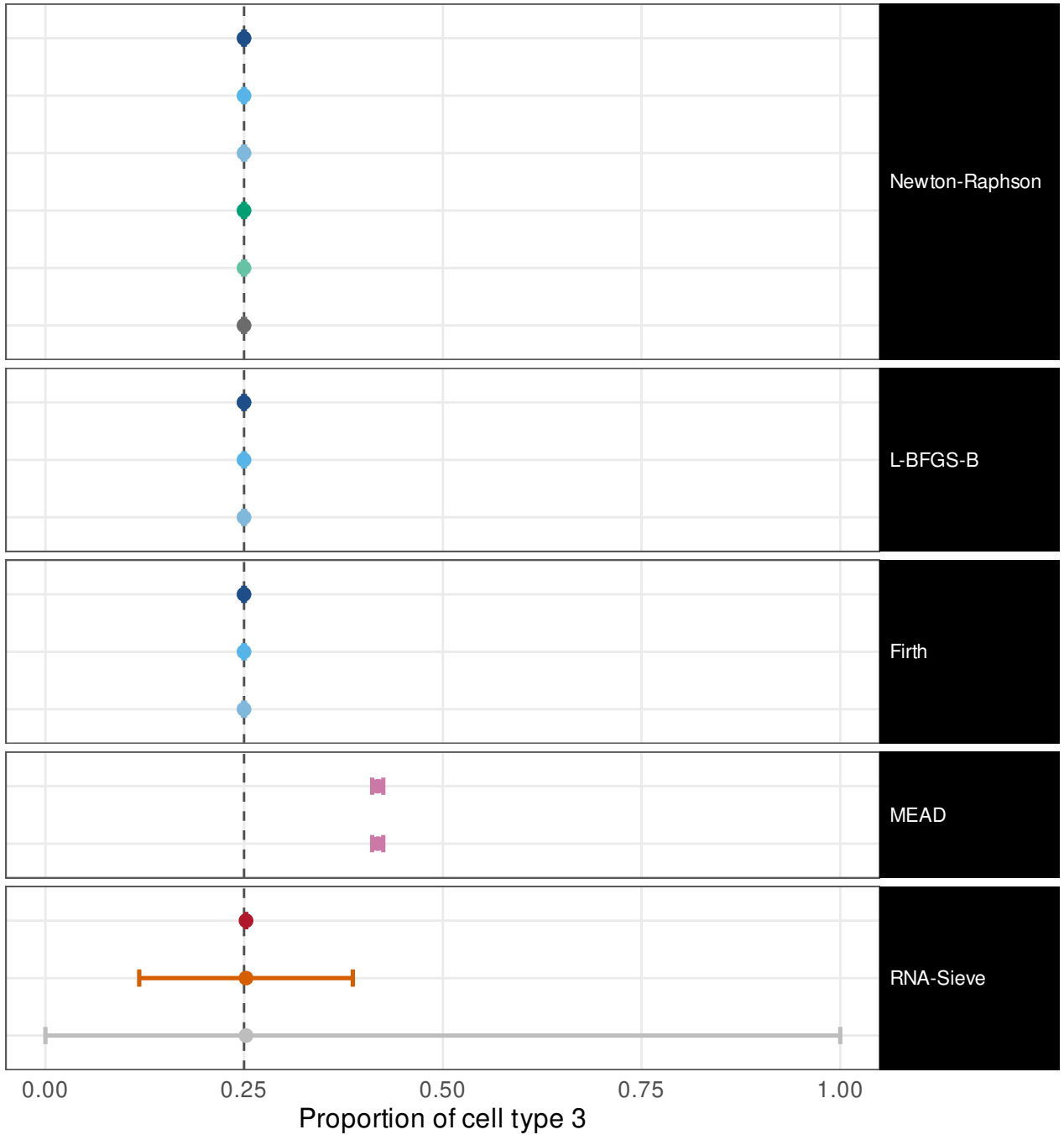
```
Wald (delta)                        0.88 [0.83, 0.92]   w = 0.00195
ILR, mapped back                    0.89 [0.84, 0.93]   w = 0.00194
ILR, REML                           0.94 [ 0.9, 0.97]   w = 0.00235
Profile likelihood                  0.9  [0.86, 0.94]   w = 0.00205
Profile, REML                       0.95 [0.91, 0.97]   w = 0.00236
Oracle (Fisher at p*)                 1  [0.98,    1]   w = 0.00236

Wald (delta)                        0.88 [0.83, 0.92]   w = 0.00195
ILR, mapped back                    0.88 [0.82, 0.91]   w = 0.00194
ILR, REML                           0.94 [ 0.9, 0.97]   w = 0.00236

Wald (delta)                        0.88 [0.83, 0.92]   w = 0.00195
ILR, mapped back                    0.88 [0.83, 0.92]   w = 0.00195
ILR, REML                           0.94 [ 0.9, 0.97]   w = 0.00235

Method's own interval                 0  [   0, 0.019]   w = 0.0139
Own interval, transcript fractions    1  [0.98,    1]   w = 0.0139

Fisher information (Wald)             0  [   0, 0.019]   w = 3.04e-05
Godambe sandwich                      1  [0.98,    1]   w = 0.269
Released code, as published           1  [0.98,    1]   w =    1
```

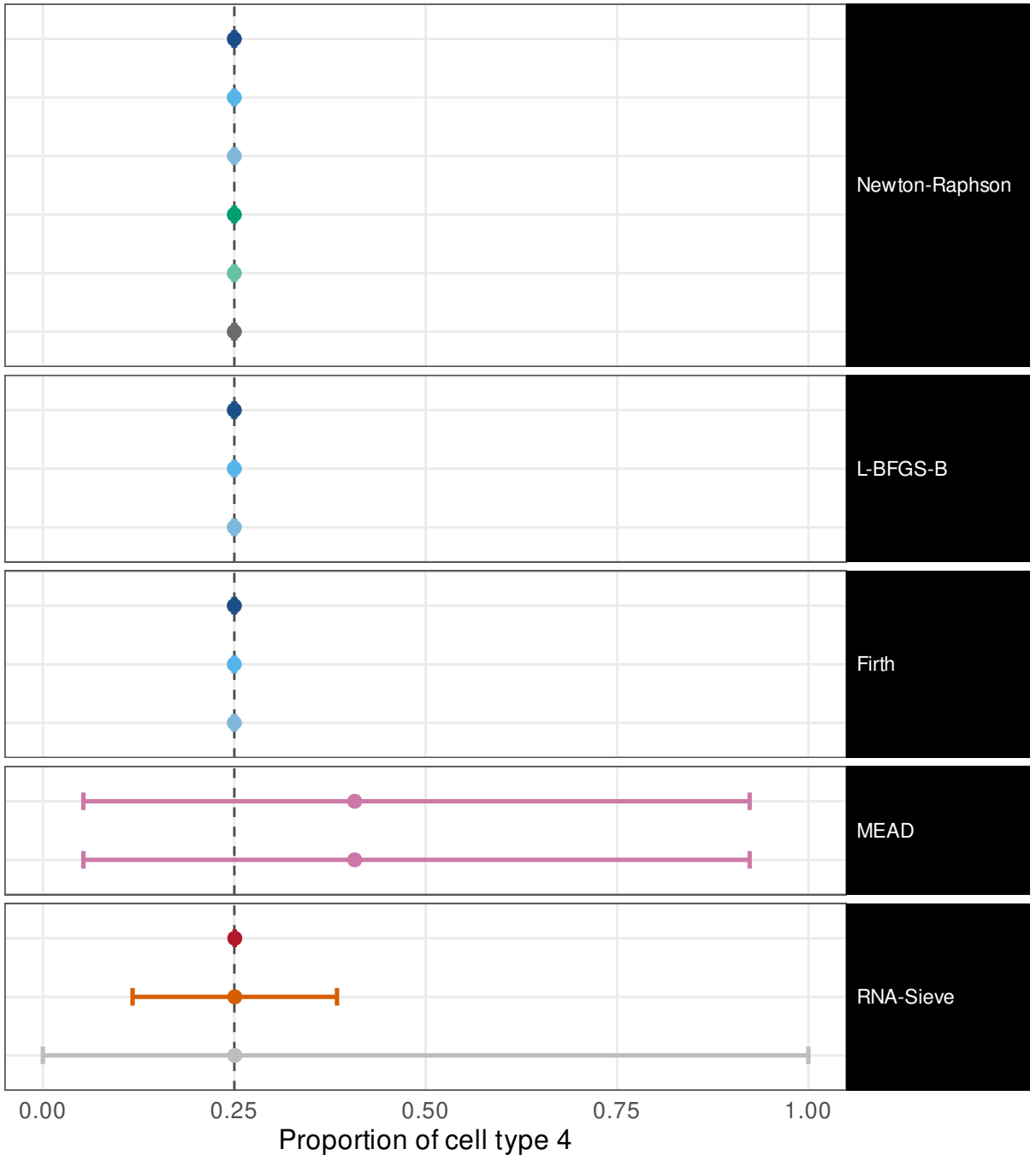


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / balanced / low sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.9 [0.85, 0.93]	w = 0.00212
ILR, mapped back	0.9 [0.86, 0.94]	w = 0.00212
ILR, REML	0.94 [0.9, 0.97]	w = 0.00256
Profile likelihood	0.92 [0.87, 0.95]	w = 0.00223
Profile, REML	0.96 [0.92, 0.98]	w = 0.00257
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.00254
Wald (delta)	0.9 [0.85, 0.93]	w = 0.00212
ILR, mapped back	0.9 [0.85, 0.93]	w = 0.00211
ILR, REML	0.94 [0.9, 0.97]	w = 0.00257
Wald (delta)	0.9 [0.85, 0.93]	w = 0.00212
ILR, mapped back	0.9 [0.85, 0.93]	w = 0.00212
ILR, REML	0.95 [0.91, 0.97]	w = 0.00255
Method's own interval	0.87 [0.82, 0.91]	w = 0.87
Own interval, transcript fractions	0.87 [0.82, 0.91]	w = 0.87
Fisher information (Wald)	0.005 [0.00088, 0.028]	w = 3.06e-05
Godambe sandwich	1 [0.98, 1]	w = 0.267
Released code, as published	1 [0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / balanced / high sibling overlap / a = 1 / equal sizes

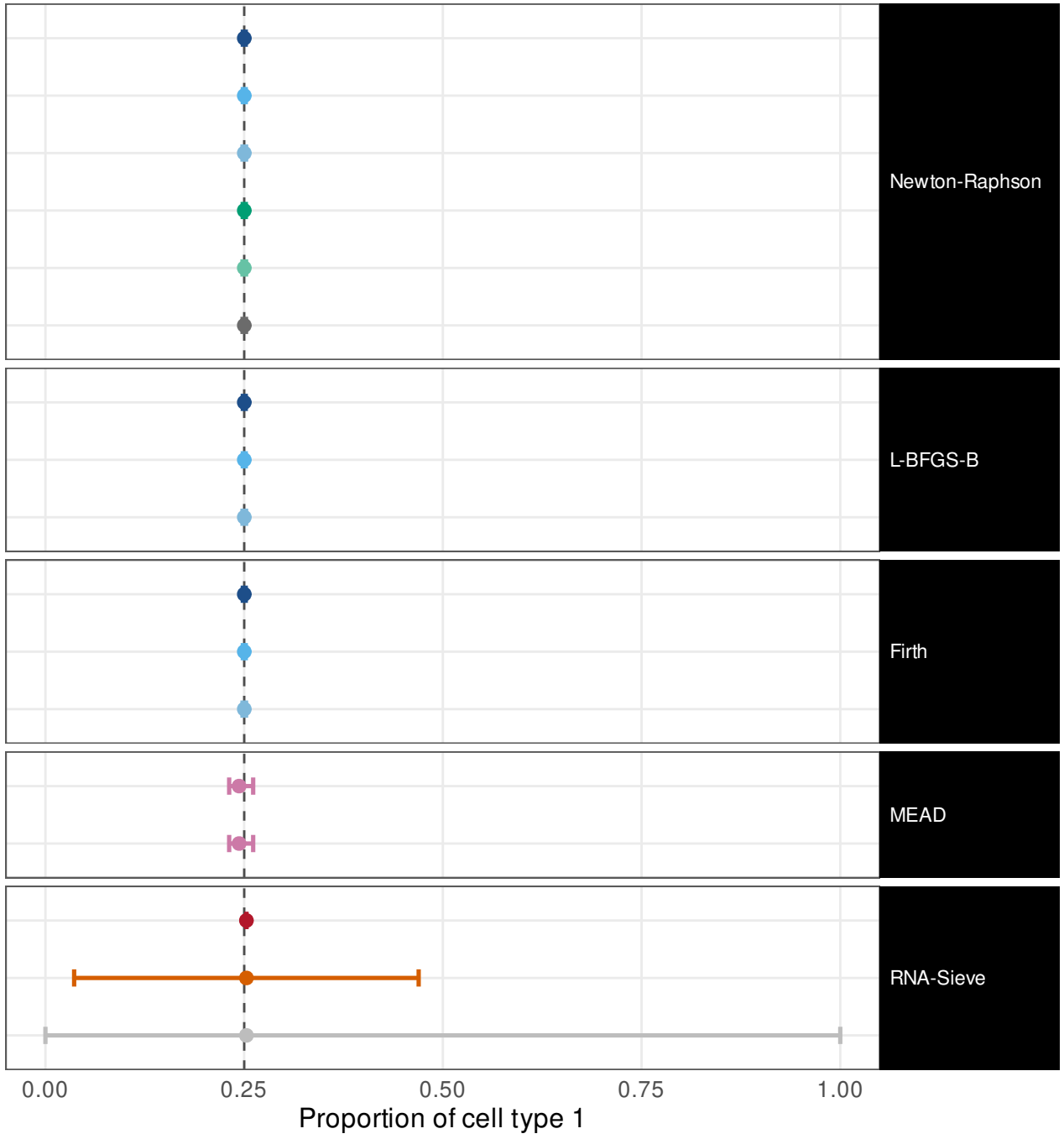
```
Wald (delta)                          0.94 [ 0.9, 0.97]   w = 0.00365
ILR, mapped back                      0.94 [0.89, 0.96]   w = 0.00366
ILR, REML                             0.97 [0.94, 0.99]   w = 0.00441
Profile likelihood                    0.94 [ 0.9, 0.97]   w = 0.00384
Profile, REML                         0.97 [0.94, 0.99]   w = 0.00442
Oracle (Fisher at p*)                    1 [0.98,    1]   w = 0.00429

Wald (delta)                          0.94 [ 0.9, 0.97]   w = 0.00365
ILR, mapped back                      0.94 [ 0.9, 0.97]   w = 0.00365
ILR, REML                             0.97 [0.94, 0.99]   w = 0.0044

Wald (delta)                          0.94 [ 0.9, 0.97]   w = 0.00365
ILR, mapped back                      0.94 [ 0.9, 0.97]   w = 0.00364
ILR, REML                             0.96 [0.92, 0.98]   w = 0.00441

Method's own interval                 0.92 [0.87, 0.95]   w = 0.0302
Own interval, transcript fractions    0.92 [0.87, 0.95]   w = 0.0302

Fisher information (Wald)                0 [   0, 0.019]   w = 0.000159
Godambe sandwich                         1 [0.98,    1]   w = 0.433
Released code, as published              1 [0.98,    1]   w =     1
```

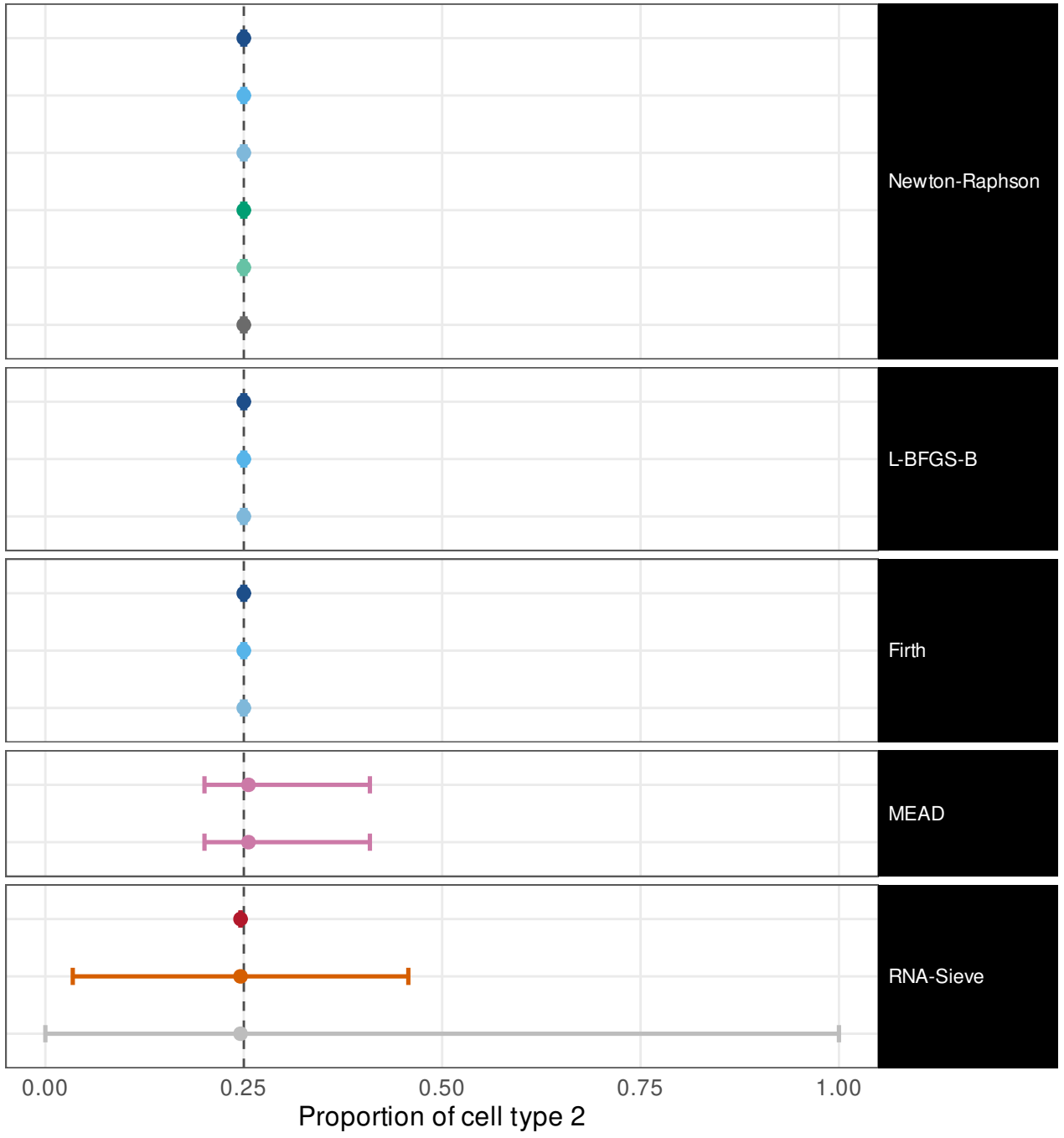


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / balanced / high sibling overlap / a = 1 / equal sizes

Wald (delta)	0.96 [0.92, 0.98]	w = 0.00368
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.00369
ILR, REML	0.98 [0.95, 0.99]	w = 0.00444
Profile likelihood	0.96 [0.92, 0.98]	w = 0.00387
Profile, REML	0.98 [0.94, 0.99]	w = 0.00445
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.00435
Wald (delta)	0.96 [0.92, 0.98]	w = 0.00368
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.00367
ILR, REML	0.98 [0.94, 0.99]	w = 0.00444
Wald (delta)	0.96 [0.92, 0.98]	w = 0.00368
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.00367
ILR, REML	0.98 [0.94, 0.99]	w = 0.00444
Method's own interval	0.53 [0.46, 0.6]	w = 0.208
Own interval, transcript fractions	0.53 [0.46, 0.6]	w = 0.208
Fisher information (Wald)	0 [0, 0.019]	w = 0.000147
Godambe sandwich	1 [0.98, 1]	w = 0.423
Released code, as published	1 [0.98, 1]	w = 1

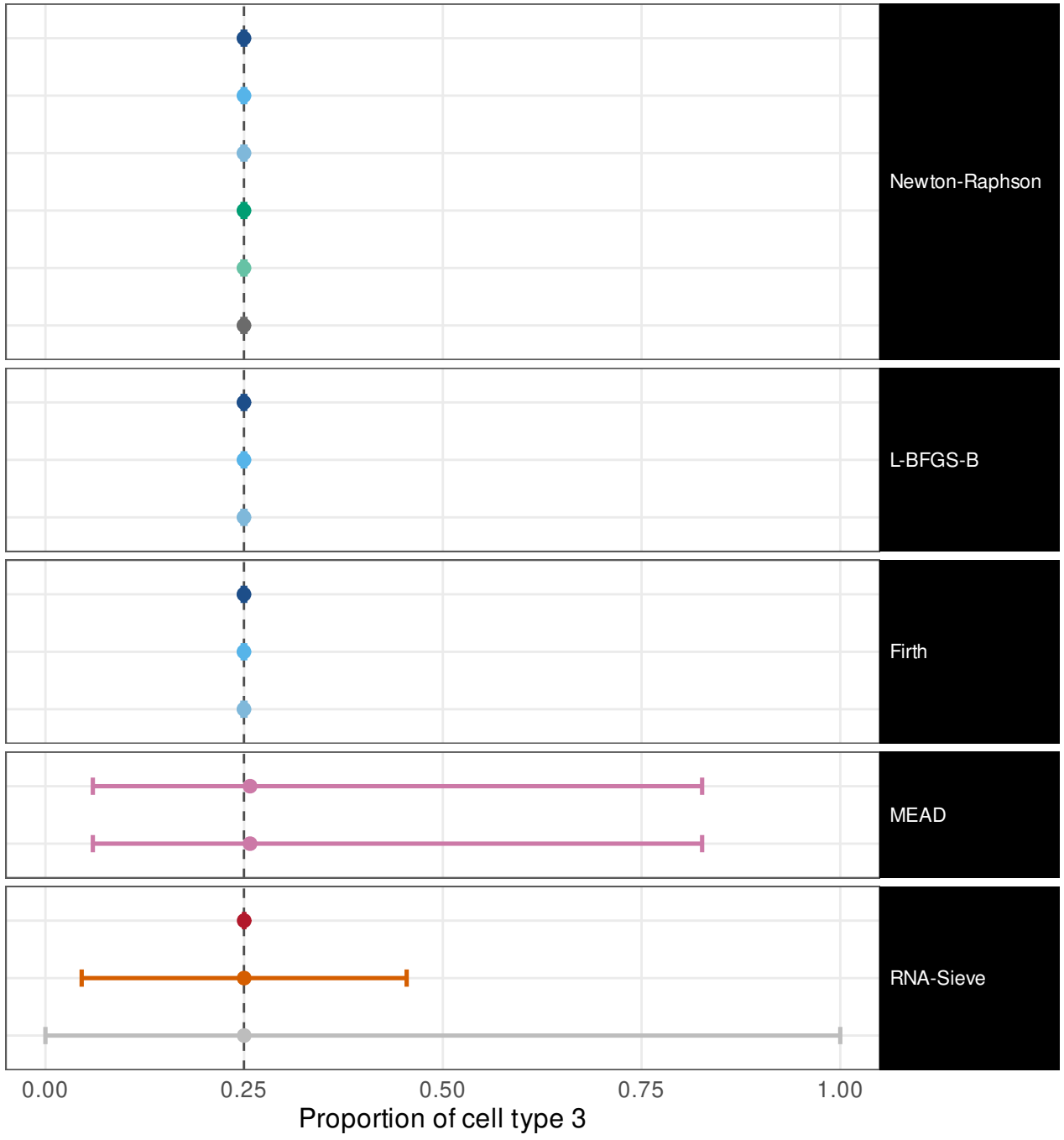


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / balanced / high sibling overlap / a = 1 / equal sizes

Wald (delta)	0.92 [0.88, 0.95]	w = 0.00322
ILR, mapped back	0.92 [0.88, 0.95]	w = 0.00322
ILR, REML	0.97 [0.94, 0.99]	w = 0.00388
Profile likelihood	0.94 [0.9, 0.97]	w = 0.00338
Profile, REML	0.97 [0.94, 0.99]	w = 0.00389
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.00385
Wald (delta)	0.92 [0.88, 0.95]	w = 0.00322
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.00321
ILR, REML	0.97 [0.94, 0.99]	w = 0.00389
Wald (delta)	0.92 [0.88, 0.95]	w = 0.00322
ILR, mapped back	0.92 [0.88, 0.95]	w = 0.0032
ILR, REML	0.96 [0.93, 0.98]	w = 0.00387
Method's own interval	0.76 [0.7, 0.82]	w = 0.767
Own interval, transcript fractions	0.76 [0.7, 0.82]	w = 0.767
Fisher information (Wald)	0.05 [0.027, 0.09]	w = 0.000144
Godambe sandwich	1 [0.98, 1]	w = 0.409
Released code, as published	1 [0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / balanced / high sibling overlap / a = 1 / equal sizes

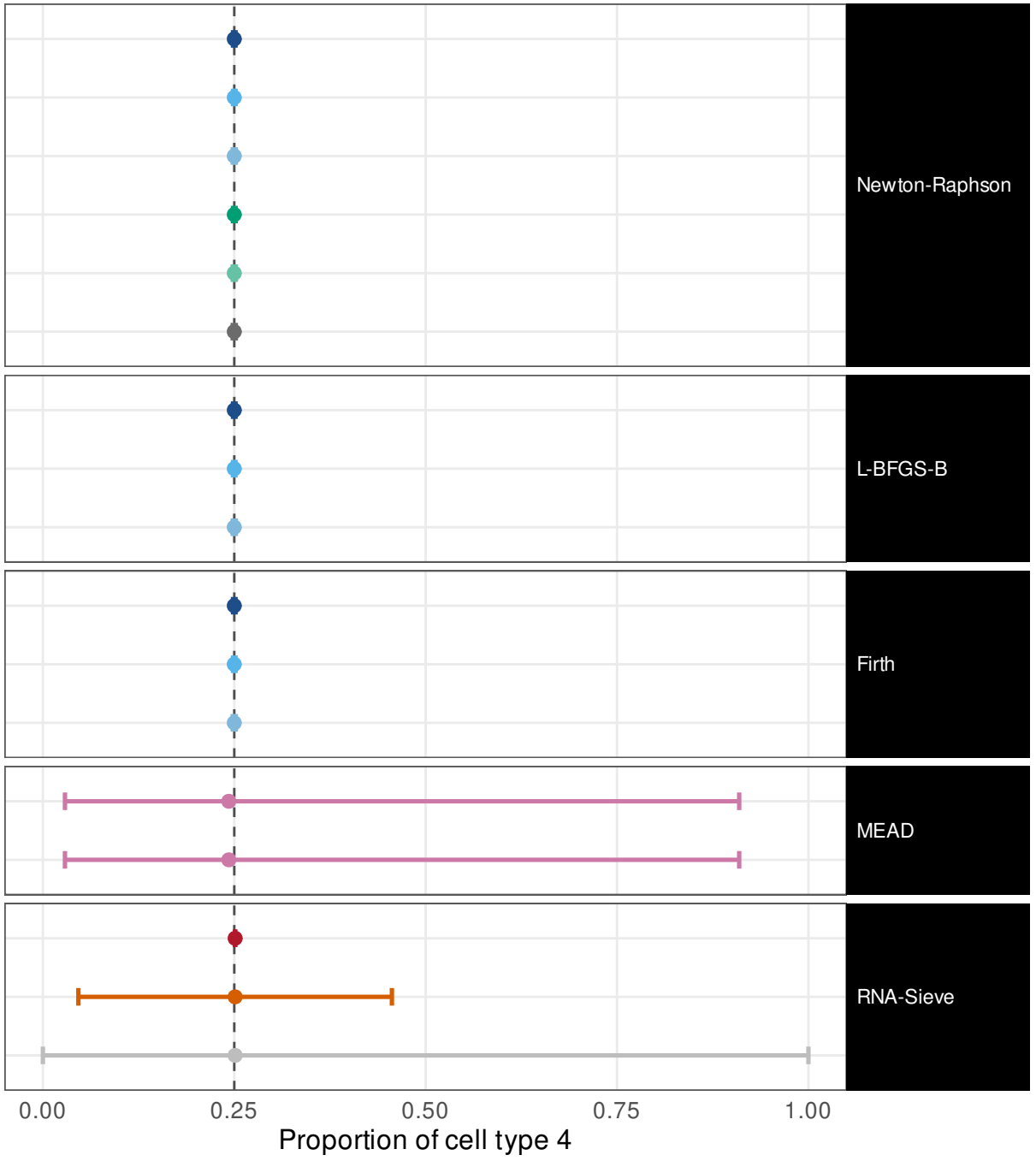
```
Wald (delta)                          0.9  [0.86, 0.94]   w = 0.00328
ILR, mapped back                      0.9  [0.86, 0.94]   w = 0.00327
ILR, REML                             0.96 [0.93, 0.98]   w = 0.00396
Profile likelihood                    0.91 [0.86, 0.94]   w = 0.00345
Profile, REML                         0.96 [0.92, 0.98]   w = 0.00397
Oracle (Fisher at p*)                    1 [0.98,    1]   w = 0.00399

Wald (delta)                          0.9  [0.86, 0.94]   w = 0.00328
ILR, mapped back                      0.9  [0.86, 0.94]   w = 0.00328
ILR, REML                             0.96 [0.93, 0.98]   w = 0.00397

Wald (delta)                          0.9  [0.86, 0.94]   w = 0.00328
ILR, mapped back                      0.9  [0.86, 0.94]   w = 0.00328
ILR, REML                             0.96 [0.93, 0.98]   w = 0.00396

Method's own interval                 0.88 [0.83, 0.92]   w = 0.881
Own interval, transcript fractions    0.88 [0.83, 0.92]   w = 0.881

Fisher information (Wald)             0.005 [0.00088, 0.028]   w = 0.000154
Godambe sandwich                         1 [0.98,    1]   w = 0.41
Released code, as published              1 [0.98,    1]   w =    1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / balanced / high sibling overlap / a = 1 / sizes 1:1:5:5

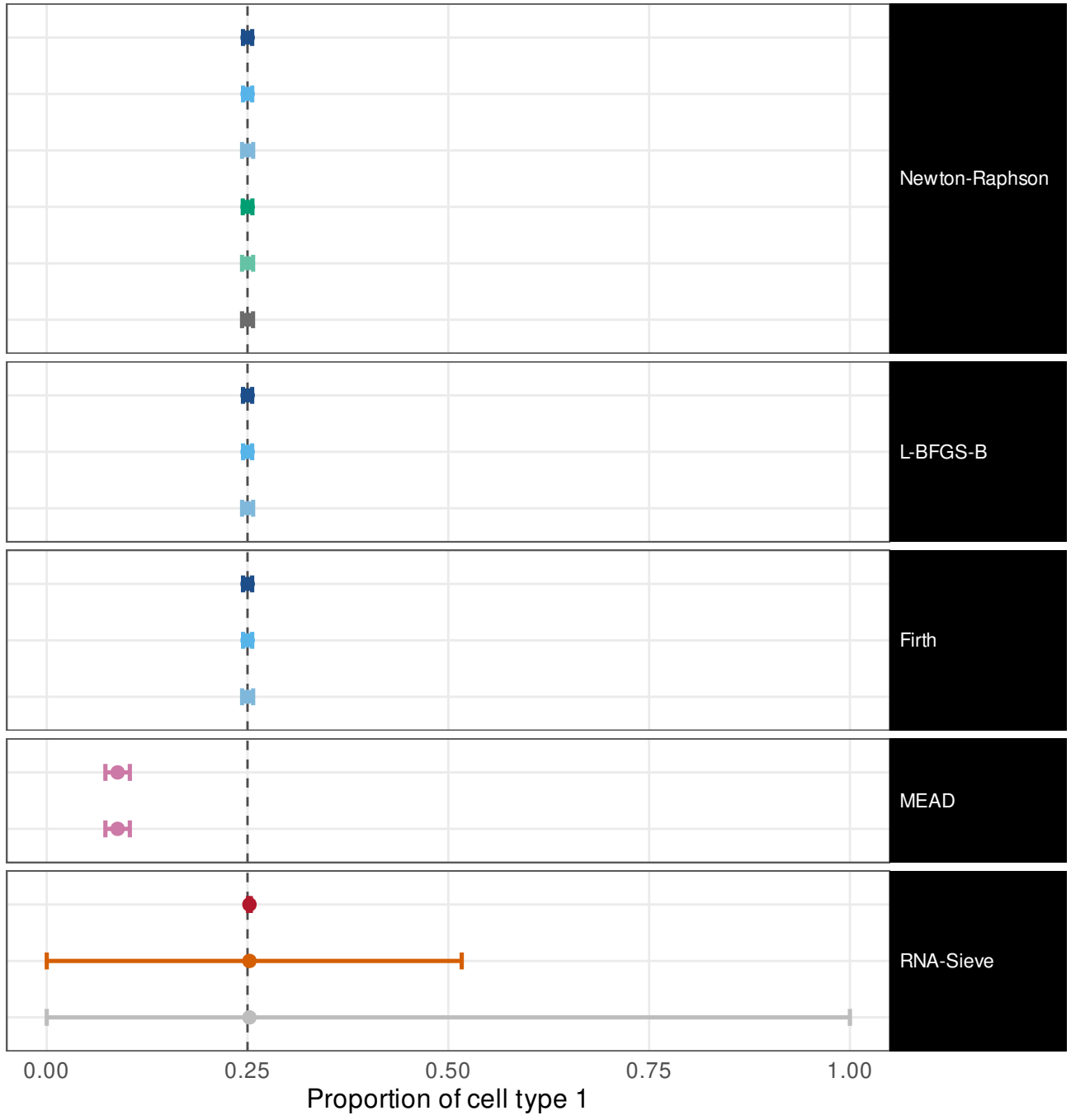
```
Wald (delta)                        0.92 [0.87, 0.95]   w = 0.0114
ILR, mapped back                    0.91 [0.86, 0.94]   w = 0.0114
ILR, REML                           0.96 [0.92, 0.98]   w = 0.0137
Profile likelihood                  0.94 [0.89, 0.96]   w = 0.012
Profile, REML                       0.96 [0.92, 0.98]   w = 0.0138
Oracle (Fisher at p*)                 1 [0.98,    1]    w = 0.0136

Wald (delta)                        0.92 [0.87, 0.95]   w = 0.0114
ILR, mapped back                    0.91 [0.86, 0.94]   w = 0.0113
ILR, REML                           0.96 [0.92, 0.98]   w = 0.0137

Wald (delta)                        0.92 [0.87, 0.95]   w = 0.0114
ILR, mapped back                    0.92 [0.87, 0.95]   w = 0.0113
ILR, REML                           0.96 [0.92, 0.98]   w = 0.0137

Method's own interval                 0 [   0, 0.019]   w = 0.0304
Own interval, transcript fractions    1 [0.98,    1]    w = 0.0304

Fisher information (Wald)           0.32 [0.25, 0.38]   w = 0.00344
Godambe sandwich                      1 [0.98,    1]    w = 0.517
Released code, as published           1 [0.98,    1]    w =     1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / balanced / high sibling overlap / a = 1 / sizes 1:1:5:5

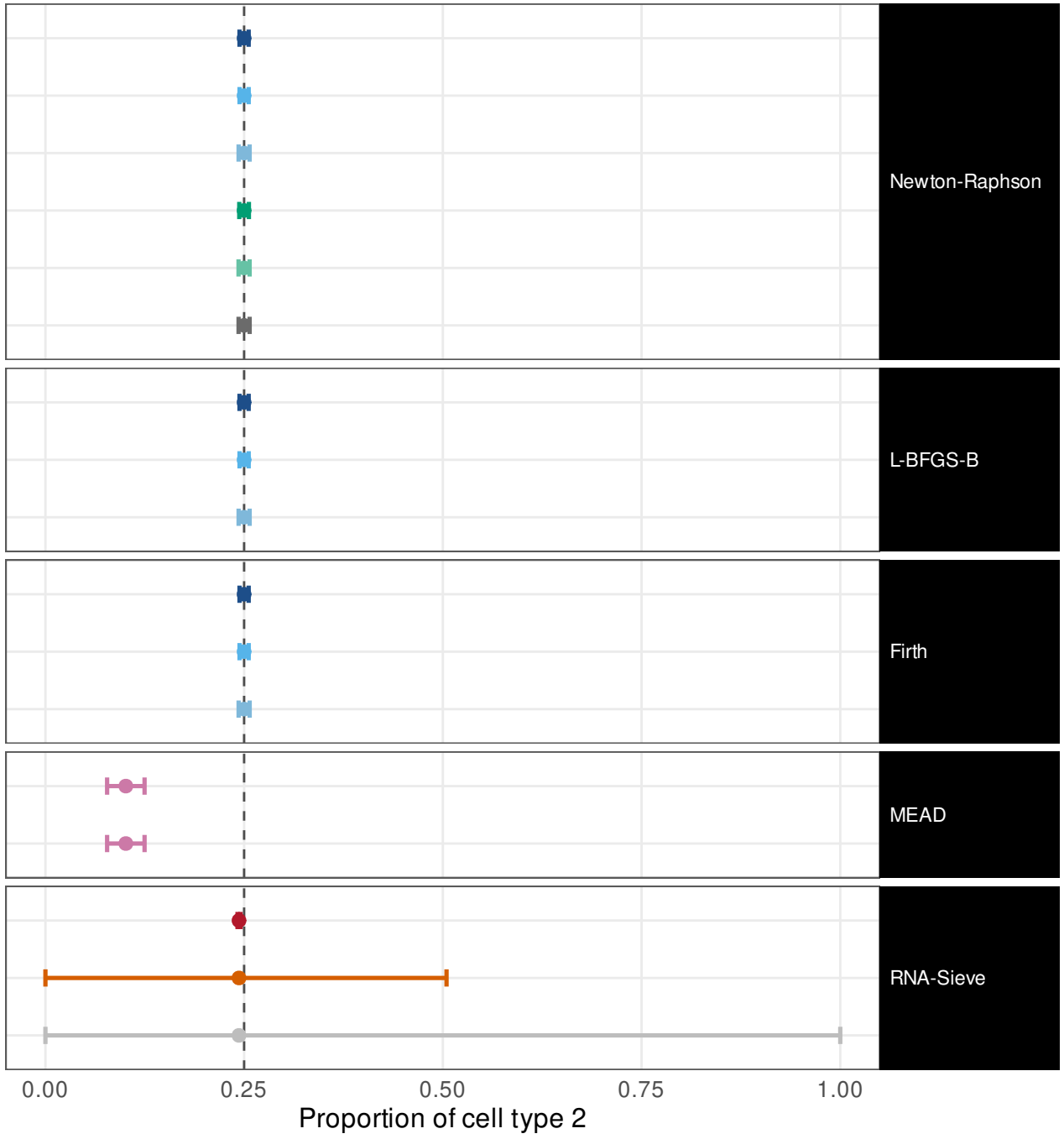
```
Wald (delta)                        0.92 [0.88, 0.95]   w = 0.0115
ILR, mapped back                    0.92 [0.88, 0.95]   w = 0.0115
ILR, REML                           0.97 [0.94, 0.99]   w = 0.0139
Profile likelihood                  0.94 [0.89, 0.96]   w = 0.0121
Profile, REML                       0.96 [0.93, 0.98]   w = 0.0139
Oracle (Fisher at p*)                 1 [0.98,    1]    w = 0.0139

Wald (delta)                        0.92 [0.88, 0.95]   w = 0.0115
ILR, mapped back                    0.92 [0.87, 0.95]   w = 0.0115
ILR, REML                           0.96 [0.93, 0.98]   w = 0.0139

Wald (delta)                        0.92 [0.88, 0.95]   w = 0.0115
ILR, mapped back                    0.92 [0.87, 0.95]   w = 0.0115
ILR, REML                           0.97 [0.94, 0.99]   w = 0.0139

Method's own interval                 0 [   0, 0.019]   w = 0.0471
Own interval, transcript fractions    1 [0.98,    1]    w = 0.0471

Fisher information (Wald)           0.085 [0.054, 0.13]   w = 0.00368
Godambe sandwich                      1 [0.98,    1]    w = 0.505
Released code, as published           1 [0.98,    1]    w =    1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / balanced / high sibling overlap / a = 1 / sizes 1:1:5:5

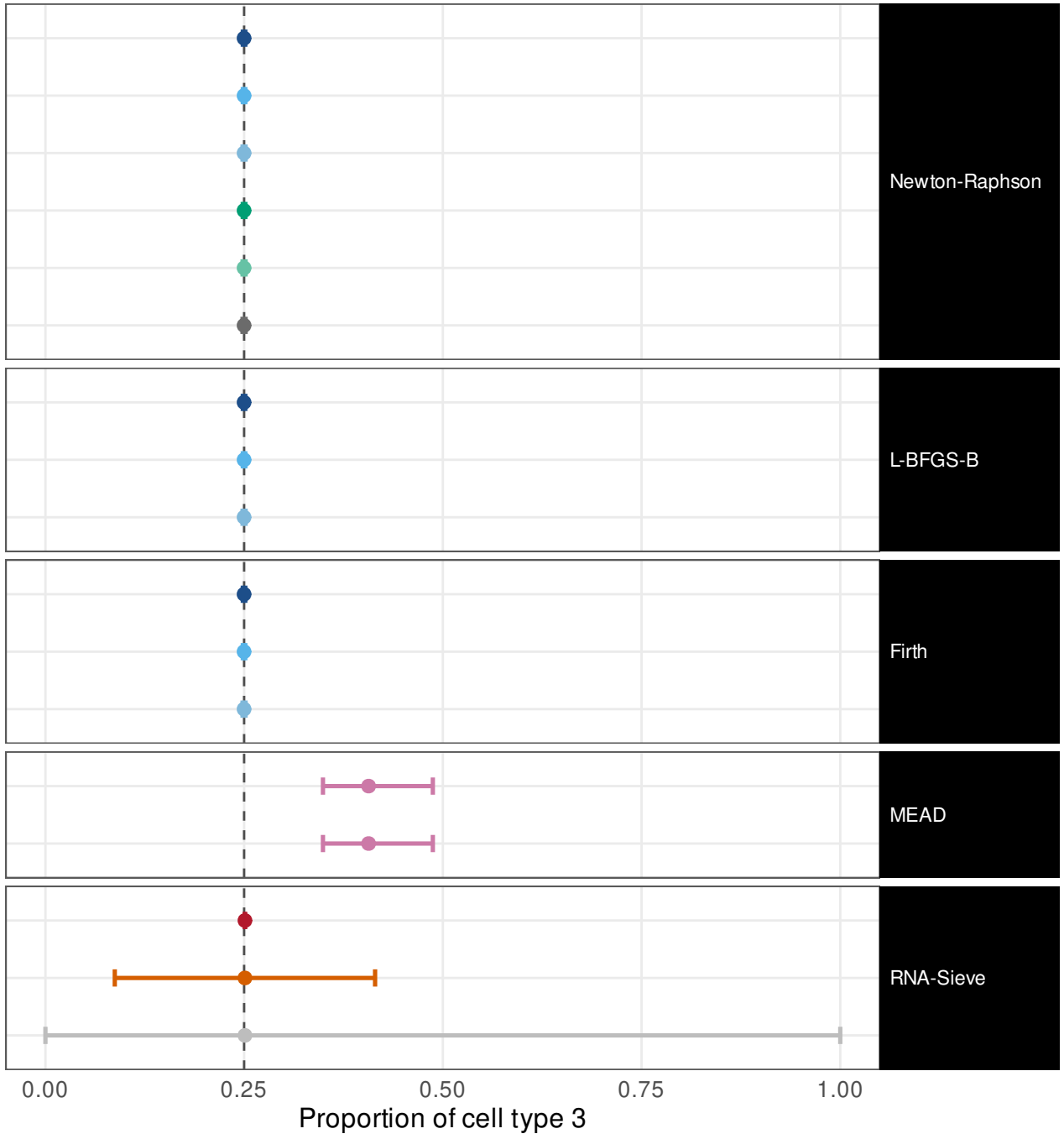
```
Wald (delta)                          0.9  [0.86, 0.94]   w = 0.00306
ILR, mapped back                      0.9  [0.85, 0.93]   w = 0.00305
ILR, REML                            0.94  [ 0.9, 0.97]   w = 0.0037
Profile likelihood                   0.92  [0.87, 0.95]   w = 0.00322
Profile, REML                        0.94  [ 0.9, 0.97]   w = 0.0037
Oracle (Fisher at p*)                   1  [0.98,    1]   w = 0.00377

Wald (delta)                          0.9  [0.86, 0.94]   w = 0.00306
ILR, mapped back                      0.9  [0.86, 0.94]   w = 0.00306
ILR, REML                            0.94  [ 0.9, 0.97]   w = 0.0037

Wald (delta)                          0.9  [0.86, 0.94]   w = 0.00306
ILR, mapped back                      0.9  [0.86, 0.94]   w = 0.00305
ILR, REML                            0.94  [ 0.9, 0.97]   w = 0.0037

Method's own interval                0.12  [0.086, 0.18]   w = 0.138
Own interval, transcript fractions   0.32  [0.25, 0.38]   w = 0.138

Fisher information (Wald)            0.12  [0.078, 0.17]   w = 0.000532
Godambe sandwich                        1  [0.98,    1]   w = 0.328
Released code, as published             1  [0.98,    1]   w =     1
```

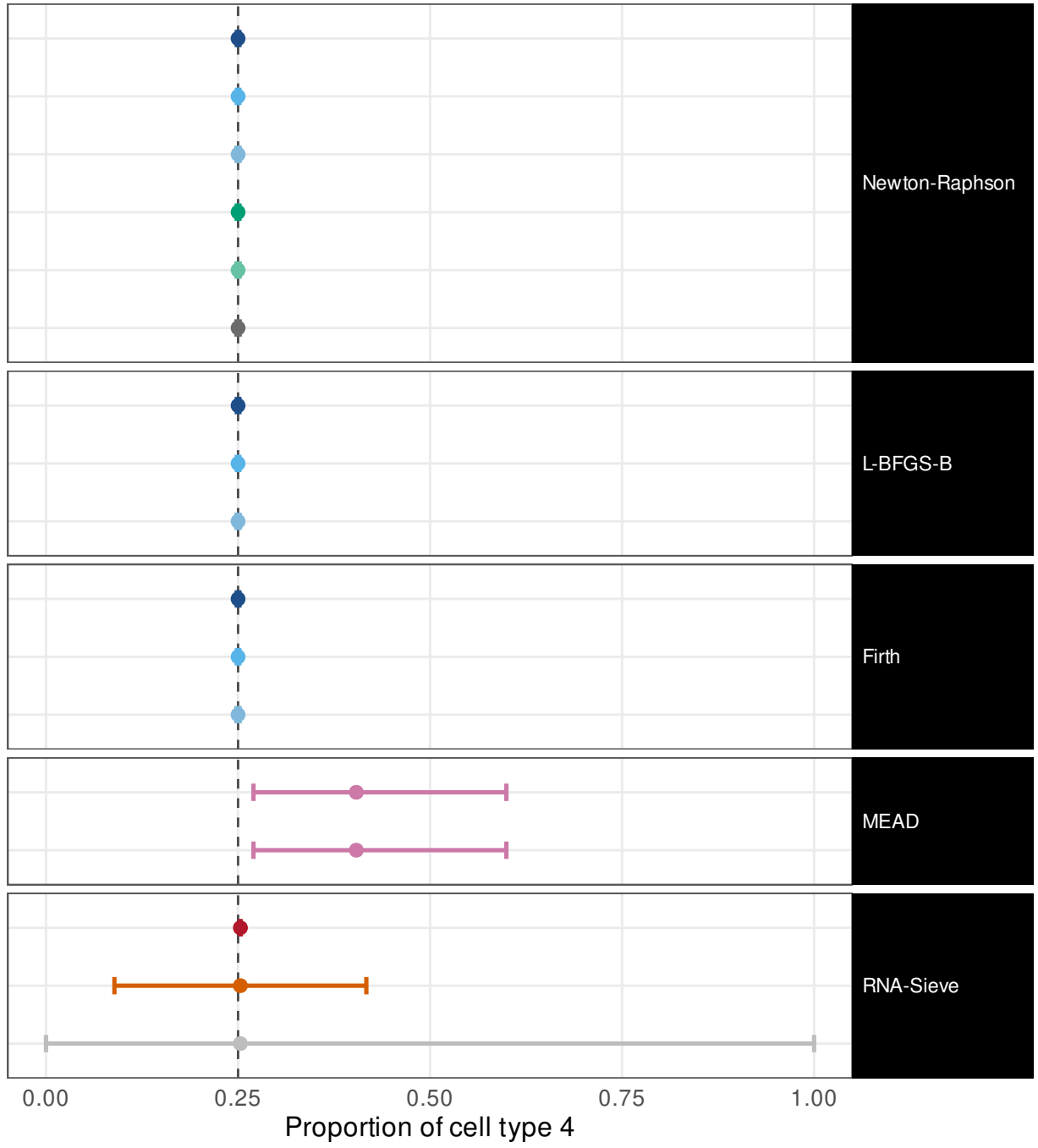


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / balanced / high sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.9 [0.84, 0.93]	w = 0.00333
ILR, mapped back	0.88 [0.83, 0.92]	w = 0.00333
ILR, REML	0.95 [0.91, 0.97]	w = 0.00403
Profile likelihood	0.91 [0.86, 0.94]	w = 0.0035
Profile, REML	0.95 [0.91, 0.97]	w = 0.00403
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.00413
Wald (delta)	0.9 [0.84, 0.93]	w = 0.00333
ILR, mapped back	0.88 [0.83, 0.92]	w = 0.00332
ILR, REML	0.95 [0.91, 0.97]	w = 0.00403
Wald (delta)	0.9 [0.84, 0.93]	w = 0.00333
ILR, mapped back	0.89 [0.84, 0.93]	w = 0.00332
ILR, REML	0.94 [0.9, 0.97]	w = 0.00402
Method's own interval	0.32 [0.26, 0.39]	w = 0.329
Own interval, transcript fractions	0.34 [0.27, 0.4]	w = 0.329
Fisher information (Wald)	0.015 [0.0051, 0.043]	w = 0.000542
Godambe sandwich	1 [0.98, 1]	w = 0.328
Released code, as published	1 [0.98, 1]	w = 1

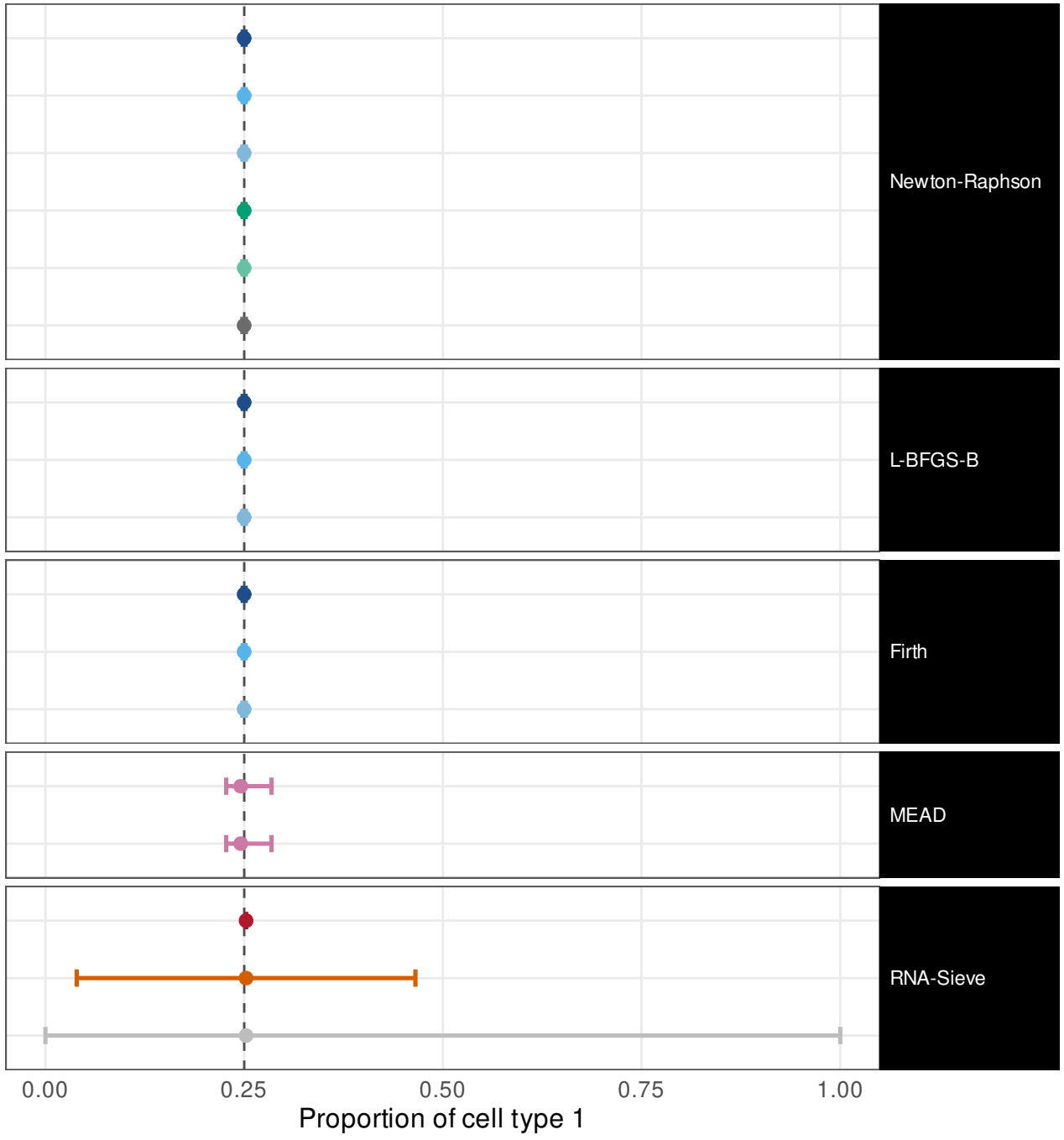


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / balanced / high sibling overlap / a = 5 / equal sizes

Wald (delta)	0.86 [0.81, 0.91]	w = 0.00363
ILR, mapped back	0.87 [0.82, 0.91]	w = 0.00362
ILR, REML	0.96 [0.92, 0.98]	w = 0.00437
Profile likelihood	0.88 [0.83, 0.92]	w = 0.00381
Profile, REML	0.96 [0.92, 0.98]	w = 0.00439
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.00434
Wald (delta)	0.86 [0.81, 0.91]	w = 0.00363
ILR, mapped back	0.86 [0.81, 0.91]	w = 0.00362
ILR, REML	0.96 [0.93, 0.98]	w = 0.00437
Wald (delta)	0.86 [0.81, 0.91]	w = 0.00363
ILR, mapped back	0.87 [0.82, 0.91]	w = 0.00362
ILR, REML	0.94 [0.9, 0.97]	w = 0.00438
Method's own interval	0.93 [0.89, 0.96]	w = 0.057
Own interval, transcript fractions	0.93 [0.89, 0.96]	w = 0.057
Fisher information (Wald)	0 [0, 0.019]	w = 3.35e-05
Godambe sandwich	1 [0.98, 1]	w = 0.426
Released code, as published	1 [0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / balanced / high sibling overlap / a = 5 / equal sizes

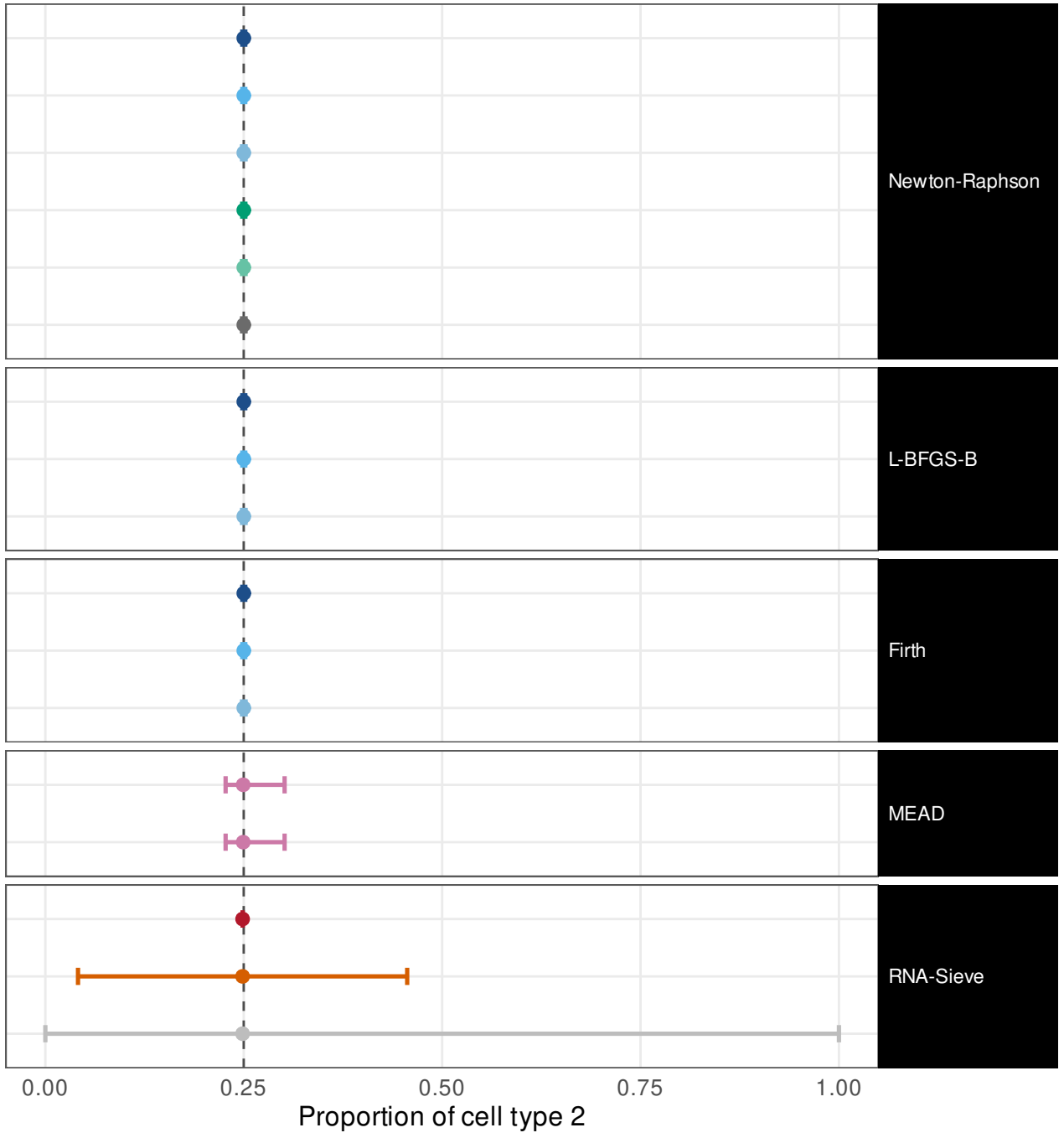
```
Wald (delta)                          0.89 [0.84, 0.93]   w = 0.00366
ILR, mapped back                      0.89 [0.84, 0.93]   w = 0.00366
ILR, REML                             0.94 [ 0.9, 0.97]   w = 0.0044
Profile likelihood                    0.9  [0.85, 0.93]   w = 0.00384
Profile, REML                         0.94 [0.89, 0.96]   w = 0.00442
Oracle (Fisher at p*)                   1  [0.98,    1]   w = 0.00436

Wald (delta)                          0.89 [0.84, 0.93]   w = 0.00366
ILR, mapped back                      0.88 [0.83, 0.92]   w = 0.00365
ILR, REML                             0.93 [0.89, 0.96]   w = 0.0044

Wald (delta)                          0.89 [0.84, 0.93]   w = 0.00366
ILR, mapped back                      0.9  [0.85, 0.93]   w = 0.00364
ILR, REML                             0.94 [0.89, 0.96]   w = 0.00442

Method's own interval                 0.98 [0.96, 0.99]   w = 0.0743
Own interval, transcript fractions    0.98 [0.96, 0.99]   w = 0.0743

Fisher information (Wald)               0  [   0, 0.019]   w = 3.09e-05
Godambe sandwich                        1  [0.98,    1]   w = 0.415
Released code, as published             1  [0.98,    1]   w =    1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / balanced / high sibling overlap / a = 5 / equal sizes

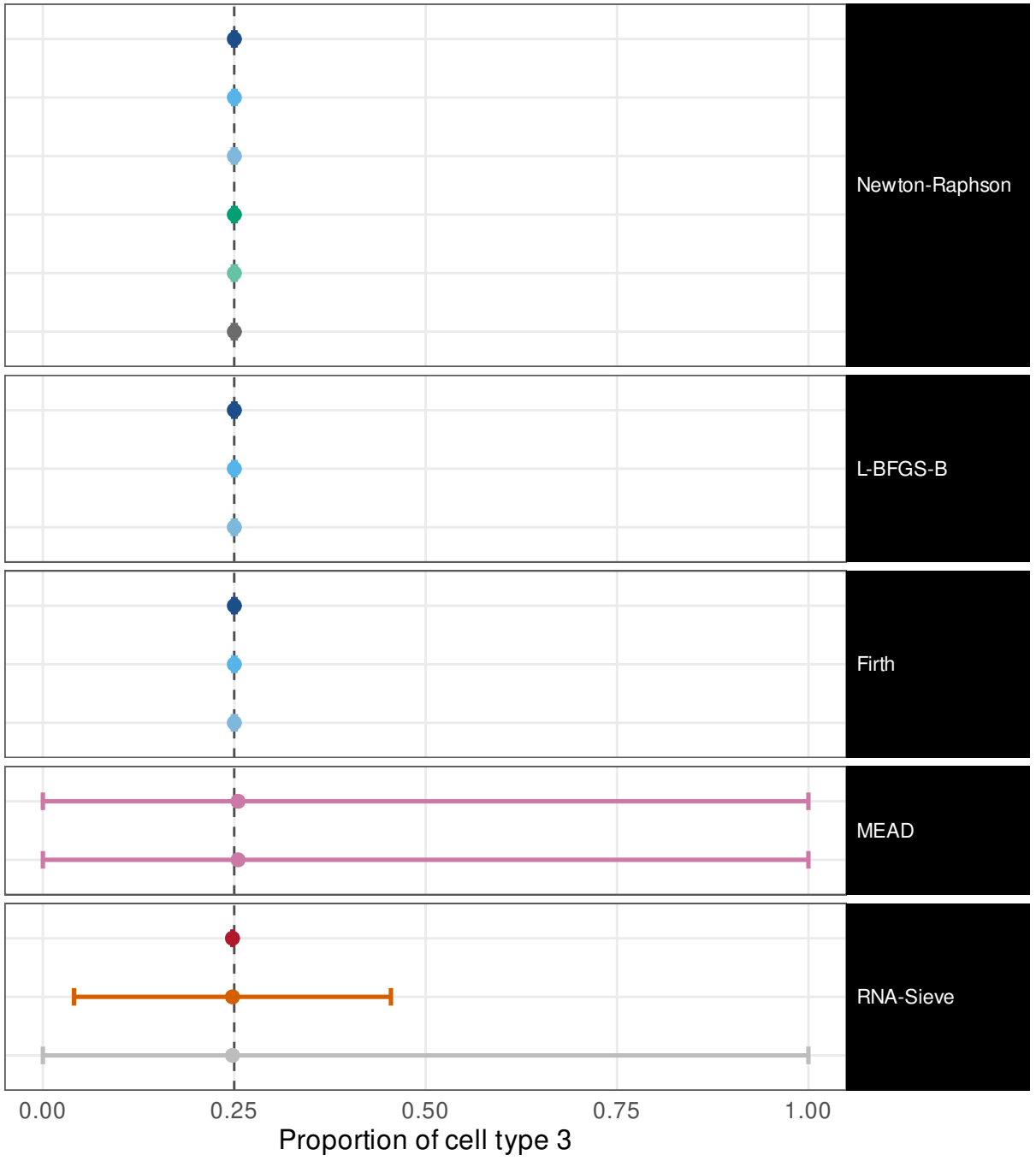
```
Wald (delta)                        0.92 [0.87, 0.95]   w = 0.0032
ILR, mapped back                    0.91 [0.86, 0.94]   w = 0.00319
ILR, REML                           0.94 [ 0.9, 0.97]   w = 0.00385
Profile likelihood                  0.92 [0.87, 0.95]   w = 0.00336
Profile, REML                       0.96 [0.92, 0.98]   w = 0.00387
Oracle (Fisher at p*)                  1 [0.98,    1]   w = 0.00383

Wald (delta)                        0.92 [0.87, 0.95]   w = 0.0032
ILR, mapped back                    0.9  [0.85, 0.93]   w = 0.00319
ILR, REML                           0.96 [0.92, 0.98]   w = 0.00386

Wald (delta)                        0.92 [0.87, 0.95]   w = 0.0032
ILR, mapped back                    0.92 [0.87, 0.95]   w = 0.00319
ILR, REML                           0.96 [0.92, 0.98]   w = 0.00385

Method's own interval                  1 [0.98,    1]   w =    1
Own interval, transcript fractions     1 [0.98,    1]   w =    1

Fisher information (Wald)           0.005 [0.00088, 0.028]   w = 2.96e-05
Godambe sandwich                       1 [0.98,    1]   w = 0.414
Released code, as published            1 [0.98,    1]   w =    1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / balanced / high sibling overlap / a = 5 / equal sizes

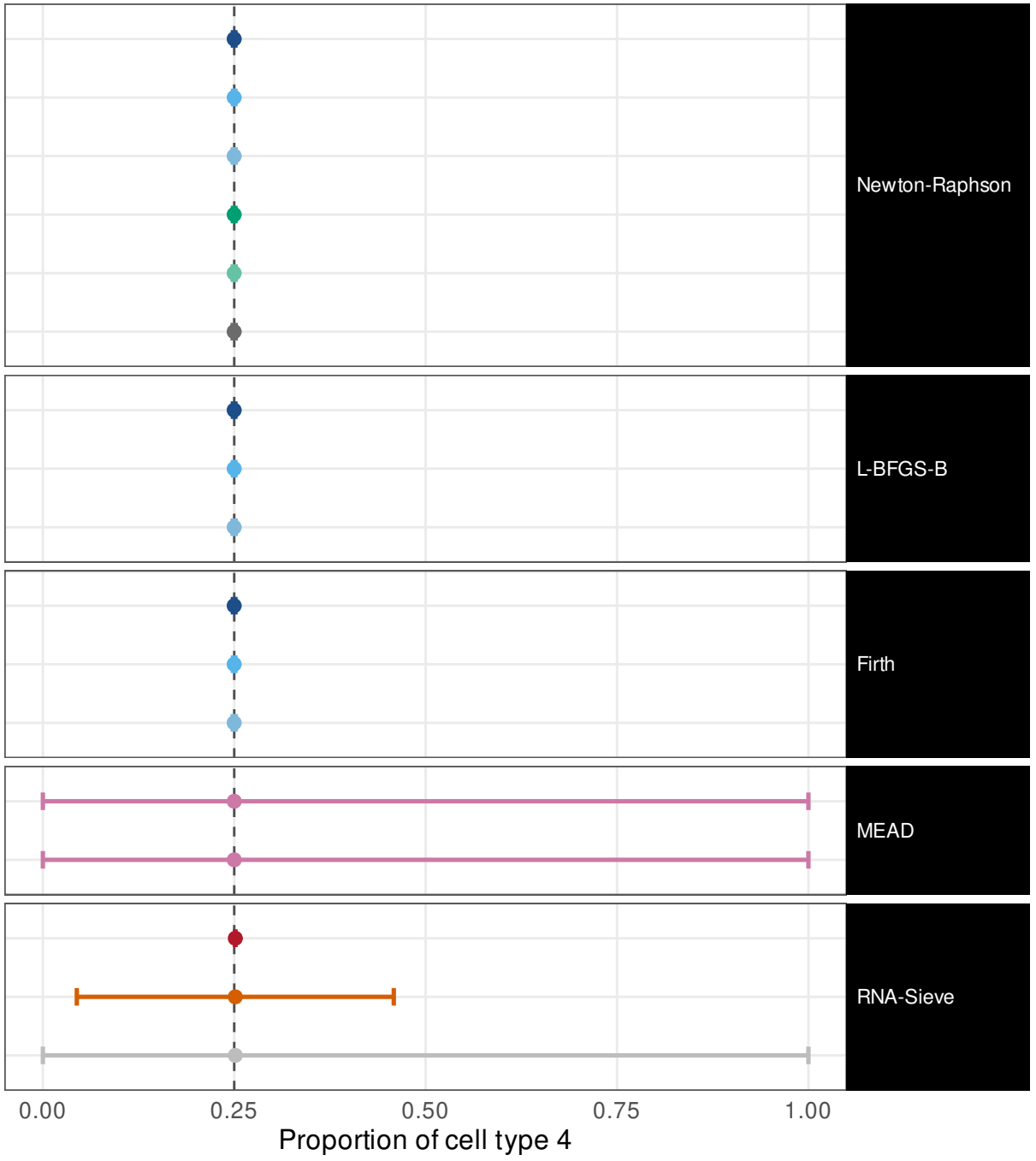
```
Wald (delta)                          0.9  [0.85, 0.93]   w = 0.00326
ILR, mapped back                      0.89 [0.84, 0.93]   w = 0.00325
ILR, REML                             0.96 [0.92, 0.98]   w = 0.00393
Profile likelihood                    0.92 [0.87, 0.95]   w = 0.00342
Profile, REML                         0.96 [0.92, 0.98]   w = 0.00394
Oracle (Fisher at p*)                   1  [0.98,    1]   w = 0.00397

Wald (delta)                          0.9  [0.85, 0.93]   w = 0.00326
ILR, mapped back                      0.9  [0.85, 0.93]   w = 0.00326
ILR, REML                             0.96 [0.92, 0.98]   w = 0.00394

Wald (delta)                          0.9  [0.85, 0.93]   w = 0.00326
ILR, mapped back                      0.9  [0.85, 0.93]   w = 0.00326
ILR, REML                             0.96 [0.93, 0.98]   w = 0.00393

Method's own interval                   1  [0.98,    1]   w =    1
Own interval, transcript fractions      1  [0.98,    1]   w =    1

Fisher information (Wald)             0.005 [0.00088, 0.028]   w = 3.27e-05
Godambe sandwich                        1  [0.98,    1]   w = 0.414
Released code, as published             1  [0.98,    1]   w =    1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / balanced / high sibling overlap / a = 5 / sizes 1:1:5:5

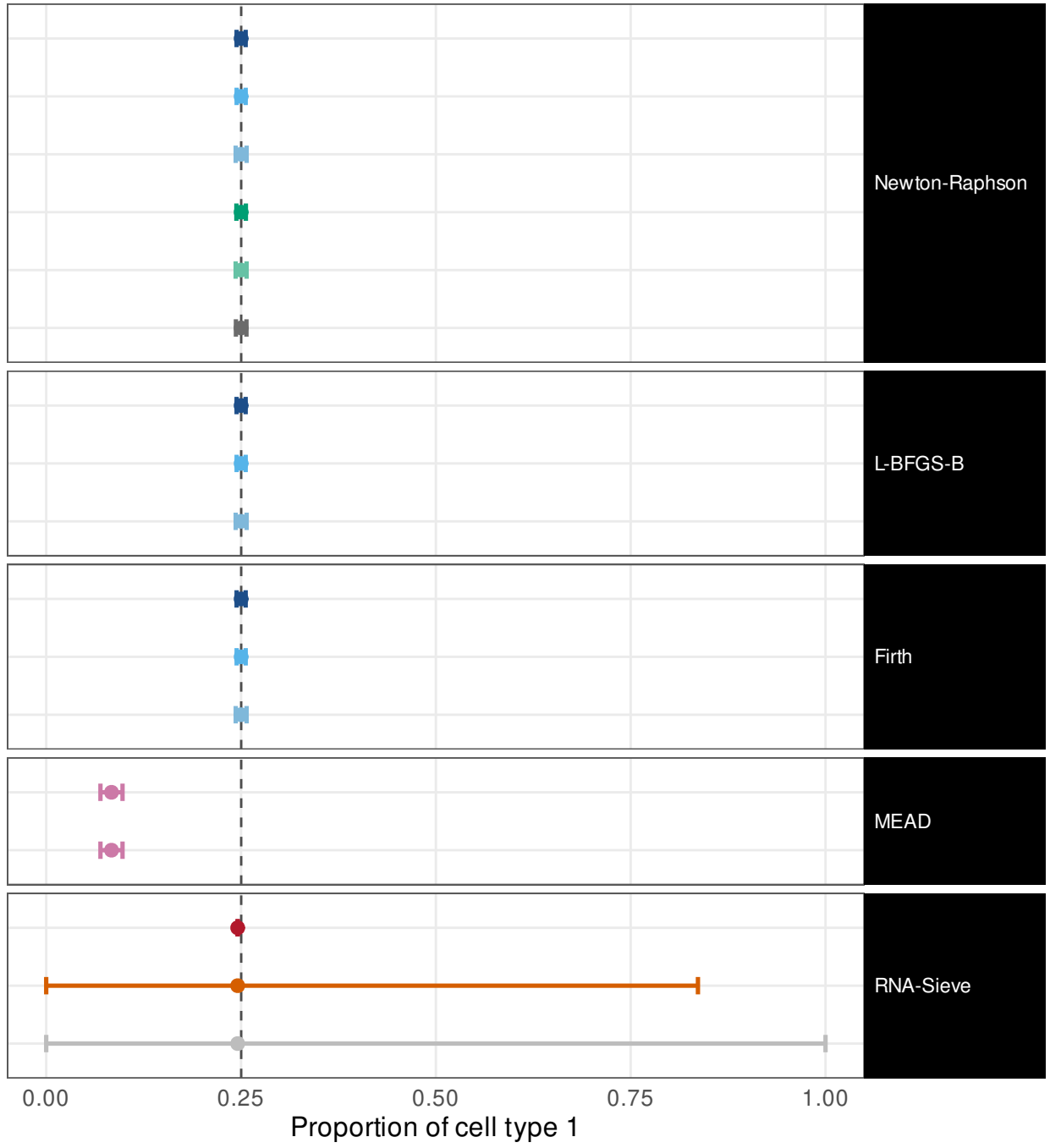
```
Wald (delta)                        0.94 [ 0.9, 0.97]   w = 0.0116
ILR, mapped back                    0.94 [ 0.9, 0.97]   w = 0.0116
ILR, REML                           0.96 [0.93, 0.98]   w = 0.014
Profile likelihood                  0.94 [ 0.9, 0.97]   w = 0.0122
Profile, REML                       0.97 [0.94, 0.99]   w = 0.014
Oracle (Fisher at p*)                 1  [0.98,    1]   w = 0.0135

Wald (delta)                        0.94 [ 0.9, 0.97]   w = 0.0116
ILR, mapped back                    0.94 [0.89, 0.96]   w = 0.0116
ILR, REML                           0.98 [0.95, 0.99]   w = 0.014

Wald (delta)                        0.94 [ 0.9, 0.97]   w = 0.0116
ILR, mapped back                    0.94 [ 0.9, 0.97]   w = 0.0116
ILR, REML                           0.97 [0.94, 0.99]   w = 0.014

Method's own interval                 0  [   0, 0.019]   w = 0.0283
Own interval, transcript fractions    1  [0.98,    1]   w = 0.0283

Fisher information (Wald)           0.025 [0.011, 0.057]   w = 0.000297
Godambe sandwich                      1  [0.98,    1]   w = 0.836
Released code, as published           1  [0.98,    1]   w =    1
```

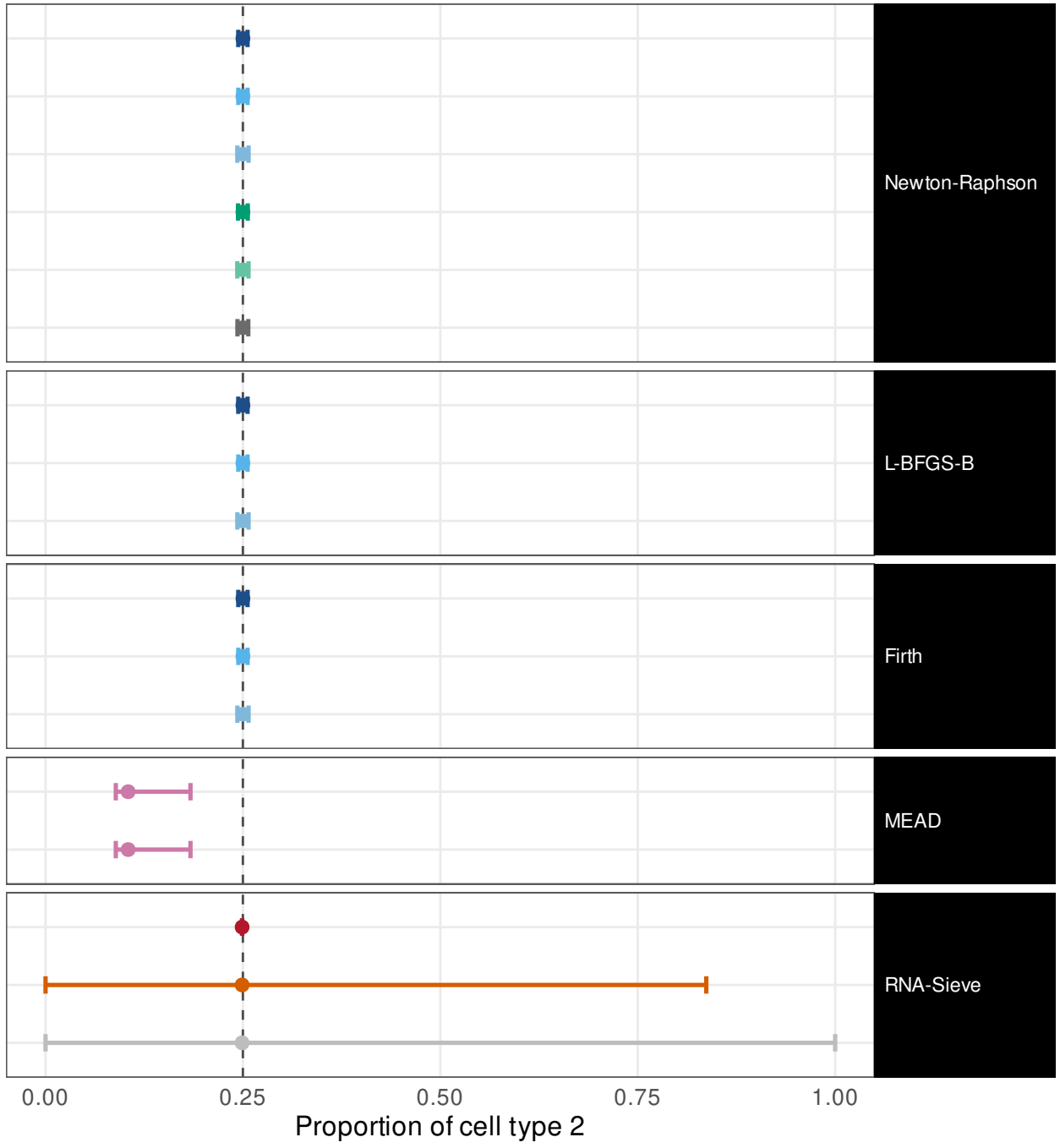


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / balanced / high sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.96 [0.92, 0.98]	w = 0.0117
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.0117
ILR, REML	0.98 [0.95, 0.99]	w = 0.0142
Profile likelihood	0.96 [0.92, 0.98]	w = 0.0123
Profile, REML	0.98 [0.94, 0.99]	w = 0.0142
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0137
Wald (delta)	0.96 [0.92, 0.98]	w = 0.0117
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.0117
ILR, REML	0.98 [0.94, 0.99]	w = 0.0142
Wald (delta)	0.96 [0.92, 0.98]	w = 0.0117
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.0117
ILR, REML	0.97 [0.94, 0.99]	w = 0.0141
Method's own interval	0.08 [0.05, 0.13]	w = 0.0946
Own interval, transcript fractions	0.08 [0.05, 0.13]	w = 0.0946
Fisher information (Wald)	0.01 [0.0027, 0.036]	w = 0.00032
Godambe sandwich	1 [0.98, 1]	w = 0.837
Released code, as published	1 [0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / balanced / high sibling overlap / a = 5 / sizes 1:1:5:5

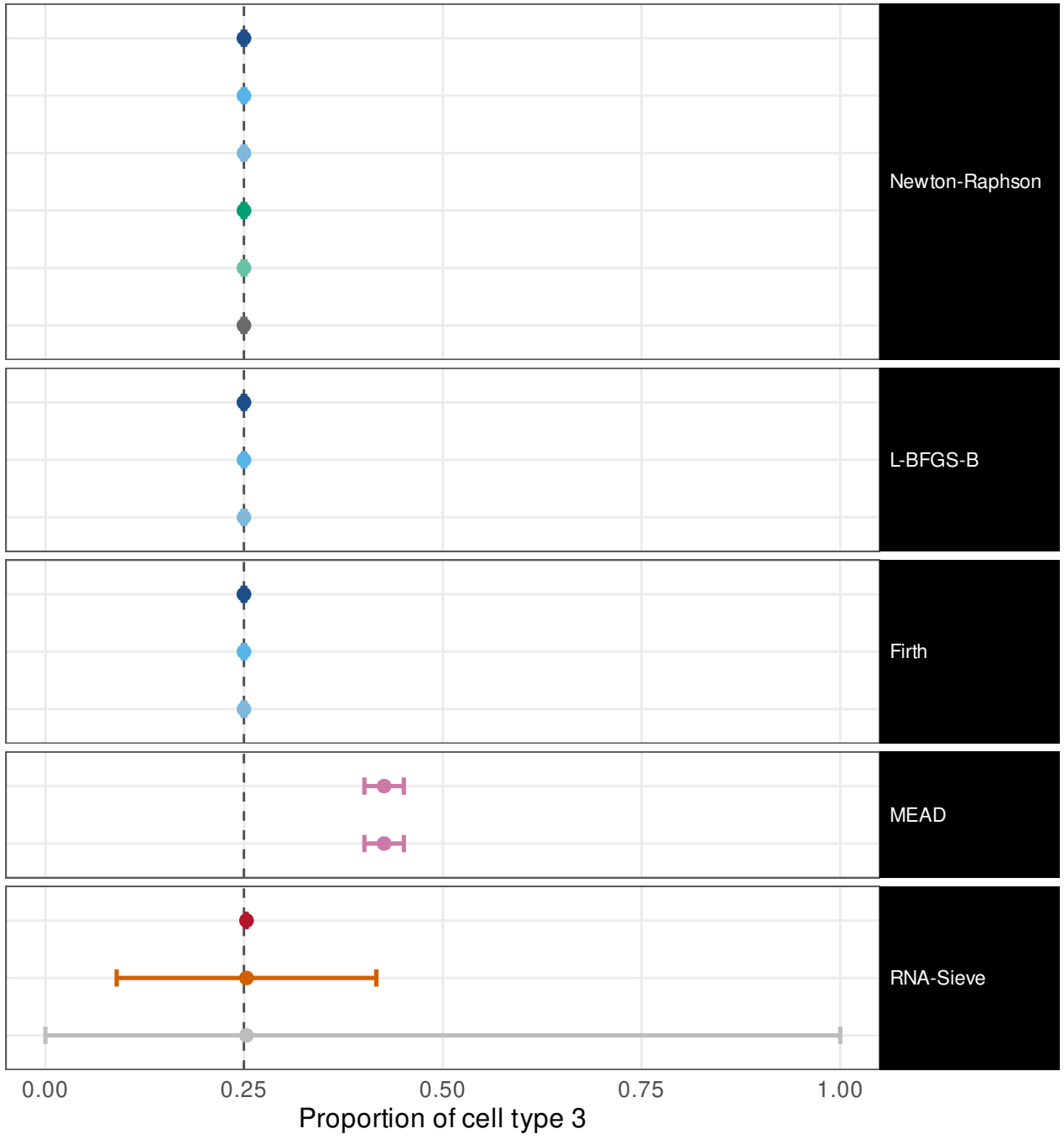
```
Wald (delta)                        0.92 [0.88, 0.95]   w = 0.00312
ILR, mapped back                    0.92 [0.88, 0.95]   w = 0.00311
ILR, REML                           0.96 [0.92, 0.98]   w = 0.00376
Profile likelihood                  0.94 [0.89, 0.96]   w = 0.00327
Profile, REML                       0.95 [0.91, 0.97]   w = 0.00377
Oracle (Fisher at p*)                  1 [0.98,    1]   w = 0.00369

Wald (delta)                        0.92 [0.88, 0.95]   w = 0.00312
ILR, mapped back                    0.93 [0.89, 0.96]   w = 0.00311
ILR, REML                           0.95 [0.91, 0.97]   w = 0.00377

Wald (delta)                        0.92 [0.88, 0.95]   w = 0.00312
ILR, mapped back                    0.92 [0.88, 0.95]   w = 0.00312
ILR, REML                           0.96 [0.92, 0.98]   w = 0.00377

Method's own interval                  0 [   0, 0.019]   w = 0.0496
Own interval, transcript fractions     1 [0.98,    1]   w = 0.0496

Fisher information (Wald)              0 [   0, 0.019]   w = 0.000101
Godambe sandwich                       1 [0.98,    1]   w = 0.327
Released code, as published            1 [0.98,    1]   w =    1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / balanced / high sibling overlap / a = 5 / sizes 1:1:5:5

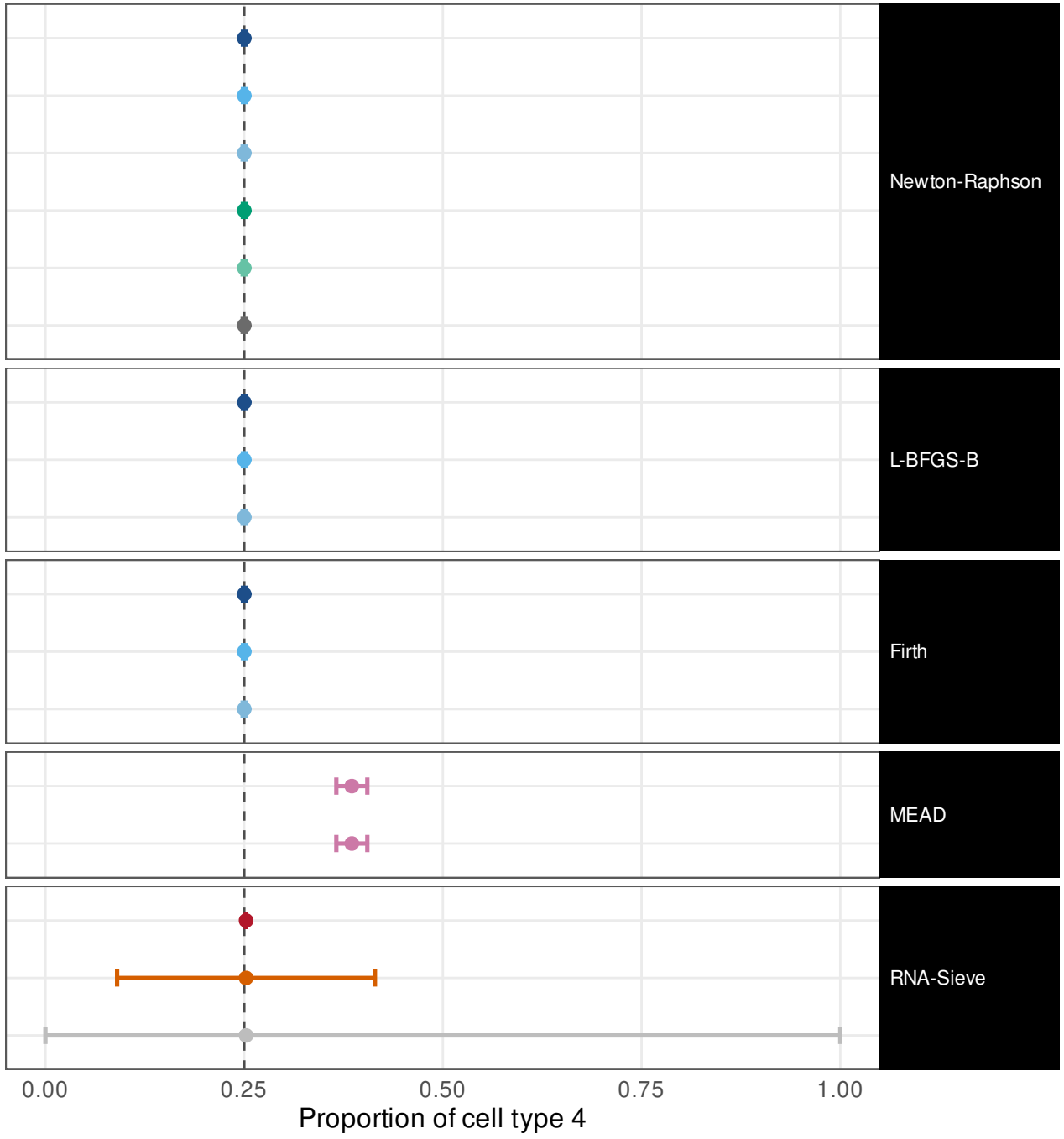
```
Wald (delta)                        0.91 [0.86, 0.94]   w = 0.0034
ILR, mapped back                    0.92 [0.87, 0.95]   w = 0.00339
ILR, REML                           0.96 [0.92, 0.98]   w = 0.00411
Profile likelihood                  0.94 [0.89, 0.96]   w = 0.00357
Profile, REML                       0.96 [0.92, 0.98]   w = 0.00411
Oracle (Fisher at p*)                  1 [0.98,    1]   w = 0.00401

Wald (delta)                        0.91 [0.86, 0.94]   w = 0.0034
ILR, mapped back                     0.9 [0.85, 0.93]   w = 0.00339
ILR, REML                           0.96 [0.92, 0.98]   w = 0.00409

Wald (delta)                        0.91 [0.86, 0.94]   w = 0.0034
ILR, mapped back                    0.92 [0.87, 0.95]   w = 0.00339
ILR, REML                           0.96 [0.92, 0.98]   w = 0.0041

Method's own interval                  0 [   0, 0.019]   w = 0.039
Own interval, transcript fractions     0 [   0, 0.019]   w = 0.039

Fisher information (Wald)              0 [   0, 0.019]   w = 0.000102
Godambe sandwich                       1 [0.98,    1]   w = 0.324
Released code, as published            1 [0.98,    1]   w =    1
```

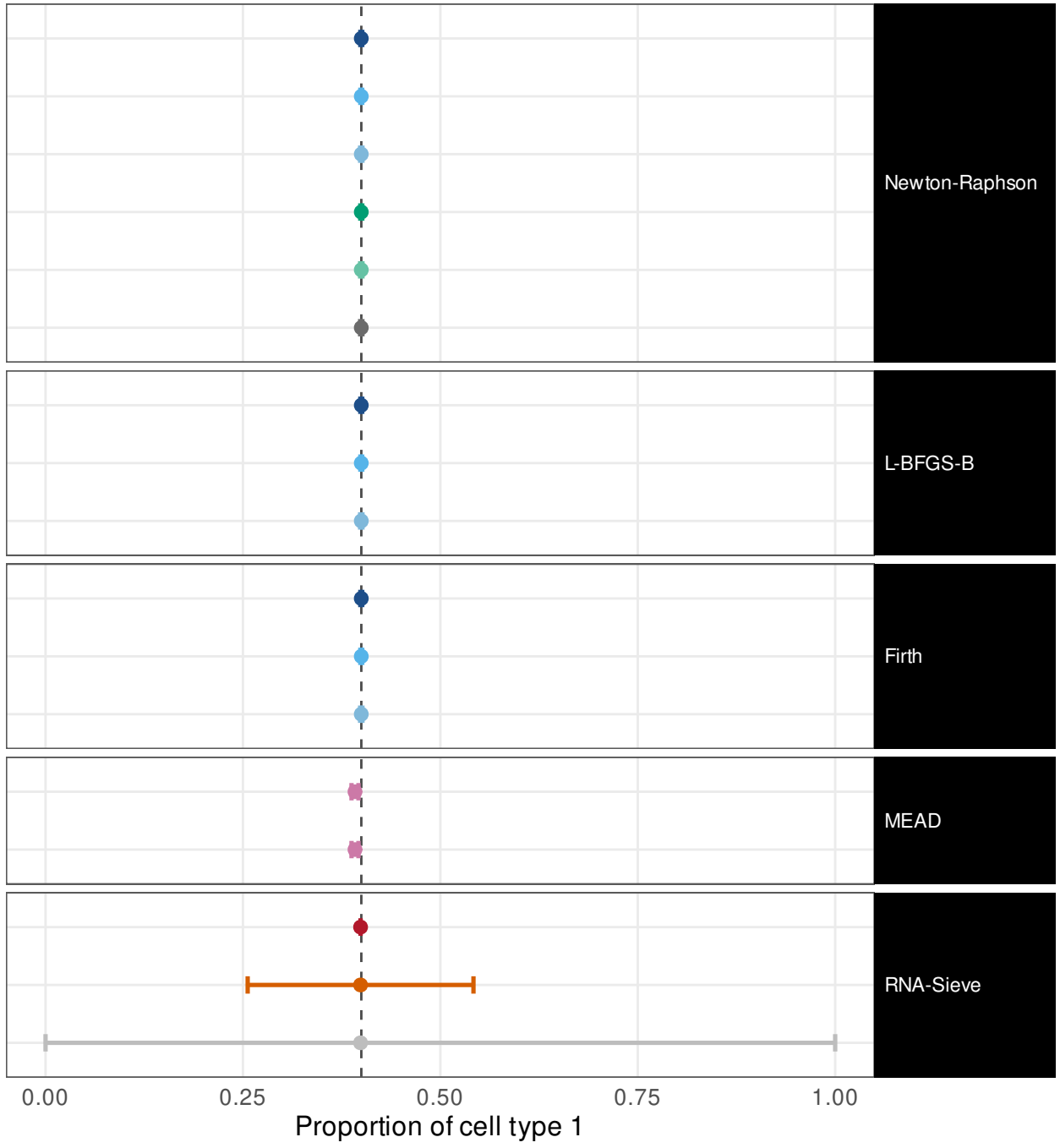


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / unbalanced / low sibling overlap / a = 1 / equal sizes

Wald (delta)	0.88 [0.83, 0.92]	w = 0.00238
ILR, mapped back	0.9 [0.84, 0.93]	w = 0.00237
ILR, REML	0.94 [0.9, 0.97]	w = 0.00286
Profile likelihood	0.92 [0.87, 0.95]	w = 0.0025
Profile, REML	0.94 [0.9, 0.97]	w = 0.00288
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.00284
Wald (delta)	0.88 [0.83, 0.92]	w = 0.00238
ILR, mapped back	0.89 [0.84, 0.93]	w = 0.00237
ILR, REML	0.94 [0.9, 0.97]	w = 0.00287
Wald (delta)	0.88 [0.83, 0.92]	w = 0.00238
ILR, mapped back	0.88 [0.83, 0.92]	w = 0.00238
ILR, REML	0.94 [0.9, 0.97]	w = 0.00288
Method's own interval	0 [0, 0.019]	w = 0.00883
Own interval, transcript fractions	0 [0, 0.019]	w = 0.00883
Fisher information (Wald)	0.035 [0.017, 0.07]	w = 0.000114
Godambe sandwich	1 [0.98, 1]	w = 0.286
Released code, as published	1 [0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / unbalanced / low sibling overlap / a = 1 / equal sizes

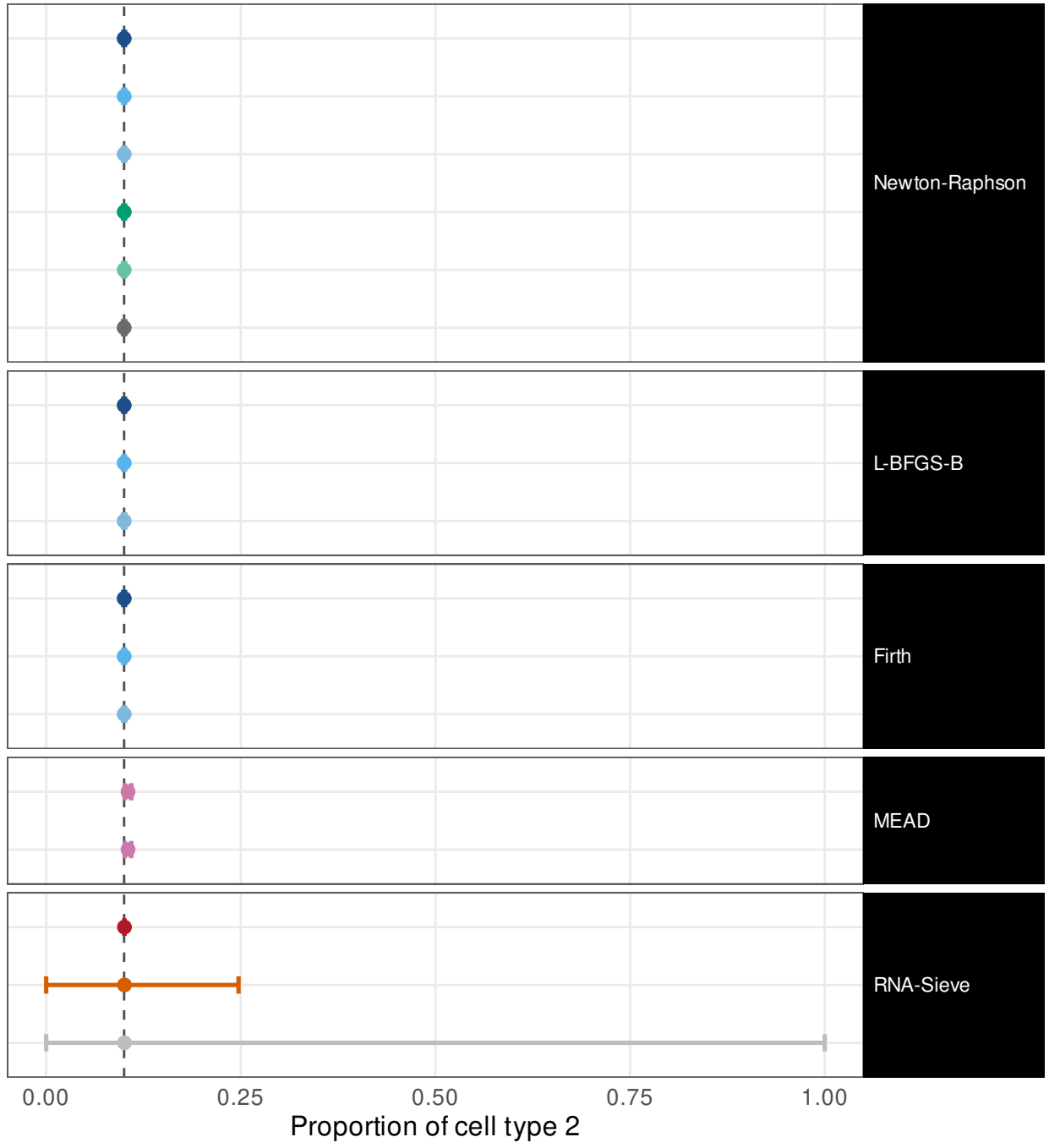
```
Wald (delta)                        0.91 [0.86, 0.94]   w = 0.0024
ILR, mapped back                    0.91 [0.86, 0.94]   w = 0.00239
ILR, REML                           0.96 [0.93, 0.98]   w = 0.00289
Profile likelihood                  0.92 [0.87, 0.95]   w = 0.00252
Profile, REML                       0.96 [0.93, 0.98]   w = 0.0029
Oracle (Fisher at p*)                 1  [0.98,    1]   w = 0.00286

Wald (delta)                        0.91 [0.86, 0.94]   w = 0.0024
ILR, mapped back                    0.91 [0.86, 0.94]   w = 0.00239
ILR, REML                           0.96 [0.93, 0.98]   w = 0.0029

Wald (delta)                        0.91 [0.86, 0.94]   w = 0.0024
ILR, mapped back                    0.92 [0.87, 0.95]   w = 0.00239
ILR, REML                           0.96 [0.92, 0.98]   w = 0.00289

Method's own interval               0.18 [0.14, 0.24]   w = 0.00849
Own interval, transcript fractions  0.18 [0.14, 0.24]   w = 0.00849

Fisher information (Wald)           0.01 [0.0027, 0.036]   w = 5.68e-05
Godambe sandwich                      1  [0.98,    1]   w = 0.247
Released code, as published           1  [0.98,    1]   w =    1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / unbalanced / low sibling overlap / a = 1 / equal sizes

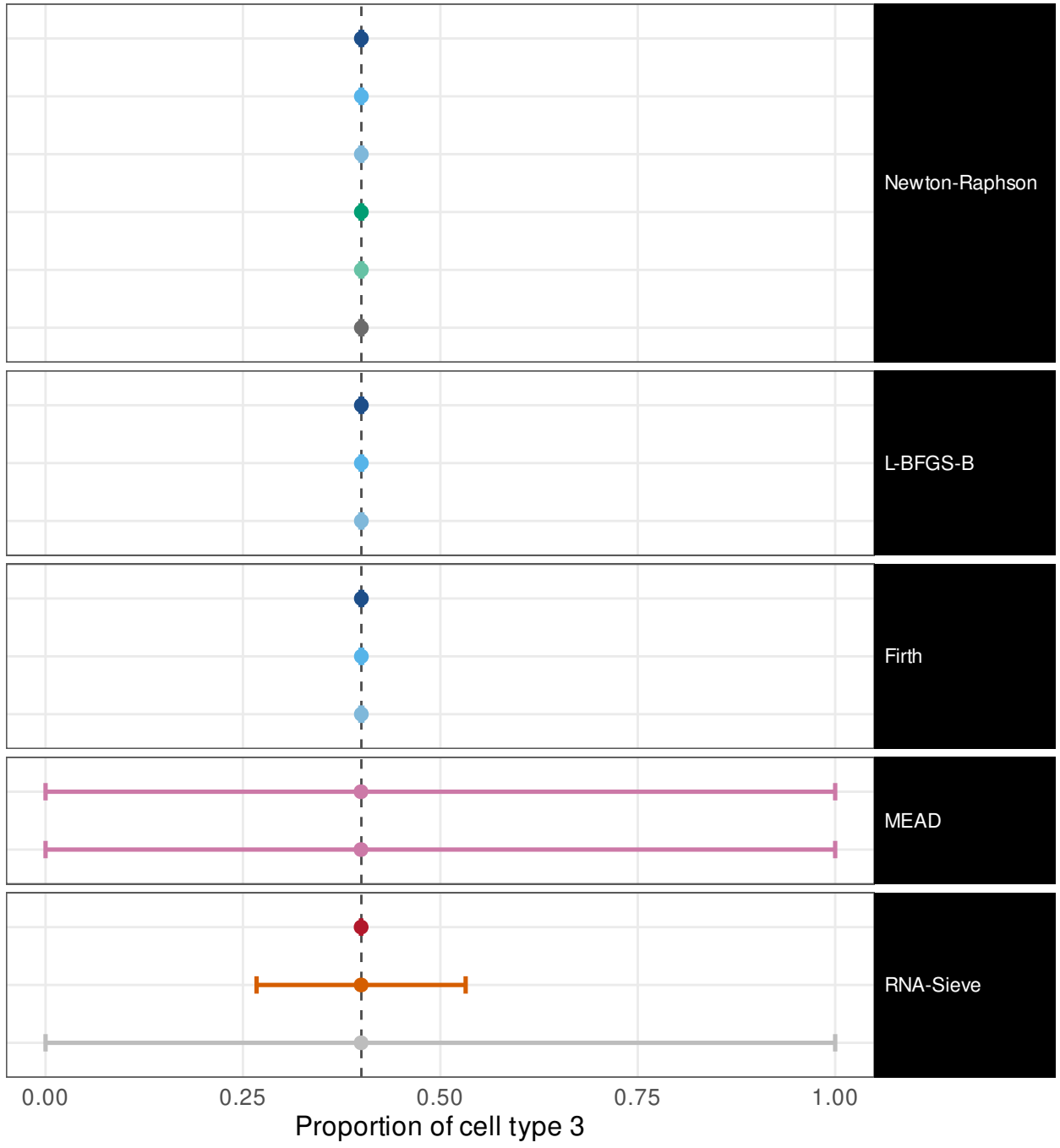
```
Wald (delta)                        0.9  [0.85, 0.93]   w = 0.00207
ILR, mapped back                    0.91 [0.86, 0.94]   w = 0.00207
ILR, REML                           0.94 [0.89, 0.96]   w = 0.00249
Profile likelihood                  0.92 [0.88, 0.95]   w = 0.00217
Profile, REML                       0.94 [0.89, 0.96]   w = 0.0025
Oracle (Fisher at p*)                 1  [0.98,    1]   w = 0.00251

Wald (delta)                        0.9  [0.85, 0.93]   w = 0.00207
ILR, mapped back                    0.9  [0.85, 0.93]   w = 0.00207
ILR, REML                           0.94 [ 0.9, 0.97]   w = 0.0025

Wald (delta)                        0.9  [0.85, 0.93]   w = 0.00207
ILR, mapped back                    0.9  [0.85, 0.93]   w = 0.00206
ILR, REML                           0.94 [0.89, 0.96]   w = 0.00249

Method's own interval                 1  [0.98,    1]   w =    1
Own interval, transcript fractions    1  [0.98,    1]   w =    1

Fisher information (Wald)           0.075 [0.046, 0.12]   w = 0.000145
Godambe sandwich                      1  [0.98,    1]   w = 0.265
Released code, as published           1  [0.98,    1]   w =    1
```

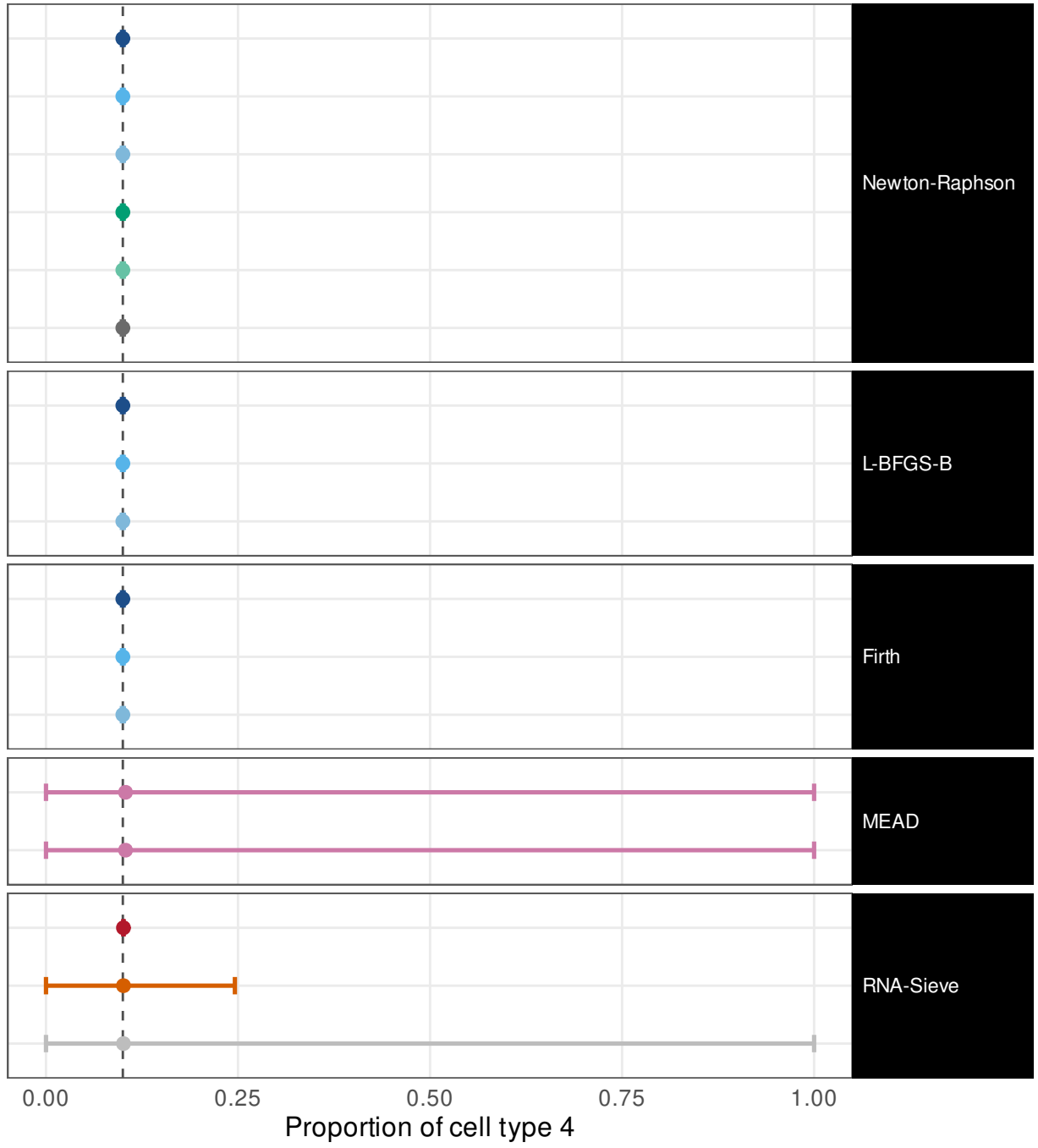


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / unbalanced / low sibling overlap / a = 1 / equal sizes

Wald (delta)	0.92 [0.87, 0.95]	w = 0.0021
ILR, mapped back	0.9 [0.85, 0.93]	w = 0.0021
ILR, REML	0.96 [0.92, 0.98]	w = 0.00253
Profile likelihood	0.93 [0.89, 0.96]	w = 0.0022
Profile, REML	0.96 [0.92, 0.98]	w = 0.00254
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.00253
Wald (delta)	0.92 [0.87, 0.95]	w = 0.0021
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.00209
ILR, REML	0.95 [0.91, 0.97]	w = 0.00253
Wald (delta)	0.92 [0.87, 0.95]	w = 0.0021
ILR, mapped back	0.91 [0.86, 0.94]	w = 0.0021
ILR, REML	0.94 [0.9, 0.97]	w = 0.00253
Method's own interval	1 [0.98, 1]	w = 1
Own interval, transcript fractions	1 [0.98, 1]	w = 1
Fisher information (Wald)	0.015 [0.0051, 0.043]	w = 4.12e-05
Godambe sandwich	1 [0.98, 1]	w = 0.246
Released code, as published	1 [0.98, 1]	w = 1

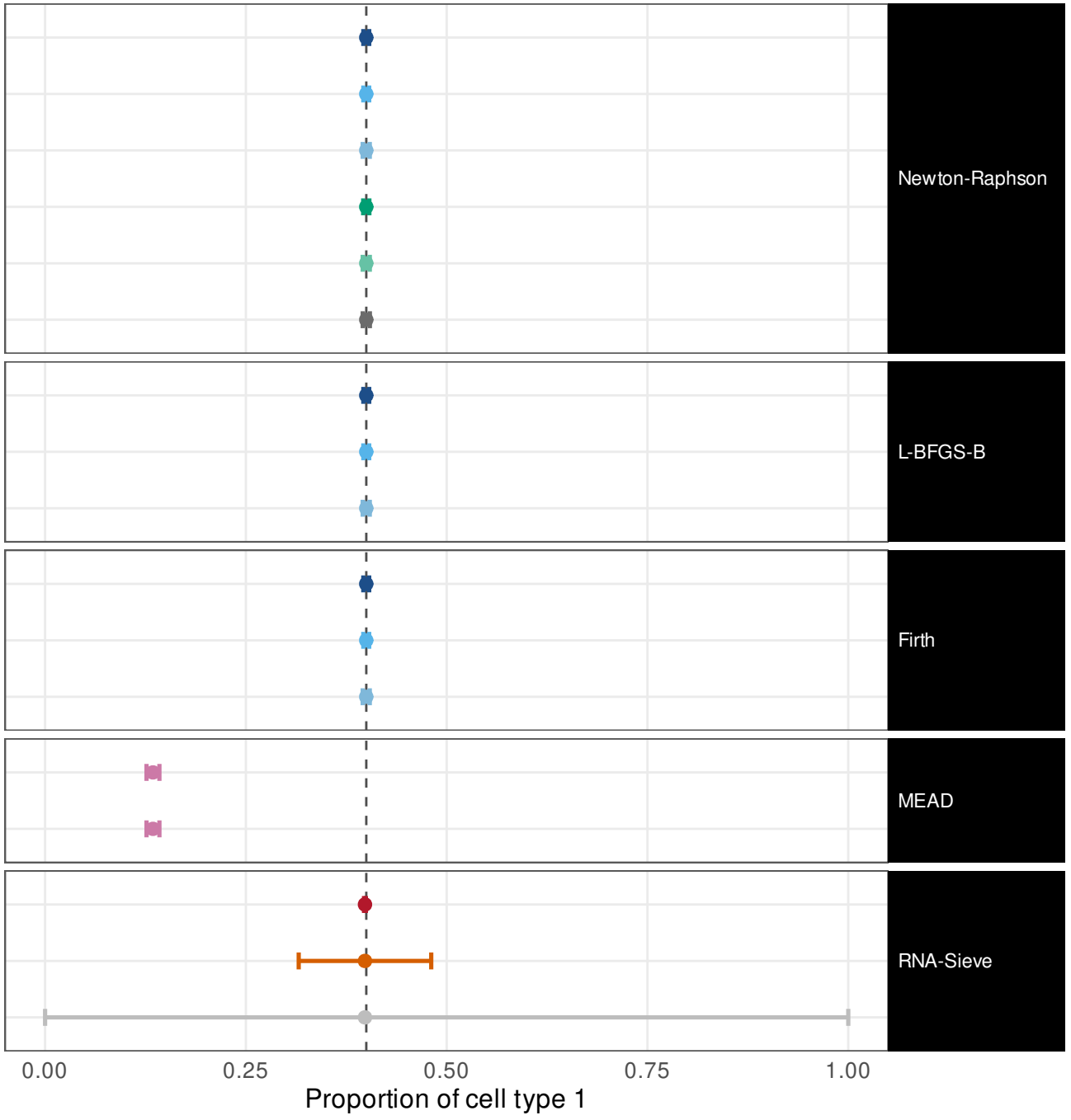


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / unbalanced / low sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.9	[0.84, 0.93]	w = 0.00731
ILR, mapped back	0.9	[0.84, 0.93]	w = 0.0073
ILR, REML	0.96	[0.92, 0.98]	w = 0.0088
Profile likelihood	0.91	[0.86, 0.94]	w = 0.00767
Profile, REML	0.96	[0.92, 0.98]	w = 0.00883
Oracle (Fisher at p*)	1	[0.98, 1]	w = 0.00891
Wald (delta)	0.9	[0.84, 0.93]	w = 0.00731
ILR, mapped back	0.9	[0.84, 0.93]	w = 0.00731
ILR, REML	0.96	[0.92, 0.98]	w = 0.0088
Wald (delta)	0.9	[0.84, 0.93]	w = 0.0073
ILR, mapped back	0.9	[0.85, 0.93]	w = 0.00729
ILR, REML	0.96	[0.92, 0.98]	w = 0.00883
Method's own interval	0	[0, 0.019]	w = 0.016
Own interval, transcript fractions	1	[0.98, 1]	w = 0.016
Fisher information (Wald)	0.28	[0.22, 0.34]	w = 0.00231
Godambe sandwich	1	[0.98, 1]	w = 0.165
Released code, as published	1	[0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / unbalanced / low sibling overlap / a = 1 / sizes 1:1:5:5

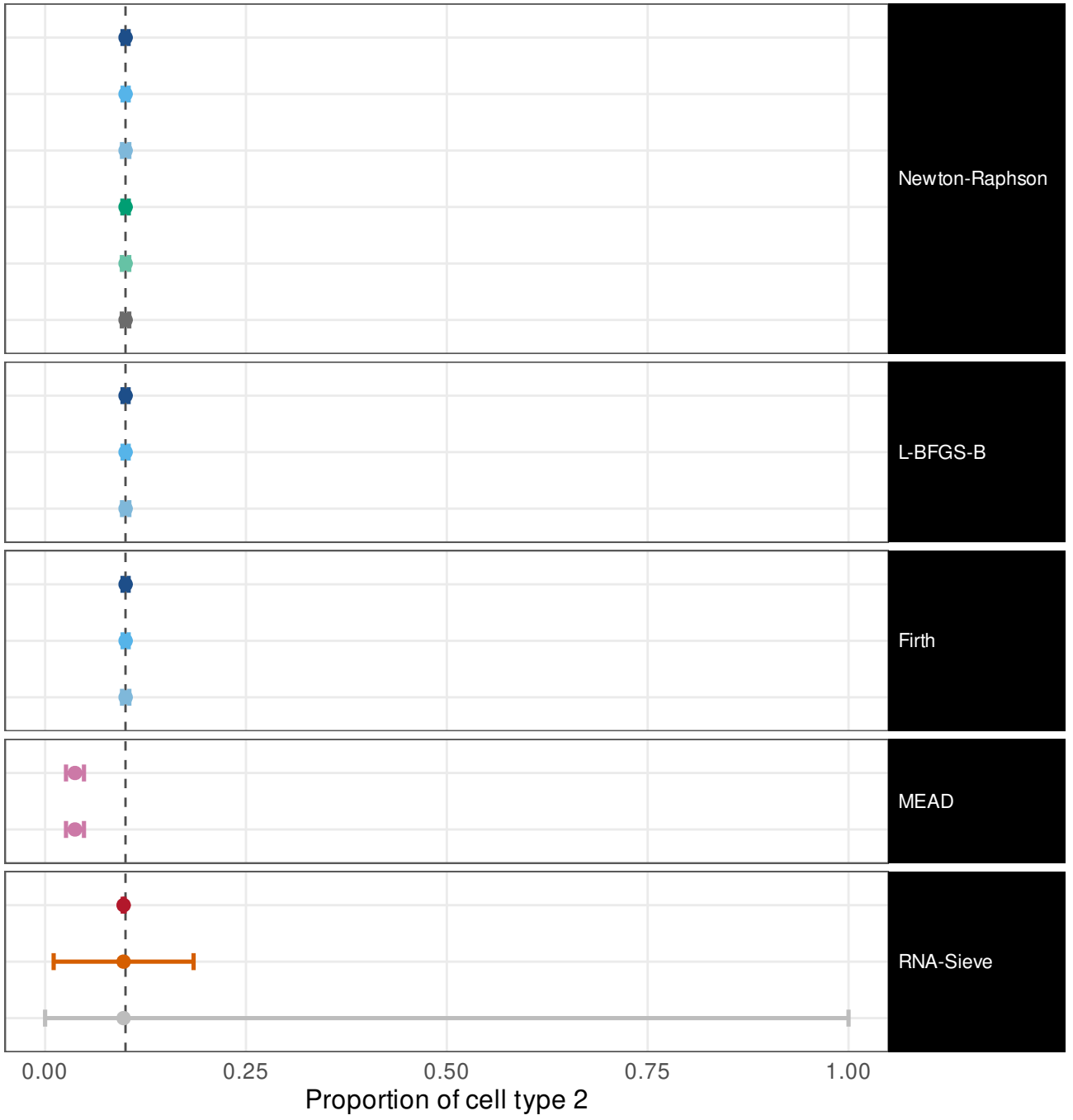
```
Wald (delta)                        0.9  [0.84, 0.93]   w = 0.00754
ILR, mapped back                    0.9  [0.84, 0.93]   w = 0.00753
ILR, REML                          0.94  [ 0.9, 0.97]   w = 0.00909
Profile likelihood                 0.91  [0.86, 0.94]   w = 0.00792
Profile, REML                      0.94  [ 0.9, 0.97]   w = 0.00912
Oracle (Fisher at p*)                 1  [0.98,    1]   w = 0.00922

Wald (delta)                        0.9  [0.84, 0.93]   w = 0.00754
ILR, mapped back                   0.89  [0.84, 0.93]   w = 0.00753
ILR, REML                          0.94  [ 0.9, 0.97]   w = 0.00909

Wald (delta)                        0.9  [0.85, 0.93]   w = 0.00754
ILR, mapped back                   0.89  [0.84, 0.93]   w = 0.00752
ILR, REML                          0.94  [ 0.9, 0.97]   w = 0.0091

Method's own interval                 0  [   0, 0.019]   w = 0.0224
Own interval, transcript fractions    1  [0.98,    1]   w = 0.0224

Fisher information (Wald)          0.22  [0.16, 0.28]   w = 0.0022
Godambe sandwich                      1  [0.98,    1]   w = 0.174
Released code, as published           1  [0.98,    1]   w =     1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / unbalanced / low sibling overlap / a = 1 / sizes 1:1:5:5

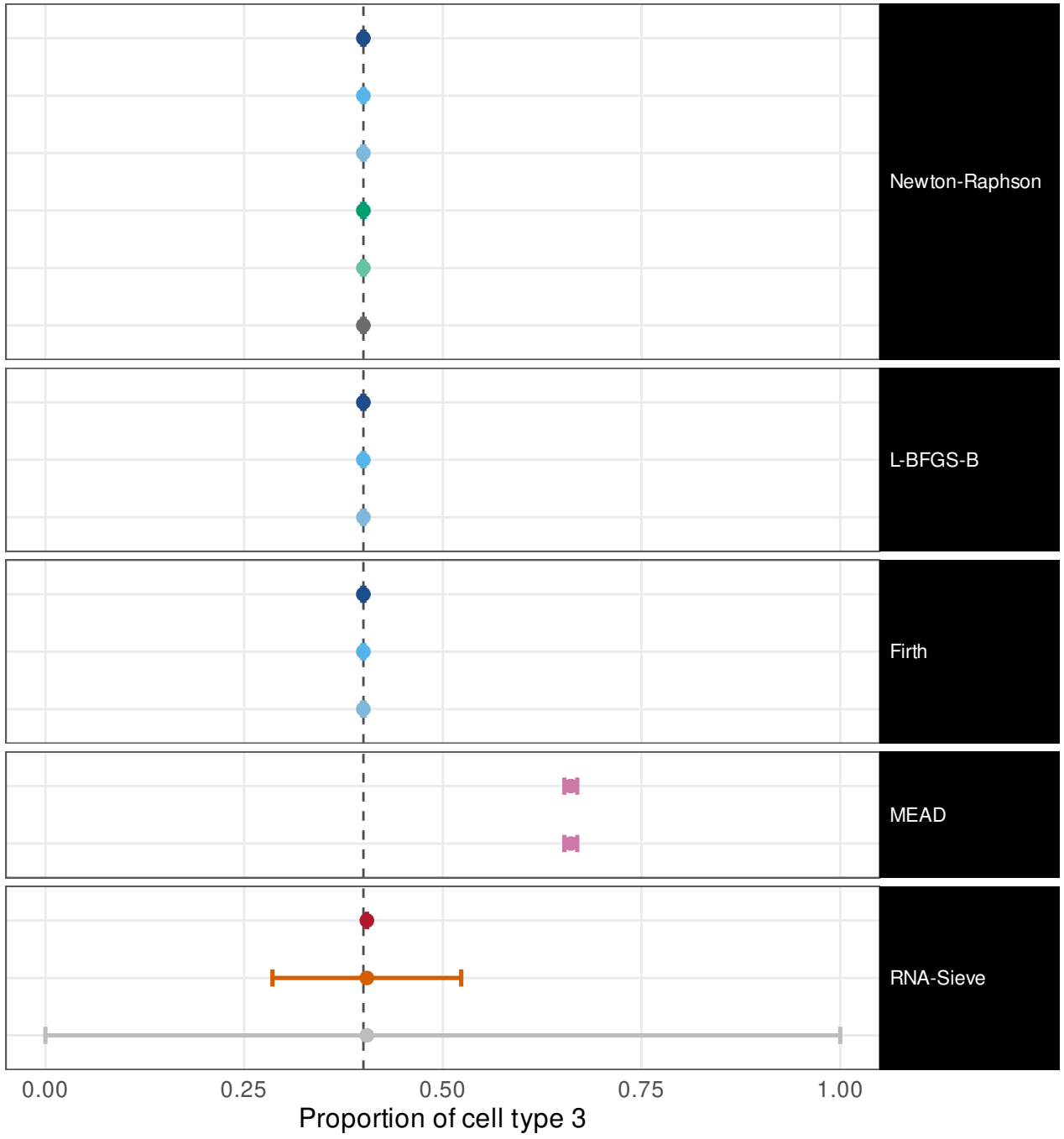
```
Wald (delta)                        0.88 [0.82, 0.91]   w = 0.00252
ILR, mapped back                    0.87 [0.82, 0.91]   w = 0.00252
ILR, REML                           0.96 [0.93, 0.98]   w = 0.00304
Profile likelihood                  0.91 [0.86, 0.94]   w = 0.00264
Profile, REML                       0.97 [0.94, 0.99]   w = 0.00305
Oracle (Fisher at p*)                  1 [0.98,    1]   w = 0.00302

Wald (delta)                        0.88 [0.82, 0.91]   w = 0.00252
ILR, mapped back                    0.88 [0.83, 0.92]   w = 0.00251
ILR, REML                           0.97 [0.94, 0.99]   w = 0.00304

Wald (delta)                        0.87 [0.82, 0.91]   w = 0.00252
ILR, mapped back                    0.88 [0.82, 0.91]   w = 0.00251
ILR, REML                           0.96 [0.93, 0.98]   w = 0.00304

Method's own interval                  0 [   0, 0.019]   w = 0.0161
Own interval, transcript fractions  0.96 [0.92, 0.98]   w = 0.0161

Fisher information (Wald)              0 [   0, 0.019]   w = 0.000192
Godambe sandwich                       1 [0.98,    1]   w = 0.237
Released code, as published            1 [0.98,    1]   w =     1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / unbalanced / low sibling overlap / a = 1 / sizes 1:1:5:5

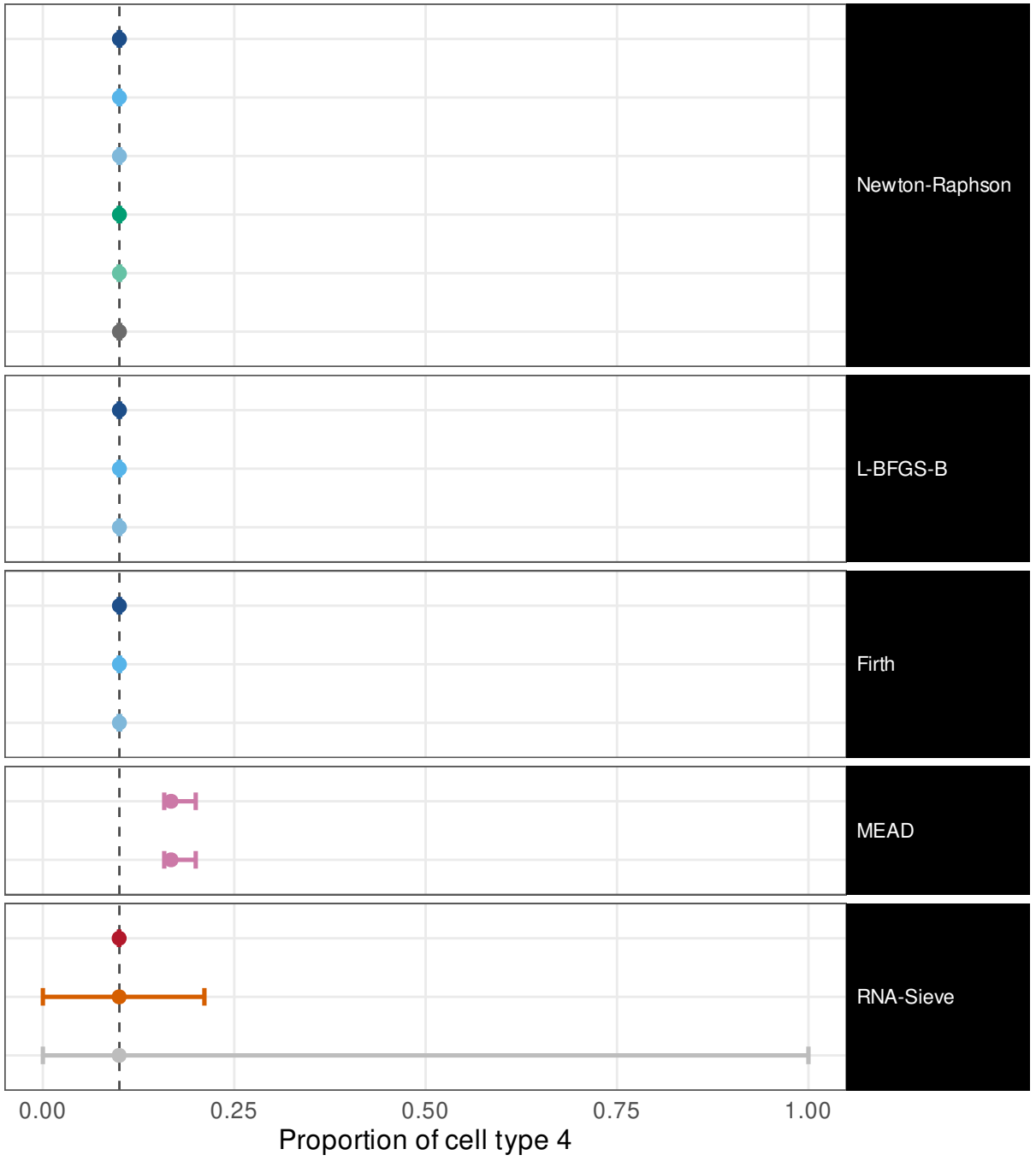
```
Wald (delta)                        0.88 [0.83, 0.92]   w = 0.00171
ILR, mapped back                    0.88 [0.83, 0.92]   w = 0.00171
ILR, REML                           0.94 [ 0.9, 0.97]   w = 0.00207
Profile likelihood                  0.9  [0.85, 0.93]   w = 0.00179
Profile, REML                       0.95 [0.91, 0.97]   w = 0.00207
Oracle (Fisher at p*)                 1  [0.98,    1]   w = 0.00206

Wald (delta)                        0.88 [0.83, 0.92]   w = 0.00171
ILR, mapped back                    0.88 [0.83, 0.92]   w = 0.0017
ILR, REML                           0.94 [ 0.9, 0.97]   w = 0.00206

Wald (delta)                        0.88 [0.83, 0.92]   w = 0.00171
ILR, mapped back                    0.88 [0.83, 0.92]   w = 0.0017
ILR, REML                           0.95 [0.91, 0.97]   w = 0.00207

Method's own interval               0.035 [0.017, 0.07]   w = 0.0411
Own interval, transcript fractions  0.94 [ 0.9, 0.97]   w = 0.0411

Fisher information (Wald)           0.005 [0.00088, 0.028]   w = 4.35e-05
Godambe sandwich                      1  [0.98,    1]   w = 0.211
Released code, as published           1  [0.98,    1]   w =    1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / unbalanced / low sibling overlap / a = 5 / equal sizes

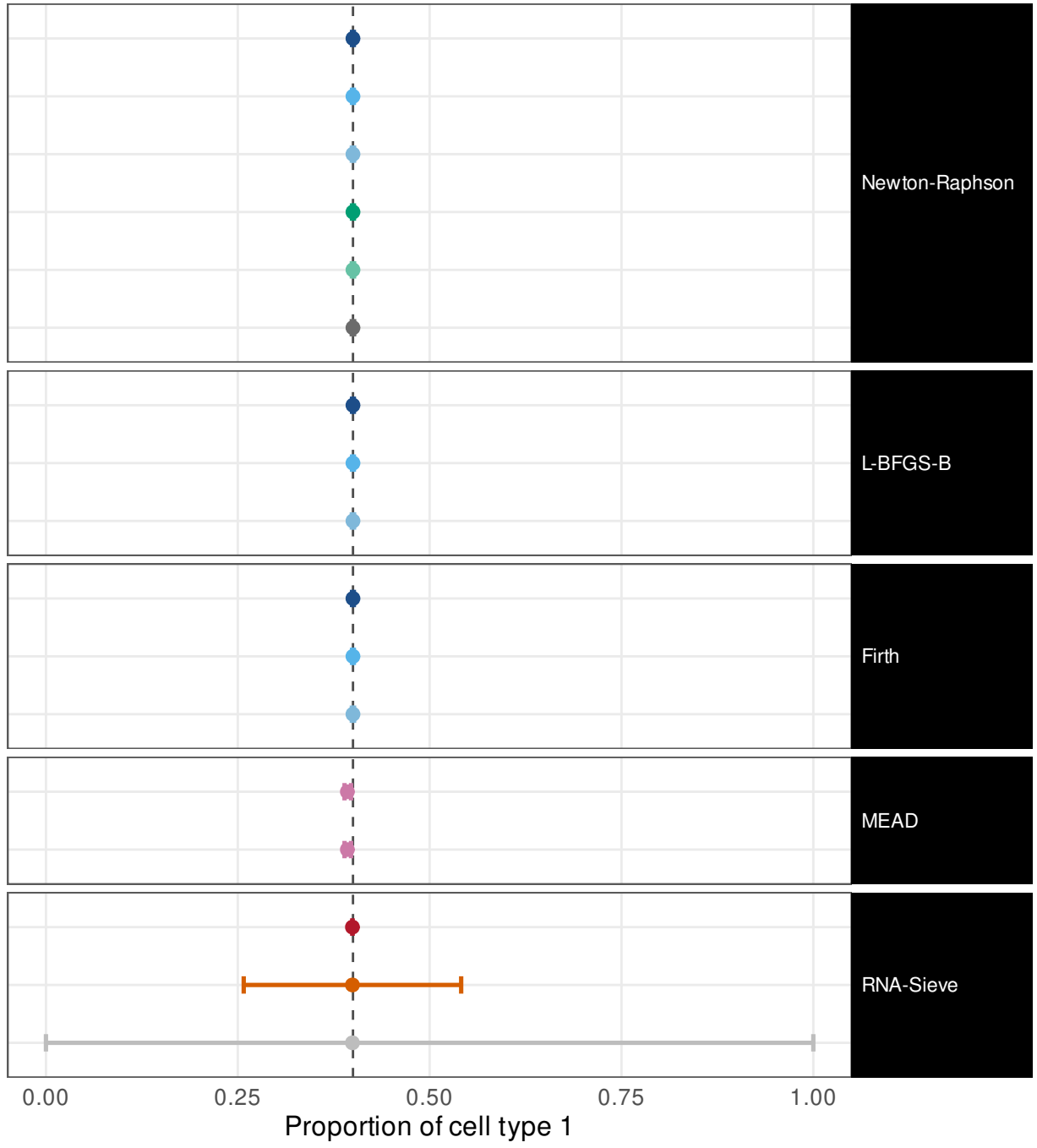
```
Wald (delta)                          0.86 [0.81,  0.9]   w = 0.00238
ILR, mapped back                      0.86 [0.81, 0.91]   w = 0.00238
ILR, REML                             0.94 [ 0.9, 0.97]   w = 0.00286
Profile likelihood                   0.89 [0.84, 0.93]   w = 0.0025
Profile, REML                        0.94 [0.89, 0.96]   w = 0.00287
Oracle (Fisher at p*)                   1  [0.98,    1]   w = 0.00302

Wald (delta)                          0.86 [0.81,  0.9]   w = 0.00238
ILR, mapped back                      0.87 [0.82, 0.91]   w = 0.00237
ILR, REML                             0.94 [0.89, 0.96]   w = 0.00286

Wald (delta)                          0.86 [0.81,  0.9]   w = 0.00238
ILR, mapped back                      0.86 [0.81,  0.9]   w = 0.00238
ILR, REML                             0.93 [0.89, 0.96]   w = 0.00287

Method's own interval                0.005 [0.00088, 0.028]   w = 0.00847
Own interval, transcript fractions   0.005 [0.00088, 0.028]   w = 0.00847

Fisher information (Wald)             0.01 [0.0027, 0.036]   w = 2.54e-05
Godambe sandwich                        1  [0.98,    1]   w = 0.283
Released code, as published             1  [0.98,    1]   w =    1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / unbalanced / low sibling overlap / a = 5 / equal sizes

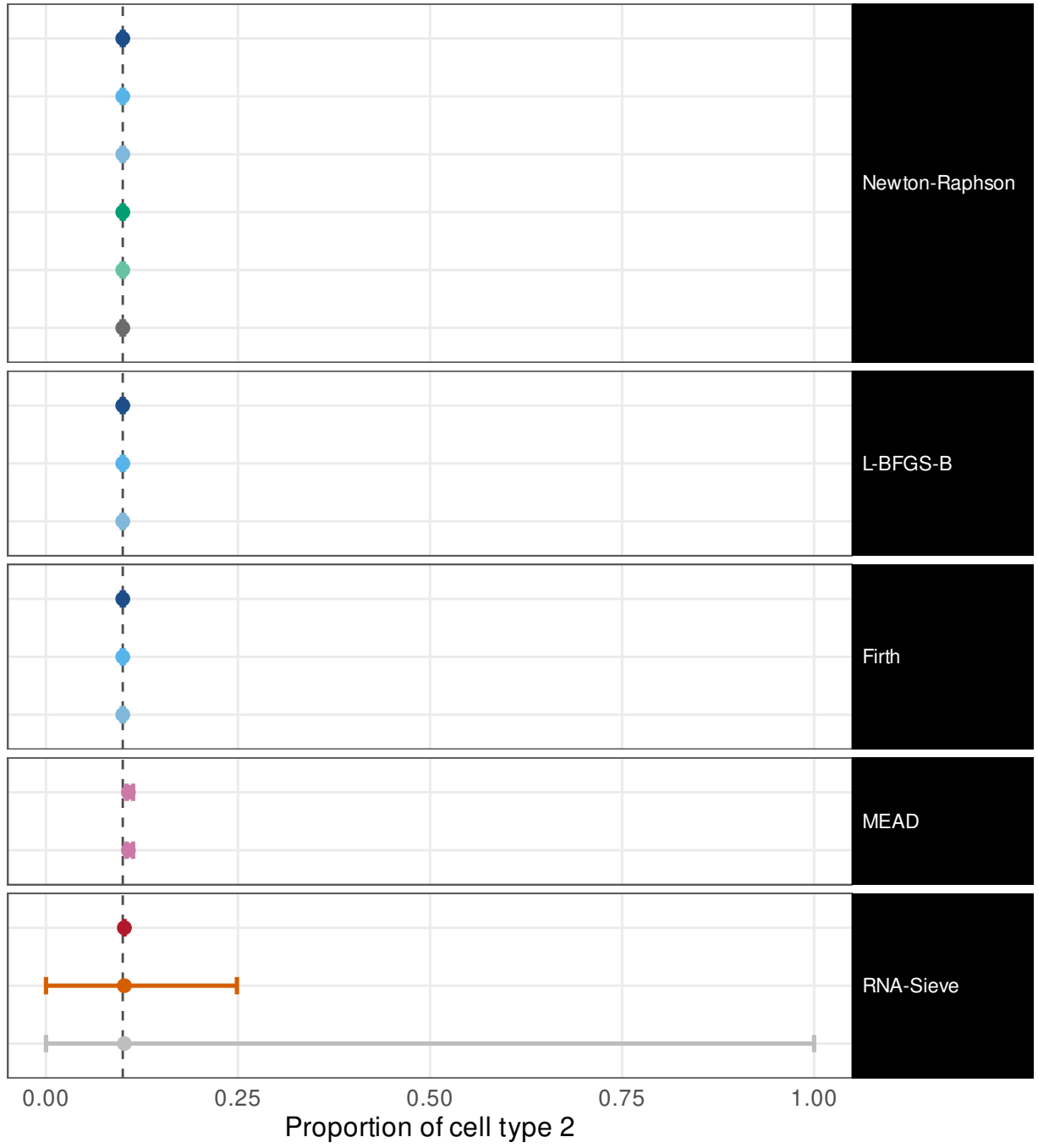
```
Wald (delta)                        0.86 [0.81,  0.9]   w = 0.0024
ILR, mapped back                    0.86 [ 0.8,  0.9]   w = 0.00239
ILR, REML                           0.91 [0.86, 0.94]   w = 0.00289
Profile likelihood                  0.88 [0.83, 0.92]   w = 0.00252
Profile, REML                       0.91 [0.86, 0.94]   w = 0.0029
Oracle (Fisher at p*)                 1  [0.98,    1]   w = 0.00307

Wald (delta)                        0.86 [0.81,  0.9]   w = 0.0024
ILR, mapped back                    0.86 [ 0.8,  0.9]   w = 0.00238
ILR, REML                           0.91 [0.86, 0.94]   w = 0.00288

Wald (delta)                        0.86 [0.81,  0.9]   w = 0.0024
ILR, mapped back                    0.85 [0.79, 0.89]   w = 0.00239
ILR, REML                           0.92 [0.87, 0.95]   w = 0.00289

Method's own interval               0.005 [0.00088, 0.028]   w = 0.00837
Own interval, transcript fractions  0.005 [0.00088, 0.028]   w = 0.00837

Fisher information (Wald)             0  [   0, 0.019]   w = 1.12e-05
Godambe sandwich                      1  [0.98,    1]   w = 0.249
Released code, as published           1  [0.98,    1]   w =    1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / unbalanced / low sibling overlap / a = 5 / equal sizes

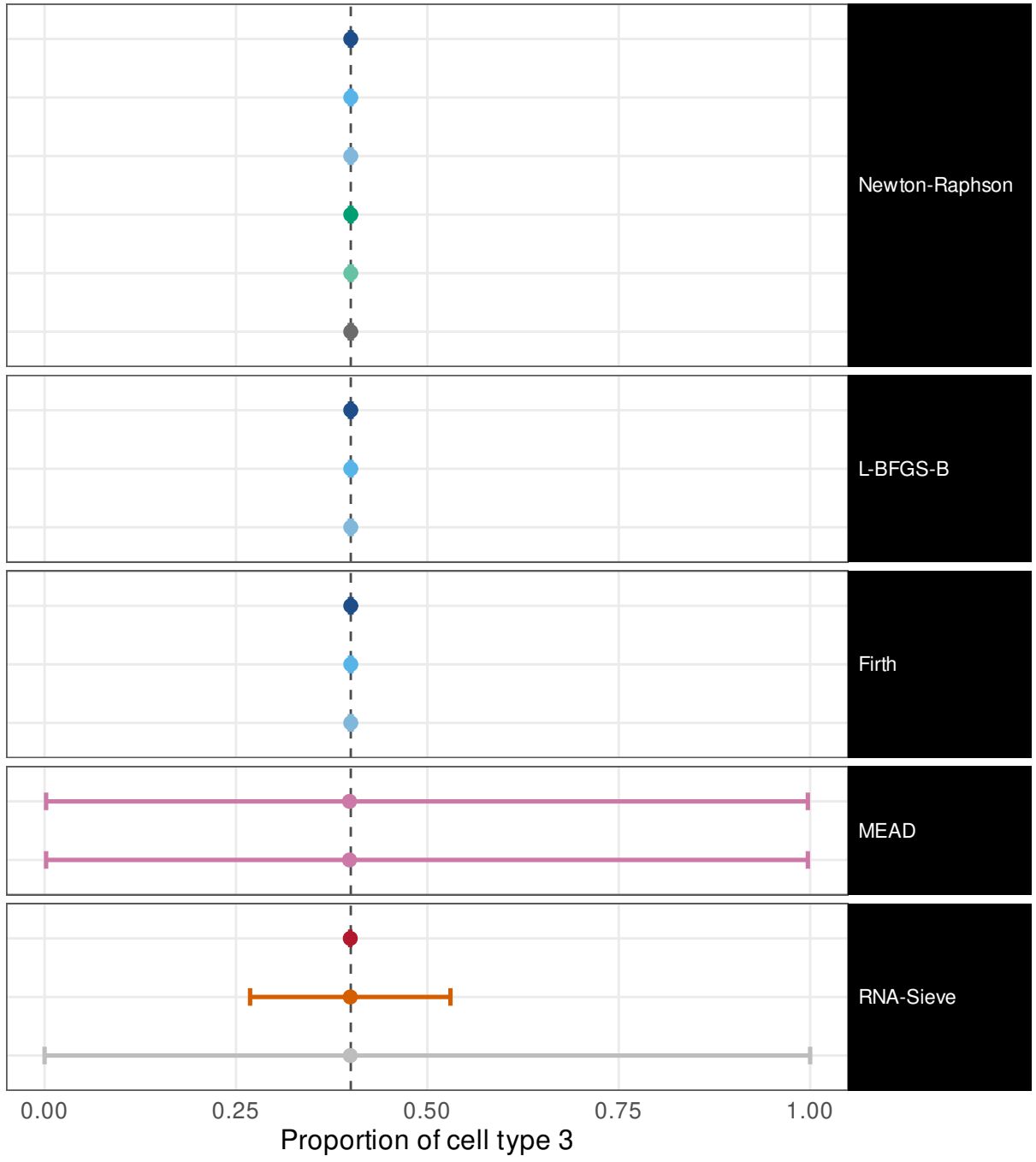
```
Wald (delta)                        0.9  [0.85, 0.93]   w = 0.00207
ILR, mapped back                    0.9  [0.85, 0.93]   w = 0.00207
ILR, REML                           0.94 [ 0.9, 0.97]   w = 0.0025
Profile likelihood                  0.9  [0.86, 0.94]   w = 0.00217
Profile, REML                       0.94 [0.89, 0.96]   w = 0.0025
Oracle (Fisher at p*)                 1  [0.98,    1]   w = 0.00258

Wald (delta)                        0.9  [0.85, 0.93]   w = 0.00207
ILR, mapped back                    0.9  [0.84, 0.93]   w = 0.00205
ILR, REML                           0.94 [ 0.9, 0.97]   w = 0.0025

Wald (delta)                        0.9  [0.85, 0.93]   w = 0.00207
ILR, mapped back                    0.9  [0.85, 0.93]   w = 0.00207
ILR, REML                           0.92 [0.88, 0.95]   w = 0.00249

Method's own interval                 1  [0.97,    1]   w = 0.995
Own interval, transcript fractions    1  [0.97,    1]   w = 0.995

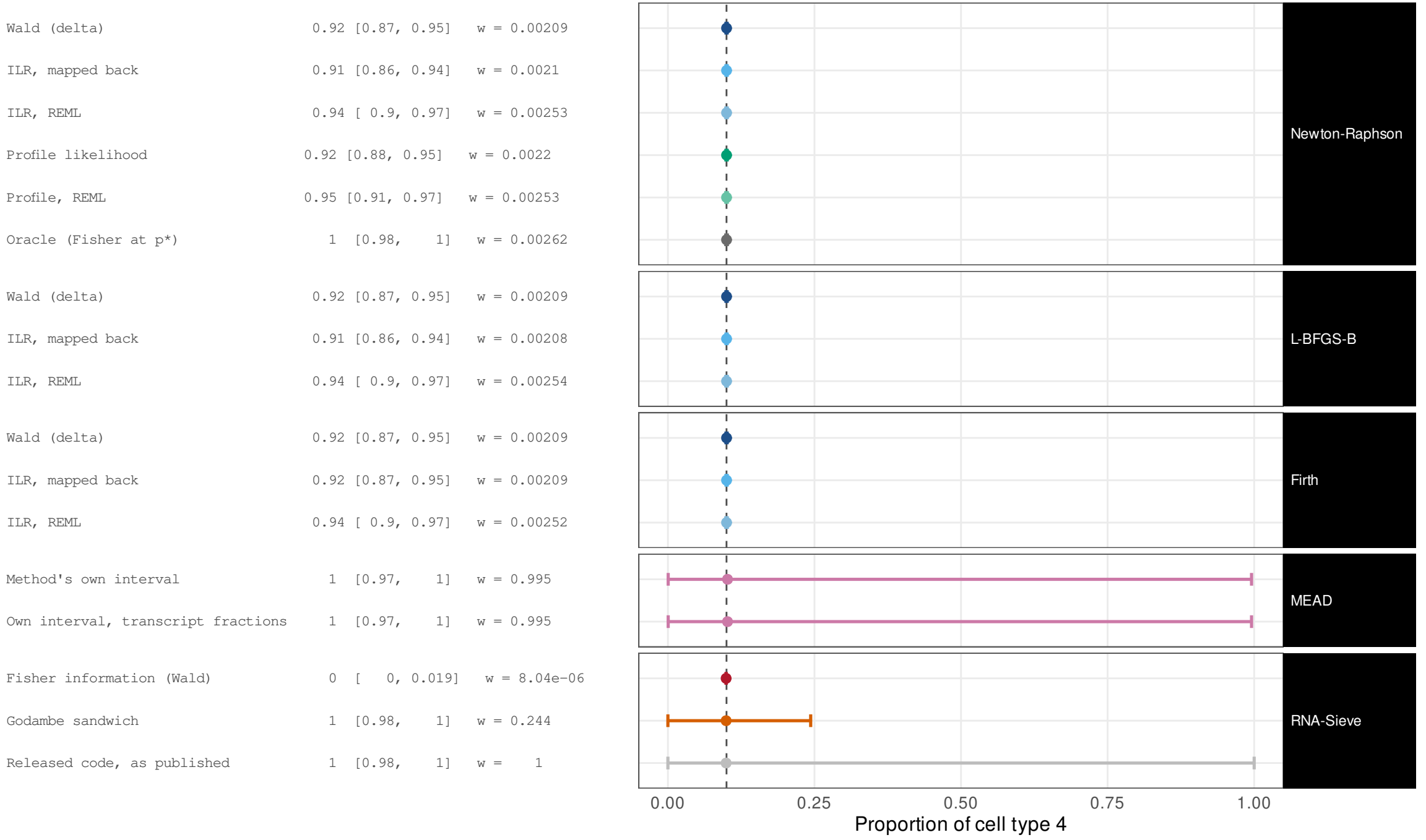
Fisher information (Wald)           0.005 [0.00088, 0.028]   w = 3.12e-05
Godambe sandwich                      1  [0.98,    1]   w = 0.262
Released code, as published           1  [0.98,    1]   w =    1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / unbalanced / low sibling overlap / a = 5 / equal sizes

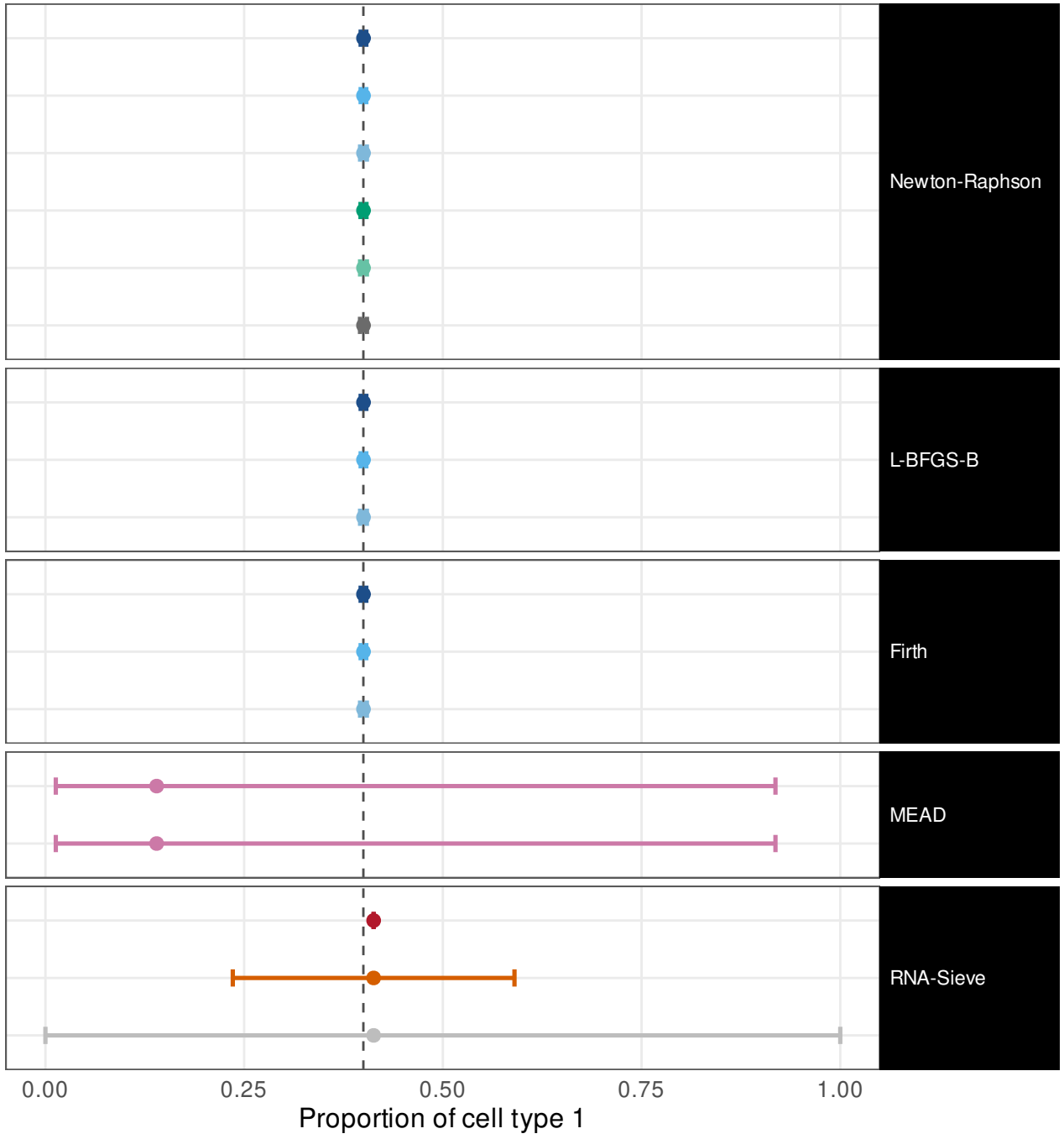


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / unbalanced / low sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.89 [0.84, 0.93]	w = 0.00708
ILR, mapped back	0.88 [0.83, 0.92]	w = 0.00706
ILR, REML	0.94 [0.9, 0.97]	w = 0.00854
Profile likelihood	0.91 [0.86, 0.94]	w = 0.00744
Profile, REML	0.94 [0.9, 0.97]	w = 0.00856
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.00871
Wald (delta)	0.89 [0.84, 0.93]	w = 0.00708
ILR, mapped back	0.89 [0.84, 0.93]	w = 0.00707
ILR, REML	0.94 [0.9, 0.97]	w = 0.00854
Wald (delta)	0.9 [0.84, 0.93]	w = 0.00708
ILR, mapped back	0.88 [0.83, 0.92]	w = 0.00707
ILR, REML	0.96 [0.92, 0.98]	w = 0.00854
Method's own interval	0.9 [0.86, 0.94]	w = 0.905
Own interval, transcript fractions	0.9 [0.86, 0.94]	w = 0.905
Fisher information (Wald)	0 [0, 0.019]	w = 0.000181
Godambe sandwich	1 [0.98, 1]	w = 0.354
Released code, as published	1 [0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / unbalanced / low sibling overlap / a = 5 / sizes 1:1:5:5

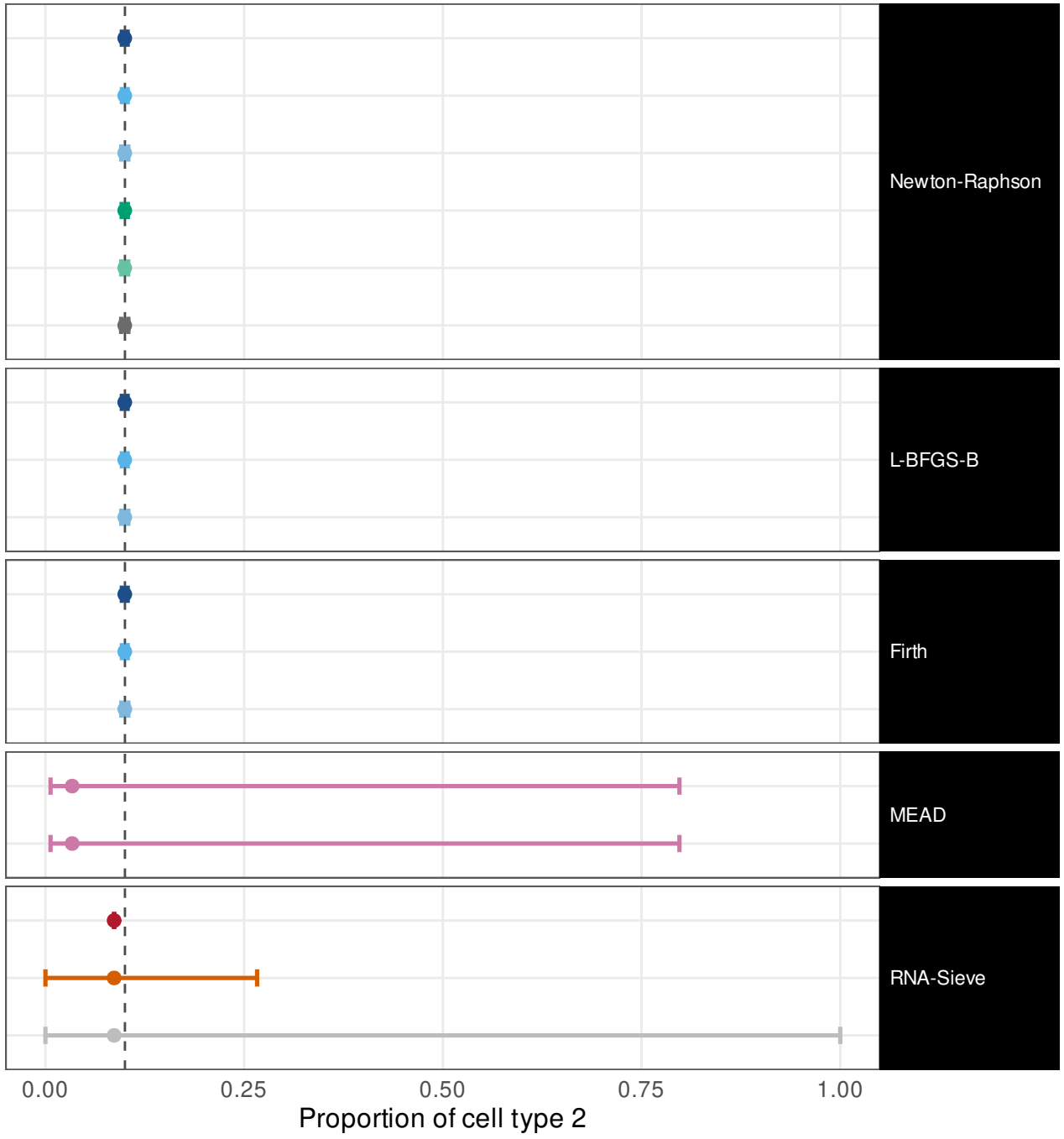
```
Wald (delta)                          0.88 [0.83, 0.92]   w = 0.00731
ILR, mapped back                      0.88 [0.83, 0.92]   w = 0.00731
ILR, REML                             0.94 [ 0.9, 0.97]   w = 0.00883
Profile likelihood                    0.9  [0.84, 0.93]   w = 0.00767
Profile, REML                         0.94 [ 0.9, 0.97]   w = 0.00884
Oracle (Fisher at p*)                   1  [0.98,    1]   w = 0.00905

Wald (delta)                          0.88 [0.83, 0.92]   w = 0.00731
ILR, mapped back                      0.89 [0.84, 0.93]   w = 0.00731
ILR, REML                             0.94 [ 0.9, 0.97]   w = 0.0088

Wald (delta)                          0.88 [0.83, 0.92]   w = 0.00731
ILR, mapped back                      0.88 [0.82, 0.91]   w = 0.0073
ILR, REML                             0.94 [ 0.9, 0.97]   w = 0.00881

Method's own interval                 0.79 [0.73, 0.84]   w = 0.791
Own interval, transcript fractions    0.98 [0.95, 0.99]   w = 0.791

Fisher information (Wald)               0 [   0, 0.019]   w = 0.000171
Godambe sandwich                        1 [0.98,    1]   w = 0.266
Released code, as published             1 [0.98,    1]   w =     1
```

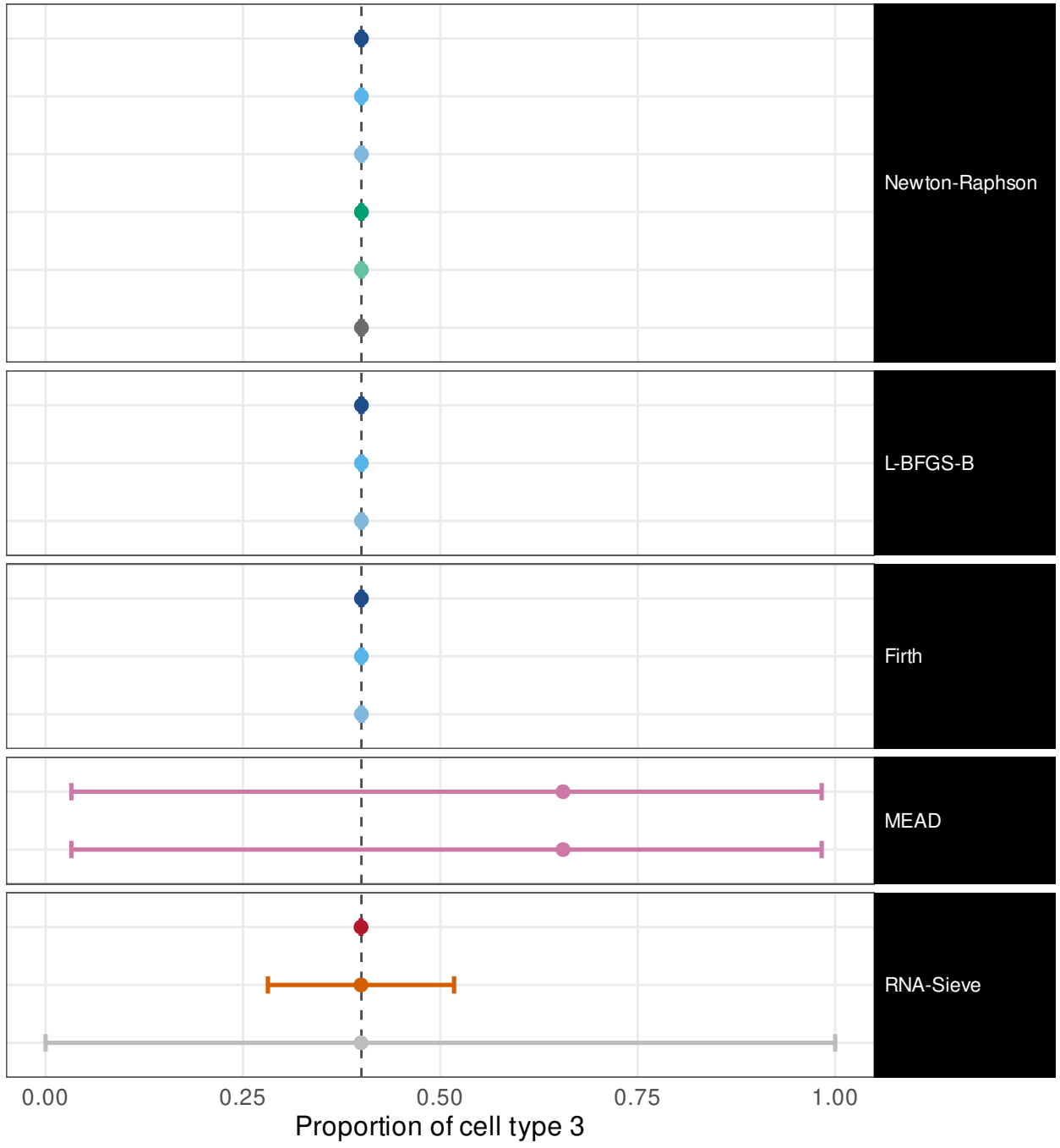


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / unbalanced / low sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.88 [0.83, 0.92]	w = 0.00244
ILR, mapped back	0.88 [0.83, 0.92]	w = 0.00243
ILR, REML	0.94 [0.89, 0.96]	w = 0.00294
Profile likelihood	0.91 [0.86, 0.94]	w = 0.00256
Profile, REML	0.94 [0.89, 0.96]	w = 0.00295
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.00304
Wald (delta)	0.88 [0.83, 0.92]	w = 0.00244
ILR, mapped back	0.9 [0.85, 0.93]	w = 0.00242
ILR, REML	0.94 [0.89, 0.96]	w = 0.00294
Wald (delta)	0.9 [0.84, 0.93]	w = 0.00244
ILR, mapped back	0.88 [0.83, 0.92]	w = 0.00244
ILR, REML	0.94 [0.9, 0.97]	w = 0.00294
Method's own interval	0.95 [0.91, 0.97]	w = 0.95
Own interval, transcript fractions	0.95 [0.91, 0.97]	w = 0.95
Fisher information (Wald)	0.02 [0.0078, 0.05]	w = 3.45e-05
Godambe sandwich	1 [0.98, 1]	w = 0.236
Released code, as published	1 [0.98, 1]	w = 1

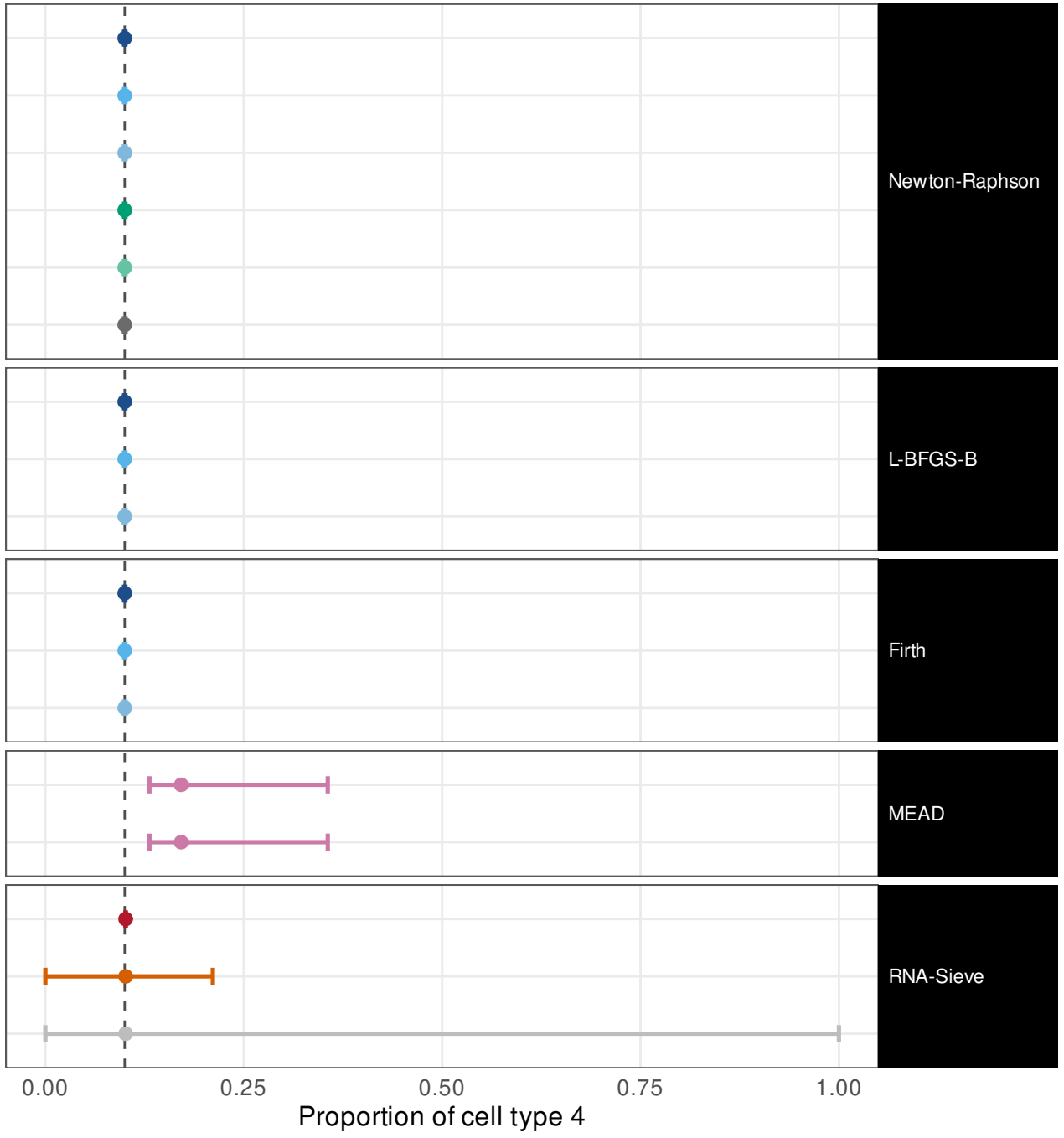


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / unbalanced / low sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.88 [0.83, 0.92]	w = 0.00166
ILR, mapped back	0.88 [0.82, 0.91]	w = 0.00165
ILR, REML	0.92 [0.88, 0.95]	w = 0.00199
Profile likelihood	0.89 [0.84, 0.93]	w = 0.00174
Profile, REML	0.92 [0.88, 0.95]	w = 0.002
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.00208
Wald (delta)	0.88 [0.83, 0.92]	w = 0.00166
ILR, mapped back	0.88 [0.83, 0.92]	w = 0.00166
ILR, REML	0.92 [0.87, 0.95]	w = 0.002
Wald (delta)	0.88 [0.83, 0.92]	w = 0.00166
ILR, mapped back	0.88 [0.83, 0.92]	w = 0.00165
ILR, REML	0.92 [0.87, 0.95]	w = 0.002
Method's own interval	0.22 [0.17, 0.28]	w = 0.225
Own interval, transcript fractions	0.3 [0.24, 0.36]	w = 0.225
Fisher information (Wald)	0 [0, 0.019]	w = 7.54e-06
Godambe sandwich	1 [0.98, 1]	w = 0.211
Released code, as published	1 [0.98, 1]	w = 1

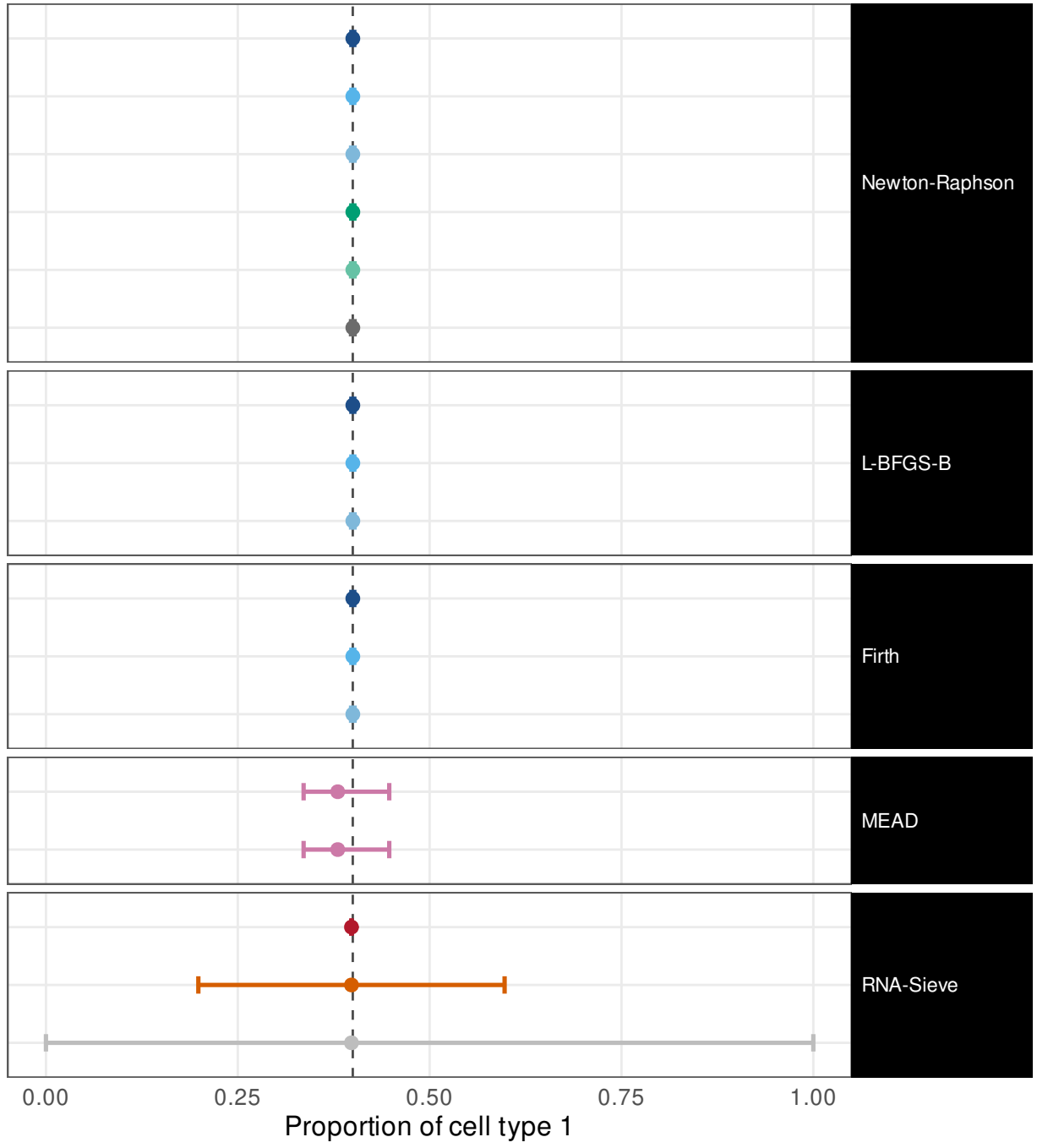


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / unbalanced / high sibling overlap / a = 1 / equal sizes

Wald (delta)	0.9 [0.84, 0.93]	w = 0.00376
ILR, mapped back	0.9 [0.86, 0.94]	w = 0.00375
ILR, REML	0.96 [0.92, 0.98]	w = 0.00454
Profile likelihood	0.92 [0.87, 0.95]	w = 0.00394
Profile, REML	0.96 [0.93, 0.98]	w = 0.00454
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.00466
Wald (delta)	0.9 [0.84, 0.93]	w = 0.00376
ILR, mapped back	0.9 [0.85, 0.93]	w = 0.00374
ILR, REML	0.96 [0.92, 0.98]	w = 0.00454
Wald (delta)	0.9 [0.84, 0.93]	w = 0.00376
ILR, mapped back	0.9 [0.84, 0.93]	w = 0.00375
ILR, REML	0.96 [0.93, 0.98]	w = 0.00453
Method's own interval	0.095 [0.062, 0.14]	w = 0.112
Own interval, transcript fractions	0.095 [0.062, 0.14]	w = 0.112
Fisher information (Wald)	0.015 [0.0051, 0.043]	w = 0.000226
Godambe sandwich	1 [0.98, 1]	w = 0.399
Released code, as published	1 [0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / unbalanced / high sibling overlap / a = 1 / equal sizes

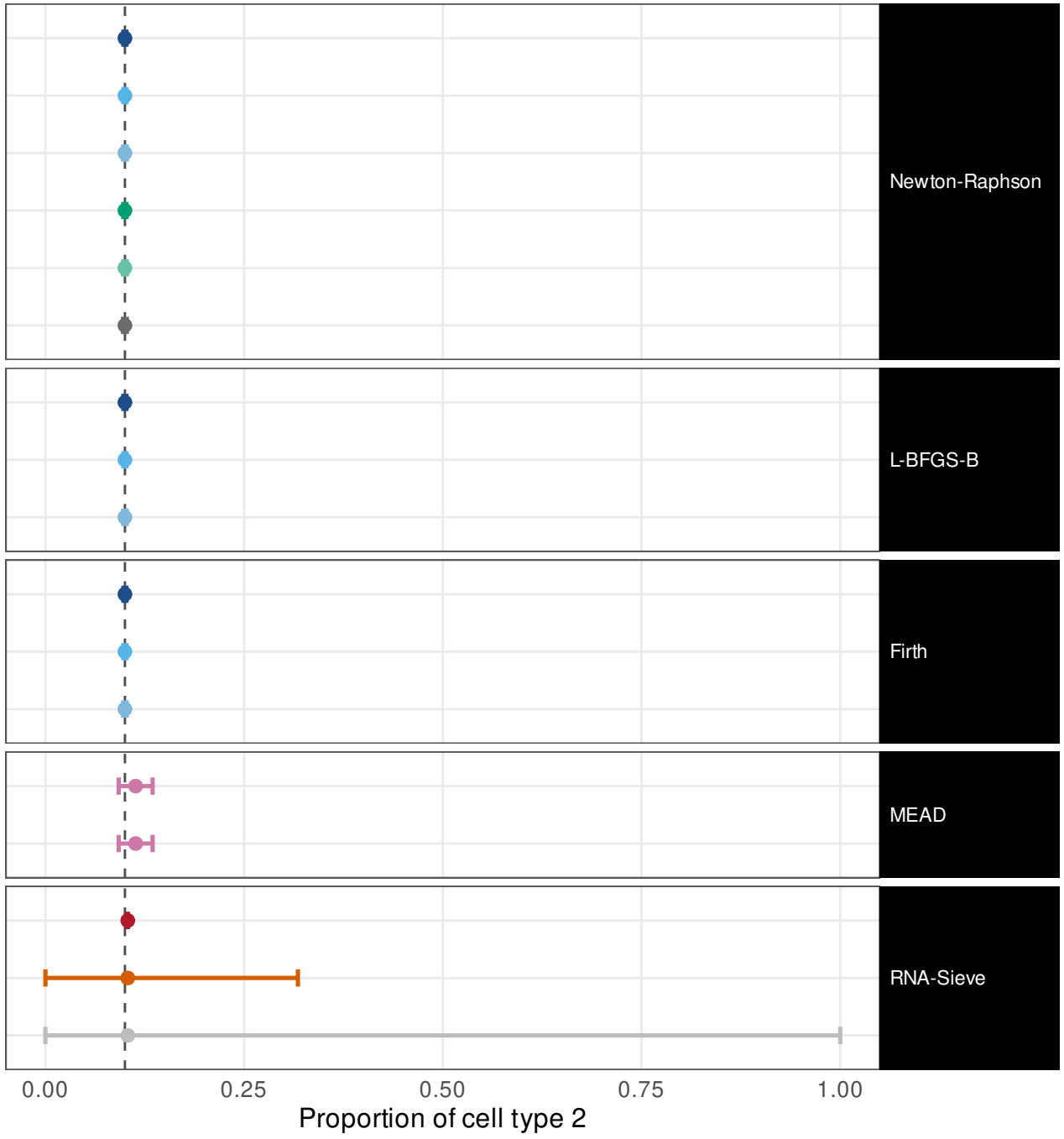
```
Wald (delta)                      0.9  [0.86, 0.94]   w = 0.00378
ILR, mapped back                  0.9  [0.85, 0.93]   w = 0.00377
ILR, REML                         0.96 [0.92, 0.98]   w = 0.00457
Profile likelihood                0.92 [0.87, 0.95]   w = 0.00397
Profile, REML                     0.96 [0.92, 0.98]   w = 0.00458
Oracle (Fisher at p*)               1  [0.98,    1]   w = 0.00468

Wald (delta)                      0.9  [0.86, 0.94]   w = 0.00378
ILR, mapped back                  0.9  [0.84, 0.93]   w = 0.00378
ILR, REML                         0.96 [0.92, 0.98]   w = 0.00456

Wald (delta)                      0.9  [0.86, 0.94]   w = 0.00378
ILR, mapped back                  0.88 [0.83, 0.92]   w = 0.00378
ILR, REML                         0.96 [0.92, 0.98]   w = 0.00458

Method's own interval               1  [0.98,    1]   w = 0.0426
Own interval, transcript fractions  1  [0.98,    1]   w = 0.0426

Fisher information (Wald)           0  [   0, 0.019]   w = 0.000143
Godambe sandwich                    1  [0.98,    1]   w = 0.318
Released code, as published         1  [0.98,    1]   w =    1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / unbalanced / high sibling overlap / a = 1 / equal sizes

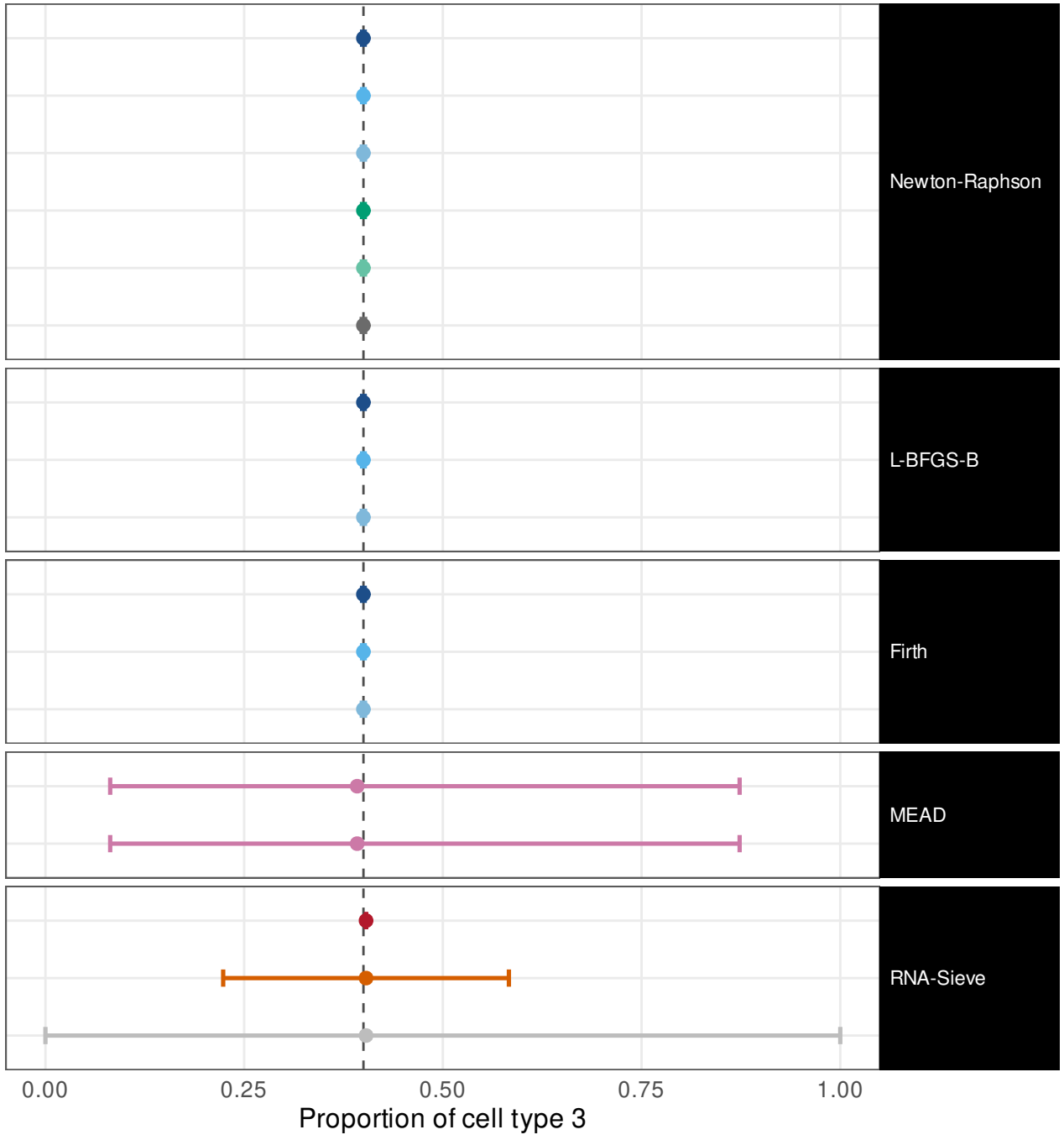
```
Wald (delta)                        0.91 [0.86, 0.94]   w = 0.00327
ILR, mapped back                    0.9  [0.85, 0.93]   w = 0.00326
ILR, REML                           0.96 [0.92, 0.98]   w = 0.00394
Profile likelihood                  0.93 [0.89, 0.96]   w = 0.00343
Profile, REML                       0.95 [0.91, 0.97]   w = 0.00395
Oracle (Fisher at p*)                  1 [0.98,    1]   w = 0.00428

Wald (delta)                        0.91 [0.86, 0.94]   w = 0.00327
ILR, mapped back                    0.91 [0.86, 0.94]   w = 0.00326
ILR, REML                           0.96 [0.92, 0.98]   w = 0.00395

Wald (delta)                        0.91 [0.86, 0.94]   w = 0.00327
ILR, mapped back                    0.92 [0.88, 0.95]   w = 0.00327
ILR, REML                           0.95 [0.91, 0.97]   w = 0.00394

Method's own interval               0.82 [0.77, 0.87]   w = 0.792
Own interval, transcript fractions  0.82 [0.77, 0.87]   w = 0.792

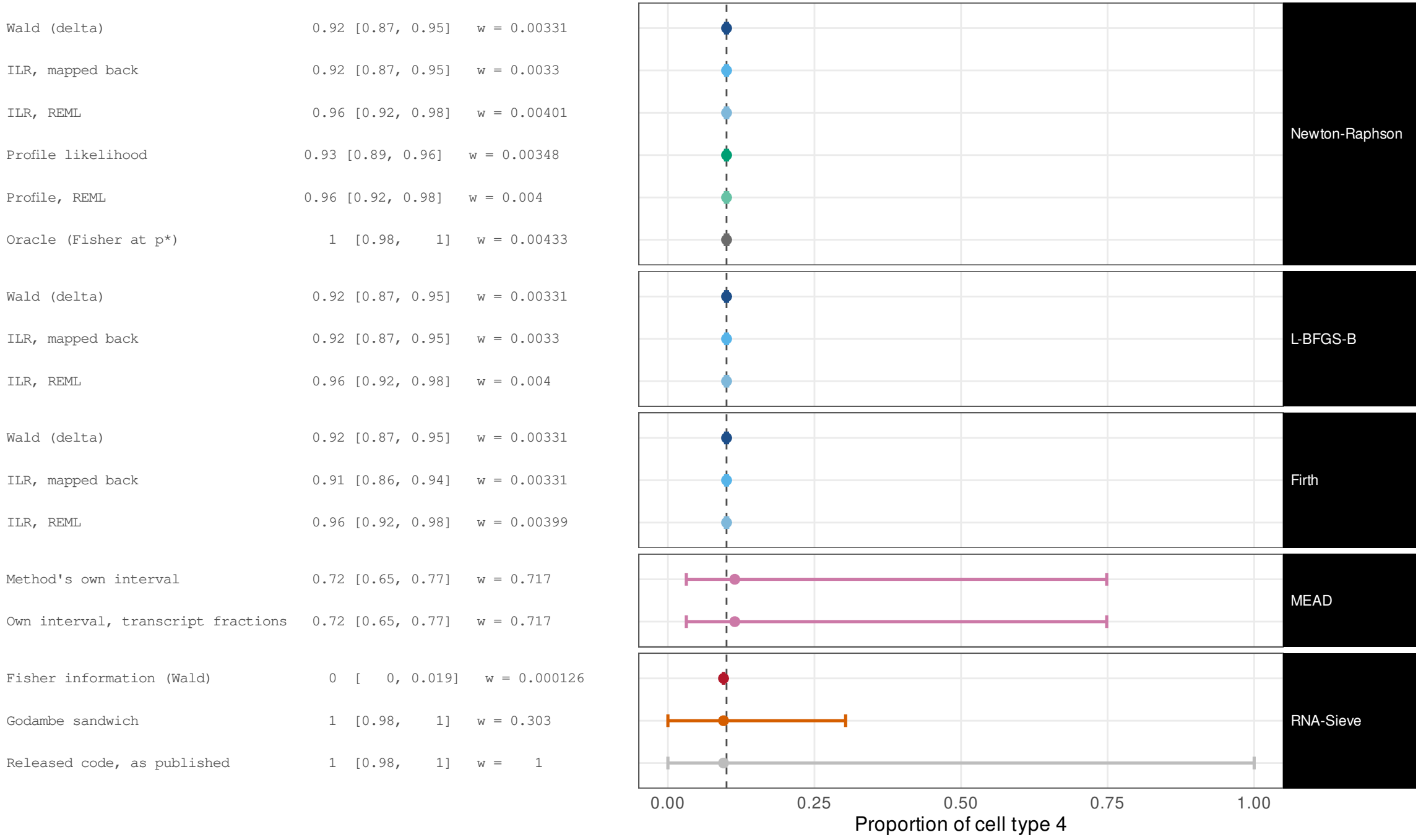
Fisher information (Wald)              0 [   0, 0.019]   w = 0.000274
Godambe sandwich                       1 [0.98,    1]   w = 0.359
Released code, as published            1 [0.98,    1]   w =    1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / unbalanced / high sibling overlap / a = 1 / equal sizes

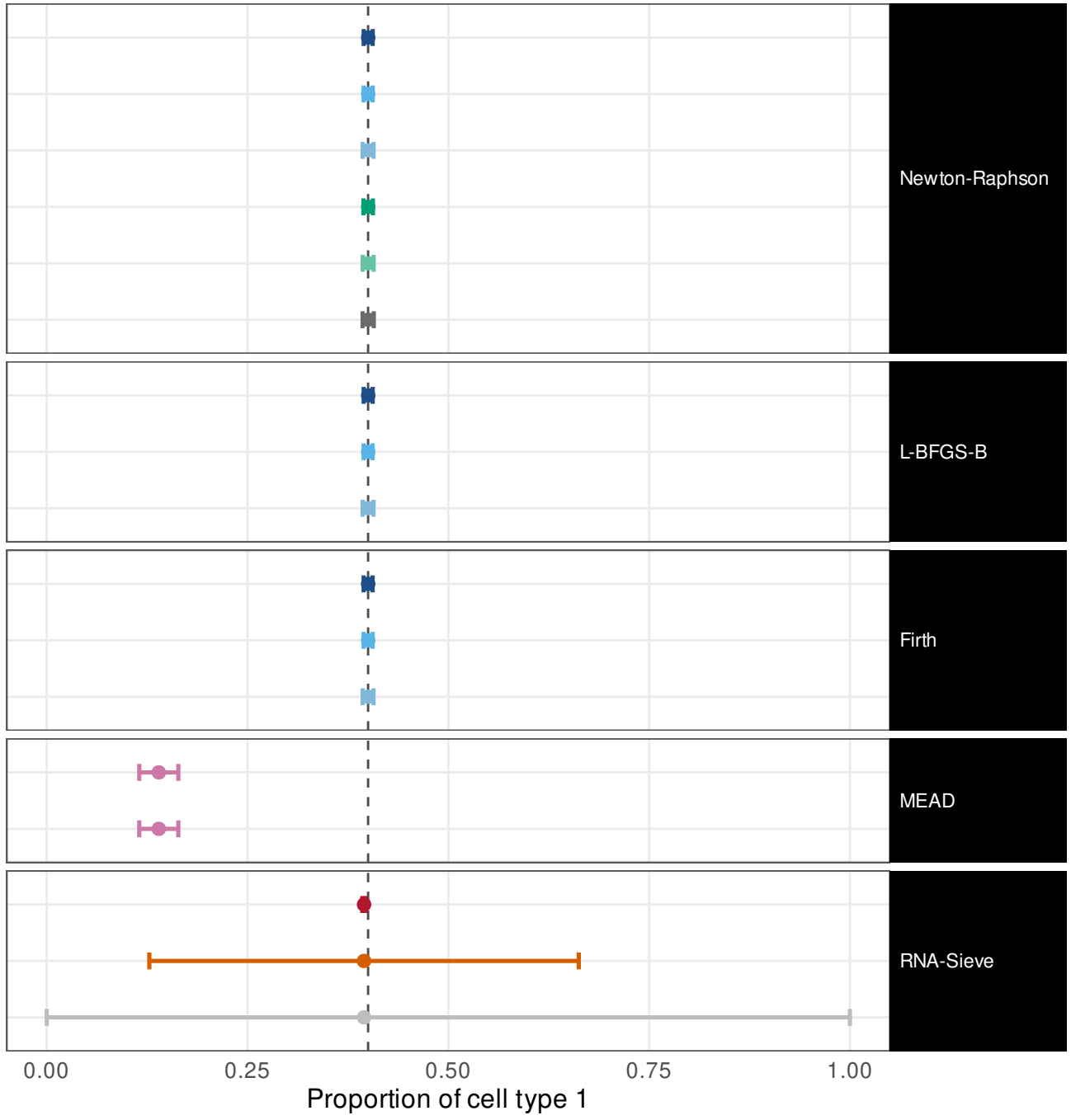


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / unbalanced / high sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.9	[0.86, 0.94]	w = 0.0112
ILR, mapped back	0.91	[0.86, 0.94]	w = 0.0112
ILR, REML	0.94	[0.9, 0.97]	w = 0.0135
Profile likelihood	0.92	[0.87, 0.95]	w = 0.0117
Profile, REML	0.94	[0.9, 0.97]	w = 0.0135
Oracle (Fisher at p*)	1	[0.98, 1]	w = 0.0139
Wald (delta)	0.9	[0.86, 0.94]	w = 0.0112
ILR, mapped back	0.9	[0.86, 0.94]	w = 0.0112
ILR, REML	0.94	[0.9, 0.97]	w = 0.0135
Wald (delta)	0.9	[0.86, 0.94]	w = 0.0112
ILR, mapped back	0.9	[0.86, 0.94]	w = 0.0112
ILR, REML	0.94	[0.9, 0.97]	w = 0.0135
Method's own interval	0	[0, 0.019]	w = 0.0487
Own interval, transcript fractions	1	[0.98, 1]	w = 0.0487
Fisher information (Wald)	0.18	[0.13, 0.23]	w = 0.00424
Godambe sandwich	1	[0.98, 1]	w = 0.535
Released code, as published	1	[0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / unbalanced / high sibling overlap / a = 1 / sizes 1:1:5:5

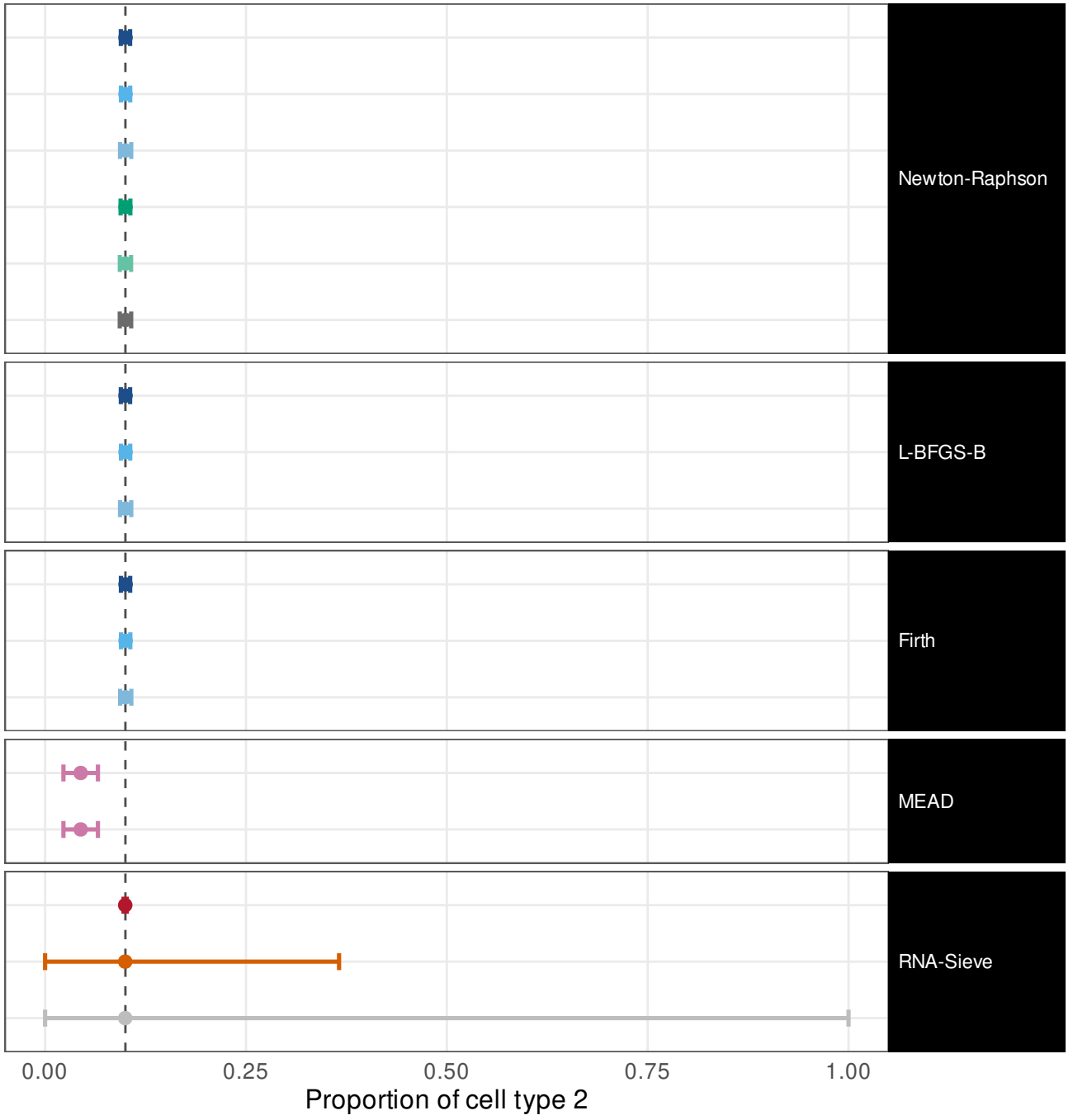
```
Wald (delta)                        0.94 [0.89, 0.96]   w = 0.0115
ILR, mapped back                    0.94 [0.89, 0.96]   w = 0.0115
ILR, REML                           0.96 [0.92, 0.98]   w = 0.0139
Profile likelihood                  0.94 [ 0.9, 0.97]   w = 0.0121
Profile, REML                       0.96 [0.92, 0.98]   w = 0.014
Oracle (Fisher at p*)                 1  [0.98,    1]   w = 0.0142

Wald (delta)                        0.94 [0.89, 0.96]   w = 0.0115
ILR, mapped back                    0.94 [0.89, 0.96]   w = 0.0115
ILR, REML                           0.96 [0.93, 0.98]   w = 0.014

Wald (delta)                        0.94 [0.89, 0.96]   w = 0.0115
ILR, mapped back                    0.92 [0.88, 0.95]   w = 0.0116
ILR, REML                           0.96 [0.92, 0.98]   w = 0.0139

Method's own interval                 0  [   0, 0.019]   w = 0.043
Own interval, transcript fractions    1  [0.98,    1]   w = 0.043

Fisher information (Wald)           0.38 [0.31, 0.44]   w = 0.00404
Godambe sandwich                      1  [0.98,    1]   w = 0.366
Released code, as published           1  [0.98,    1]   w =    1
```

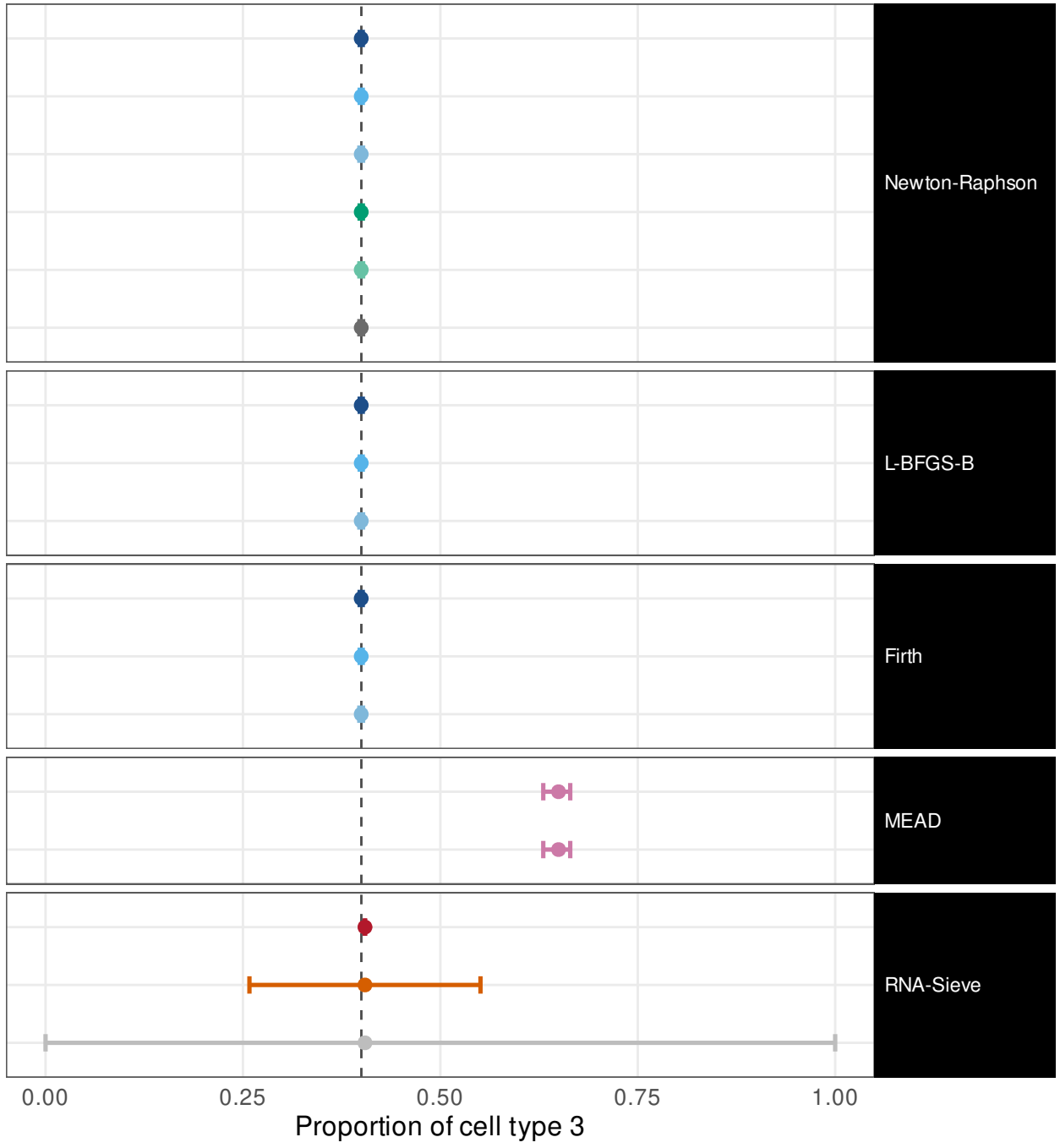


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / unbalanced / high sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.89 [0.84, 0.93]	w = 0.00386
ILR, mapped back	0.9 [0.84, 0.93]	w = 0.00385
ILR, REML	0.94 [0.9, 0.97]	w = 0.00466
Profile likelihood	0.91 [0.86, 0.94]	w = 0.00405
Profile, REML	0.94 [0.9, 0.97]	w = 0.00466
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.00467
Wald (delta)	0.89 [0.84, 0.93]	w = 0.00386
ILR, mapped back	0.89 [0.84, 0.93]	w = 0.00385
ILR, REML	0.94 [0.9, 0.97]	w = 0.00466
Wald (delta)	0.89 [0.84, 0.93]	w = 0.00386
ILR, mapped back	0.89 [0.84, 0.93]	w = 0.00385
ILR, REML	0.94 [0.9, 0.97]	w = 0.00466
Method's own interval	0.015 [0.0051, 0.043]	w = 0.0342
Own interval, transcript fractions	0.015 [0.0051, 0.043]	w = 0.0342
Fisher information (Wald)	0 [0, 0.019]	w = 0.000601
Godambe sandwich	1 [0.98, 1]	w = 0.293
Released code, as published	1 [0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / unbalanced / high sibling overlap / a = 1 / sizes 1:1:5:5

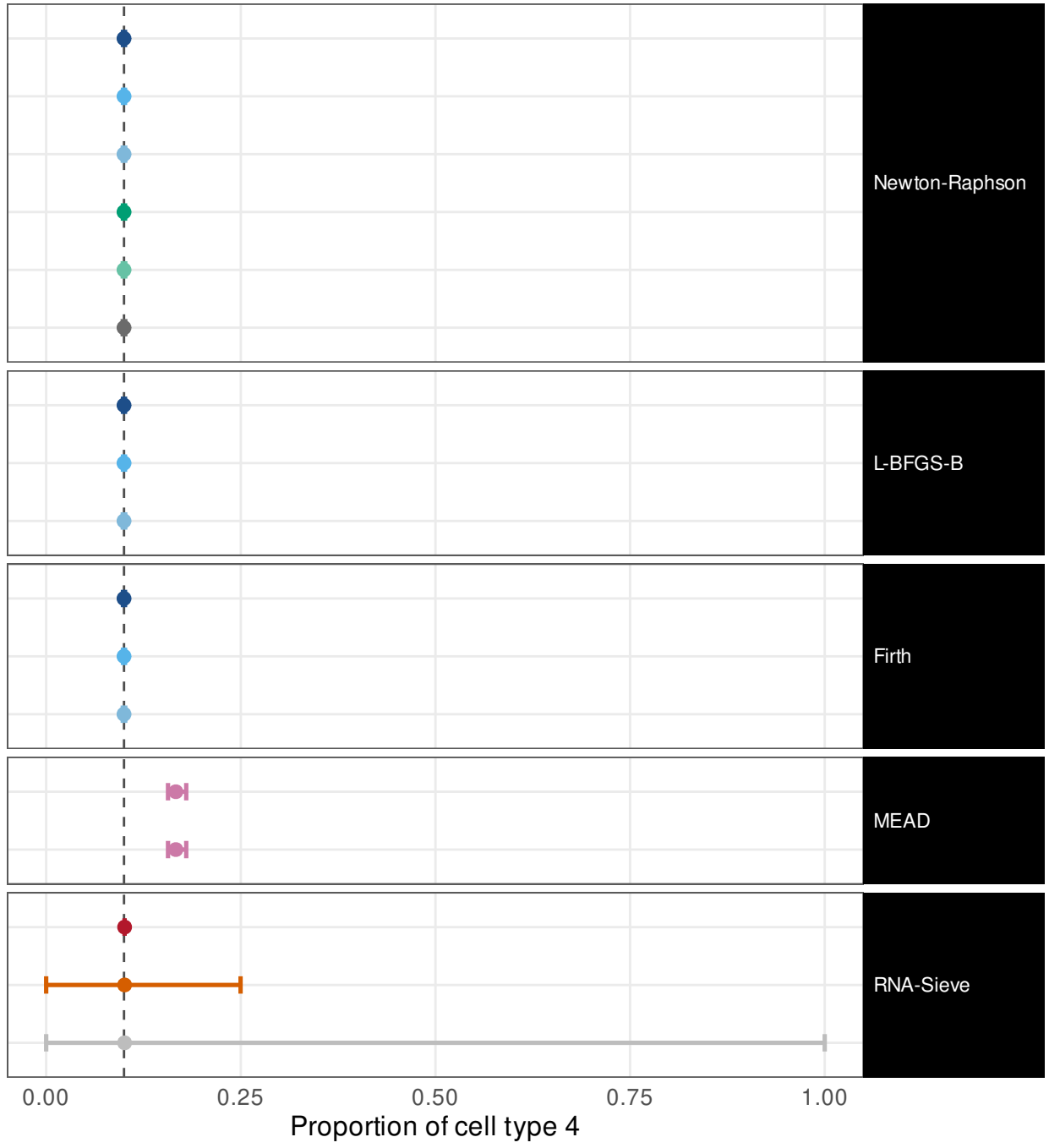
```
Wald (delta)                        0.92 [0.87, 0.95]   w = 0.00262
ILR, mapped back                    0.92 [0.87, 0.95]   w = 0.00261
ILR, REML                           0.97 [0.94, 0.99]   w = 0.00316
Profile likelihood                  0.92 [0.88, 0.95]   w = 0.00275
Profile, REML                       0.97 [0.94, 0.99]   w = 0.00317
Oracle (Fisher at p*)                 1  [0.98,    1]   w = 0.00322

Wald (delta)                        0.92 [0.87, 0.95]   w = 0.00262
ILR, mapped back                    0.92 [0.87, 0.95]   w = 0.00261
ILR, REML                           0.95 [0.91, 0.97]   w = 0.00315

Wald (delta)                        0.92 [0.87, 0.95]   w = 0.00262
ILR, mapped back                    0.92 [0.87, 0.95]   w = 0.00261
ILR, REML                           0.98 [0.94, 0.99]   w = 0.00316

Method's own interval               0.005 [0.00088, 0.028]   w = 0.0234
Own interval, transcript fractions    1  [0.97,    1]   w = 0.0234

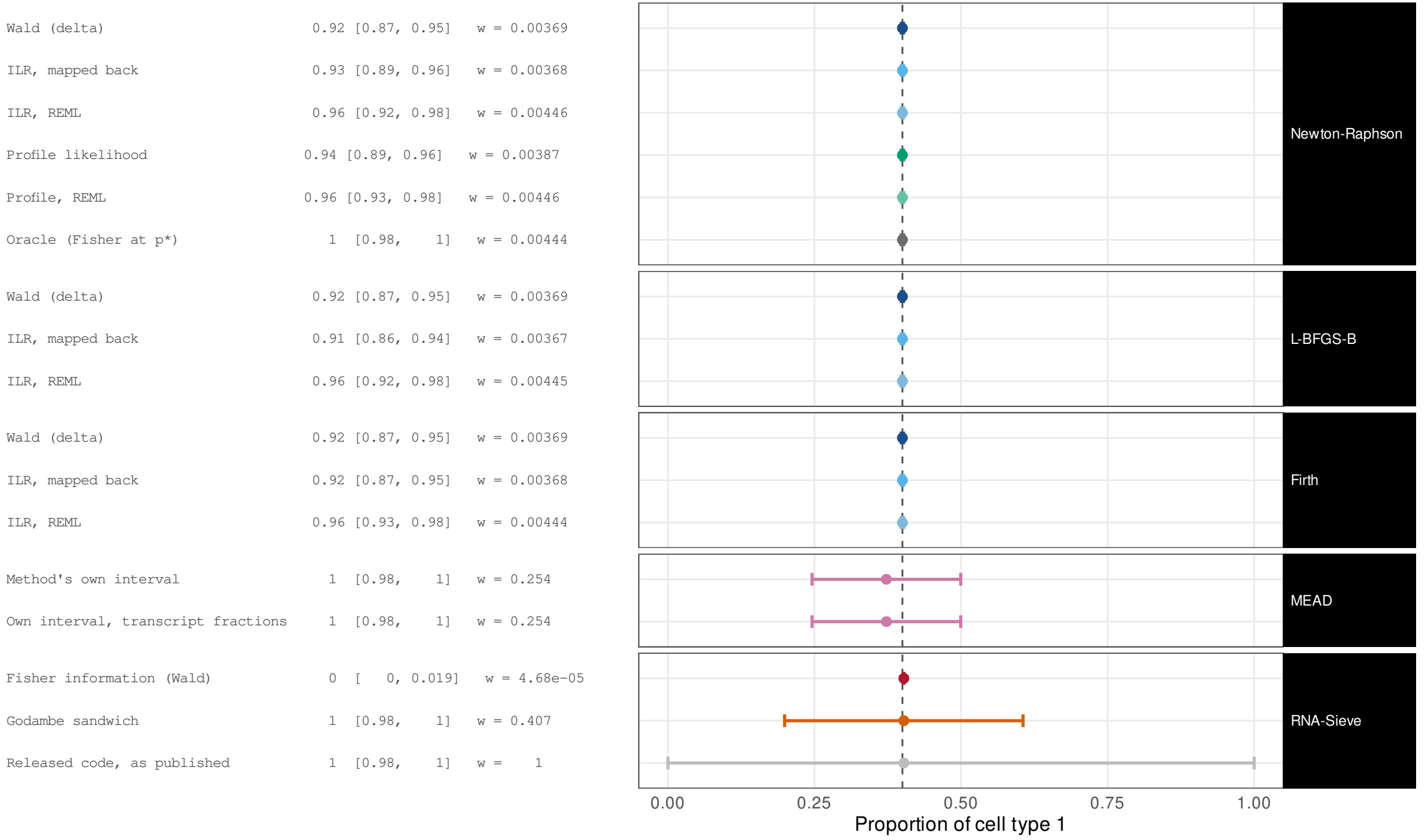
Fisher information (Wald)           0.025 [0.011, 0.057]   w = 8.42e-05
Godambe sandwich                      1  [0.98,    1]   w = 0.25
Released code, as published           1  [0.98,    1]   w =    1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / unbalanced / high sibling overlap / a = 5 / equal sizes

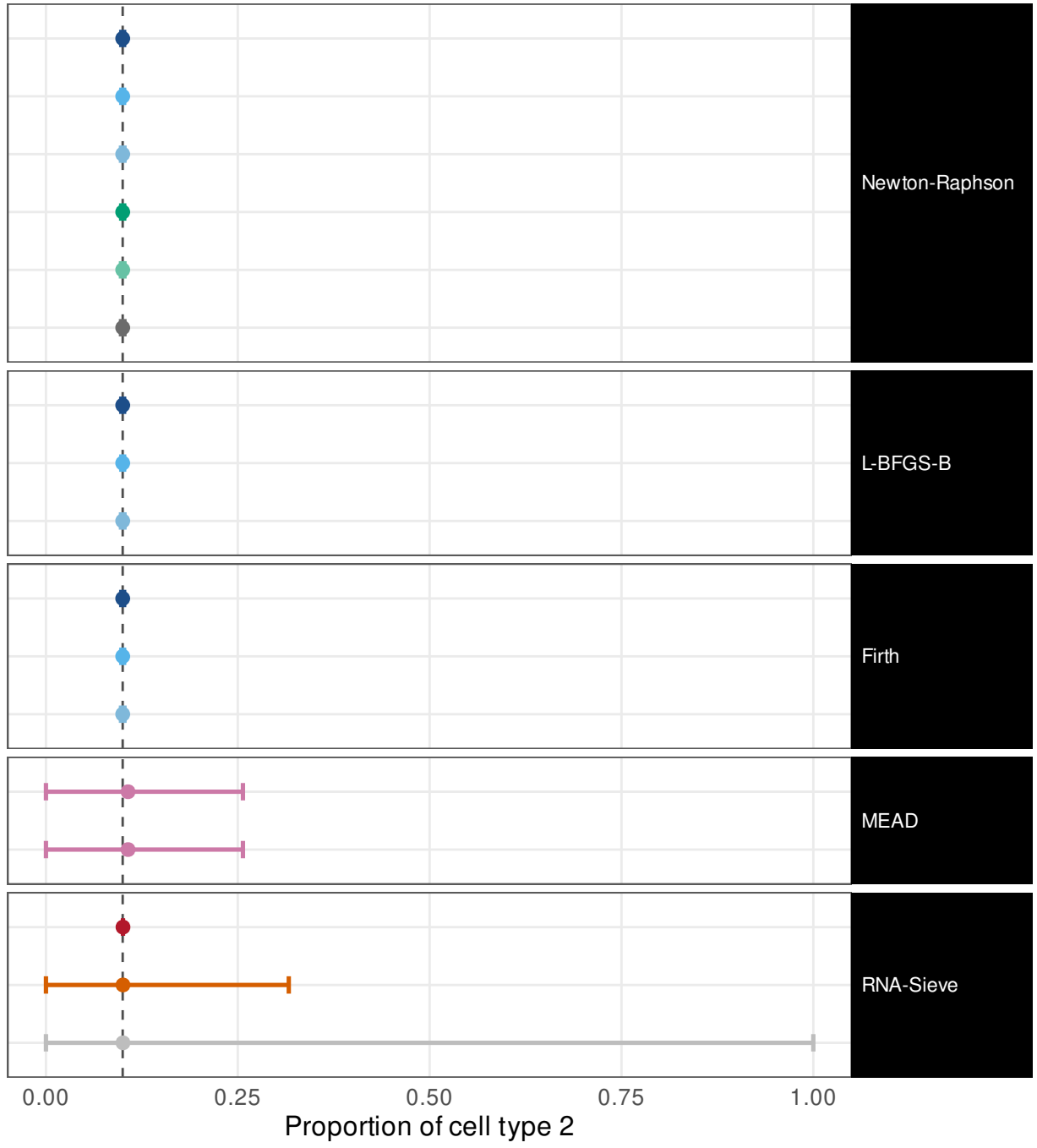


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / unbalanced / high sibling overlap / a = 5 / equal sizes

Wald (delta)	0.92 [0.87, 0.95]	w = 0.00371
ILR, mapped back	0.9 [0.85, 0.93]	w = 0.00371
ILR, REML	0.98 [0.96, 0.99]	w = 0.0045
Profile likelihood	0.93 [0.89, 0.96]	w = 0.0039
Profile, REML	0.98 [0.96, 0.99]	w = 0.00449
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.00441
Wald (delta)	0.92 [0.87, 0.95]	w = 0.00371
ILR, mapped back	0.9 [0.86, 0.94]	w = 0.0037
ILR, REML	0.98 [0.95, 0.99]	w = 0.00448
Wald (delta)	0.92 [0.87, 0.95]	w = 0.00371
ILR, mapped back	0.9 [0.86, 0.94]	w = 0.00371
ILR, REML	0.98 [0.96, 0.99]	w = 0.00448
Method's own interval	1 [0.98, 1]	w = 0.257
Own interval, transcript fractions	1 [0.98, 1]	w = 0.257
Fisher information (Wald)	0.015 [0.0051, 0.043]	w = 2.93e-05
Godambe sandwich	1 [0.98, 1]	w = 0.316
Released code, as published	1 [0.98, 1]	w = 1

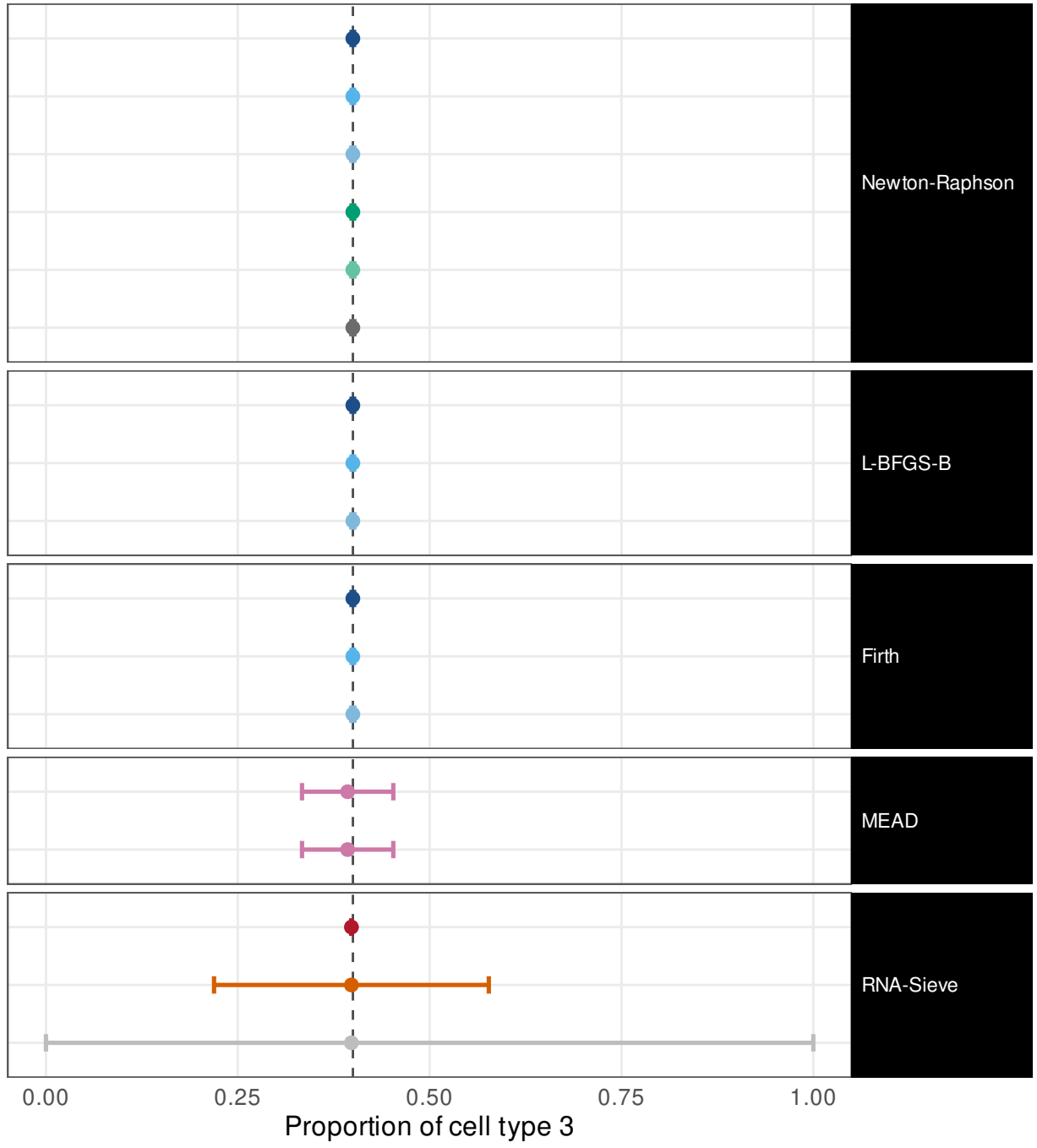


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / unbalanced / high sibling overlap / a = 5 / equal sizes

Wald (delta)	0.91 [0.86, 0.94]	w = 0.00321
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.0032
ILR, REML	0.94 [0.9, 0.97]	w = 0.00388
Profile likelihood	0.93 [0.89, 0.96]	w = 0.00337
Profile, REML	0.94 [0.9, 0.97]	w = 0.00388
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0039
Wald (delta)	0.91 [0.86, 0.94]	w = 0.00321
ILR, mapped back	0.91 [0.86, 0.94]	w = 0.00319
ILR, REML	0.96 [0.92, 0.98]	w = 0.00387
Wald (delta)	0.91 [0.86, 0.94]	w = 0.00321
ILR, mapped back	0.91 [0.86, 0.94]	w = 0.0032
ILR, REML	0.95 [0.91, 0.97]	w = 0.00388
Method's own interval	1 [0.98, 1]	w = 0.119
Own interval, transcript fractions	1 [0.98, 1]	w = 0.119
Fisher information (Wald)	0.015 [0.0051, 0.043]	w = 5.75e-05
Godambe sandwich	1 [0.98, 1]	w = 0.358
Released code, as published	1 [0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / unbalanced / high sibling overlap / a = 5 / equal sizes

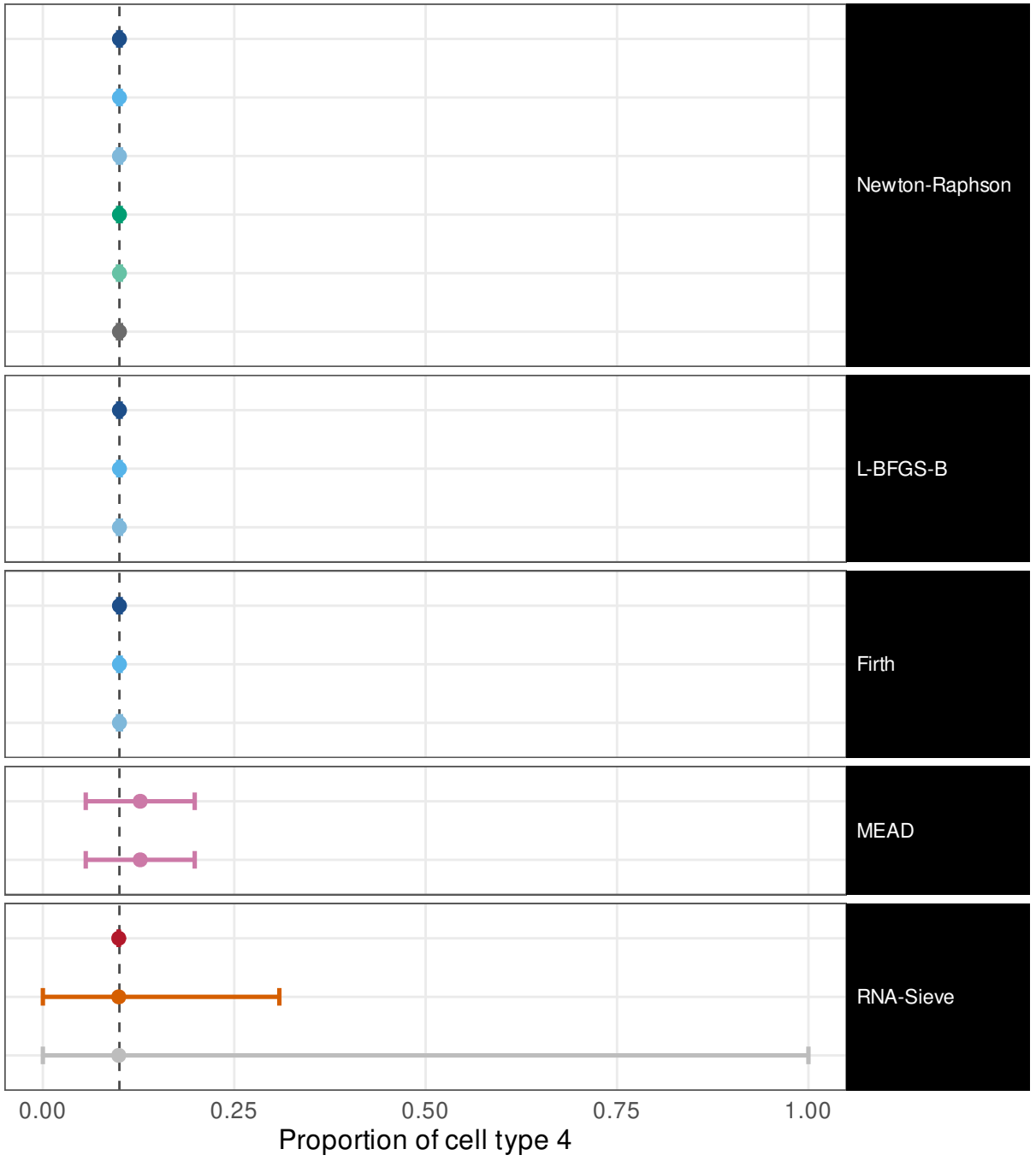
```
Wald (delta)                        0.9  [0.84, 0.93]   w = 0.00325
ILR, mapped back                    0.9  [0.84, 0.93]   w = 0.00324
ILR, REML                           0.96 [0.92, 0.98]   w = 0.00392
Profile likelihood                  0.92 [0.87, 0.95]   w = 0.00341
Profile, REML                       0.96 [0.92, 0.98]   w = 0.00393
Oracle (Fisher at p*)                 1  [0.98,    1]   w = 0.00391

Wald (delta)                        0.9  [0.84, 0.93]   w = 0.00325
ILR, mapped back                    0.89 [0.84, 0.93]   w = 0.00324
ILR, REML                           0.95 [0.91, 0.97]   w = 0.00393

Wald (delta)                        0.9  [0.84, 0.93]   w = 0.00325
ILR, mapped back                    0.89 [0.84, 0.93]   w = 0.00324
ILR, REML                           0.96 [0.92, 0.98]   w = 0.00392

Method's own interval                 1  [0.98,    1]   w = 0.142
Own interval, transcript fractions    1  [0.98,    1]   w = 0.142

Fisher information (Wald)           0.005 [0.00088, 0.028]   w = 2.69e-05
Godambe sandwich                      1  [0.98,    1]   w = 0.309
Released code, as published           1  [0.98,    1]   w =    1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / unbalanced / high sibling overlap / a = 5 / sizes 1:1:5:5

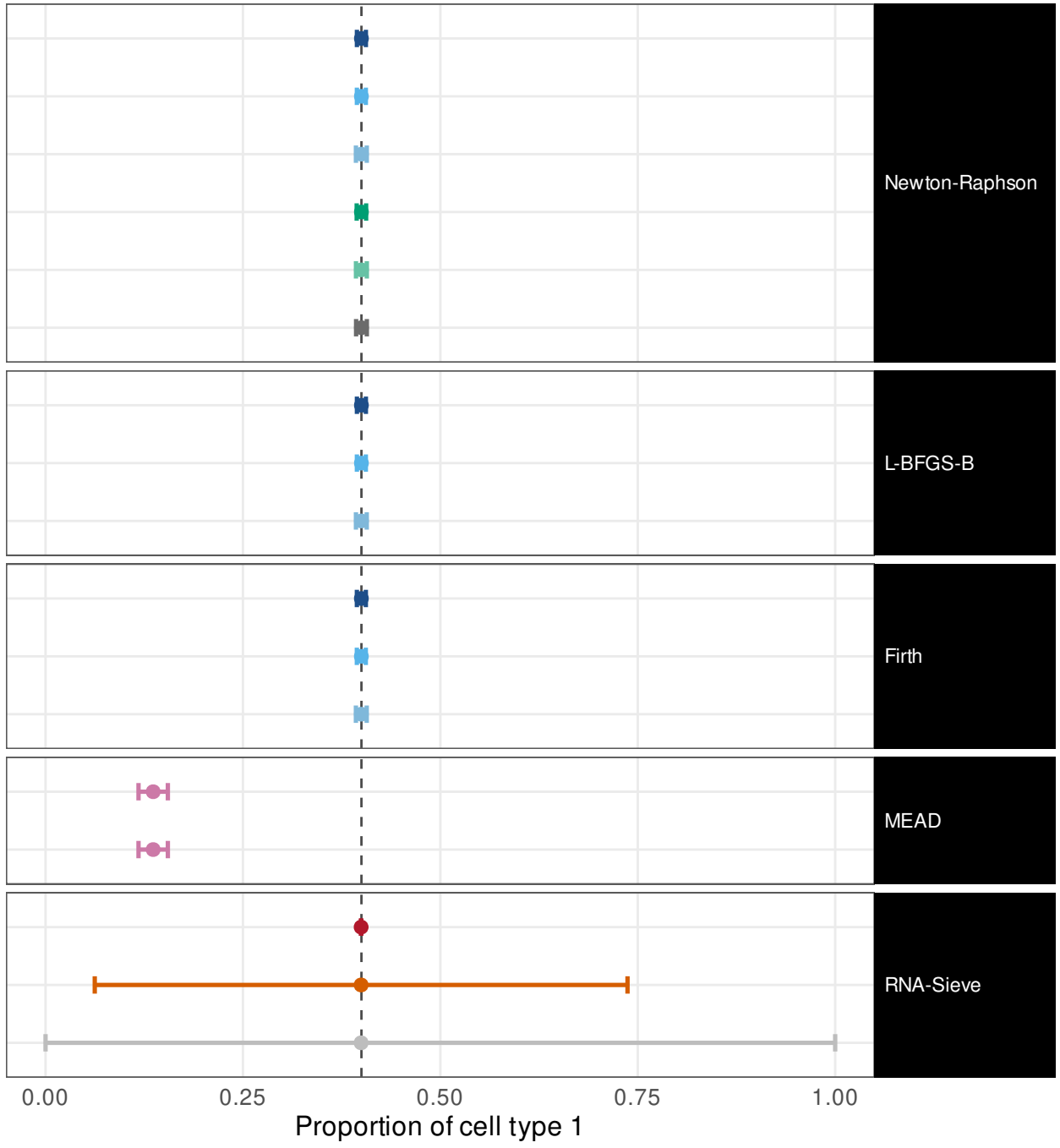
```
Wald (delta)                        0.91 [0.86, 0.94]   w = 0.0114
ILR, mapped back                    0.92 [0.87, 0.95]   w = 0.0113
ILR, REML                           0.96 [0.92, 0.98]   w = 0.0137
Profile likelihood                  0.92 [0.88, 0.95]   w = 0.0119
Profile, REML                       0.96 [0.92, 0.98]   w = 0.0138
Oracle (Fisher at p*)                  1 [0.98,    1]   w = 0.0136

Wald (delta)                        0.91 [0.86, 0.94]   w = 0.0114
ILR, mapped back                    0.9  [0.86, 0.94]   w = 0.0113
ILR, REML                           0.95 [0.91, 0.97]   w = 0.0137

Wald (delta)                        0.92 [0.87, 0.95]   w = 0.0114
ILR, mapped back                    0.9  [0.85, 0.93]   w = 0.0113
ILR, REML                           0.96 [0.92, 0.98]   w = 0.0137

Method's own interval                  0 [   0, 0.019]   w = 0.0372
Own interval, transcript fractions     1 [0.98,    1]   w = 0.0372

Fisher information (Wald)           0.075 [0.046, 0.12]   w = 0.000543
Godambe sandwich                       1 [0.98,    1]   w = 0.675
Released code, as published            1 [0.98,    1]   w =     1
```

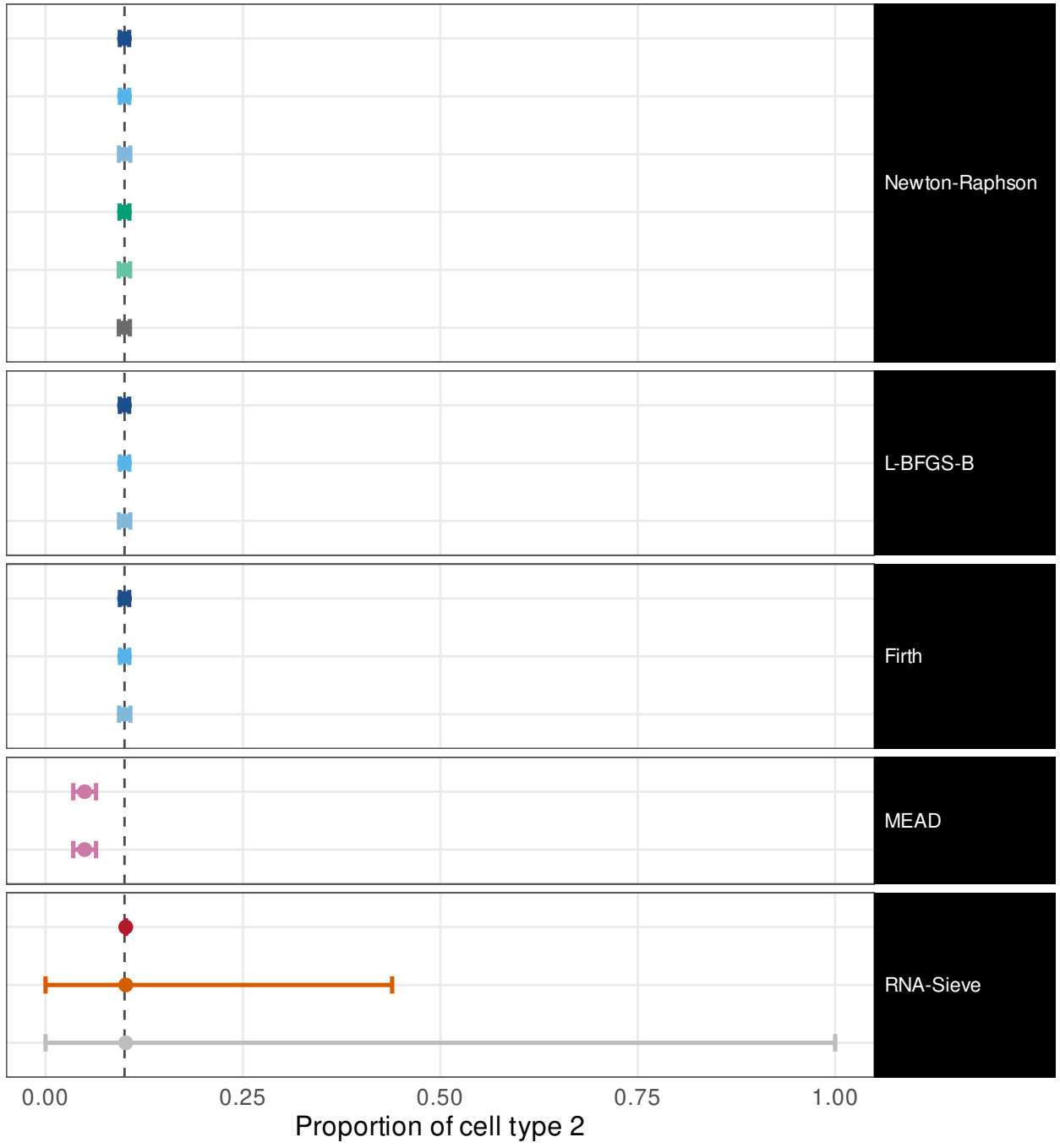


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / unbalanced / high sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.9 [0.85, 0.93]	w = 0.0117
ILR, mapped back	0.9 [0.86, 0.94]	w = 0.0117
ILR, REML	0.96 [0.92, 0.98]	w = 0.0142
Profile likelihood	0.92 [0.88, 0.95]	w = 0.0123
Profile, REML	0.96 [0.93, 0.98]	w = 0.0142
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0139
Wald (delta)	0.9 [0.85, 0.93]	w = 0.0117
ILR, mapped back	0.9 [0.85, 0.93]	w = 0.0117
ILR, REML	0.96 [0.93, 0.98]	w = 0.0141
Wald (delta)	0.9 [0.85, 0.93]	w = 0.0117
ILR, mapped back	0.9 [0.84, 0.93]	w = 0.0117
ILR, REML	0.96 [0.93, 0.98]	w = 0.0141
Method's own interval	0 [0, 0.019]	w = 0.0289
Own interval, transcript fractions	0.18 [0.14, 0.24]	w = 0.0289
Fisher information (Wald)	0.065 [0.038, 0.11]	w = 0.000506
Godambe sandwich	1 [0.98, 1]	w = 0.439
Released code, as published	1 [0.98, 1]	w = 1

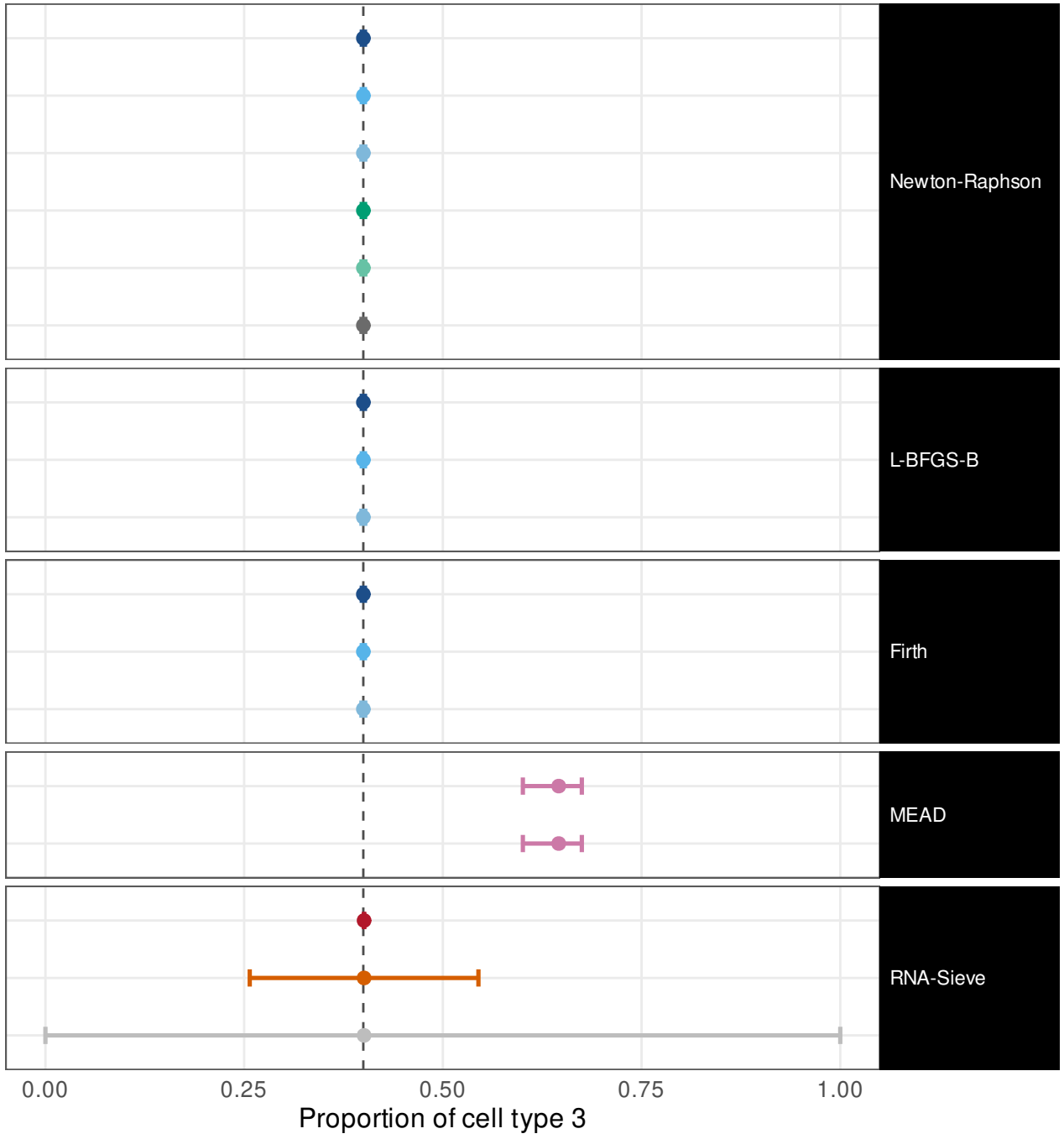


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / unbalanced / high sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.86 [0.81, 0.91]	w = 0.00392
ILR, mapped back	0.88 [0.83, 0.92]	w = 0.00391
ILR, REML	0.94 [0.9, 0.97]	w = 0.00474
Profile likelihood	0.9 [0.84, 0.93]	w = 0.00412
Profile, REML	0.94 [0.9, 0.97]	w = 0.00474
Oracle (Fisher at p*)	1 [0.98, 1]	w = 0.0047
Wald (delta)	0.86 [0.81, 0.91]	w = 0.00392
ILR, mapped back	0.87 [0.82, 0.91]	w = 0.00392
ILR, REML	0.94 [0.9, 0.97]	w = 0.00472
Wald (delta)	0.88 [0.82, 0.91]	w = 0.00392
ILR, mapped back	0.86 [0.81, 0.91]	w = 0.0039
ILR, REML	0.95 [0.91, 0.97]	w = 0.00474
Method's own interval	0.055 [0.031, 0.096]	w = 0.0742
Own interval, transcript fractions	0.055 [0.031, 0.096]	w = 0.0742
Fisher information (Wald)	0.04 [0.02, 0.077]	w = 0.000127
Godambe sandwich	1 [0.98, 1]	w = 0.288
Released code, as published	1 [0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 20 / 1e+05 cells per bulk / unbalanced / high sibling overlap / a = 5 / sizes 1:1:5:5

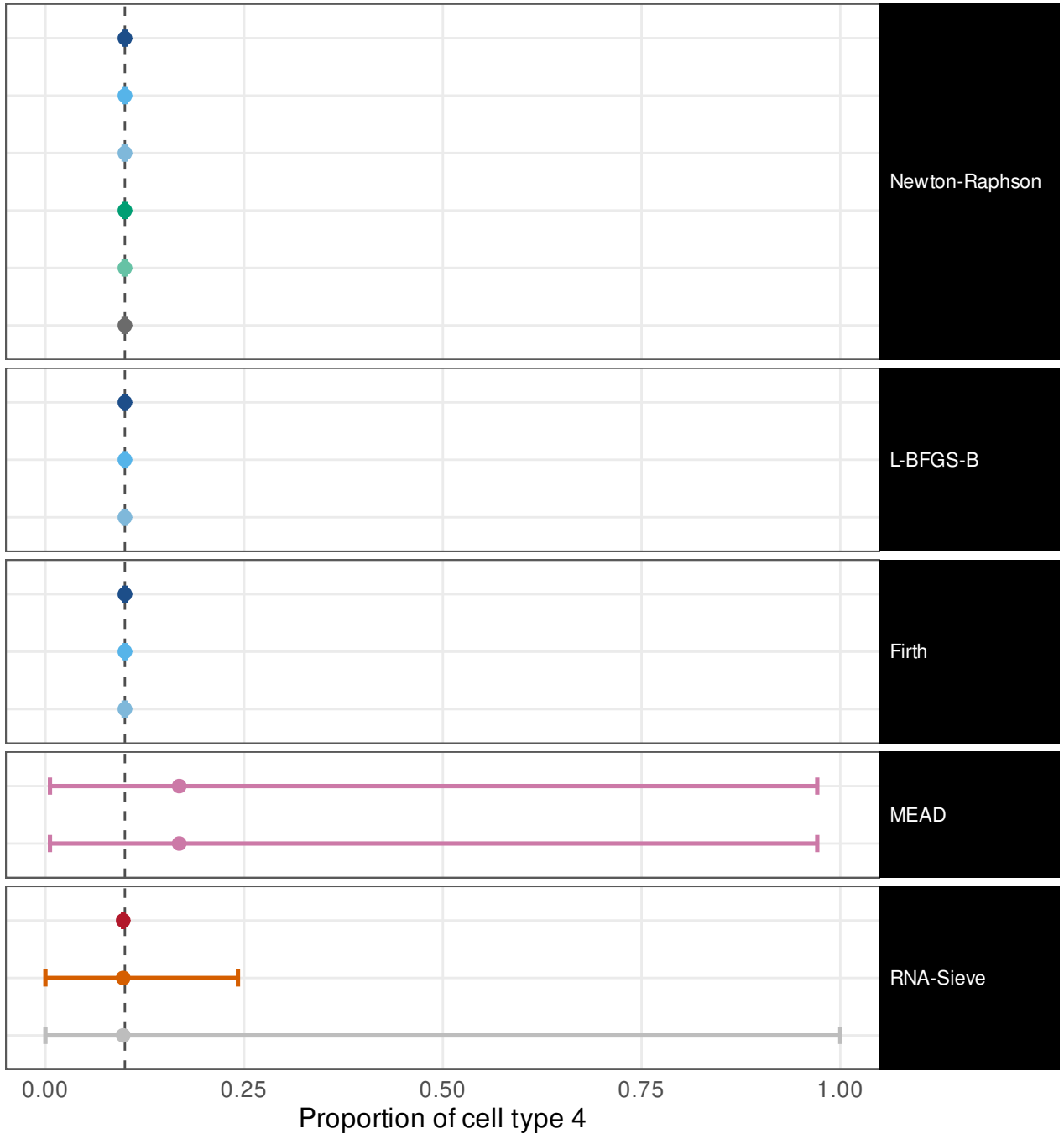
```
Wald (delta)                      0.92 [0.87, 0.95]   w = 0.00266
ILR, mapped back                  0.92 [0.87, 0.95]   w = 0.00266
ILR, REML                         0.97 [0.94, 0.99]   w = 0.00321
Profile likelihood                0.94 [ 0.9, 0.97]   w = 0.00279
Profile, REML                     0.96 [0.93, 0.98]   w = 0.00322
Oracle (Fisher at p*)                1 [0.98,    1]   w = 0.00316

Wald (delta)                      0.92 [0.87, 0.95]   w = 0.00266
ILR, mapped back                  0.92 [0.88, 0.95]   w = 0.00266
ILR, REML                         0.98 [0.94, 0.99]   w = 0.00322

Wald (delta)                      0.92 [0.87, 0.95]   w = 0.00266
ILR, mapped back                  0.92 [0.87, 0.95]   w = 0.00266
ILR, REML                         0.96 [0.93, 0.98]   w = 0.00321

Method's own interval             0.96 [0.93, 0.98]   w = 0.965
Own interval, transcript fractions   1 [0.97,    1]   w = 0.965

Fisher information (Wald)            0 [   0, 0.019]   w = 1.68e-05
Godambe sandwich                     1 [0.98,    1]   w = 0.242
Released code, as published          1 [0.98,    1]   w =     1
```

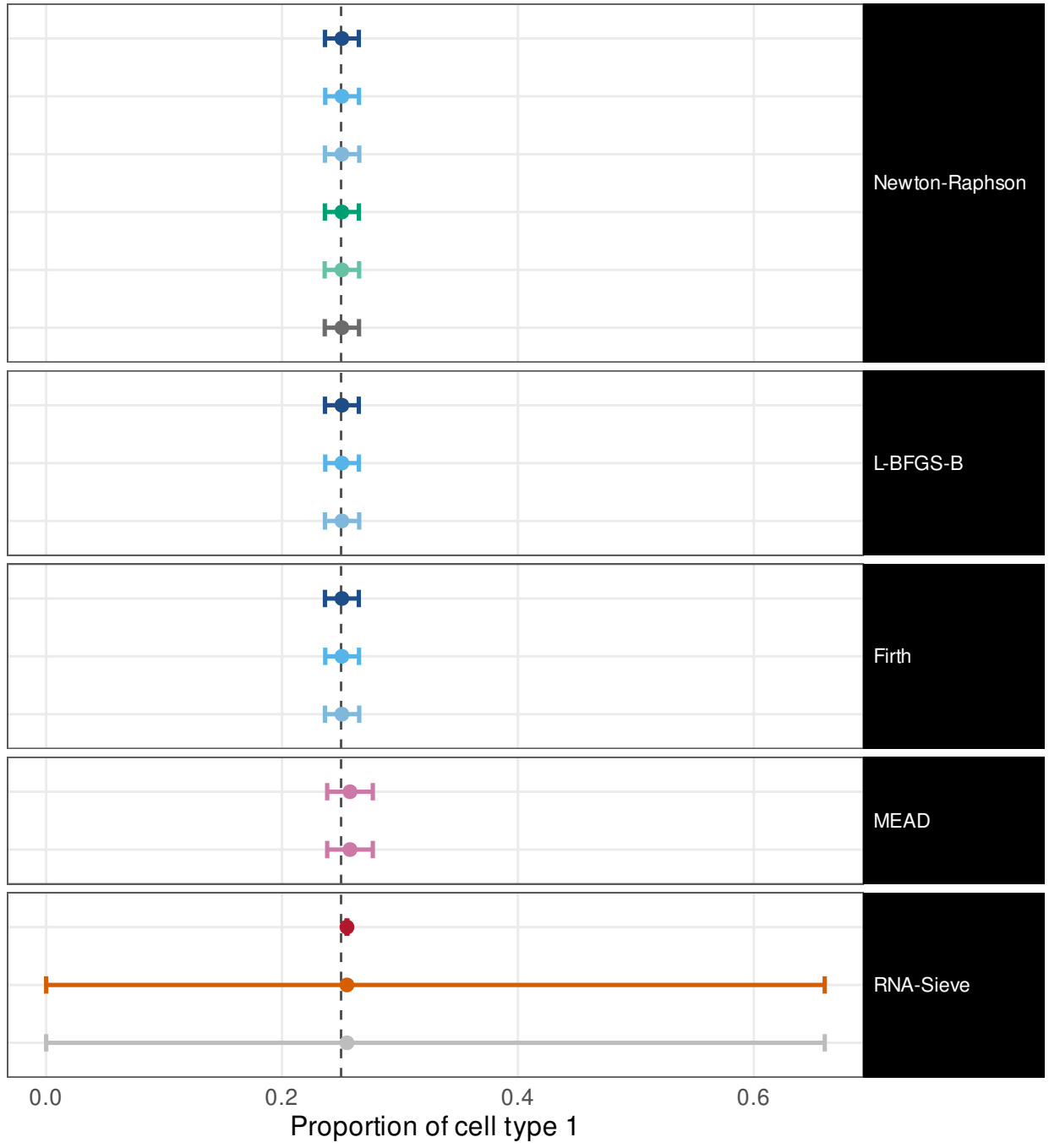


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / balanced / low sibling overlap / a = 1 / equal sizes

Wald (delta)	0.96 [0.92, 0.98]	w = 0.0287
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.0286
ILR, REML	0.94 [0.9, 0.97]	w = 0.0291
Profile likelihood	0.96 [0.92, 0.98]	w = 0.0288
Profile, REML	0.96 [0.92, 0.98]	w = 0.0292
Oracle (Fisher at p*)	0.97 [0.94, 0.99]	w = 0.029
Wald (delta)	0.96 [0.92, 0.98]	w = 0.0287
ILR, mapped back	0.94 [0.89, 0.96]	w = 0.0285
ILR, REML	0.96 [0.92, 0.98]	w = 0.029
Wald (delta)	0.96 [0.92, 0.98]	w = 0.0287
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.0286
ILR, REML	0.96 [0.93, 0.98]	w = 0.0291
Method's own interval	0.92 [0.88, 0.95]	w = 0.0387
Own interval, transcript fractions	0.92 [0.88, 0.95]	w = 0.0387
Fisher information (Wald)	0.045 [0.024, 0.083]	w = 0.000776
Godambe sandwich	1 [0.98, 1]	w = 0.66
Released code, as published	1 [0.98, 1]	w = 0.66

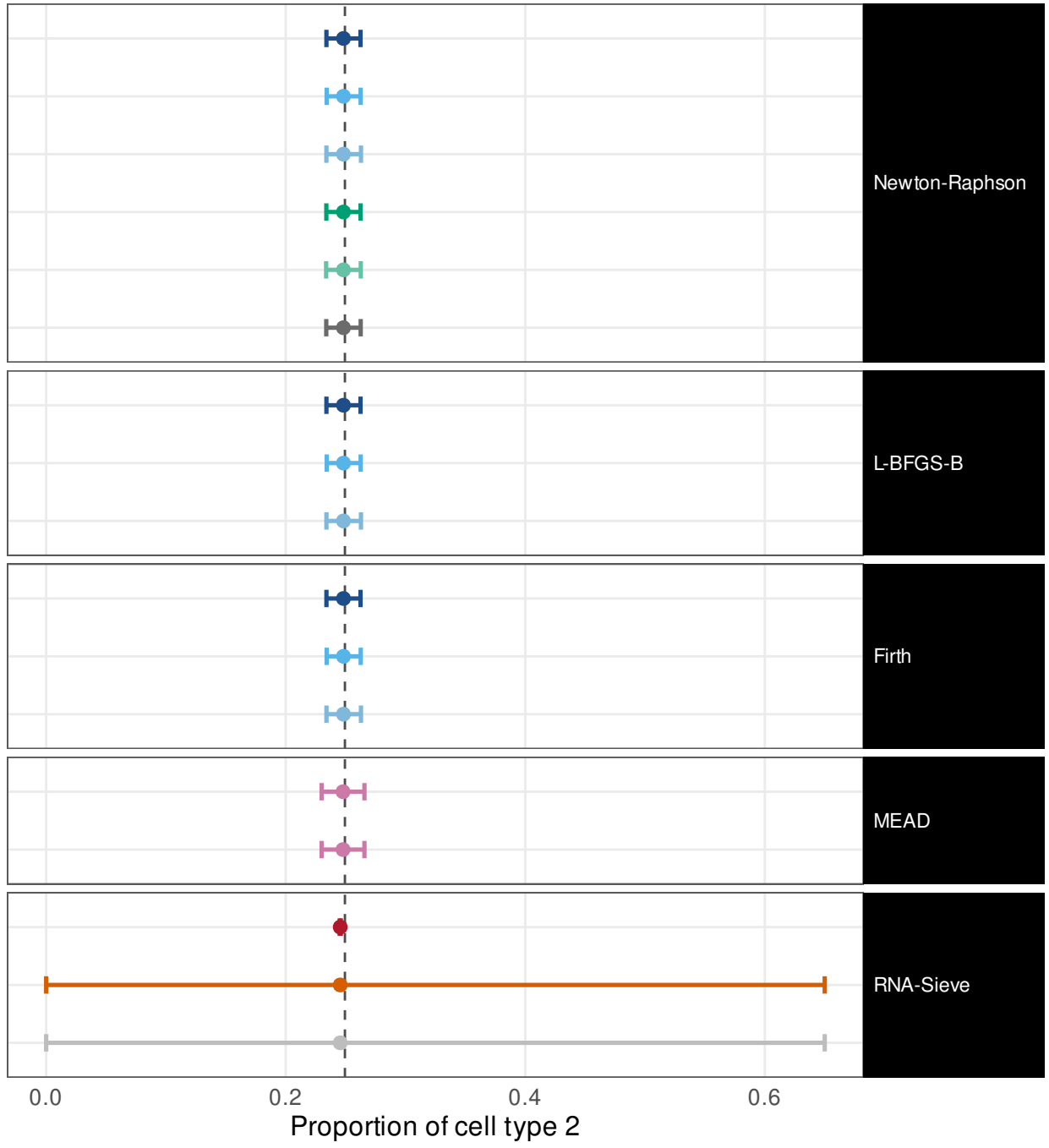


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / balanced / low sibling overlap / a = 1 / equal sizes

Wald (delta)	0.94 [0.89, 0.96]	w = 0.0285
ILR, mapped back	0.94 [0.89, 0.96]	w = 0.0284
ILR, REML	0.94 [0.9, 0.97]	w = 0.0289
Profile likelihood	0.94 [0.89, 0.96]	w = 0.0286
Profile, REML	0.94 [0.9, 0.97]	w = 0.0289
Oracle (Fisher at p*)	0.94 [0.89, 0.96]	w = 0.0288
Wald (delta)	0.94 [0.89, 0.96]	w = 0.0285
ILR, mapped back	0.94 [0.89, 0.96]	w = 0.0283
ILR, REML	0.94 [0.9, 0.97]	w = 0.0288
Wald (delta)	0.94 [0.89, 0.96]	w = 0.0285
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.0284
ILR, REML	0.94 [0.9, 0.97]	w = 0.0288
Method's own interval	0.96 [0.92, 0.98]	w = 0.0358
Own interval, transcript fractions	0.96 [0.92, 0.98]	w = 0.0358
Fisher information (Wald)	0.045 [0.024, 0.083]	w = 0.000891
Godambe sandwich	1 [0.98, 1]	w = 0.65
Released code, as published	1 [0.98, 1]	w = 0.65

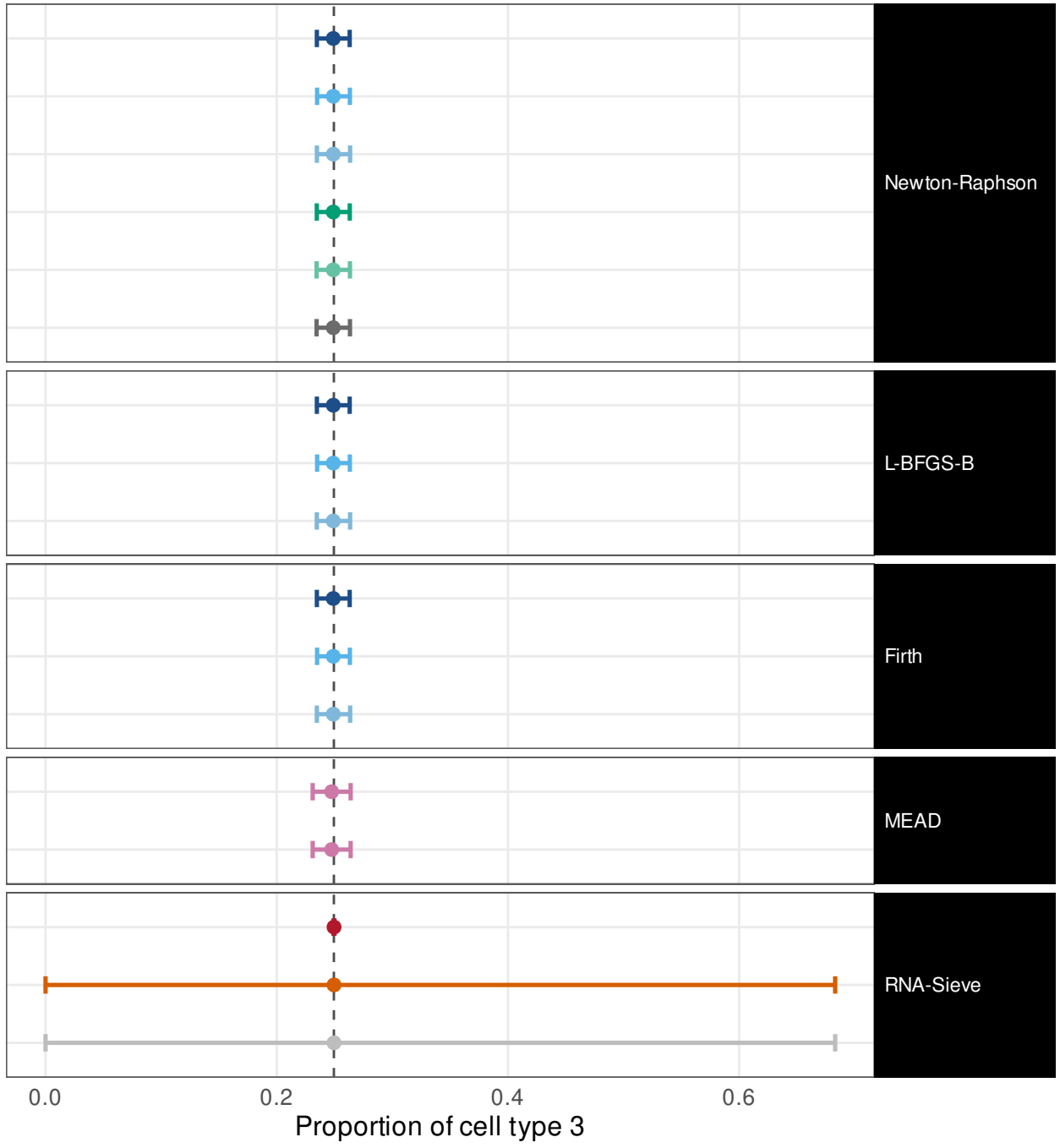


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / balanced / low sibling overlap / a = 1 / equal sizes

Wald (delta)	0.96 [0.92, 0.98]	w = 0.0284
ILR, mapped back	0.97 [0.94, 0.99]	w = 0.0284
ILR, REML	0.96 [0.93, 0.98]	w = 0.0287
Profile likelihood	0.96 [0.92, 0.98]	w = 0.0285
Profile, REML	0.96 [0.92, 0.98]	w = 0.0289
Oracle (Fisher at p*)	0.96 [0.92, 0.98]	w = 0.0289
Wald (delta)	0.96 [0.92, 0.98]	w = 0.0284
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.0283
ILR, REML	0.96 [0.93, 0.98]	w = 0.0288
Wald (delta)	0.96 [0.92, 0.98]	w = 0.0284
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.0284
ILR, REML	0.96 [0.93, 0.98]	w = 0.0288
Method's own interval	0.98 [0.94, 0.99]	w = 0.0328
Own interval, transcript fractions	0.98 [0.94, 0.99]	w = 0.0328
Fisher information (Wald)	0.03 [0.014, 0.064]	w = 0.000806
Godambe sandwich	1 [0.98, 1]	w = 0.683
Released code, as published	1 [0.98, 1]	w = 0.683

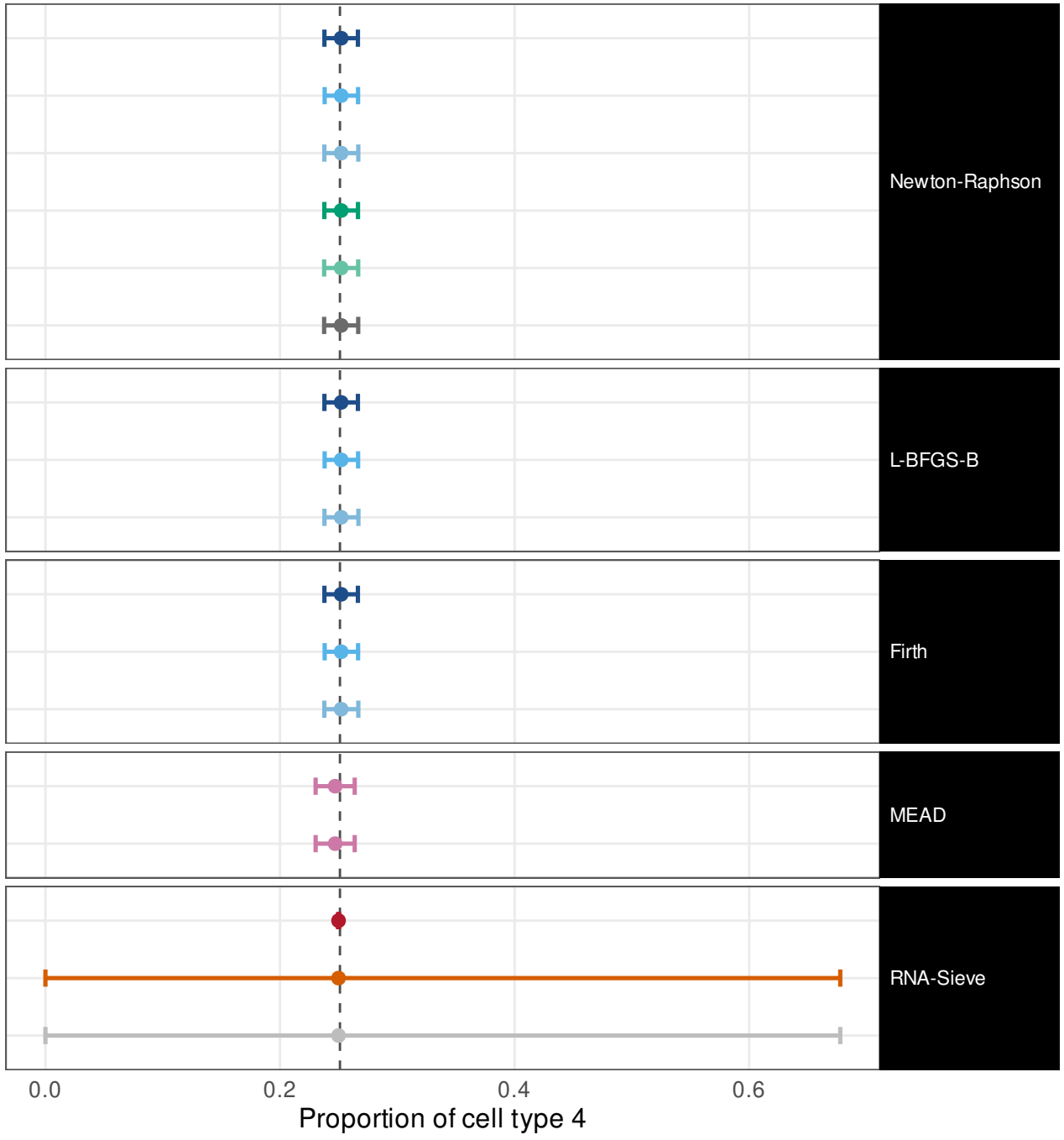


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / balanced / low sibling overlap / a = 1 / equal sizes

Wald (delta)	0.92 [0.87, 0.95]	w = 0.0285
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.0285
ILR, REML	0.92 [0.87, 0.95]	w = 0.0289
Profile likelihood	0.92 [0.87, 0.95]	w = 0.0287
Profile, REML	0.92 [0.87, 0.95]	w = 0.029
Oracle (Fisher at p*)	0.92 [0.87, 0.95]	w = 0.0291
Wald (delta)	0.92 [0.87, 0.95]	w = 0.0285
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.0285
ILR, REML	0.92 [0.87, 0.95]	w = 0.0289
Wald (delta)	0.92 [0.87, 0.95]	w = 0.0285
ILR, mapped back	0.91 [0.86, 0.94]	w = 0.0285
ILR, REML	0.92 [0.87, 0.95]	w = 0.029
Method's own interval	0.94 [0.9, 0.97]	w = 0.0333
Own interval, transcript fractions	0.94 [0.9, 0.97]	w = 0.0333
Fisher information (Wald)	0.04 [0.02, 0.077]	w = 0.000864
Godambe sandwich	1 [0.98, 1]	w = 0.678
Released code, as published	1 [0.98, 1]	w = 0.678

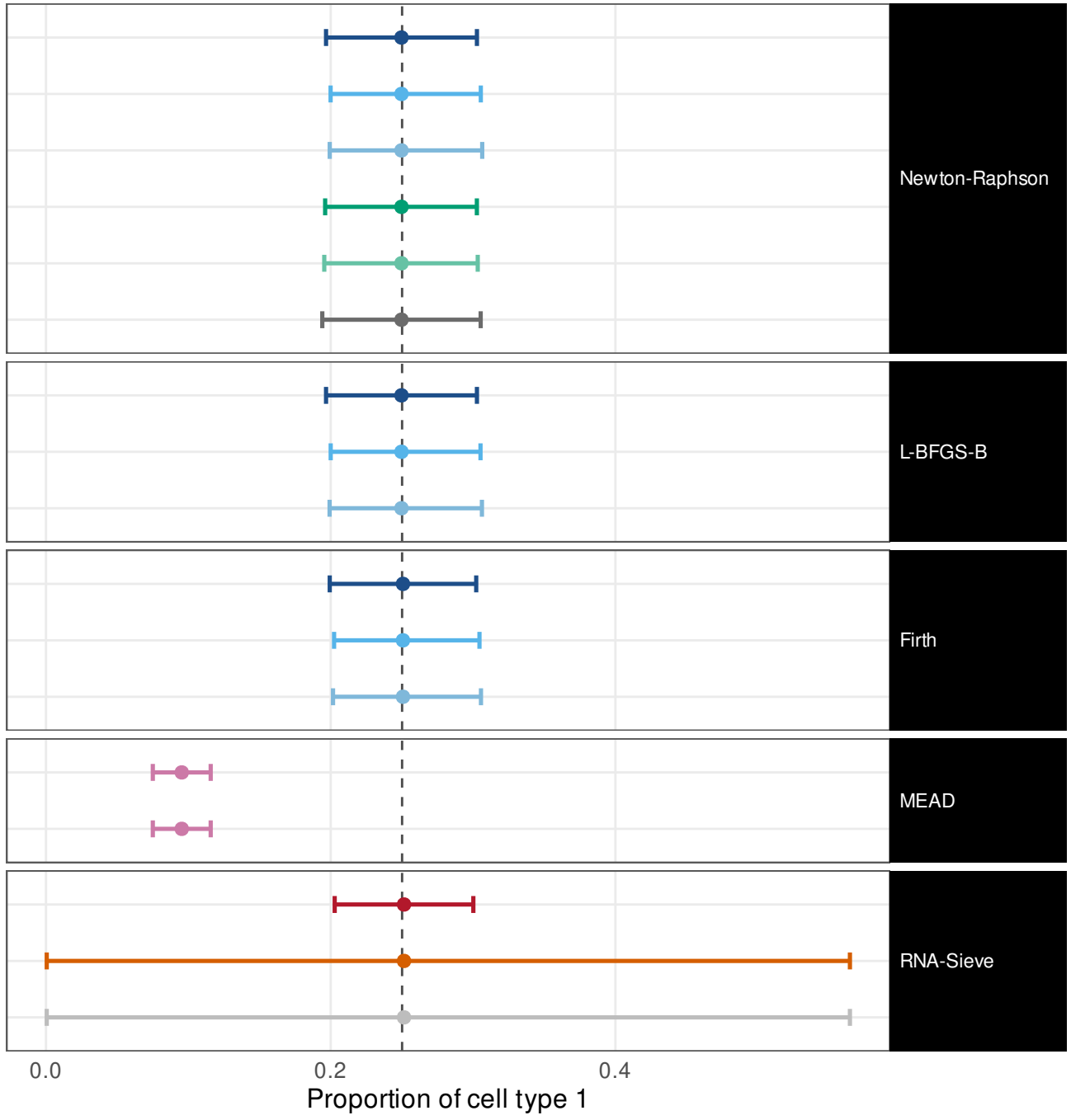


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / balanced / low sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.96 [0.92, 0.98]	w = 0.106
ILR, mapped back	0.96 [0.93, 0.98]	w = 0.106
ILR, REML	0.97 [0.94, 0.99]	w = 0.107
Profile likelihood	0.96 [0.92, 0.98]	w = 0.107
Profile, REML	0.96 [0.92, 0.98]	w = 0.108
Oracle (Fisher at p*)	0.96 [0.92, 0.98]	w = 0.111
Wald (delta)	0.96 [0.92, 0.98]	w = 0.106
ILR, mapped back	0.97 [0.94, 0.99]	w = 0.105
ILR, REML	0.96 [0.93, 0.98]	w = 0.107
Wald (delta)	0.96 [0.92, 0.98]	w = 0.103
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.102
ILR, REML	0.96 [0.93, 0.98]	w = 0.104
Method's own interval	0 [0, 0.019]	w = 0.0407
Own interval, transcript fractions	0.84 [0.78, 0.88]	w = 0.0407
Fisher information (Wald)	0.91 [0.86, 0.94]	w = 0.0974
Godambe sandwich	1 [0.98, 1]	w = 0.564
Released code, as published	1 [0.98, 1]	w = 0.564

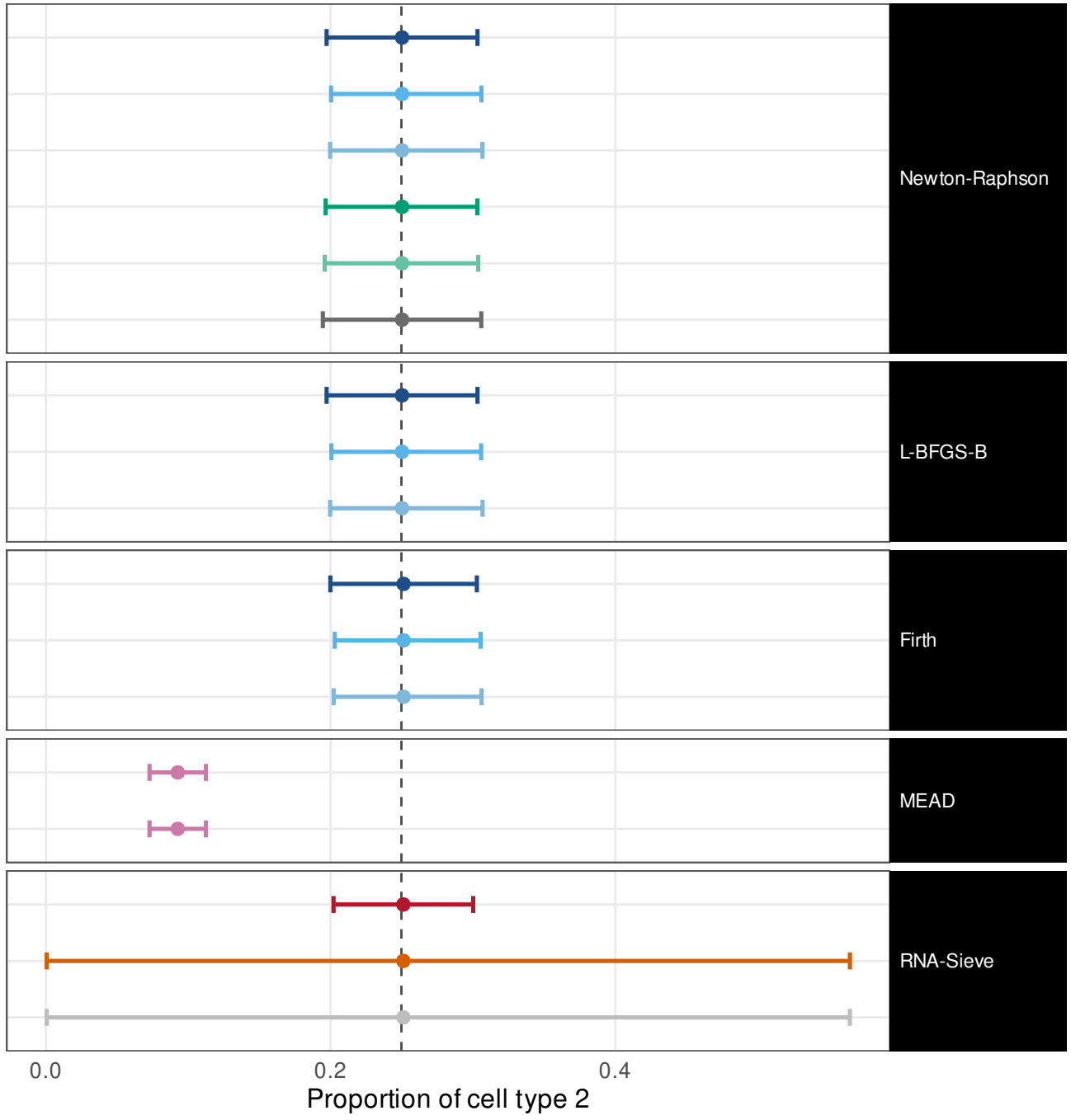


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / balanced / low sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.95 [0.91, 0.97]	w = 0.106
ILR, mapped back	0.96 [0.93, 0.98]	w = 0.106
ILR, REML	0.96 [0.93, 0.98]	w = 0.107
Profile likelihood	0.95 [0.91, 0.97]	w = 0.107
Profile, REML	0.95 [0.91, 0.97]	w = 0.108
Oracle (Fisher at p*)	0.96 [0.92, 0.98]	w = 0.111
Wald (delta)	0.95 [0.91, 0.97]	w = 0.106
ILR, mapped back	0.96 [0.93, 0.98]	w = 0.105
ILR, REML	0.96 [0.93, 0.98]	w = 0.107
Wald (delta)	0.96 [0.92, 0.98]	w = 0.103
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.102
ILR, REML	0.96 [0.93, 0.98]	w = 0.104
Method's own interval	0 [0, 0.019]	w = 0.0396
Own interval, transcript fractions	0.85 [0.79, 0.89]	w = 0.0396
Fisher information (Wald)	0.92 [0.88, 0.95]	w = 0.0981
Godambe sandwich	1 [0.98, 1]	w = 0.564
Released code, as published	1 [0.98, 1]	w = 0.564

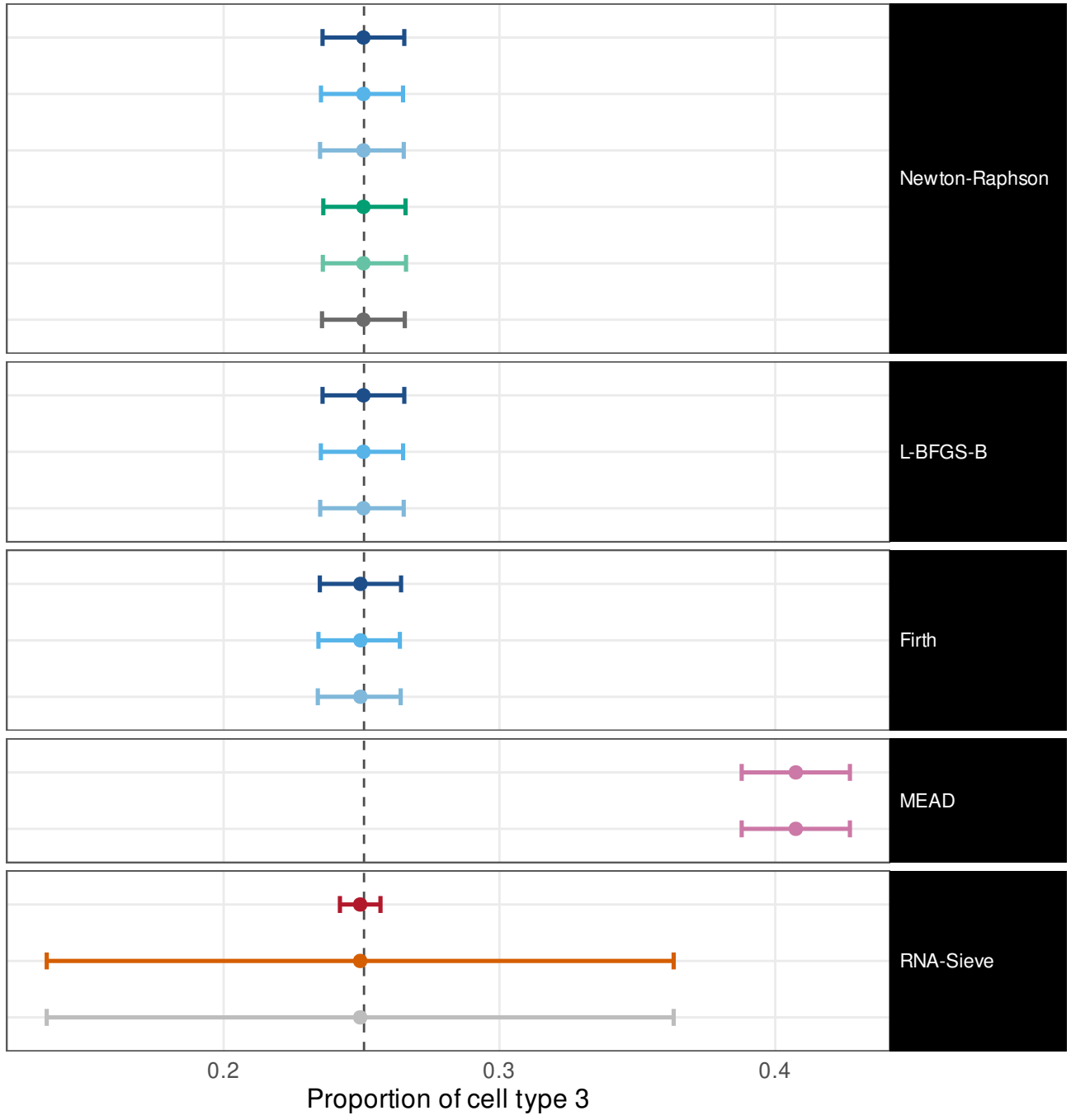


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / balanced / low sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.96 [0.92, 0.98]	w = 0.0296
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.0298
ILR, REML	0.96 [0.92, 0.98]	w = 0.0303
Profile likelihood	0.96 [0.92, 0.98]	w = 0.0298
Profile, REML	0.96 [0.92, 0.98]	w = 0.0302
Oracle (Fisher at p*)	0.96 [0.93, 0.98]	w = 0.03
Wald (delta)	0.96 [0.92, 0.98]	w = 0.0296
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.0298
ILR, REML	0.96 [0.92, 0.98]	w = 0.0303
Wald (delta)	0.94 [0.9, 0.97]	w = 0.0295
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.0295
ILR, REML	0.94 [0.9, 0.97]	w = 0.03
Method's own interval	0 [0, 0.019]	w = 0.0392
Own interval, transcript fractions	0.82 [0.76, 0.87]	w = 0.0392
Fisher information (Wald)	0.6 [0.53, 0.66]	w = 0.0147
Godambe sandwich	1 [0.98, 1]	w = 0.227
Released code, as published	1 [0.98, 1]	w = 0.227

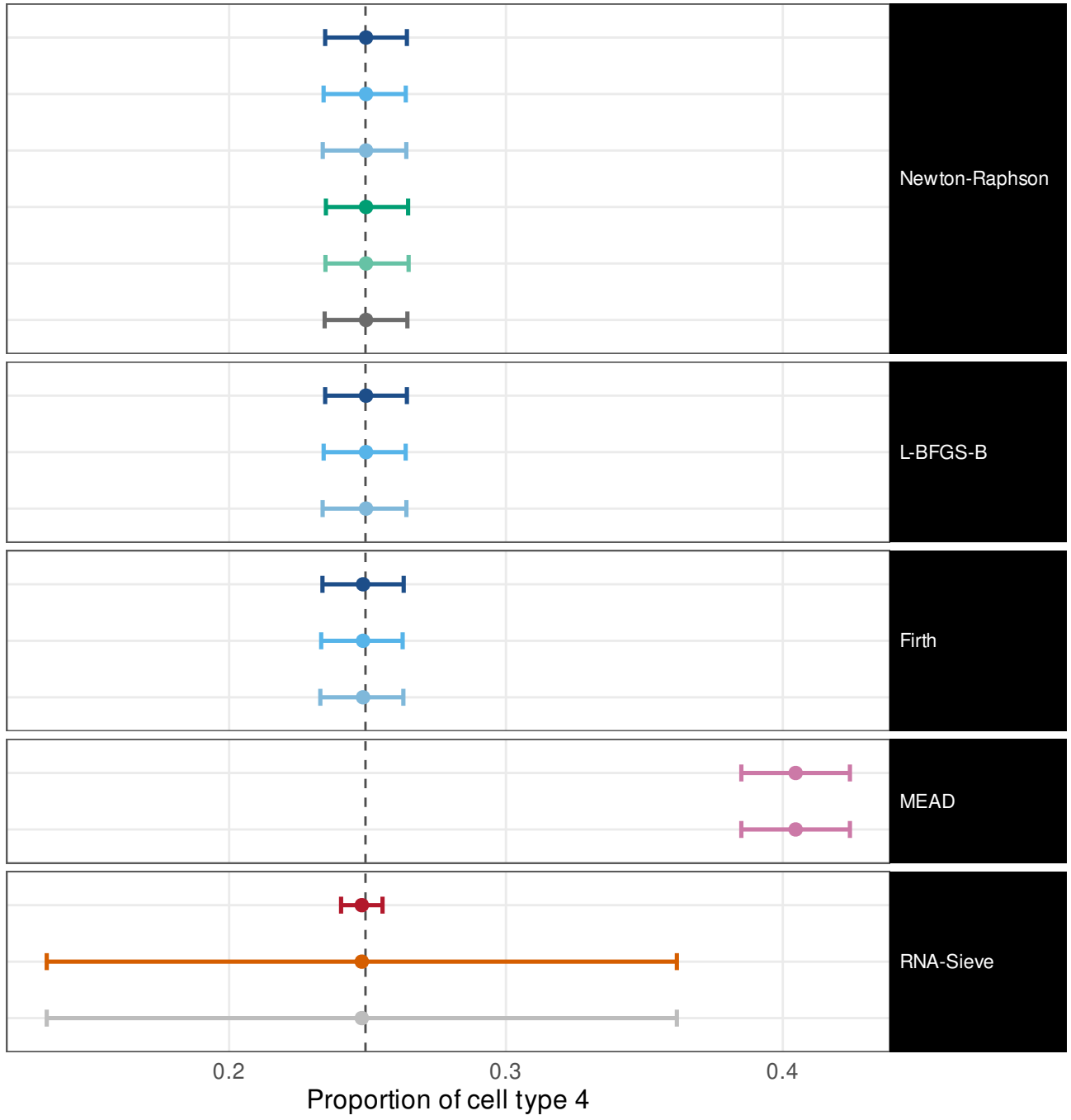


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / balanced / low sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.98 [0.94, 0.99]	w = 0.0295
ILR, mapped back	0.98 [0.94, 0.99]	w = 0.0297
ILR, REML	0.98 [0.94, 0.99]	w = 0.0302
Profile likelihood	0.97 [0.94, 0.99]	w = 0.0297
Profile, REML	0.98 [0.95, 0.99]	w = 0.03
Oracle (Fisher at p*)	0.99 [0.96, 1]	w = 0.0299
Wald (delta)	0.98 [0.94, 0.99]	w = 0.0295
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.0296
ILR, REML	0.97 [0.94, 0.99]	w = 0.0303
Wald (delta)	0.96 [0.93, 0.98]	w = 0.0293
ILR, mapped back	0.97 [0.94, 0.99]	w = 0.0294
ILR, REML	0.96 [0.93, 0.98]	w = 0.03
Method's own interval	0 [0, 0.019]	w = 0.0392
Own interval, transcript fractions	0.82 [0.77, 0.87]	w = 0.0392
Fisher information (Wald)	0.65 [0.58, 0.71]	w = 0.015
Godambe sandwich	1 [0.98, 1]	w = 0.228
Released code, as published	1 [0.98, 1]	w = 0.228

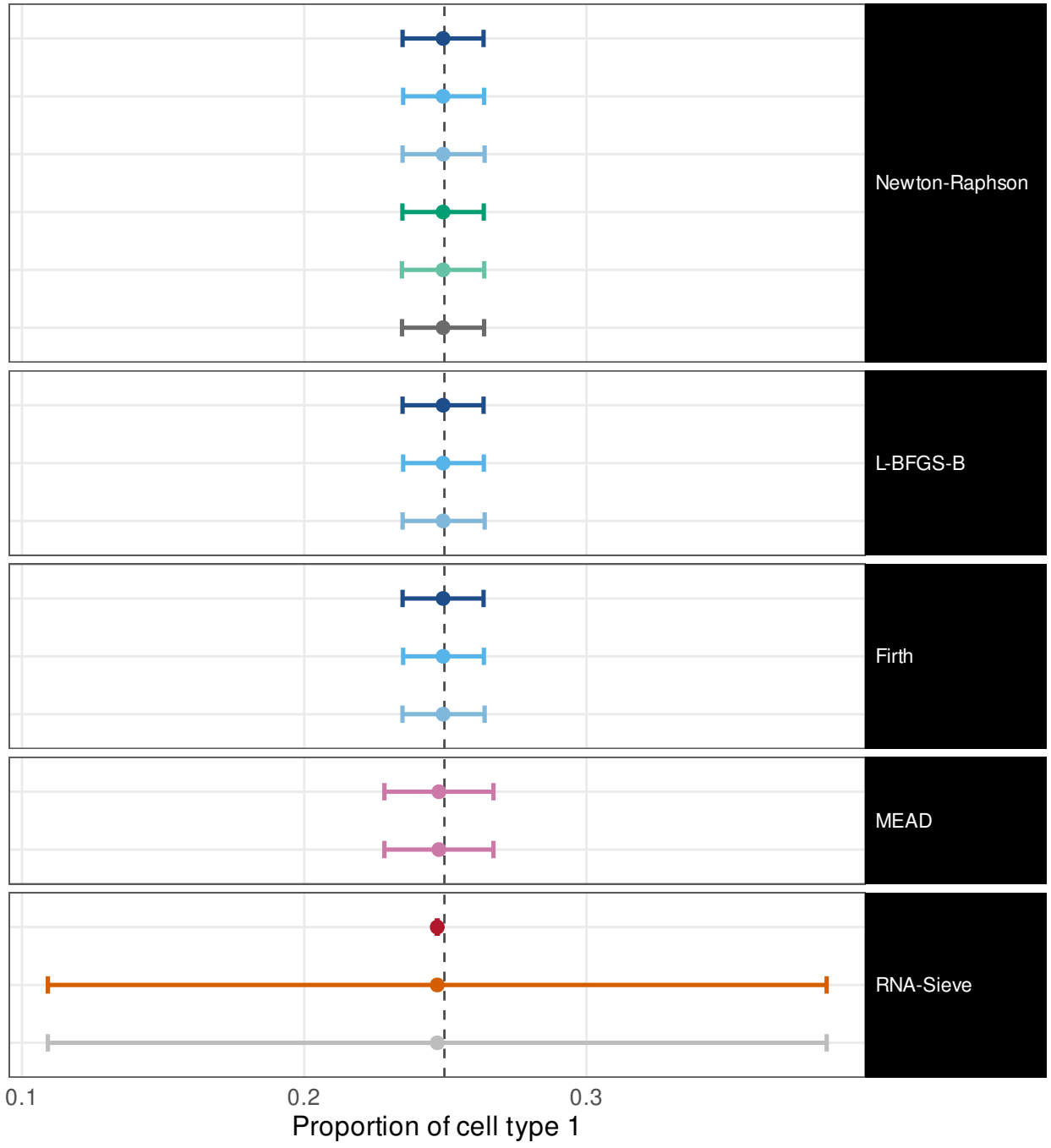


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / balanced / low sibling overlap / a = 5 / equal sizes

Wald (delta)	0.98 [0.94, 0.99]	w = 0.0287
ILR, mapped back	0.98 [0.94, 0.99]	w = 0.0287
ILR, REML	0.98 [0.95, 0.99]	w = 0.0291
Profile likelihood	0.98 [0.95, 0.99]	w = 0.0288
Profile, REML	0.98 [0.95, 0.99]	w = 0.0292
Oracle (Fisher at p*)	0.98 [0.94, 0.99]	w = 0.0291
Wald (delta)	0.98 [0.94, 0.99]	w = 0.0287
ILR, mapped back	0.98 [0.95, 0.99]	w = 0.0286
ILR, REML	0.98 [0.94, 0.99]	w = 0.0291
Wald (delta)	0.98 [0.95, 0.99]	w = 0.0287
ILR, mapped back	0.98 [0.94, 0.99]	w = 0.0286
ILR, REML	0.98 [0.95, 0.99]	w = 0.0291
Method's own interval	0.99 [0.96, 1]	w = 0.0386
Own interval, transcript fractions	0.99 [0.96, 1]	w = 0.0386
Fisher information (Wald)	0.015 [0.0051, 0.043]	w = 0.00036
Godambe sandwich	1 [0.98, 1]	w = 0.276
Released code, as published	1 [0.98, 1]	w = 0.276

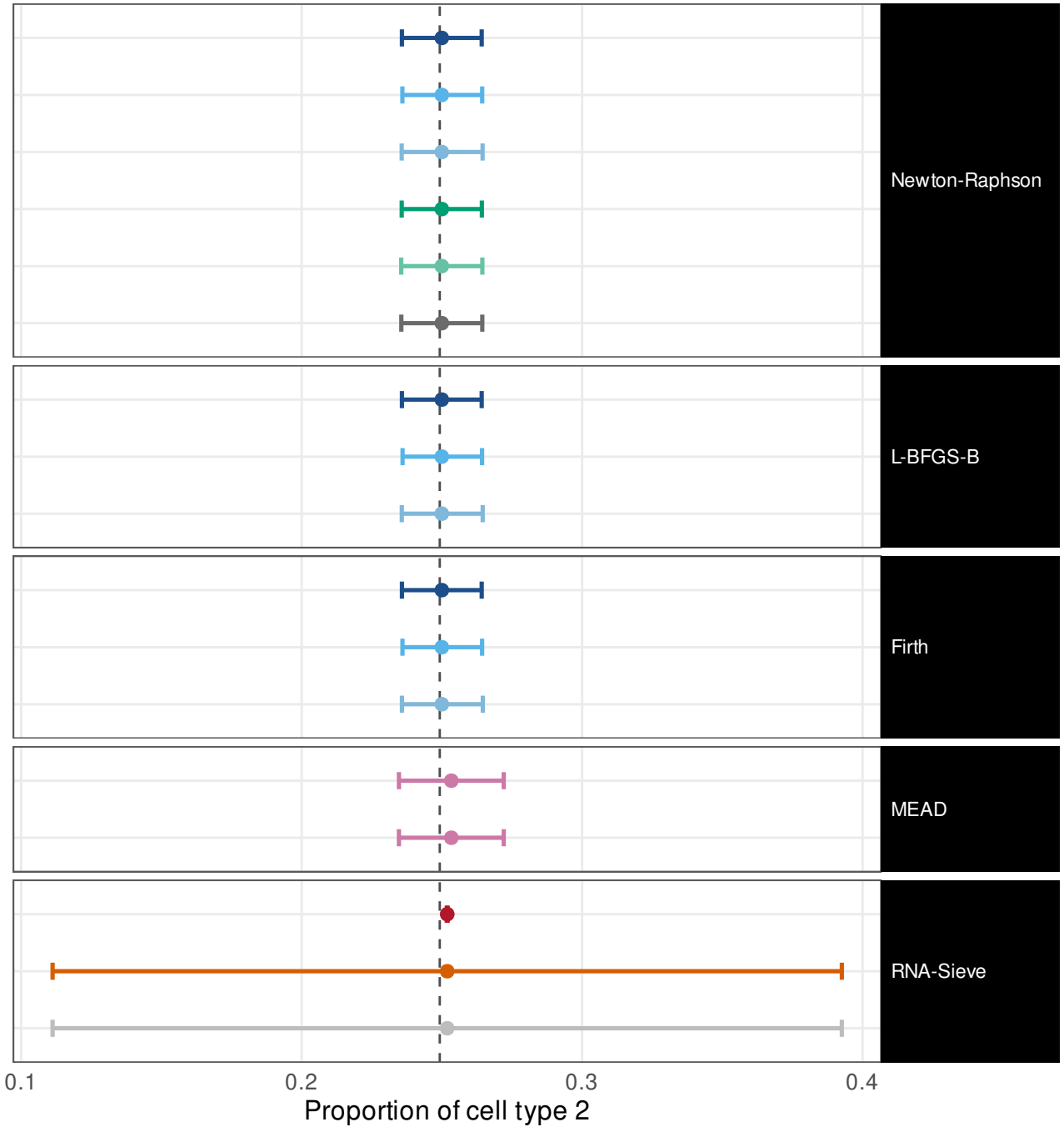


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / balanced / low sibling overlap / a = 5 / equal sizes

Wald (delta)	0.96 [0.93, 0.98]	w = 0.0284
ILR, mapped back	0.96 [0.93, 0.98]	w = 0.0285
ILR, REML	0.97 [0.94, 0.99]	w = 0.0288
Profile likelihood	0.96 [0.93, 0.98]	w = 0.0286
Profile, REML	0.96 [0.93, 0.98]	w = 0.0289
Oracle (Fisher at p*)	0.96 [0.93, 0.98]	w = 0.0289
Wald (delta)	0.96 [0.93, 0.98]	w = 0.0284
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.0284
ILR, REML	0.96 [0.93, 0.98]	w = 0.0288
Wald (delta)	0.96 [0.93, 0.98]	w = 0.0284
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.0284
ILR, REML	0.96 [0.93, 0.98]	w = 0.0288
Method's own interval	0.97 [0.94, 0.99]	w = 0.0374
Own interval, transcript fractions	0.97 [0.94, 0.99]	w = 0.0374
Fisher information (Wald)	0 [0, 0.019]	w = 0.000302
Godambe sandwich	1 [0.98, 1]	w = 0.281
Released code, as published	1 [0.98, 1]	w = 0.281

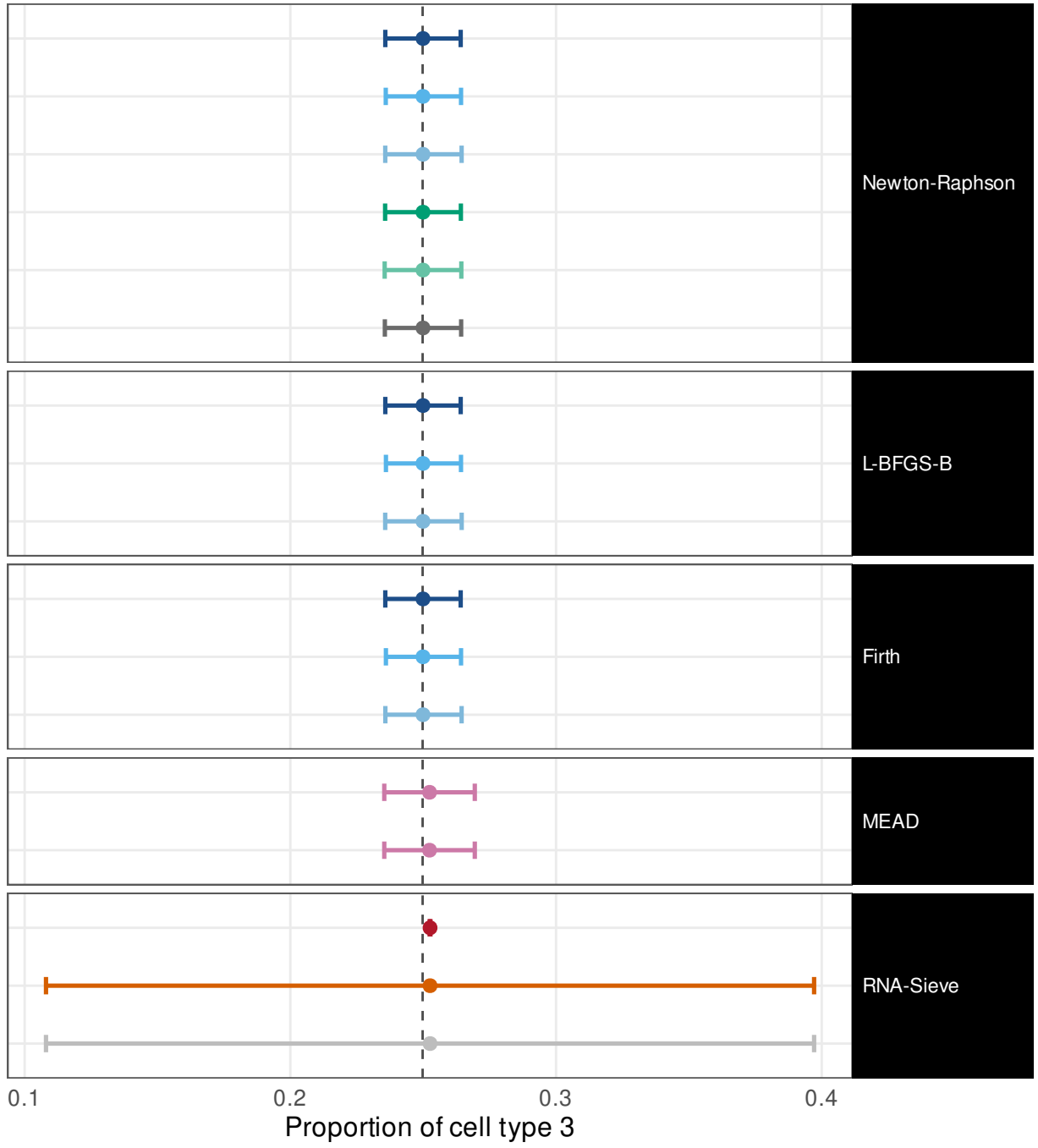


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / balanced / low sibling overlap / a = 5 / equal sizes

Wald (delta)	0.96 [0.93, 0.98]	w = 0.0284
ILR, mapped back	0.97 [0.94, 0.99]	w = 0.0283
ILR, REML	0.97 [0.94, 0.99]	w = 0.0287
Profile likelihood	0.97 [0.94, 0.99]	w = 0.0285
Profile, REML	0.97 [0.94, 0.99]	w = 0.0288
Oracle (Fisher at p*)	0.97 [0.94, 0.99]	w = 0.0287
Wald (delta)	0.96 [0.93, 0.98]	w = 0.0284
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.0283
ILR, REML	0.96 [0.92, 0.98]	w = 0.0288
Wald (delta)	0.96 [0.93, 0.98]	w = 0.0284
ILR, mapped back	0.97 [0.94, 0.99]	w = 0.0283
ILR, REML	0.96 [0.93, 0.98]	w = 0.0287
Method's own interval	0.96 [0.93, 0.98]	w = 0.0341
Own interval, transcript fractions	0.96 [0.93, 0.98]	w = 0.0341
Fisher information (Wald)	0.015 [0.0051, 0.043]	w = 0.000292
Godambe sandwich	1 [0.98, 1]	w = 0.289
Released code, as published	1 [0.98, 1]	w = 0.289

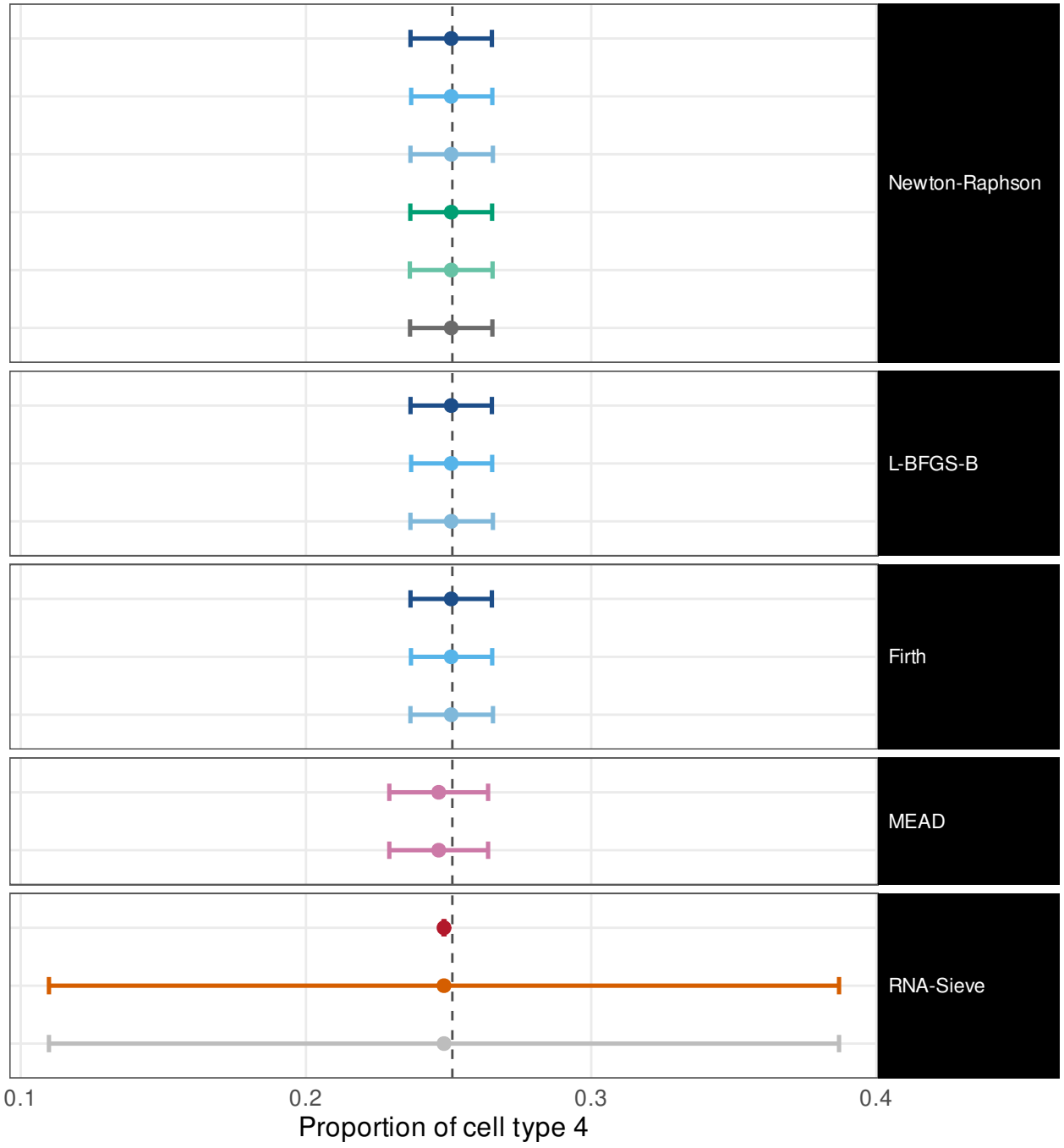


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / balanced / low sibling overlap / a = 5 / equal sizes

Wald (delta)	0.96 [0.93, 0.98]	w = 0.0285
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.0285
ILR, REML	0.96 [0.92, 0.98]	w = 0.0289
Profile likelihood	0.96 [0.93, 0.98]	w = 0.0287
Profile, REML	0.96 [0.93, 0.98]	w = 0.029
Oracle (Fisher at p*)	0.98 [0.94, 0.99]	w = 0.0289
Wald (delta)	0.96 [0.93, 0.98]	w = 0.0285
ILR, mapped back	0.96 [0.93, 0.98]	w = 0.0284
ILR, REML	0.96 [0.93, 0.98]	w = 0.0289
Wald (delta)	0.96 [0.93, 0.98]	w = 0.0285
ILR, mapped back	0.96 [0.93, 0.98]	w = 0.0285
ILR, REML	0.97 [0.94, 0.99]	w = 0.0289
Method's own interval	0.94 [0.9, 0.97]	w = 0.0346
Own interval, transcript fractions	0.94 [0.9, 0.97]	w = 0.0346
Fisher information (Wald)	0.02 [0.0078, 0.05]	w = 0.000357
Godambe sandwich	1 [0.98, 1]	w = 0.277
Released code, as published	1 [0.98, 1]	w = 0.277

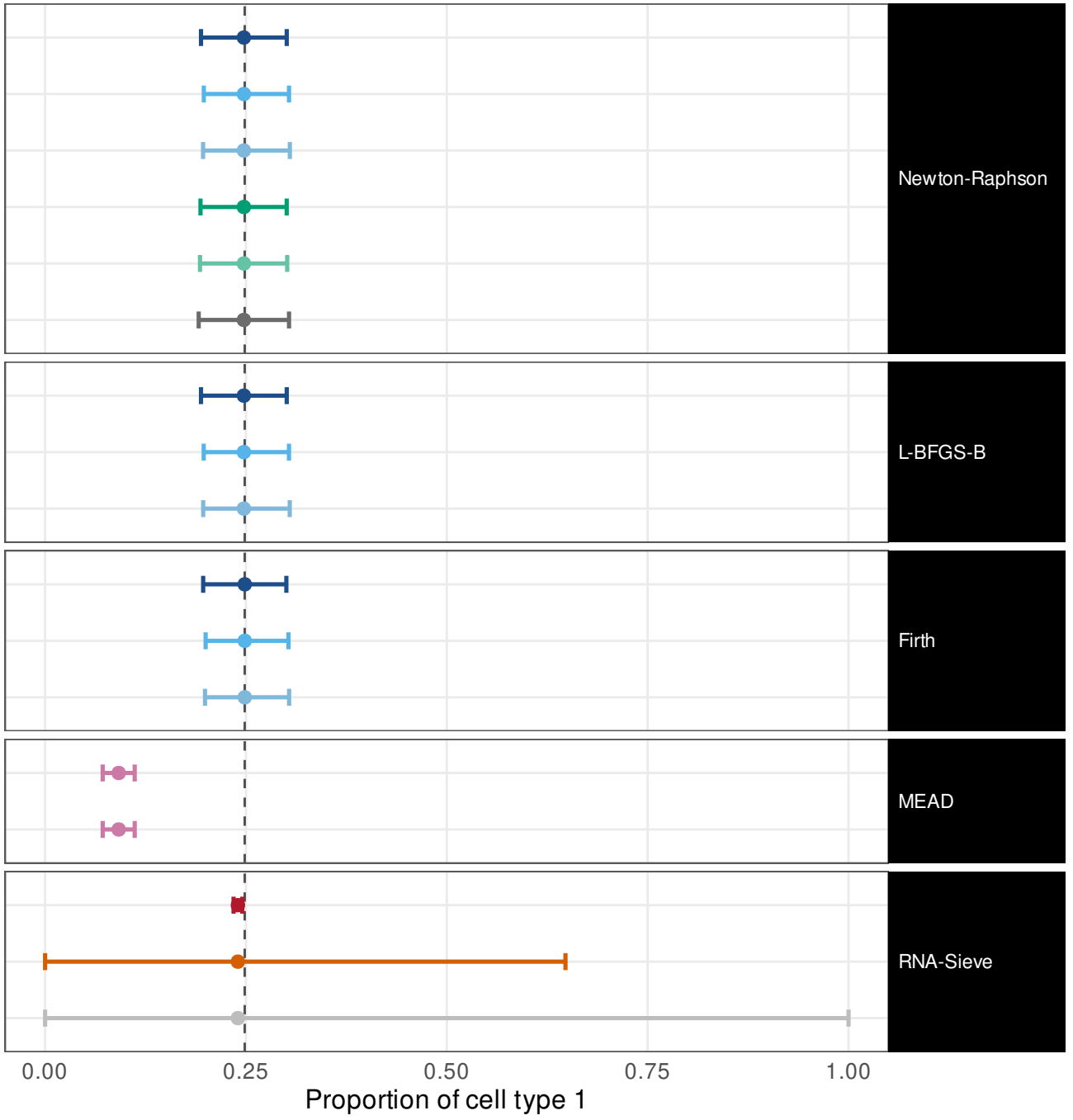


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / balanced / low sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.96 [0.93, 0.98]	w = 0.107
ILR, mapped back	0.97 [0.94, 0.99]	w = 0.106
ILR, REML	0.98 [0.94, 0.99]	w = 0.108
Profile likelihood	0.96 [0.93, 0.98]	w = 0.107
Profile, REML	0.96 [0.93, 0.98]	w = 0.109
Oracle (Fisher at p*)	0.98 [0.95, 0.99]	w = 0.112
Wald (delta)	0.96 [0.93, 0.98]	w = 0.107
ILR, mapped back	0.97 [0.94, 0.99]	w = 0.106
ILR, REML	0.97 [0.94, 0.99]	w = 0.108
Wald (delta)	0.96 [0.93, 0.98]	w = 0.104
ILR, mapped back	0.97 [0.94, 0.99]	w = 0.103
ILR, REML	0.98 [0.94, 0.99]	w = 0.105
Method's own interval	0 [0, 0.019]	w = 0.0398
Own interval, transcript fractions	0.86 [0.81, 0.9]	w = 0.0398
Fisher information (Wald)	0.12 [0.078, 0.17]	w = 0.0108
Godambe sandwich	1 [0.98, 1]	w = 0.648
Released code, as published	1 [0.98, 1]	w = 1

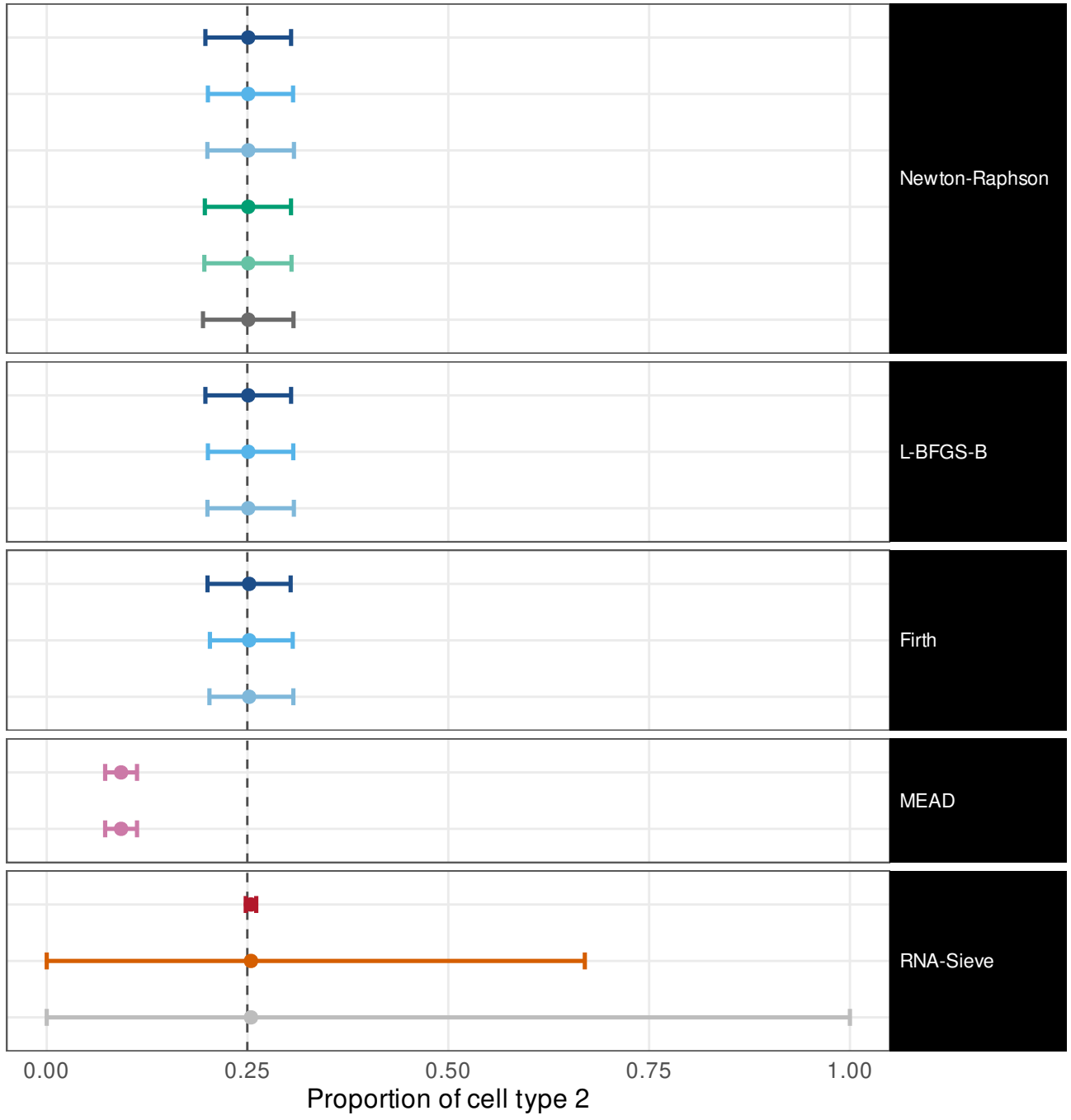


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / balanced / low sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.94 [0.9, 0.97]	w = 0.107
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.106
ILR, REML	0.94 [0.9, 0.97]	w = 0.108
Profile likelihood	0.96 [0.92, 0.98]	w = 0.107
Profile, REML	0.96 [0.92, 0.98]	w = 0.109
Oracle (Fisher at p*)	0.96 [0.92, 0.98]	w = 0.113
Wald (delta)	0.94 [0.9, 0.97]	w = 0.107
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.106
ILR, REML	0.94 [0.9, 0.97]	w = 0.108
Wald (delta)	0.94 [0.89, 0.96]	w = 0.104
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.103
ILR, REML	0.95 [0.91, 0.97]	w = 0.104
Method's own interval	0 [0, 0.019]	w = 0.0397
Own interval, transcript fractions	0.87 [0.82, 0.91]	w = 0.0397
Fisher information (Wald)	0.18 [0.13, 0.24]	w = 0.0131
Godambe sandwich	1 [0.98, 1]	w = 0.67
Released code, as published	1 [0.98, 1]	w = 1

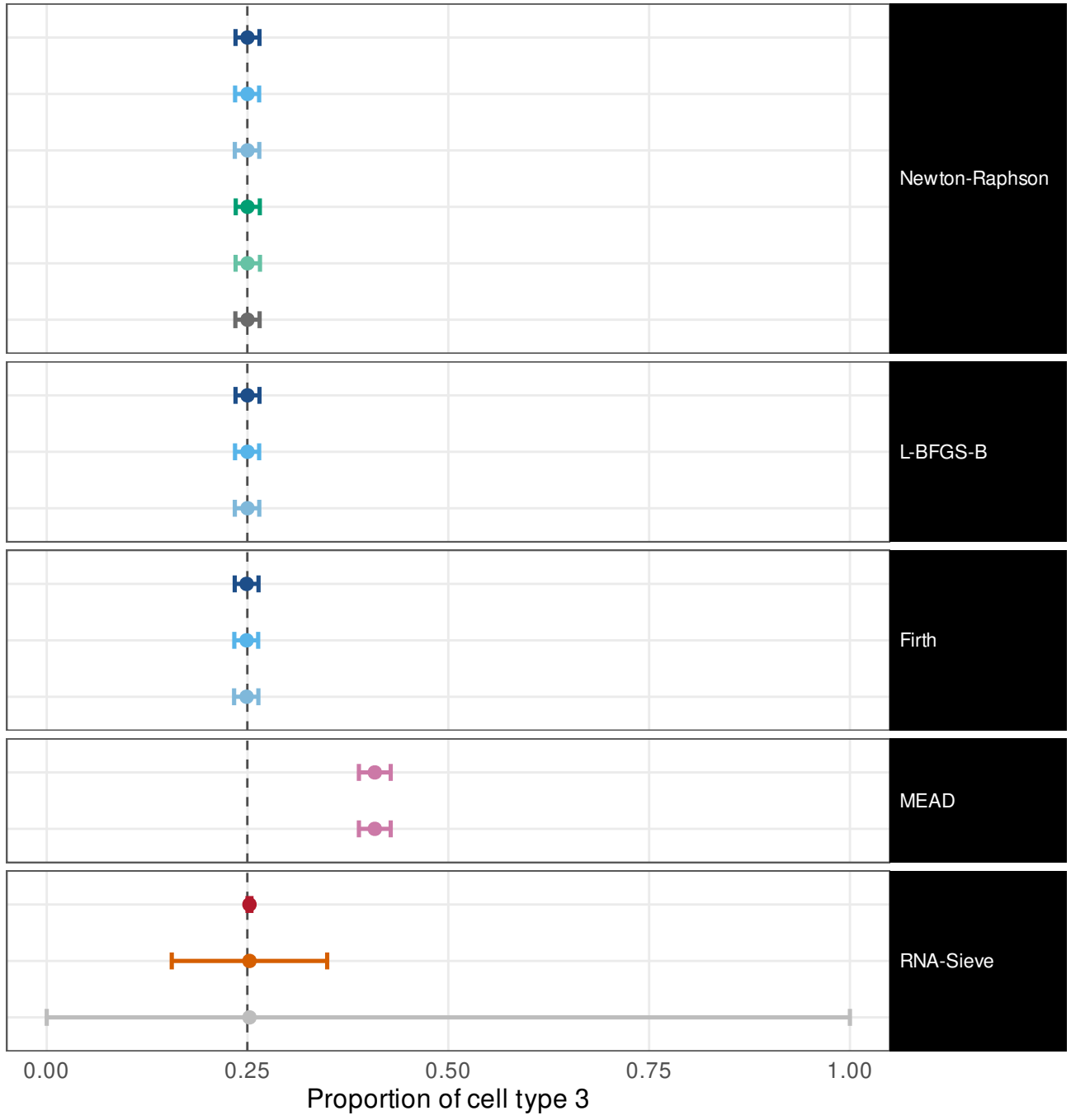


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / balanced / low sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.96 [0.92, 0.98]	w = 0.0298
ILR, mapped back	0.96 [0.93, 0.98]	w = 0.0299
ILR, REML	0.96 [0.92, 0.98]	w = 0.0304
Profile likelihood	0.96 [0.92, 0.98]	w = 0.03
Profile, REML	0.96 [0.92, 0.98]	w = 0.0303
Oracle (Fisher at p*)	0.96 [0.92, 0.98]	w = 0.0302
Wald (delta)	0.96 [0.92, 0.98]	w = 0.0298
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.03
ILR, REML	0.96 [0.93, 0.98]	w = 0.0304
Wald (delta)	0.96 [0.92, 0.98]	w = 0.0296
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.0297
ILR, REML	0.95 [0.91, 0.97]	w = 0.0302
Method's own interval	0 [0, 0.019]	w = 0.0396
Own interval, transcript fractions	0.9 [0.85, 0.93]	w = 0.0396
Fisher information (Wald)	0.22 [0.16, 0.28]	w = 0.00484
Godambe sandwich	1 [0.98, 1]	w = 0.194
Released code, as published	1 [0.98, 1]	w = 1

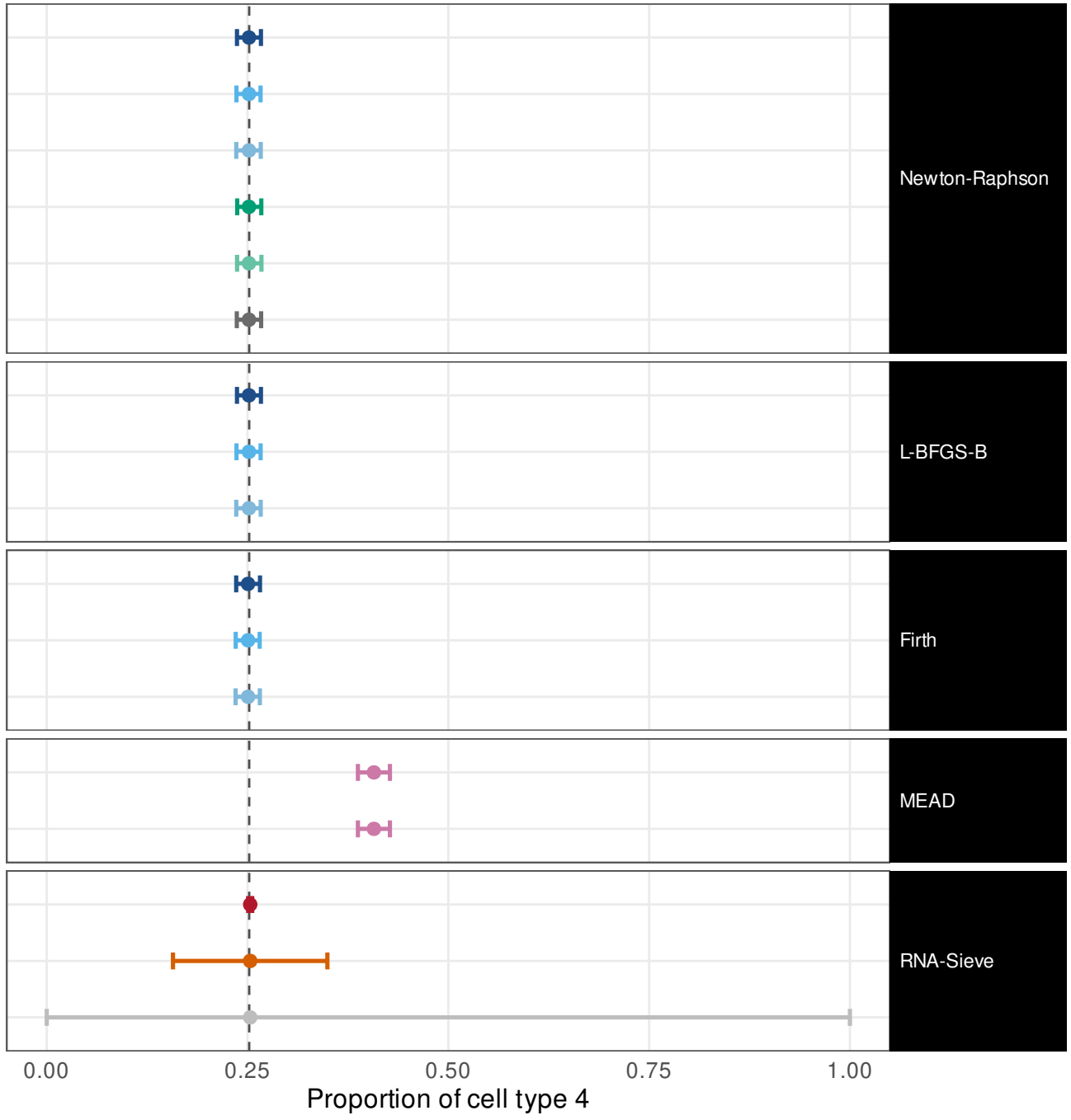


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / balanced / low sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.94 [0.9, 0.97]	w = 0.0298
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.03
ILR, REML	0.94 [0.9, 0.97]	w = 0.0305
Profile likelihood	0.94 [0.9, 0.97]	w = 0.03
Profile, REML	0.94 [0.9, 0.97]	w = 0.0303
Oracle (Fisher at p*)	0.95 [0.91, 0.97]	w = 0.0303
Wald (delta)	0.94 [0.9, 0.97]	w = 0.0298
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.03
ILR, REML	0.93 [0.89, 0.96]	w = 0.0305
Wald (delta)	0.94 [0.89, 0.96]	w = 0.0296
ILR, mapped back	0.92 [0.88, 0.95]	w = 0.0298
ILR, REML	0.94 [0.89, 0.96]	w = 0.0302
Method's own interval	0 [0, 0.019]	w = 0.0398
Own interval, transcript fractions	0.79 [0.73, 0.84]	w = 0.0398
Fisher information (Wald)	0.24 [0.19, 0.31]	w = 0.00498
Godambe sandwich	1 [0.98, 1]	w = 0.192
Released code, as published	1 [0.98, 1]	w = 1

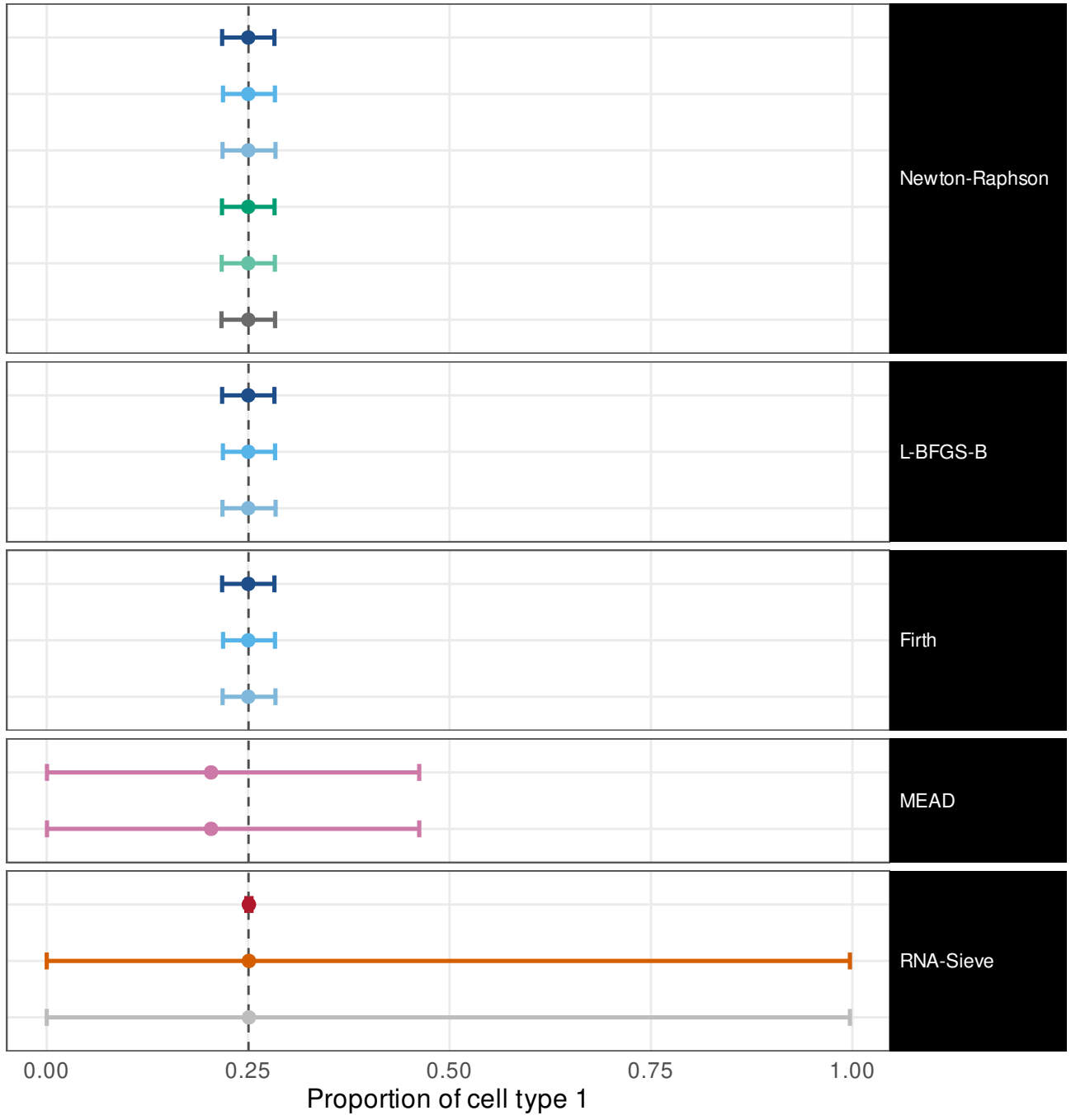


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / balanced / high sibling overlap / a = 1 / equal sizes

Wald (delta)	0.96 [0.92, 0.98]	w = 0.0649
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.0645
ILR, REML	0.96 [0.92, 0.98]	w = 0.0657
Profile likelihood	0.96 [0.92, 0.98]	w = 0.0653
Profile, REML	0.96 [0.92, 0.98]	w = 0.066
Oracle (Fisher at p*)	0.95 [0.91, 0.97]	w = 0.0665
Wald (delta)	0.96 [0.92, 0.98]	w = 0.0649
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.0648
ILR, REML	0.96 [0.92, 0.98]	w = 0.0659
Wald (delta)	0.96 [0.92, 0.98]	w = 0.0649
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.0645
ILR, REML	0.96 [0.92, 0.98]	w = 0.0656
Method's own interval	1 [0.98, 1]	w = 0.462
Own interval, transcript fractions	1 [0.98, 1]	w = 0.462
Fisher information (Wald)	0.16 [0.12, 0.22]	w = 0.00642
Godambe sandwich	1 [0.98, 1]	w = 0.997
Released code, as published	1 [0.98, 1]	w = 0.997

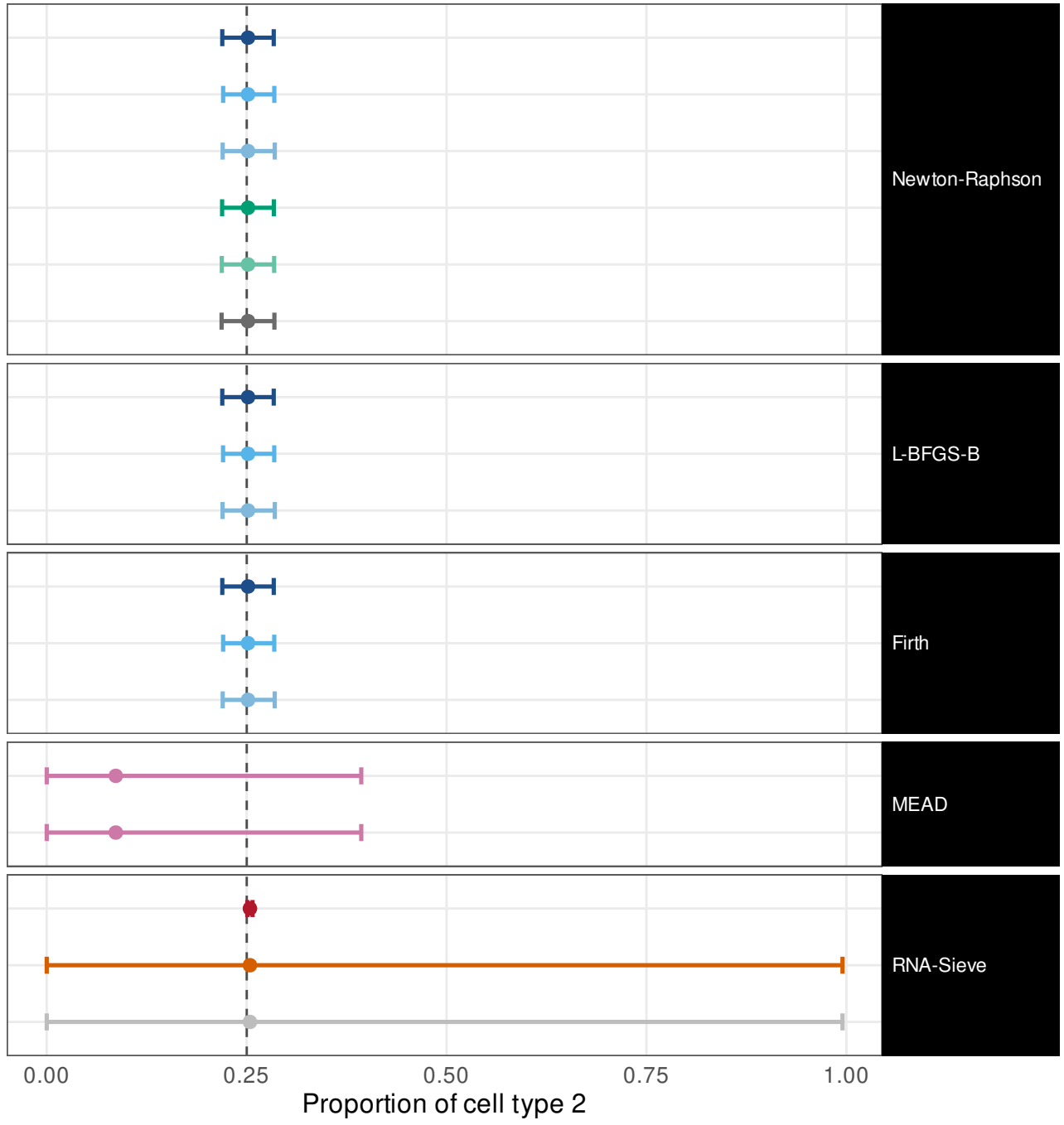


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / balanced / high sibling overlap / a = 1 / equal sizes

Wald (delta)	0.94 [0.9, 0.97]	w = 0.0644
ILR, mapped back	0.94 [0.89, 0.96]	w = 0.064
ILR, REML	0.94 [0.9, 0.97]	w = 0.0652
Profile likelihood	0.94 [0.9, 0.97]	w = 0.0647
Profile, REML	0.94 [0.9, 0.97]	w = 0.0654
Oracle (Fisher at p*)	0.96 [0.93, 0.98]	w = 0.066
Wald (delta)	0.94 [0.9, 0.97]	w = 0.0644
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.064
ILR, REML	0.94 [0.9, 0.97]	w = 0.0653
Wald (delta)	0.94 [0.9, 0.97]	w = 0.0643
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.0639
ILR, REML	0.96 [0.92, 0.98]	w = 0.0651
Method's own interval	1 [0.98, 1]	w = 0.393
Own interval, transcript fractions	1 [0.98, 1]	w = 0.393
Fisher information (Wald)	0.14 [0.099, 0.19]	w = 0.00651
Godambe sandwich	1 [0.98, 1]	w = 0.995
Released code, as published	1 [0.98, 1]	w = 0.995

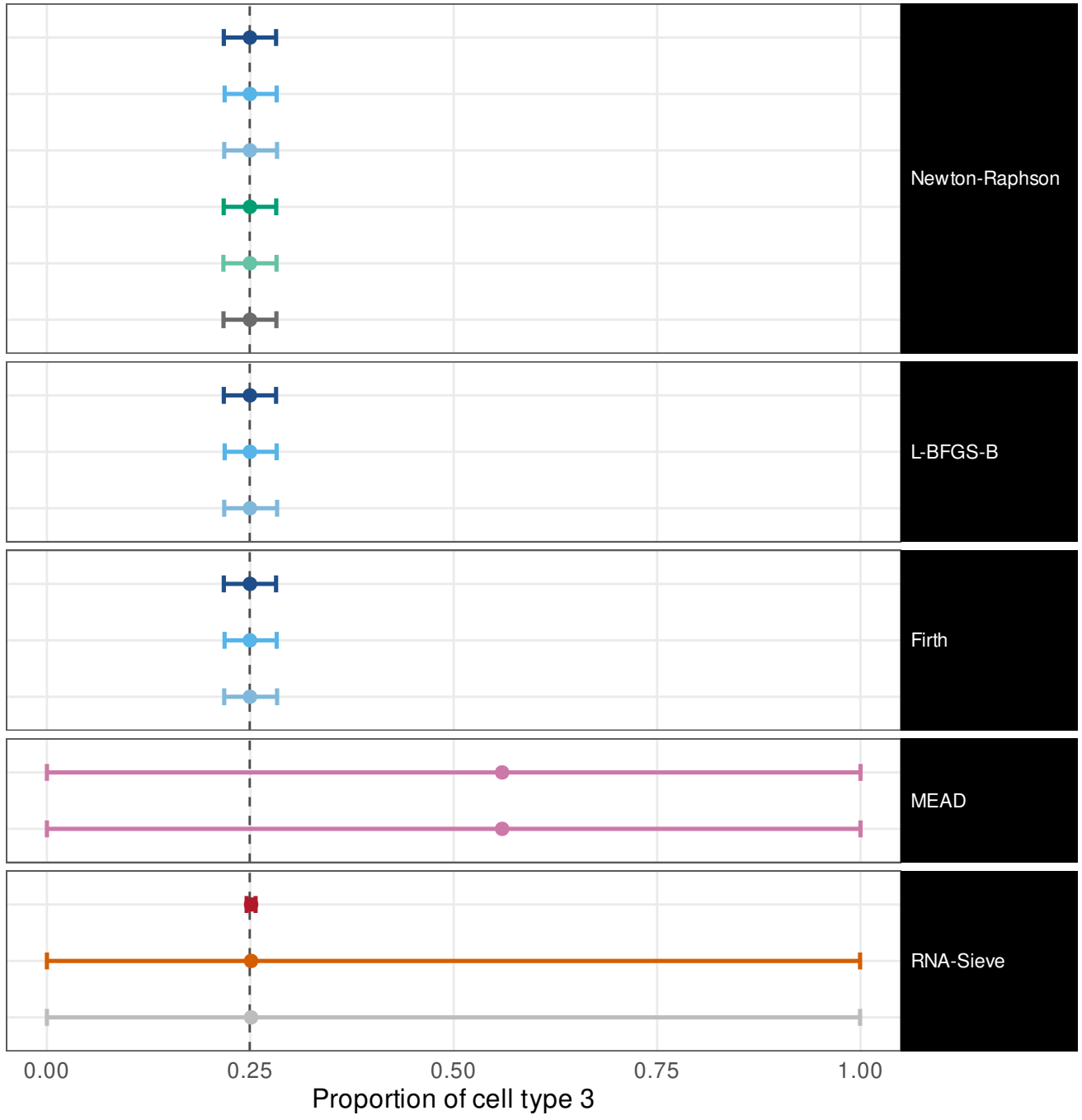


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / balanced / high sibling overlap / a = 1 / equal sizes

Wald (delta)	0.93 [0.89, 0.96]	w = 0.0642
ILR, mapped back	0.92 [0.88, 0.95]	w = 0.0639
ILR, REML	0.94 [0.89, 0.96]	w = 0.0649
Profile likelihood	0.93 [0.89, 0.96]	w = 0.0646
Profile, REML	0.94 [0.89, 0.96]	w = 0.0653
Oracle (Fisher at p*)	0.94 [0.89, 0.96]	w = 0.065
Wald (delta)	0.93 [0.89, 0.96]	w = 0.0642
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.0639
ILR, REML	0.94 [0.9, 0.97]	w = 0.065
Wald (delta)	0.93 [0.89, 0.96]	w = 0.0642
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.0641
ILR, REML	0.94 [0.9, 0.97]	w = 0.065
Method's own interval	1 [0.98, 1]	w = 1
Own interval, transcript fractions	1 [0.98, 1]	w = 1
Fisher information (Wald)	0.24 [0.19, 0.3]	w = 0.0109
Godambe sandwich	1 [0.98, 1]	w = 0.999
Released code, as published	1 [0.98, 1]	w = 0.999

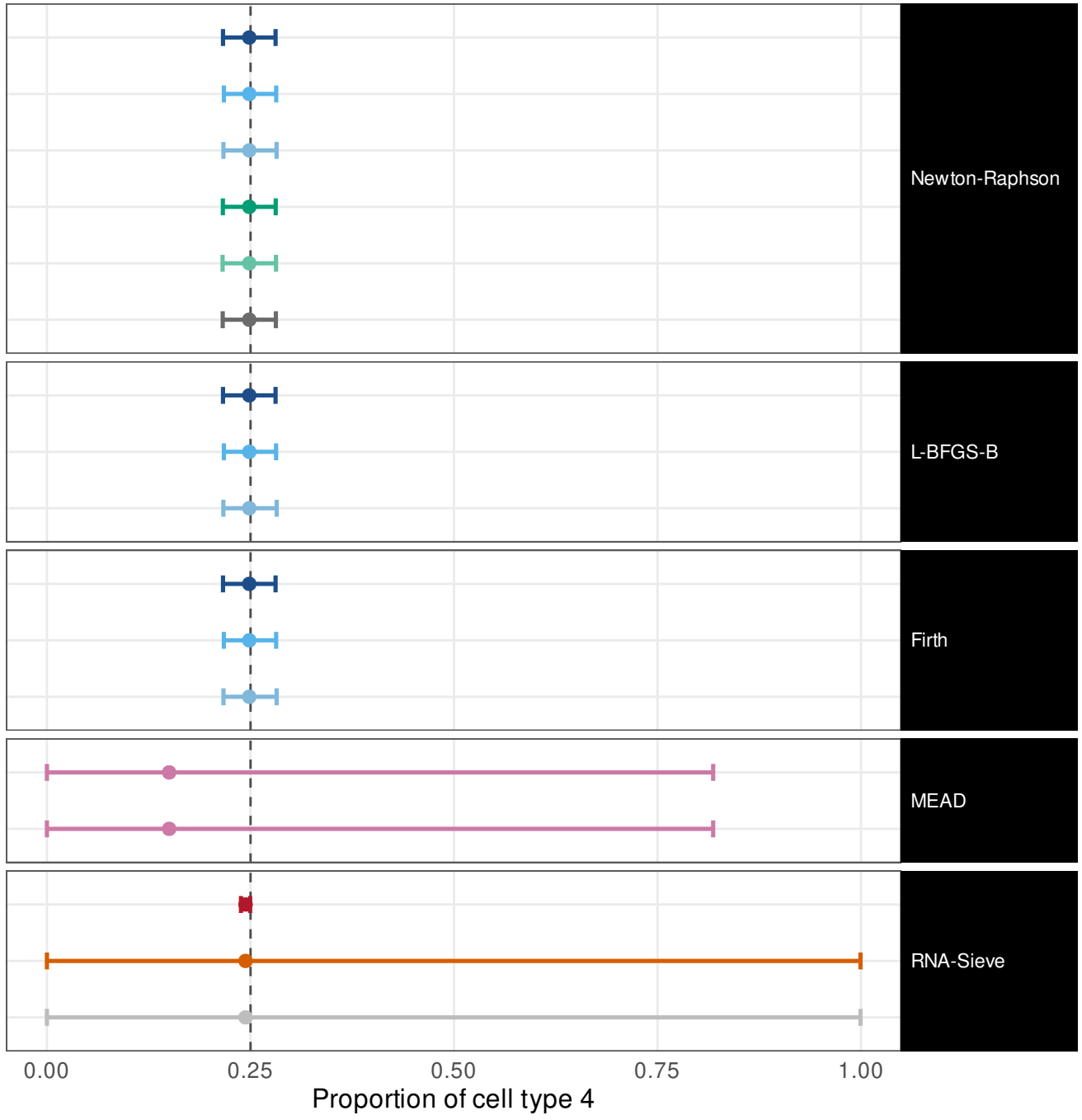


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / balanced / high sibling overlap / a = 1 / equal sizes

Wald (delta)	0.92 [0.88, 0.95]	w = 0.0646
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.0643
ILR, REML	0.93 [0.89, 0.96]	w = 0.0653
Profile likelihood	0.92 [0.88, 0.95]	w = 0.0649
Profile, REML	0.93 [0.89, 0.96]	w = 0.0657
Oracle (Fisher at p*)	0.92 [0.88, 0.95]	w = 0.0653
Wald (delta)	0.92 [0.88, 0.95]	w = 0.0646
ILR, mapped back	0.91 [0.86, 0.94]	w = 0.0641
ILR, REML	0.92 [0.88, 0.95]	w = 0.0655
Wald (delta)	0.92 [0.88, 0.95]	w = 0.0646
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.0643
ILR, REML	0.93 [0.89, 0.96]	w = 0.0653
Method's own interval	1 [0.98, 1]	w = 0.818
Own interval, transcript fractions	1 [0.98, 1]	w = 0.818
Fisher information (Wald)	0.26 [0.2, 0.32]	w = 0.0112
Godambe sandwich	1 [0.98, 1]	w = 0.999
Released code, as published	1 [0.98, 1]	w = 0.999

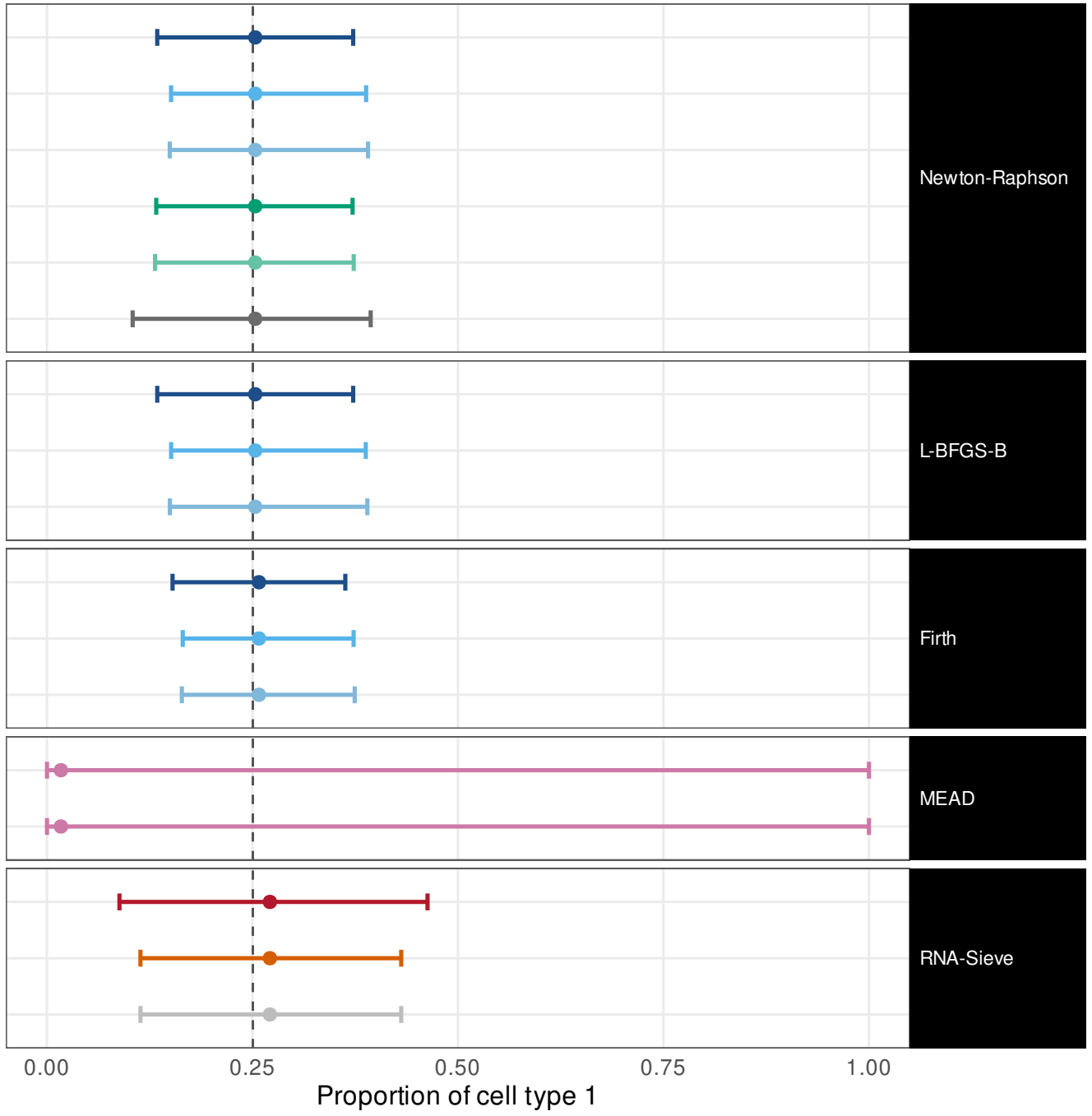


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / balanced / high sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.92 [0.88, 0.95]	w = 0.238
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.237
ILR, REML	0.96 [0.92, 0.98]	w = 0.241
Profile likelihood	0.93 [0.89, 0.96]	w = 0.239
Profile, REML	0.93 [0.89, 0.96]	w = 0.242
Oracle (Fisher at p*)	0.94 [0.9, 0.97]	w = 0.29
Wald (delta)	0.92 [0.88, 0.95]	w = 0.238
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.237
ILR, REML	0.95 [0.91, 0.97]	w = 0.24
Wald (delta)	0.92 [0.87, 0.95]	w = 0.21
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.208
ILR, REML	0.94 [0.9, 0.97]	w = 0.21
Method's own interval	1 [0.98, 1]	w = 1
Own interval, transcript fractions	1 [0.98, 1]	w = 1
Fisher information (Wald)	0.99 [0.96, 1]	w = 0.375
Godambe sandwich	0.98 [0.94, 0.99]	w = 0.317
Released code, as published	0.98 [0.94, 0.99]	w = 0.317

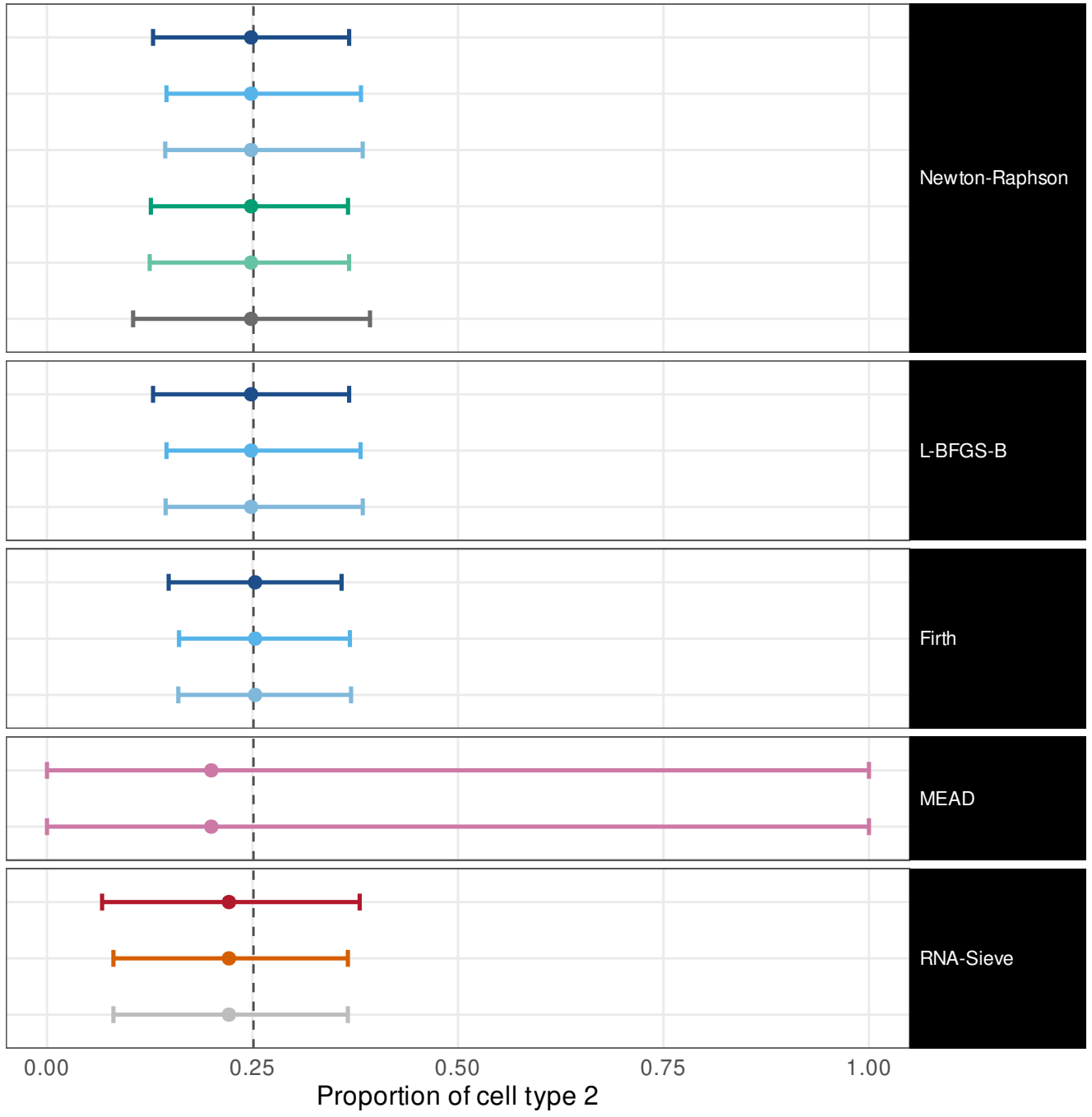


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / balanced / high sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.94 [0.9, 0.97]	w = 0.239
ILR, mapped back	0.98 [0.96, 0.99]	w = 0.237
ILR, REML	0.99 [0.96, 1]	w = 0.24
Profile likelihood	0.96 [0.92, 0.98]	w = 0.24
Profile, REML	0.96 [0.92, 0.98]	w = 0.243
Oracle (Fisher at p*)	0.96 [0.92, 0.98]	w = 0.288
Wald (delta)	0.94 [0.9, 0.97]	w = 0.239
ILR, mapped back	0.99 [0.96, 1]	w = 0.236
ILR, REML	0.98 [0.96, 0.99]	w = 0.24
Wald (delta)	0.94 [0.9, 0.97]	w = 0.21
ILR, mapped back	0.96 [0.93, 0.98]	w = 0.208
ILR, REML	0.96 [0.93, 0.98]	w = 0.21
Method's own interval	1 [0.98, 1]	w = 1
Own interval, transcript fractions	1 [0.98, 1]	w = 1
Fisher information (Wald)	0.96 [0.93, 0.98]	w = 0.313
Godambe sandwich	0.97 [0.94, 0.99]	w = 0.285
Released code, as published	0.97 [0.94, 0.99]	w = 0.285

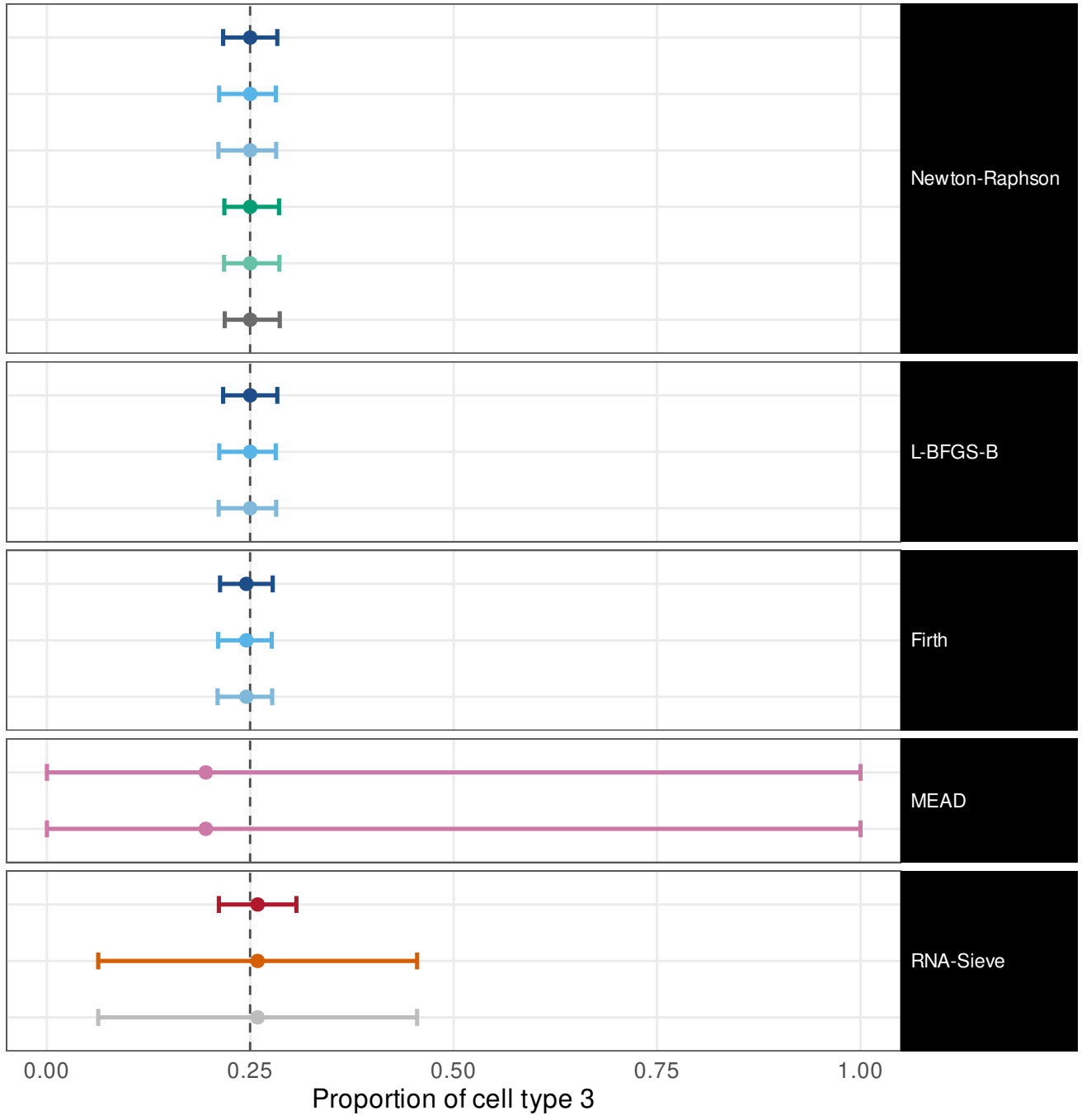


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / balanced / high sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.93 [0.89, 0.96]	w = 0.0666
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.0697
ILR, REML	0.94 [0.9, 0.97]	w = 0.071
Profile likelihood	0.92 [0.87, 0.95]	w = 0.0672
Profile, REML	0.92 [0.87, 0.95]	w = 0.068
Oracle (Fisher at p*)	0.95 [0.91, 0.97]	w = 0.0676
Wald (delta)	0.93 [0.89, 0.96]	w = 0.0666
ILR, mapped back	0.94 [0.89, 0.96]	w = 0.0694
ILR, REML	0.94 [0.89, 0.96]	w = 0.0707
Wald (delta)	0.9 [0.85, 0.93]	w = 0.0647
ILR, mapped back	0.9 [0.85, 0.93]	w = 0.0659
ILR, REML	0.9 [0.85, 0.93]	w = 0.0671
Method's own interval	1 [0.98, 1]	w = 1
Own interval, transcript fractions	1 [0.98, 1]	w = 1
Fisher information (Wald)	0.98 [0.94, 0.99]	w = 0.0953
Godambe sandwich	1 [0.98, 1]	w = 0.392
Released code, as published	1 [0.98, 1]	w = 0.392

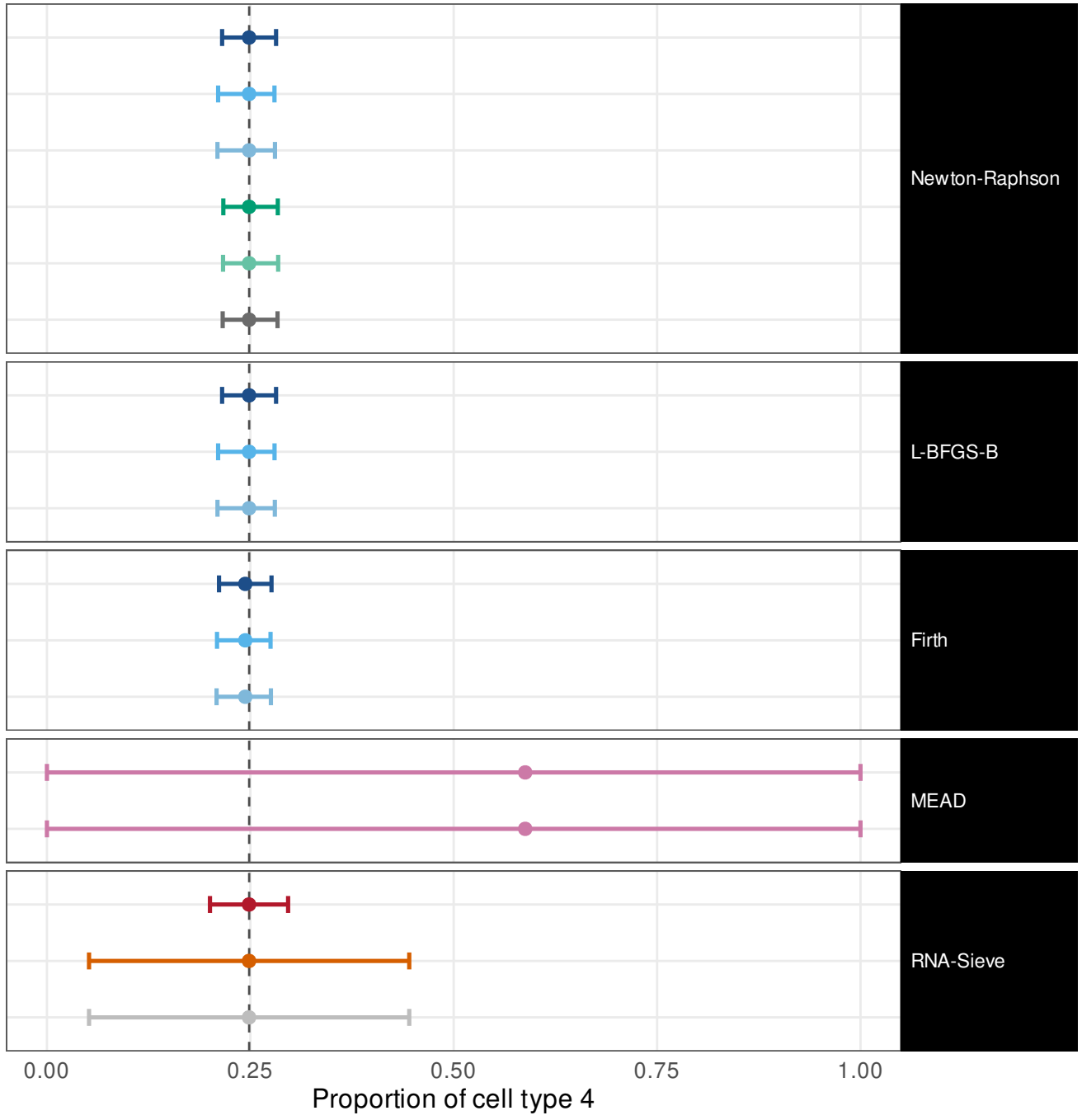


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / balanced / high sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.94 [0.9, 0.97]	w = 0.0663
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.0692
ILR, REML	0.96 [0.92, 0.98]	w = 0.0707
Profile likelihood	0.94 [0.9, 0.97]	w = 0.067
Profile, REML	0.94 [0.9, 0.97]	w = 0.0677
Oracle (Fisher at p*)	0.96 [0.92, 0.98]	w = 0.0674
Wald (delta)	0.94 [0.9, 0.97]	w = 0.0663
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.0693
ILR, REML	0.96 [0.93, 0.98]	w = 0.0705
Wald (delta)	0.94 [0.9, 0.97]	w = 0.0645
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.0658
ILR, REML	0.95 [0.91, 0.97]	w = 0.0668
Method's own interval	1 [0.98, 1]	w = 1
Own interval, transcript fractions	1 [0.98, 1]	w = 1
Fisher information (Wald)	0.99 [0.96, 1]	w = 0.0961
Godambe sandwich	1 [0.98, 1]	w = 0.394
Released code, as published	1 [0.98, 1]	w = 0.394

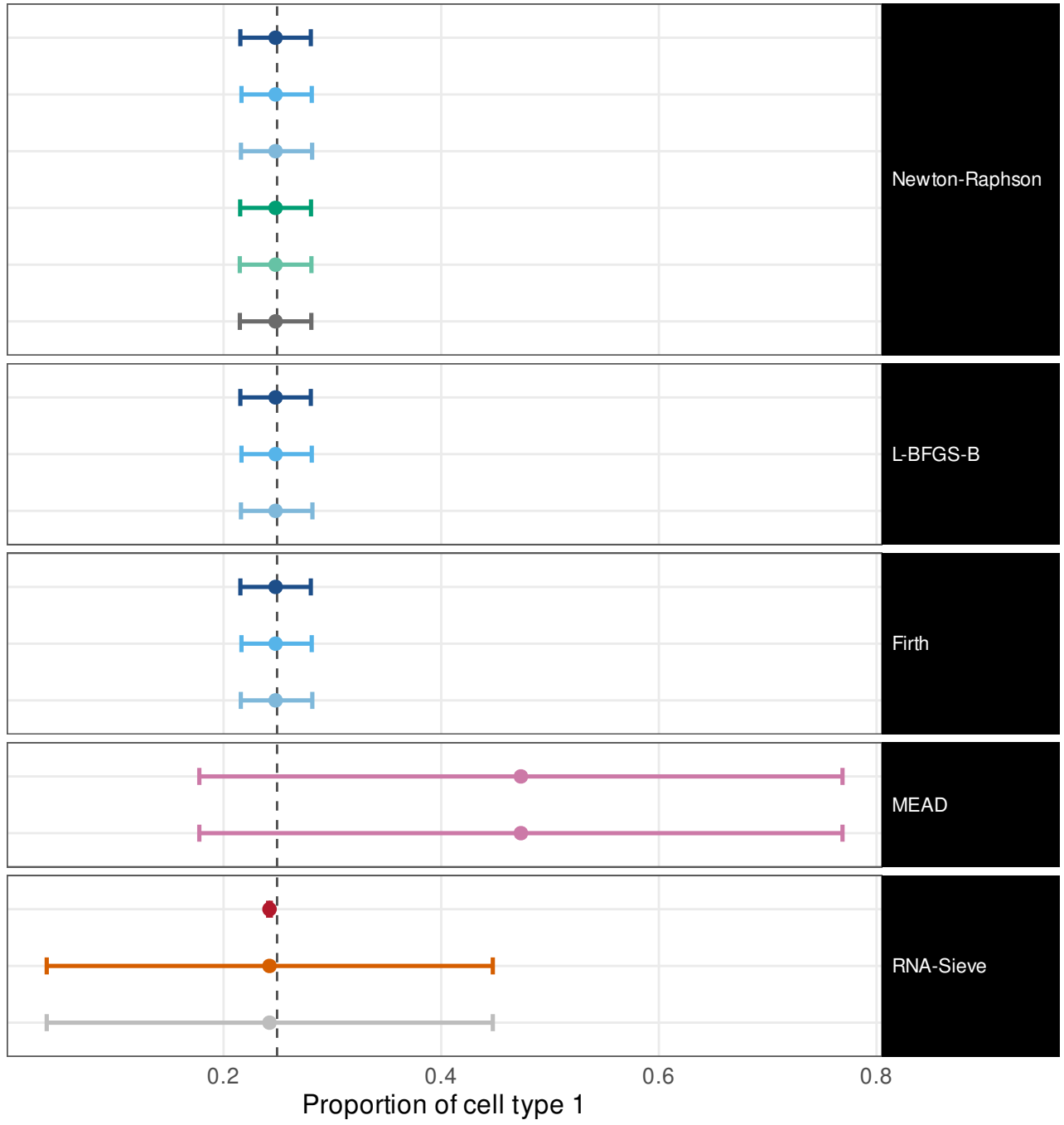


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / balanced / high sibling overlap / a = 5 / equal sizes

Wald (delta)	0.92 [0.88, 0.95]	w = 0.0647
ILR, mapped back	0.94 [0.89, 0.96]	w = 0.0646
ILR, REML	0.94 [0.9, 0.97]	w = 0.0654
Profile likelihood	0.93 [0.89, 0.96]	w = 0.0651
Profile, REML	0.93 [0.89, 0.96]	w = 0.0658
Oracle (Fisher at p*)	0.94 [0.89, 0.96]	w = 0.0657
Wald (delta)	0.92 [0.88, 0.95]	w = 0.0647
ILR, mapped back	0.92 [0.88, 0.95]	w = 0.0646
ILR, REML	0.94 [0.9, 0.97]	w = 0.0657
Wald (delta)	0.93 [0.89, 0.96]	w = 0.0647
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.0646
ILR, REML	0.93 [0.89, 0.96]	w = 0.0657
Method's own interval	1 [0.97, 1]	w = 0.591
Own interval, transcript fractions	1 [0.97, 1]	w = 0.591
Fisher information (Wald)	0.07 [0.042, 0.11]	w = 0.00243
Godambe sandwich	1 [0.98, 1]	w = 0.41
Released code, as published	1 [0.98, 1]	w = 0.41

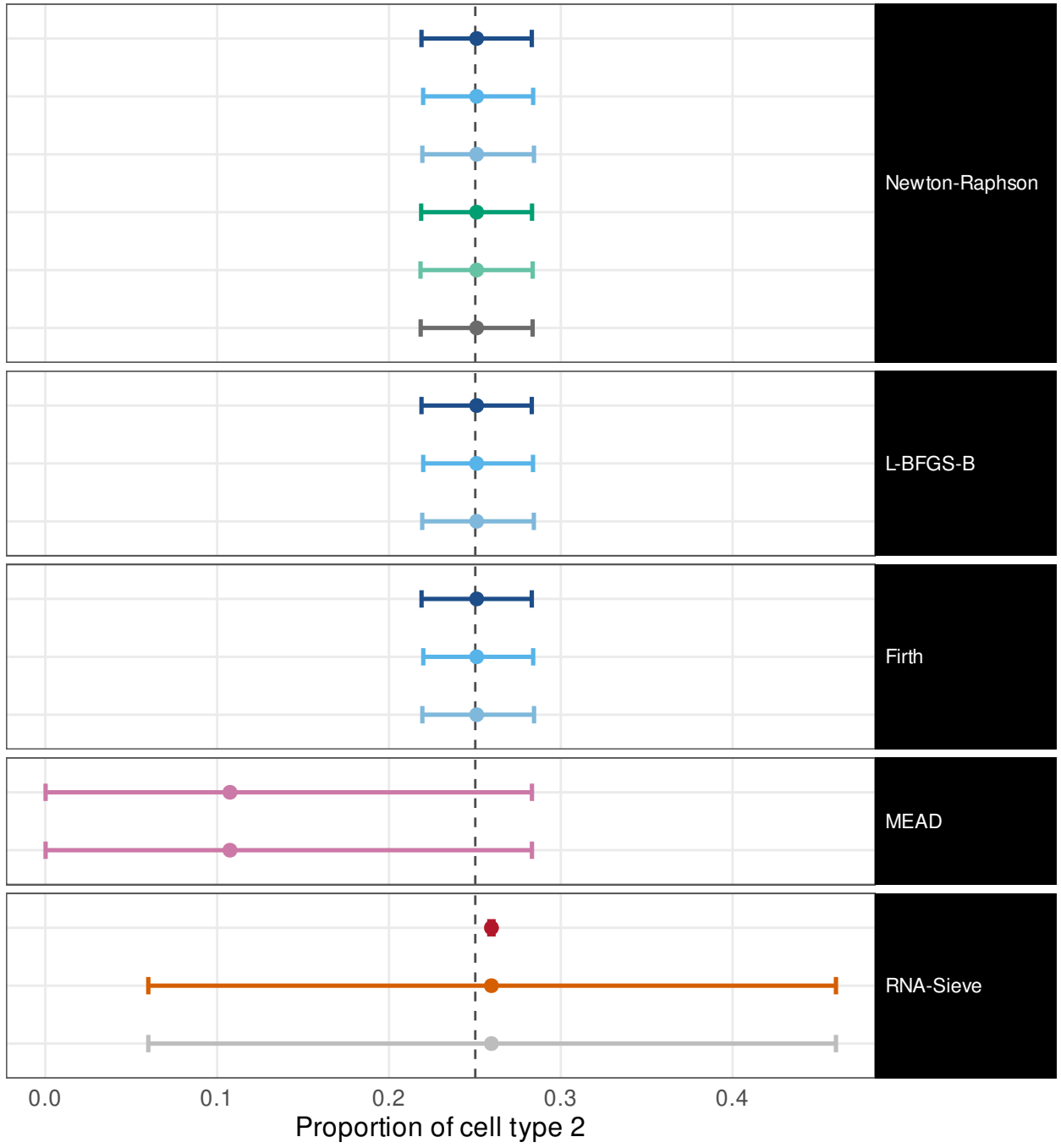


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / balanced / high sibling overlap / a = 5 / equal sizes

Wald (delta)	0.94 [0.9, 0.97]	w = 0.0642
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.064
ILR, REML	0.94 [0.9, 0.97]	w = 0.065
Profile likelihood	0.94 [0.9, 0.97]	w = 0.0645
Profile, REML	0.94 [0.9, 0.97]	w = 0.0653
Oracle (Fisher at p*)	0.96 [0.92, 0.98]	w = 0.0652
Wald (delta)	0.94 [0.9, 0.97]	w = 0.0642
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.0639
ILR, REML	0.95 [0.91, 0.97]	w = 0.0649
Wald (delta)	0.94 [0.9, 0.97]	w = 0.0642
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.064
ILR, REML	0.94 [0.9, 0.97]	w = 0.065
Method's own interval	0.96 [0.92, 0.98]	w = 0.283
Own interval, transcript fractions	0.96 [0.92, 0.98]	w = 0.283
Fisher information (Wald)	0.01 [0.0027, 0.036]	w = 0.00244
Godambe sandwich	1 [0.98, 1]	w = 0.4
Released code, as published	1 [0.98, 1]	w = 0.4

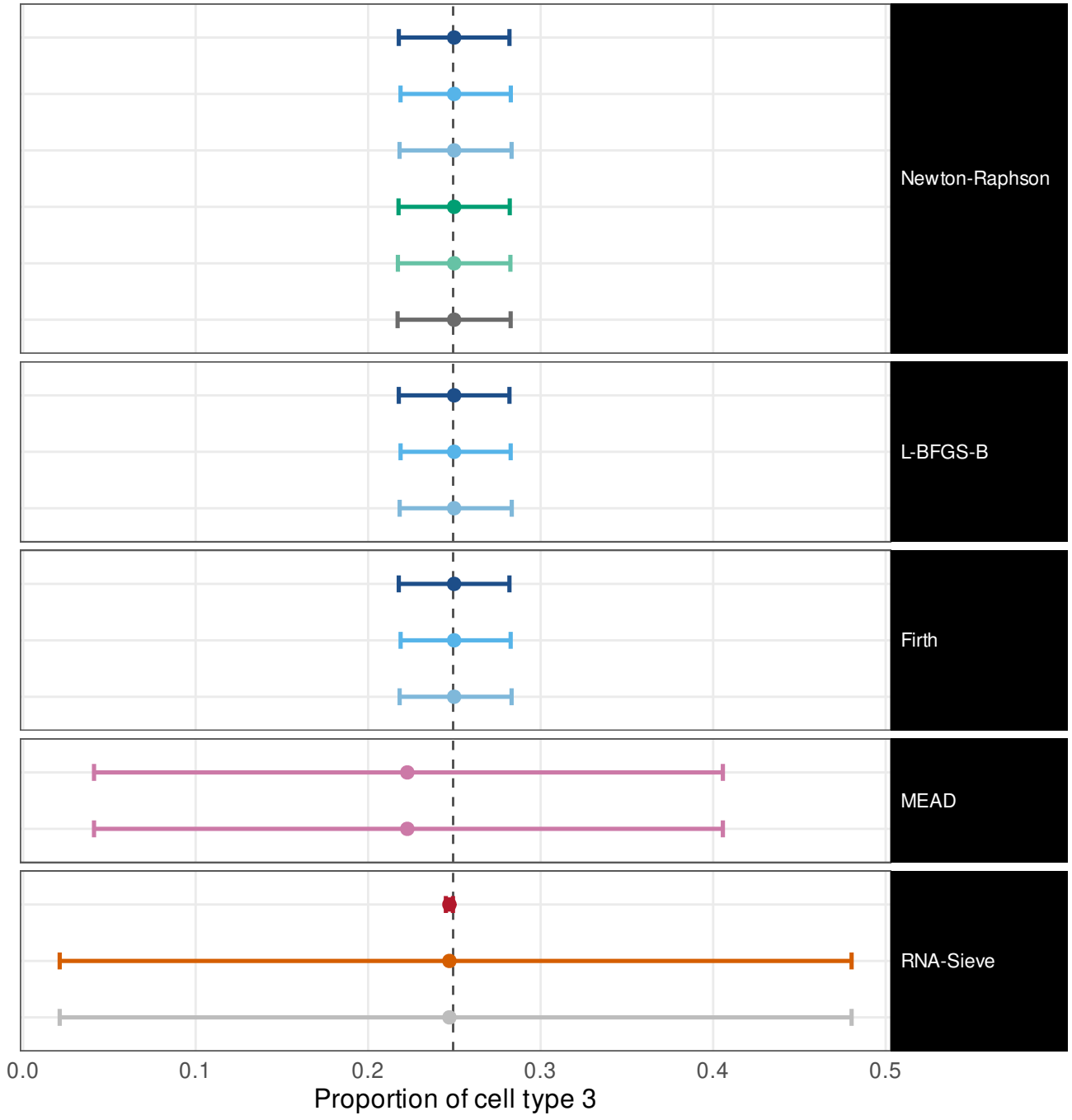


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / balanced / high sibling overlap / a = 5 / equal sizes

Wald (delta)	0.95 [0.91, 0.97]	w = 0.0641
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.0639
ILR, REML	0.96 [0.92, 0.98]	w = 0.0649
Profile likelihood	0.95 [0.91, 0.97]	w = 0.0645
Profile, REML	0.95 [0.91, 0.97]	w = 0.0652
Oracle (Fisher at p*)	0.96 [0.92, 0.98]	w = 0.0655
Wald (delta)	0.95 [0.91, 0.97]	w = 0.0641
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.0638
ILR, REML	0.95 [0.91, 0.97]	w = 0.065
Wald (delta)	0.95 [0.91, 0.97]	w = 0.0641
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.0638
ILR, REML	0.96 [0.92, 0.98]	w = 0.0649
Method's own interval	1 [0.98, 1]	w = 0.365
Own interval, transcript fractions	1 [0.98, 1]	w = 0.365
Fisher information (Wald)	0.12 [0.078, 0.17]	w = 0.00409
Godambe sandwich	1 [0.98, 1]	w = 0.459
Released code, as published	1 [0.98, 1]	w = 0.459

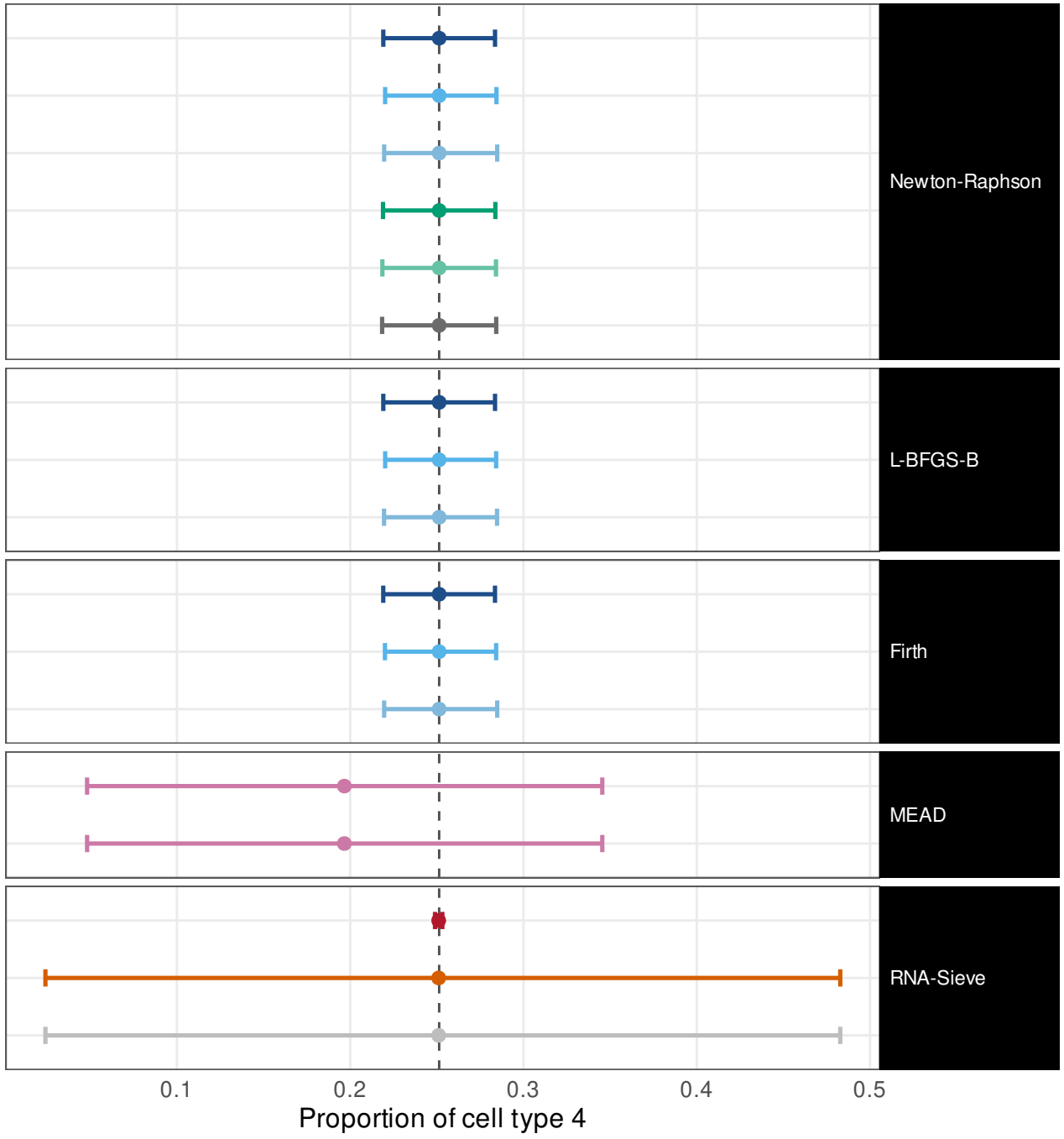


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / balanced / high sibling overlap / a = 5 / equal sizes

Wald (delta)	0.93 [0.89, 0.96]	w = 0.0645
ILR, mapped back	0.94 [0.89, 0.96]	w = 0.0643
ILR, REML	0.94 [0.9, 0.97]	w = 0.0653
Profile likelihood	0.93 [0.89, 0.96]	w = 0.0648
Profile, REML	0.93 [0.89, 0.96]	w = 0.0656
Oracle (Fisher at p*)	0.96 [0.92, 0.98]	w = 0.0658
Wald (delta)	0.93 [0.89, 0.96]	w = 0.0645
ILR, mapped back	0.94 [0.89, 0.96]	w = 0.064
ILR, REML	0.94 [0.89, 0.96]	w = 0.0652
Wald (delta)	0.93 [0.89, 0.96]	w = 0.0645
ILR, mapped back	0.94 [0.89, 0.96]	w = 0.0642
ILR, REML	0.94 [0.9, 0.97]	w = 0.0653
Method's own interval	1 [0.98, 1]	w = 0.297
Own interval, transcript fractions	1 [0.98, 1]	w = 0.297
Fisher information (Wald)	0.075 [0.046, 0.12]	w = 0.00422
Godambe sandwich	1 [0.98, 1]	w = 0.459
Released code, as published	1 [0.98, 1]	w = 0.459

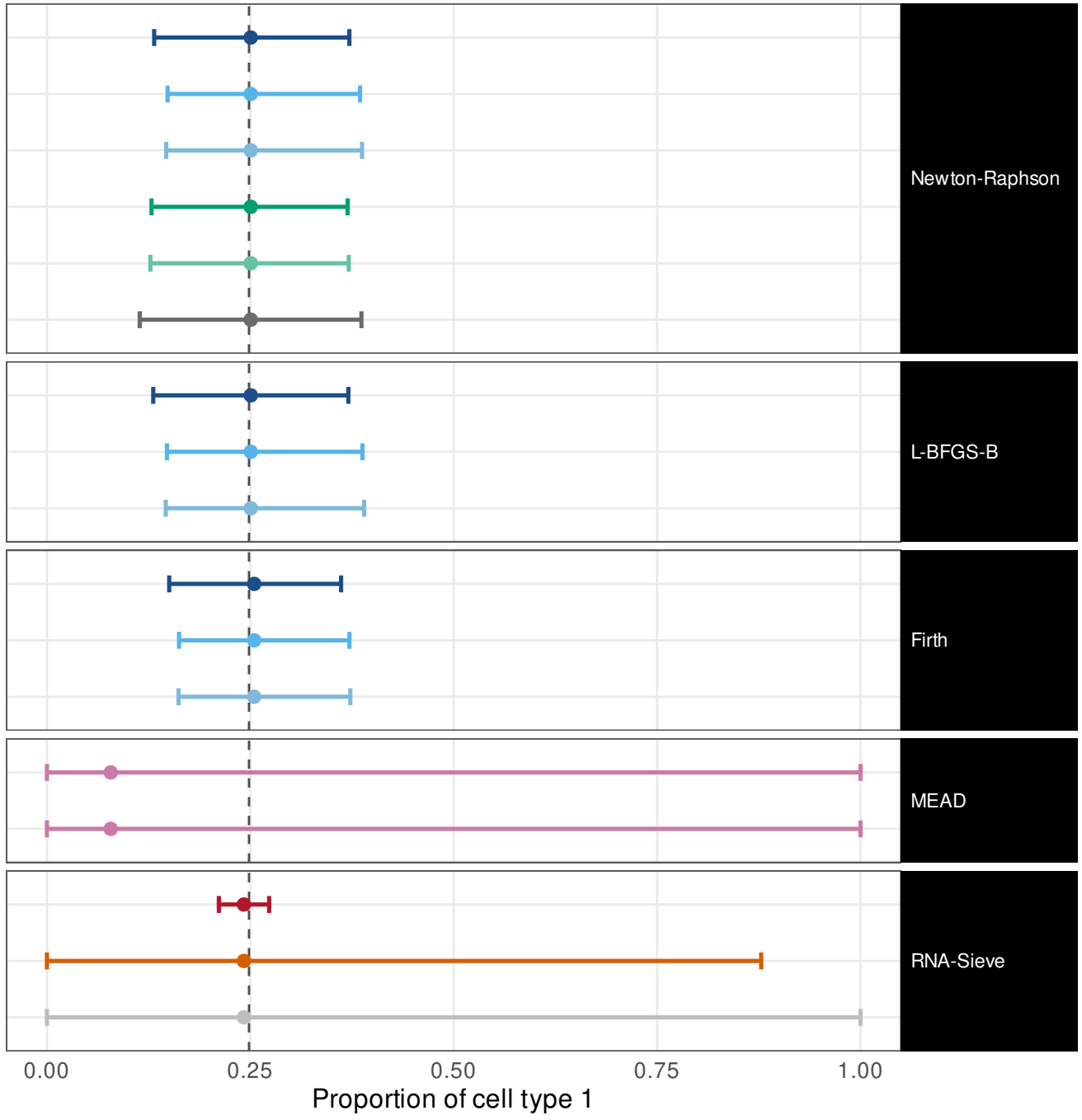


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / balanced / high sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.97 [0.94, 0.99]	w = 0.24
ILR, mapped back	0.98 [0.96, 0.99]	w = 0.237
ILR, REML	0.98 [0.96, 0.99]	w = 0.241
Profile likelihood	0.97 [0.94, 0.99]	w = 0.241
Profile, REML	0.97 [0.94, 0.99]	w = 0.244
Oracle (Fisher at p*)	0.95 [0.91, 0.97]	w = 0.273
Wald (delta)	0.97 [0.94, 0.99]	w = 0.24
ILR, mapped back	0.99 [0.96, 1]	w = 0.24
ILR, REML	0.99 [0.96, 1]	w = 0.244
Wald (delta)	0.97 [0.94, 0.99]	w = 0.211
ILR, mapped back	0.97 [0.94, 0.99]	w = 0.209
ILR, REML	0.98 [0.95, 0.99]	w = 0.211
Method's own interval	1 [0.98, 1]	w = 1
Own interval, transcript fractions	1 [0.98, 1]	w = 1
Fisher information (Wald)	0.32 [0.26, 0.39]	w = 0.0618
Godambe sandwich	1 [0.98, 1]	w = 0.878
Released code, as published	1 [0.98, 1]	w = 1

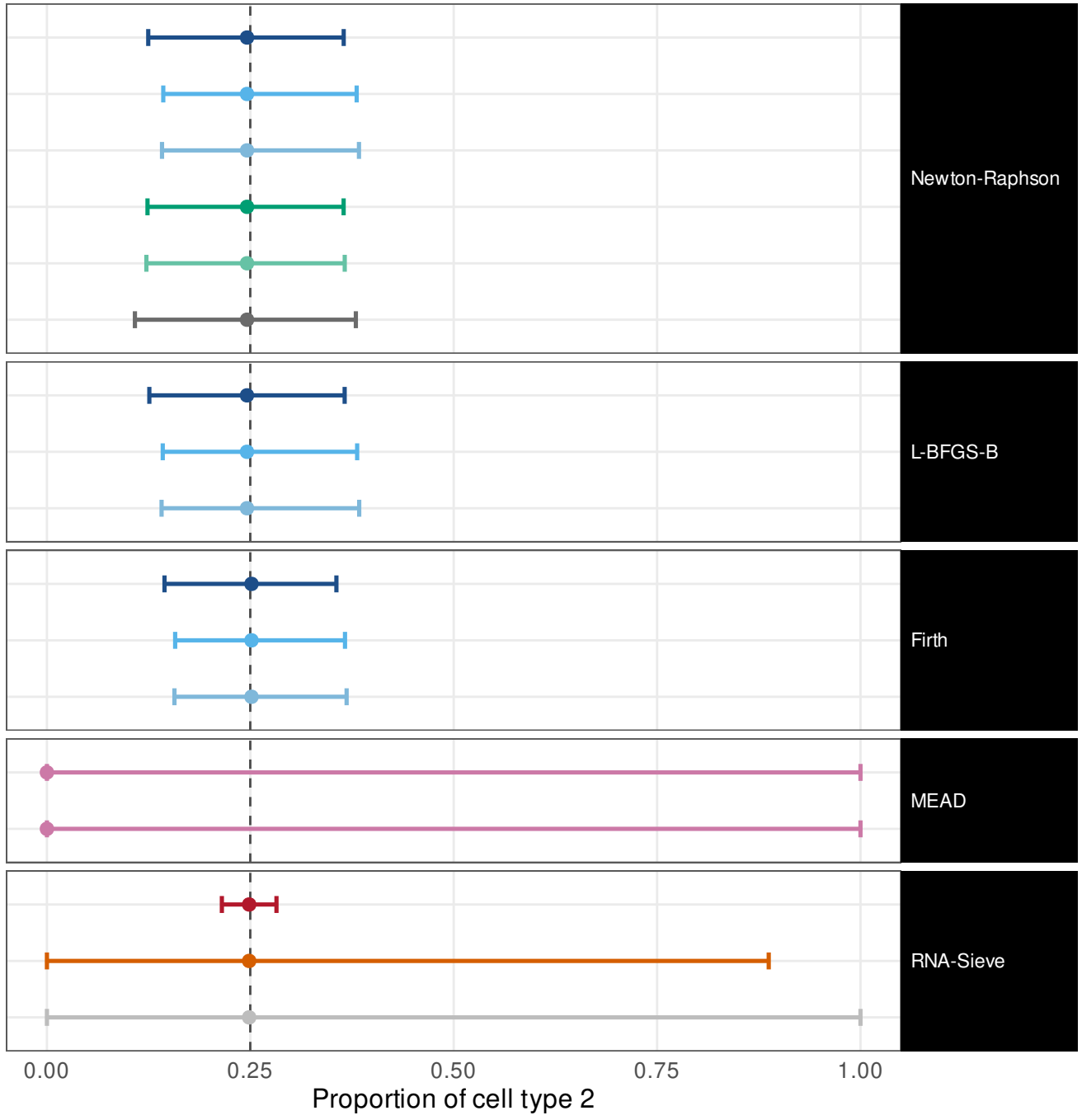


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / balanced / high sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.97 [0.94, 0.99]	w = 0.24
ILR, mapped back	0.99 [0.97, 1]	w = 0.238
ILR, REML	1 [0.98, 1]	w = 0.242
Profile likelihood	0.97 [0.94, 0.99]	w = 0.241
Profile, REML	0.97 [0.94, 0.99]	w = 0.244
Oracle (Fisher at p*)	0.95 [0.91, 0.97]	w = 0.272
Wald (delta)	0.97 [0.94, 0.99]	w = 0.24
ILR, mapped back	1 [0.97, 1]	w = 0.239
ILR, REML	1 [0.98, 1]	w = 0.243
Wald (delta)	0.96 [0.93, 0.98]	w = 0.211
ILR, mapped back	0.98 [0.96, 0.99]	w = 0.209
ILR, REML	0.99 [0.96, 1]	w = 0.212
Method's own interval	1 [0.98, 1]	w = 1
Own interval, transcript fractions	1 [0.98, 1]	w = 1
Fisher information (Wald)	0.4 [0.33, 0.47]	w = 0.0671
Godambe sandwich	1 [0.98, 1]	w = 0.887
Released code, as published	1 [0.98, 1]	w = 1

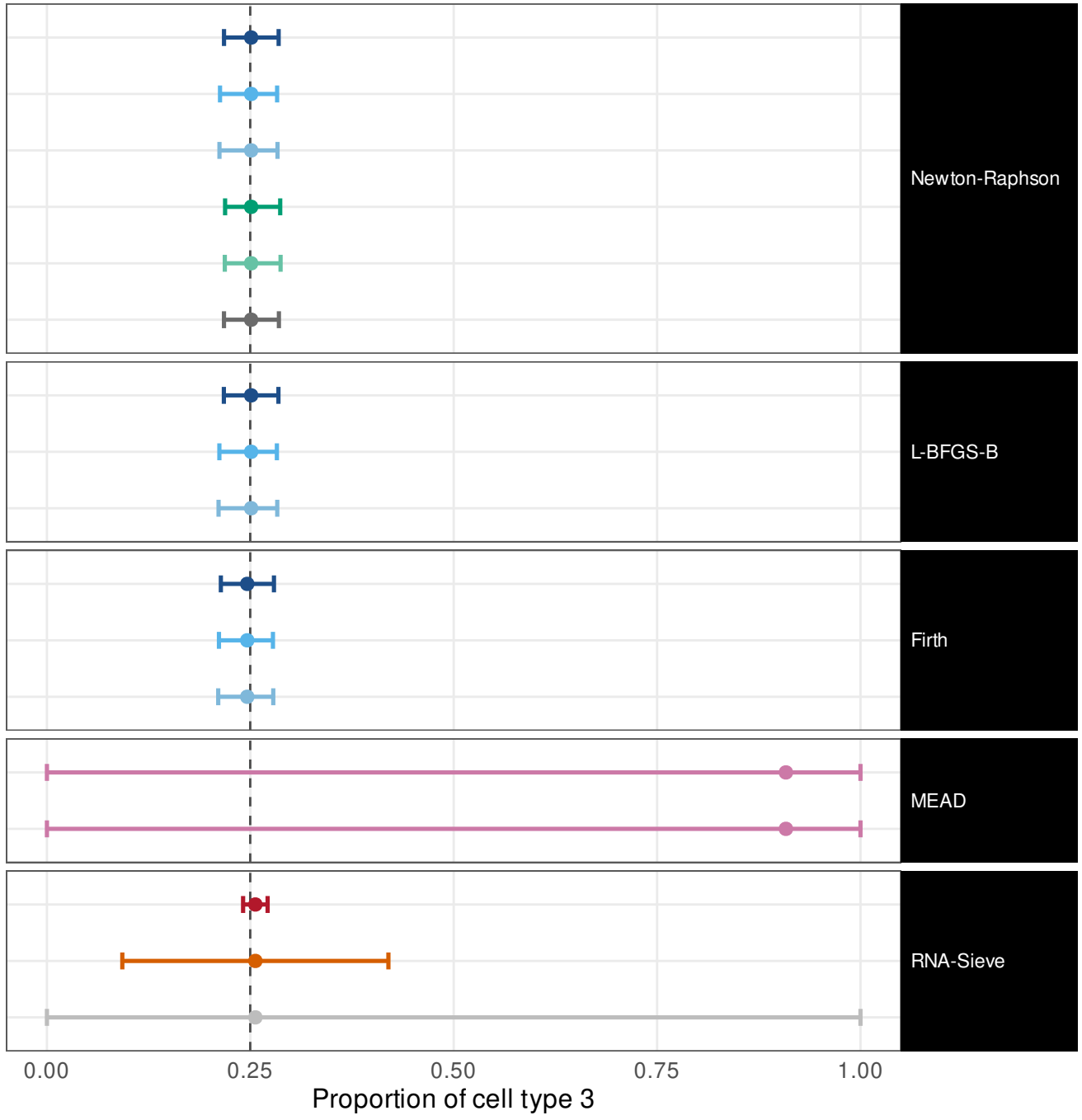


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / balanced / high sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.98 [0.95, 0.99]	w = 0.0671
ILR, mapped back	0.98 [0.95, 0.99]	w = 0.0702
ILR, REML	0.99 [0.96, 1]	w = 0.0714
Profile likelihood	0.97 [0.94, 0.99]	w = 0.0677
Profile, REML	0.97 [0.94, 0.99]	w = 0.0685
Oracle (Fisher at p*)	0.97 [0.93, 0.99]	w = 0.0675
Wald (delta)	0.98 [0.95, 0.99]	w = 0.0671
ILR, mapped back	0.98 [0.96, 0.99]	w = 0.0707
ILR, REML	0.98 [0.96, 0.99]	w = 0.0722
Wald (delta)	0.96 [0.93, 0.98]	w = 0.0653
ILR, mapped back	0.96 [0.93, 0.98]	w = 0.0664
ILR, REML	0.97 [0.94, 0.99]	w = 0.0676
Method's own interval	1 [0.98, 1]	w = 1
Own interval, transcript fractions	1 [0.98, 1]	w = 1
Fisher information (Wald)	0.6 [0.54, 0.67]	w = 0.0301
Godambe sandwich	1 [0.98, 1]	w = 0.327
Released code, as published	1 [0.98, 1]	w = 1

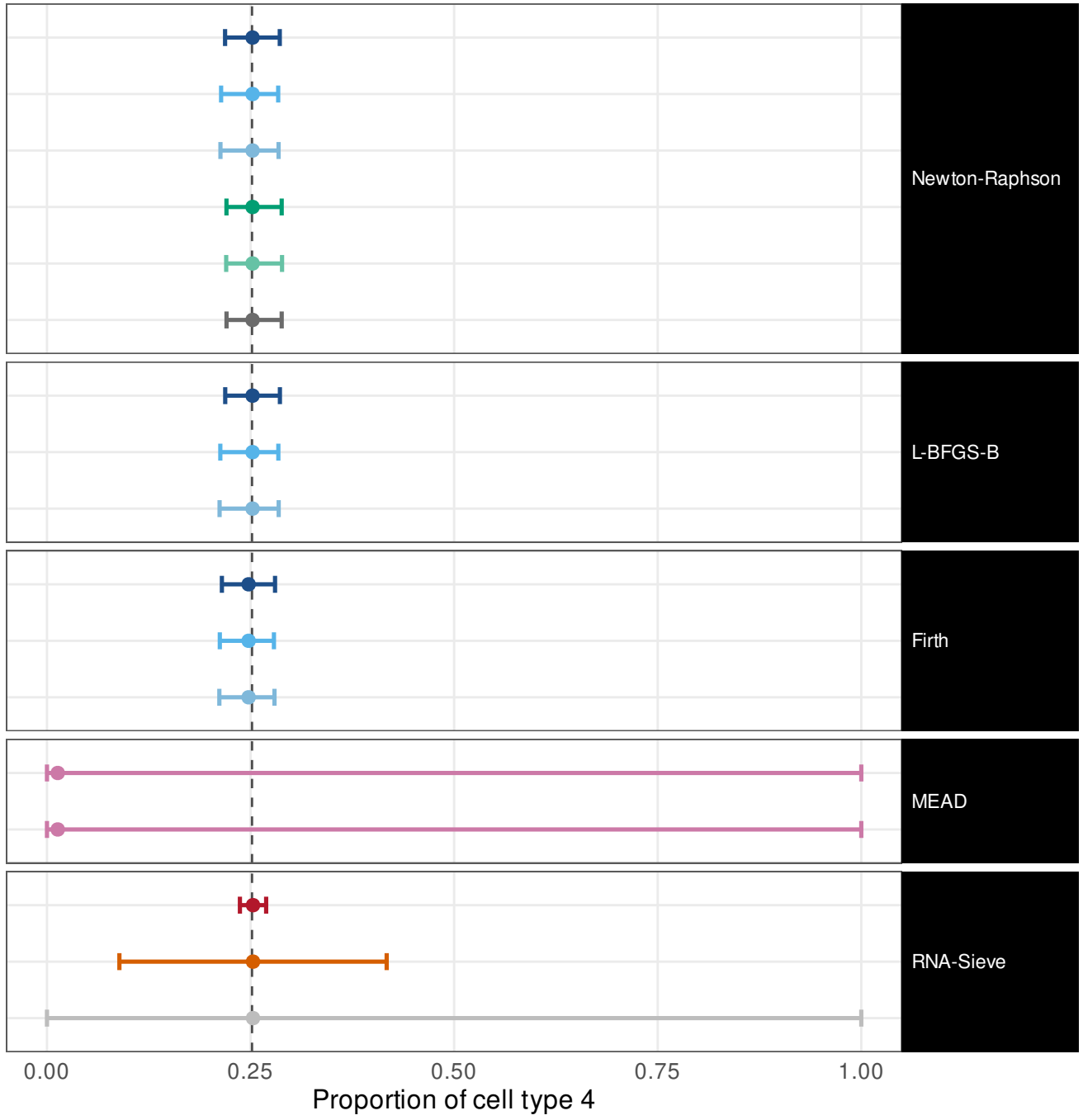


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / balanced / high sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.92 [0.87, 0.95]	w = 0.0672
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.0701
ILR, REML	0.93 [0.89, 0.96]	w = 0.0714
Profile likelihood	0.94 [0.89, 0.96]	w = 0.0678
Profile, REML	0.94 [0.89, 0.96]	w = 0.0686
Oracle (Fisher at p*)	0.96 [0.92, 0.98]	w = 0.0677
Wald (delta)	0.92 [0.87, 0.95]	w = 0.0672
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.0713
ILR, REML	0.93 [0.89, 0.96]	w = 0.0726
Wald (delta)	0.9 [0.86, 0.94]	w = 0.0653
ILR, mapped back	0.9 [0.85, 0.93]	w = 0.0665
ILR, REML	0.91 [0.87, 0.95]	w = 0.0677
Method's own interval	1 [0.98, 1]	w = 1
Own interval, transcript fractions	1 [0.98, 1]	w = 1
Fisher information (Wald)	0.61 [0.54, 0.67]	w = 0.0322
Godambe sandwich	1 [0.98, 1]	w = 0.328
Released code, as published	1 [0.98, 1]	w = 1

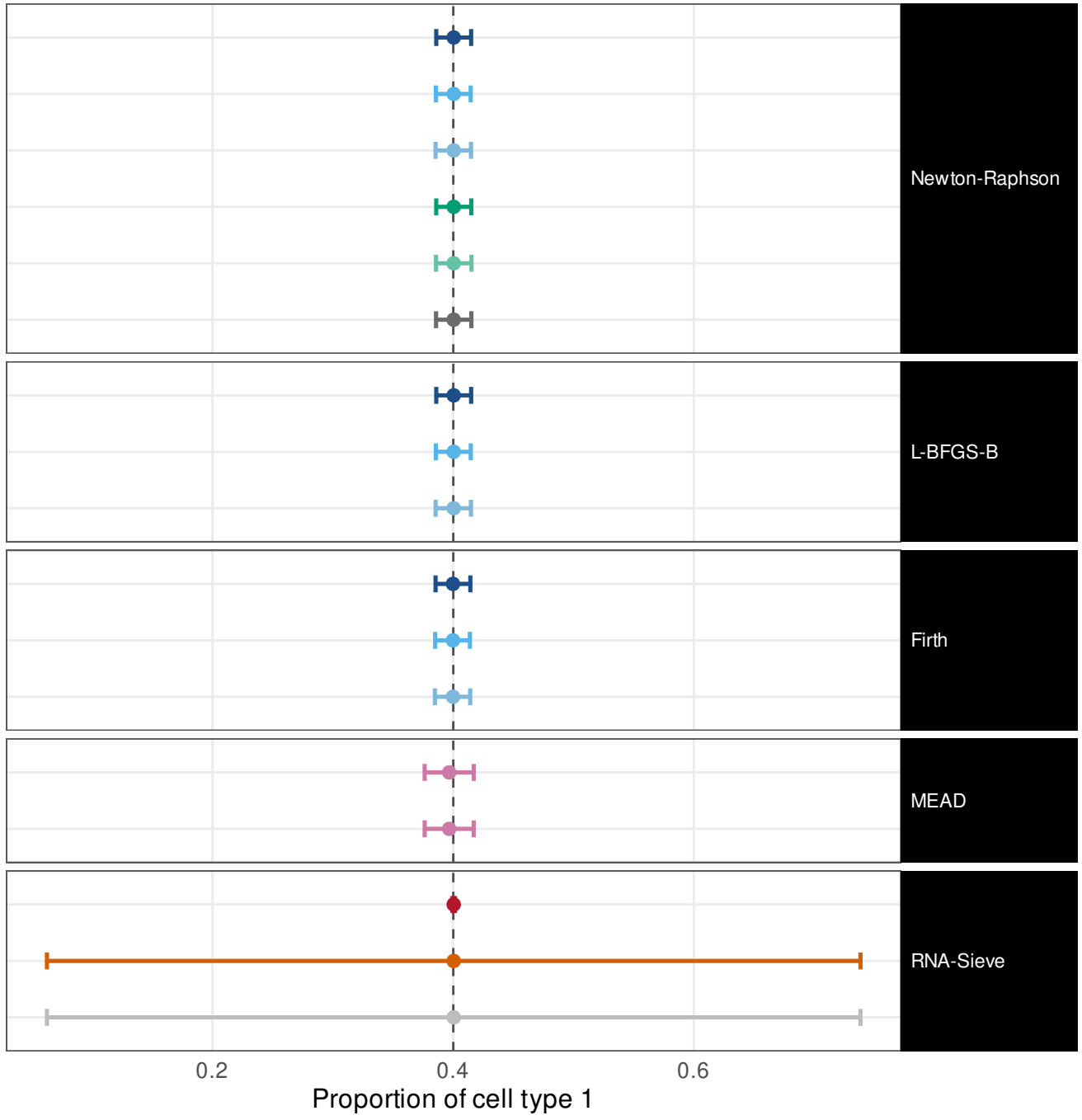


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / unbalanced / low sibling overlap / a = 1 / equal sizes

Wald (delta)	0.94 [0.89, 0.96]	w = 0.029
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.0289
ILR, REML	0.94 [0.9, 0.97]	w = 0.0295
Profile likelihood	0.94 [0.89, 0.96]	w = 0.0291
Profile, REML	0.94 [0.9, 0.97]	w = 0.0295
Oracle (Fisher at p*)	0.94 [0.9, 0.97]	w = 0.0293
Wald (delta)	0.94 [0.89, 0.96]	w = 0.029
ILR, mapped back	0.92 [0.88, 0.95]	w = 0.029
ILR, REML	0.94 [0.89, 0.96]	w = 0.0294
Wald (delta)	0.93 [0.89, 0.96]	w = 0.0289
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.0288
ILR, REML	0.94 [0.89, 0.96]	w = 0.0293
Method's own interval	0.96 [0.92, 0.98]	w = 0.0407
Own interval, transcript fractions	0.96 [0.92, 0.98]	w = 0.0407
Fisher information (Wald)	0.1 [0.07, 0.16]	w = 0.002
Godambe sandwich	1 [0.98, 1]	w = 0.676
Released code, as published	1 [0.98, 1]	w = 0.676

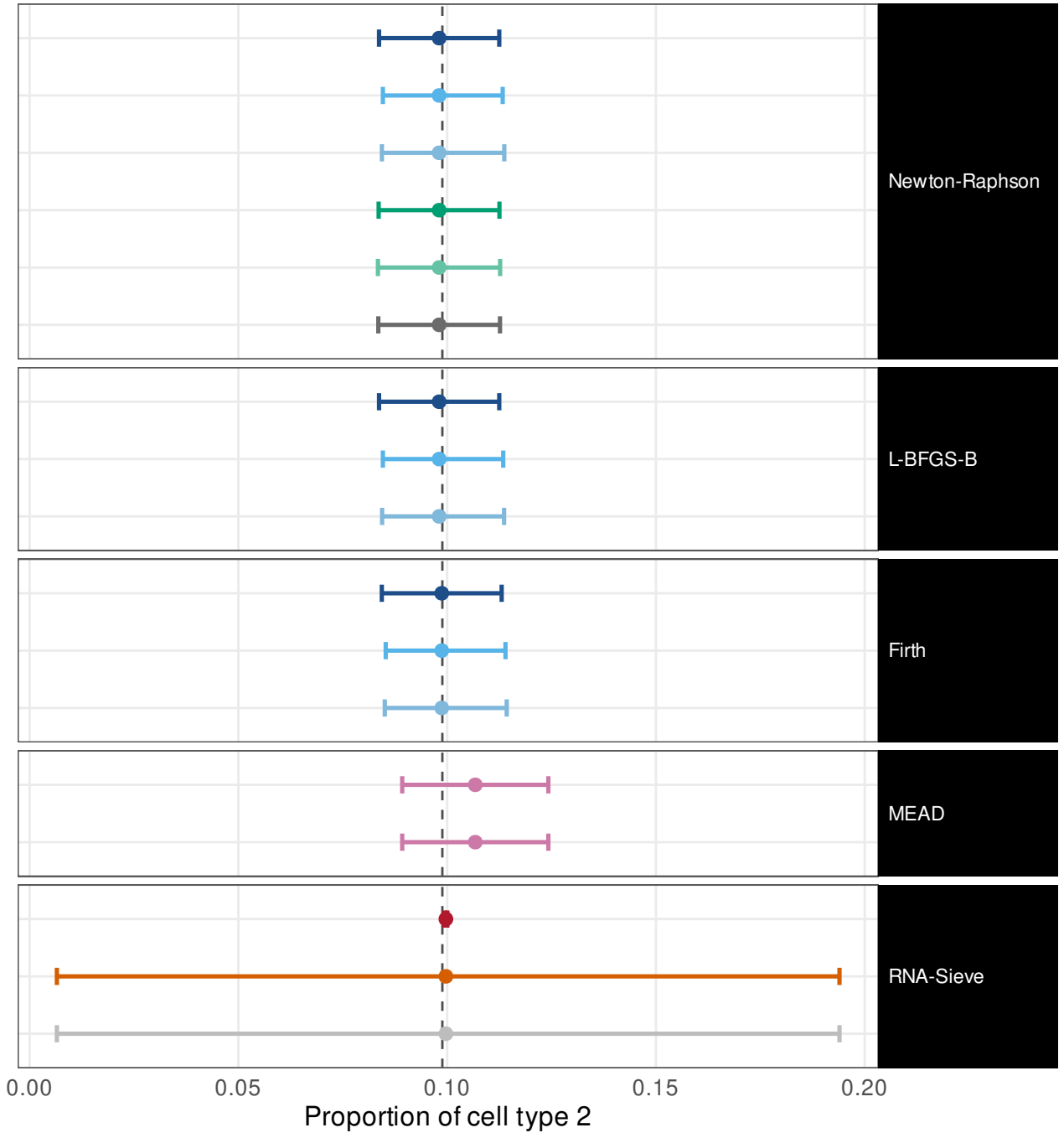


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / unbalanced / low sibling overlap / a = 1 / equal sizes

Wald (delta)	0.95 [0.91, 0.97]	w = 0.0288
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.0287
ILR, REML	0.97 [0.94, 0.99]	w = 0.0293
Profile likelihood	0.95 [0.91, 0.97]	w = 0.0289
Profile, REML	0.96 [0.93, 0.98]	w = 0.0293
Oracle (Fisher at p*)	0.96 [0.92, 0.98]	w = 0.0291
Wald (delta)	0.95 [0.91, 0.97]	w = 0.0288
ILR, mapped back	0.97 [0.94, 0.99]	w = 0.0288
ILR, REML	0.96 [0.93, 0.98]	w = 0.0292
Wald (delta)	0.96 [0.93, 0.98]	w = 0.0287
ILR, mapped back	0.96 [0.93, 0.98]	w = 0.0287
ILR, REML	0.96 [0.93, 0.98]	w = 0.0292
Method's own interval	0.84 [0.79, 0.89]	w = 0.035
Own interval, transcript fractions	0.84 [0.79, 0.89]	w = 0.035
Fisher information (Wald)	0.03 [0.014, 0.064]	w = 0.000695
Godambe sandwich	1 [0.98, 1]	w = 0.187
Released code, as published	1 [0.98, 1]	w = 0.187

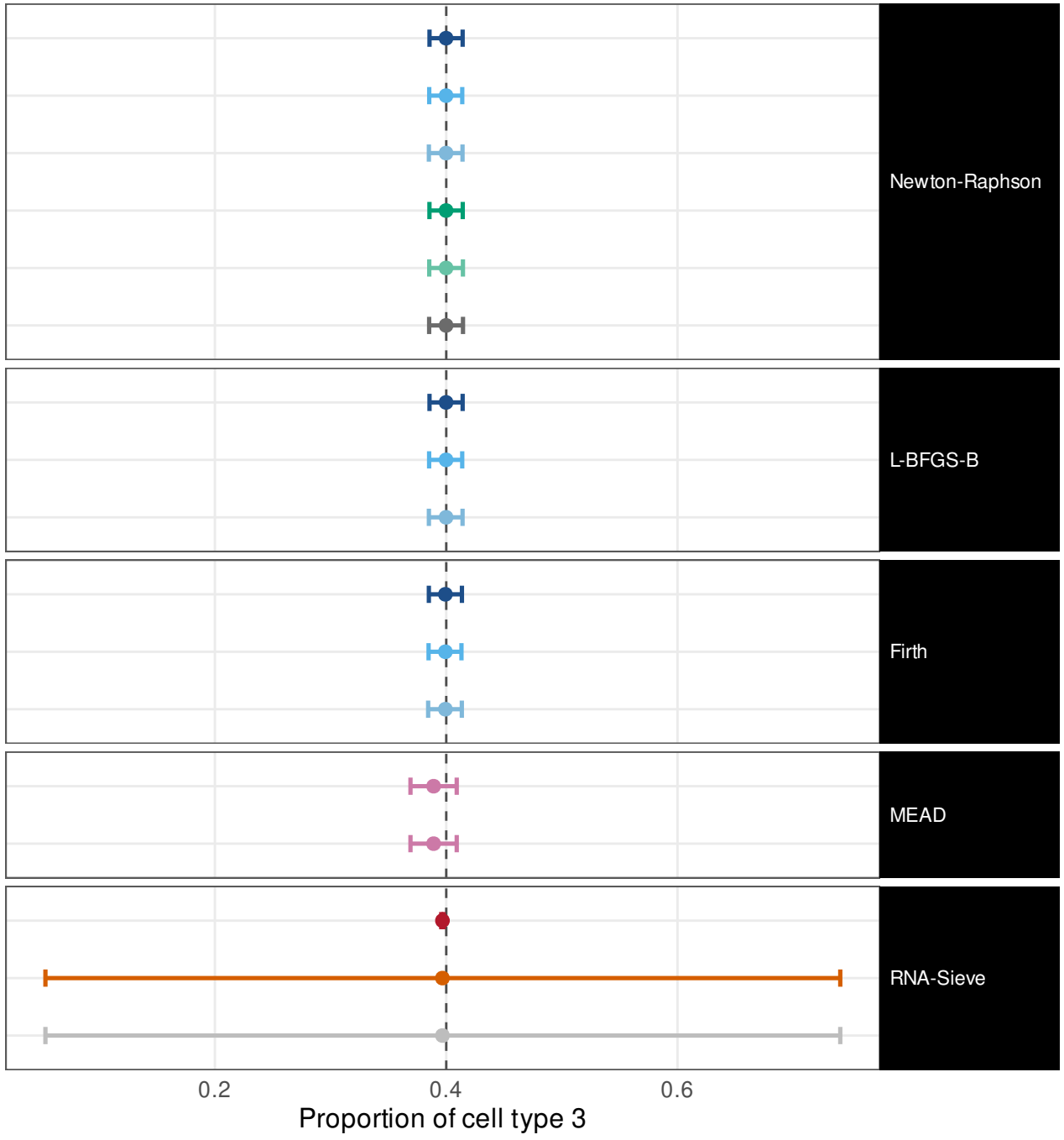


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / unbalanced / low sibling overlap / a = 1 / equal sizes

Wald (delta)	0.9 [0.86, 0.94]	w = 0.0287
ILR, mapped back	0.9 [0.86, 0.94]	w = 0.0286
ILR, REML	0.92 [0.87, 0.95]	w = 0.0291
Profile likelihood	0.9 [0.86, 0.94]	w = 0.0288
Profile, REML	0.92 [0.87, 0.95]	w = 0.0291
Oracle (Fisher at p*)	0.92 [0.87, 0.95]	w = 0.0293
Wald (delta)	0.9 [0.86, 0.94]	w = 0.0287
ILR, mapped back	0.91 [0.86, 0.94]	w = 0.0286
ILR, REML	0.92 [0.87, 0.95]	w = 0.0291
Wald (delta)	0.92 [0.87, 0.95]	w = 0.0286
ILR, mapped back	0.91 [0.86, 0.94]	w = 0.0284
ILR, REML	0.91 [0.86, 0.94]	w = 0.0291
Method's own interval	0.85 [0.79, 0.89]	w = 0.04
Own interval, transcript fractions	0.85 [0.79, 0.89]	w = 0.04
Fisher information (Wald)	0.065 [0.038, 0.11]	w = 0.00205
Godambe sandwich	1 [0.98, 1]	w = 0.687
Released code, as published	1 [0.98, 1]	w = 0.687

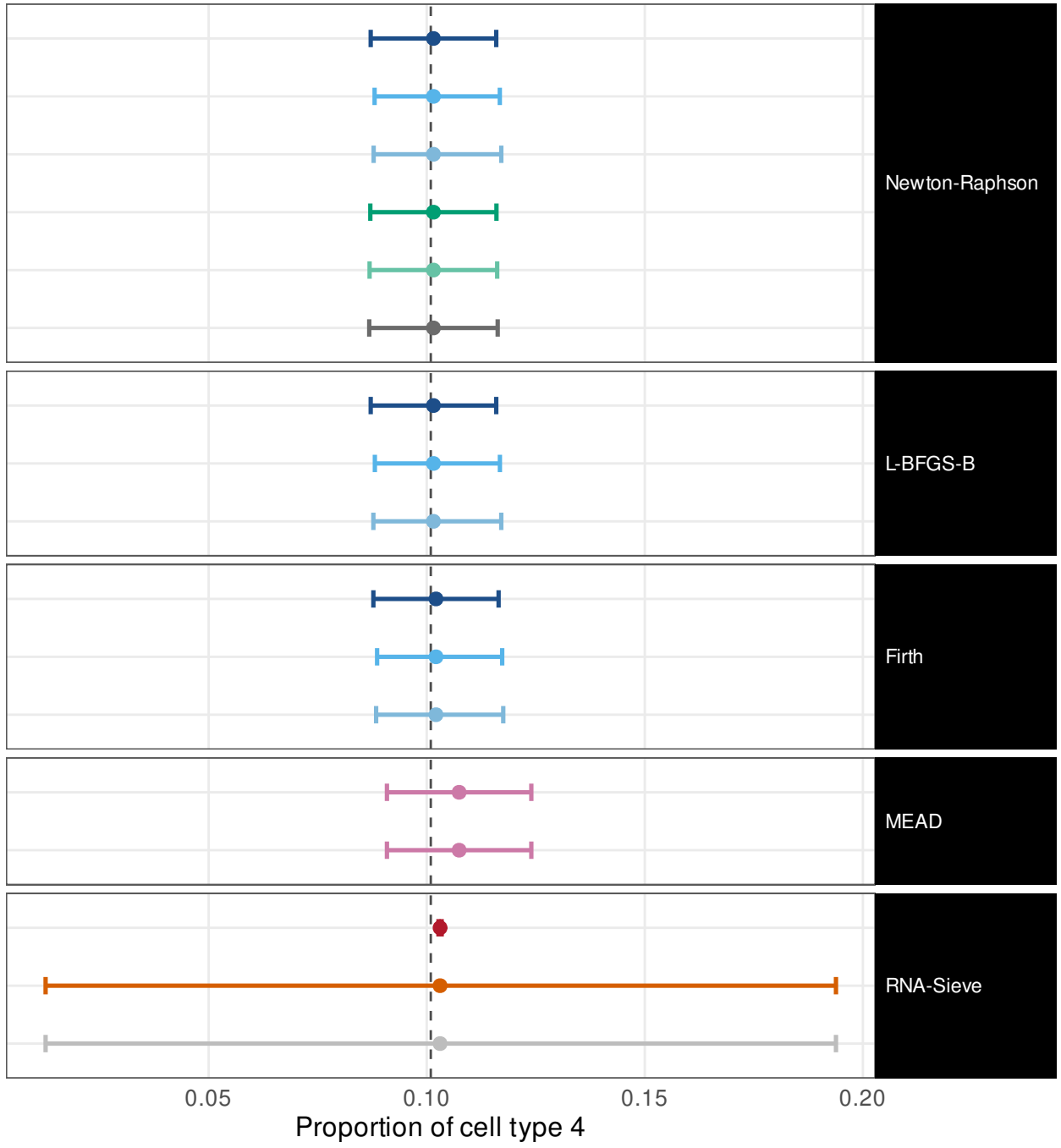


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / unbalanced / low sibling overlap / a = 1 / equal sizes

Wald (delta)	0.94 [0.89, 0.96]	w = 0.0288
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.0287
ILR, REML	0.93 [0.89, 0.96]	w = 0.0293
Profile likelihood	0.94 [0.89, 0.96]	w = 0.0289
Profile, REML	0.94 [0.89, 0.96]	w = 0.0293
Oracle (Fisher at p*)	0.94 [0.9, 0.97]	w = 0.0294
Wald (delta)	0.94 [0.89, 0.96]	w = 0.0288
ILR, mapped back	0.92 [0.88, 0.95]	w = 0.0287
ILR, REML	0.93 [0.89, 0.96]	w = 0.0293
Wald (delta)	0.94 [0.89, 0.96]	w = 0.0287
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.0287
ILR, REML	0.94 [0.9, 0.97]	w = 0.0291
Method's own interval	0.86 [0.81, 0.91]	w = 0.0331
Own interval, transcript fractions	0.86 [0.81, 0.91]	w = 0.0331
Fisher information (Wald)	0.035 [0.017, 0.07]	w = 0.000725
Godambe sandwich	1 [0.98, 1]	w = 0.181
Released code, as published	1 [0.98, 1]	w = 0.181

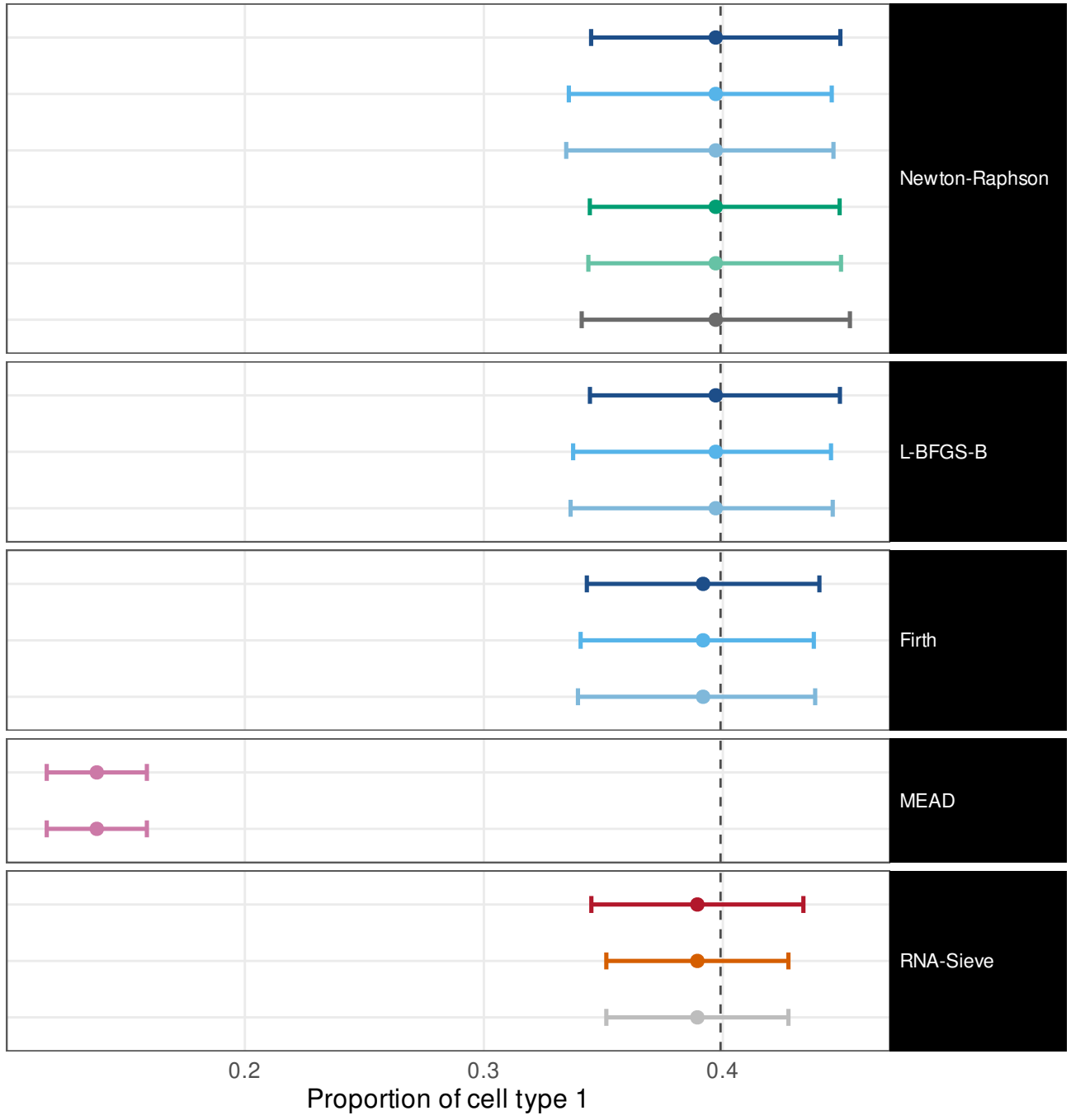


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / unbalanced / low sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.9 [0.86, 0.94]	w = 0.104
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.11
ILR, REML	0.96 [0.92, 0.98]	w = 0.112
Profile likelihood	0.9 [0.86, 0.94]	w = 0.105
Profile, REML	0.92 [0.87, 0.95]	w = 0.106
Oracle (Fisher at p*)	0.92 [0.87, 0.95]	w = 0.112
Wald (delta)	0.91 [0.86, 0.94]	w = 0.105
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.108
ILR, REML	0.95 [0.92, 0.98]	w = 0.11
Wald (delta)	0.92 [0.87, 0.95]	w = 0.0973
ILR, mapped back	0.94 [0.89, 0.96]	w = 0.0976
ILR, REML	0.94 [0.89, 0.96]	w = 0.0992
Method's own interval	0 [0, 0.019]	w = 0.0419
Own interval, transcript fractions	0.9 [0.84, 0.93]	w = 0.0419
Fisher information (Wald)	0.84 [0.78, 0.88]	w = 0.0887
Godambe sandwich	0.79 [0.73, 0.84]	w = 0.0762
Released code, as published	0.79 [0.73, 0.84]	w = 0.0762

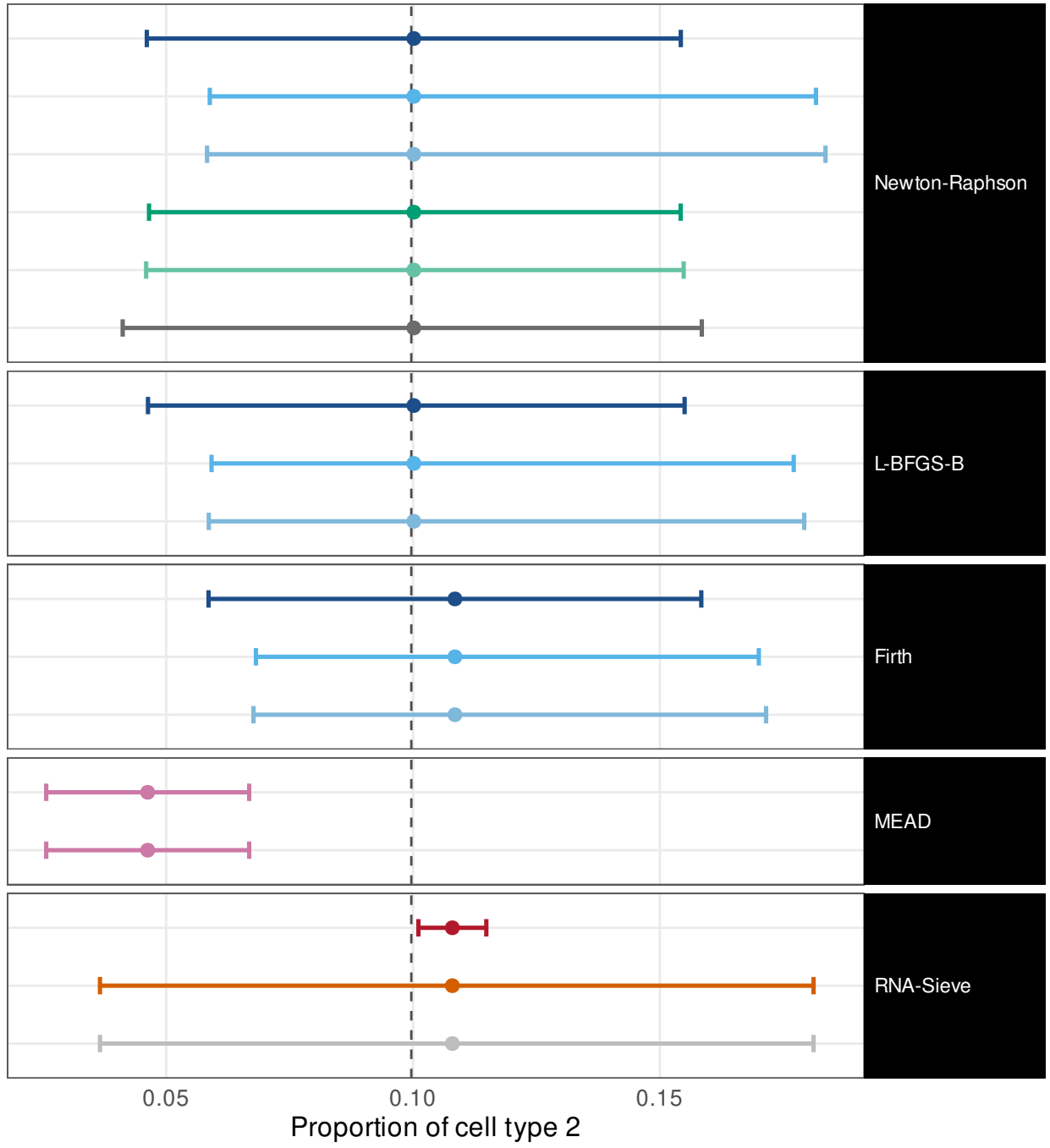


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / unbalanced / low sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.92 [0.87, 0.95]	w = 0.108
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.123
ILR, REML	0.96 [0.92, 0.98]	w = 0.125
Profile likelihood	0.92 [0.87, 0.95]	w = 0.108
Profile, REML	0.92 [0.88, 0.95]	w = 0.109
Oracle (Fisher at p*)	0.92 [0.88, 0.95]	w = 0.117
Wald (delta)	0.92 [0.87, 0.95]	w = 0.109
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.118
ILR, REML	0.95 [0.91, 0.97]	w = 0.121
Wald (delta)	0.92 [0.88, 0.95]	w = 0.0998
ILR, mapped back	0.9 [0.86, 0.94]	w = 0.102
ILR, REML	0.92 [0.87, 0.95]	w = 0.104
Method's own interval	0.005 [0.00088, 0.028]	w = 0.0411
Own interval, transcript fractions	0.77 [0.71, 0.82]	w = 0.0411
Fisher information (Wald)	0.16 [0.12, 0.22]	w = 0.0138
Godambe sandwich	0.98 [0.95, 0.99]	w = 0.145
Released code, as published	0.98 [0.95, 0.99]	w = 0.145

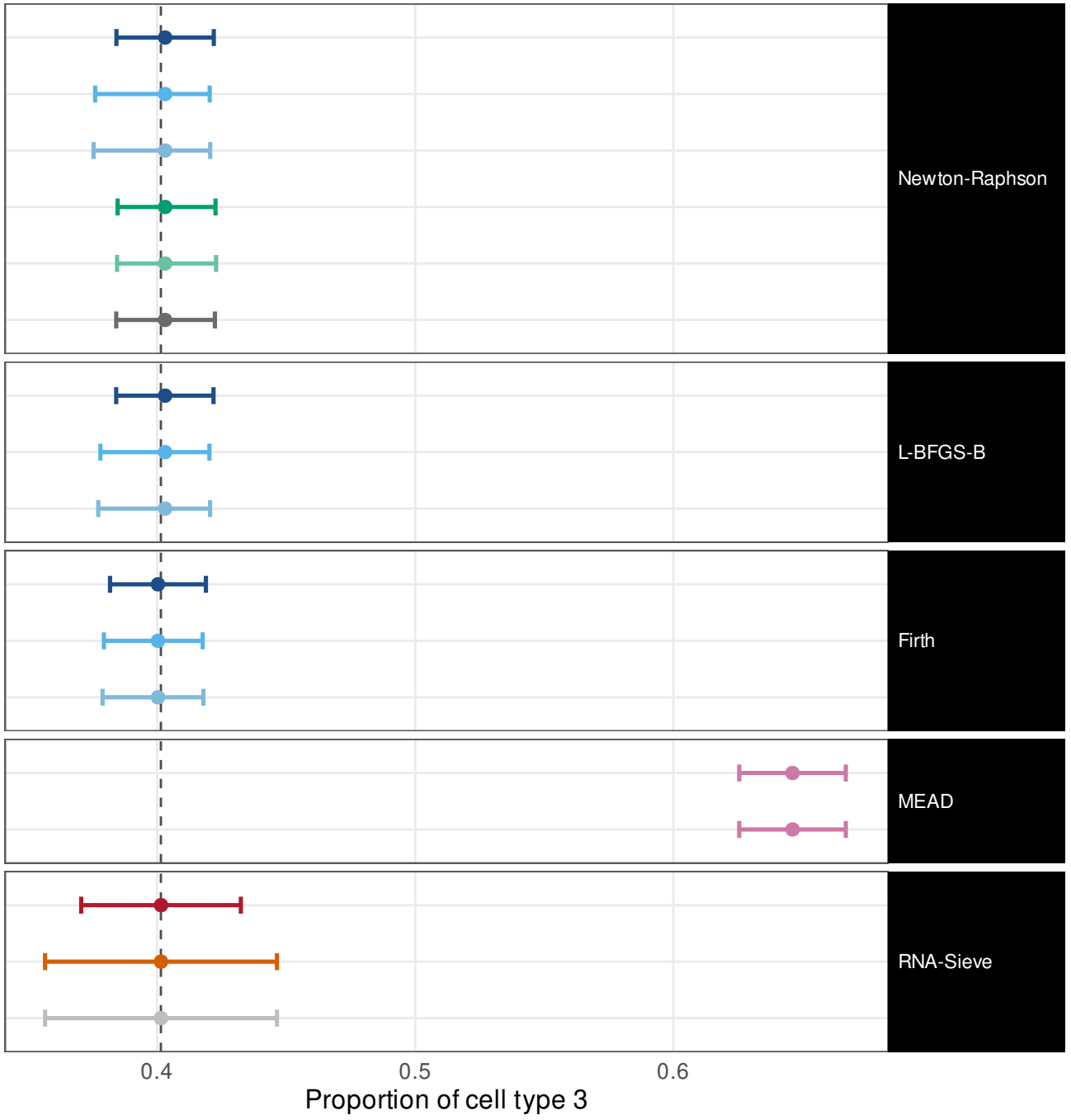


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / unbalanced / low sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.96 [0.93, 0.98]	w = 0.0377
ILR, mapped back	0.97 [0.94, 0.99]	w = 0.0444
ILR, REML	0.98 [0.94, 0.99]	w = 0.0452
Profile likelihood	0.97 [0.94, 0.99]	w = 0.0379
Profile, REML	0.97 [0.94, 0.99]	w = 0.0384
Oracle (Fisher at p*)	0.98 [0.96, 0.99]	w = 0.0382
Wald (delta)	0.96 [0.93, 0.98]	w = 0.0377
ILR, mapped back	0.98 [0.95, 0.99]	w = 0.0422
ILR, REML	0.97 [0.94, 0.99]	w = 0.0433
Wald (delta)	0.96 [0.93, 0.98]	w = 0.0371
ILR, mapped back	0.96 [0.93, 0.98]	w = 0.0383
ILR, REML	0.96 [0.93, 0.98]	w = 0.039
Method's own interval	0 [0, 0.019]	w = 0.0413
Own interval, transcript fractions	0.47 [0.4, 0.54]	w = 0.0413
Fisher information (Wald)	1 [0.97, 1]	w = 0.0618
Godambe sandwich	1 [0.98, 1]	w = 0.0898
Released code, as published	1 [0.98, 1]	w = 0.0898

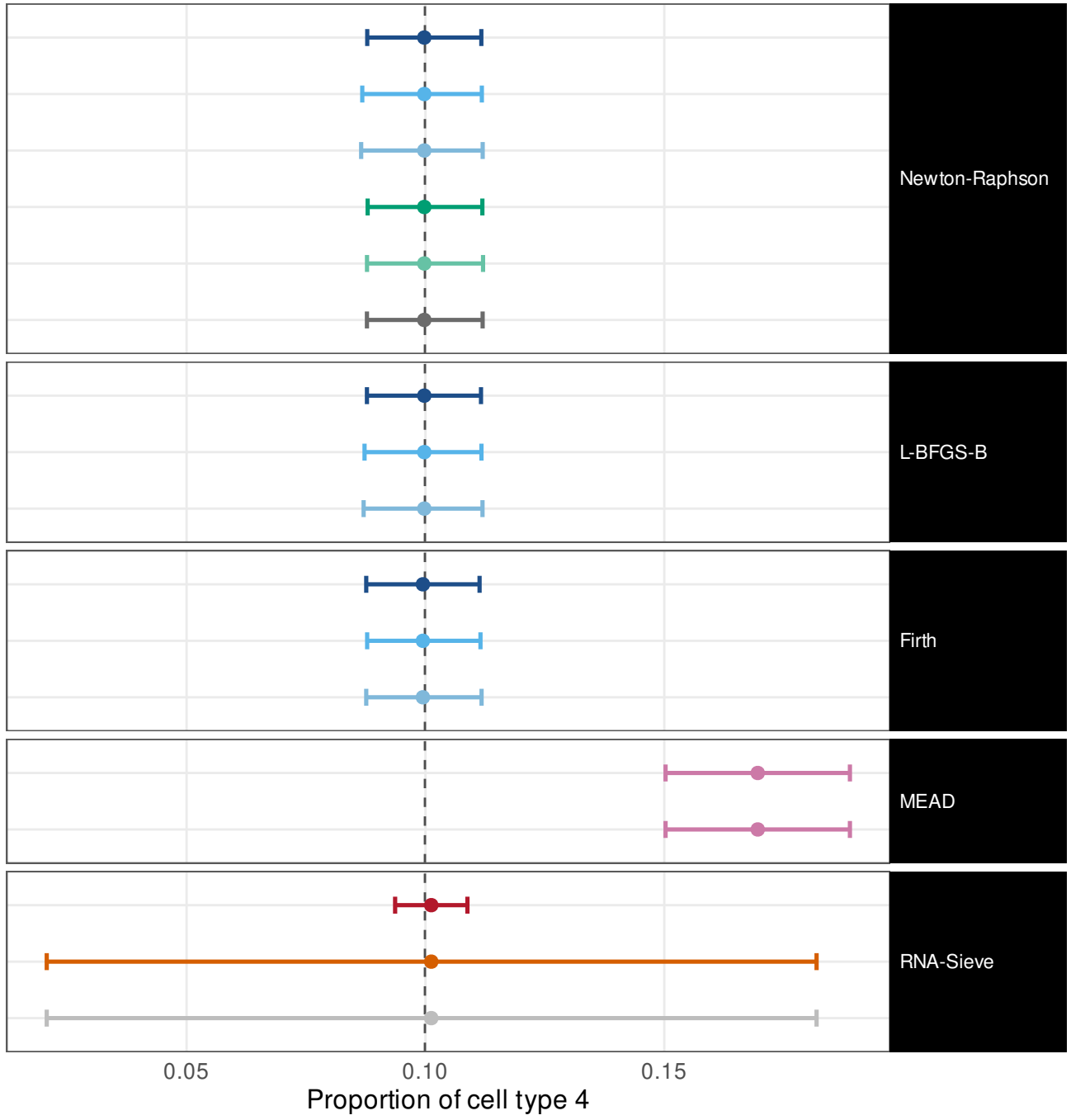


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / unbalanced / low sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.93 [0.89, 0.96]	w = 0.0239
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.025
ILR, REML	0.94 [0.9, 0.97]	w = 0.0254
Profile likelihood	0.93 [0.89, 0.96]	w = 0.024
Profile, REML	0.94 [0.89, 0.96]	w = 0.0243
Oracle (Fisher at p*)	0.94 [0.9, 0.97]	w = 0.0242
Wald (delta)	0.93 [0.89, 0.96]	w = 0.0239
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.0245
ILR, REML	0.94 [0.9, 0.97]	w = 0.0249
Wald (delta)	0.94 [0.89, 0.96]	w = 0.0237
ILR, mapped back	0.94 [0.89, 0.96]	w = 0.0237
ILR, REML	0.94 [0.9, 0.97]	w = 0.0241
Method's own interval	0 [0, 0.019]	w = 0.0386
Own interval, transcript fractions	0.92 [0.88, 0.95]	w = 0.0386
Fisher information (Wald)	0.78 [0.72, 0.83]	w = 0.0151
Godambe sandwich	1 [0.98, 1]	w = 0.161
Released code, as published	1 [0.98, 1]	w = 0.161

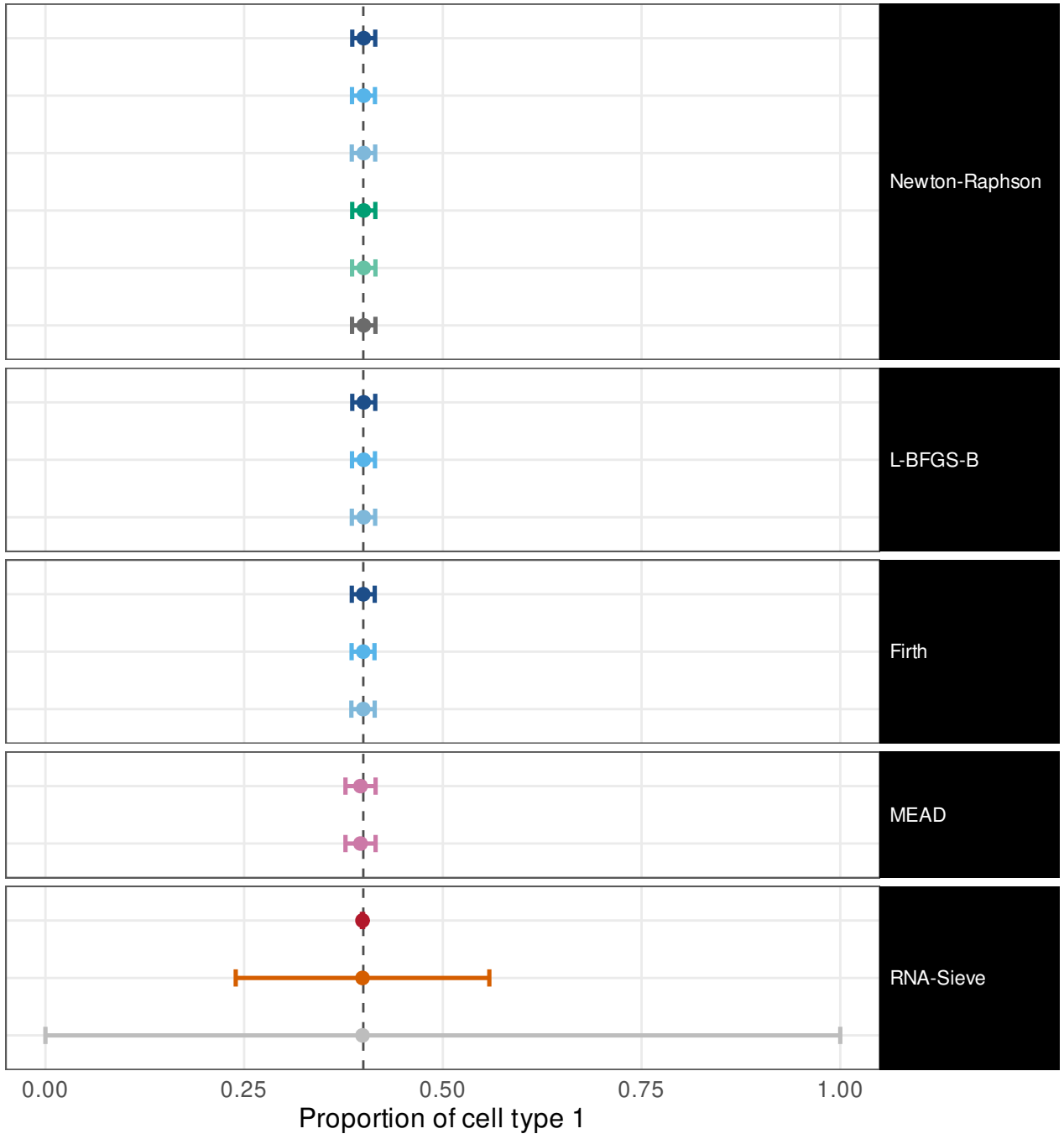


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / unbalanced / low sibling overlap / a = 5 / equal sizes

Wald (delta)	0.95 [0.91, 0.97]	w = 0.0289
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.0289
ILR, REML	0.96 [0.92, 0.98]	w = 0.0294
Profile likelihood	0.95 [0.91, 0.97]	w = 0.0291
Profile, REML	0.96 [0.92, 0.98]	w = 0.0294
Oracle (Fisher at p*)	0.96 [0.93, 0.98]	w = 0.0294
Wald (delta)	0.95 [0.91, 0.97]	w = 0.0289
ILR, mapped back	0.96 [0.93, 0.98]	w = 0.0289
ILR, REML	0.96 [0.92, 0.98]	w = 0.0293
Wald (delta)	0.96 [0.92, 0.98]	w = 0.0289
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.0288
ILR, REML	0.96 [0.92, 0.98]	w = 0.0293
Method's own interval	0.98 [0.94, 0.99]	w = 0.0379
Own interval, transcript fractions	0.98 [0.94, 0.99]	w = 0.0379
Fisher information (Wald)	0.065 [0.038, 0.11]	w = 0.00149
Godambe sandwich	1 [0.98, 1]	w = 0.319
Released code, as published	1 [0.98, 1]	w = 1

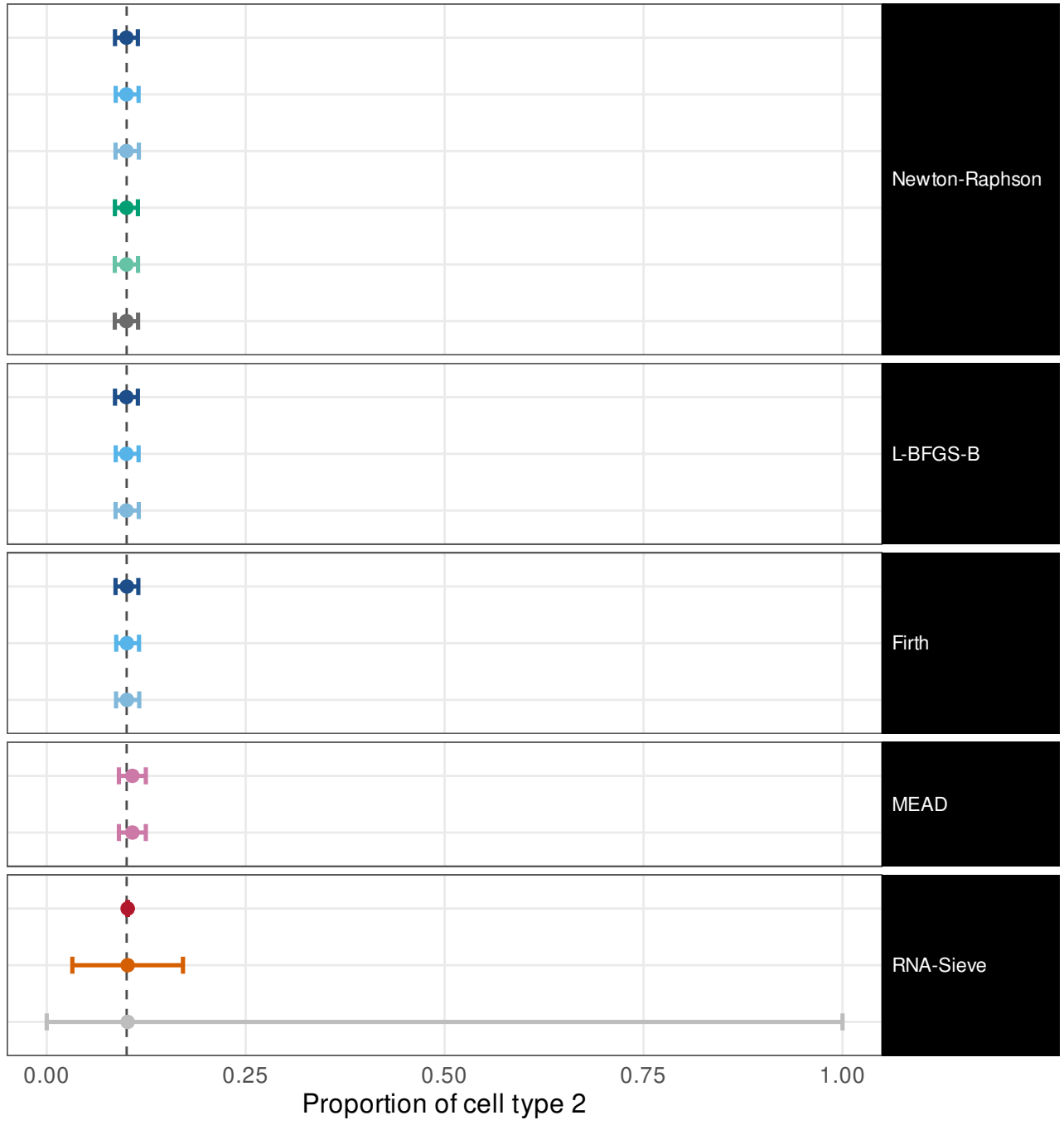


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / unbalanced / low sibling overlap / a = 5 / equal sizes

Wald (delta)	0.94 [0.9, 0.97]	w = 0.0287
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.0287
ILR, REML	0.96 [0.92, 0.98]	w = 0.0293
Profile likelihood	0.94 [0.9, 0.97]	w = 0.0289
Profile, REML	0.94 [0.9, 0.97]	w = 0.0292
Oracle (Fisher at p*)	0.94 [0.9, 0.97]	w = 0.0293
Wald (delta)	0.94 [0.9, 0.97]	w = 0.0287
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.0287
ILR, REML	0.96 [0.93, 0.98]	w = 0.0291
Wald (delta)	0.94 [0.9, 0.97]	w = 0.0286
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.0286
ILR, REML	0.96 [0.92, 0.98]	w = 0.0291
Method's own interval	0.9 [0.84, 0.93]	w = 0.0338
Own interval, transcript fractions	0.9 [0.84, 0.93]	w = 0.0338
Fisher information (Wald)	0.05 [0.027, 0.09]	w = 0.00058
Godambe sandwich	1 [0.98, 1]	w = 0.139
Released code, as published	1 [0.98, 1]	w = 1

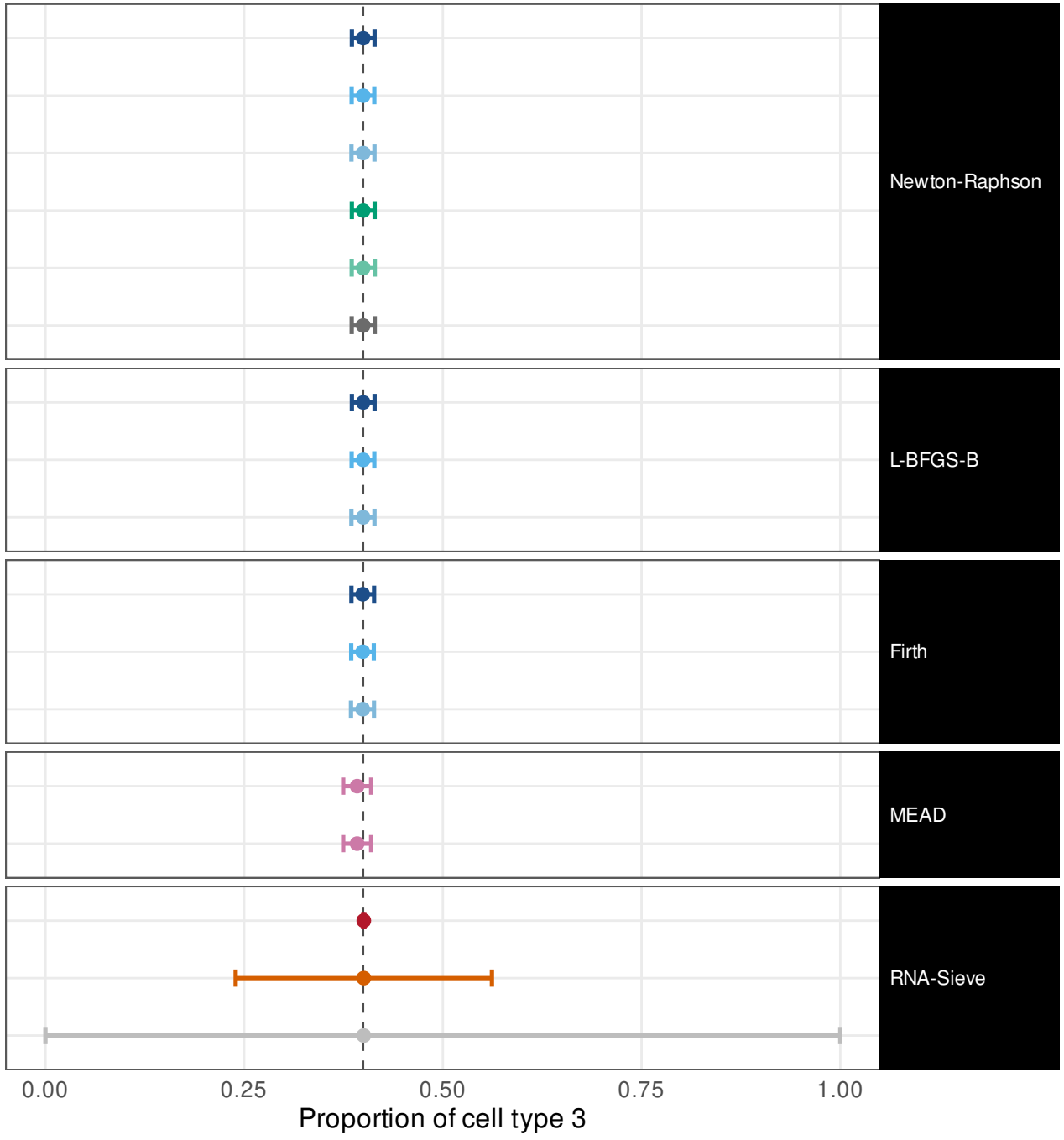


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / unbalanced / low sibling overlap / a = 5 / equal sizes

Wald (delta)	0.92 [0.87, 0.95]	w = 0.0286
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.0285
ILR, REML	0.92 [0.87, 0.95]	w = 0.029
Profile likelihood	0.92 [0.87, 0.95]	w = 0.0287
Profile, REML	0.92 [0.88, 0.95]	w = 0.0291
Oracle (Fisher at p*)	0.94 [0.89, 0.96]	w = 0.029
Wald (delta)	0.92 [0.87, 0.95]	w = 0.0286
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.0285
ILR, REML	0.93 [0.89, 0.96]	w = 0.029
Wald (delta)	0.92 [0.88, 0.95]	w = 0.0285
ILR, mapped back	0.92 [0.88, 0.95]	w = 0.0284
ILR, REML	0.92 [0.87, 0.95]	w = 0.029
Method's own interval	0.9 [0.85, 0.93]	w = 0.0352
Own interval, transcript fractions	0.9 [0.85, 0.93]	w = 0.0352
Fisher information (Wald)	0.075 [0.046, 0.12]	w = 0.00151
Godambe sandwich	1 [0.98, 1]	w = 0.322
Released code, as published	1 [0.98, 1]	w = 1

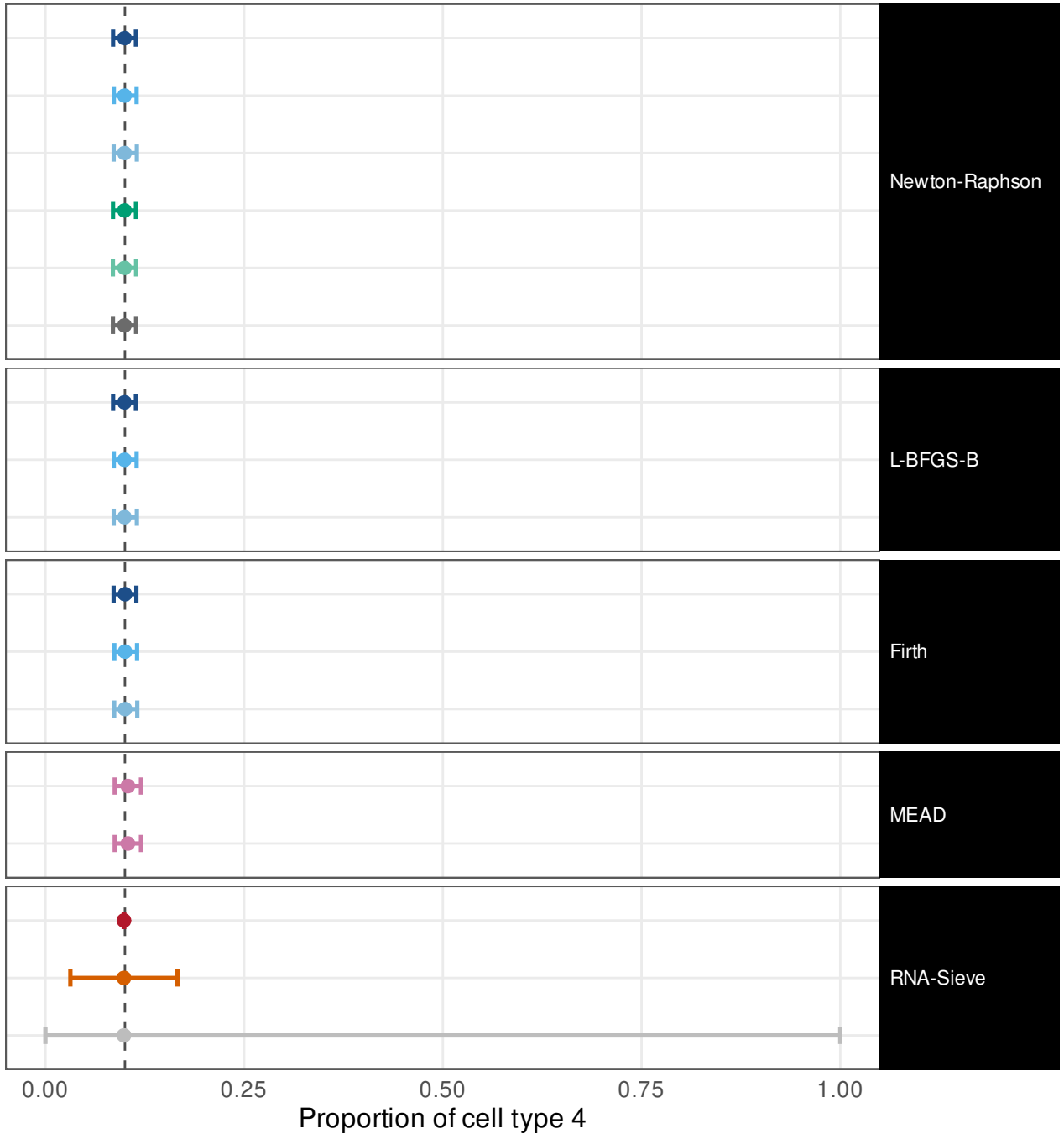


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / unbalanced / low sibling overlap / a = 5 / equal sizes

Wald (delta)	0.93 [0.89, 0.96]	w = 0.0287
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.0287
ILR, REML	0.92 [0.88, 0.95]	w = 0.0292
Profile likelihood	0.93 [0.89, 0.96]	w = 0.0289
Profile, REML	0.94 [0.89, 0.96]	w = 0.0292
Oracle (Fisher at p*)	0.95 [0.91, 0.97]	w = 0.0292
Wald (delta)	0.93 [0.89, 0.96]	w = 0.0287
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.0287
ILR, REML	0.94 [0.9, 0.97]	w = 0.0292
Wald (delta)	0.93 [0.89, 0.96]	w = 0.0287
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.0287
ILR, REML	0.93 [0.89, 0.96]	w = 0.0291
Method's own interval	0.92 [0.88, 0.95]	w = 0.033
Own interval, transcript fractions	0.92 [0.88, 0.95]	w = 0.033
Fisher information (Wald)	0.04 [0.02, 0.077]	w = 0.000564
Godambe sandwich	1 [0.98, 1]	w = 0.135
Released code, as published	1 [0.98, 1]	w = 1

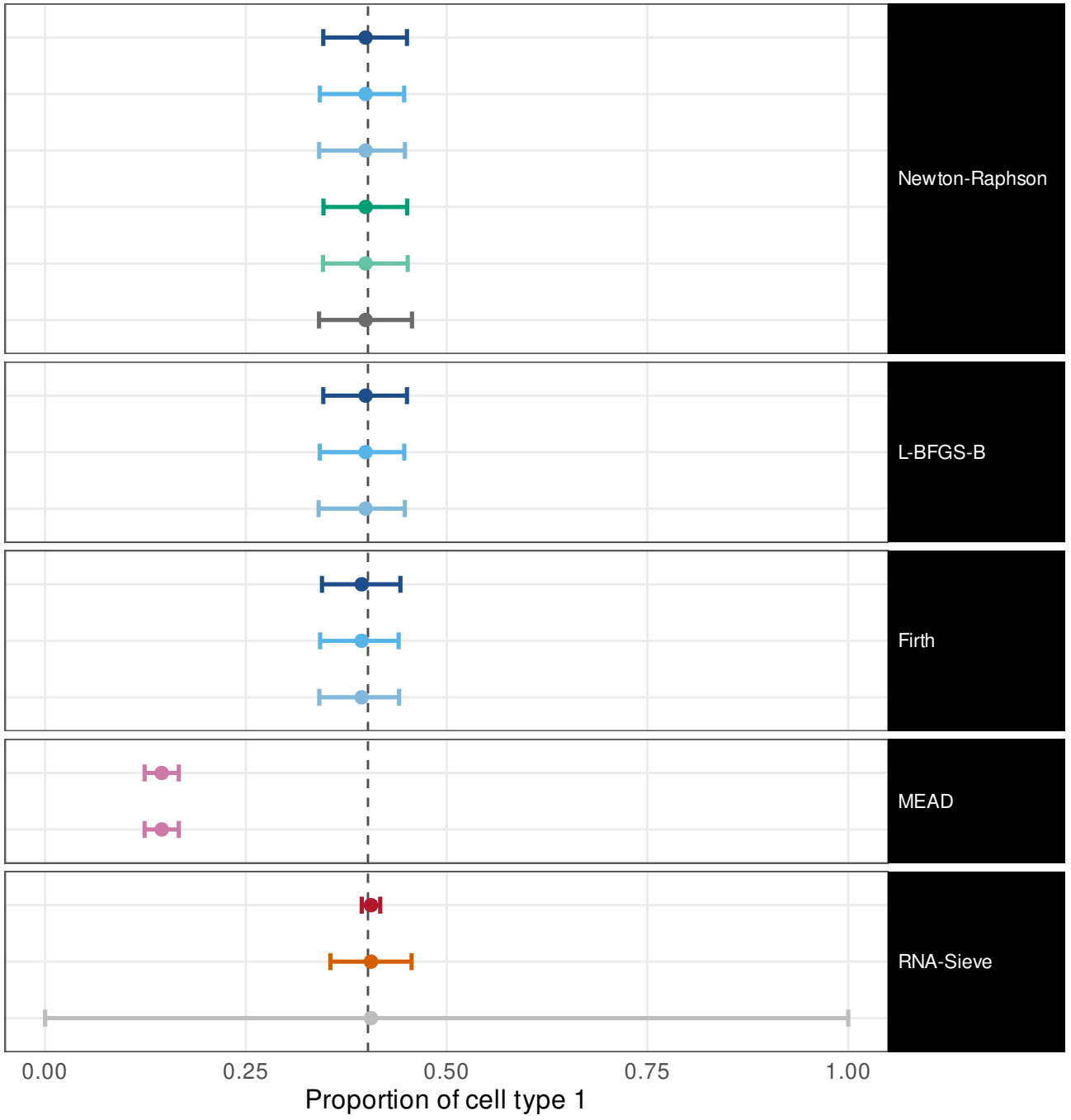


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / unbalanced / low sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.92 [0.88, 0.95]	w = 0.104
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.105
ILR, REML	0.94 [0.9, 0.97]	w = 0.107
Profile likelihood	0.92 [0.88, 0.95]	w = 0.104
Profile, REML	0.93 [0.89, 0.96]	w = 0.106
Oracle (Fisher at p*)	0.94 [0.9, 0.97]	w = 0.116
Wald (delta)	0.92 [0.88, 0.95]	w = 0.104
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.105
ILR, REML	0.95 [0.92, 0.98]	w = 0.107
Wald (delta)	0.92 [0.88, 0.95]	w = 0.0976
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.0978
ILR, REML	0.94 [0.9, 0.97]	w = 0.0994
Method's own interval	0 [0, 0.019]	w = 0.0425
Own interval, transcript fractions	0.86 [0.8, 0.9]	w = 0.0425
Fisher information (Wald)	0.36 [0.29, 0.42]	w = 0.023
Godambe sandwich	0.92 [0.87, 0.95]	w = 0.101
Released code, as published	1 [0.98, 1]	w = 1

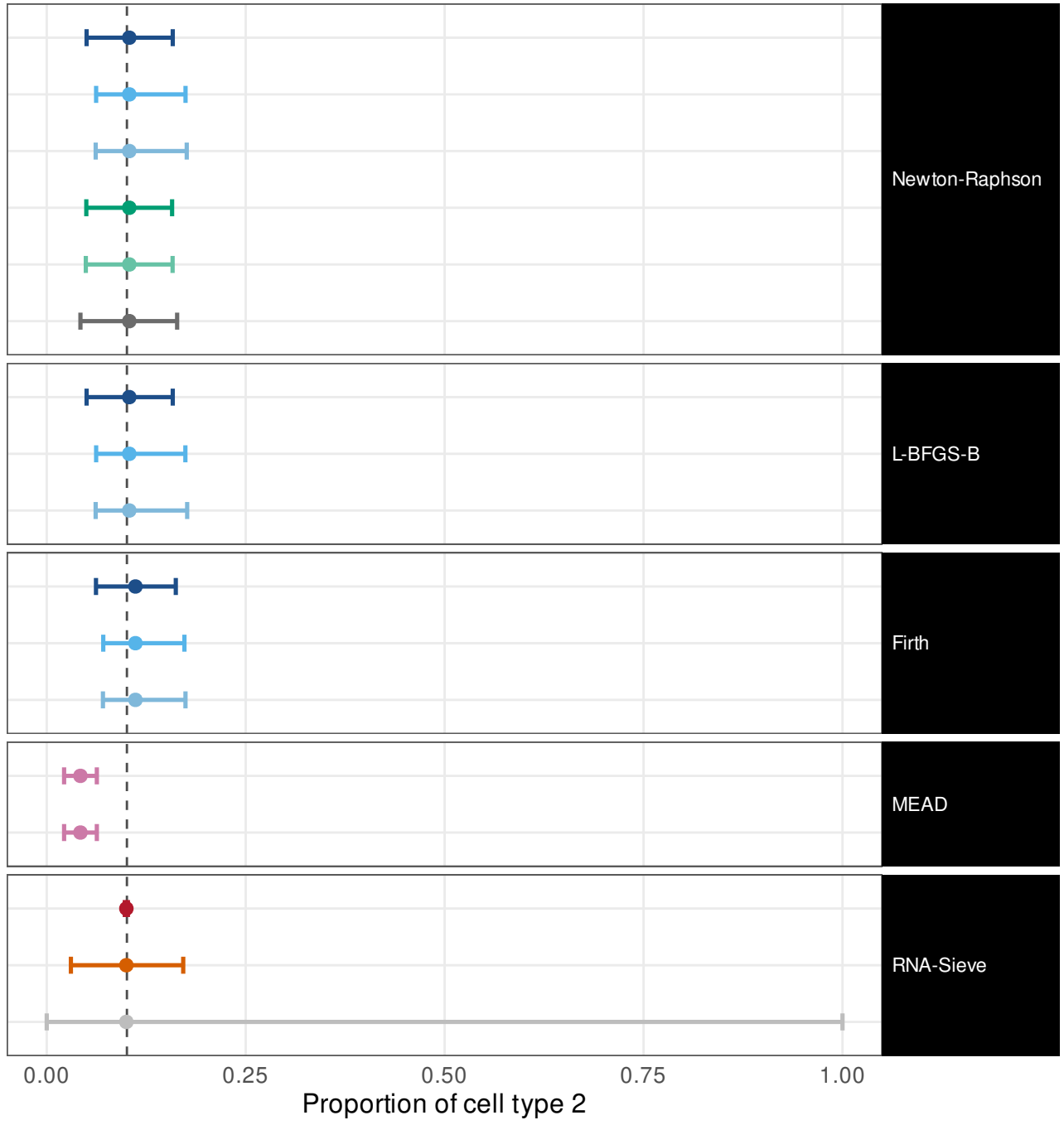


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / unbalanced / low sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.93 [0.89, 0.96]	w = 0.108
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.112
ILR, REML	0.94 [0.9, 0.97]	w = 0.114
Profile likelihood	0.94 [0.89, 0.96]	w = 0.108
Profile, REML	0.94 [0.9, 0.97]	w = 0.109
Oracle (Fisher at p*)	0.95 [0.91, 0.97]	w = 0.122
Wald (delta)	0.93 [0.89, 0.96]	w = 0.108
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.112
ILR, REML	0.94 [0.9, 0.97]	w = 0.115
Wald (delta)	0.94 [0.9, 0.97]	w = 0.1
ILR, mapped back	0.91 [0.86, 0.94]	w = 0.102
ILR, REML	0.9 [0.85, 0.93]	w = 0.104
Method's own interval	0 [0, 0.019]	w = 0.0411
Own interval, transcript fractions	0.89 [0.84, 0.93]	w = 0.0411
Fisher information (Wald)	0.07 [0.042, 0.11]	w = 0.00387
Godambe sandwich	1 [0.97, 1]	w = 0.141
Released code, as published	1 [0.98, 1]	w = 1

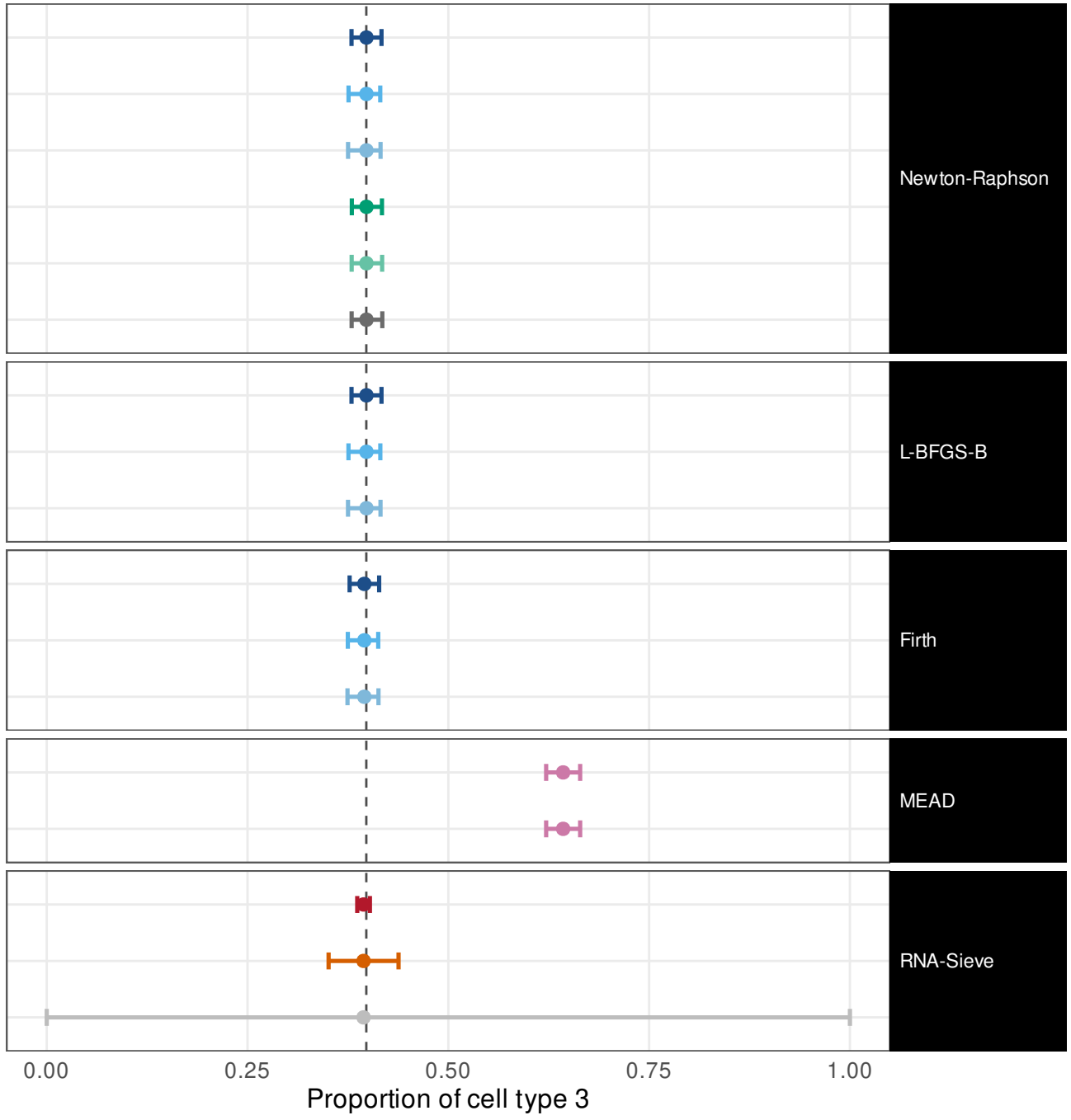


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / unbalanced / low sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.94 [0.9, 0.97]	w = 0.0373
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.0395
ILR, REML	0.95 [0.91, 0.97]	w = 0.0404
Profile likelihood	0.93 [0.89, 0.96]	w = 0.0375
Profile, REML	0.94 [0.9, 0.97]	w = 0.0379
Oracle (Fisher at p*)	0.94 [0.9, 0.97]	w = 0.0382
Wald (delta)	0.94 [0.9, 0.97]	w = 0.0373
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.0397
ILR, REML	0.94 [0.9, 0.97]	w = 0.0404
Wald (delta)	0.93 [0.89, 0.96]	w = 0.0368
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.0378
ILR, REML	0.93 [0.89, 0.96]	w = 0.0386
Method's own interval	0 [0, 0.019]	w = 0.0423
Own interval, transcript fractions	0.42 [0.36, 0.49]	w = 0.0423
Fisher information (Wald)	0.5 [0.43, 0.57]	w = 0.0158
Godambe sandwich	1 [0.98, 1]	w = 0.0871
Released code, as published	1 [0.98, 1]	w = 1

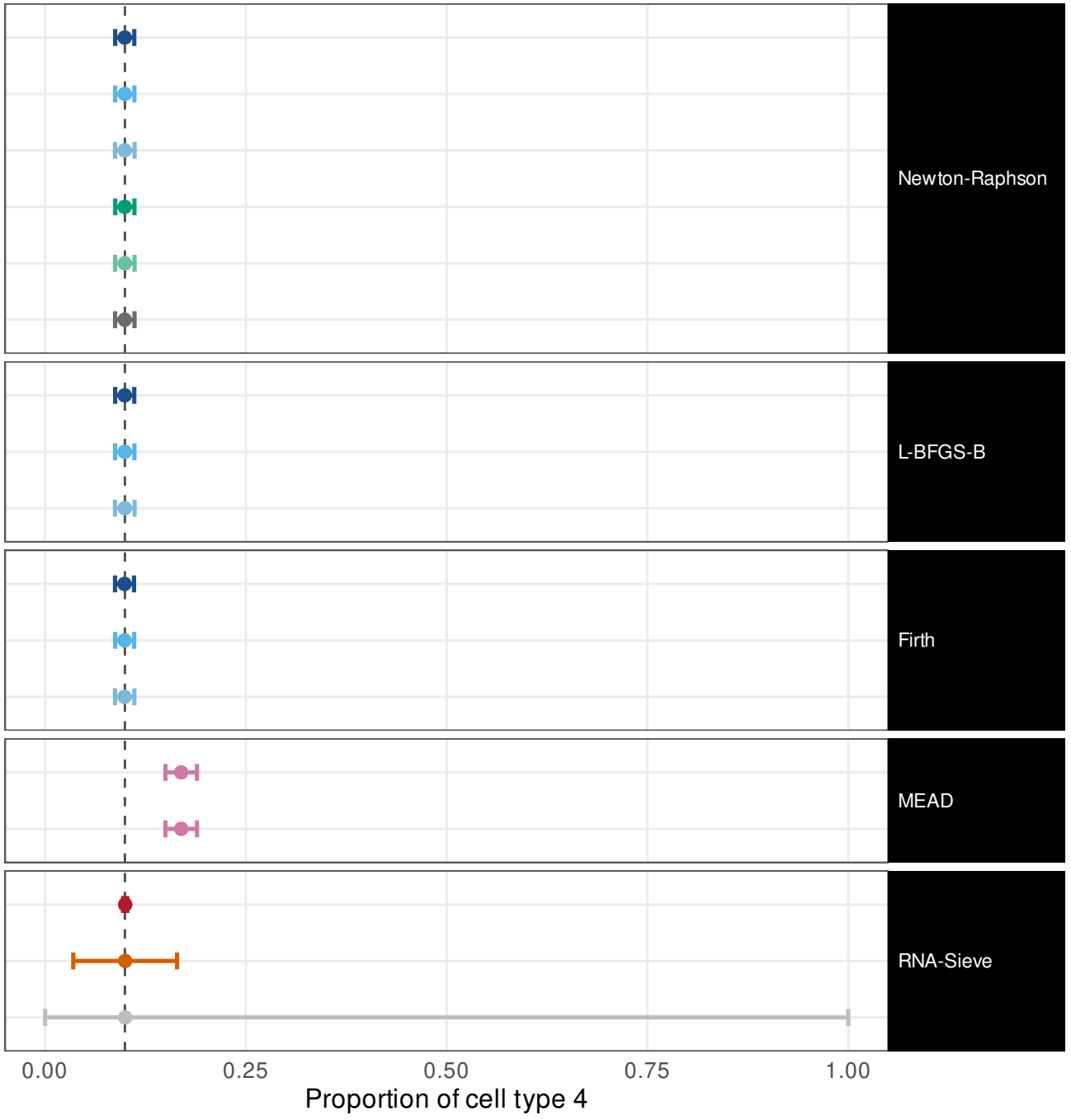


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / unbalanced / low sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.96 [0.93, 0.98]	w = 0.0238
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.0239
ILR, REML	0.97 [0.94, 0.99]	w = 0.0244
Profile likelihood	0.96 [0.93, 0.98]	w = 0.0239
Profile, REML	0.96 [0.93, 0.98]	w = 0.0242
Oracle (Fisher at p*)	0.97 [0.93, 0.99]	w = 0.0242
Wald (delta)	0.96 [0.93, 0.98]	w = 0.0238
ILR, mapped back	0.97 [0.94, 0.99]	w = 0.024
ILR, REML	0.97 [0.94, 0.99]	w = 0.0244
Wald (delta)	0.96 [0.92, 0.98]	w = 0.0236
ILR, mapped back	0.96 [0.93, 0.98]	w = 0.0236
ILR, REML	0.96 [0.92, 0.98]	w = 0.0241
Method's own interval	0 [0, 0.019]	w = 0.0393
Own interval, transcript fractions	0.96 [0.93, 0.98]	w = 0.0393
Fisher information (Wald)	0.3 [0.25, 0.37]	w = 0.00519
Godambe sandwich	1 [0.98, 1]	w = 0.129
Released code, as published	1 [0.98, 1]	w = 1

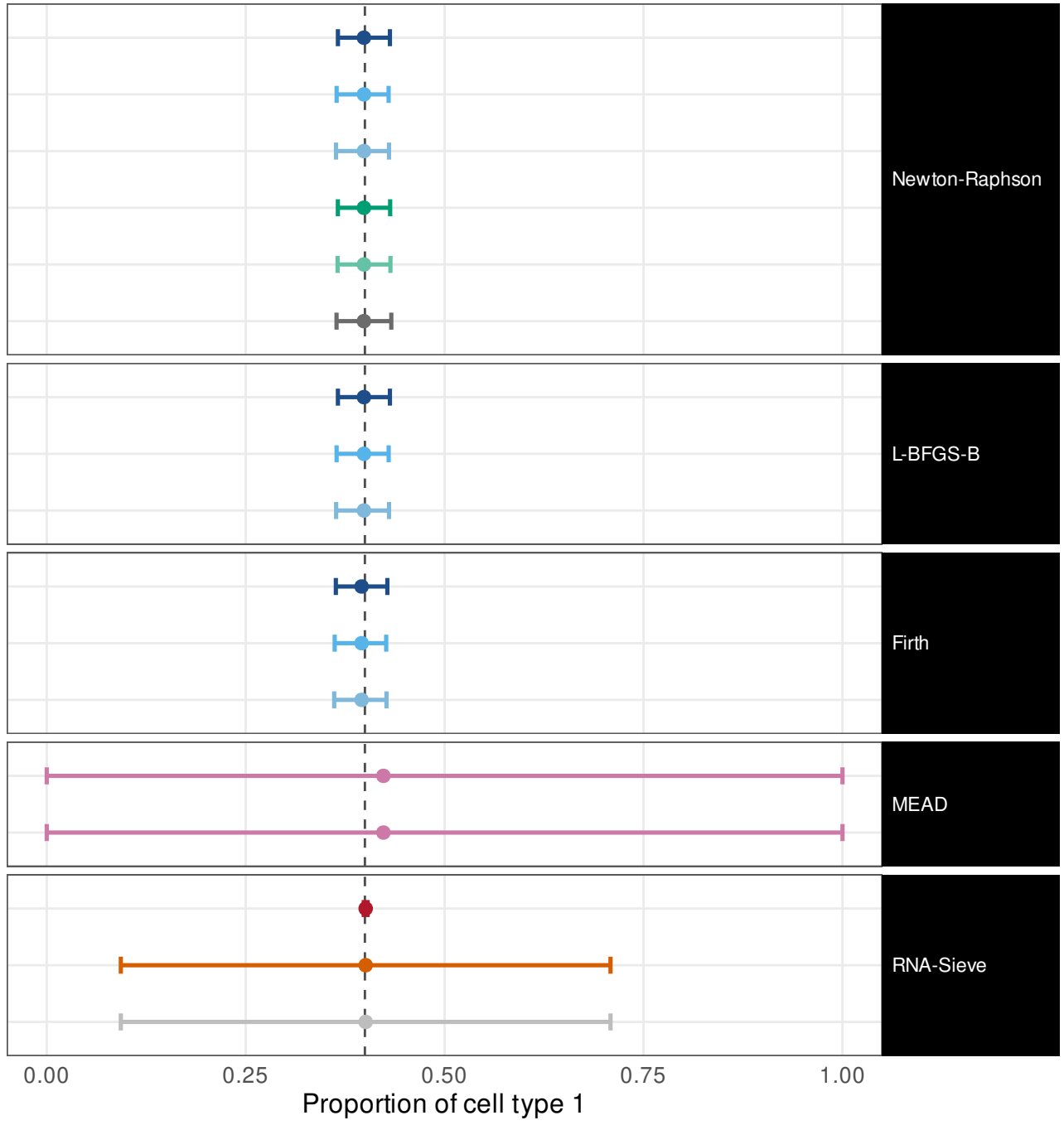


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / unbalanced / high sibling overlap / a = 1 / equal sizes

Wald (delta)	0.95 [0.91, 0.97]	w = 0.0654
ILR, mapped back	0.96 [0.93, 0.98]	w = 0.0653
ILR, REML	0.96 [0.92, 0.98]	w = 0.0666
Profile likelihood	0.96 [0.92, 0.98]	w = 0.0657
Profile, REML	0.96 [0.92, 0.98]	w = 0.0665
Oracle (Fisher at p*)	0.98 [0.94, 0.99]	w = 0.0688
Wald (delta)	0.95 [0.91, 0.97]	w = 0.0654
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.0654
ILR, REML	0.96 [0.92, 0.98]	w = 0.0664
Wald (delta)	0.94 [0.9, 0.97]	w = 0.0647
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.0647
ILR, REML	0.93 [0.89, 0.96]	w = 0.0657
Method's own interval	1 [0.98, 1]	w = 1
Own interval, transcript fractions	1 [0.98, 1]	w = 1
Fisher information (Wald)	0.11 [0.074, 0.16]	w = 0.00477
Godambe sandwich	1 [0.98, 1]	w = 0.615
Released code, as published	1 [0.98, 1]	w = 0.615

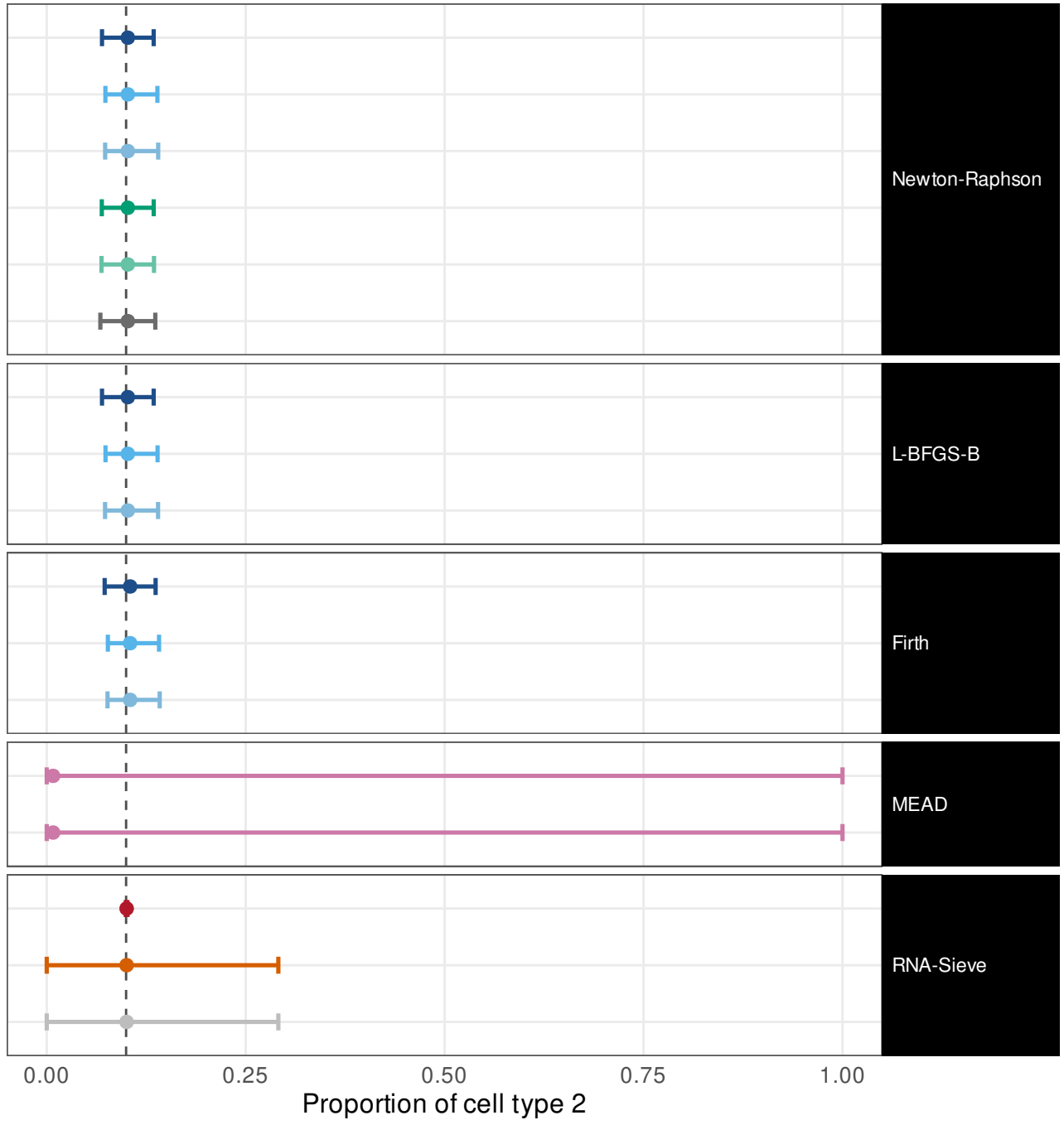


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / unbalanced / high sibling overlap / a = 1 / equal sizes

Wald (delta)	0.94 [0.9, 0.97]	w = 0.0649
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.0654
ILR, REML	0.94 [0.89, 0.96]	w = 0.0666
Profile likelihood	0.94 [0.9, 0.97]	w = 0.0652
Profile, REML	0.94 [0.9, 0.97]	w = 0.066
Oracle (Fisher at p*)	0.97 [0.94, 0.99]	w = 0.069
Wald (delta)	0.94 [0.9, 0.97]	w = 0.0649
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.0654
ILR, REML	0.93 [0.89, 0.96]	w = 0.0666
Wald (delta)	0.94 [0.89, 0.96]	w = 0.064
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.0643
ILR, REML	0.93 [0.89, 0.96]	w = 0.0656
Method's own interval	1 [0.98, 1]	w = 1
Own interval, transcript fractions	1 [0.98, 1]	w = 1
Fisher information (Wald)	0.04 [0.02, 0.077]	w = 0.00216
Godambe sandwich	1 [0.98, 1]	w = 0.291
Released code, as published	1 [0.98, 1]	w = 0.291

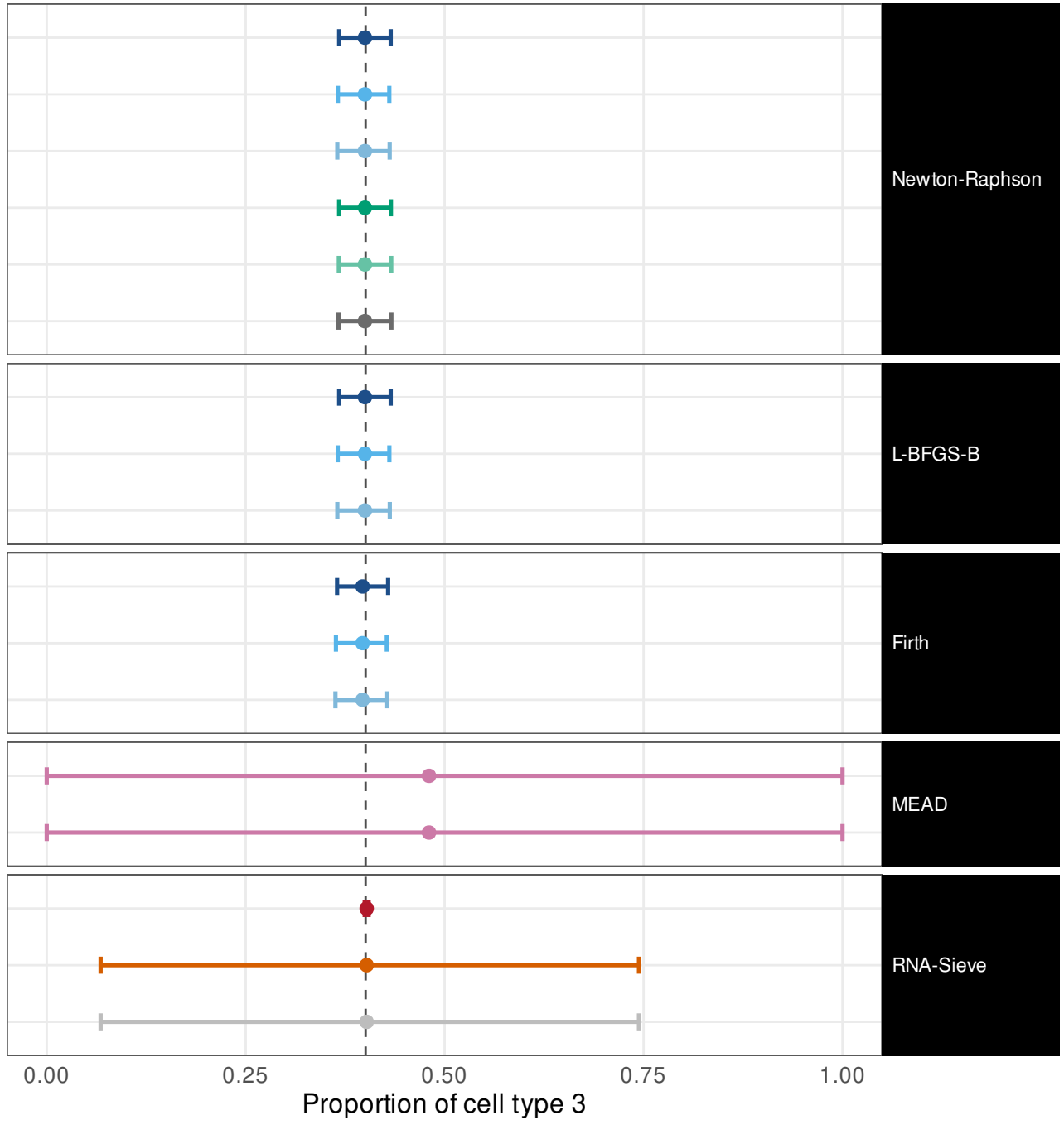


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / unbalanced / high sibling overlap / a = 1 / equal sizes

Wald (delta)	0.95 [0.91, 0.97]	w = 0.0647
ILR, mapped back	0.96 [0.93, 0.98]	w = 0.0647
ILR, REML	0.97 [0.94, 0.99]	w = 0.0658
Profile likelihood	0.95 [0.91, 0.97]	w = 0.0651
Profile, REML	0.96 [0.92, 0.98]	w = 0.0658
Oracle (Fisher at p*)	0.96 [0.92, 0.98]	w = 0.0664
Wald (delta)	0.95 [0.91, 0.97]	w = 0.0647
ILR, mapped back	0.97 [0.94, 0.99]	w = 0.0649
ILR, REML	0.96 [0.93, 0.98]	w = 0.0659
Wald (delta)	0.96 [0.93, 0.98]	w = 0.0641
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.064
ILR, REML	0.96 [0.92, 0.98]	w = 0.0652
Method's own interval	1 [0.98, 1]	w = 1
Own interval, transcript fractions	1 [0.98, 1]	w = 1
Fisher information (Wald)	0.14 [0.094, 0.19]	w = 0.00491
Godambe sandwich	1 [0.98, 1]	w = 0.677
Released code, as published	1 [0.98, 1]	w = 0.677

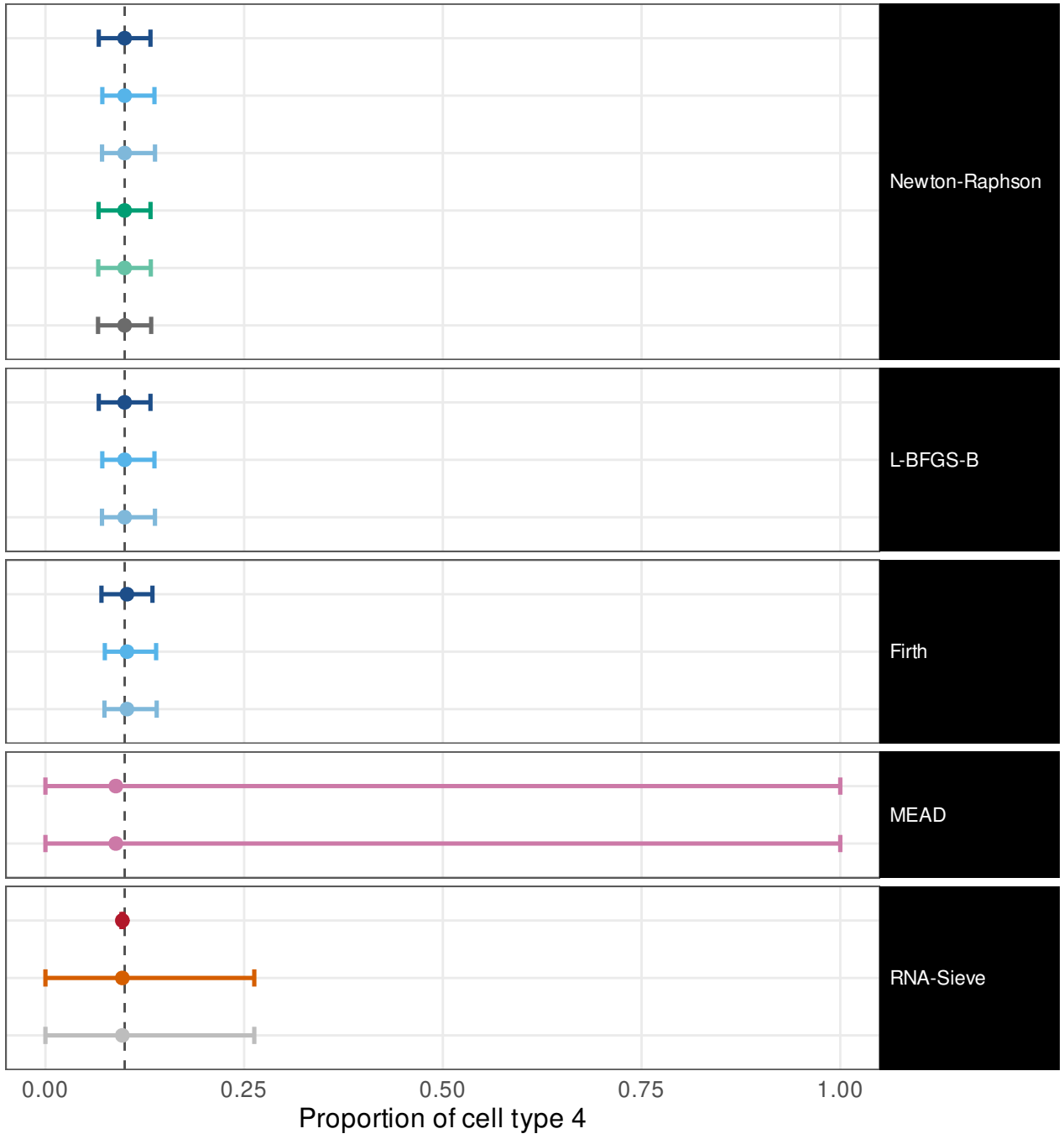


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / unbalanced / high sibling overlap / a = 1 / equal sizes

Wald (delta)	0.92 [0.88, 0.95]	w = 0.0651
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.0656
ILR, REML	0.96 [0.92, 0.98]	w = 0.0666
Profile likelihood	0.93 [0.89, 0.96]	w = 0.0654
Profile, REML	0.94 [0.9, 0.97]	w = 0.0661
Oracle (Fisher at p*)	0.94 [0.9, 0.97]	w = 0.0669
Wald (delta)	0.92 [0.88, 0.95]	w = 0.0651
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.0656
ILR, REML	0.96 [0.93, 0.98]	w = 0.0666
Wald (delta)	0.94 [0.89, 0.96]	w = 0.0642
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.0645
ILR, REML	0.94 [0.9, 0.97]	w = 0.0657
Method's own interval	1 [0.98, 1]	w = 1
Own interval, transcript fractions	1 [0.98, 1]	w = 1
Fisher information (Wald)	0.055 [0.031, 0.096]	w = 0.0022
Godambe sandwich	1 [0.98, 1]	w = 0.263
Released code, as published	1 [0.98, 1]	w = 0.263

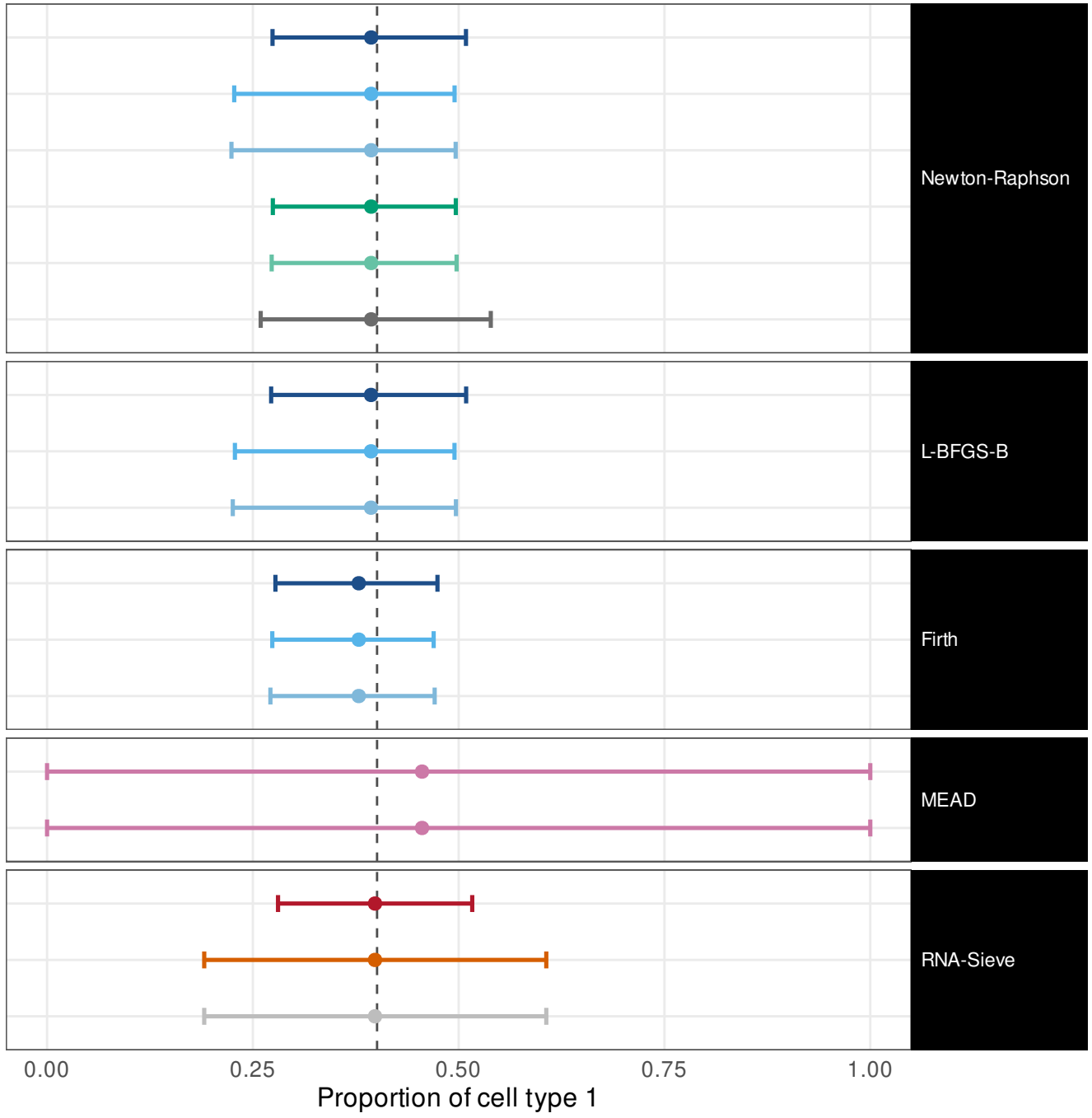


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / unbalanced / high sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.97 [0.93, 0.99]	w = 0.235
ILR, mapped back	0.98 [0.96, 0.99]	w = 0.268
ILR, REML	0.98 [0.96, 0.99]	w = 0.272
Profile likelihood	0.97 [0.94, 0.99]	w = 0.222
Profile, REML	0.98 [0.94, 0.99]	w = 0.225
Oracle (Fisher at p*)	0.95 [0.9, 0.97]	w = 0.28
Wald (delta)	0.98 [0.95, 0.99]	w = 0.237
ILR, mapped back	0.98 [0.96, 0.99]	w = 0.267
ILR, REML	0.98 [0.96, 0.99]	w = 0.271
Wald (delta)	0.96 [0.92, 0.98]	w = 0.197
ILR, mapped back	0.97 [0.94, 0.99]	w = 0.196
ILR, REML	0.98 [0.95, 0.99]	w = 0.2
Method's own interval	1 [0.98, 1]	w = 1
Own interval, transcript fractions	1 [0.98, 1]	w = 1
Fisher information (Wald)	0.92 [0.87, 0.95]	w = 0.236
Godambe sandwich	1 [0.98, 1]	w = 0.416
Released code, as published	1 [0.98, 1]	w = 0.416

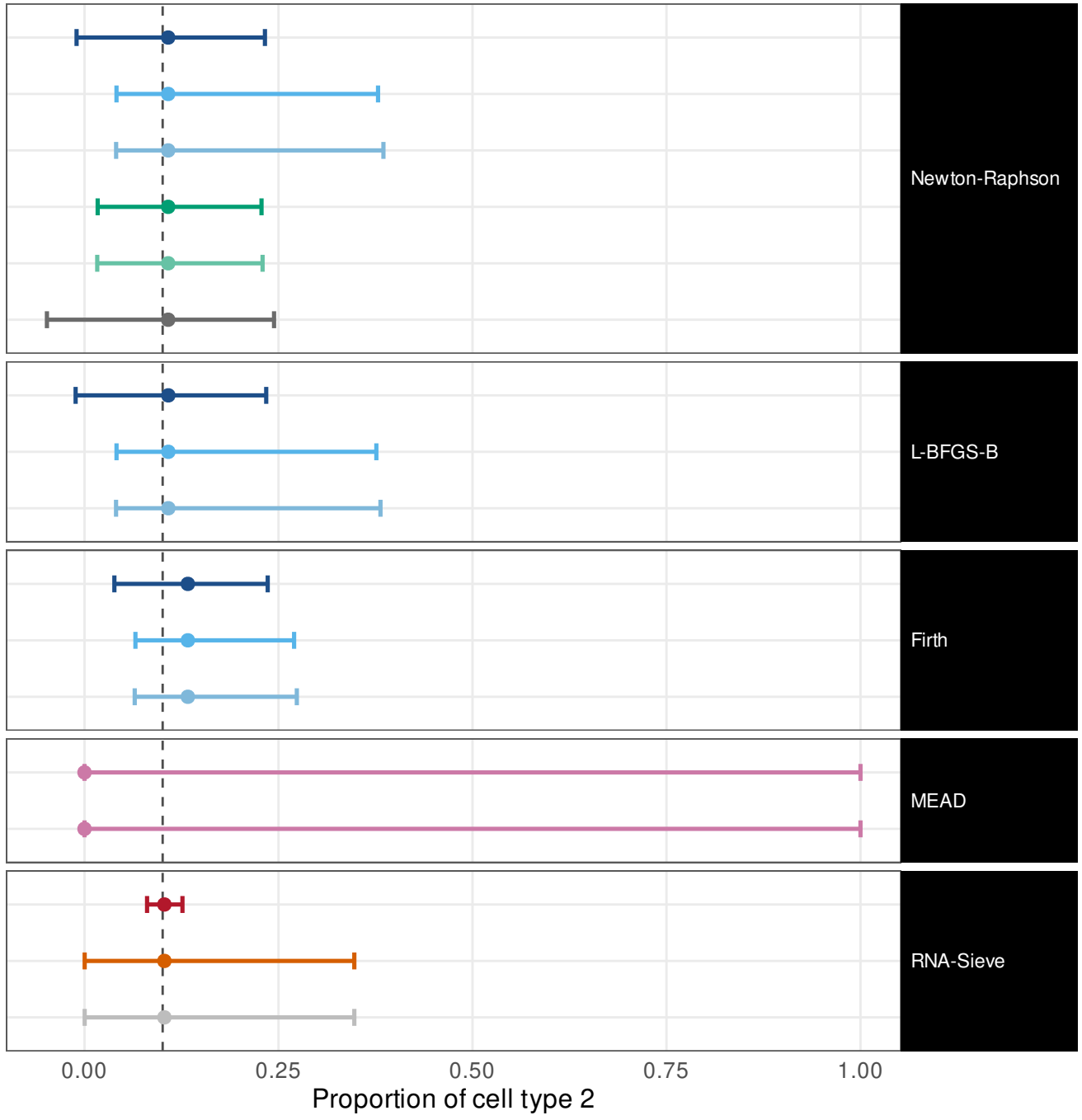


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / unbalanced / high sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.95 [0.91, 0.98]	w = 0.243
ILR, mapped back	0.94 [0.9, 0.96]	w = 0.337
ILR, REML	0.94 [0.9, 0.97]	w = 0.344
Profile likelihood	0.96 [0.92, 0.98]	w = 0.211
Profile, REML	0.96 [0.92, 0.98]	w = 0.213
Oracle (Fisher at p*)	0.95 [0.9, 0.97]	w = 0.293
Wald (delta)	0.96 [0.93, 0.98]	w = 0.246
ILR, mapped back	0.94 [0.9, 0.96]	w = 0.335
ILR, REML	0.94 [0.9, 0.96]	w = 0.341
Wald (delta)	0.94 [0.9, 0.96]	w = 0.198
ILR, mapped back	0.86 [0.8, 0.9]	w = 0.204
ILR, REML	0.87 [0.81, 0.91]	w = 0.209
Method's own interval	1 [0.98, 1]	w = 1
Own interval, transcript fractions	1 [0.98, 1]	w = 1
Fisher information (Wald)	0.25 [0.2, 0.31]	w = 0.0456
Godambe sandwich	1 [0.98, 1]	w = 0.347
Released code, as published	1 [0.98, 1]	w = 0.347

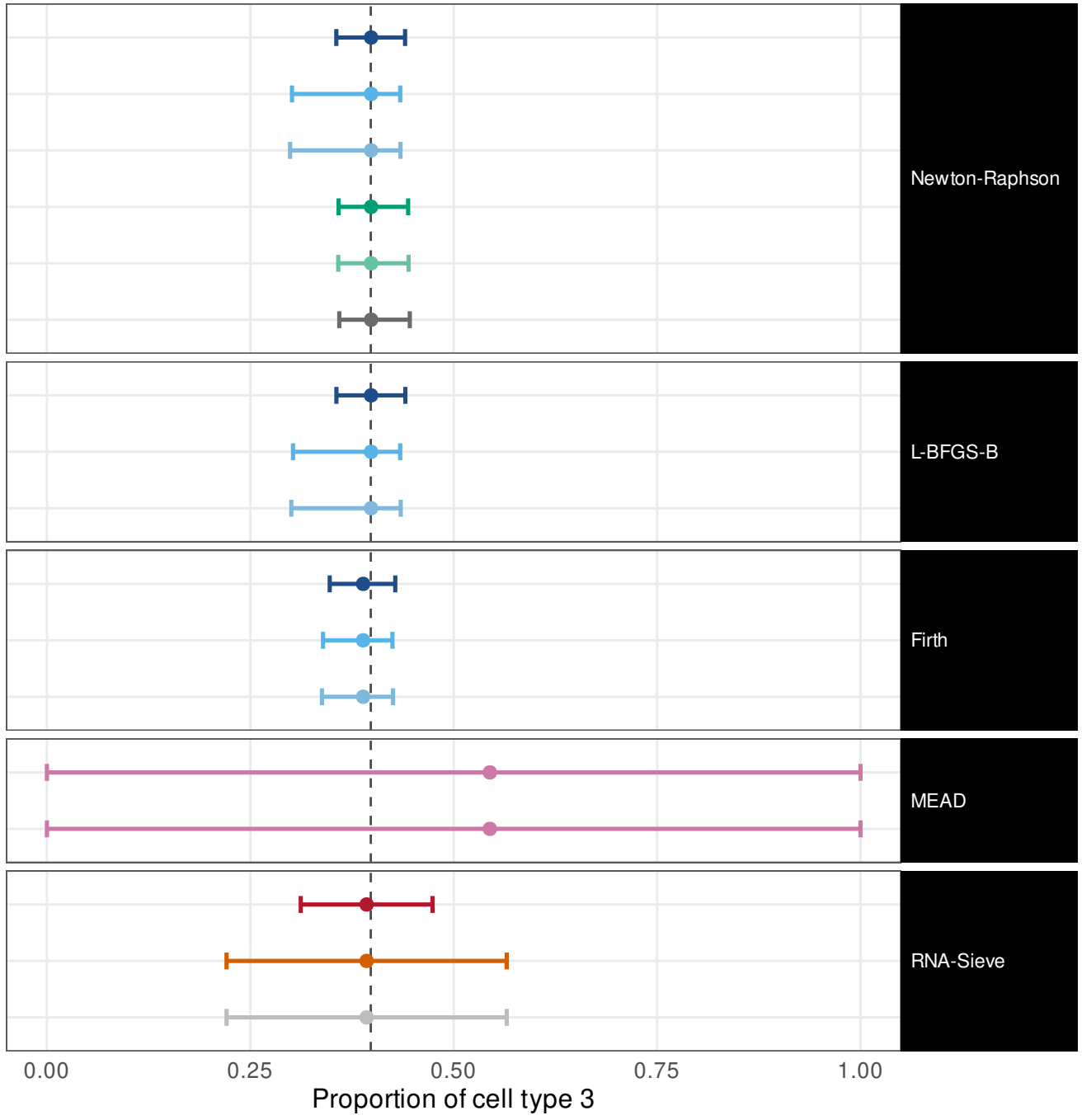


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / unbalanced / high sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.95 [0.91, 0.98]	w = 0.0845
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.133
ILR, REML	0.94 [0.9, 0.97]	w = 0.136
Profile likelihood	0.96 [0.92, 0.98]	w = 0.0855
Profile, REML	0.96 [0.92, 0.98]	w = 0.0865
Oracle (Fisher at p*)	0.97 [0.93, 0.99]	w = 0.0867
Wald (delta)	0.95 [0.91, 0.98]	w = 0.0847
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.132
ILR, REML	0.95 [0.91, 0.98]	w = 0.134
Wald (delta)	0.93 [0.88, 0.96]	w = 0.0806
ILR, mapped back	0.92 [0.88, 0.95]	w = 0.0853
ILR, REML	0.92 [0.88, 0.95]	w = 0.0874
Method's own interval	1 [0.98, 1]	w = 1
Own interval, transcript fractions	1 [0.98, 1]	w = 1
Fisher information (Wald)	1 [0.97, 1]	w = 0.162
Godambe sandwich	1 [0.98, 1]	w = 0.344
Released code, as published	1 [0.98, 1]	w = 0.344

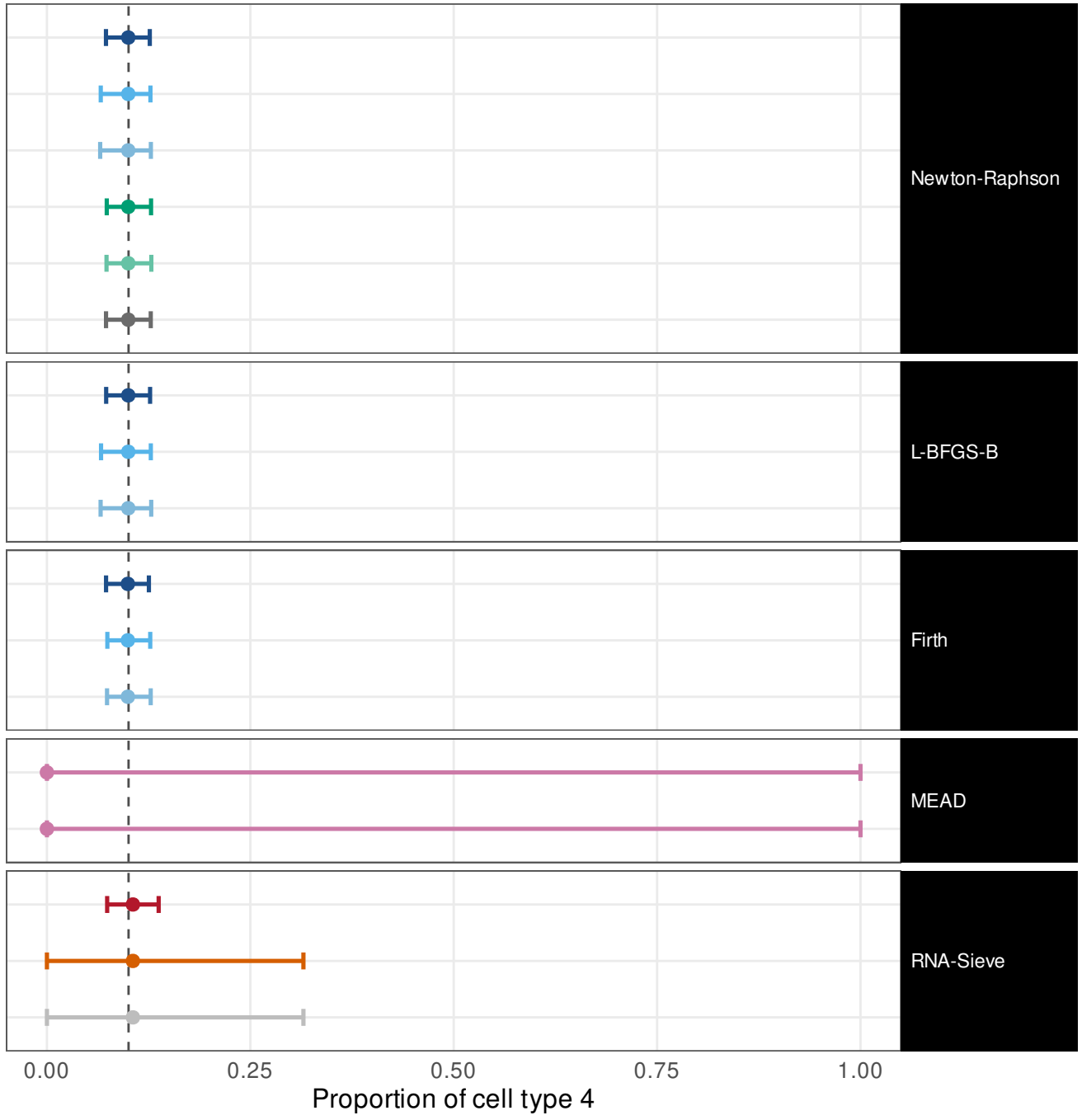


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / unbalanced / high sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.96 [0.92, 0.98]	w = 0.0539
ILR, mapped back	0.97 [0.94, 0.99]	w = 0.0611
ILR, REML	0.97 [0.94, 0.99]	w = 0.0623
Profile likelihood	0.95 [0.91, 0.97]	w = 0.0544
Profile, REML	0.96 [0.92, 0.98]	w = 0.0551
Oracle (Fisher at p*)	0.95 [0.9, 0.97]	w = 0.055
Wald (delta)	0.95 [0.91, 0.98]	w = 0.054
ILR, mapped back	0.98 [0.95, 0.99]	w = 0.0612
ILR, REML	0.98 [0.95, 0.99]	w = 0.0622
Wald (delta)	0.96 [0.93, 0.98]	w = 0.0526
ILR, mapped back	0.97 [0.93, 0.99]	w = 0.0527
ILR, REML	0.98 [0.96, 0.99]	w = 0.0537
Method's own interval	1 [0.98, 1]	w = 1
Own interval, transcript fractions	1 [0.98, 1]	w = 1
Fisher information (Wald)	0.98 [0.94, 0.99]	w = 0.0633
Godambe sandwich	1 [0.98, 1]	w = 0.315
Released code, as published	1 [0.98, 1]	w = 0.315

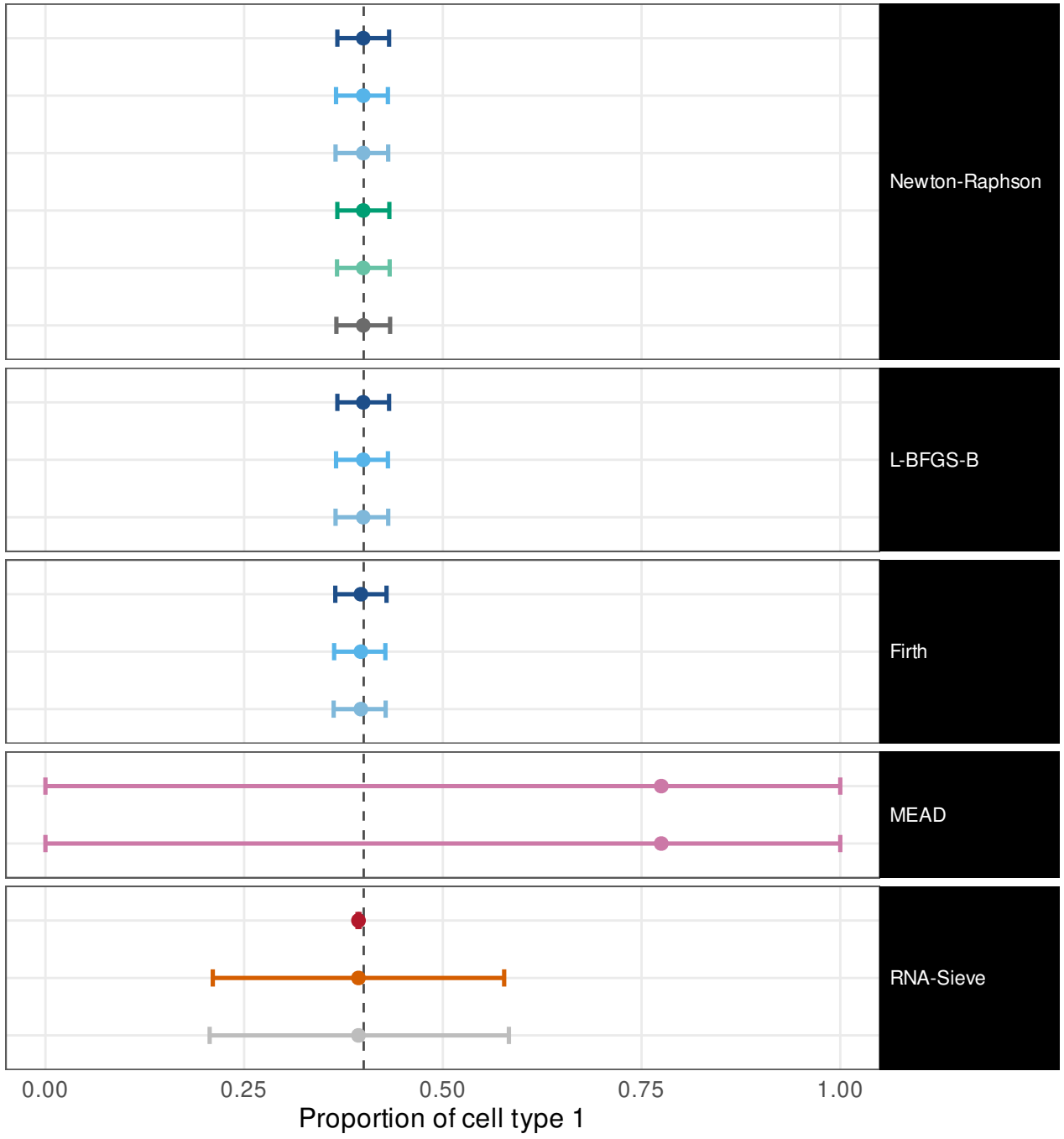


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / unbalanced / high sibling overlap / a = 5 / equal sizes

Wald (delta)	0.94 [0.9, 0.97]	w = 0.0651
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.0652
ILR, REML	0.96 [0.92, 0.98]	w = 0.0662
Profile likelihood	0.94 [0.9, 0.97]	w = 0.0654
Profile, REML	0.94 [0.9, 0.97]	w = 0.0662
Oracle (Fisher at p*)	0.96 [0.92, 0.98]	w = 0.0676
Wald (delta)	0.94 [0.9, 0.97]	w = 0.0651
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.0653
ILR, REML	0.95 [0.91, 0.97]	w = 0.0663
Wald (delta)	0.94 [0.89, 0.96]	w = 0.0645
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.0644
ILR, REML	0.93 [0.89, 0.96]	w = 0.0654
Method's own interval	1 [0.98, 1]	w = 1
Own interval, transcript fractions	1 [0.98, 1]	w = 1
Fisher information (Wald)	0.02 [0.0078, 0.05]	w = 0.00243
Godambe sandwich	1 [0.98, 1]	w = 0.367
Released code, as published	1 [0.98, 1]	w = 0.376

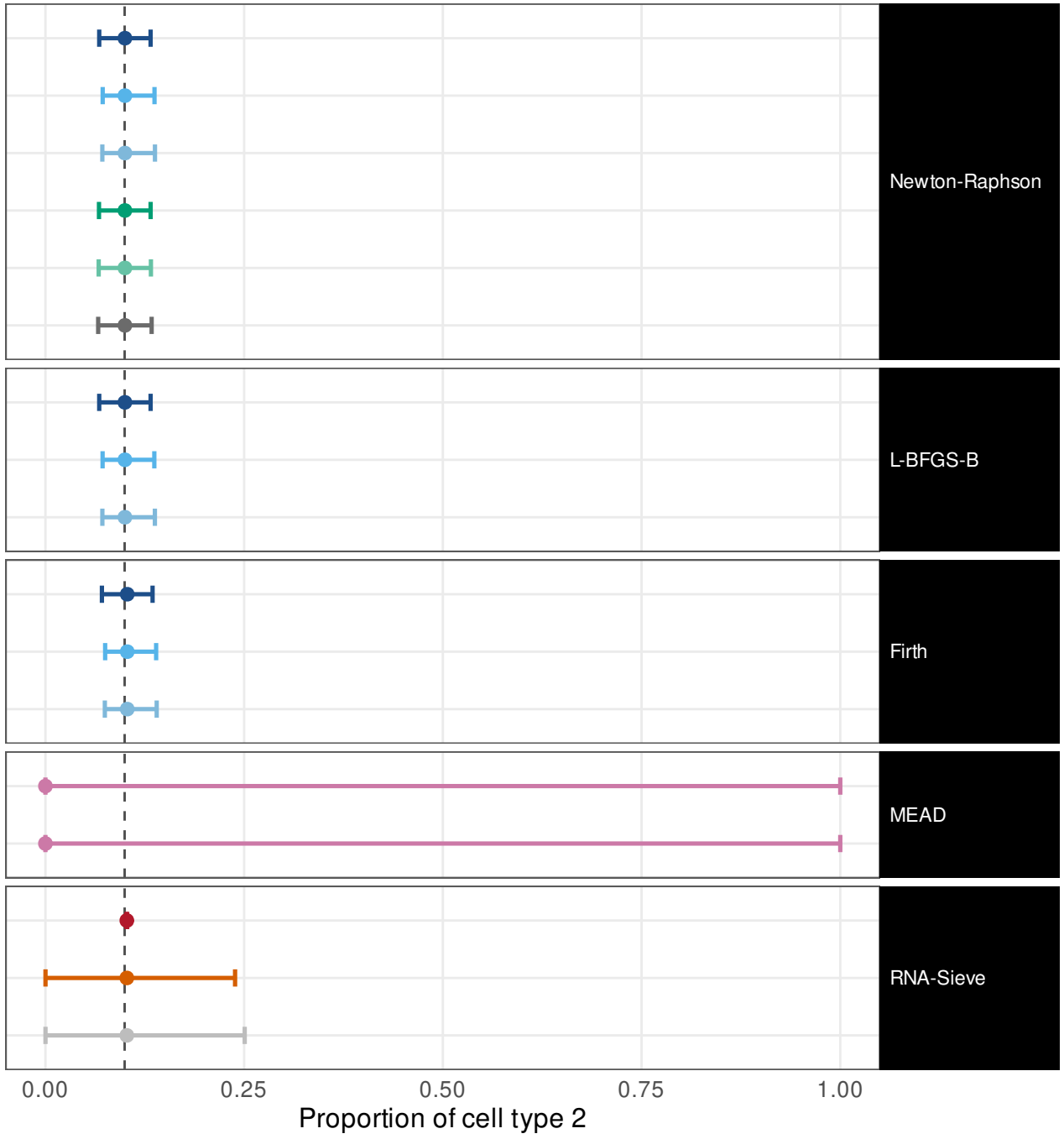


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / unbalanced / high sibling overlap / a = 5 / equal sizes

Wald (delta)	0.96 [0.92, 0.98]	w = 0.0646
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.0652
ILR, REML	0.96 [0.92, 0.98]	w = 0.0662
Profile likelihood	0.96 [0.92, 0.98]	w = 0.065
Profile, REML	0.96 [0.92, 0.98]	w = 0.0657
Oracle (Fisher at p*)	0.96 [0.93, 0.98]	w = 0.0673
Wald (delta)	0.96 [0.92, 0.98]	w = 0.0646
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.0651
ILR, REML	0.96 [0.92, 0.98]	w = 0.0661
Wald (delta)	0.94 [0.9, 0.97]	w = 0.0637
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.0642
ILR, REML	0.96 [0.92, 0.98]	w = 0.0652
Method's own interval	1 [0.98, 1]	w = 1
Own interval, transcript fractions	1 [0.98, 1]	w = 1
Fisher information (Wald)	0 [0, 0.019]	w = 0.000954
Godambe sandwich	1 [0.98, 1]	w = 0.239
Released code, as published	1 [0.98, 1]	w = 0.251

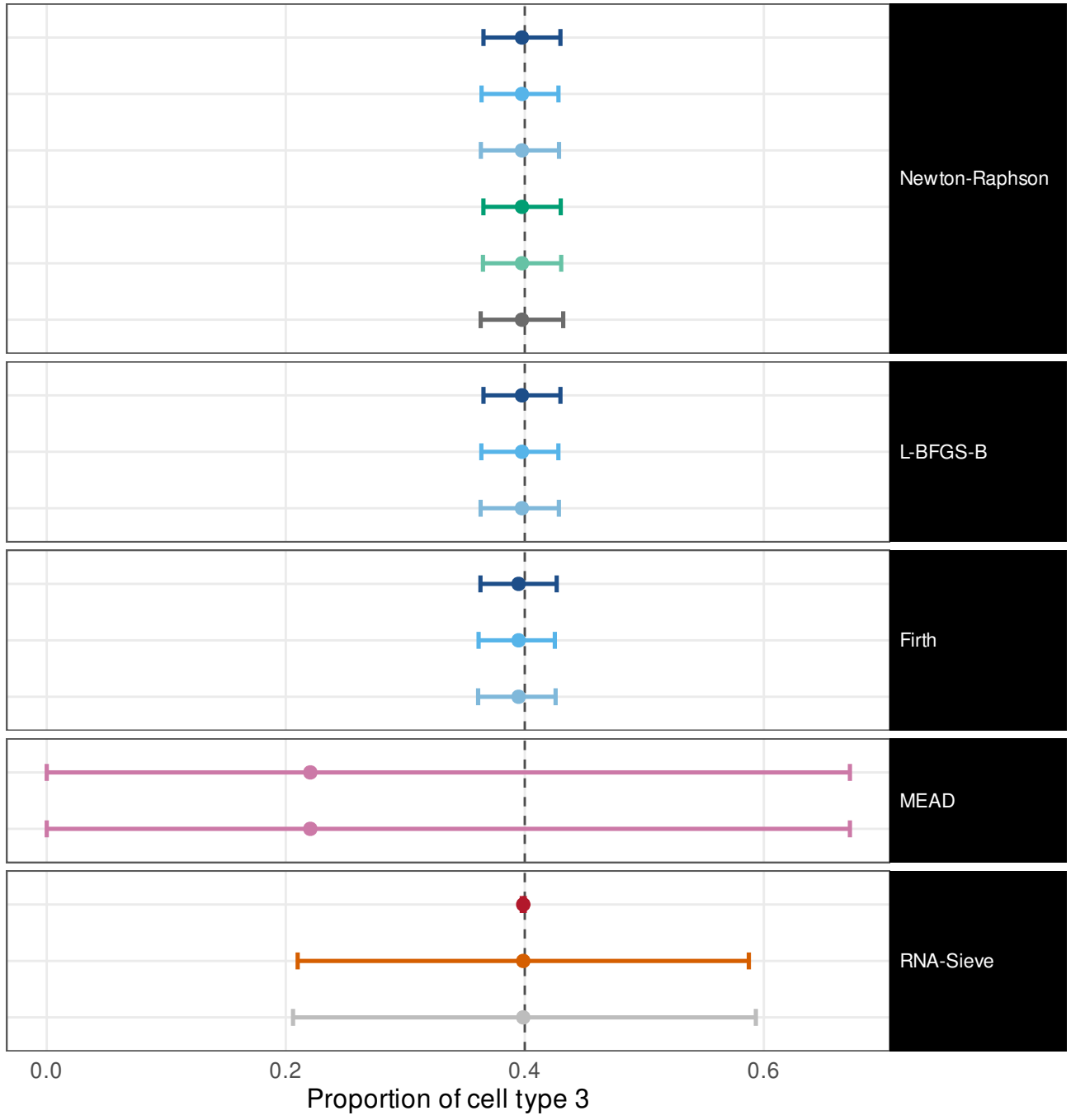


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / unbalanced / high sibling overlap / a = 5 / equal sizes

Wald (delta)	0.91 [0.86, 0.94]	w = 0.0644
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.0644
ILR, REML	0.93 [0.89, 0.96]	w = 0.0654
Profile likelihood	0.91 [0.86, 0.94]	w = 0.0647
Profile, REML	0.92 [0.87, 0.95]	w = 0.0655
Oracle (Fisher at p*)	0.92 [0.87, 0.95]	w = 0.0691
Wald (delta)	0.91 [0.86, 0.94]	w = 0.0644
ILR, mapped back	0.91 [0.86, 0.94]	w = 0.0645
ILR, REML	0.92 [0.87, 0.95]	w = 0.0655
Wald (delta)	0.91 [0.86, 0.94]	w = 0.0638
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.0638
ILR, REML	0.92 [0.87, 0.95]	w = 0.0648
Method's own interval	1 [0.98, 1]	w = 0.672
Own interval, transcript fractions	1 [0.98, 1]	w = 0.672
Fisher information (Wald)	0.08 [0.05, 0.13]	w = 0.00246
Godambe sandwich	1 [0.98, 1]	w = 0.377
Released code, as published	1 [0.98, 1]	w = 0.387

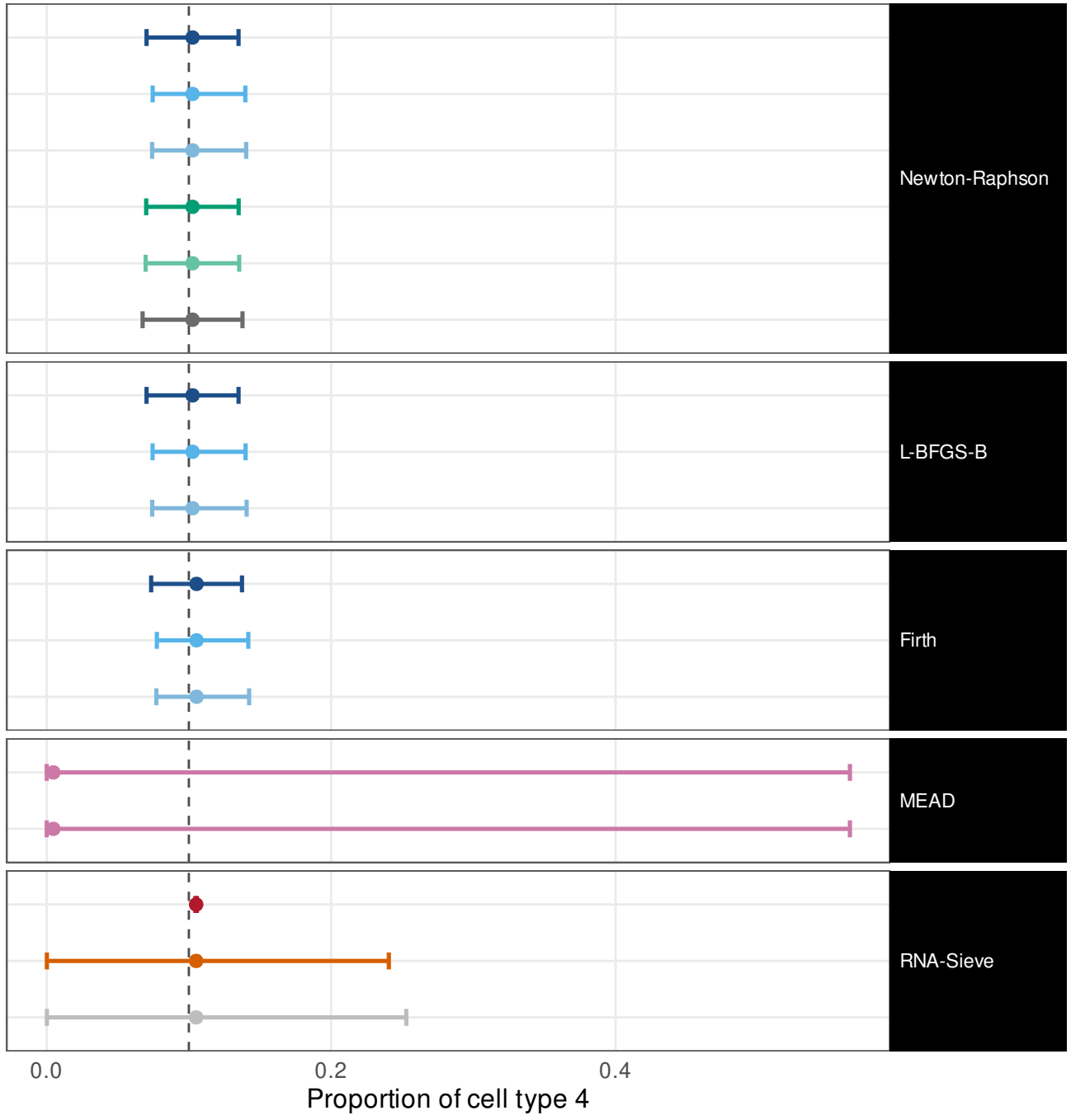


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / unbalanced / high sibling overlap / a = 5 / equal sizes

Wald (delta)	0.92 [0.87, 0.95]	w = 0.0647
ILR, mapped back	0.94 [0.89, 0.96]	w = 0.0651
ILR, REML	0.94 [0.9, 0.97]	w = 0.0662
Profile likelihood	0.92 [0.87, 0.95]	w = 0.065
Profile, REML	0.92 [0.88, 0.95]	w = 0.0658
Oracle (Fisher at p*)	0.94 [0.9, 0.97]	w = 0.0702
Wald (delta)	0.92 [0.87, 0.95]	w = 0.0647
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.0654
ILR, REML	0.94 [0.9, 0.97]	w = 0.0664
Wald (delta)	0.92 [0.87, 0.95]	w = 0.0639
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.0643
ILR, REML	0.93 [0.89, 0.96]	w = 0.0653
Method's own interval	1 [0.98, 1]	w = 0.565
Own interval, transcript fractions	1 [0.98, 1]	w = 0.565
Fisher information (Wald)	0.035 [0.017, 0.07]	w = 0.001
Godambe sandwich	1 [0.98, 1]	w = 0.24
Released code, as published	1 [0.98, 1]	w = 0.253

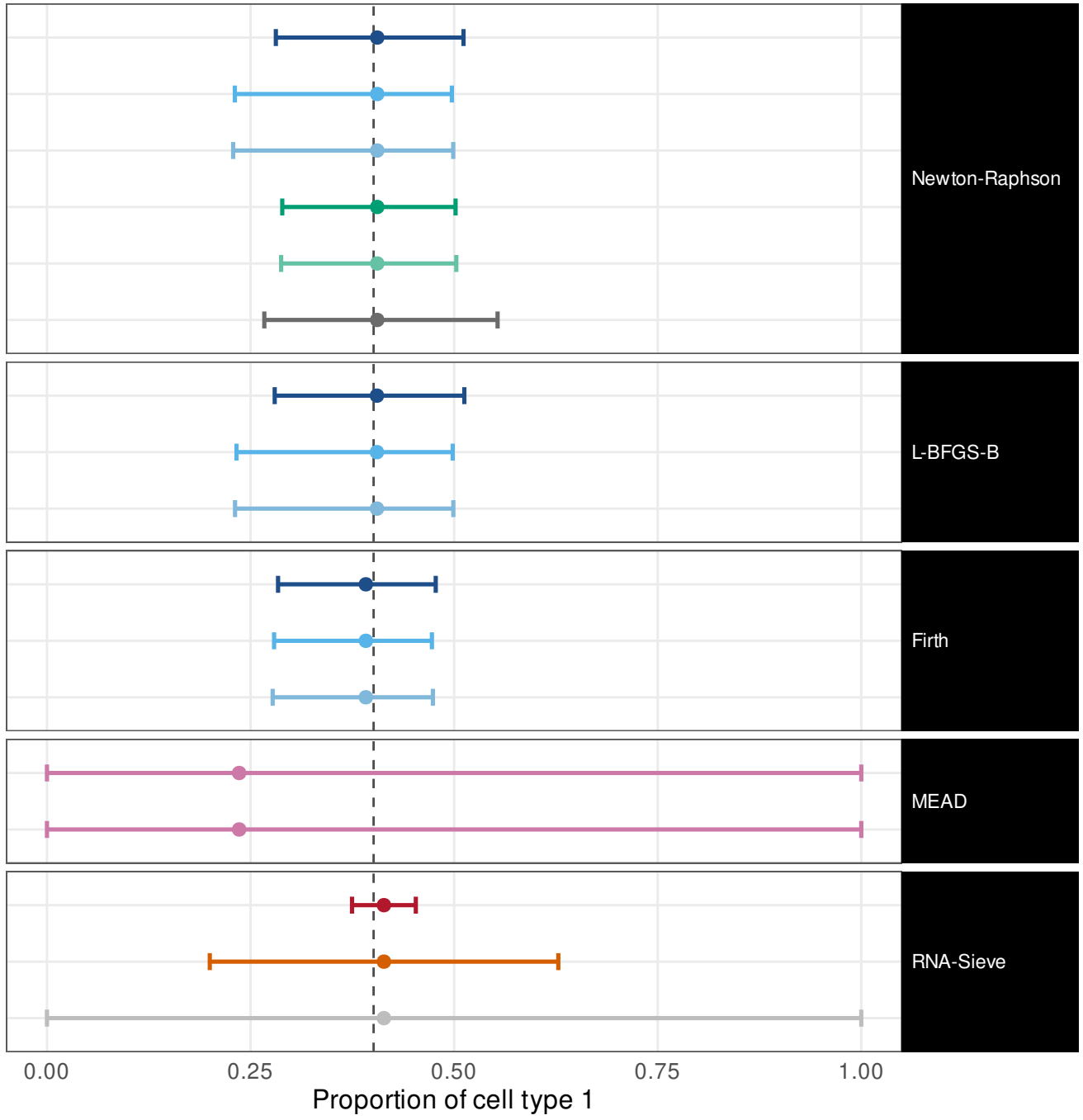


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / unbalanced / high sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.95 [0.91, 0.97]	w = 0.23
ILR, mapped back	0.98 [0.94, 0.99]	w = 0.267
ILR, REML	0.98 [0.95, 0.99]	w = 0.27
Profile likelihood	0.94 [0.9, 0.97]	w = 0.213
Profile, REML	0.94 [0.9, 0.97]	w = 0.215
Oracle (Fisher at p*)	0.94 [0.89, 0.96]	w = 0.286
Wald (delta)	0.97 [0.93, 0.98]	w = 0.233
ILR, mapped back	0.98 [0.94, 0.99]	w = 0.265
ILR, REML	0.97 [0.94, 0.99]	w = 0.268
Wald (delta)	0.94 [0.89, 0.97]	w = 0.194
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.194
ILR, REML	0.96 [0.92, 0.98]	w = 0.197
Method's own interval	1 [0.98, 1]	w = 1
Own interval, transcript fractions	1 [0.98, 1]	w = 1
Fisher information (Wald)	0.39 [0.33, 0.46]	w = 0.0783
Godambe sandwich	1 [0.98, 1]	w = 0.428
Released code, as published	1 [0.98, 1]	w = 1

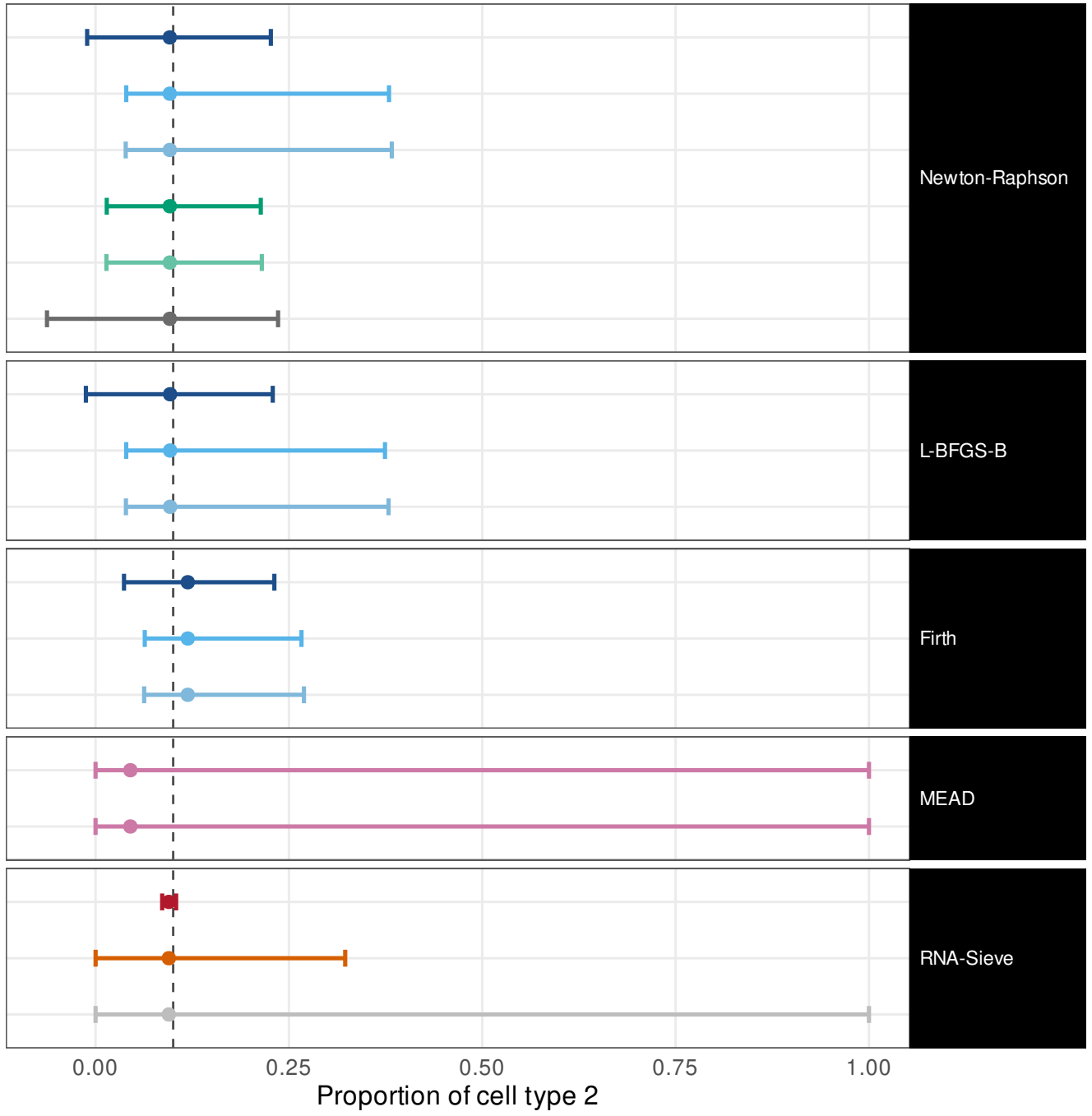


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / unbalanced / high sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.96 [0.92, 0.98]	w = 0.238
ILR, mapped back	0.93 [0.88, 0.96]	w = 0.34
ILR, REML	0.93 [0.89, 0.96]	w = 0.344
Profile likelihood	0.94 [0.9, 0.97]	w = 0.199
Profile, REML	0.94 [0.9, 0.97]	w = 0.201
Oracle (Fisher at p*)	0.85 [0.79, 0.9]	w = 0.299
Wald (delta)	0.98 [0.94, 0.99]	w = 0.242
ILR, mapped back	0.93 [0.88, 0.96]	w = 0.335
ILR, REML	0.93 [0.89, 0.96]	w = 0.34
Wald (delta)	0.93 [0.88, 0.96]	w = 0.194
ILR, mapped back	0.85 [0.79, 0.9]	w = 0.203
ILR, REML	0.88 [0.82, 0.92]	w = 0.207
Method's own interval	1 [0.98, 1]	w = 1
Own interval, transcript fractions	1 [0.98, 1]	w = 1
Fisher information (Wald)	0.14 [0.1, 0.2]	w = 0.018
Godambe sandwich	1 [0.98, 1]	w = 0.323
Released code, as published	1 [0.98, 1]	w = 1

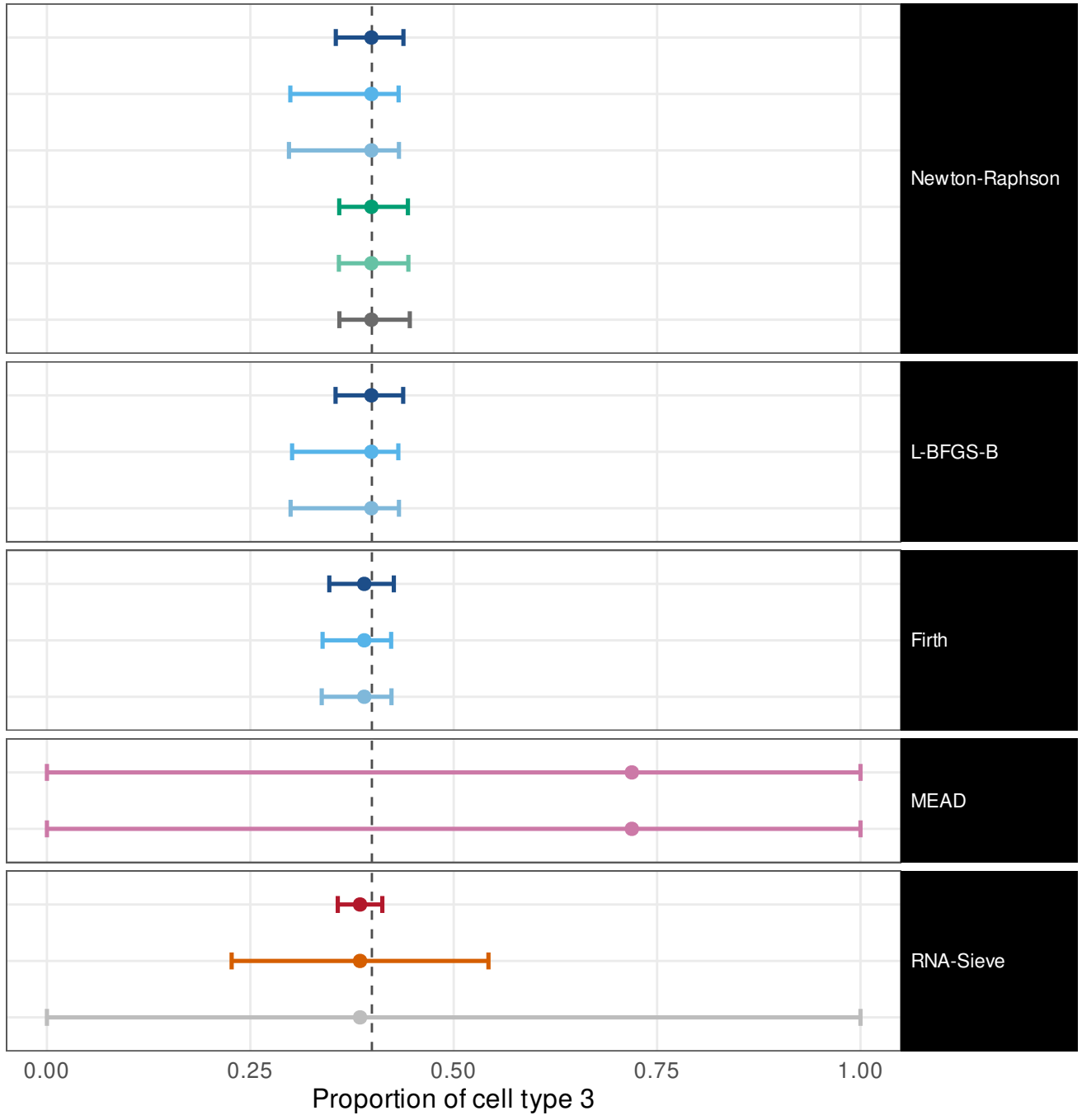


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / unbalanced / high sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.99 [0.96, 1]	w = 0.0831
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.133
ILR, REML	0.97 [0.93, 0.98]	w = 0.135
Profile likelihood	0.98 [0.94, 0.99]	w = 0.0844
Profile, REML	0.98 [0.94, 0.99]	w = 0.0854
Oracle (Fisher at p*)	0.97 [0.93, 0.99]	w = 0.0865
Wald (delta)	0.99 [0.96, 1]	w = 0.0831
ILR, mapped back	0.97 [0.94, 0.99]	w = 0.131
ILR, REML	0.97 [0.94, 0.99]	w = 0.133
Wald (delta)	0.92 [0.87, 0.95]	w = 0.0793
ILR, mapped back	0.89 [0.84, 0.93]	w = 0.0843
ILR, REML	0.89 [0.84, 0.93]	w = 0.0857
Method's own interval	1 [0.98, 1]	w = 1
Own interval, transcript fractions	1 [0.98, 1]	w = 1
Fisher information (Wald)	0.68 [0.61, 0.74]	w = 0.0547
Godambe sandwich	1 [0.98, 1]	w = 0.316
Released code, as published	1 [0.98, 1]	w = 1

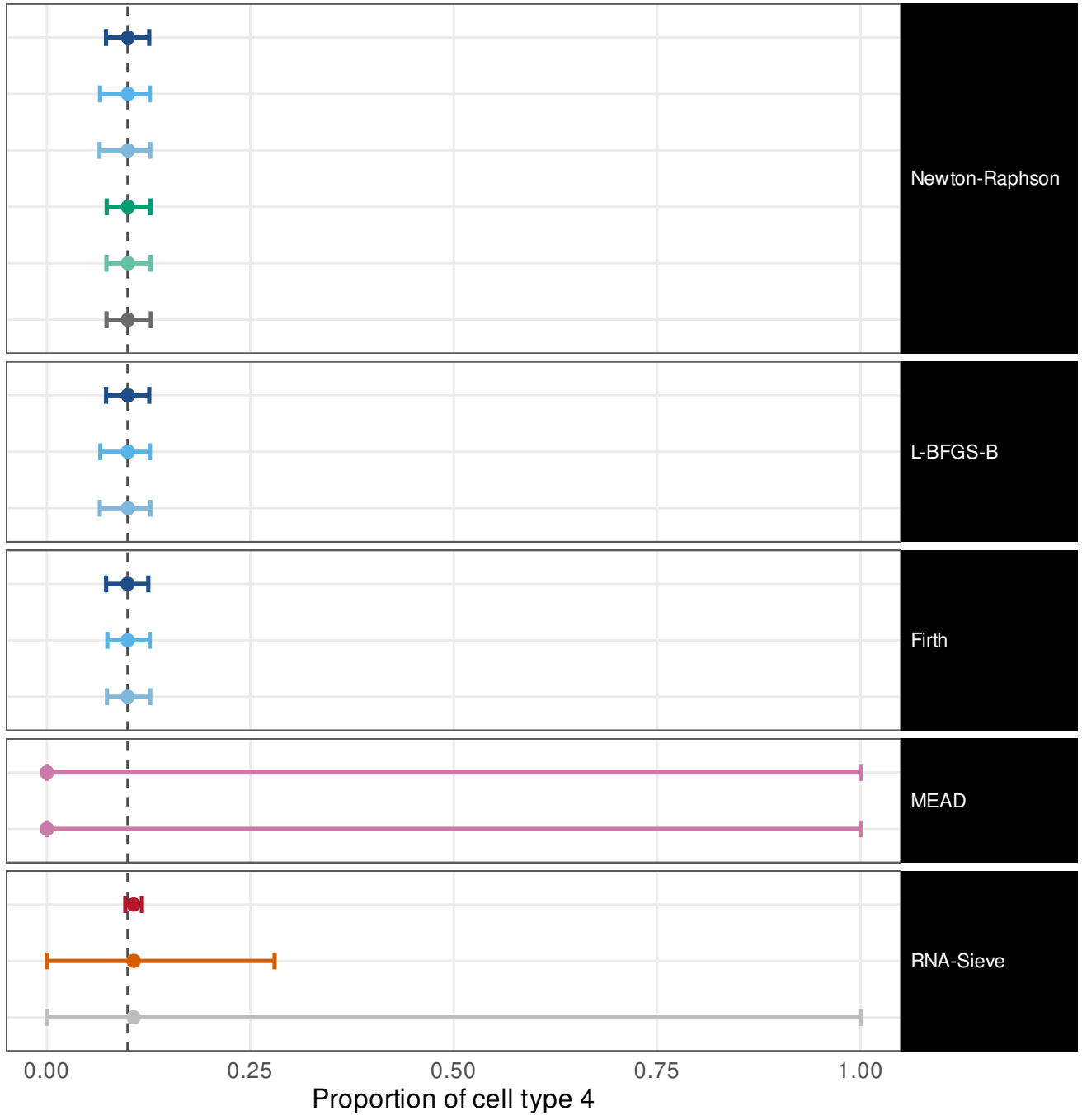


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1000 cells per bulk / unbalanced / high sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.95 [0.91, 0.97]	w = 0.0531
ILR, mapped back	0.98 [0.95, 0.99]	w = 0.0613
ILR, REML	0.98 [0.95, 0.99]	w = 0.0623
Profile likelihood	0.95 [0.91, 0.97]	w = 0.0538
Profile, REML	0.95 [0.91, 0.97]	w = 0.0545
Oracle (Fisher at p*)	0.96 [0.92, 0.98]	w = 0.0546
Wald (delta)	0.95 [0.91, 0.97]	w = 0.0531
ILR, mapped back	0.99 [0.96, 1]	w = 0.061
ILR, REML	0.98 [0.95, 0.99]	w = 0.0621
Wald (delta)	0.96 [0.92, 0.98]	w = 0.0519
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.0521
ILR, REML	0.97 [0.94, 0.99]	w = 0.0531
Method's own interval	1 [0.98, 1]	w = 1
Own interval, transcript fractions	1 [0.98, 1]	w = 1
Fisher information (Wald)	0.47 [0.4, 0.54]	w = 0.0204
Godambe sandwich	1 [0.98, 1]	w = 0.28
Released code, as published	1 [0.98, 1]	w = 1

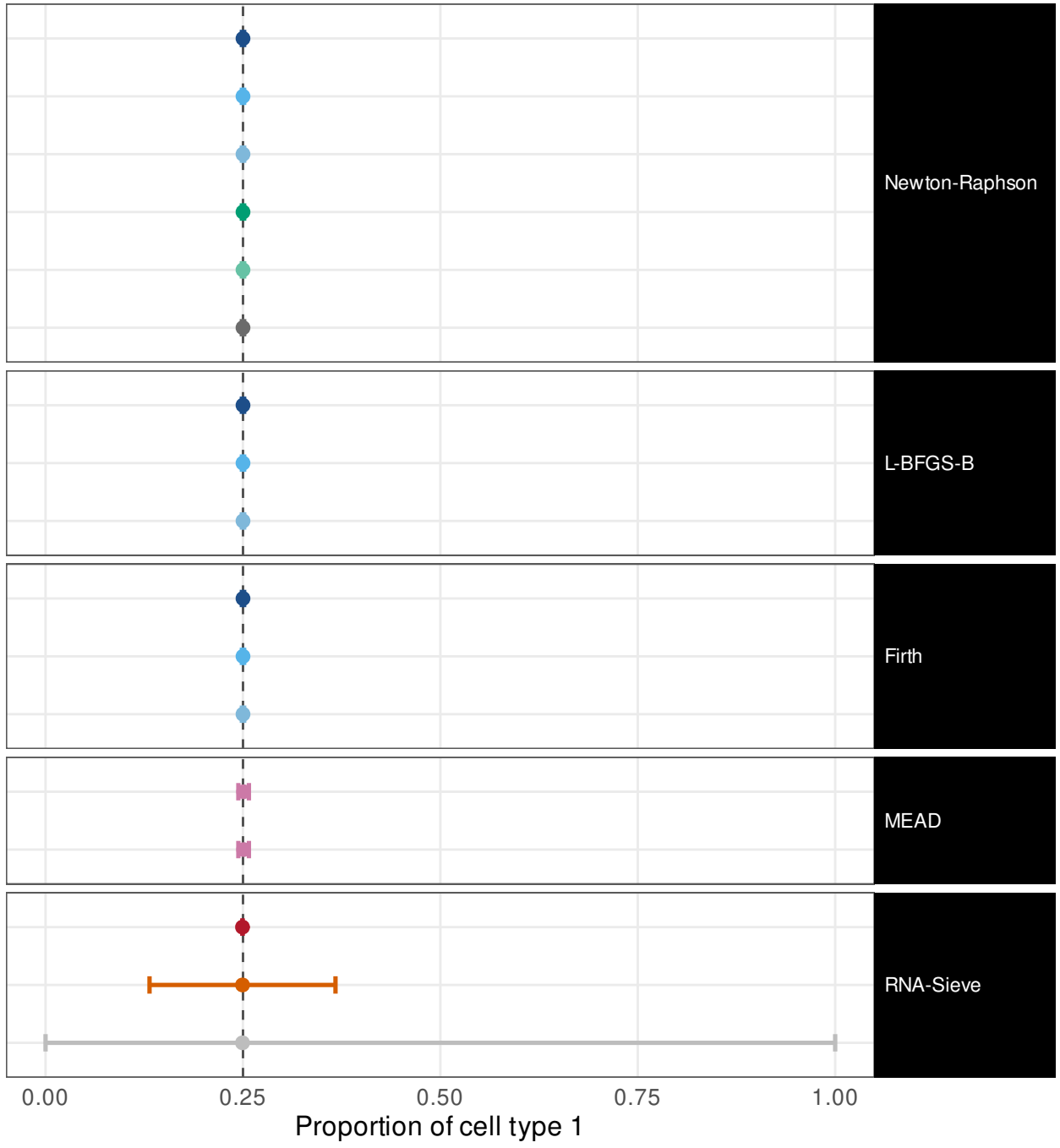


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / balanced / low sibling overlap / a = 1 / equal sizes

Wald (delta)	0.94 [0.89, 0.96]	w = 0.00286
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.00286
ILR, REML	0.92 [0.88, 0.95]	w = 0.00291
Profile likelihood	0.94 [0.89, 0.96]	w = 0.00288
Profile, REML	0.94 [0.89, 0.96]	w = 0.00291
Oracle (Fisher at p*)	0.94 [0.89, 0.96]	w = 0.0029
Wald (delta)	0.94 [0.89, 0.96]	w = 0.00286
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.00286
ILR, REML	0.94 [0.89, 0.96]	w = 0.0029
Wald (delta)	0.94 [0.89, 0.96]	w = 0.00286
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.00286
ILR, REML	0.93 [0.89, 0.96]	w = 0.0029
Method's own interval	1 [0.98, 1]	w = 0.0134
Own interval, transcript fractions	1 [0.98, 1]	w = 0.0134
Fisher information (Wald)	0.03 [0.014, 0.064]	w = 0.000115
Godambe sandwich	1 [0.98, 1]	w = 0.236
Released code, as published	1 [0.98, 1]	w = 1

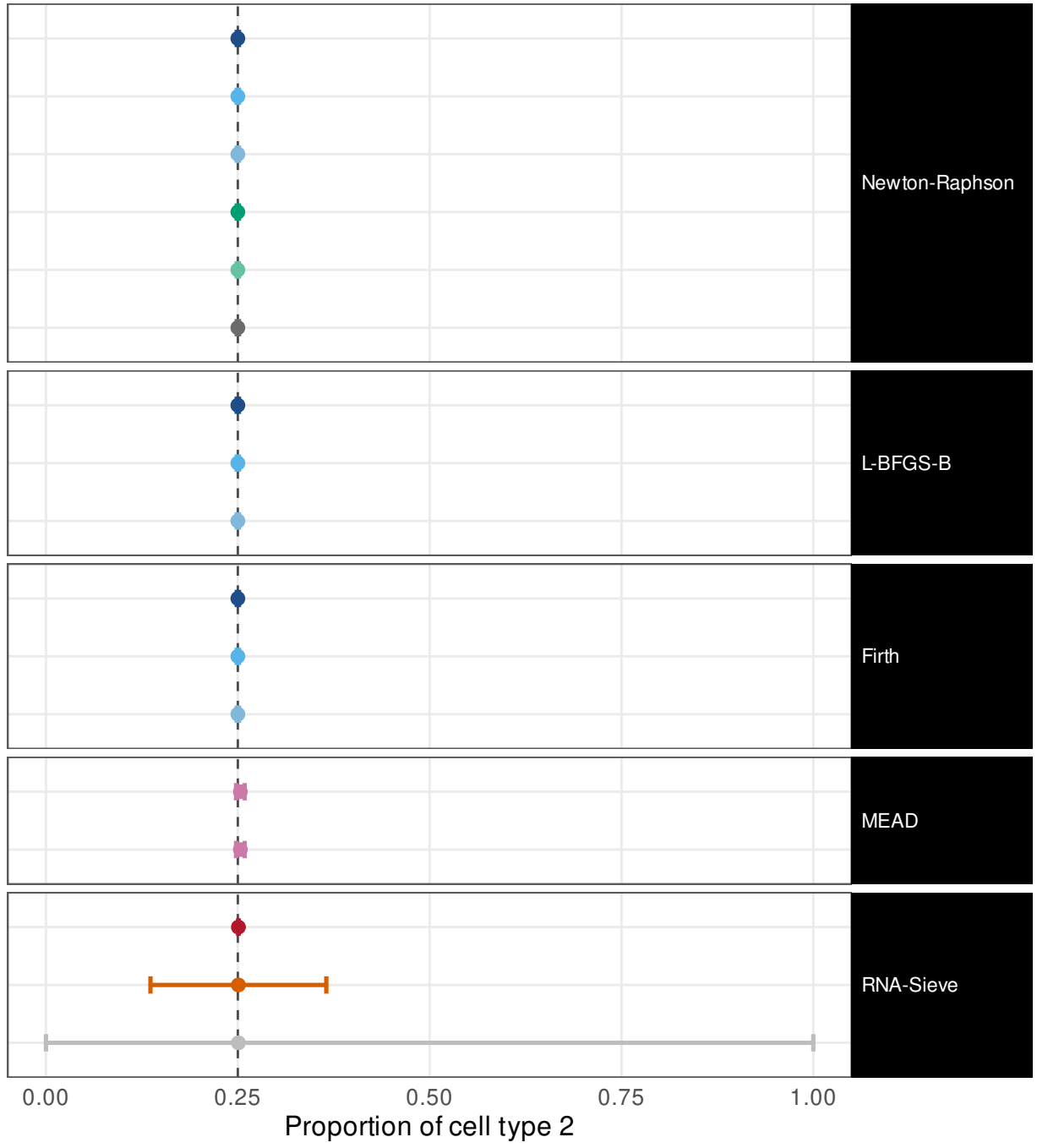


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / balanced / low sibling overlap / a = 1 / equal sizes

Wald (delta)	0.96 [0.92, 0.98]	w = 0.00284
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.00284
ILR, REML	0.96 [0.92, 0.98]	w = 0.00288
Profile likelihood	0.96 [0.92, 0.98]	w = 0.00285
Profile, REML	0.96 [0.92, 0.98]	w = 0.00289
Oracle (Fisher at p*)	0.96 [0.92, 0.98]	w = 0.00288
Wald (delta)	0.96 [0.92, 0.98]	w = 0.00284
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.00284
ILR, REML	0.95 [0.91, 0.97]	w = 0.00288
Wald (delta)	0.96 [0.92, 0.98]	w = 0.00284
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.00283
ILR, REML	0.96 [0.92, 0.98]	w = 0.00289
Method's own interval	1 [0.98, 1]	w = 0.0112
Own interval, transcript fractions	1 [0.98, 1]	w = 0.0112
Fisher information (Wald)	0.015 [0.0051, 0.043]	w = 0.000108
Godambe sandwich	1 [0.98, 1]	w = 0.229
Released code, as published	1 [0.98, 1]	w = 1

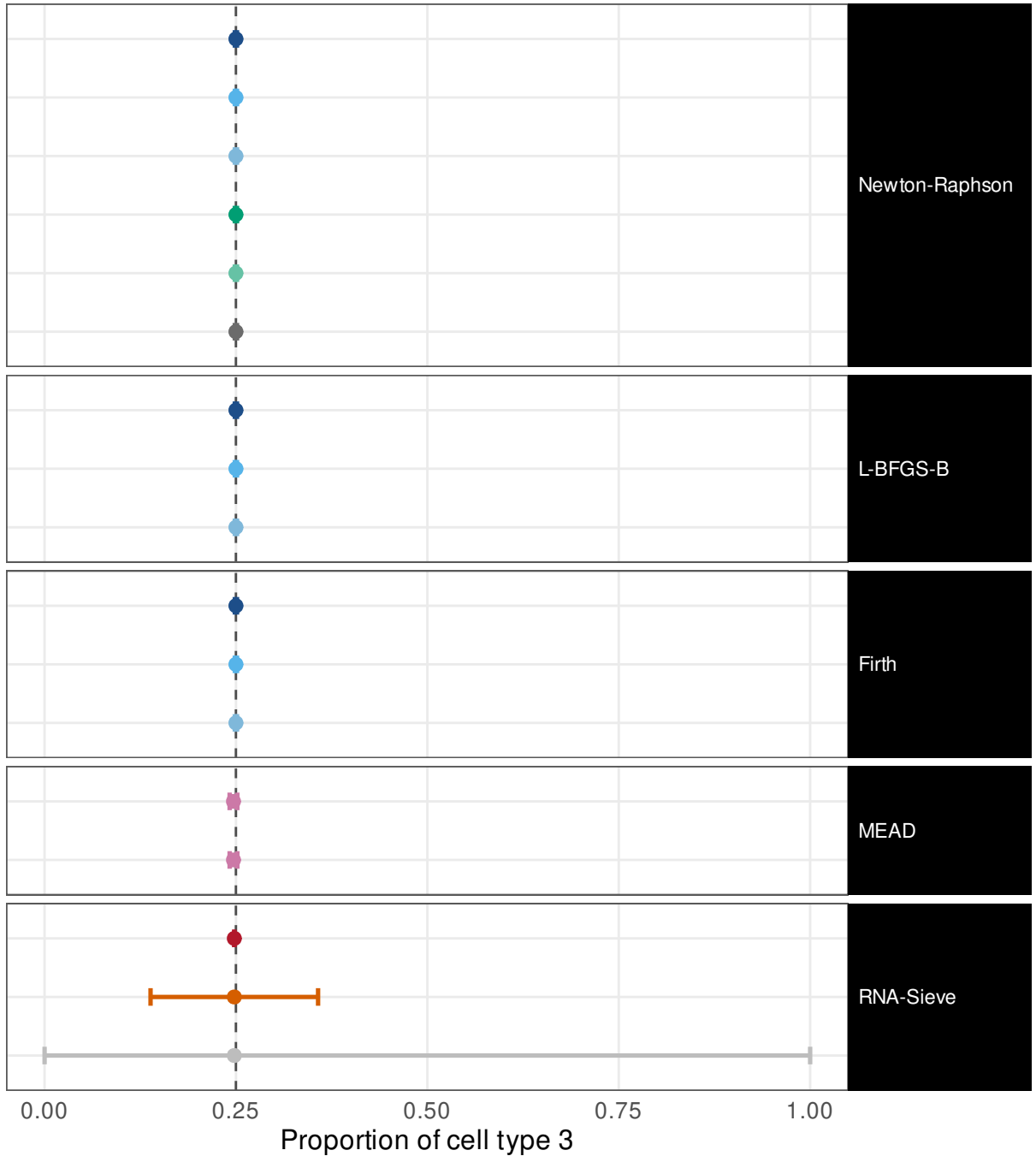


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / balanced / low sibling overlap / a = 1 / equal sizes

Wald (delta)	0.94 [0.9, 0.97]	w = 0.00283
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.00282
ILR, REML	0.94 [0.9, 0.97]	w = 0.00288
Profile likelihood	0.94 [0.9, 0.97]	w = 0.00285
Profile, REML	0.94 [0.9, 0.97]	w = 0.00288
Oracle (Fisher at p*)	0.94 [0.9, 0.97]	w = 0.00287
Wald (delta)	0.94 [0.9, 0.97]	w = 0.00283
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.00282
ILR, REML	0.94 [0.9, 0.97]	w = 0.00287
Wald (delta)	0.94 [0.9, 0.97]	w = 0.00283
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.00282
ILR, REML	0.94 [0.9, 0.97]	w = 0.00288
Method's own interval	0.99 [0.96, 1]	w = 0.0101
Own interval, transcript fractions	0.99 [0.96, 1]	w = 0.0101
Fisher information (Wald)	0.005 [0.00088, 0.028]	w = 0.000103
Godambe sandwich	1 [0.98, 1]	w = 0.219
Released code, as published	1 [0.98, 1]	w = 1

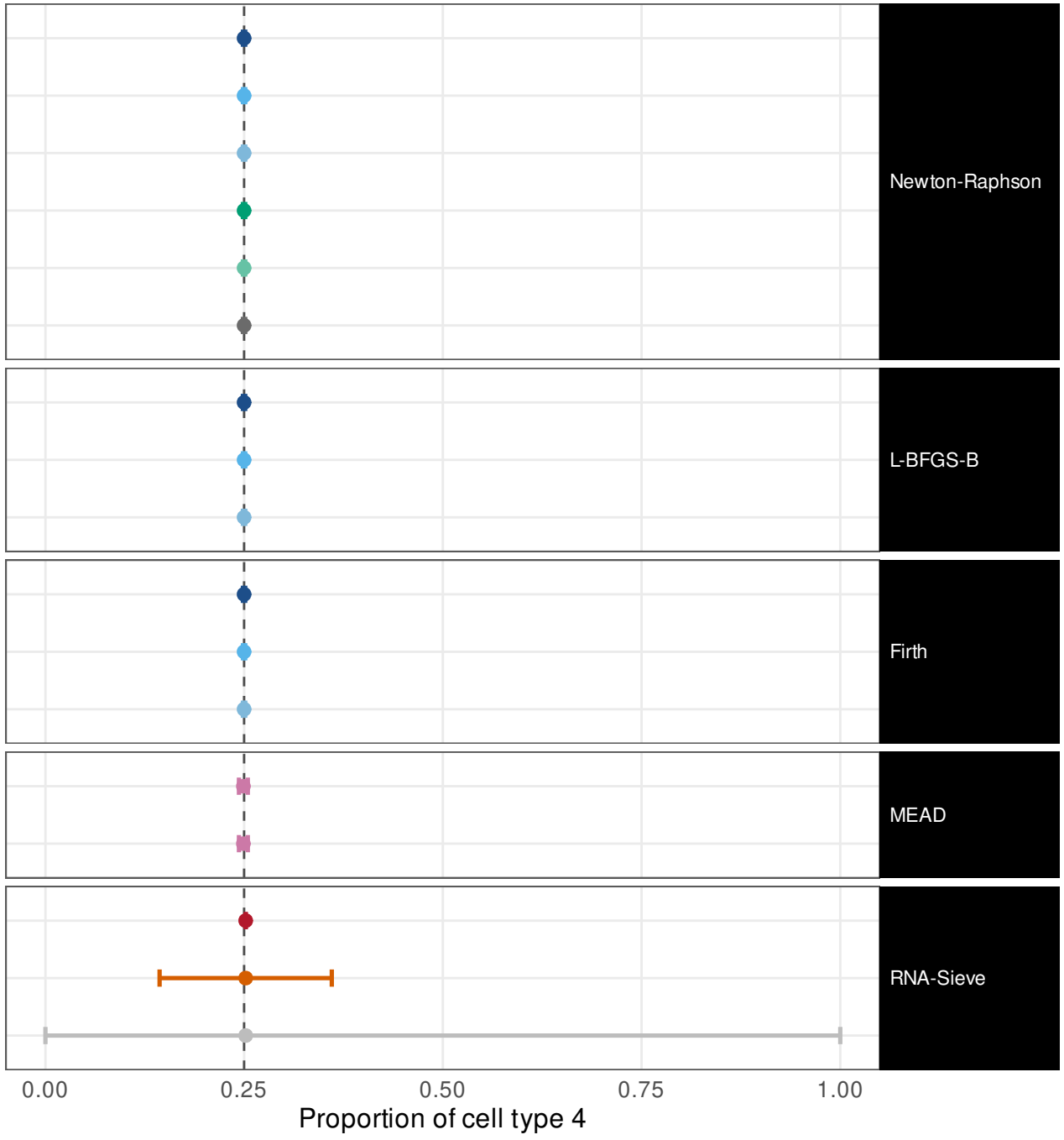


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / balanced / low sibling overlap / a = 1 / equal sizes

Wald (delta)	0.95 [0.91, 0.97]	w = 0.00285
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.00283
ILR, REML	0.96 [0.92, 0.98]	w = 0.00289
Profile likelihood	0.96 [0.92, 0.98]	w = 0.00286
Profile, REML	0.96 [0.92, 0.98]	w = 0.00289
Oracle (Fisher at p*)	0.96 [0.92, 0.98]	w = 0.00288
Wald (delta)	0.95 [0.91, 0.97]	w = 0.00285
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.00283
ILR, REML	0.95 [0.91, 0.97]	w = 0.00288
Wald (delta)	0.95 [0.91, 0.97]	w = 0.00285
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.00284
ILR, REML	0.94 [0.9, 0.97]	w = 0.00288
Method's own interval	1 [0.98, 1]	w = 0.0112
Own interval, transcript fractions	1 [0.98, 1]	w = 0.0112
Fisher information (Wald)	0 [0, 0.019]	w = 0.000112
Godambe sandwich	1 [0.98, 1]	w = 0.217
Released code, as published	1 [0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / balanced / low sibling overlap / a = 1 / sizes 1:1:5:5

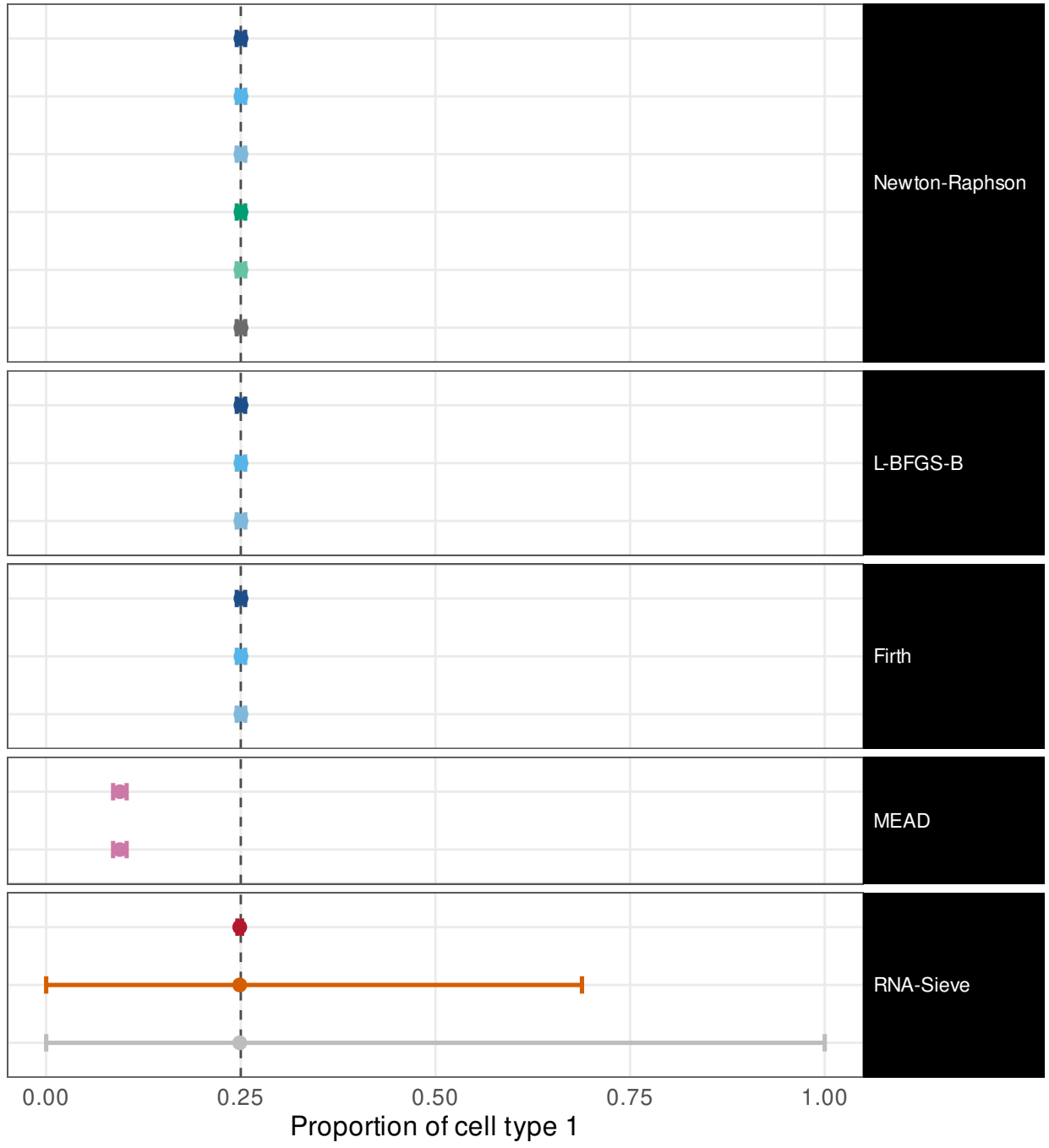
```
Wald (delta)                          0.96 [0.93, 0.98]   w = 0.0106
ILR, mapped back                      0.96 [0.93, 0.98]   w = 0.0106
ILR, REML                             0.96 [0.93, 0.98]   w = 0.0107
Profile likelihood                    0.96 [0.93, 0.98]   w = 0.0106
Profile, REML                         0.96 [0.93, 0.98]   w = 0.0108
Oracle (Fisher at p*)                 0.97 [0.94, 0.99]   w = 0.0107

Wald (delta)                          0.96 [0.93, 0.98]   w = 0.0106
ILR, mapped back                      0.96 [0.93, 0.98]   w = 0.0105
ILR, REML                             0.96 [0.93, 0.98]   w = 0.0108

Wald (delta)                          0.96 [0.93, 0.98]   w = 0.0106
ILR, mapped back                      0.96 [0.93, 0.98]   w = 0.0106
ILR, REML                             0.96 [0.93, 0.98]   w = 0.0107

Method's own interval                    0 [   0, 0.019]   w = 0.0173
Own interval, transcript fractions   0.005 [0.00088, 0.028]   w = 0.0173

Fisher information (Wald)             0.64 [0.57,  0.7]   w = 0.00587
Godambe sandwich                         1 [0.98,    1]   w = 0.688
Released code, as published              1 [0.98,    1]   w =    1
```

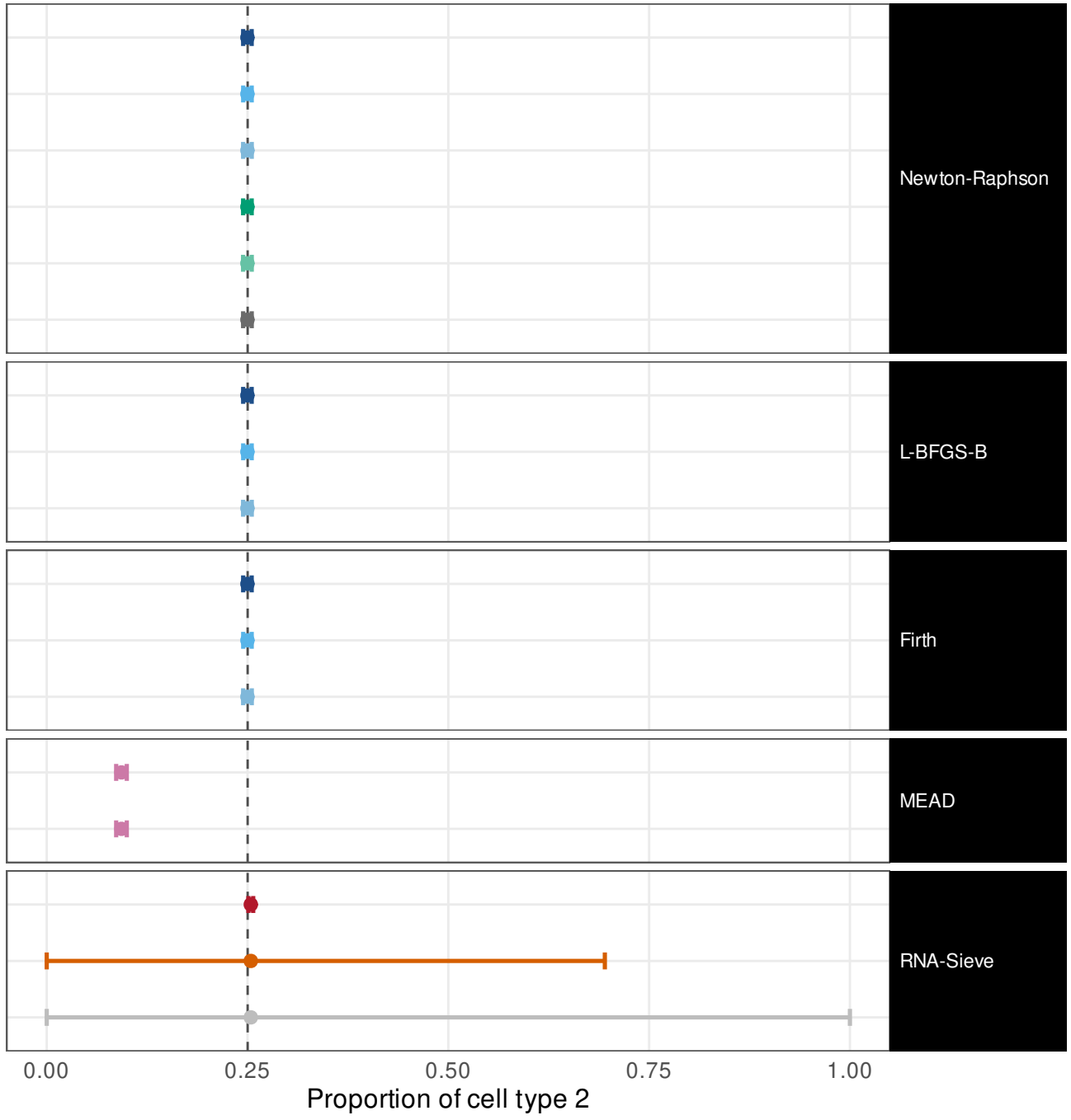


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / balanced / low sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.95 [0.91, 0.97]	w = 0.0106
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.0106
ILR, REML	0.96 [0.92, 0.98]	w = 0.0107
Profile likelihood	0.95 [0.91, 0.97]	w = 0.0106
Profile, REML	0.95 [0.91, 0.97]	w = 0.0108
Oracle (Fisher at p*)	0.96 [0.92, 0.98]	w = 0.0107
Wald (delta)	0.95 [0.91, 0.97]	w = 0.0106
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.0106
ILR, REML	0.96 [0.92, 0.98]	w = 0.0108
Wald (delta)	0.95 [0.91, 0.97]	w = 0.0106
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.0106
ILR, REML	0.95 [0.91, 0.97]	w = 0.0107
Method's own interval	0 [0, 0.019]	w = 0.0132
Own interval, transcript fractions	0 [0, 0.019]	w = 0.0132
Fisher information (Wald)	0.33 [0.27, 0.4]	w = 0.00595
Godambe sandwich	1 [0.98, 1]	w = 0.695
Released code, as published	1 [0.98, 1]	w = 1

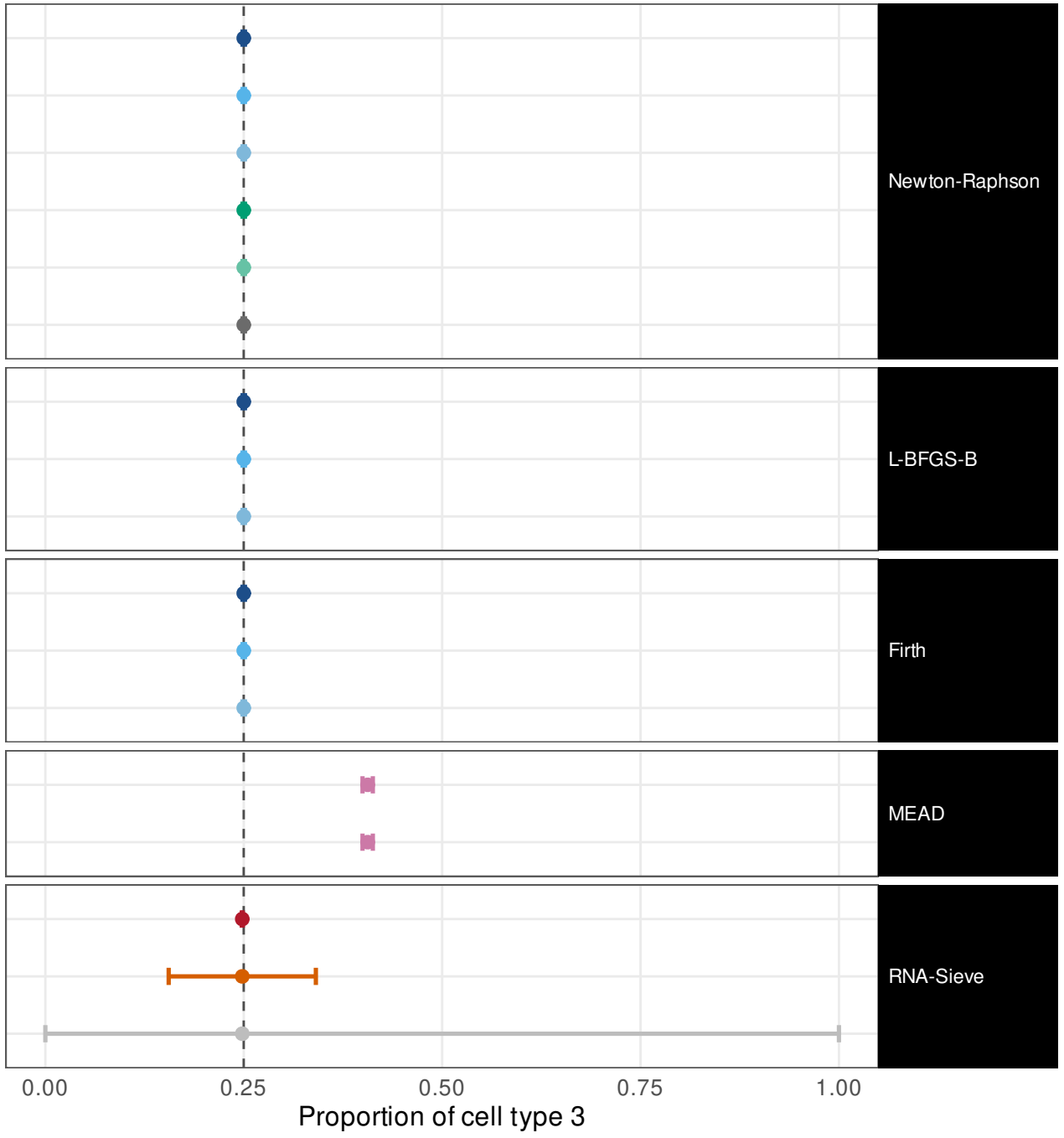


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / balanced / low sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.97 [0.94, 0.99]	w = 0.00296
ILR, mapped back	0.97 [0.94, 0.99]	w = 0.00295
ILR, REML	0.97 [0.94, 0.99]	w = 0.00299
Profile likelihood	0.97 [0.94, 0.99]	w = 0.00297
Profile, REML	0.97 [0.94, 0.99]	w = 0.00301
Oracle (Fisher at p*)	0.97 [0.94, 0.99]	w = 0.00299
Wald (delta)	0.97 [0.94, 0.99]	w = 0.00296
ILR, mapped back	0.96 [0.93, 0.98]	w = 0.00295
ILR, REML	0.97 [0.94, 0.99]	w = 0.003
Wald (delta)	0.97 [0.94, 0.99]	w = 0.00296
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.00295
ILR, REML	0.97 [0.94, 0.99]	w = 0.003
Method's own interval	0 [0, 0.019]	w = 0.013
Own interval, transcript fractions	0 [0, 0.019]	w = 0.013
Fisher information (Wald)	0.095 [0.062, 0.14]	w = 0.00117
Godambe sandwich	1 [0.98, 1]	w = 0.185
Released code, as published	1 [0.98, 1]	w = 1

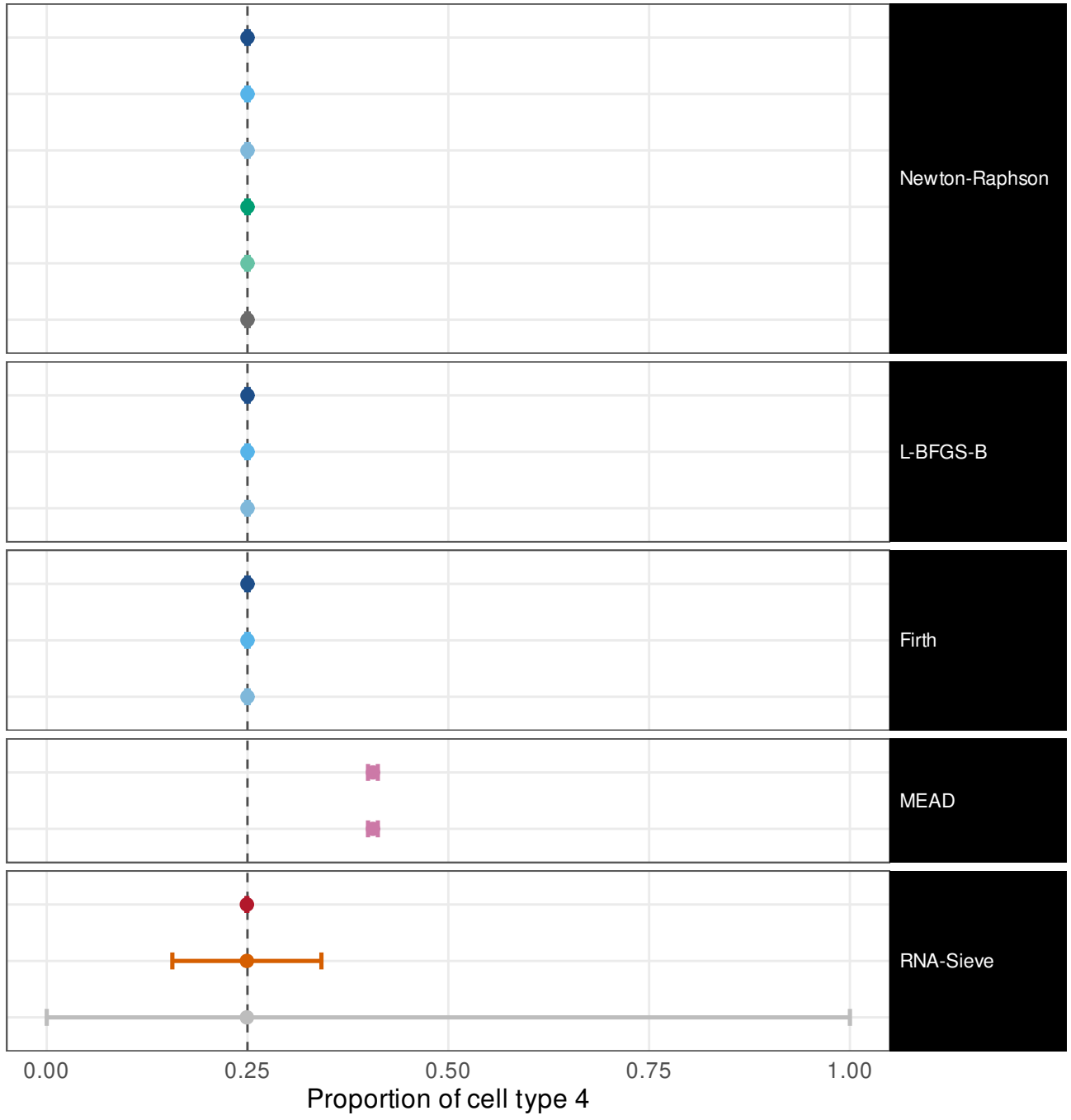


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / balanced / low sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.94 [0.9, 0.97]	w = 0.00295
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.00295
ILR, REML	0.95 [0.91, 0.97]	w = 0.00299
Profile likelihood	0.94 [0.9, 0.97]	w = 0.00297
Profile, REML	0.94 [0.9, 0.97]	w = 0.003
Oracle (Fisher at p*)	0.96 [0.92, 0.98]	w = 0.00299
Wald (delta)	0.94 [0.9, 0.97]	w = 0.00295
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.00294
ILR, REML	0.94 [0.9, 0.97]	w = 0.00299
Wald (delta)	0.94 [0.9, 0.97]	w = 0.00295
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.00294
ILR, REML	0.95 [0.91, 0.97]	w = 0.00299
Method's own interval	0 [0, 0.019]	w = 0.0122
Own interval, transcript fractions	0 [0, 0.019]	w = 0.0122
Fisher information (Wald)	0.32 [0.26, 0.39]	w = 0.00121
Godambe sandwich	1 [0.98, 1]	w = 0.186
Released code, as published	1 [0.98, 1]	w = 1

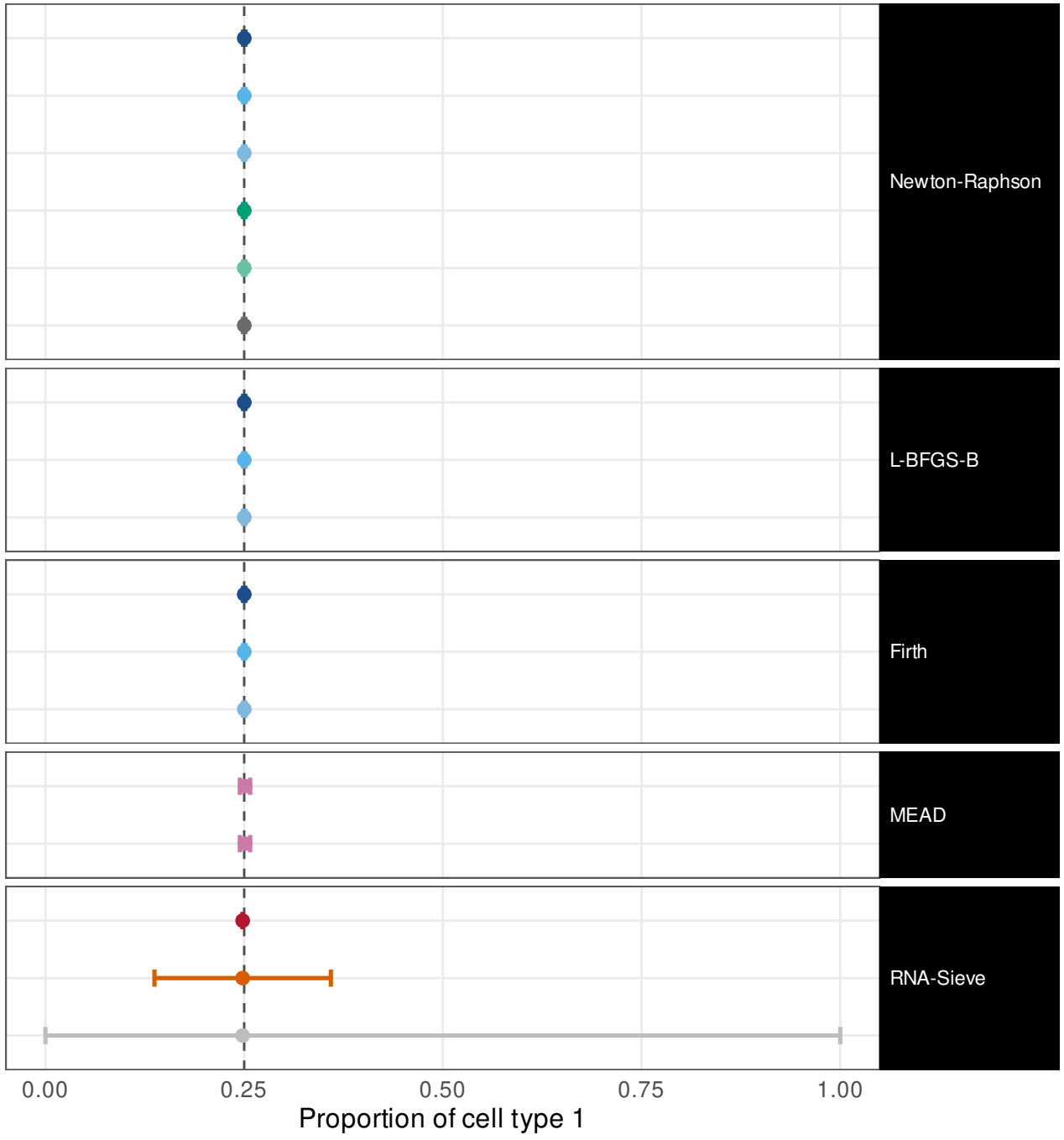


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / balanced / low sibling overlap / a = 5 / equal sizes

Wald (delta)	0.96 [0.92, 0.98]	w = 0.00291
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.00291
ILR, REML	0.96 [0.93, 0.98]	w = 0.00296
Profile likelihood	0.96 [0.92, 0.98]	w = 0.00292
Profile, REML	0.96 [0.93, 0.98]	w = 0.00296
Oracle (Fisher at p*)	0.96 [0.93, 0.98]	w = 0.00295
Wald (delta)	0.96 [0.92, 0.98]	w = 0.00291
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.0029
ILR, REML	0.96 [0.92, 0.98]	w = 0.00294
Wald (delta)	0.96 [0.92, 0.98]	w = 0.00291
ILR, mapped back	0.96 [0.93, 0.98]	w = 0.0029
ILR, REML	0.96 [0.92, 0.98]	w = 0.00295
Method's own interval	1 [0.98, 1]	w = 0.0129
Own interval, transcript fractions	1 [0.98, 1]	w = 0.0129
Fisher information (Wald)	0 [0, 0.019]	w = 5.22e-05
Godambe sandwich	1 [0.98, 1]	w = 0.222
Released code, as published	1 [0.98, 1]	w = 1

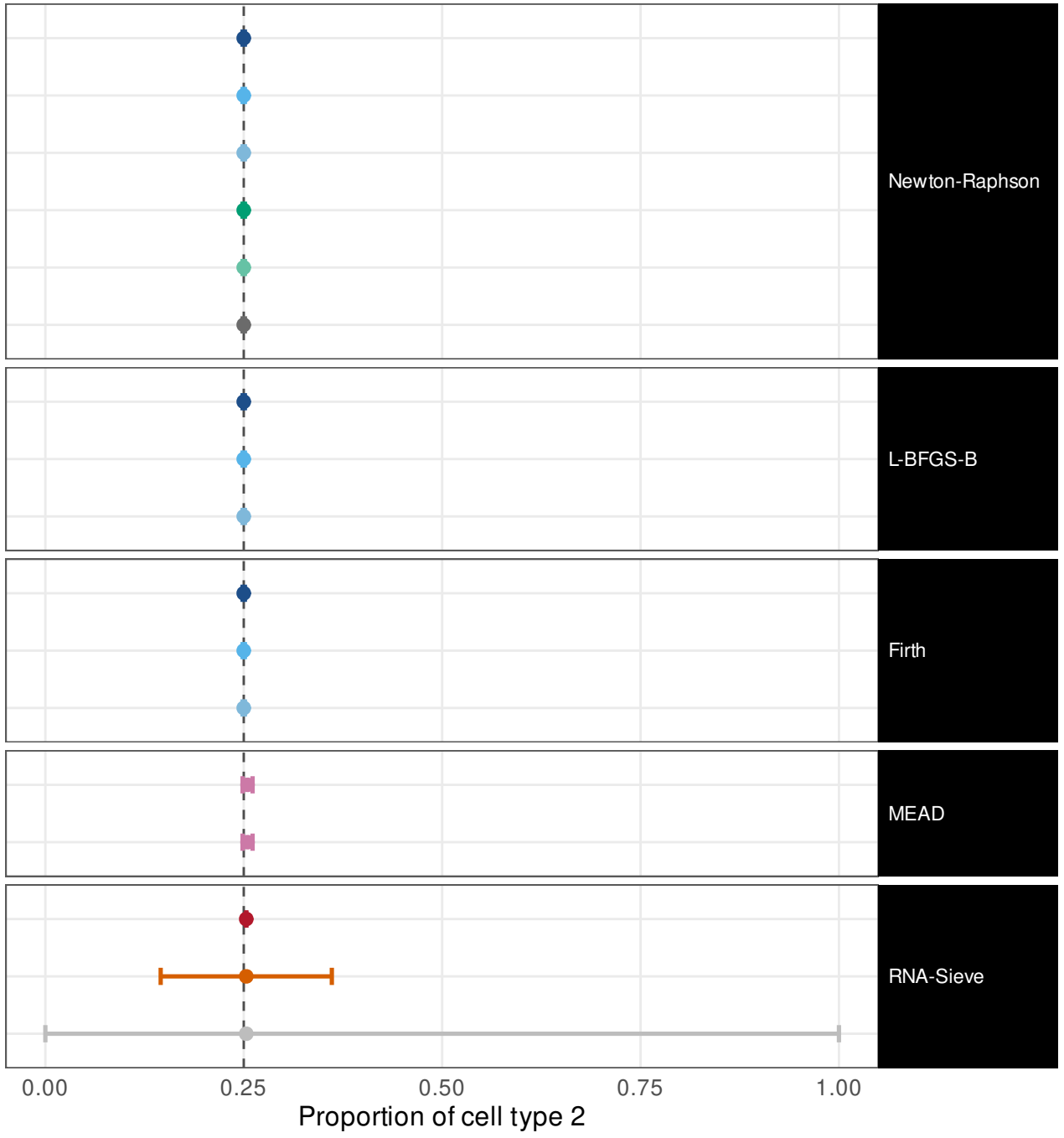


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / balanced / low sibling overlap / a = 5 / equal sizes

Wald (delta)	0.96 [0.93, 0.98]	w = 0.00288
ILR, mapped back	0.96 [0.93, 0.98]	w = 0.00288
ILR, REML	0.96 [0.93, 0.98]	w = 0.00292
Profile likelihood	0.96 [0.93, 0.98]	w = 0.0029
Profile, REML	0.96 [0.93, 0.98]	w = 0.00293
Oracle (Fisher at p*)	0.96 [0.93, 0.98]	w = 0.00292
Wald (delta)	0.96 [0.93, 0.98]	w = 0.00288
ILR, mapped back	0.96 [0.93, 0.98]	w = 0.00288
ILR, REML	0.96 [0.93, 0.98]	w = 0.00292
Wald (delta)	0.96 [0.93, 0.98]	w = 0.00288
ILR, mapped back	0.96 [0.93, 0.98]	w = 0.00288
ILR, REML	0.96 [0.93, 0.98]	w = 0.00292
Method's own interval	0.98 [0.95, 0.99]	w = 0.0126
Own interval, transcript fractions	0.98 [0.95, 0.99]	w = 0.0126
Fisher information (Wald)	0 [0, 0.019]	w = 4.89e-05
Godambe sandwich	1 [0.98, 1]	w = 0.216
Released code, as published	1 [0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / balanced / low sibling overlap / a = 5 / equal sizes

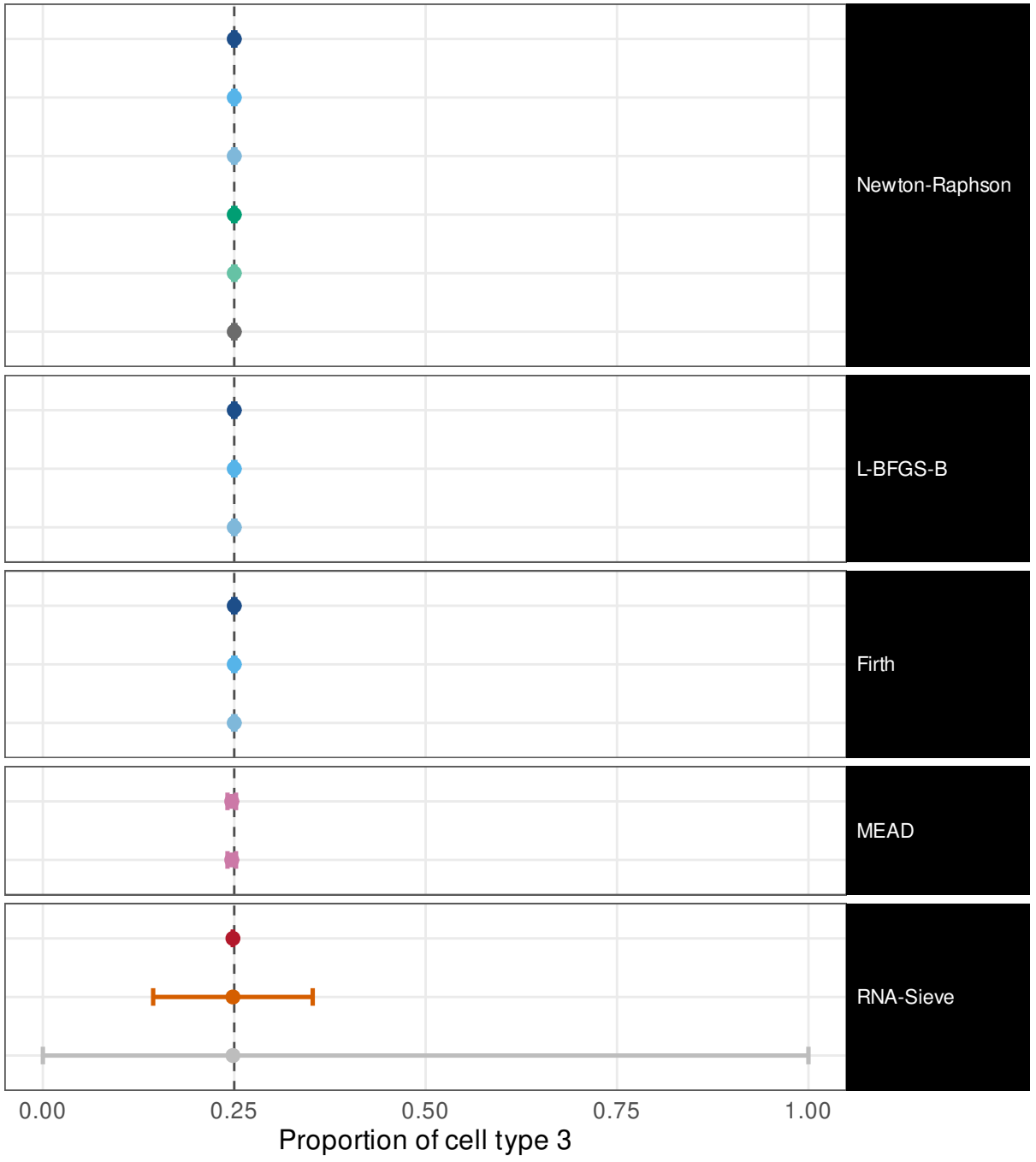
```
Wald (delta)                        0.96 [0.92, 0.98]   w = 0.00288
ILR, mapped back                    0.96 [0.92, 0.98]   w = 0.00287
ILR, REML                           0.95 [0.91, 0.97]   w = 0.00292
Profile likelihood                  0.96 [0.92, 0.98]   w = 0.00289
Profile, REML                       0.96 [0.92, 0.98]   w = 0.00292
Oracle (Fisher at p*)               0.96 [0.92, 0.98]   w = 0.00291

Wald (delta)                        0.96 [0.92, 0.98]   w = 0.00288
ILR, mapped back                    0.94 [0.89, 0.96]   w = 0.00287
ILR, REML                           0.96 [0.92, 0.98]   w = 0.00293

Wald (delta)                        0.96 [0.92, 0.98]   w = 0.00288
ILR, mapped back                    0.95 [0.91, 0.97]   w = 0.00287
ILR, REML                           0.94 [ 0.9, 0.97]   w = 0.00292

Method's own interval                  1 [0.98,    1]   w = 0.0111
Own interval, transcript fractions     1 [0.98,    1]   w = 0.0111

Fisher information (Wald)           0.005 [0.00088, 0.028]   w = 4.65e-05
Godambe sandwich                       1 [0.98,    1]   w = 0.208
Released code, as published            1 [0.98,    1]   w =    1
```

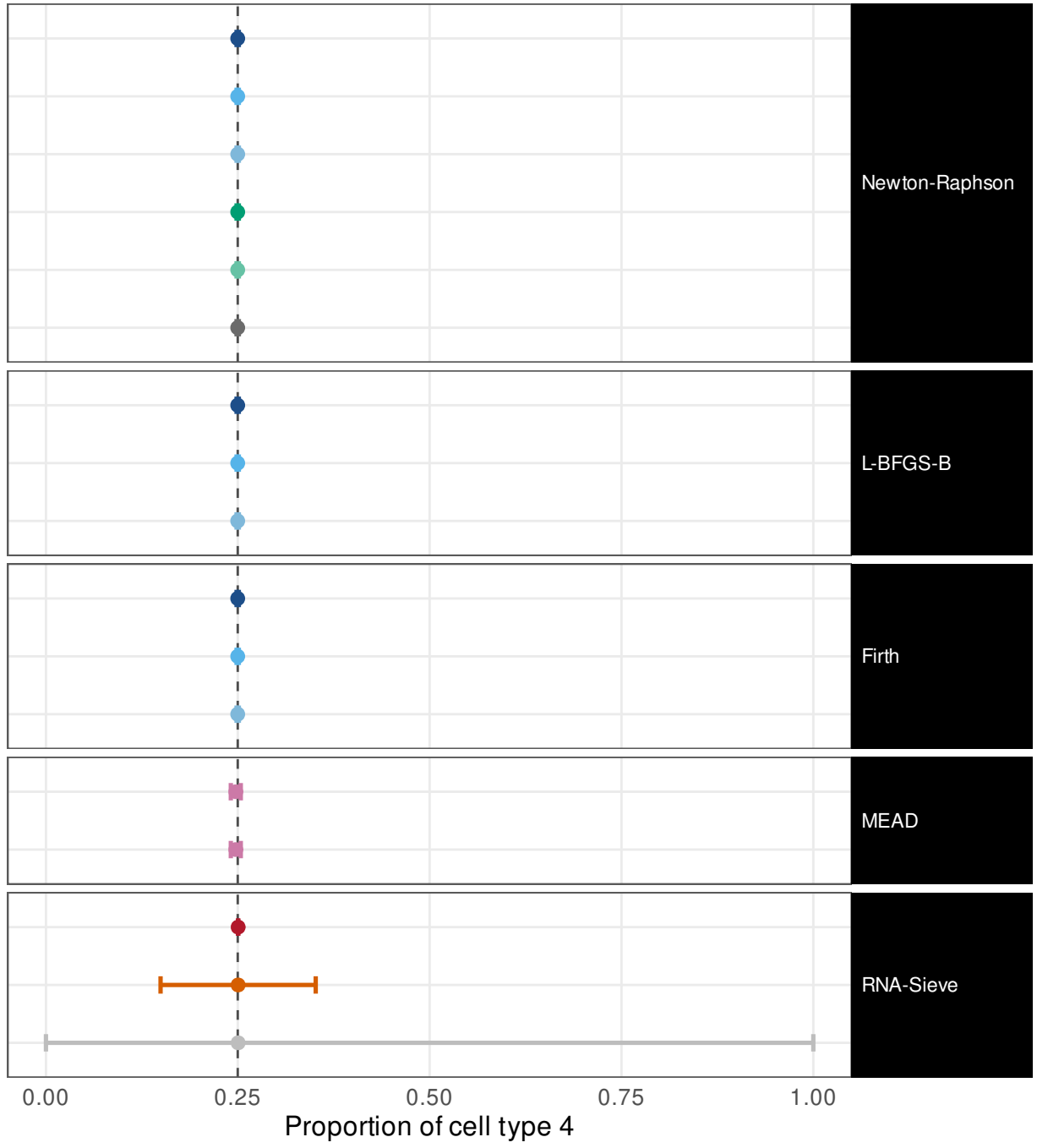


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / balanced / low sibling overlap / a = 5 / equal sizes

Wald (delta)	0.94 [0.9, 0.97]	w = 0.00289
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.00289
ILR, REML	0.96 [0.92, 0.98]	w = 0.00293
Profile likelihood	0.94 [0.9, 0.97]	w = 0.00291
Profile, REML	0.94 [0.9, 0.97]	w = 0.00294
Oracle (Fisher at p*)	0.95 [0.91, 0.97]	w = 0.00293
Wald (delta)	0.94 [0.9, 0.97]	w = 0.00289
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.00289
ILR, REML	0.94 [0.9, 0.97]	w = 0.00293
Wald (delta)	0.94 [0.9, 0.97]	w = 0.00289
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.00288
ILR, REML	0.94 [0.9, 0.97]	w = 0.00294
Method's own interval	1 [0.98, 1]	w = 0.013
Own interval, transcript fractions	1 [0.98, 1]	w = 0.013
Fisher information (Wald)	0.015 [0.0051, 0.043]	w = 5.19e-05
Godambe sandwich	1 [0.98, 1]	w = 0.202
Released code, as published	1 [0.98, 1]	w = 1

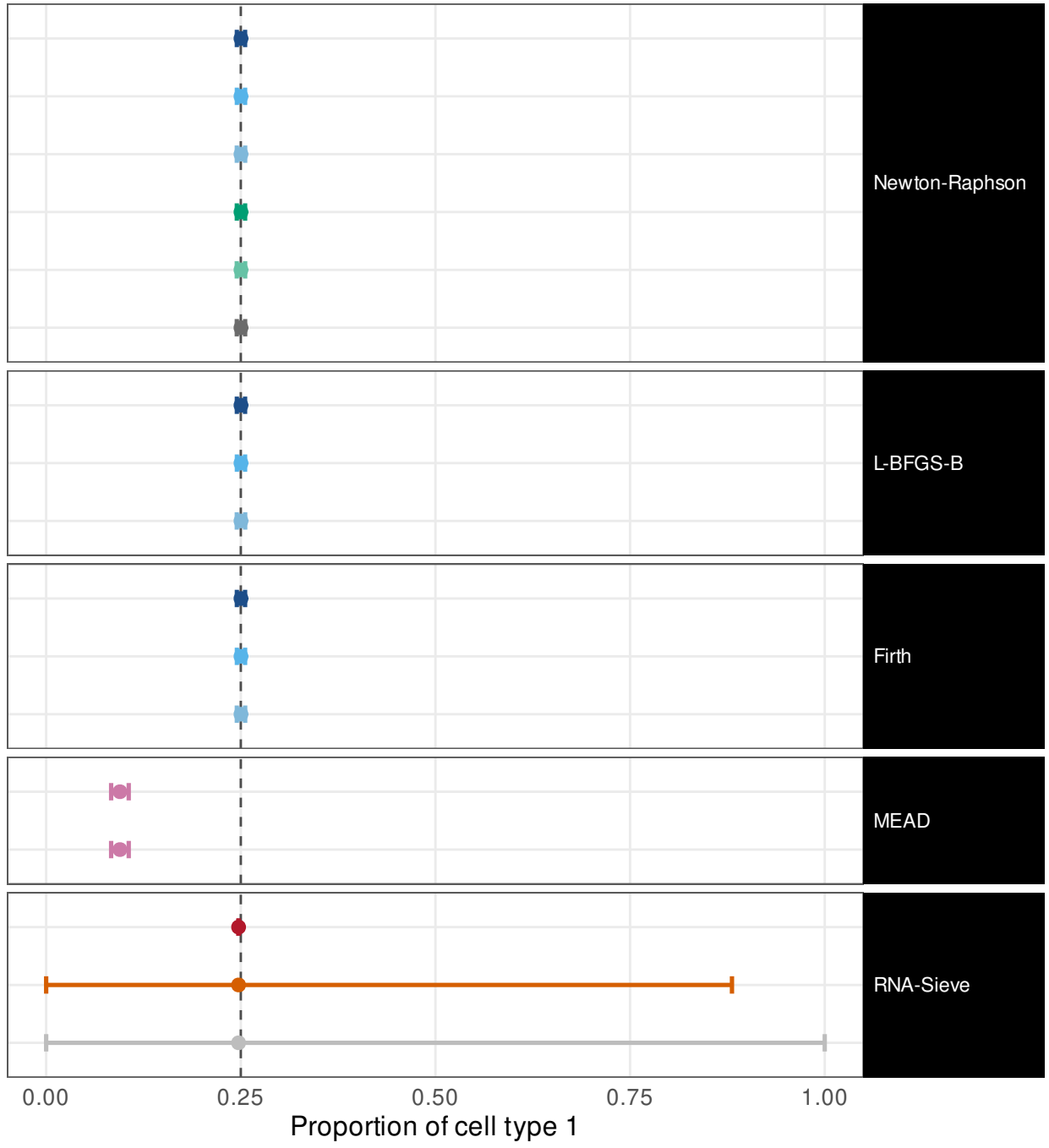


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / balanced / low sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.94 [0.9, 0.97]	w = 0.0106
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.0105
ILR, REML	0.96 [0.92, 0.98]	w = 0.0107
Profile likelihood	0.94 [0.9, 0.97]	w = 0.0106
Profile, REML	0.96 [0.92, 0.98]	w = 0.0107
Oracle (Fisher at p*)	0.96 [0.93, 0.98]	w = 0.0107
Wald (delta)	0.94 [0.9, 0.97]	w = 0.0106
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.0105
ILR, REML	0.96 [0.92, 0.98]	w = 0.0107
Wald (delta)	0.94 [0.9, 0.97]	w = 0.0106
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.0105
ILR, REML	0.94 [0.9, 0.97]	w = 0.0107
Method's own interval	0 [0, 0.019]	w = 0.0226
Own interval, transcript fractions	0.49 [0.42, 0.56]	w = 0.0226
Fisher information (Wald)	0.055 [0.031, 0.096]	w = 0.000661
Godambe sandwich	1 [0.98, 1]	w = 0.881
Released code, as published	1 [0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / balanced / low sibling overlap / a = 5 / sizes 1:1:5:5

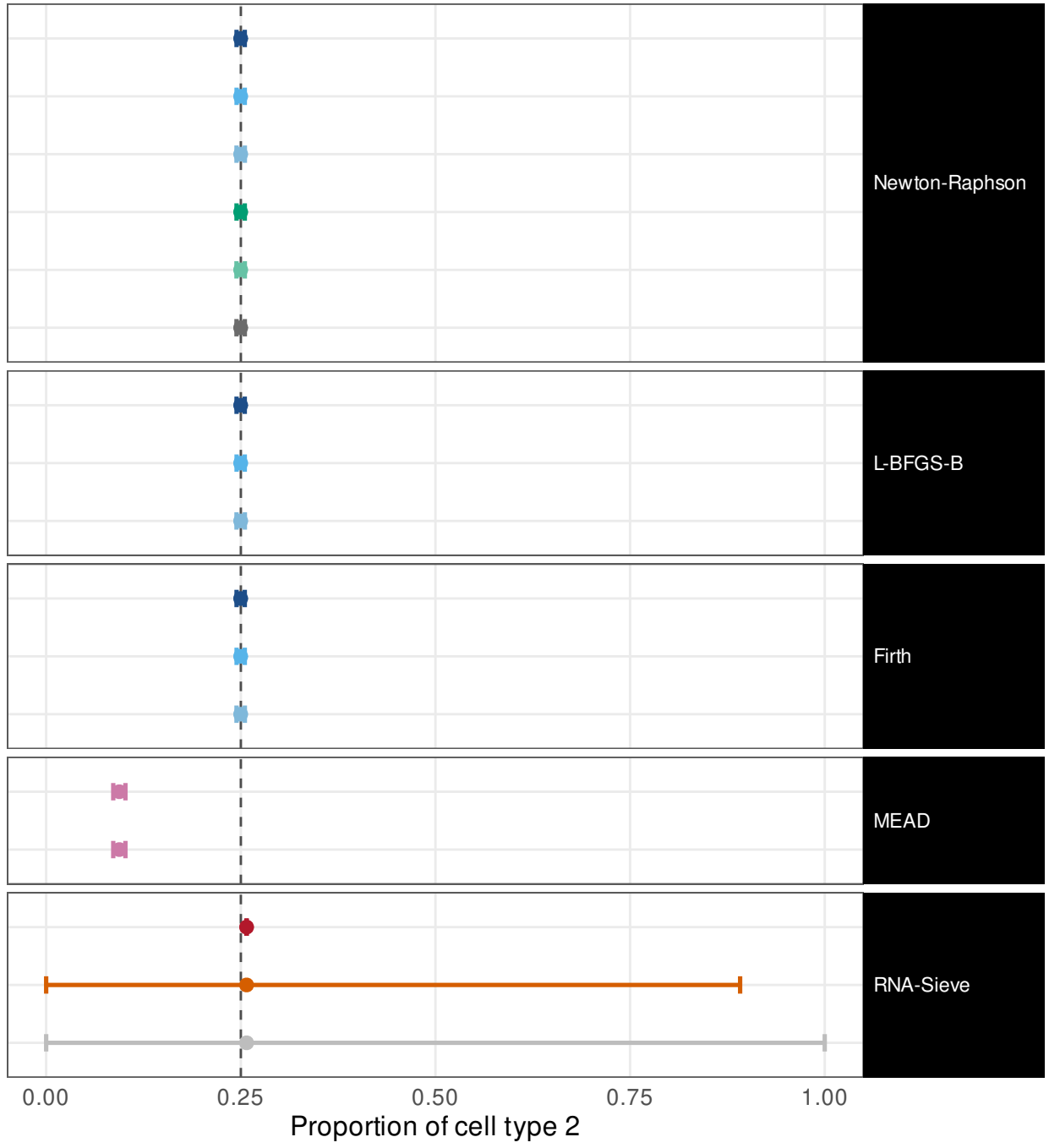
```
Wald (delta)                          0.96 [0.93, 0.98]   w = 0.0106
ILR, mapped back                      0.96 [0.93, 0.98]   w = 0.0105
ILR, REML                             0.97 [0.94, 0.99]   w = 0.0107
Profile likelihood                    0.96 [0.93, 0.98]   w = 0.0106
Profile, REML                         0.96 [0.93, 0.98]   w = 0.0107
Oracle (Fisher at p*)                 0.96 [0.93, 0.98]   w = 0.0107

Wald (delta)                          0.96 [0.93, 0.98]   w = 0.0106
ILR, mapped back                      0.96 [0.93, 0.98]   w = 0.0105
ILR, REML                             0.96 [0.93, 0.98]   w = 0.0107

Wald (delta)                          0.96 [0.93, 0.98]   w = 0.0106
ILR, mapped back                      0.96 [0.92, 0.98]   w = 0.0105
ILR, REML                             0.96 [0.93, 0.98]   w = 0.0107

Method's own interval                    0 [   0, 0.019]   w = 0.0156
Own interval, transcript fractions   0.005 [0.00088, 0.028]   w = 0.0156

Fisher information (Wald)             0.01 [0.0027, 0.036]   w = 0.000659
Godambe sandwich                         1 [0.98,    1]   w = 0.891
Released code, as published              1 [0.98,    1]   w =    1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / balanced / low sibling overlap / a = 5 / sizes 1:1:5:5

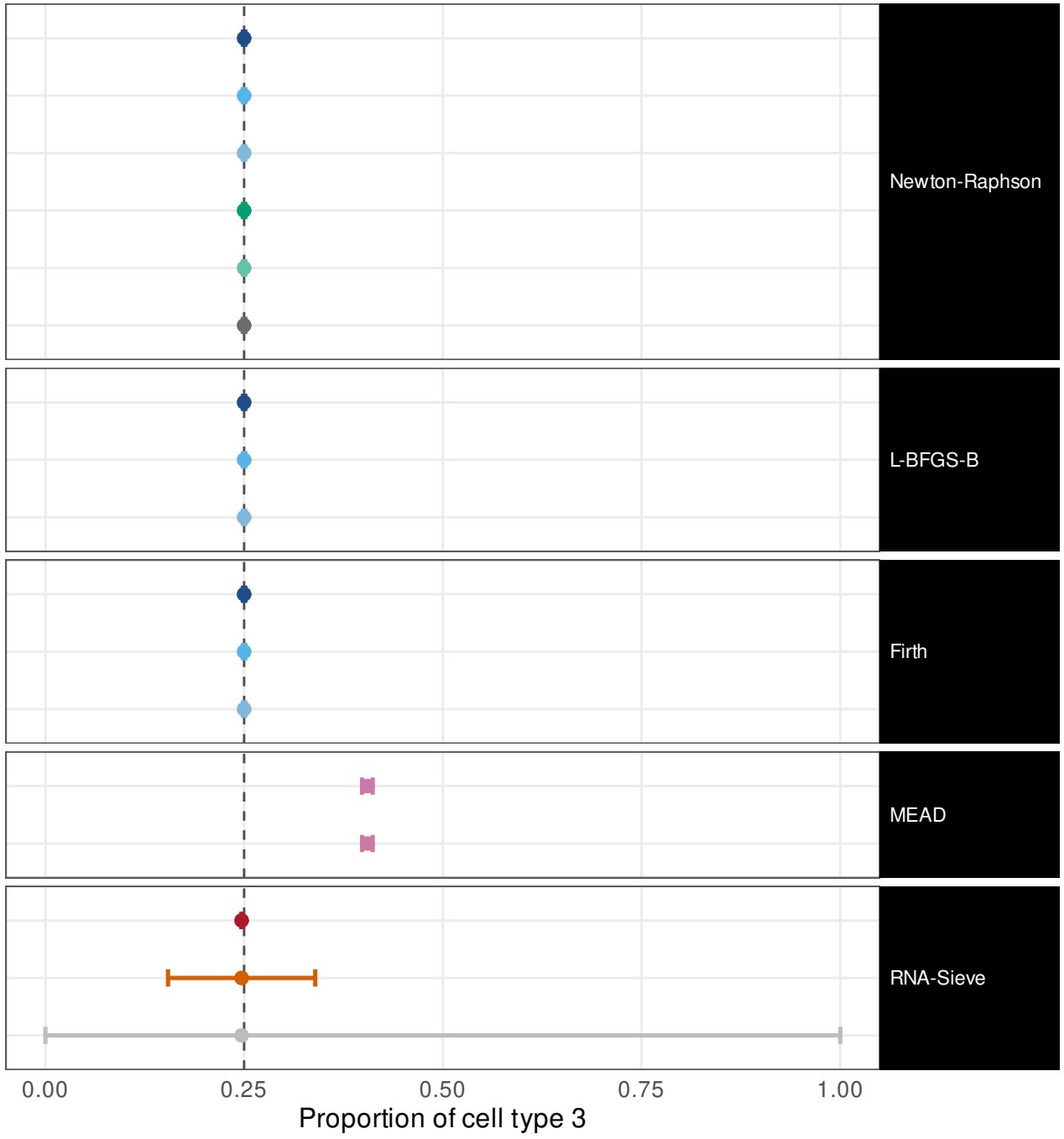
```
Wald (delta)                        0.94 [ 0.9, 0.97]   w = 0.00295
ILR, mapped back                    0.94 [0.89, 0.96]   w = 0.00294
ILR, REML                           0.94 [0.89, 0.96]   w = 0.003
Profile likelihood                  0.94 [ 0.9, 0.97]   w = 0.00296
Profile, REML                       0.94 [ 0.9, 0.97]   w = 0.003
Oracle (Fisher at p*)               0.94 [ 0.9, 0.97]   w = 0.00299

Wald (delta)                        0.94 [ 0.9, 0.97]   w = 0.00295
ILR, mapped back                    0.94 [ 0.9, 0.97]   w = 0.00294
ILR, REML                           0.94 [ 0.9, 0.97]   w = 0.00299

Wald (delta)                        0.94 [0.89, 0.96]   w = 0.00295
ILR, mapped back                    0.94 [0.89, 0.96]   w = 0.00294
ILR, REML                           0.94 [0.89, 0.96]   w = 0.00299

Method's own interval                  0 [   0, 0.019]   w = 0.0132
Own interval, transcript fractions     0 [   0, 0.019]   w = 0.0132

Fisher information (Wald)              0 [   0, 0.019]   w = 0.000261
Godambe sandwich                       1 [0.98,    1]   w = 0.185
Released code, as published            1 [0.98,    1]   w =    1
```

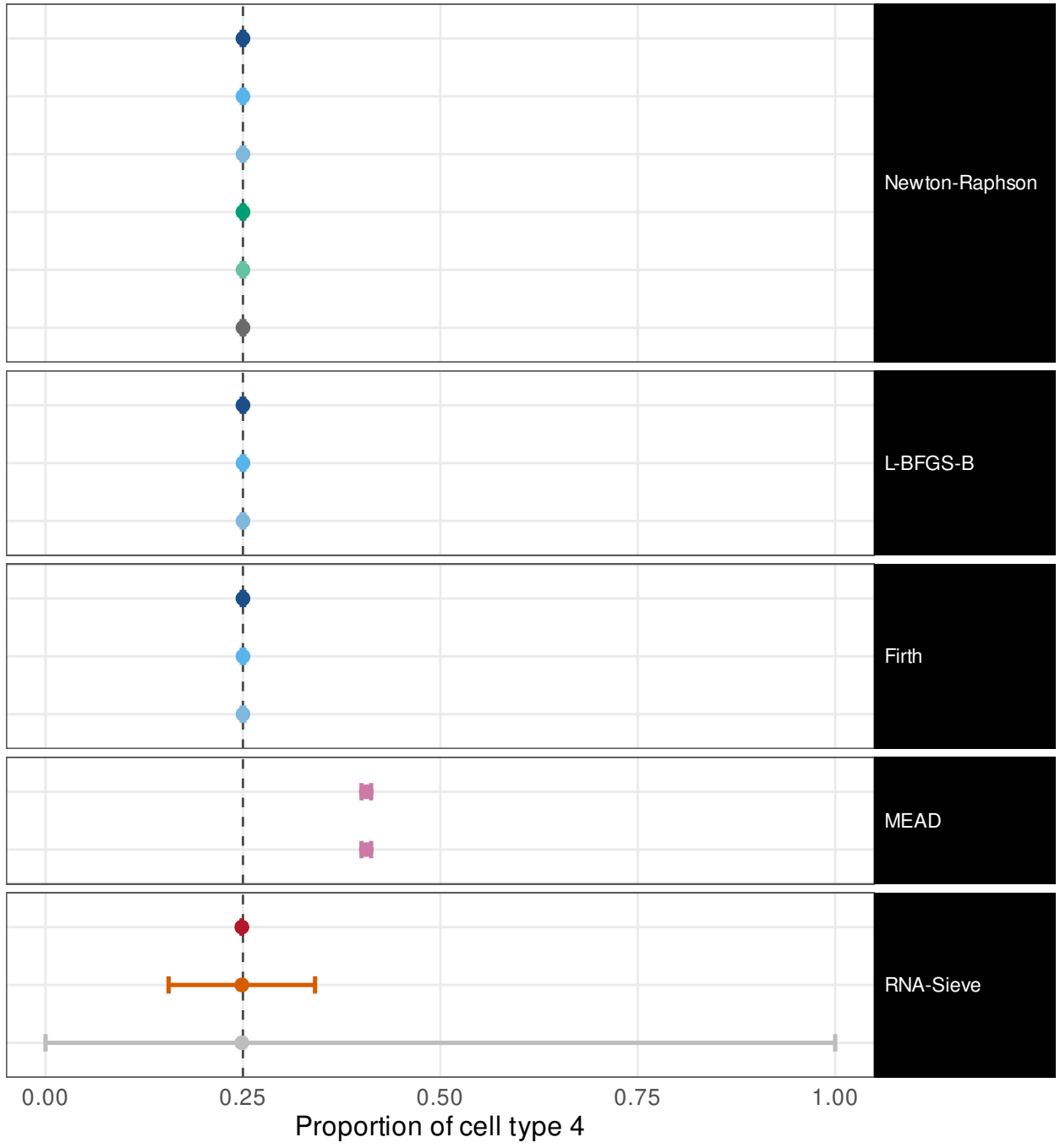


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / balanced / low sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.98 [0.94, 0.99]	w = 0.00294
ILR, mapped back	0.98 [0.94, 0.99]	w = 0.00294
ILR, REML	0.98 [0.94, 0.99]	w = 0.00298
Profile likelihood	0.98 [0.94, 0.99]	w = 0.00296
Profile, REML	0.98 [0.94, 0.99]	w = 0.00299
Oracle (Fisher at p*)	0.98 [0.94, 0.99]	w = 0.00298
Wald (delta)	0.98 [0.94, 0.99]	w = 0.00294
ILR, mapped back	0.98 [0.94, 0.99]	w = 0.00294
ILR, REML	0.98 [0.94, 0.99]	w = 0.00298
Wald (delta)	0.98 [0.94, 0.99]	w = 0.00294
ILR, mapped back	0.98 [0.94, 0.99]	w = 0.00293
ILR, REML	0.98 [0.94, 0.99]	w = 0.00298
Method's own interval	0 [0, 0.019]	w = 0.012
Own interval, transcript fractions	0 [0, 0.019]	w = 0.012
Fisher information (Wald)	0.035 [0.017, 0.07]	w = 0.000272
Godambe sandwich	1 [0.98, 1]	w = 0.185
Released code, as published	1 [0.98, 1]	w = 1

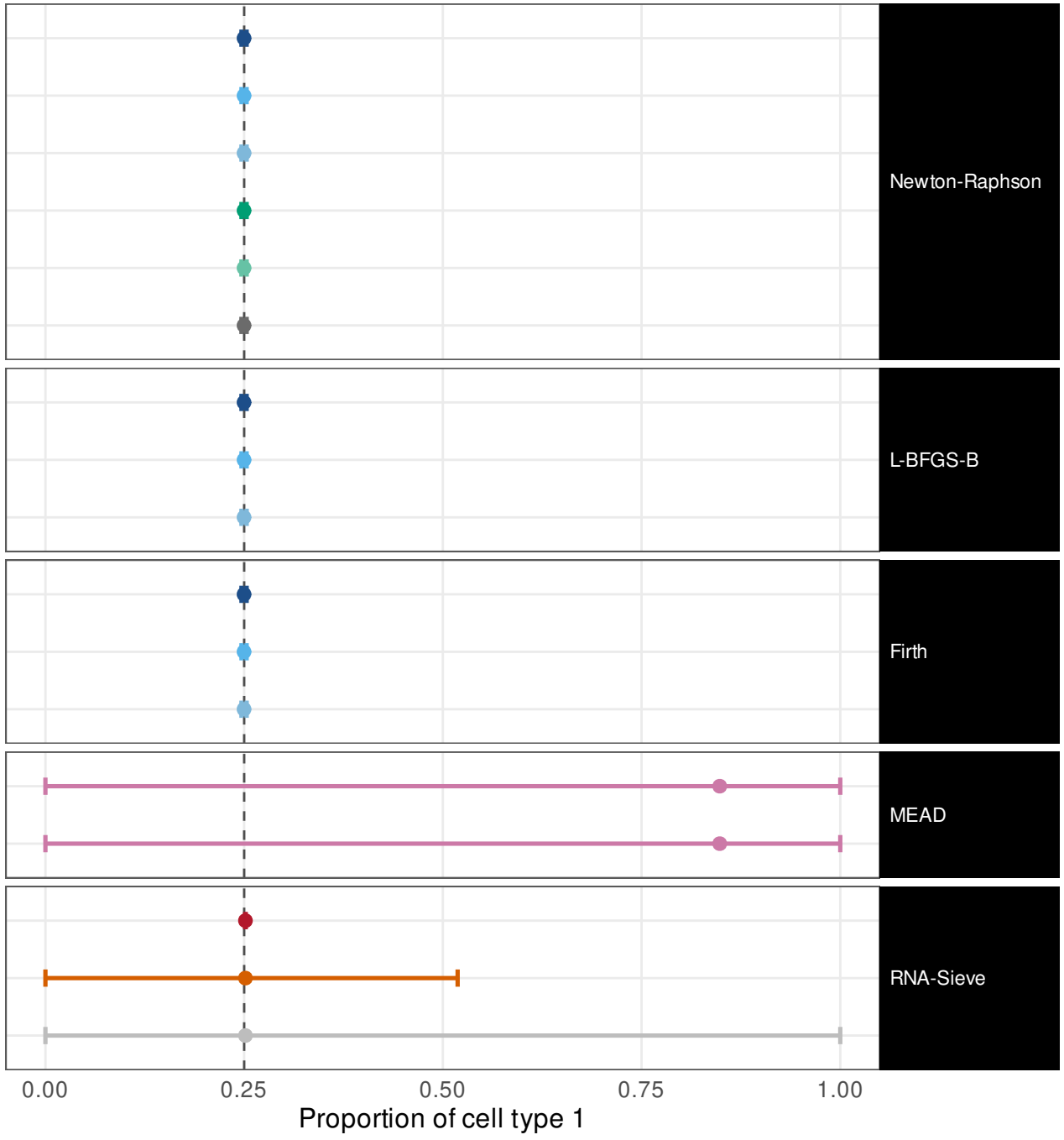


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / balanced / high sibling overlap / a = 1 / equal sizes

Wald (delta)	0.94 [0.9, 0.97]	w = 0.00646
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.00644
ILR, REML	0.94 [0.9, 0.97]	w = 0.00655
Profile likelihood	0.96 [0.92, 0.98]	w = 0.00649
Profile, REML	0.96 [0.92, 0.98]	w = 0.00657
Oracle (Fisher at p*)	0.96 [0.92, 0.98]	w = 0.00655
Wald (delta)	0.94 [0.9, 0.97]	w = 0.00646
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.00644
ILR, REML	0.94 [0.9, 0.97]	w = 0.00655
Wald (delta)	0.94 [0.9, 0.97]	w = 0.00646
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.00644
ILR, REML	0.95 [0.91, 0.97]	w = 0.00656
Method's own interval	1 [0.98, 1]	w = 1
Own interval, transcript fractions	1 [0.98, 1]	w = 1
Fisher information (Wald)	0.12 [0.082, 0.17]	w = 0.000852
Godambe sandwich	1 [0.98, 1]	w = 0.519
Released code, as published	1 [0.98, 1]	w = 1

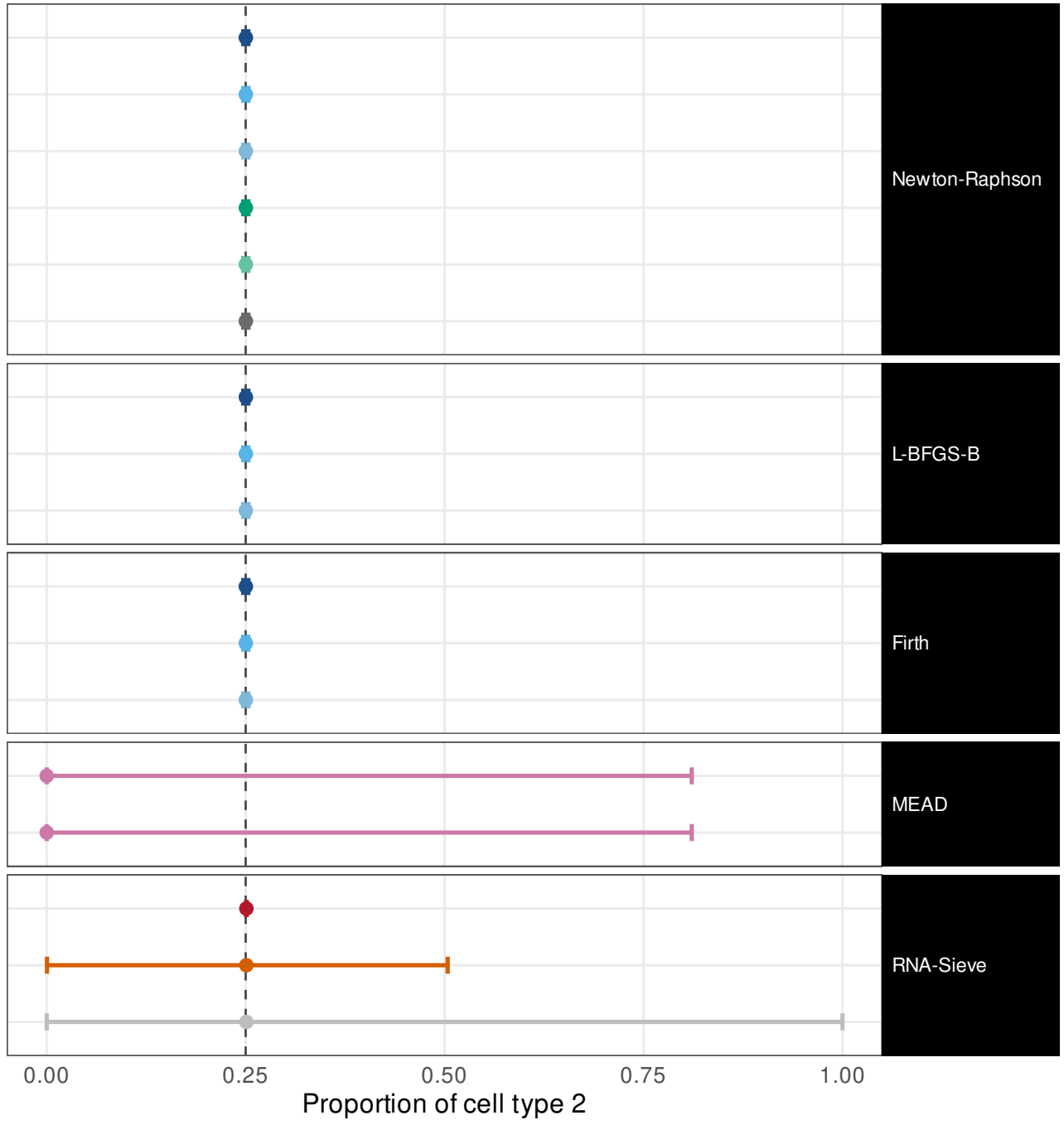


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / balanced / high sibling overlap / a = 1 / equal sizes

Wald (delta)	0.92 [0.87, 0.95]	w = 0.00641
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.00638
ILR, REML	0.94 [0.89, 0.96]	w = 0.0065
Profile likelihood	0.92 [0.88, 0.95]	w = 0.00644
Profile, REML	0.92 [0.88, 0.95]	w = 0.00651
Oracle (Fisher at p*)	0.94 [0.9, 0.97]	w = 0.00649
Wald (delta)	0.92 [0.87, 0.95]	w = 0.00641
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.0064
ILR, REML	0.92 [0.88, 0.95]	w = 0.0065
Wald (delta)	0.92 [0.87, 0.95]	w = 0.00641
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.00638
ILR, REML	0.94 [0.89, 0.96]	w = 0.0065
Method's own interval	1 [0.98, 1]	w = 0.811
Own interval, transcript fractions	1 [0.98, 1]	w = 0.811
Fisher information (Wald)	0.15 [0.11, 0.21]	w = 0.000889
Godambe sandwich	1 [0.98, 1]	w = 0.503
Released code, as published	1 [0.98, 1]	w = 1

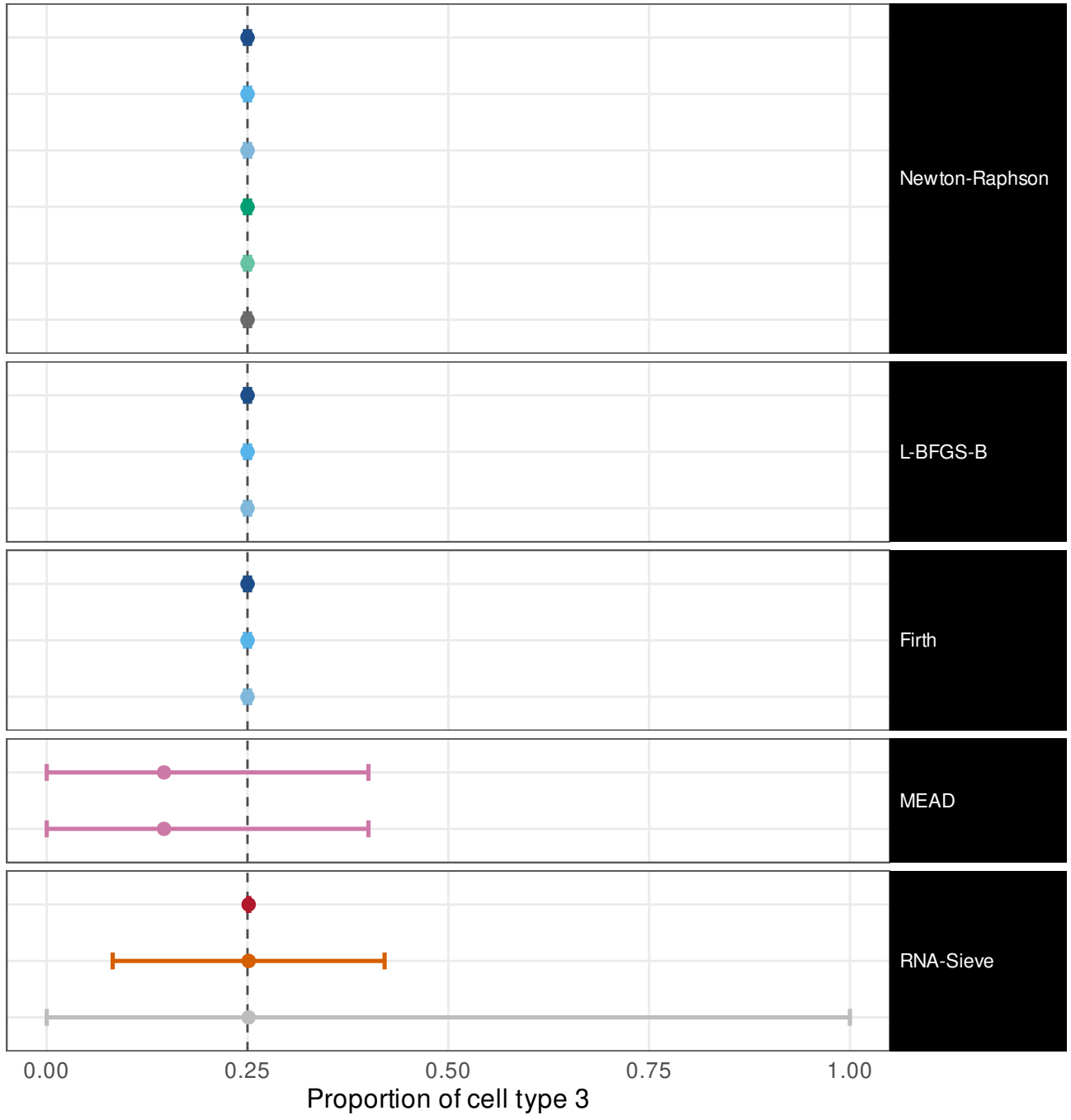


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / balanced / high sibling overlap / a = 1 / equal sizes

Wald (delta)	0.96 [0.92, 0.98]	w = 0.00639
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.00635
ILR, REML	0.96 [0.92, 0.98]	w = 0.00647
Profile likelihood	0.96 [0.92, 0.98]	w = 0.00642
Profile, REML	0.96 [0.92, 0.98]	w = 0.00649
Oracle (Fisher at p*)	0.96 [0.92, 0.98]	w = 0.00648
Wald (delta)	0.96 [0.92, 0.98]	w = 0.00639
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.00637
ILR, REML	0.96 [0.92, 0.98]	w = 0.00649
Wald (delta)	0.96 [0.92, 0.98]	w = 0.00639
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.00637
ILR, REML	0.96 [0.92, 0.98]	w = 0.00648
Method's own interval	1 [0.98, 1]	w = 0.401
Own interval, transcript fractions	1 [0.98, 1]	w = 0.401
Fisher information (Wald)	0.32 [0.26, 0.39]	w = 0.00154
Godambe sandwich	1 [0.98, 1]	w = 0.339
Released code, as published	1 [0.98, 1]	w = 1

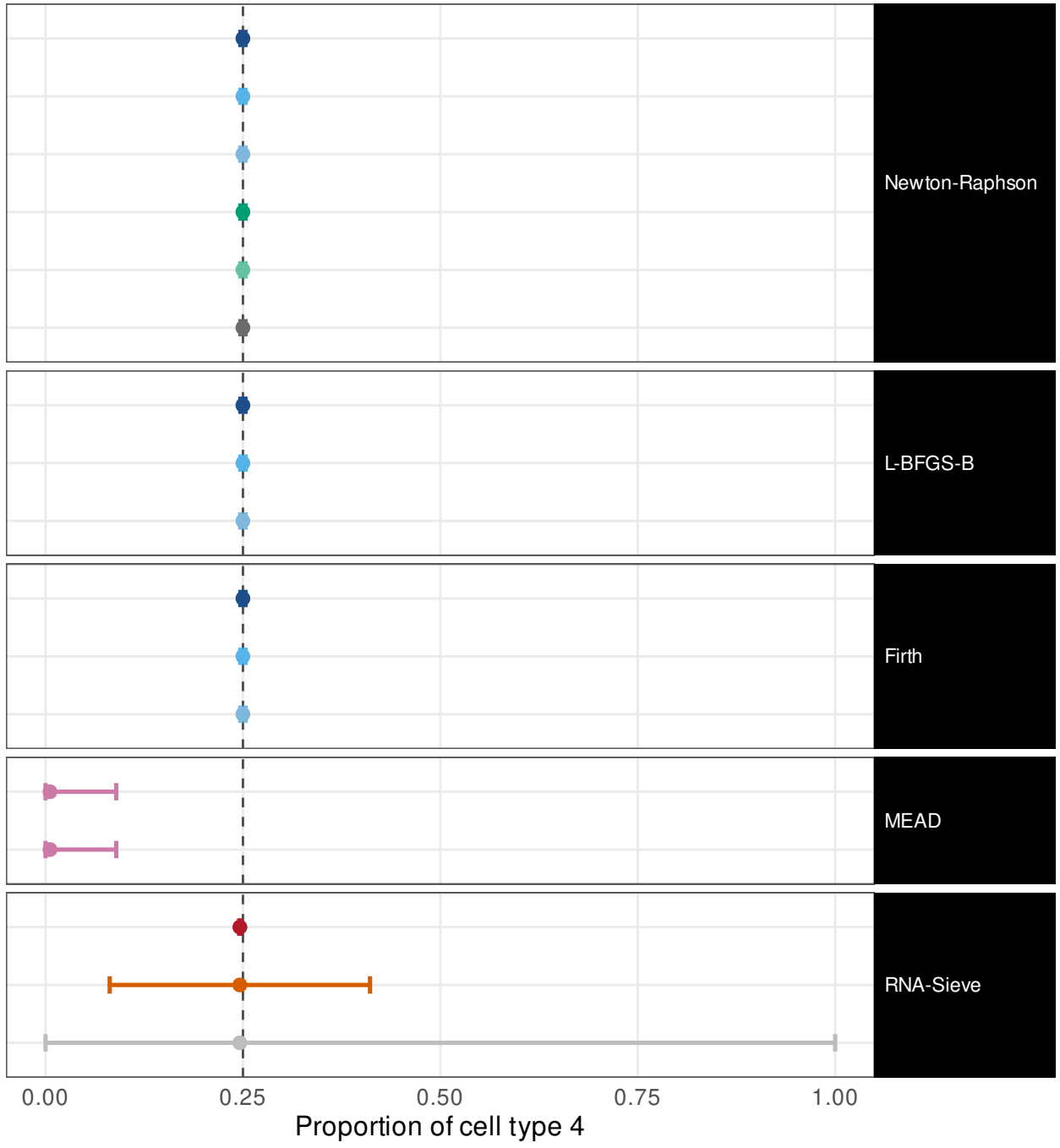


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / balanced / high sibling overlap / a = 1 / equal sizes

Wald (delta)	0.94 [0.89, 0.96]	w = 0.00642
ILR, mapped back	0.94 [0.89, 0.96]	w = 0.0064
ILR, REML	0.94 [0.9, 0.97]	w = 0.00651
Profile likelihood	0.94 [0.9, 0.97]	w = 0.00646
Profile, REML	0.94 [0.9, 0.97]	w = 0.00653
Oracle (Fisher at p*)	0.94 [0.9, 0.97]	w = 0.00651
Wald (delta)	0.94 [0.89, 0.96]	w = 0.00642
ILR, mapped back	0.94 [0.89, 0.96]	w = 0.00643
ILR, REML	0.94 [0.9, 0.97]	w = 0.00651
Wald (delta)	0.94 [0.89, 0.96]	w = 0.00642
ILR, mapped back	0.94 [0.89, 0.96]	w = 0.00639
ILR, REML	0.95 [0.91, 0.97]	w = 0.00654
Method's own interval	0 [0, 0.019]	w = 0.0895
Own interval, transcript fractions	0 [0, 0.019]	w = 0.0895
Fisher information (Wald)	0.025 [0.011, 0.057]	w = 0.00158
Godambe sandwich	1 [0.98, 1]	w = 0.33
Released code, as published	1 [0.98, 1]	w = 1

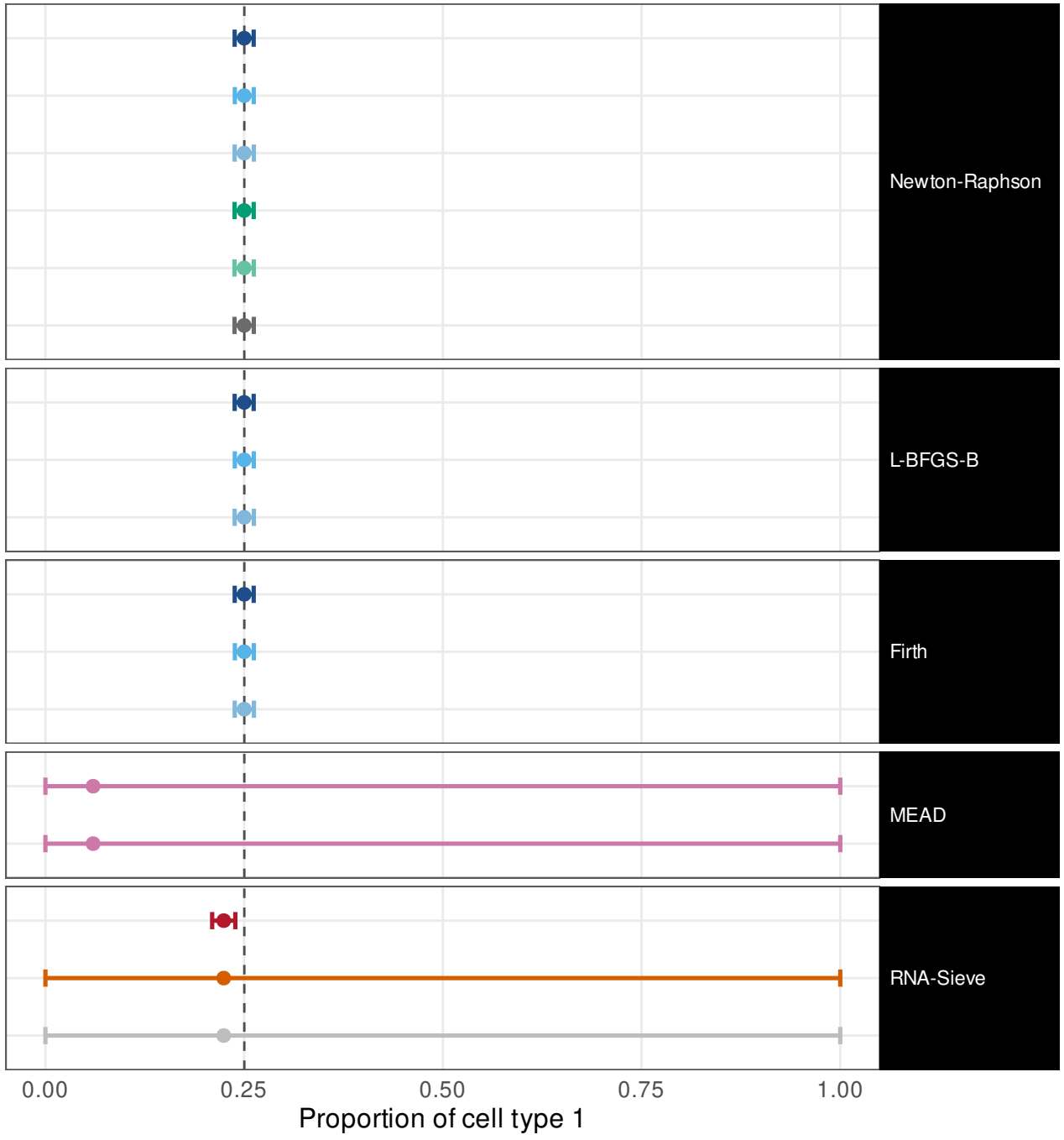


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / balanced / high sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.94 [0.9, 0.97]	w = 0.0239
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.0238
ILR, REML	0.95 [0.91, 0.97]	w = 0.0242
Profile likelihood	0.94 [0.9, 0.97]	w = 0.024
Profile, REML	0.96 [0.92, 0.98]	w = 0.0243
Oracle (Fisher at p*)	0.95 [0.91, 0.97]	w = 0.0242
Wald (delta)	0.94 [0.9, 0.97]	w = 0.0239
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.0239
ILR, REML	0.96 [0.92, 0.98]	w = 0.0242
Wald (delta)	0.94 [0.9, 0.97]	w = 0.0239
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.0239
ILR, REML	0.93 [0.89, 0.96]	w = 0.0241
Method's own interval	1 [0.98, 1]	w = 1
Own interval, transcript fractions	1 [0.98, 1]	w = 1
Fisher information (Wald)	0.055 [0.031, 0.096]	w = 0.0291
Godambe sandwich	1 [0.98, 1]	w = 1
Released code, as published	1 [0.98, 1]	w = 1

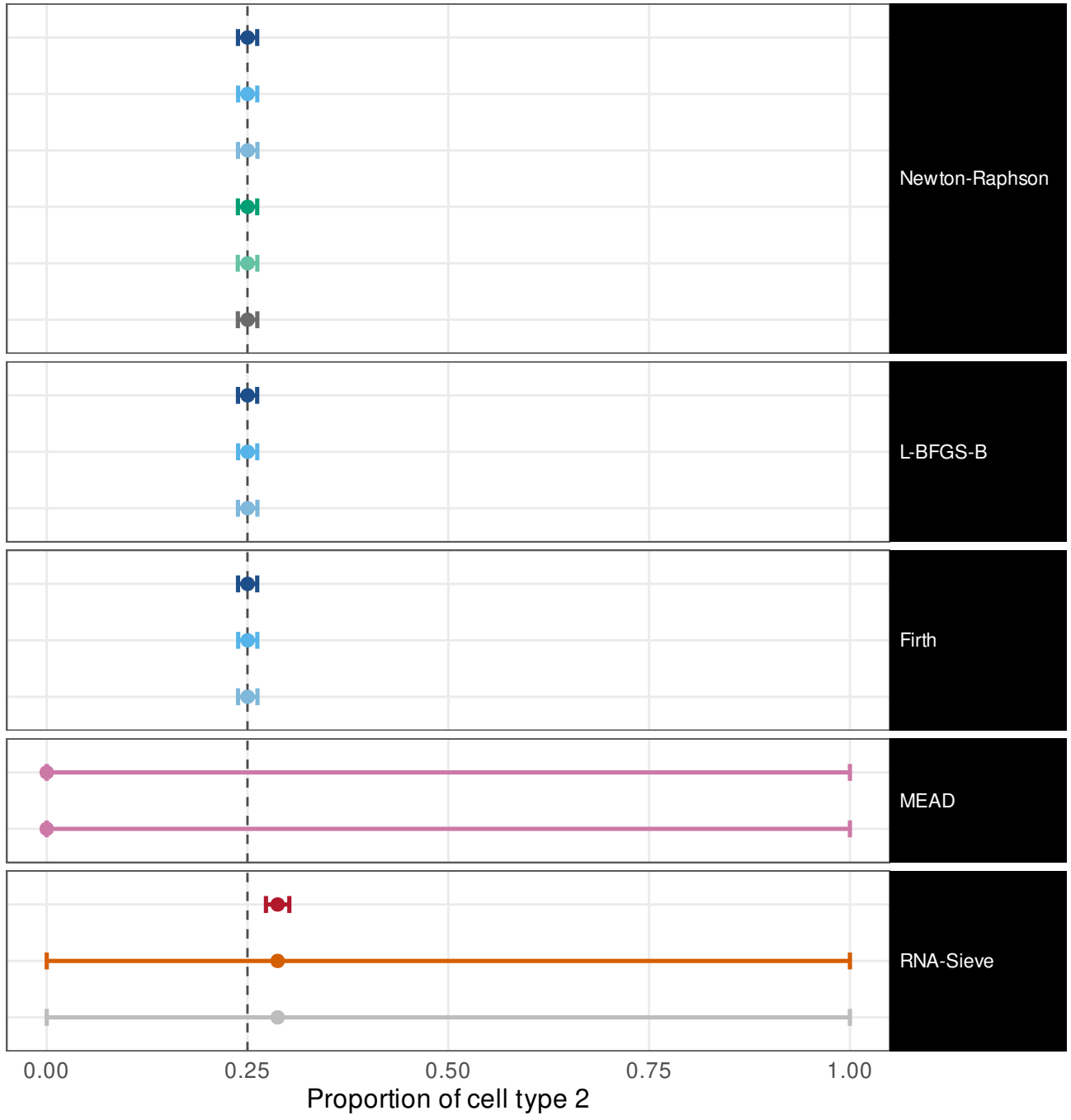


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / balanced / high sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.94 [0.89, 0.96]	w = 0.0239
ILR, mapped back	0.94 [0.89, 0.96]	w = 0.0238
ILR, REML	0.95 [0.91, 0.97]	w = 0.0242
Profile likelihood	0.94 [0.9, 0.97]	w = 0.024
Profile, REML	0.94 [0.9, 0.97]	w = 0.0243
Oracle (Fisher at p*)	0.94 [0.9, 0.97]	w = 0.0242
Wald (delta)	0.94 [0.89, 0.96]	w = 0.0239
ILR, mapped back	0.92 [0.88, 0.95]	w = 0.0239
ILR, REML	0.94 [0.9, 0.97]	w = 0.0242
Wald (delta)	0.94 [0.89, 0.96]	w = 0.0239
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.0238
ILR, REML	0.96 [0.92, 0.98]	w = 0.0242
Method's own interval	1 [0.98, 1]	w = 1
Own interval, transcript fractions	1 [0.98, 1]	w = 1
Fisher information (Wald)	0 [0, 0.019]	w = 0.0291
Godambe sandwich	1 [0.98, 1]	w = 1
Released code, as published	1 [0.98, 1]	w = 1

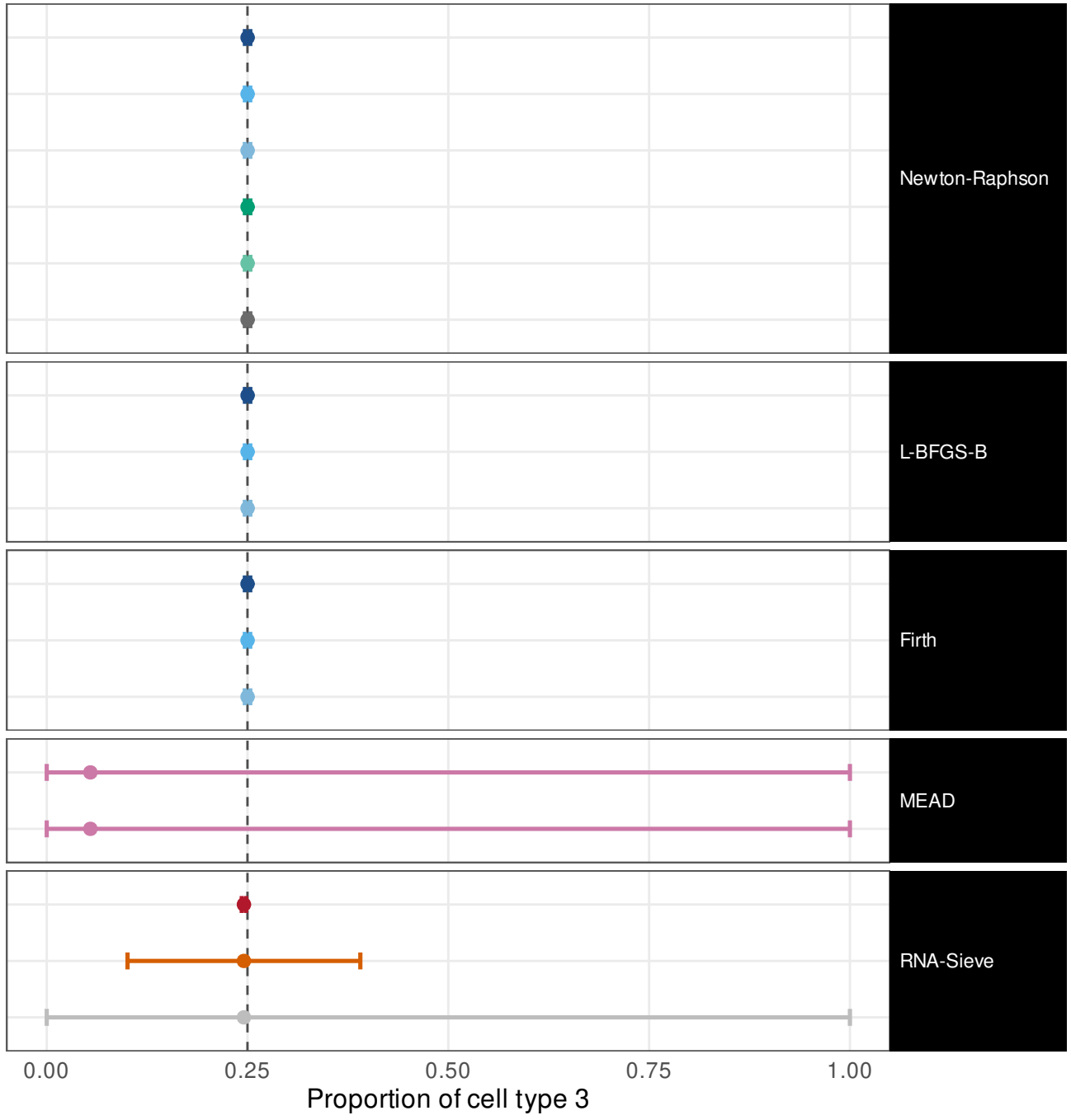


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / balanced / high sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.94 [0.89, 0.96]	w = 0.00668
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.00668
ILR, REML	0.95 [0.91, 0.97]	w = 0.00677
Profile likelihood	0.94 [0.9, 0.97]	w = 0.00671
Profile, REML	0.94 [0.9, 0.97]	w = 0.00679
Oracle (Fisher at p*)	0.96 [0.92, 0.98]	w = 0.00676
Wald (delta)	0.94 [0.89, 0.96]	w = 0.00668
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.00667
ILR, REML	0.95 [0.91, 0.97]	w = 0.00676
Wald (delta)	0.94 [0.9, 0.97]	w = 0.00668
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.00665
ILR, REML	0.94 [0.9, 0.97]	w = 0.00678
Method's own interval	1 [0.98, 1]	w = 1
Own interval, transcript fractions	1 [0.98, 1]	w = 1
Fisher information (Wald)	0.16 [0.12, 0.22]	w = 0.00544
Godambe sandwich	1 [0.98, 1]	w = 0.29
Released code, as published	1 [0.98, 1]	w = 1

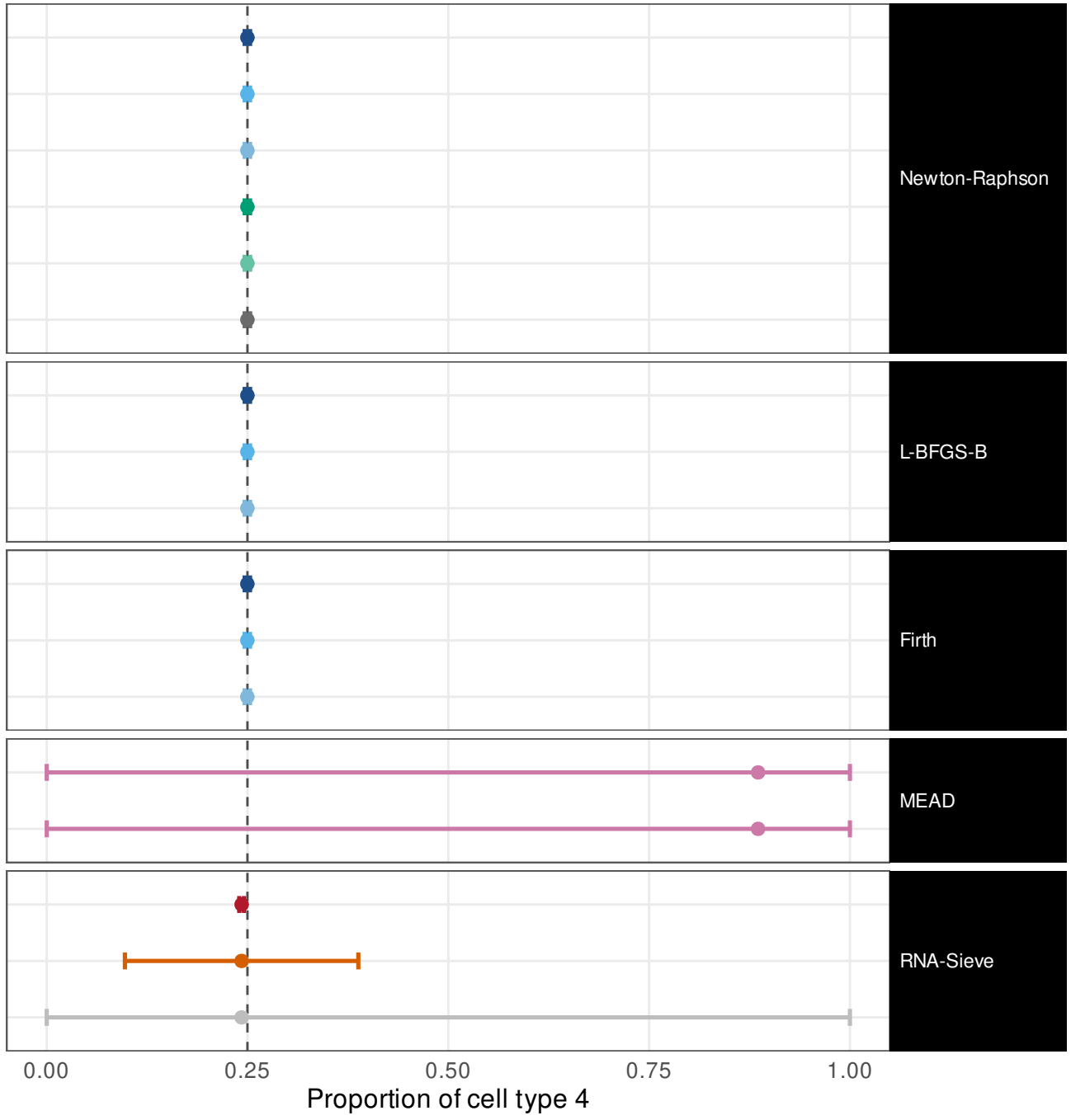


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / balanced / high sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.97 [0.94, 0.99]	w = 0.00666
ILR, mapped back	0.96 [0.93, 0.98]	w = 0.00666
ILR, REML	0.97 [0.94, 0.99]	w = 0.00675
Profile likelihood	0.97 [0.94, 0.99]	w = 0.0067
Profile, REML	0.97 [0.94, 0.99]	w = 0.00677
Oracle (Fisher at p*)	0.97 [0.94, 0.99]	w = 0.00674
Wald (delta)	0.97 [0.94, 0.99]	w = 0.00666
ILR, mapped back	0.97 [0.94, 0.99]	w = 0.00665
ILR, REML	0.96 [0.93, 0.98]	w = 0.00675
Wald (delta)	0.97 [0.94, 0.99]	w = 0.00666
ILR, mapped back	0.97 [0.94, 0.99]	w = 0.00664
ILR, REML	0.96 [0.93, 0.98]	w = 0.00675
Method's own interval	1 [0.98, 1]	w = 1
Own interval, transcript fractions	1 [0.98, 1]	w = 1
Fisher information (Wald)	0 [0, 0.019]	w = 0.0059
Godambe sandwich	1 [0.98, 1]	w = 0.291
Released code, as published	1 [0.98, 1]	w = 1

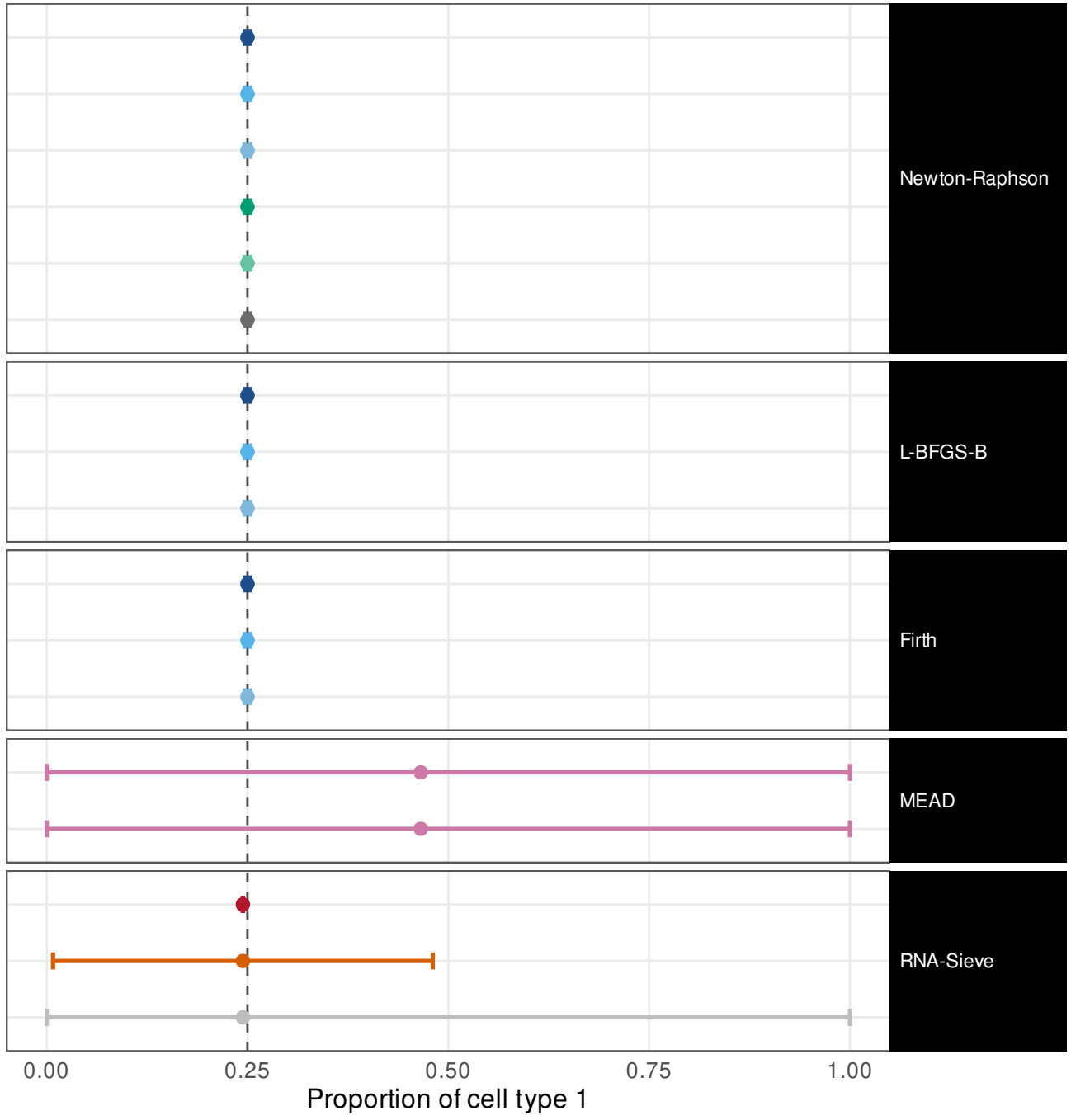


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / balanced / high sibling overlap / a = 5 / equal sizes

Wald (delta)	0.96 [0.92, 0.98]	w = 0.00643
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.0064
ILR, REML	0.96 [0.92, 0.98]	w = 0.00652
Profile likelihood	0.96 [0.92, 0.98]	w = 0.00646
Profile, REML	0.96 [0.92, 0.98]	w = 0.00653
Oracle (Fisher at p*)	0.96 [0.93, 0.98]	w = 0.0065
Wald (delta)	0.96 [0.92, 0.98]	w = 0.00643
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.00642
ILR, REML	0.96 [0.92, 0.98]	w = 0.00651
Wald (delta)	0.96 [0.92, 0.98]	w = 0.00643
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.00643
ILR, REML	0.96 [0.93, 0.98]	w = 0.00653
Method's own interval	1 [0.98, 1]	w = 1
Own interval, transcript fractions	1 [0.98, 1]	w = 1
Fisher information (Wald)	0 [0, 0.019]	w = 0.0004
Godambe sandwich	1 [0.98, 1]	w = 0.473
Released code, as published	1 [0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / balanced / high sibling overlap / a = 5 / equal sizes

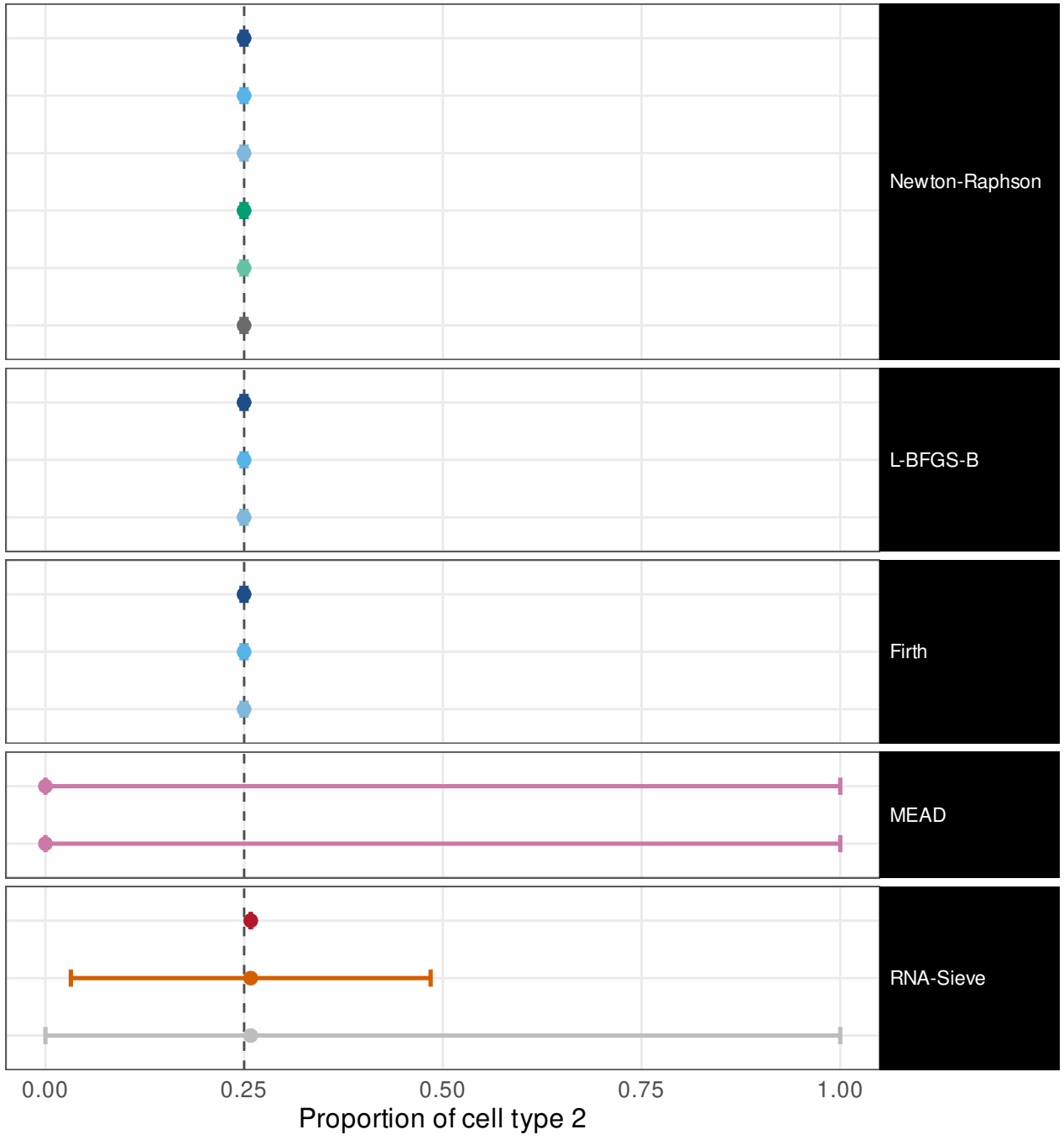
```
Wald (delta)                        0.95 [0.91, 0.97]   w = 0.00637
ILR, mapped back                    0.96 [0.92, 0.98]   w = 0.00635
ILR, REML                           0.96 [0.92, 0.98]   w = 0.00646
Profile likelihood                  0.95 [0.91, 0.97]   w = 0.0064
Profile, REML                       0.96 [0.92, 0.98]   w = 0.00648
Oracle (Fisher at p*)               0.96 [0.93, 0.98]   w = 0.00644

Wald (delta)                        0.95 [0.91, 0.97]   w = 0.00637
ILR, mapped back                    0.96 [0.92, 0.98]   w = 0.00637
ILR, REML                           0.96 [0.92, 0.98]   w = 0.00646

Wald (delta)                        0.95 [0.91, 0.97]   w = 0.00637
ILR, mapped back                    0.96 [0.92, 0.98]   w = 0.00636
ILR, REML                           0.96 [0.92, 0.98]   w = 0.00647

Method's own interval                  1 [0.98,    1]   w =    1
Own interval, transcript fractions     1 [0.98,    1]   w =    1

Fisher information (Wald)              0 [   0, 0.019]   w = 0.000411
Godambe sandwich                       1 [0.98,    1]   w = 0.453
Released code, as published            1 [0.98,    1]   w =    1
```

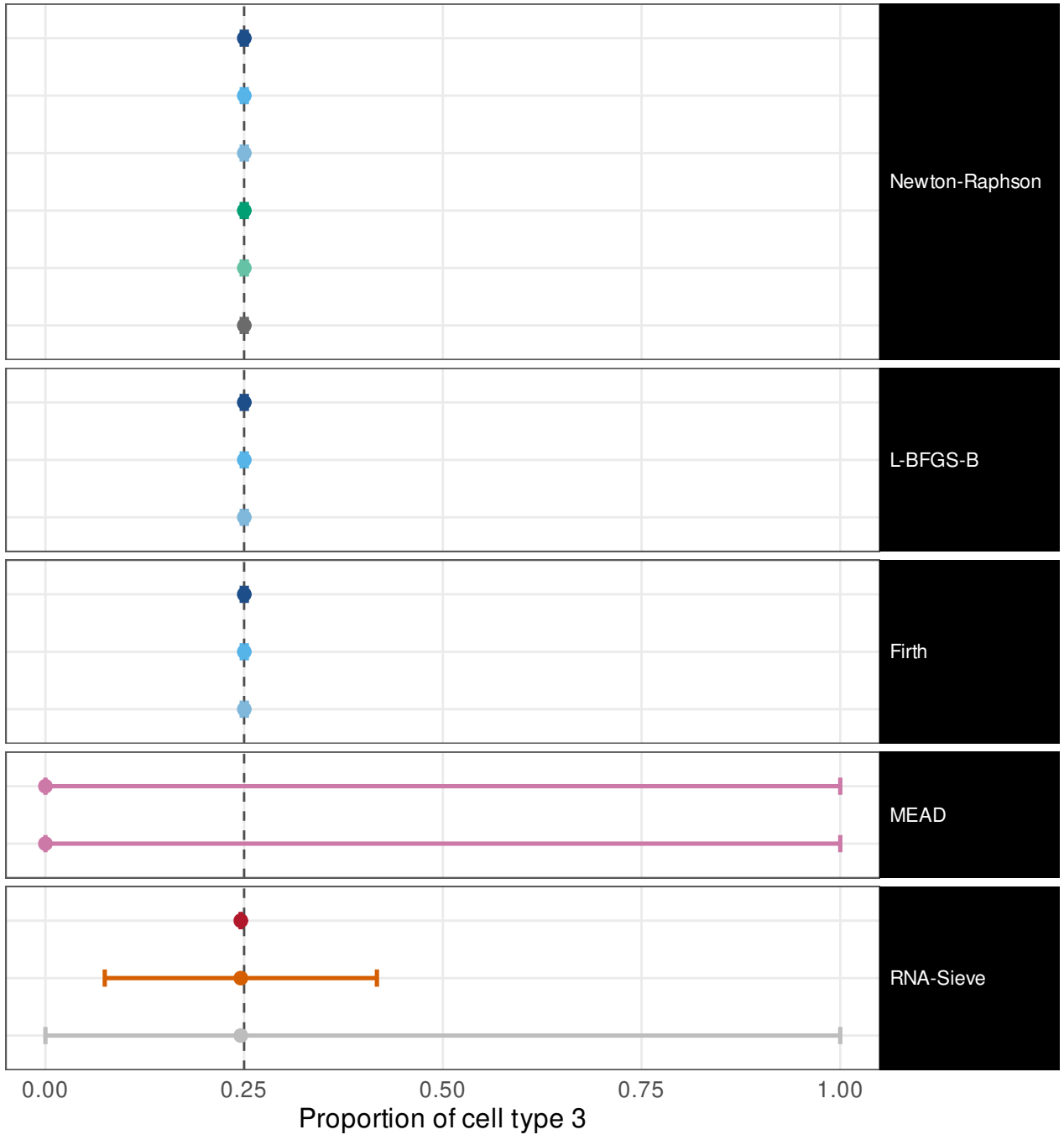


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / balanced / high sibling overlap / a = 5 / equal sizes

Wald (delta)	0.94 [0.9, 0.97]	w = 0.00635
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.00633
ILR, REML	0.94 [0.9, 0.97]	w = 0.00646
Profile likelihood	0.94 [0.9, 0.97]	w = 0.00638
Profile, REML	0.94 [0.9, 0.97]	w = 0.00646
Oracle (Fisher at p*)	0.94 [0.9, 0.97]	w = 0.00644
Wald (delta)	0.94 [0.9, 0.97]	w = 0.00635
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.00634
ILR, REML	0.94 [0.9, 0.97]	w = 0.00644
Wald (delta)	0.94 [0.9, 0.97]	w = 0.00635
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.00633
ILR, REML	0.94 [0.9, 0.97]	w = 0.00643
Method's own interval	1 [0.98, 1]	w = 1
Own interval, transcript fractions	1 [0.98, 1]	w = 1
Fisher information (Wald)	0.02 [0.0078, 0.05]	w = 0.00059
Godambe sandwich	1 [0.98, 1]	w = 0.342
Released code, as published	1 [0.98, 1]	w = 1

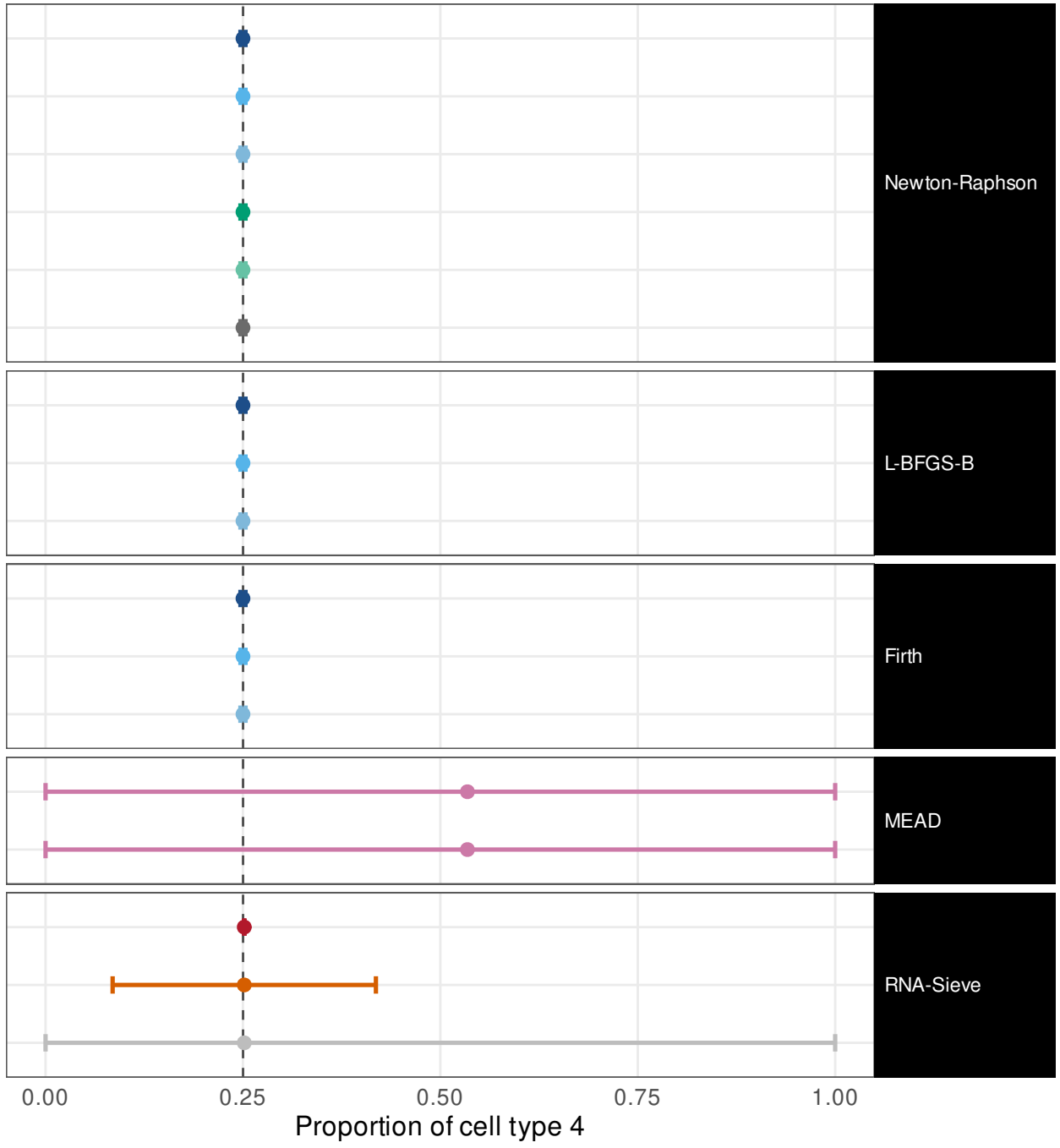


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / balanced / high sibling overlap / a = 5 / equal sizes

Wald (delta)	0.93 [0.89, 0.96]	w = 0.00639
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.00636
ILR, REML	0.94 [0.9, 0.97]	w = 0.0065
Profile likelihood	0.94 [0.89, 0.96]	w = 0.00642
Profile, REML	0.94 [0.9, 0.97]	w = 0.00649
Oracle (Fisher at p*)	0.95 [0.91, 0.97]	w = 0.00647
Wald (delta)	0.93 [0.89, 0.96]	w = 0.00639
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.00635
ILR, REML	0.94 [0.9, 0.97]	w = 0.00648
Wald (delta)	0.93 [0.89, 0.96]	w = 0.00639
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.00637
ILR, REML	0.94 [0.9, 0.97]	w = 0.00647
Method's own interval	1 [0.98, 1]	w = 1
Own interval, transcript fractions	1 [0.98, 1]	w = 1
Fisher information (Wald)	0.065 [0.038, 0.11]	w = 0.000608
Godambe sandwich	1 [0.98, 1]	w = 0.333
Released code, as published	1 [0.98, 1]	w = 1

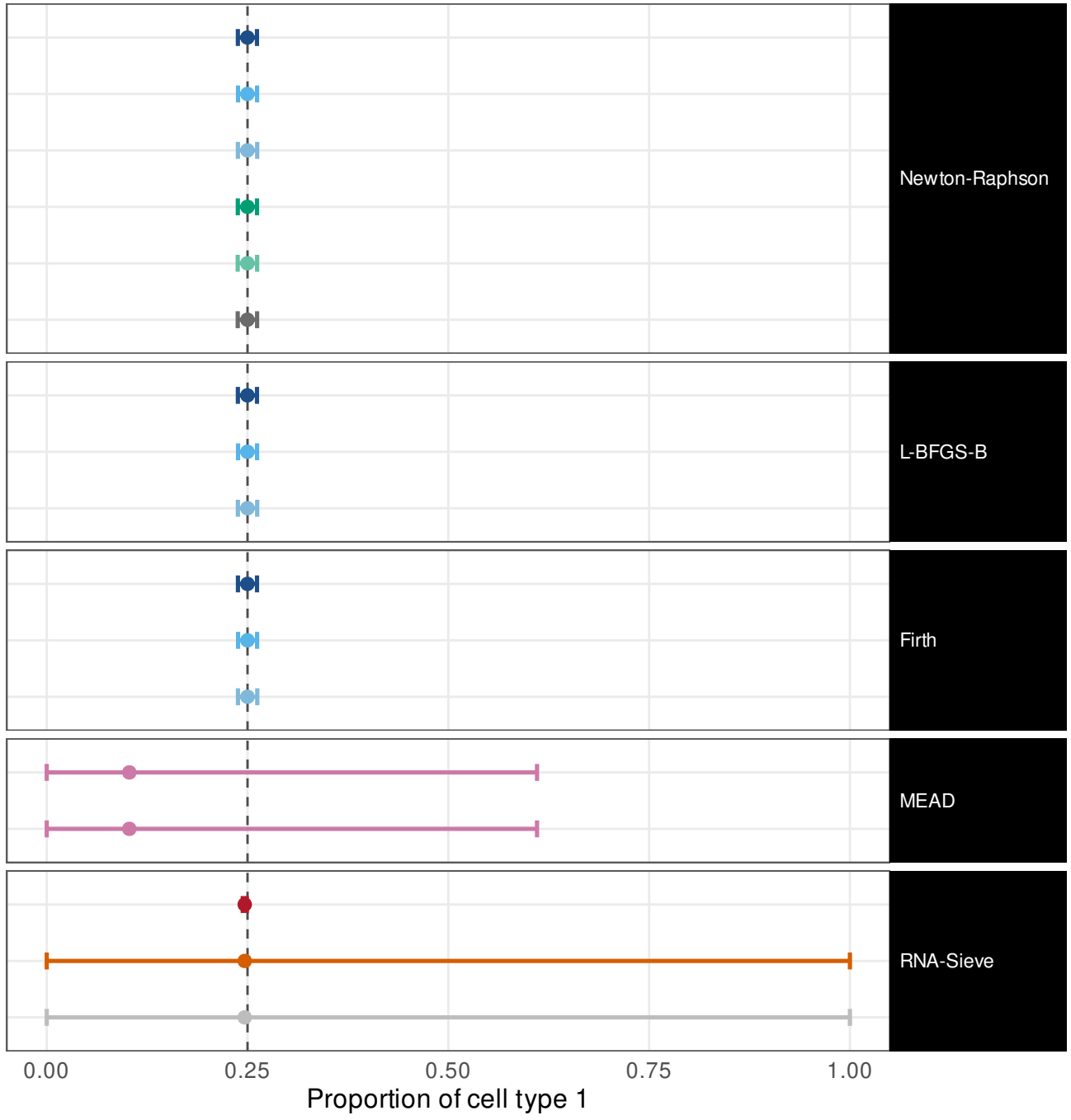


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / balanced / high sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.94 [0.9, 0.97]	w = 0.0238
ILR, mapped back	0.94 [0.89, 0.96]	w = 0.0237
ILR, REML	0.93 [0.89, 0.96]	w = 0.024
Profile likelihood	0.94 [0.9, 0.97]	w = 0.0239
Profile, REML	0.94 [0.9, 0.97]	w = 0.0242
Oracle (Fisher at p*)	0.94 [0.9, 0.97]	w = 0.0241
Wald (delta)	0.94 [0.9, 0.97]	w = 0.0238
ILR, mapped back	0.94 [0.89, 0.96]	w = 0.0237
ILR, REML	0.94 [0.89, 0.96]	w = 0.024
Wald (delta)	0.93 [0.89, 0.96]	w = 0.0237
ILR, mapped back	0.94 [0.89, 0.96]	w = 0.0237
ILR, REML	0.94 [0.89, 0.96]	w = 0.0241
Method's own interval	1 [0.98, 1]	w = 0.61
Own interval, transcript fractions	1 [0.98, 1]	w = 0.61
Fisher information (Wald)	0.22 [0.16, 0.28]	w = 0.00489
Godambe sandwich	1 [0.98, 1]	w = 1
Released code, as published	1 [0.98, 1]	w = 1

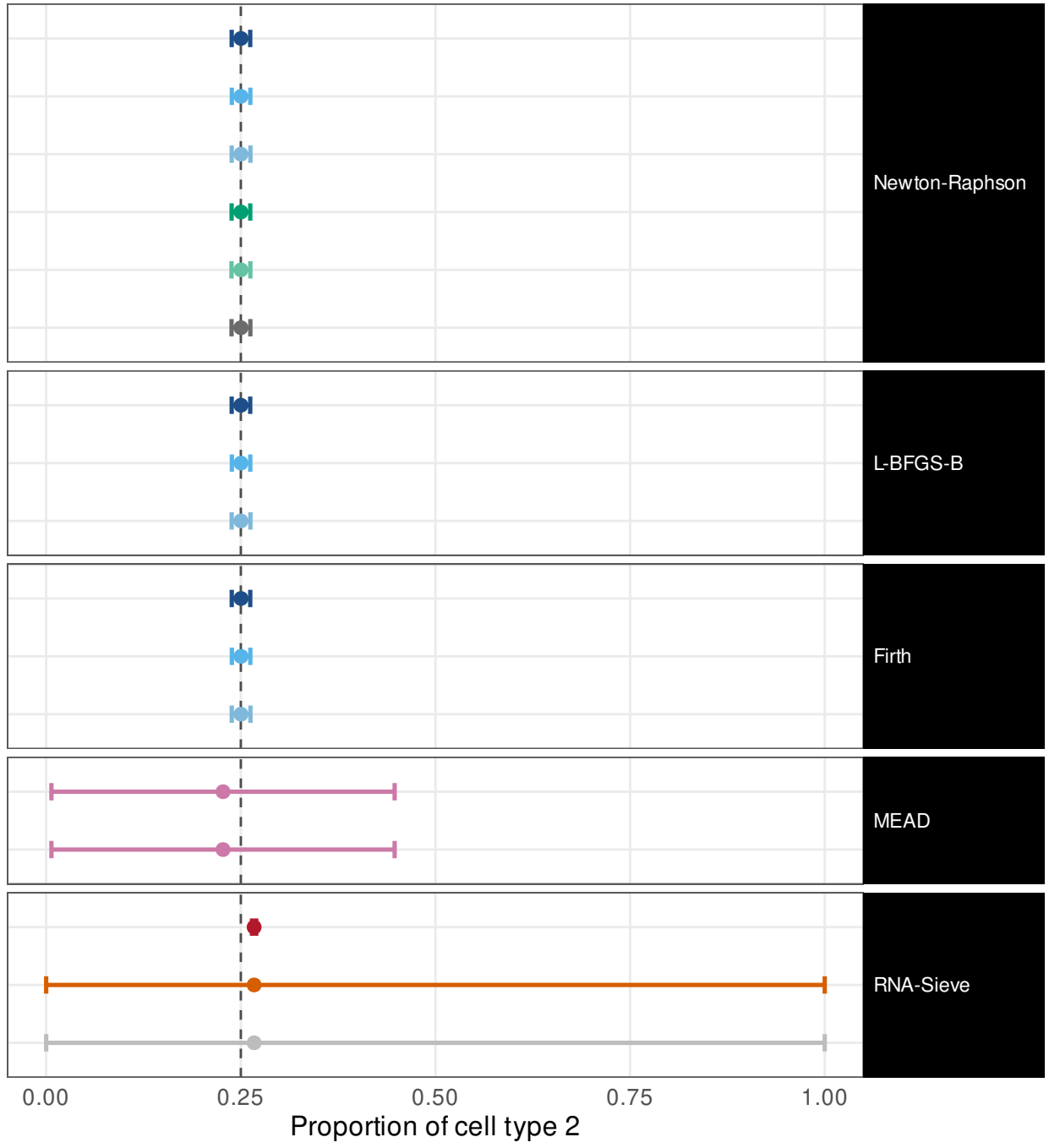


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / balanced / high sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.94 [0.89, 0.96]	w = 0.0238
ILR, mapped back	0.94 [0.89, 0.96]	w = 0.0237
ILR, REML	0.93 [0.89, 0.96]	w = 0.0241
Profile likelihood	0.94 [0.89, 0.96]	w = 0.0239
Profile, REML	0.94 [0.9, 0.97]	w = 0.0242
Oracle (Fisher at p*)	0.94 [0.9, 0.97]	w = 0.0241
Wald (delta)	0.94 [0.89, 0.96]	w = 0.0238
ILR, mapped back	0.94 [0.89, 0.96]	w = 0.0237
ILR, REML	0.94 [0.89, 0.96]	w = 0.0241
Wald (delta)	0.94 [0.89, 0.96]	w = 0.0237
ILR, mapped back	0.94 [0.89, 0.96]	w = 0.0237
ILR, REML	0.93 [0.89, 0.96]	w = 0.024
Method's own interval	1 [0.98, 1]	w = 0.441
Own interval, transcript fractions	1 [0.98, 1]	w = 0.441
Fisher information (Wald)	0.015 [0.0051, 0.043]	w = 0.00532
Godambe sandwich	1 [0.98, 1]	w = 1
Released code, as published	1 [0.98, 1]	w = 1

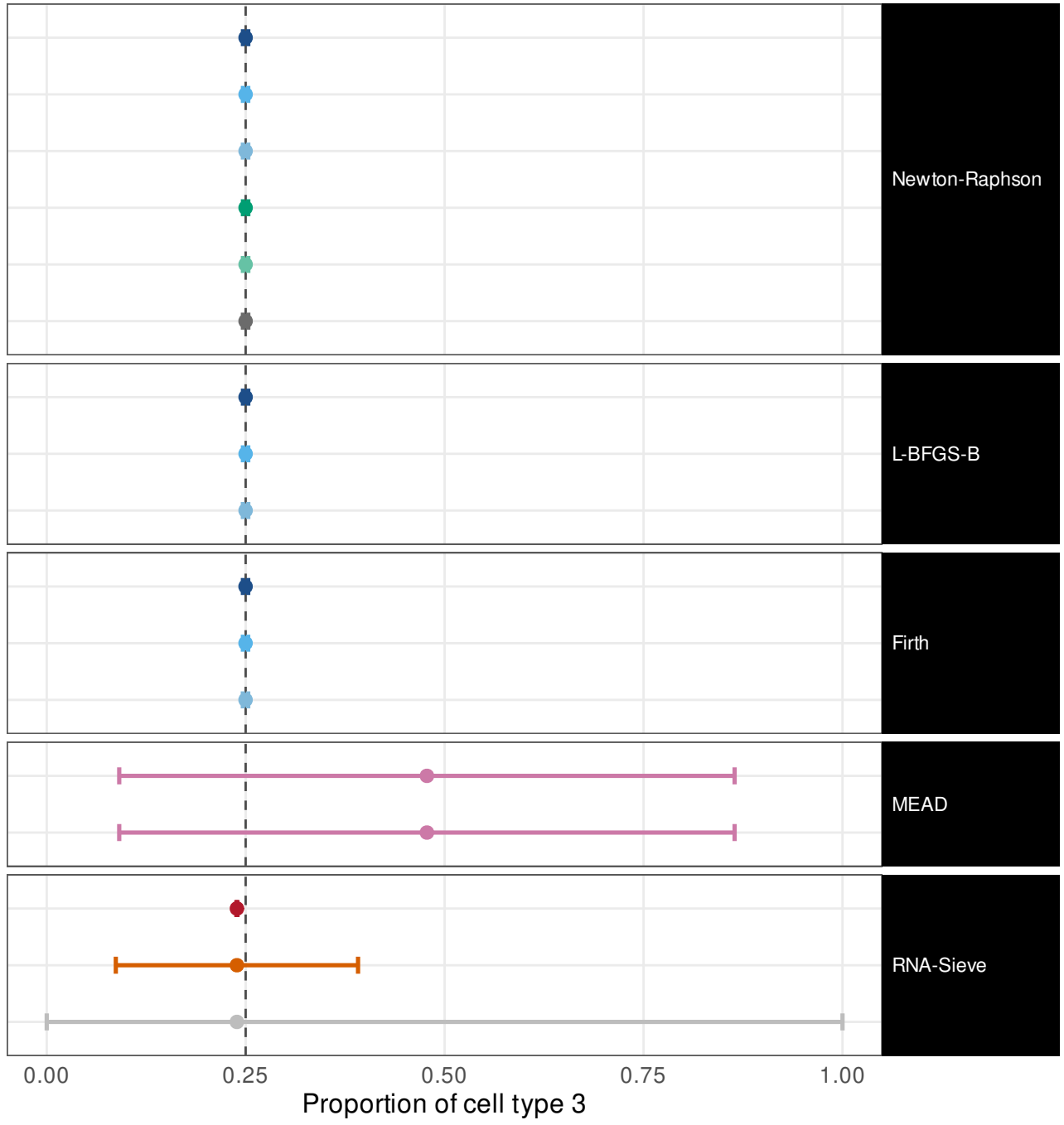


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / balanced / high sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.94 [0.9, 0.97]	w = 0.00664
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.00662
ILR, REML	0.95 [0.91, 0.97]	w = 0.00674
Profile likelihood	0.94 [0.9, 0.97]	w = 0.00667
Profile, REML	0.95 [0.91, 0.97]	w = 0.00675
Oracle (Fisher at p*)	0.95 [0.91, 0.97]	w = 0.00672
Wald (delta)	0.94 [0.9, 0.97]	w = 0.00664
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.00664
ILR, REML	0.95 [0.91, 0.97]	w = 0.00674
Wald (delta)	0.94 [0.9, 0.97]	w = 0.00663
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.00661
ILR, REML	0.94 [0.9, 0.97]	w = 0.00672
Method's own interval	1 [0.98, 1]	w = 0.773
Own interval, transcript fractions	1 [0.98, 1]	w = 0.773
Fisher information (Wald)	0 [0, 0.019]	w = 0.00128
Godambe sandwich	1 [0.98, 1]	w = 0.304
Released code, as published	1 [0.98, 1]	w = 1

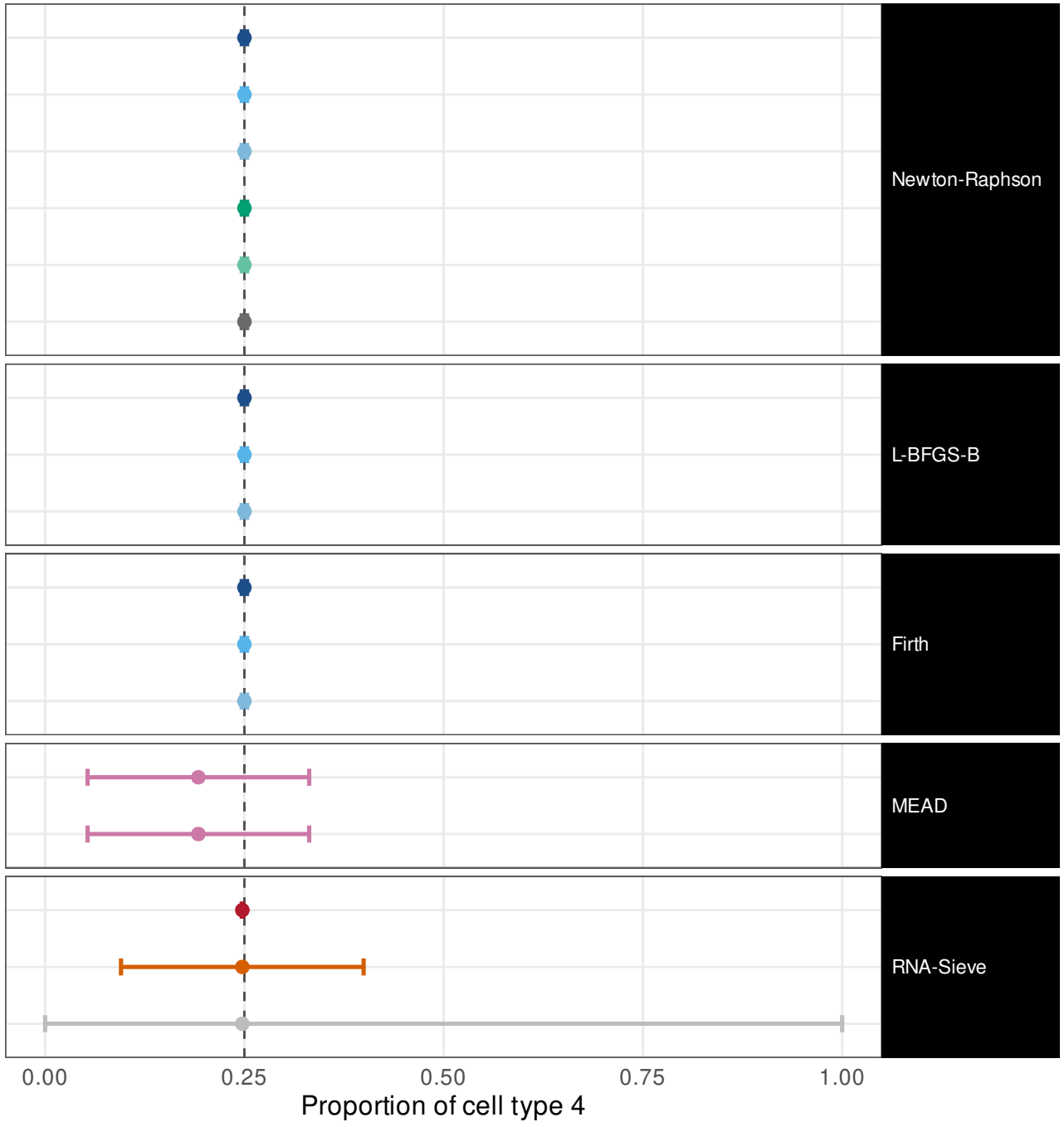


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / balanced / high sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.96 [0.92, 0.98]	w = 0.00663
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.00663
ILR, REML	0.96 [0.92, 0.98]	w = 0.00672
Profile likelihood	0.96 [0.92, 0.98]	w = 0.00666
Profile, REML	0.96 [0.92, 0.98]	w = 0.00674
Oracle (Fisher at p*)	0.96 [0.93, 0.98]	w = 0.00671
Wald (delta)	0.96 [0.92, 0.98]	w = 0.00663
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.00661
ILR, REML	0.96 [0.92, 0.98]	w = 0.00671
Wald (delta)	0.96 [0.92, 0.98]	w = 0.00662
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.00662
ILR, REML	0.96 [0.92, 0.98]	w = 0.00671
Method's own interval	1 [0.98, 1]	w = 0.278
Own interval, transcript fractions	0 [0, 0.019]	w = 0.278
Fisher information (Wald)	0.12 [0.086, 0.18]	w = 0.00143
Godambe sandwich	1 [0.98, 1]	w = 0.304
Released code, as published	1 [0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / unbalanced / low sibling overlap / a = 1 / equal sizes

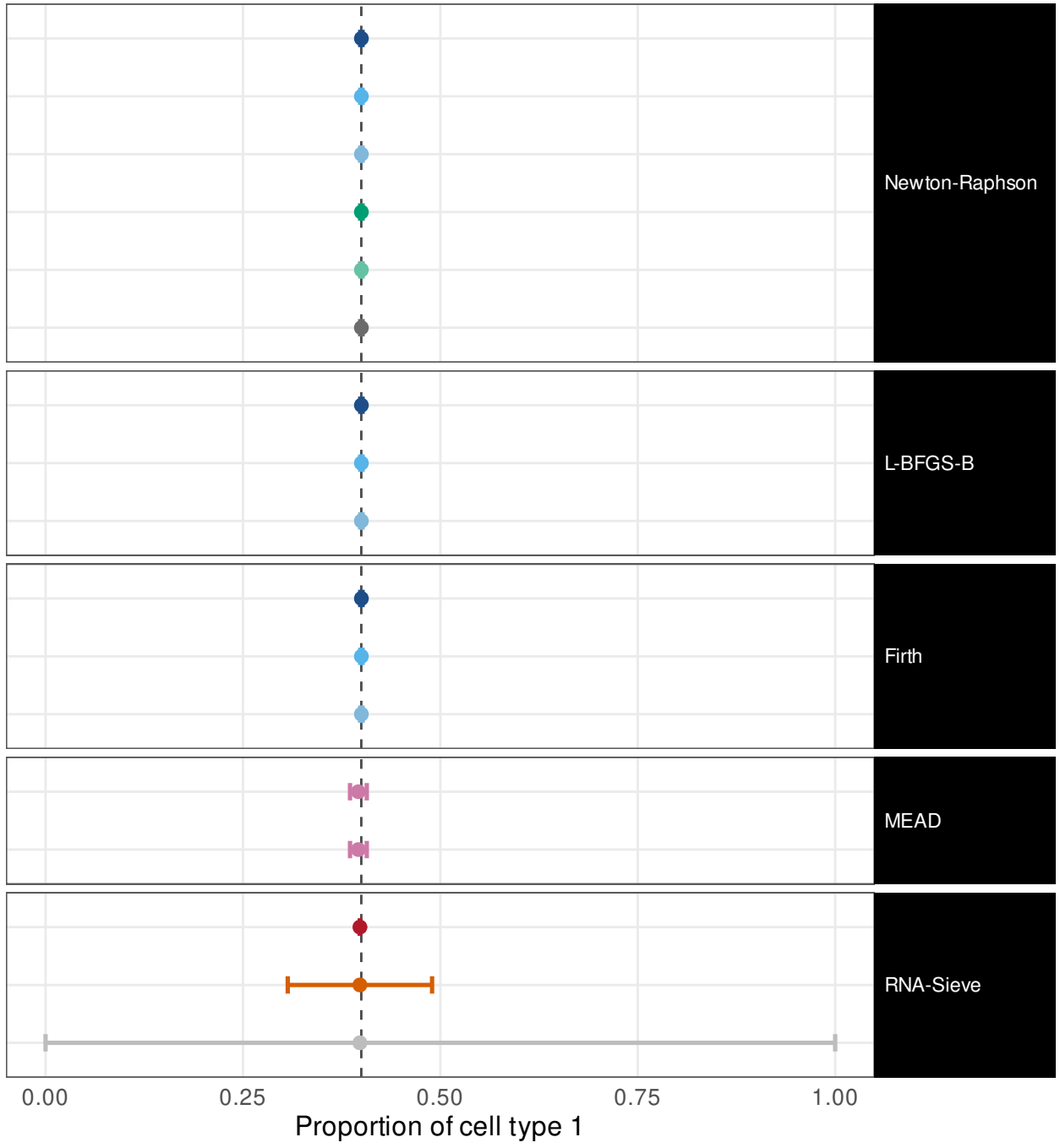
```
Wald (delta)                        0.95 [0.91, 0.97]   w = 0.00292
ILR, mapped back                    0.94 [ 0.9, 0.97]   w = 0.00291
ILR, REML                           0.94 [ 0.9, 0.97]   w = 0.00296
Profile likelihood                  0.95 [0.91, 0.97]   w = 0.00293
Profile, REML                       0.95 [0.91, 0.97]   w = 0.00297
Oracle (Fisher at p*)               0.95 [0.91, 0.97]   w = 0.00296

Wald (delta)                        0.95 [0.91, 0.97]   w = 0.00292
ILR, mapped back                    0.94 [ 0.9, 0.97]   w = 0.00291
ILR, REML                           0.95 [0.91, 0.97]   w = 0.00296

Wald (delta)                        0.95 [0.91, 0.97]   w = 0.00292
ILR, mapped back                    0.94 [ 0.9, 0.97]   w = 0.00291
ILR, REML                           0.94 [ 0.9, 0.97]   w = 0.00295

Method's own interval                  1 [0.98,    1]   w = 0.0212
Own interval, transcript fractions     1 [0.98,    1]   w = 0.0212

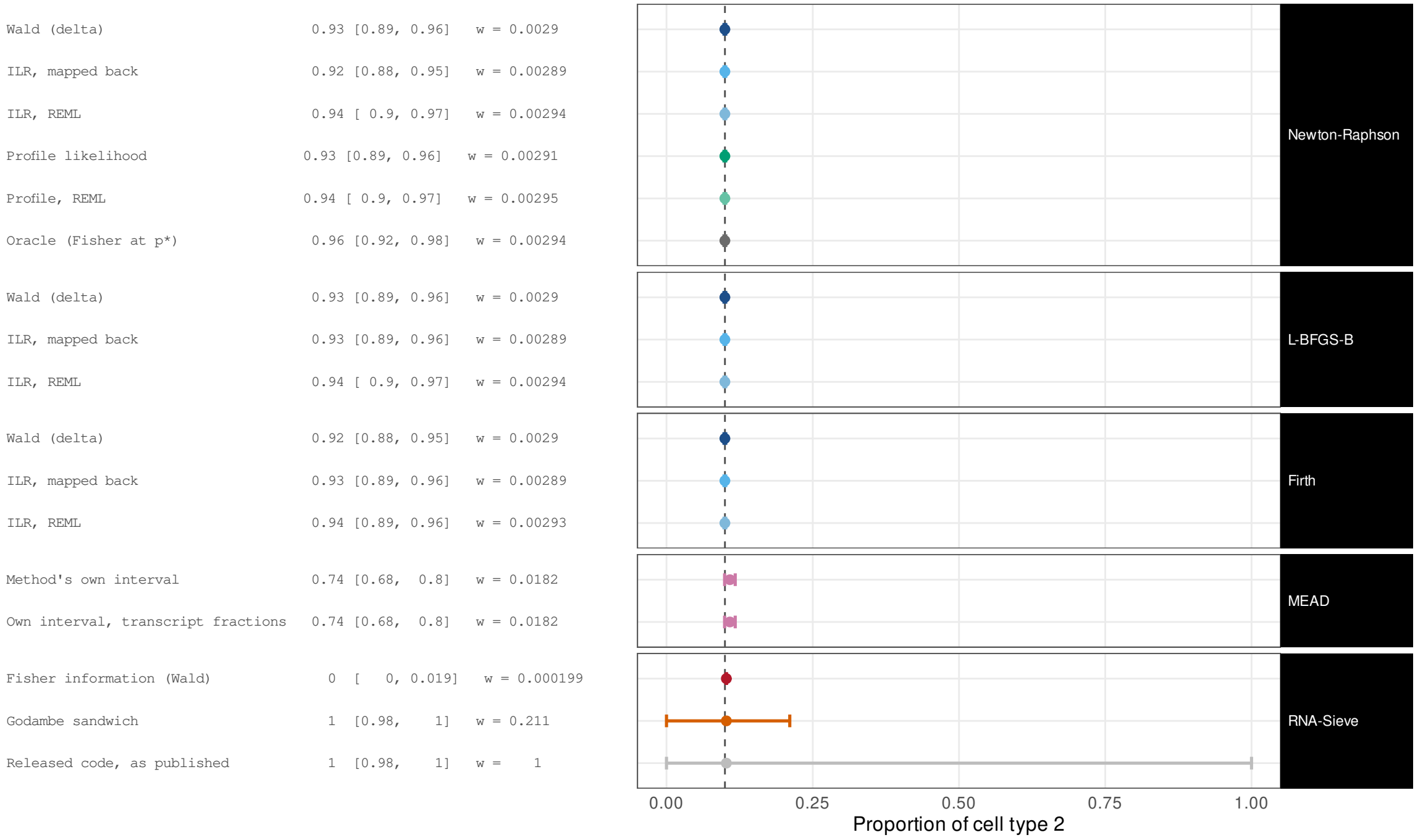
Fisher information (Wald)           0.02 [0.0078, 0.05]   w = 0.000201
Godambe sandwich                       1 [0.98,    1]   w = 0.183
Released code, as published            1 [0.98,    1]   w =     1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / unbalanced / low sibling overlap / a = 1 / equal sizes



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / unbalanced / low sibling overlap / a = 1 / equal sizes

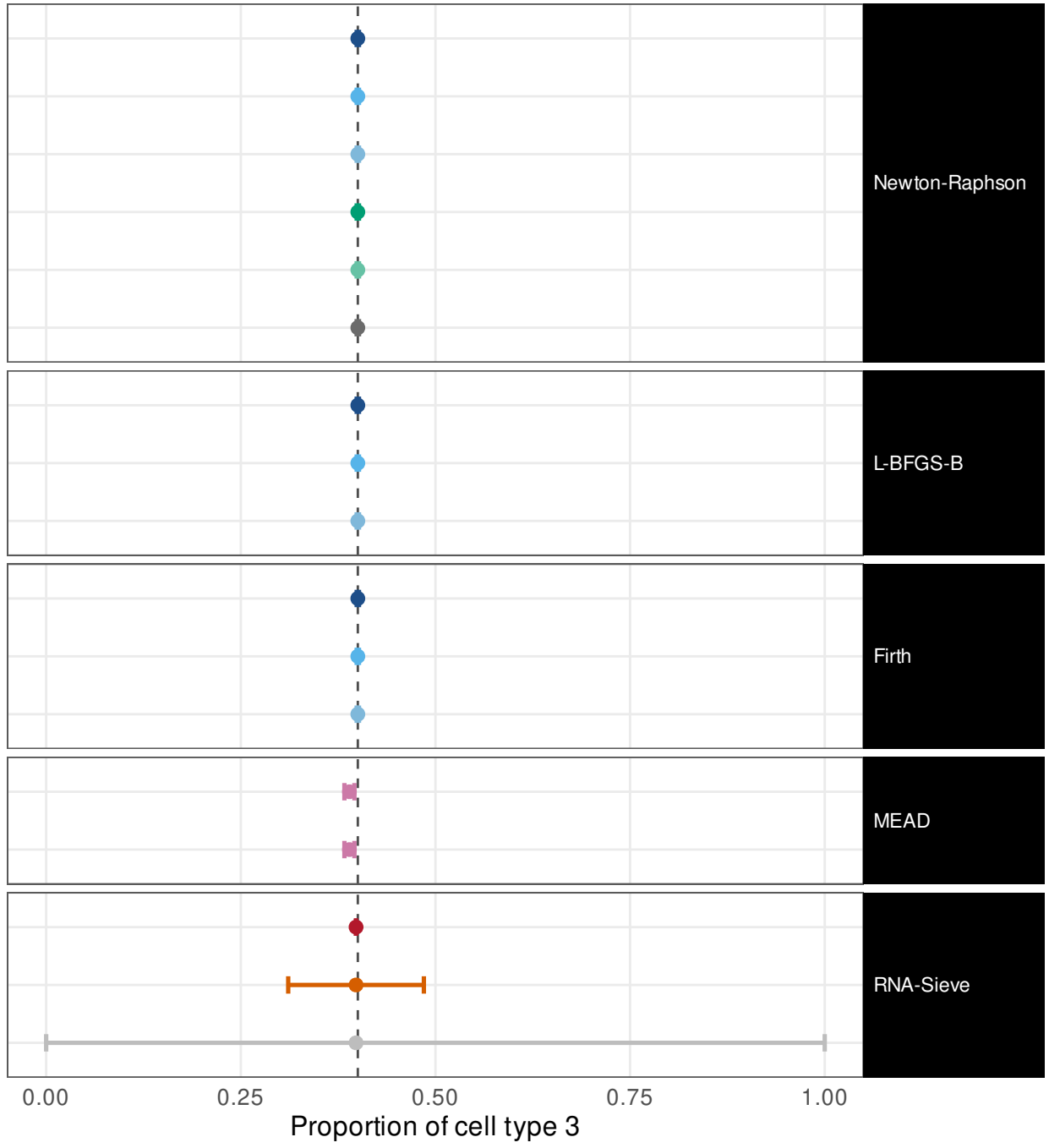
```
Wald (delta)                        0.9  [0.86, 0.94]   w = 0.00288
ILR, mapped back                    0.9  [0.84, 0.93]   w = 0.00287
ILR, REML                           0.92 [0.87, 0.95]   w = 0.00292
Profile likelihood                  0.9  [0.86, 0.94]   w = 0.0029
Profile, REML                       0.91 [0.86, 0.94]   w = 0.00293
Oracle (Fisher at p*)               0.92 [0.87, 0.95]   w = 0.00293

Wald (delta)                        0.9  [0.86, 0.94]   w = 0.00288
ILR, mapped back                    0.9  [0.86, 0.94]   w = 0.00288
ILR, REML                           0.92 [0.87, 0.95]   w = 0.00292

Wald (delta)                        0.9  [0.85, 0.93]   w = 0.00288
ILR, mapped back                    0.92 [0.87, 0.95]   w = 0.00288
ILR, REML                           0.92 [0.87, 0.95]   w = 0.00292

Method's own interval               0    [   0, 0.019]   w = 0.0129
Own interval, transcript fractions  0    [   0, 0.019]   w = 0.0129

Fisher information (Wald)           0.01 [0.0027, 0.036]   w = 0.000212
Godambe sandwich                    1    [0.98,    1]   w = 0.174
Released code, as published         1    [0.98,    1]   w =    1
```

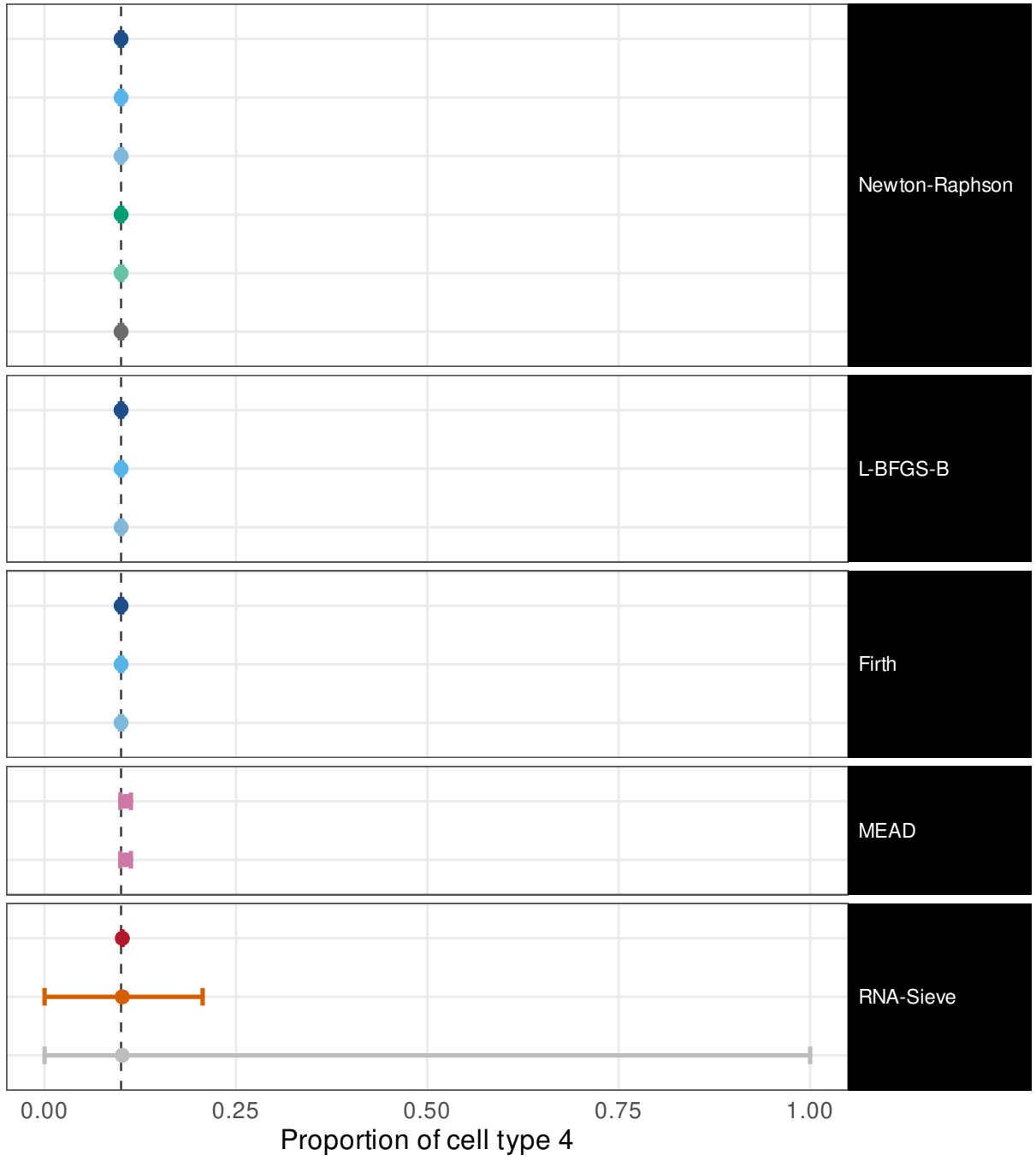


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / unbalanced / low sibling overlap / a = 1 / equal sizes

Wald (delta)	0.96 [0.92, 0.98]	w = 0.0029
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.00289
ILR, REML	0.96 [0.92, 0.98]	w = 0.00293
Profile likelihood	0.96 [0.92, 0.98]	w = 0.00291
Profile, REML	0.96 [0.92, 0.98]	w = 0.00294
Oracle (Fisher at p*)	0.96 [0.92, 0.98]	w = 0.00294
Wald (delta)	0.96 [0.92, 0.98]	w = 0.0029
ILR, mapped back	0.95 [0.91, 0.97]	w = 0.00289
ILR, REML	0.95 [0.91, 0.97]	w = 0.00293
Wald (delta)	0.96 [0.92, 0.98]	w = 0.0029
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.00289
ILR, REML	0.96 [0.92, 0.98]	w = 0.00294
Method's own interval	0.86 [0.81, 0.9]	w = 0.0137
Own interval, transcript fractions	0.86 [0.81, 0.9]	w = 0.0137
Fisher information (Wald)	0.005 [0.00088, 0.028]	w = 0.000192
Godambe sandwich	1 [0.98, 1]	w = 0.206
Released code, as published	1 [0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / unbalanced / low sibling overlap / a = 1 / sizes 1:1:5:5

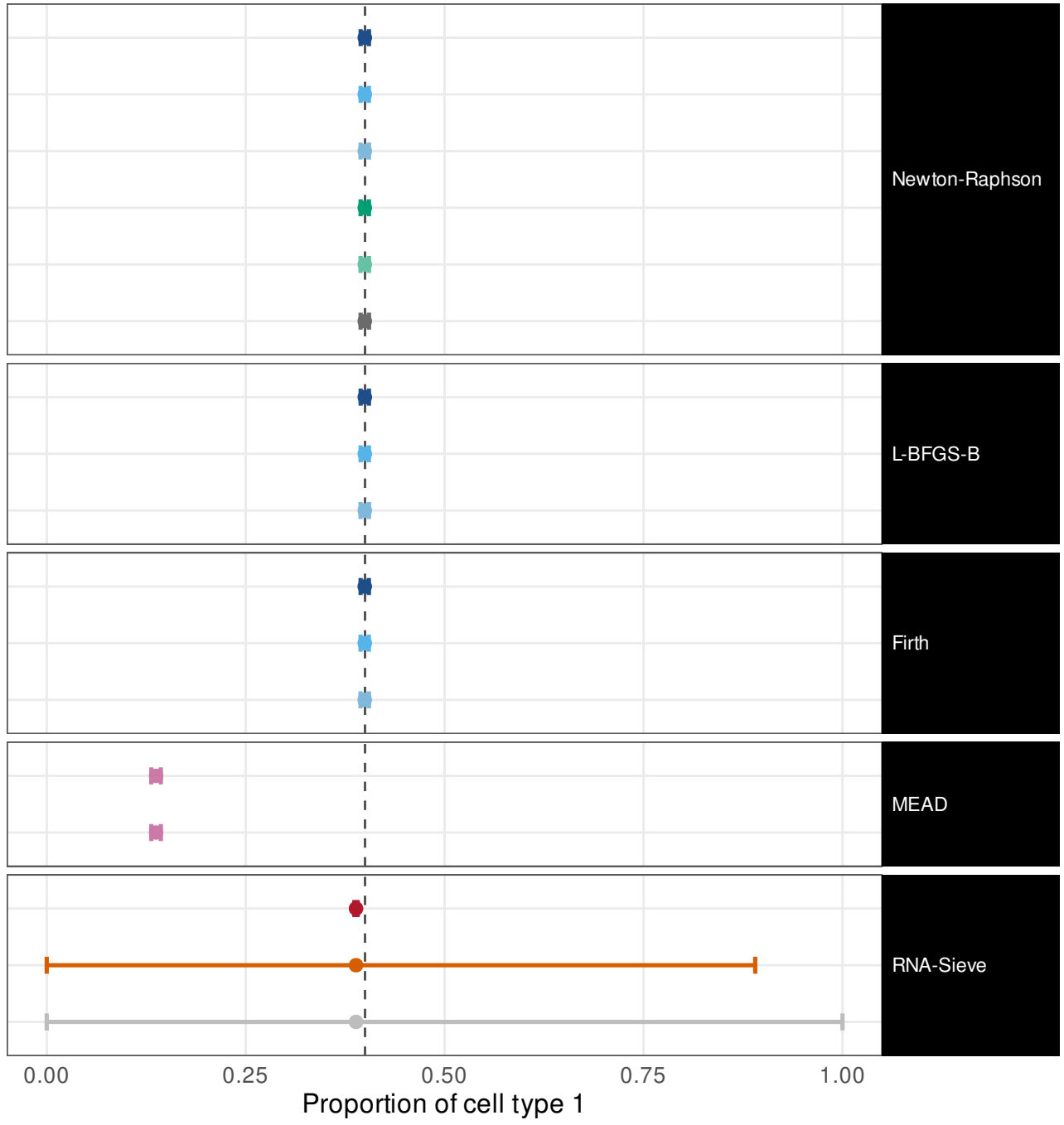
```
Wald (delta)                        0.98 [0.96, 0.99]   w = 0.0104
ILR, mapped back                    0.98 [0.96, 0.99]   w = 0.0104
ILR, REML                           0.98 [0.96, 0.99]   w = 0.0106
Profile likelihood                  0.98 [0.96, 0.99]   w = 0.0105
Profile, REML                       0.98 [0.96, 0.99]   w = 0.0106
Oracle (Fisher at p*)               0.98 [0.96, 0.99]   w = 0.0105

Wald (delta)                        0.98 [0.96, 0.99]   w = 0.0104
ILR, mapped back                    0.98 [0.96, 0.99]   w = 0.0104
ILR, REML                           0.98 [0.96, 0.99]   w = 0.0106

Wald (delta)                        0.98 [0.96, 0.99]   w = 0.0104
ILR, mapped back                    0.98 [0.96, 0.99]   w = 0.0104
ILR, REML                           0.98 [0.96, 0.99]   w = 0.0105

Method's own interval                  0 [   0, 0.019]   w = 0.012
Own interval, transcript fractions  0.98 [0.95, 0.99]   w = 0.012

Fisher information (Wald)              0 [   0, 0.019]   w = 0.00412
Godambe sandwich                       1 [0.98,    1]   w = 0.89
Released code, as published            1 [0.98,    1]   w =    1
```

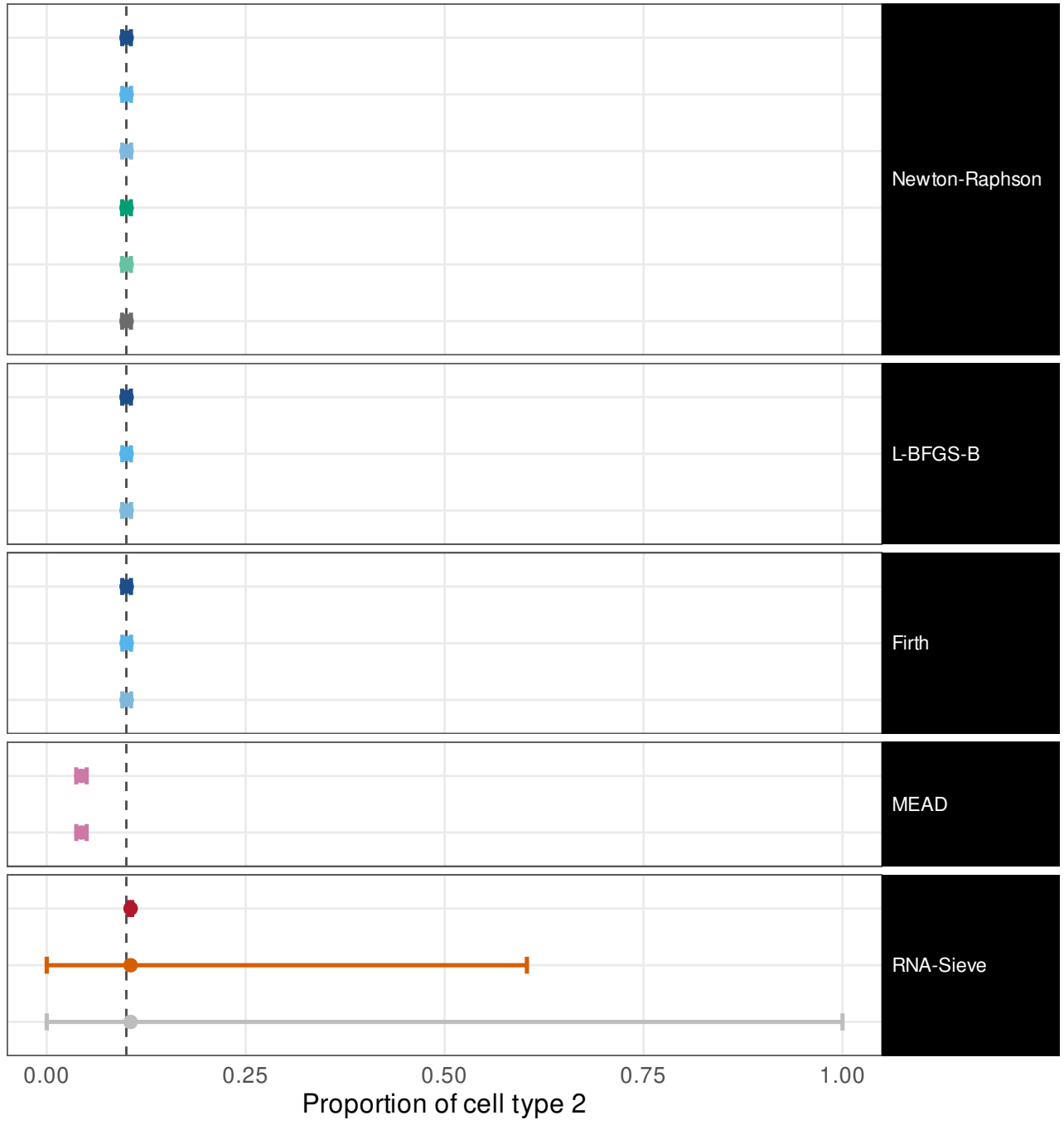


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / unbalanced / low sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.98 [0.94, 0.99]	w = 0.0108
ILR, mapped back	0.97 [0.94, 0.99]	w = 0.0108
ILR, REML	0.98 [0.96, 0.99]	w = 0.011
Profile likelihood	0.98 [0.94, 0.99]	w = 0.0109
Profile, REML	0.98 [0.94, 0.99]	w = 0.011
Oracle (Fisher at p*)	0.99 [0.96, 1]	w = 0.011
Wald (delta)	0.98 [0.94, 0.99]	w = 0.0108
ILR, mapped back	0.97 [0.94, 0.99]	w = 0.0108
ILR, REML	0.97 [0.94, 0.99]	w = 0.011
Wald (delta)	0.98 [0.94, 0.99]	w = 0.0108
ILR, mapped back	0.97 [0.94, 0.99]	w = 0.0108
ILR, REML	0.98 [0.95, 0.99]	w = 0.011
Method's own interval	0 [0, 0.019]	w = 0.013
Own interval, transcript fractions	0 [0, 0.019]	w = 0.013
Fisher information (Wald)	0.12 [0.086, 0.18]	w = 0.00395
Godambe sandwich	1 [0.98, 1]	w = 0.603
Released code, as published	1 [0.98, 1]	w = 1

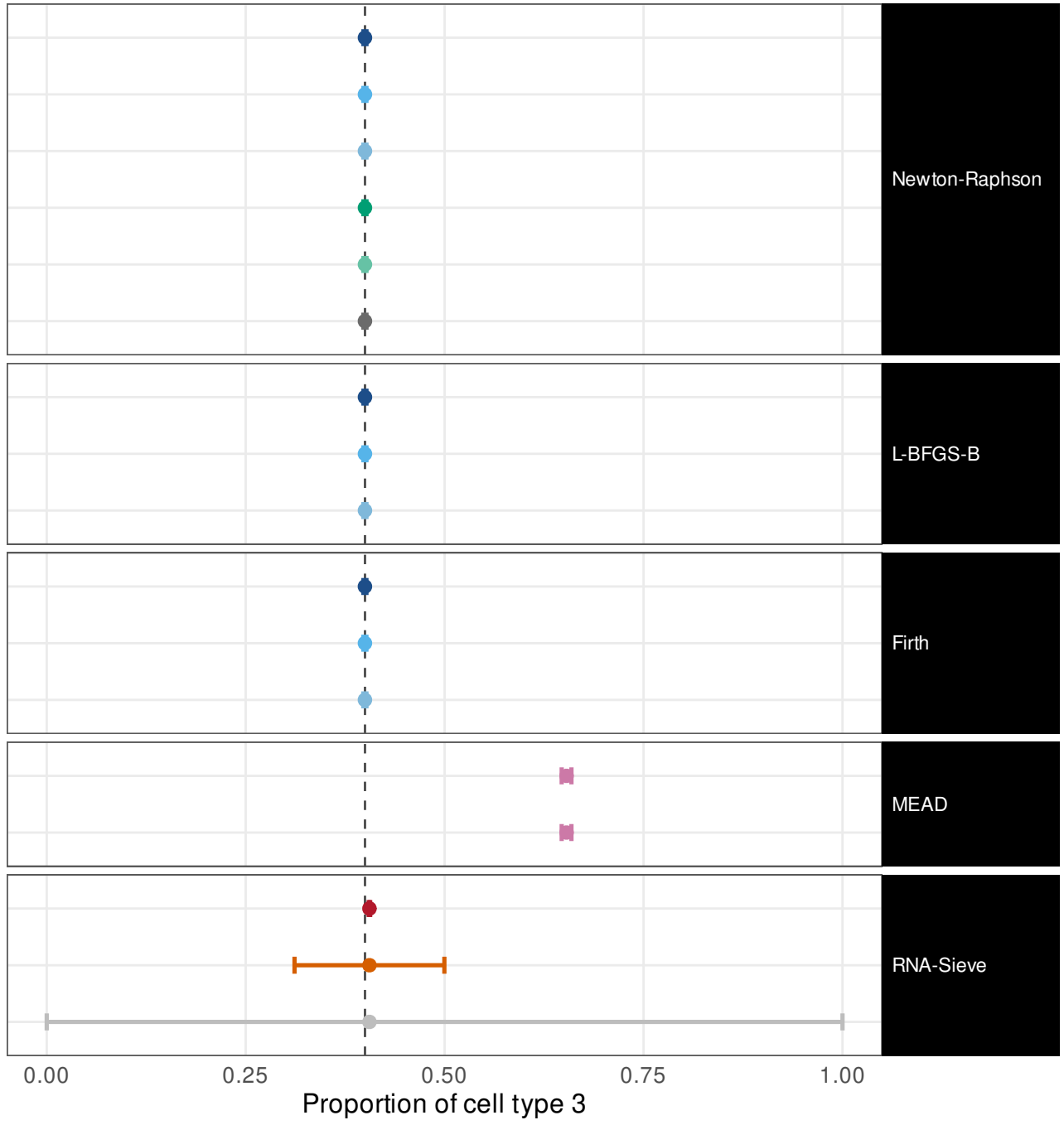


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / unbalanced / low sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.92 [0.87, 0.95]	w = 0.00373
ILR, mapped back	0.9 [0.86, 0.94]	w = 0.00372
ILR, REML	0.94 [0.89, 0.96]	w = 0.00379
Profile likelihood	0.92 [0.88, 0.95]	w = 0.00375
Profile, REML	0.93 [0.89, 0.96]	w = 0.00379
Oracle (Fisher at p*)	0.93 [0.89, 0.96]	w = 0.00378
Wald (delta)	0.92 [0.87, 0.95]	w = 0.00373
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.00373
ILR, REML	0.92 [0.87, 0.95]	w = 0.00379
Wald (delta)	0.92 [0.87, 0.95]	w = 0.00373
ILR, mapped back	0.91 [0.86, 0.94]	w = 0.00372
ILR, REML	0.92 [0.87, 0.95]	w = 0.00379
Method's own interval	0 [0, 0.019]	w = 0.0123
Own interval, transcript fractions	0 [0, 0.019]	w = 0.0123
Fisher information (Wald)	0 [0, 0.019]	w = 0.00173
Godambe sandwich	1 [0.98, 1]	w = 0.188
Released code, as published	1 [0.98, 1]	w = 1

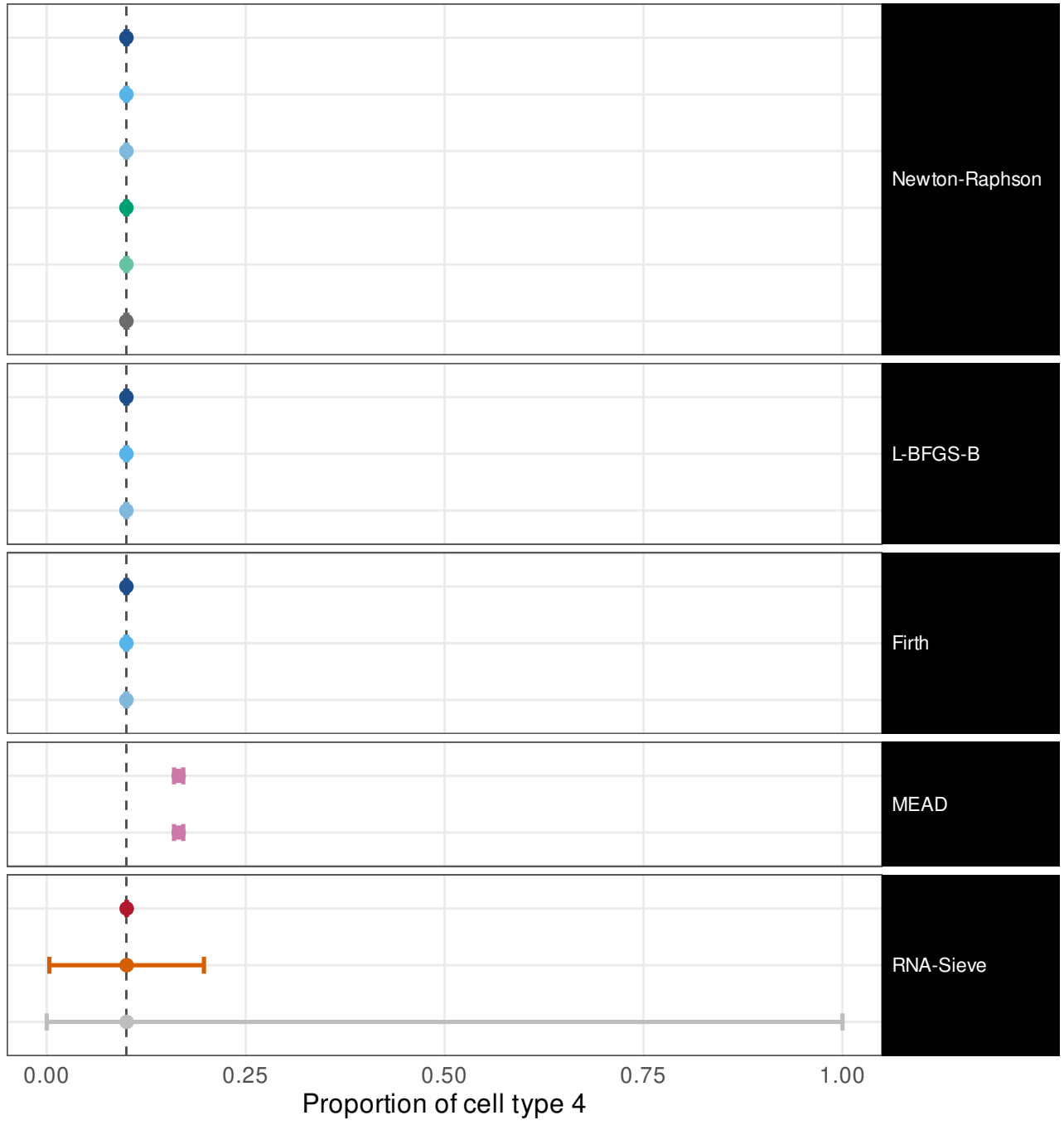


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / unbalanced / low sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.98 [0.95, 0.99]	w = 0.00238
ILR, mapped back	0.98 [0.95, 0.99]	w = 0.00237
ILR, REML	0.98 [0.95, 0.99]	w = 0.00241
Profile likelihood	0.98 [0.95, 0.99]	w = 0.00239
Profile, REML	0.98 [0.95, 0.99]	w = 0.00242
Oracle (Fisher at p*)	0.98 [0.95, 0.99]	w = 0.0024
Wald (delta)	0.98 [0.95, 0.99]	w = 0.00238
ILR, mapped back	0.98 [0.95, 0.99]	w = 0.00237
ILR, REML	0.98 [0.96, 0.99]	w = 0.00242
Wald (delta)	0.98 [0.95, 0.99]	w = 0.00238
ILR, mapped back	0.98 [0.95, 0.99]	w = 0.00237
ILR, REML	0.98 [0.95, 0.99]	w = 0.00241
Method's own interval	0 [0, 0.019]	w = 0.0115
Own interval, transcript fractions	1 [0.98, 1]	w = 0.0115
Fisher information (Wald)	0.1 [0.07, 0.16]	w = 0.000214
Godambe sandwich	1 [0.98, 1]	w = 0.194
Released code, as published	1 [0.98, 1]	w = 1

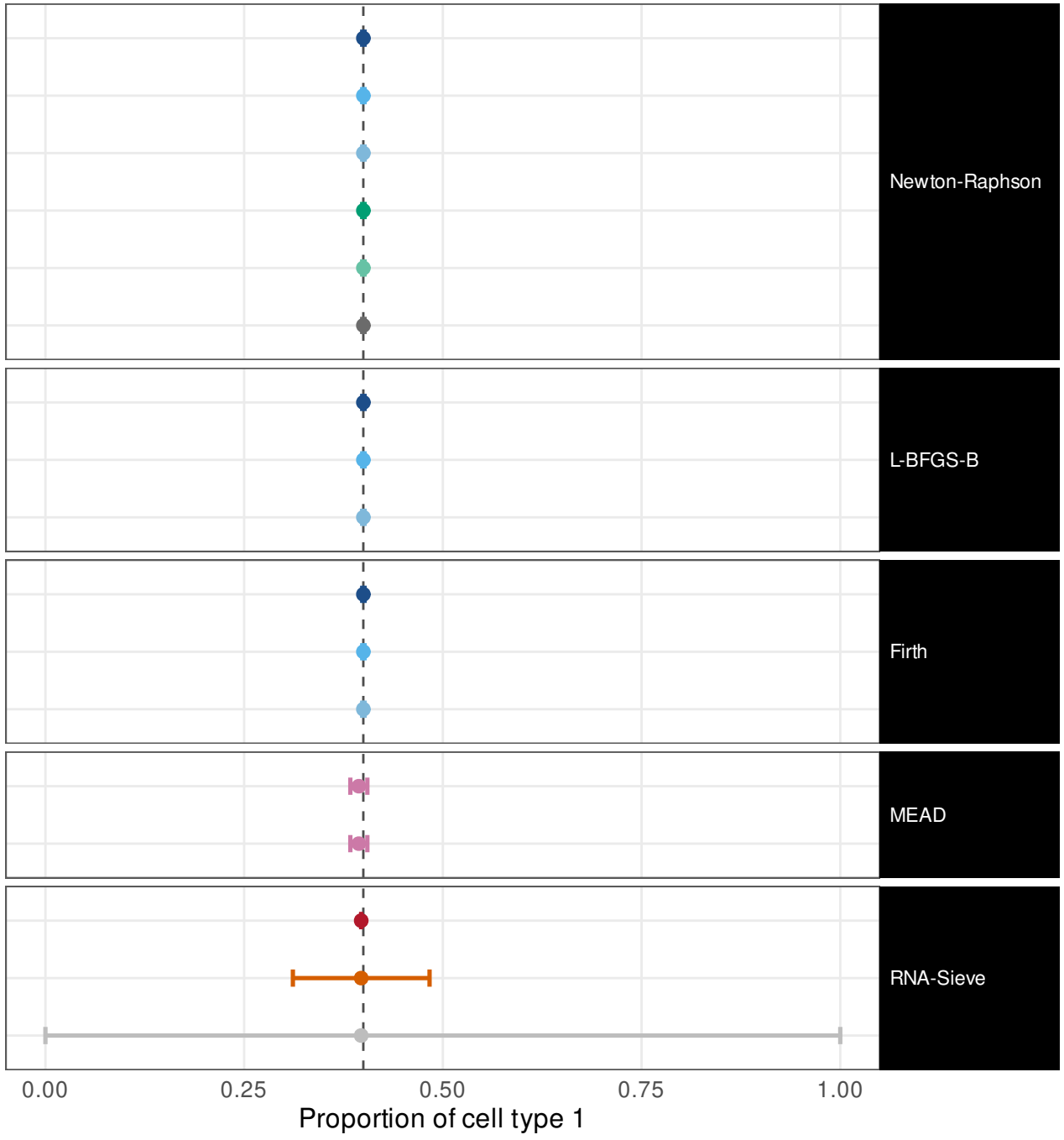


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / unbalanced / low sibling overlap / a = 5 / equal sizes

Wald (delta)	0.92	[0.87, 0.95]	w = 0.00292
ILR, mapped back	0.92	[0.88, 0.95]	w = 0.00291
ILR, REML	0.92	[0.87, 0.95]	w = 0.00296
Profile likelihood	0.92	[0.87, 0.95]	w = 0.00293
Profile, REML	0.92	[0.87, 0.95]	w = 0.00297
Oracle (Fisher at p*)	0.92	[0.88, 0.95]	w = 0.00296
Wald (delta)	0.92	[0.87, 0.95]	w = 0.00292
ILR, mapped back	0.92	[0.87, 0.95]	w = 0.00291
ILR, REML	0.92	[0.88, 0.95]	w = 0.00296
Wald (delta)	0.92	[0.87, 0.95]	w = 0.00292
ILR, mapped back	0.92	[0.87, 0.95]	w = 0.00291
ILR, REML	0.92	[0.87, 0.95]	w = 0.00296
Method's own interval	1	[0.98, 1]	w = 0.0214
Own interval, transcript fractions	1	[0.98, 1]	w = 0.0214
Fisher information (Wald)	0	[0, 0.019]	w = 7.86e-05
Godambe sandwich	1	[0.98, 1]	w = 0.172
Released code, as published	1	[0.98, 1]	w = 1

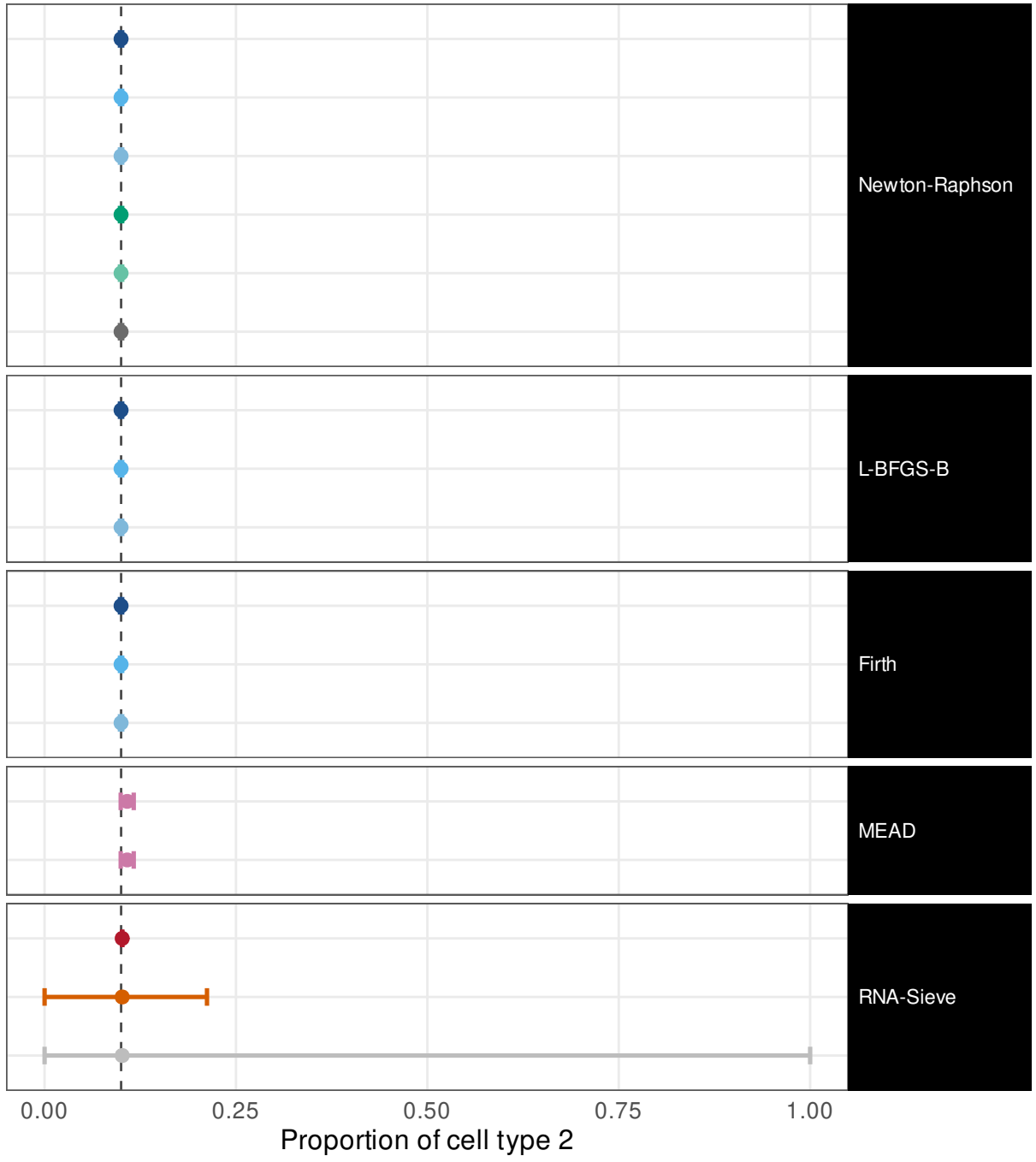


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / unbalanced / low sibling overlap / a = 5 / equal sizes

Wald (delta)	0.94 [0.89, 0.96]	w = 0.0029
ILR, mapped back	0.94 [0.89, 0.96]	w = 0.00289
ILR, REML	0.94 [0.9, 0.97]	w = 0.00294
Profile likelihood	0.94 [0.9, 0.97]	w = 0.00291
Profile, REML	0.94 [0.9, 0.97]	w = 0.00295
Oracle (Fisher at p*)	0.96 [0.92, 0.98]	w = 0.00294
Wald (delta)	0.94 [0.9, 0.97]	w = 0.0029
ILR, mapped back	0.94 [0.89, 0.96]	w = 0.00289
ILR, REML	0.94 [0.9, 0.97]	w = 0.00294
Wald (delta)	0.94 [0.9, 0.97]	w = 0.0029
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.00289
ILR, REML	0.96 [0.92, 0.98]	w = 0.00293
Method's own interval	0.74 [0.68, 0.8]	w = 0.017
Own interval, transcript fractions	0.74 [0.68, 0.8]	w = 0.017
Fisher information (Wald)	0.005 [0.00088, 0.028]	w = 7.97e-05
Godambe sandwich	1 [0.98, 1]	w = 0.212
Released code, as published	1 [0.98, 1]	w = 1

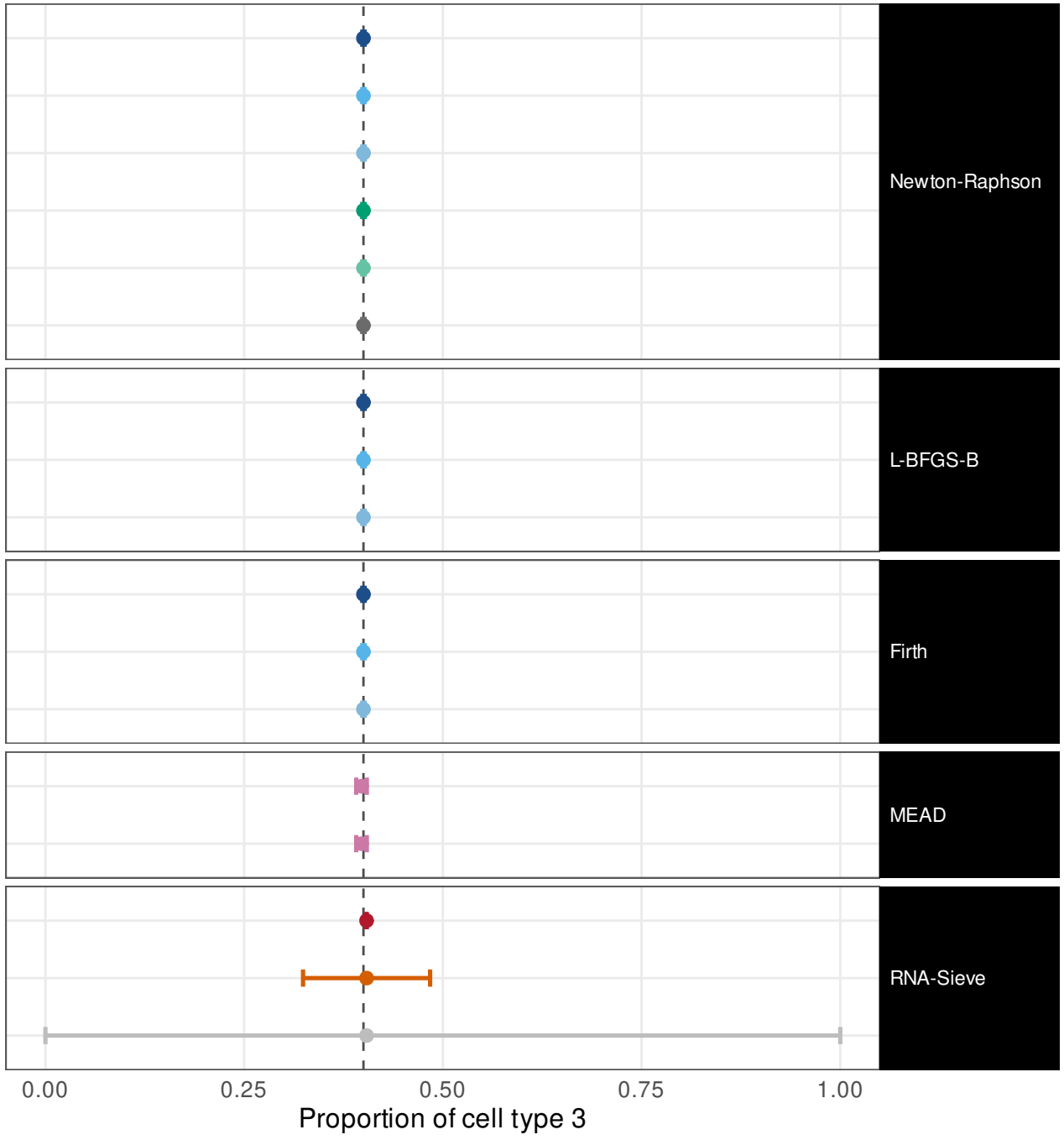


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / unbalanced / low sibling overlap / a = 5 / equal sizes

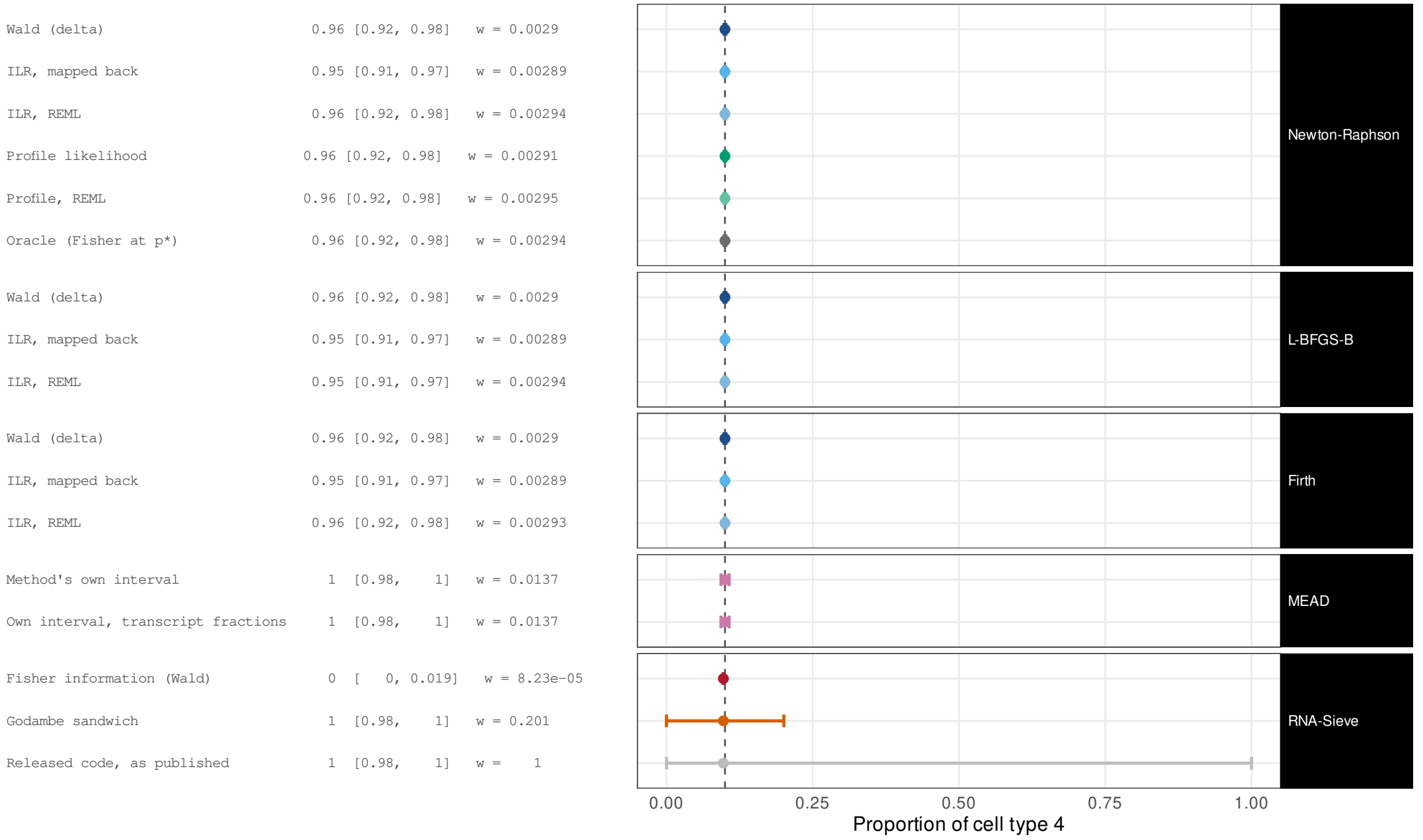
Wald (delta)	0.93 [0.89, 0.96]	w = 0.00288
ILR, mapped back	0.94 [0.89, 0.96]	w = 0.00288
ILR, REML	0.94 [0.9, 0.97]	w = 0.00292
Profile likelihood	0.93 [0.89, 0.96]	w = 0.0029
Profile, REML	0.94 [0.9, 0.97]	w = 0.00293
Oracle (Fisher at p*)	0.96 [0.92, 0.98]	w = 0.00292
Wald (delta)	0.93 [0.89, 0.96]	w = 0.00288
ILR, mapped back	0.92 [0.88, 0.95]	w = 0.00287
ILR, REML	0.94 [0.9, 0.97]	w = 0.00292
Wald (delta)	0.93 [0.89, 0.96]	w = 0.00288
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.00288
ILR, REML	0.94 [0.9, 0.97]	w = 0.00293
Method's own interval	1 [0.98, 1]	w = 0.0133
Own interval, transcript fractions	1 [0.98, 1]	w = 0.0133
Fisher information (Wald)	0 [0, 0.019]	w = 9.31e-05
Godambe sandwich	1 [0.98, 1]	w = 0.16
Released code, as published	1 [0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / unbalanced / low sibling overlap / a = 5 / equal sizes



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / unbalanced / low sibling overlap / a = 5 / sizes 1:1:5:5

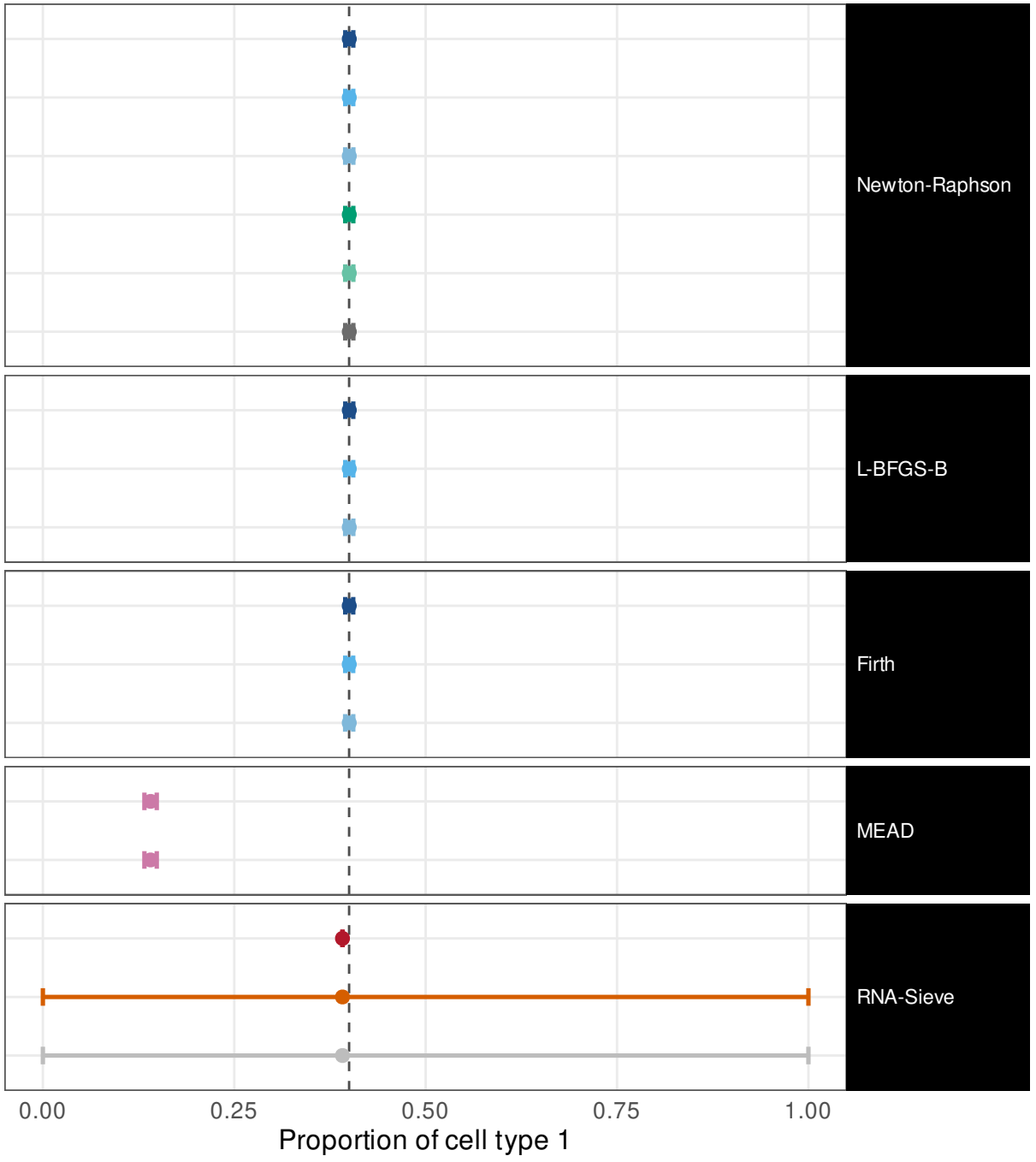
```
Wald (delta)                        0.93 [0.89, 0.96]   w = 0.0104
ILR, mapped back                    0.94 [0.89, 0.96]   w = 0.0104
ILR, REML                           0.94 [0.89, 0.96]   w = 0.0106
Profile likelihood                  0.93 [0.89, 0.96]   w = 0.0105
Profile, REML                       0.94 [0.89, 0.96]   w = 0.0106
Oracle (Fisher at p*)               0.94 [0.89, 0.96]   w = 0.0106

Wald (delta)                        0.93 [0.89, 0.96]   w = 0.0104
ILR, mapped back                    0.93 [0.89, 0.96]   w = 0.0104
ILR, REML                           0.94 [0.89, 0.96]   w = 0.0106

Wald (delta)                        0.93 [0.89, 0.96]   w = 0.0104
ILR, mapped back                    0.93 [0.89, 0.96]   w = 0.0104
ILR, REML                           0.94 [0.89, 0.96]   w = 0.0106

Method's own interval                 0  [   0, 0.019]   w = 0.0161
Own interval, transcript fractions  0.74 [0.68,  0.8]   w = 0.0161

Fisher information (Wald)           0.005 [0.00088, 0.028]   w = 0.000545
Godambe sandwich                      1  [0.98,    1]   w =    1
Released code, as published           1  [0.98,    1]   w =    1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / unbalanced / low sibling overlap / a = 5 / sizes 1:1:5:5

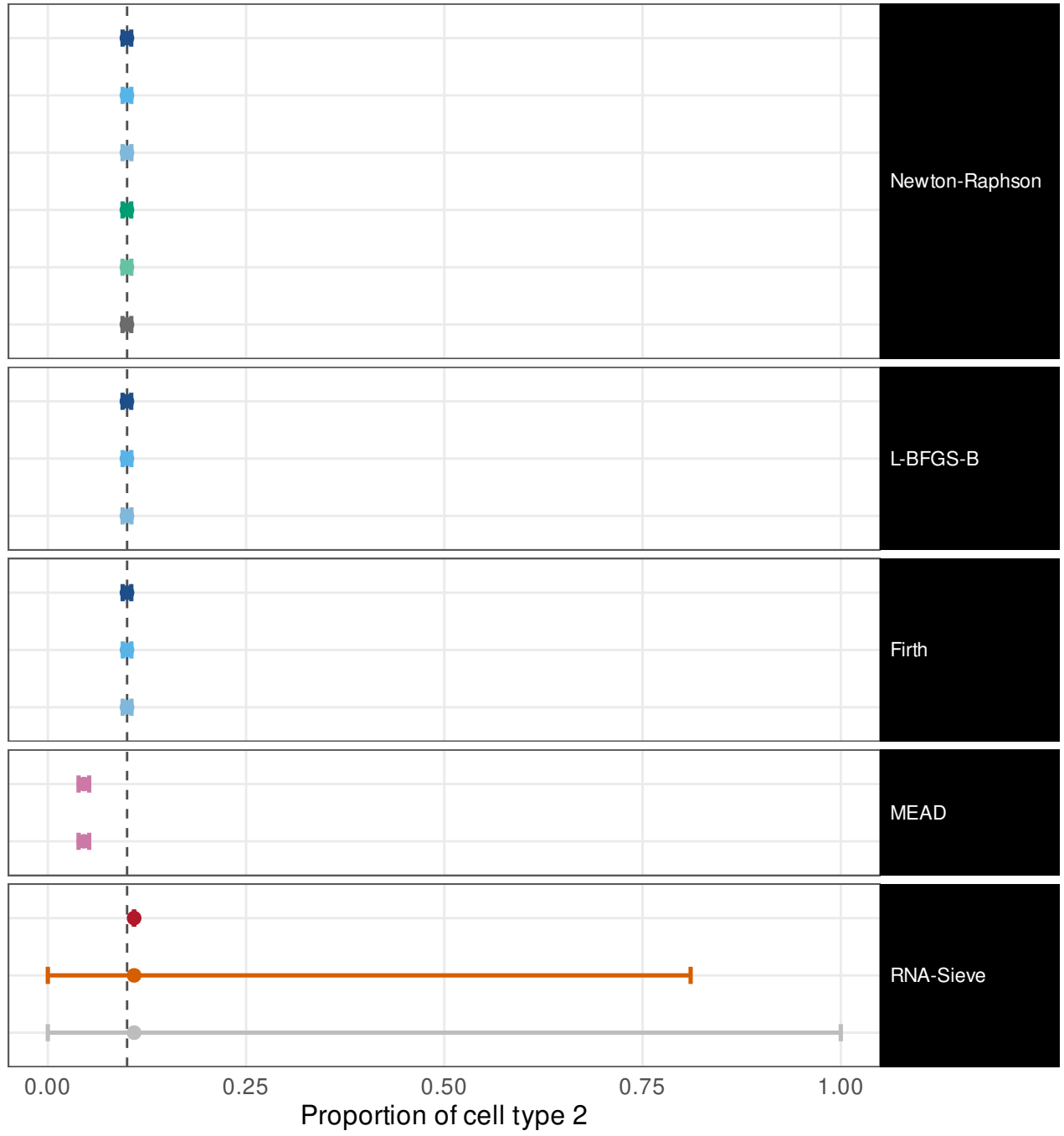
```
Wald (delta)                      0.94 [ 0.9, 0.97]   w = 0.0109
ILR, mapped back                  0.96 [0.92, 0.98]   w = 0.0108
ILR, REML                         0.96 [0.93, 0.98]   w = 0.011
Profile likelihood                0.94 [ 0.9, 0.97]   w = 0.0109
Profile, REML                     0.94 [ 0.9, 0.97]   w = 0.011
Oracle (Fisher at p*)             0.96 [0.92, 0.98]   w = 0.011

Wald (delta)                      0.94 [ 0.9, 0.97]   w = 0.0109
ILR, mapped back                  0.94 [ 0.9, 0.97]   w = 0.0108
ILR, REML                         0.96 [0.92, 0.98]   w = 0.011

Wald (delta)                      0.94 [0.89, 0.96]   w = 0.0108
ILR, mapped back                  0.94 [ 0.9, 0.97]   w = 0.0108
ILR, REML                         0.95 [0.91, 0.97]   w = 0.011

Method's own interval               0 [   0, 0.019]   w = 0.0134
Own interval, transcript fractions  0 [   0, 0.019]   w = 0.0134

Fisher information (Wald)           0 [   0, 0.019]   w = 0.000578
Godambe sandwich                    1 [0.98,    1]   w = 0.811
Released code, as published         1 [0.98,    1]   w =    1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / unbalanced / low sibling overlap / a = 5 / sizes 1:1:5:5

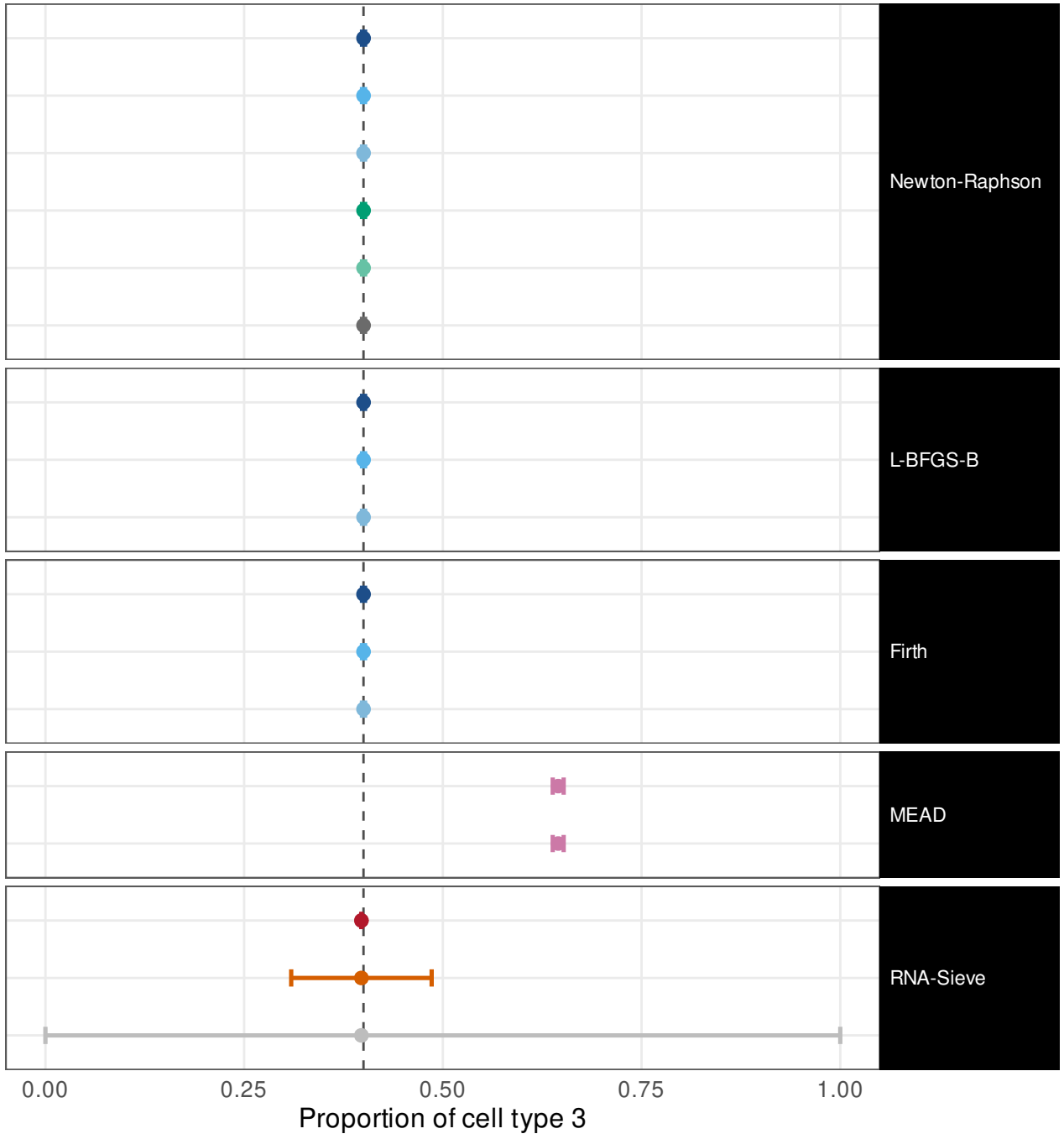
```
Wald (delta)                        0.94 [0.89, 0.96]   w = 0.00375
ILR, mapped back                    0.92 [0.88, 0.95]   w = 0.00375
ILR, REML                           0.94 [0.89, 0.96]   w = 0.00379
Profile likelihood                  0.94 [0.89, 0.96]   w = 0.00377
Profile, REML                       0.94 [0.89, 0.96]   w = 0.00381
Oracle (Fisher at p*)               0.94 [0.89, 0.96]   w = 0.0038

Wald (delta)                        0.94 [0.89, 0.96]   w = 0.00375
ILR, mapped back                    0.94 [0.89, 0.96]   w = 0.00374
ILR, REML                           0.94 [0.89, 0.96]   w = 0.0038

Wald (delta)                        0.93 [0.89, 0.96]   w = 0.00375
ILR, mapped back                    0.93 [0.89, 0.96]   w = 0.00374
ILR, REML                           0.94 [0.89, 0.96]   w = 0.00381

Method's own interval                  0 [   0, 0.019]   w = 0.0138
Own interval, transcript fractions     0 [   0, 0.019]   w = 0.0138

Fisher information (Wald)              0 [   0, 0.019]   w = 0.000393
Godambe sandwich                       1 [0.98,    1]   w = 0.177
Released code, as published            1 [0.98,    1]   w =    1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / unbalanced / low sibling overlap / a = 5 / sizes 1:1:5:5

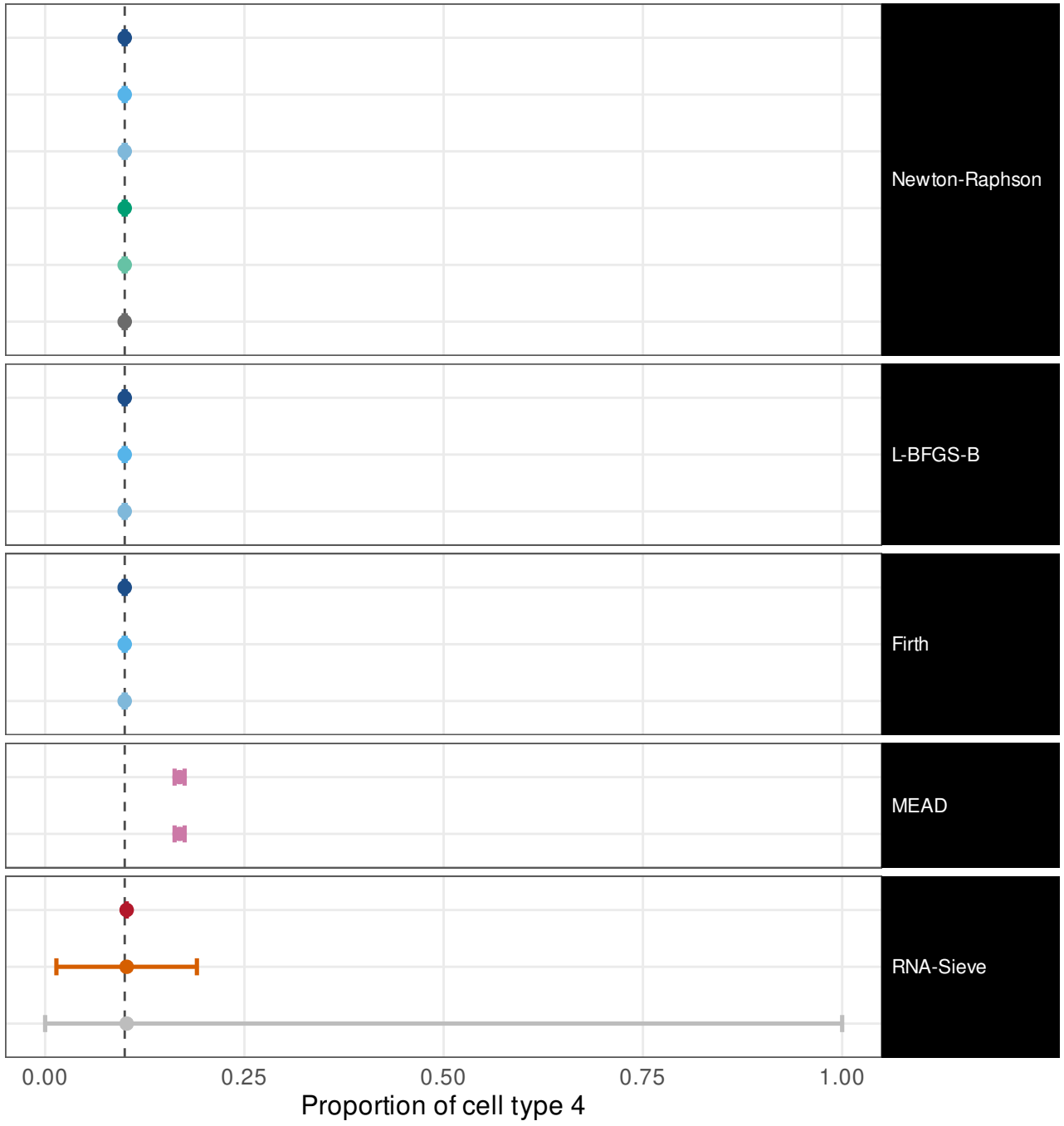
```
Wald (delta)                        0.95 [0.91, 0.97]   w = 0.00238
ILR, mapped back                    0.94 [ 0.9, 0.97]   w = 0.00237
ILR, REML                           0.95 [0.91, 0.97]   w = 0.00242
Profile likelihood                  0.95 [0.91, 0.97]   w = 0.00239
Profile, REML                       0.95 [0.91, 0.97]   w = 0.00242
Oracle (Fisher at p*)               0.95 [0.91, 0.97]   w = 0.00241

Wald (delta)                        0.95 [0.91, 0.97]   w = 0.00238
ILR, mapped back                    0.95 [0.91, 0.97]   w = 0.00238
ILR, REML                           0.94 [ 0.9, 0.97]   w = 0.00242

Wald (delta)                        0.95 [0.91, 0.97]   w = 0.00238
ILR, mapped back                    0.95 [0.91, 0.97]   w = 0.00238
ILR, REML                           0.95 [0.91, 0.97]   w = 0.00241

Method's own interval                  0 [   0, 0.019]   w = 0.0126
Own interval, transcript fractions     1 [0.98,    1]   w = 0.0126

Fisher information (Wald)              0 [   0, 0.019]   w = 3.3e-05
Godambe sandwich                       1 [0.98,    1]   w = 0.176
Released code, as published            1 [0.98,    1]   w =    1
```

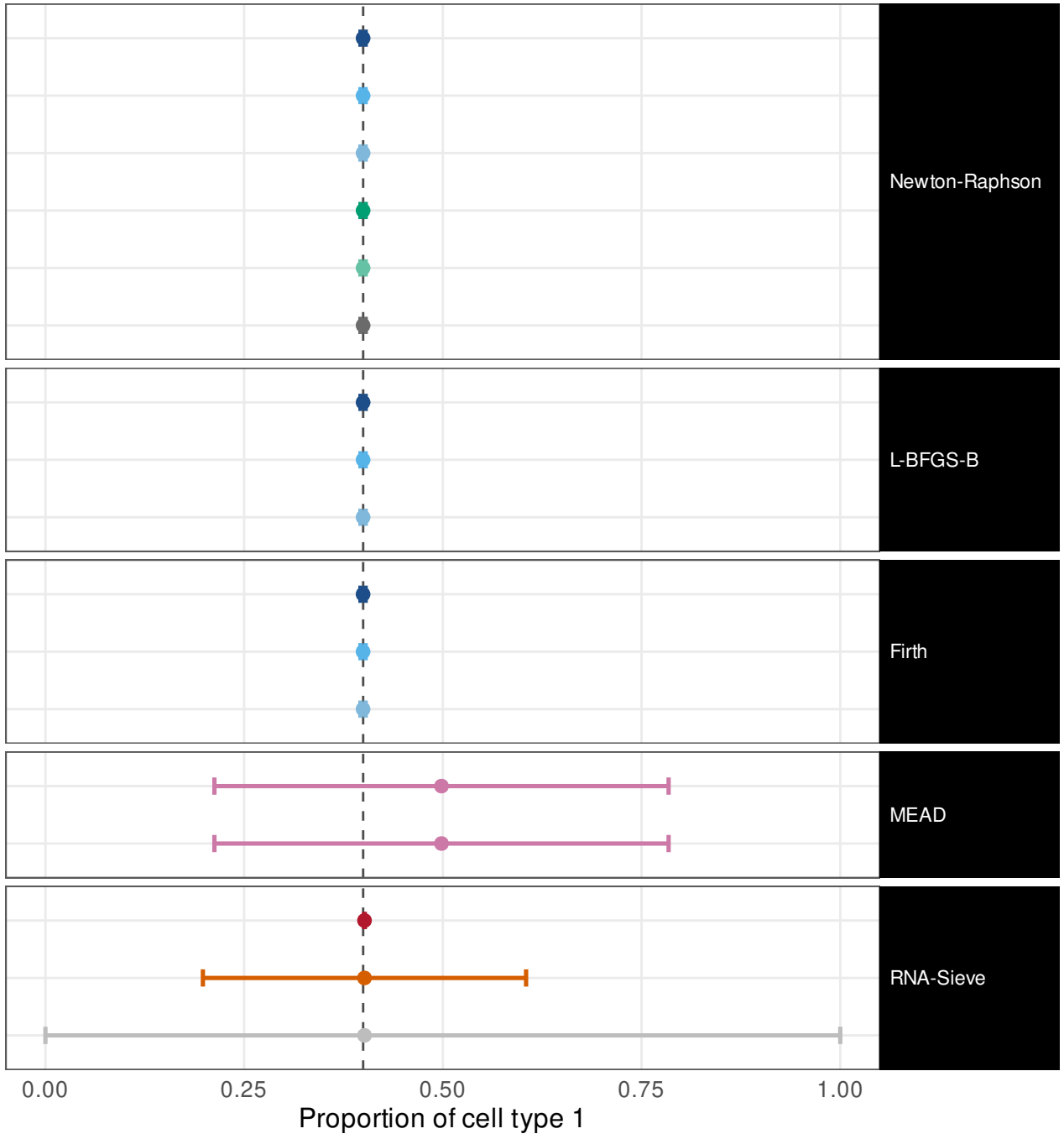


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / unbalanced / high sibling overlap / a = 1 / equal sizes

Wald (delta)	0.92 [0.87, 0.95]	w = 0.00652
ILR, mapped back	0.91 [0.86, 0.94]	w = 0.0065
ILR, REML	0.91 [0.86, 0.94]	w = 0.00662
Profile likelihood	0.92 [0.87, 0.95]	w = 0.00655
Profile, REML	0.92 [0.87, 0.95]	w = 0.00662
Oracle (Fisher at p*)	0.93 [0.89, 0.96]	w = 0.00661
Wald (delta)	0.92 [0.87, 0.95]	w = 0.00652
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.00651
ILR, REML	0.92 [0.87, 0.95]	w = 0.00661
Wald (delta)	0.91 [0.86, 0.94]	w = 0.00652
ILR, mapped back	0.91 [0.86, 0.94]	w = 0.00649
ILR, REML	0.92 [0.87, 0.95]	w = 0.00661
Method's own interval	1 [0.98, 1]	w = 0.571
Own interval, transcript fractions	1 [0.98, 1]	w = 0.571
Fisher information (Wald)	0.12 [0.086, 0.18]	w = 0.000875
Godambe sandwich	1 [0.98, 1]	w = 0.407
Released code, as published	1 [0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / unbalanced / high sibling overlap / a = 1 / equal sizes

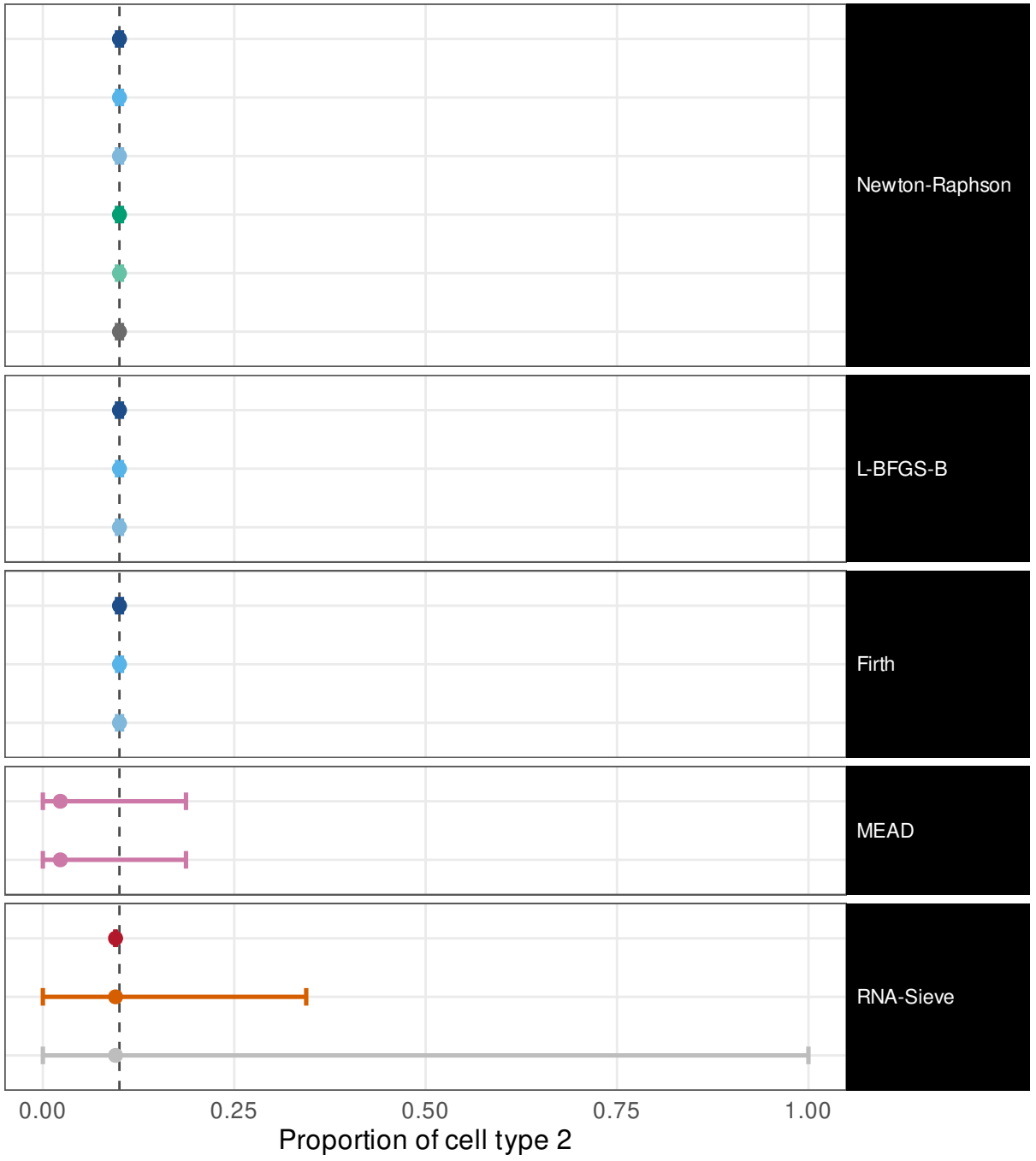
```
Wald (delta)                        0.9  [0.86, 0.94]   w = 0.00647
ILR, mapped back                    0.92 [0.87, 0.95]   w = 0.00647
ILR, REML                           0.91 [0.86, 0.94]   w = 0.00658
Profile likelihood                  0.91 [0.86, 0.94]   w = 0.0065
Profile, REML                       0.91 [0.86, 0.94]   w = 0.00658
Oracle (Fisher at p*)               0.92 [0.87, 0.95]   w = 0.00657

Wald (delta)                        0.9  [0.86, 0.94]   w = 0.00647
ILR, mapped back                    0.9  [0.86, 0.94]   w = 0.00645
ILR, REML                           0.9  [0.86, 0.94]   w = 0.00657

Wald (delta)                        0.9  [0.86, 0.94]   w = 0.00647
ILR, mapped back                    0.9  [0.84, 0.93]   w = 0.00646
ILR, REML                           0.92 [0.87, 0.95]   w = 0.00656

Method's own interval                 1  [0.98,    1]   w = 0.187
Own interval, transcript fractions    1  [0.98,    1]   w = 0.187

Fisher information (Wald)           0.005 [0.00088, 0.028]   w = 0.000847
Godambe sandwich                      1  [0.98,    1]   w = 0.344
Released code, as published           1  [0.98,    1]   w =    1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / unbalanced / high sibling overlap / a = 1 / equal sizes

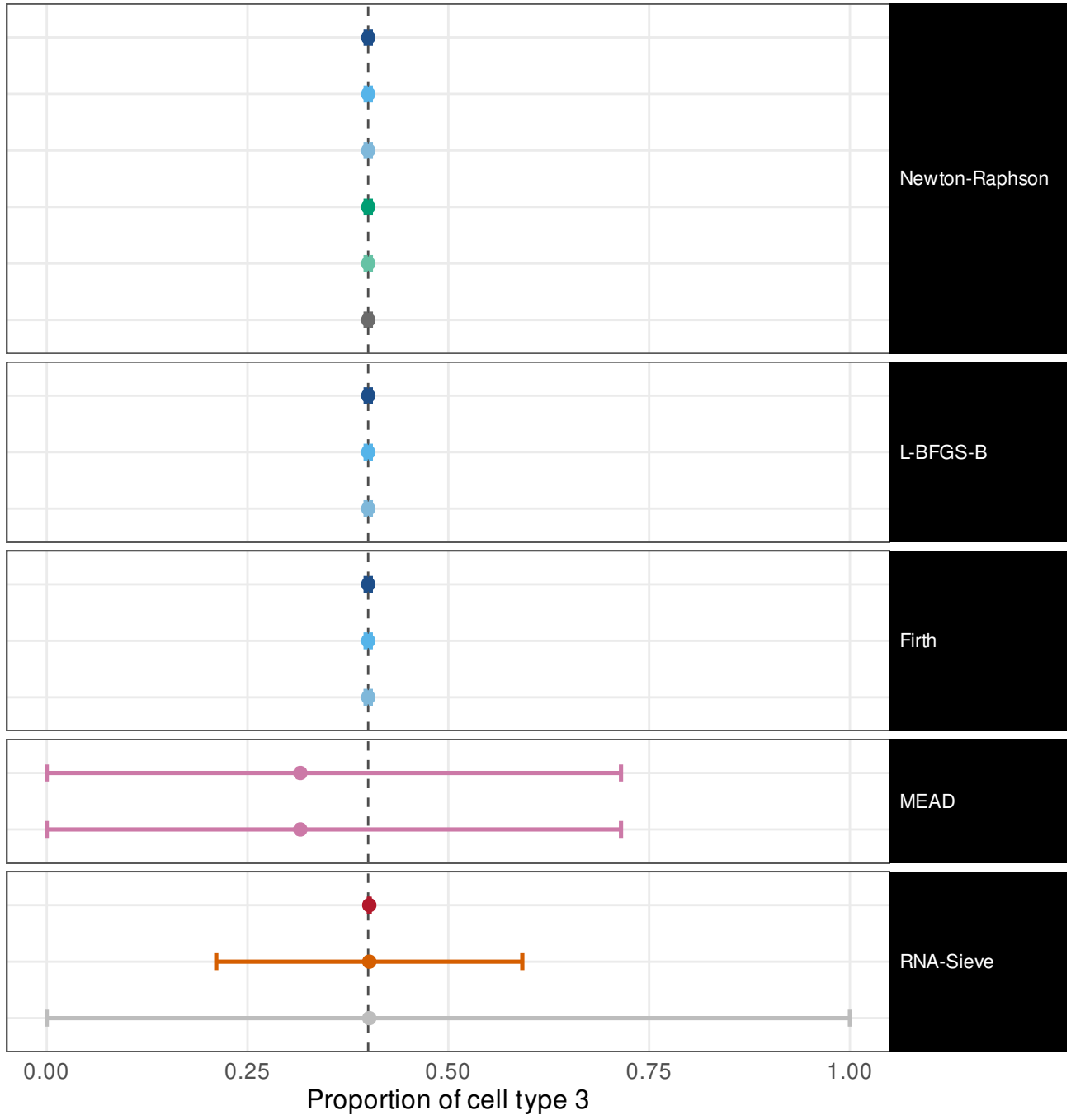
```
Wald (delta)                        0.92 [0.87, 0.95]   w = 0.00644
ILR, mapped back                    0.92 [0.87, 0.95]   w = 0.00642
ILR, REML                           0.92 [0.87, 0.95]   w = 0.00652
Profile likelihood                  0.92 [0.87, 0.95]   w = 0.00647
Profile, REML                       0.92 [0.87, 0.95]   w = 0.00655
Oracle (Fisher at p*)               0.92 [0.87, 0.95]   w = 0.00654

Wald (delta)                        0.92 [0.87, 0.95]   w = 0.00644
ILR, mapped back                    0.92 [0.87, 0.95]   w = 0.00643
ILR, REML                           0.92 [0.87, 0.95]   w = 0.00654

Wald (delta)                        0.92 [0.87, 0.95]   w = 0.00644
ILR, mapped back                    0.92 [0.87, 0.95]   w = 0.00642
ILR, REML                           0.92 [0.88, 0.95]   w = 0.00652

Method's own interval                  1 [0.98,    1]   w = 0.715
Own interval, transcript fractions     1 [0.98,    1]   w = 0.715

Fisher information (Wald)           0.19 [0.14, 0.25]   w = 0.00111
Godambe sandwich                       1 [0.98,    1]   w = 0.381
Released code, as published            1 [0.98,    1]   w =     1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / unbalanced / high sibling overlap / a = 1 / equal sizes

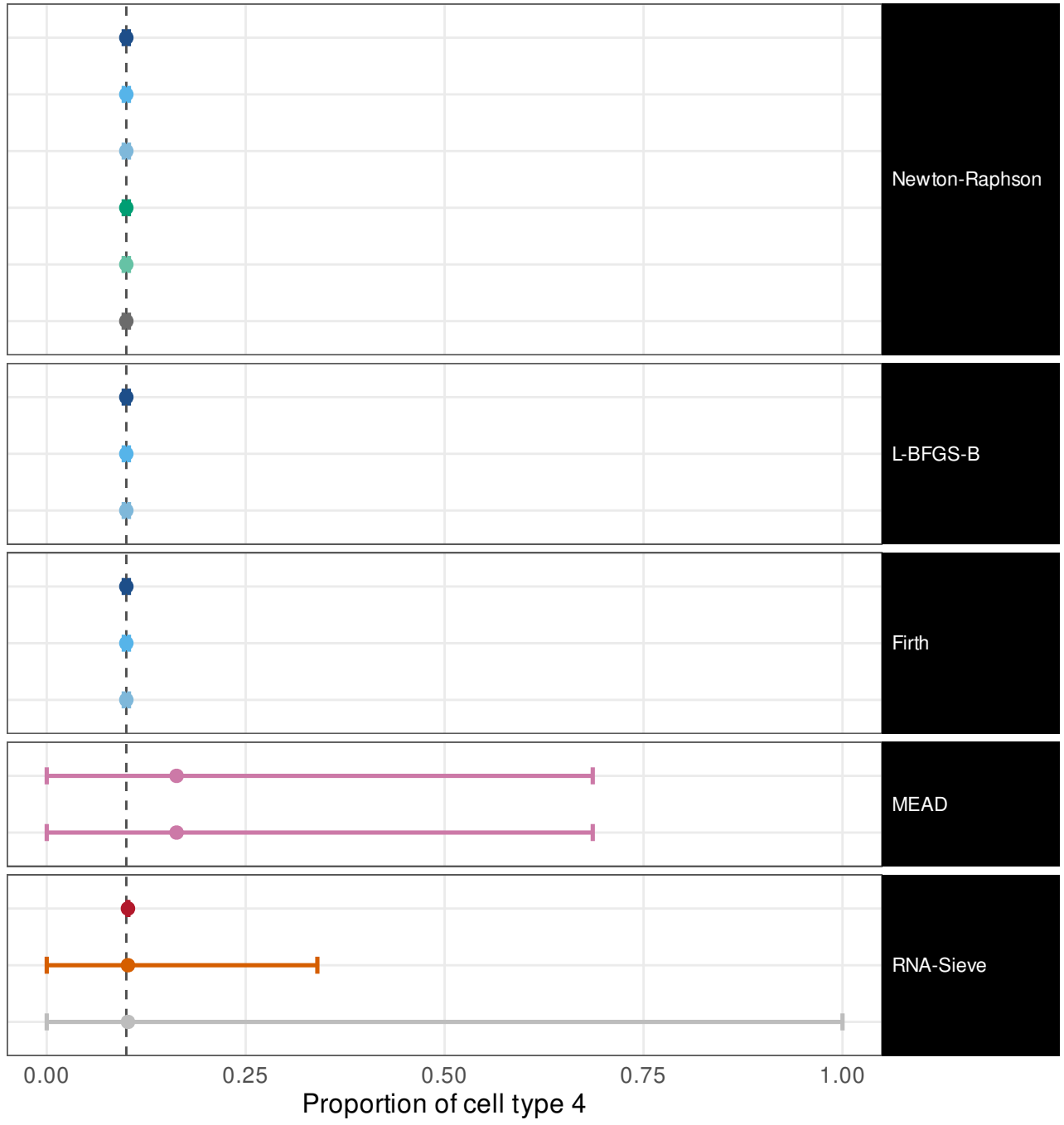
```
Wald (delta)                        0.94 [0.89, 0.96]   w = 0.00647
ILR, mapped back                    0.93 [0.89, 0.96]   w = 0.00646
ILR, REML                           0.93 [0.89, 0.96]   w = 0.00656
Profile likelihood                  0.94 [0.89, 0.96]   w = 0.0065
Profile, REML                       0.94 [0.89, 0.96]   w = 0.00658
Oracle (Fisher at p*)               0.94 [0.89, 0.96]   w = 0.00657

Wald (delta)                        0.94 [0.89, 0.96]   w = 0.00647
ILR, mapped back                    0.93 [0.89, 0.96]   w = 0.00647
ILR, REML                           0.94 [ 0.9, 0.97]   w = 0.00657

Wald (delta)                        0.94 [0.89, 0.96]   w = 0.00647
ILR, mapped back                    0.93 [0.89, 0.96]   w = 0.00646
ILR, REML                           0.94 [0.89, 0.96]   w = 0.00655

Method's own interval                  1 [0.98,    1]   w = 0.686
Own interval, transcript fractions     1 [0.98,    1]   w = 0.686

Fisher information (Wald)           0.14 [0.094, 0.19]   w = 0.00105
Godambe sandwich                       1 [0.98,    1]   w = 0.34
Released code, as published            1 [0.98,    1]   w =    1
```

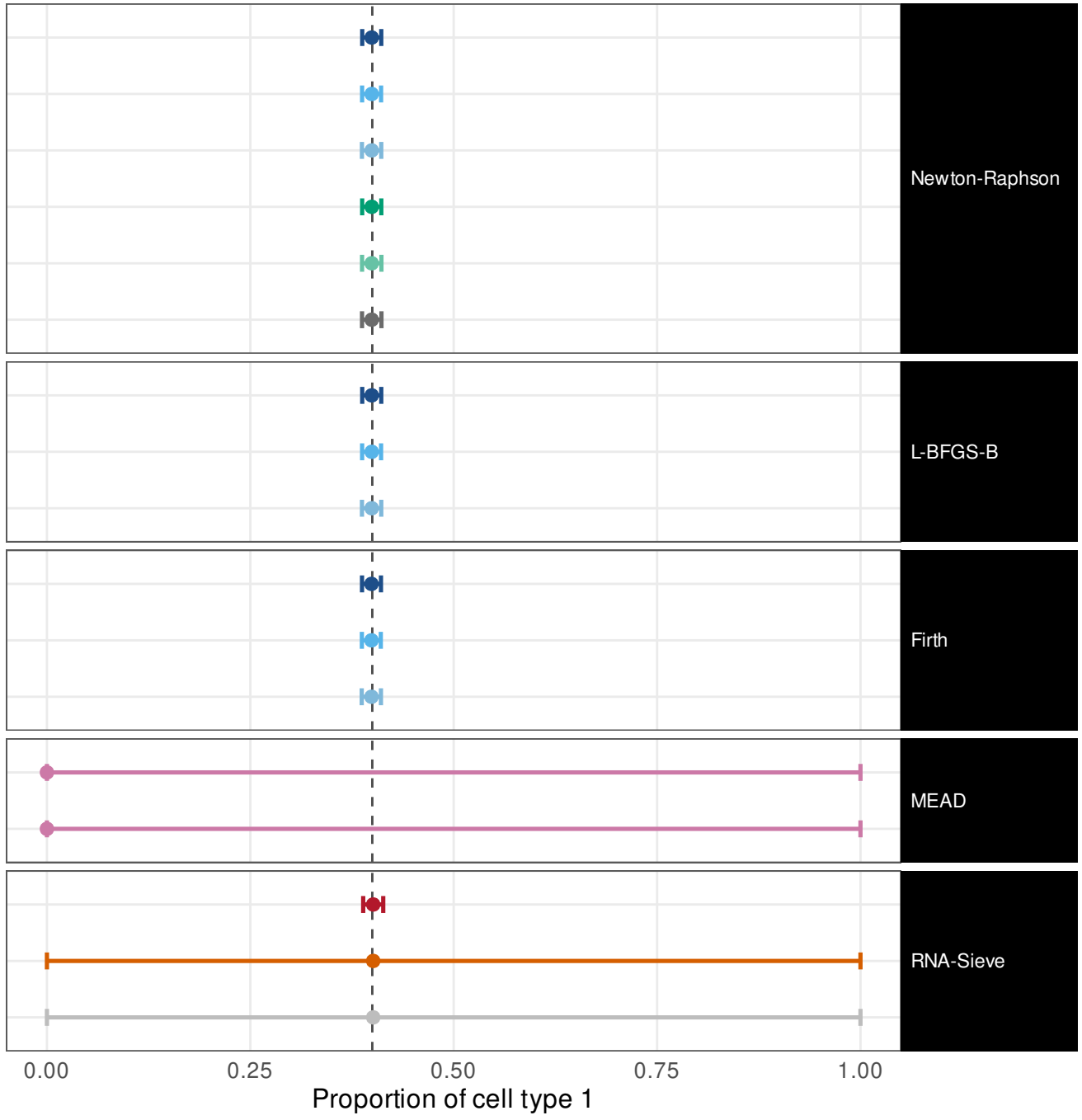


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / unbalanced / high sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.91 [0.86, 0.94]	w = 0.0234
ILR, mapped back	0.91 [0.86, 0.94]	w = 0.0233
ILR, REML	0.92 [0.87, 0.95]	w = 0.0238
Profile likelihood	0.91 [0.86, 0.94]	w = 0.0235
Profile, REML	0.91 [0.86, 0.94]	w = 0.0238
Oracle (Fisher at p*)	0.92 [0.87, 0.95]	w = 0.0239
Wald (delta)	0.91 [0.86, 0.94]	w = 0.0234
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.0233
ILR, REML	0.92 [0.87, 0.95]	w = 0.0237
Wald (delta)	0.92 [0.87, 0.95]	w = 0.0233
ILR, mapped back	0.92 [0.87, 0.95]	w = 0.0233
ILR, REML	0.92 [0.87, 0.95]	w = 0.0236
Method's own interval	1 [0.98, 1]	w = 1
Own interval, transcript fractions	1 [0.98, 1]	w = 1
Fisher information (Wald)	0.88 [0.83, 0.92]	w = 0.0246
Godambe sandwich	1 [0.98, 1]	w = 1
Released code, as published	1 [0.98, 1]	w = 1

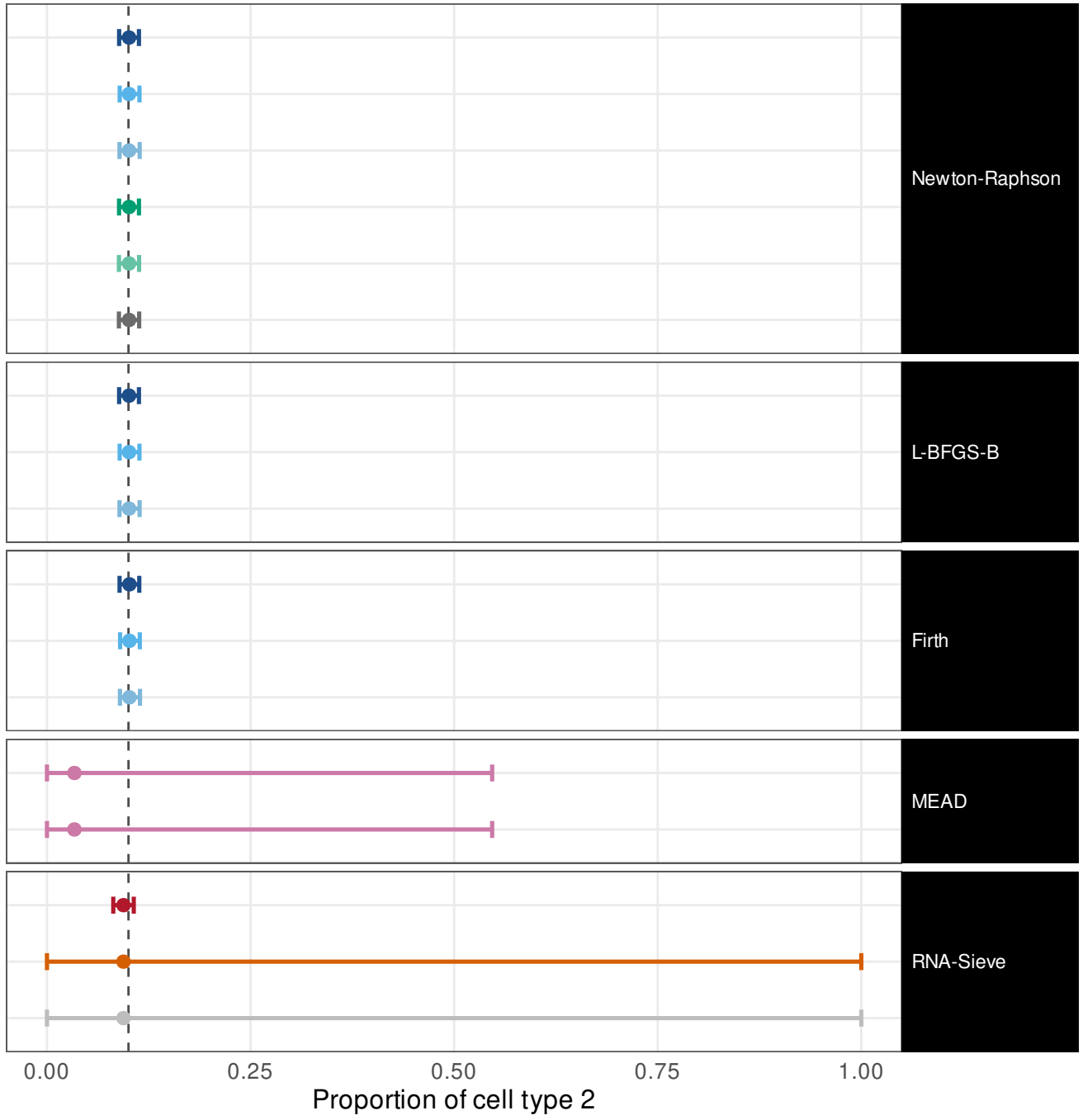


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / unbalanced / high sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.89	[0.84, 0.93]	w = 0.0243
ILR, mapped back	0.9	[0.85, 0.93]	w = 0.0243
ILR, REML	0.91	[0.86, 0.94]	w = 0.0247
Profile likelihood	0.9	[0.85, 0.93]	w = 0.0244
Profile, REML	0.9	[0.85, 0.93]	w = 0.0247
Oracle (Fisher at p*)	0.92	[0.88, 0.95]	w = 0.0248
Wald (delta)	0.89	[0.84, 0.93]	w = 0.0243
ILR, mapped back	0.9	[0.84, 0.93]	w = 0.0243
ILR, REML	0.9	[0.86, 0.94]	w = 0.0247
Wald (delta)	0.9	[0.84, 0.93]	w = 0.0242
ILR, mapped back	0.9	[0.86, 0.94]	w = 0.0242
ILR, REML	0.9	[0.85, 0.93]	w = 0.0246
Method's own interval	1	[0.98, 1]	w = 0.547
Own interval, transcript fractions	1	[0.98, 1]	w = 0.547
Fisher information (Wald)	0.81	[0.75, 0.86]	w = 0.0251
Godambe sandwich	1	[0.98, 1]	w = 1
Released code, as published	1	[0.98, 1]	w = 1

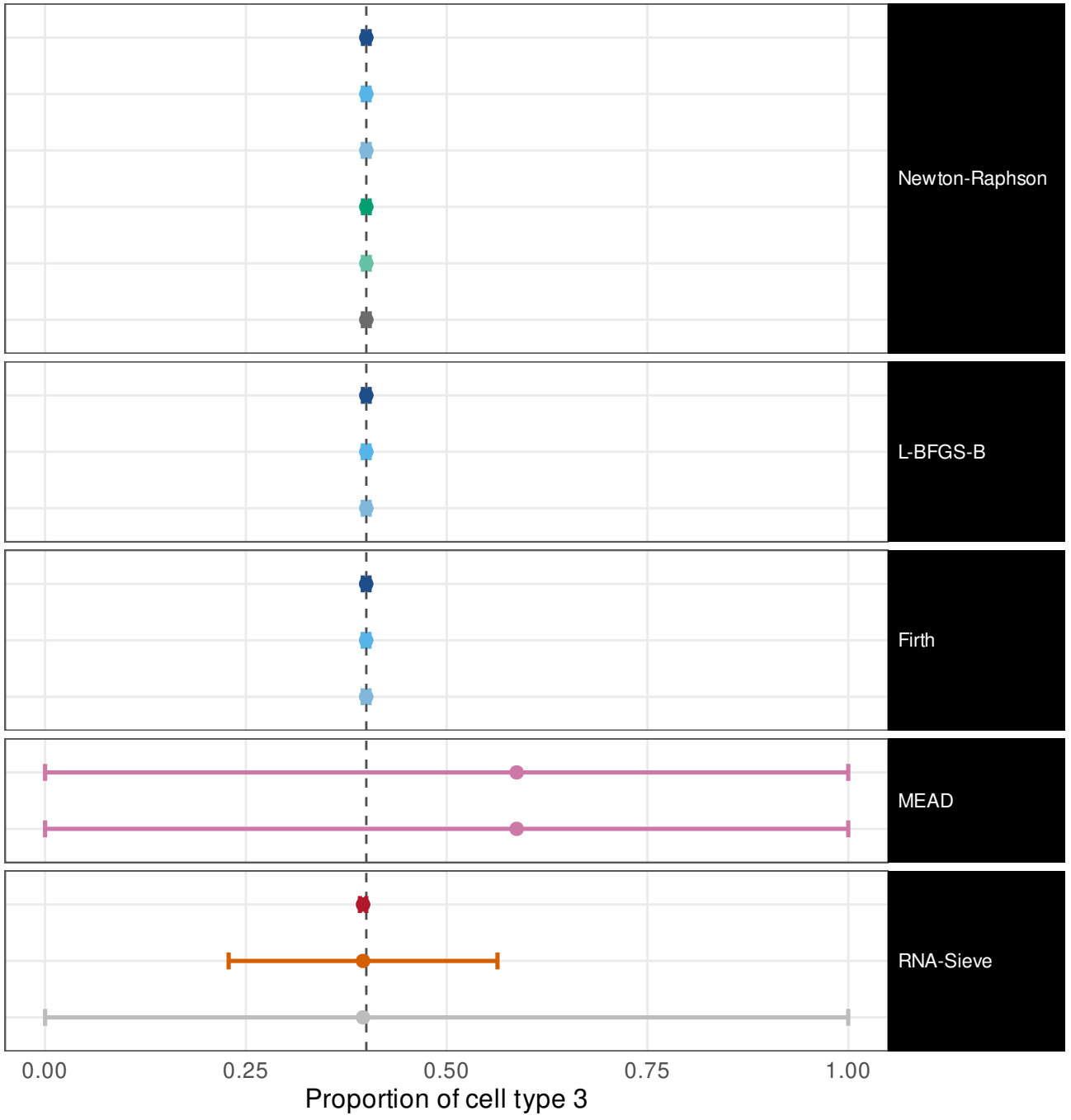


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / unbalanced / high sibling overlap / a = 1 / sizes 1:1:5:5

Wald (delta)	0.96 [0.92, 0.98]	w = 0.00839
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.0084
ILR, REML	0.96 [0.93, 0.98]	w = 0.00854
Profile likelihood	0.96 [0.92, 0.98]	w = 0.00843
Profile, REML	0.96 [0.92, 0.98]	w = 0.00853
Oracle (Fisher at p*)	0.96 [0.92, 0.98]	w = 0.00851
Wald (delta)	0.96 [0.92, 0.98]	w = 0.00839
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.00839
ILR, REML	0.96 [0.93, 0.98]	w = 0.00851
Wald (delta)	0.96 [0.92, 0.98]	w = 0.00839
ILR, mapped back	0.96 [0.93, 0.98]	w = 0.00839
ILR, REML	0.96 [0.92, 0.98]	w = 0.00851
Method's own interval	1 [0.98, 1]	w = 1
Own interval, transcript fractions	1 [0.98, 1]	w = 1
Fisher information (Wald)	0.5 [0.44, 0.57]	w = 0.00784
Godambe sandwich	1 [0.98, 1]	w = 0.335
Released code, as published	1 [0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / unbalanced / high sibling overlap / a = 1 / sizes 1:1:5:5

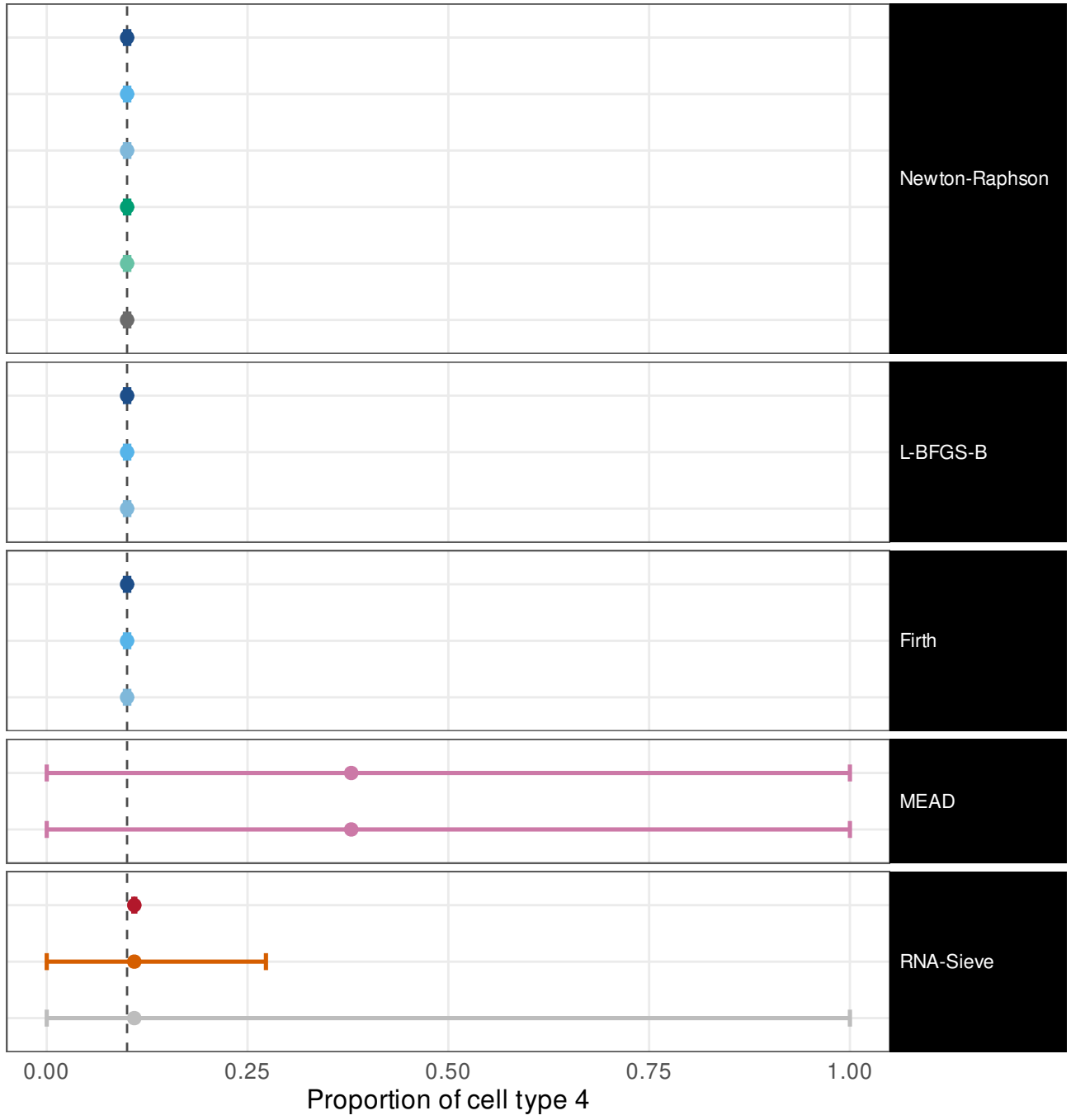
```
Wald (delta)                        0.92 [0.88, 0.95]   w = 0.00534
ILR, mapped back                    0.92 [0.87, 0.95]   w = 0.00533
ILR, REML                           0.93 [0.89, 0.96]   w = 0.00543
Profile likelihood                  0.92 [0.88, 0.95]   w = 0.00537
Profile, REML                       0.93 [0.89, 0.96]   w = 0.00543
Oracle (Fisher at p*)               0.93 [0.89, 0.96]   w = 0.00542

Wald (delta)                        0.92 [0.88, 0.95]   w = 0.00534
ILR, mapped back                    0.92 [0.88, 0.95]   w = 0.00533
ILR, REML                           0.93 [0.89, 0.96]   w = 0.00541

Wald (delta)                        0.92 [0.88, 0.95]   w = 0.00534
ILR, mapped back                    0.92 [0.88, 0.95]   w = 0.00531
ILR, REML                           0.92 [0.88, 0.95]   w = 0.00541

Method's own interval                  1 [0.98,    1]   w =    1
Own interval, transcript fractions     1 [0.98,    1]   w =    1

Fisher information (Wald)              0 [   0, 0.019]   w = 0.0025
Godambe sandwich                       1 [0.98,    1]   w = 0.273
Released code, as published            1 [0.98,    1]   w =    1
```

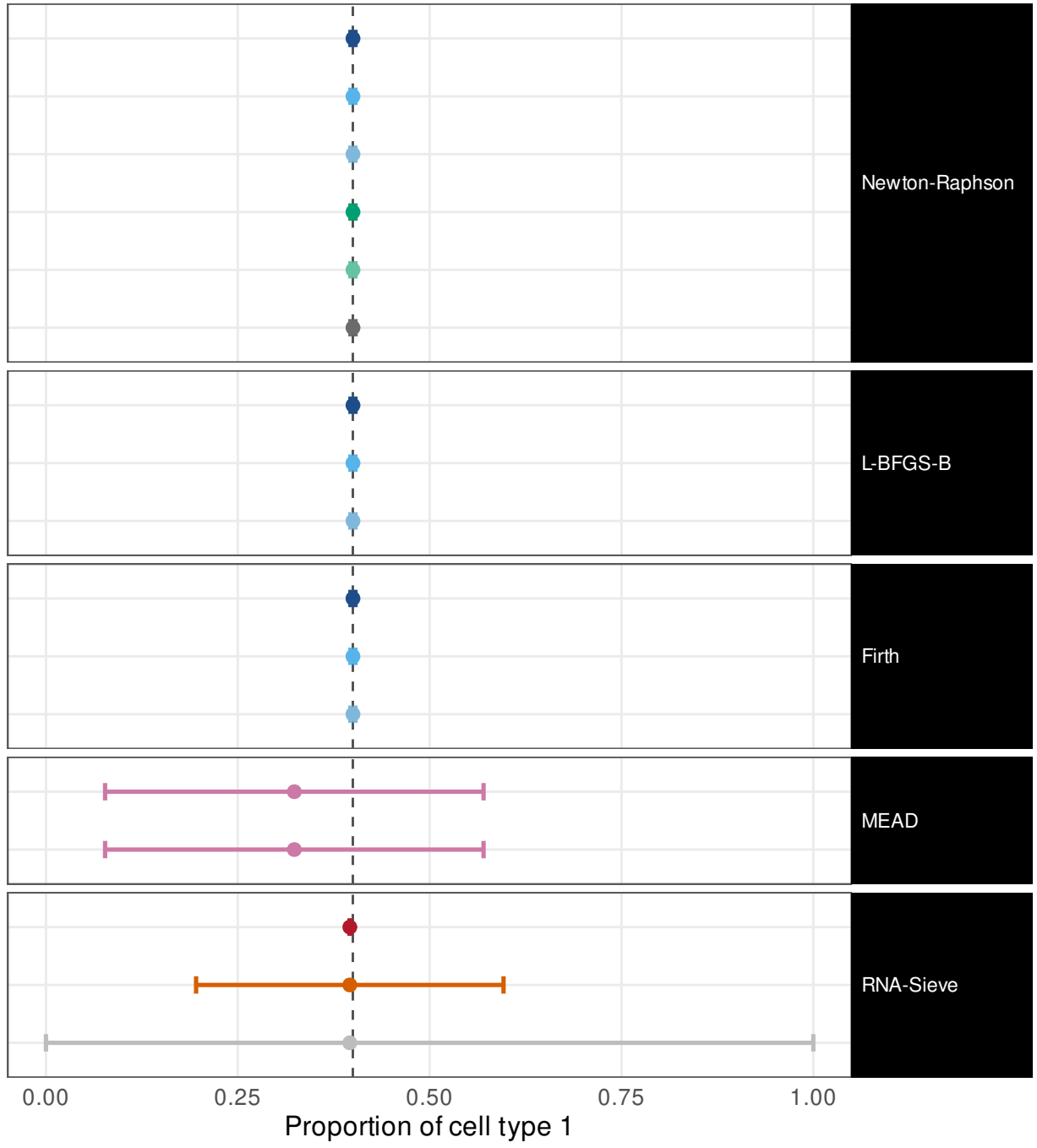


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / unbalanced / high sibling overlap / a = 5 / equal sizes

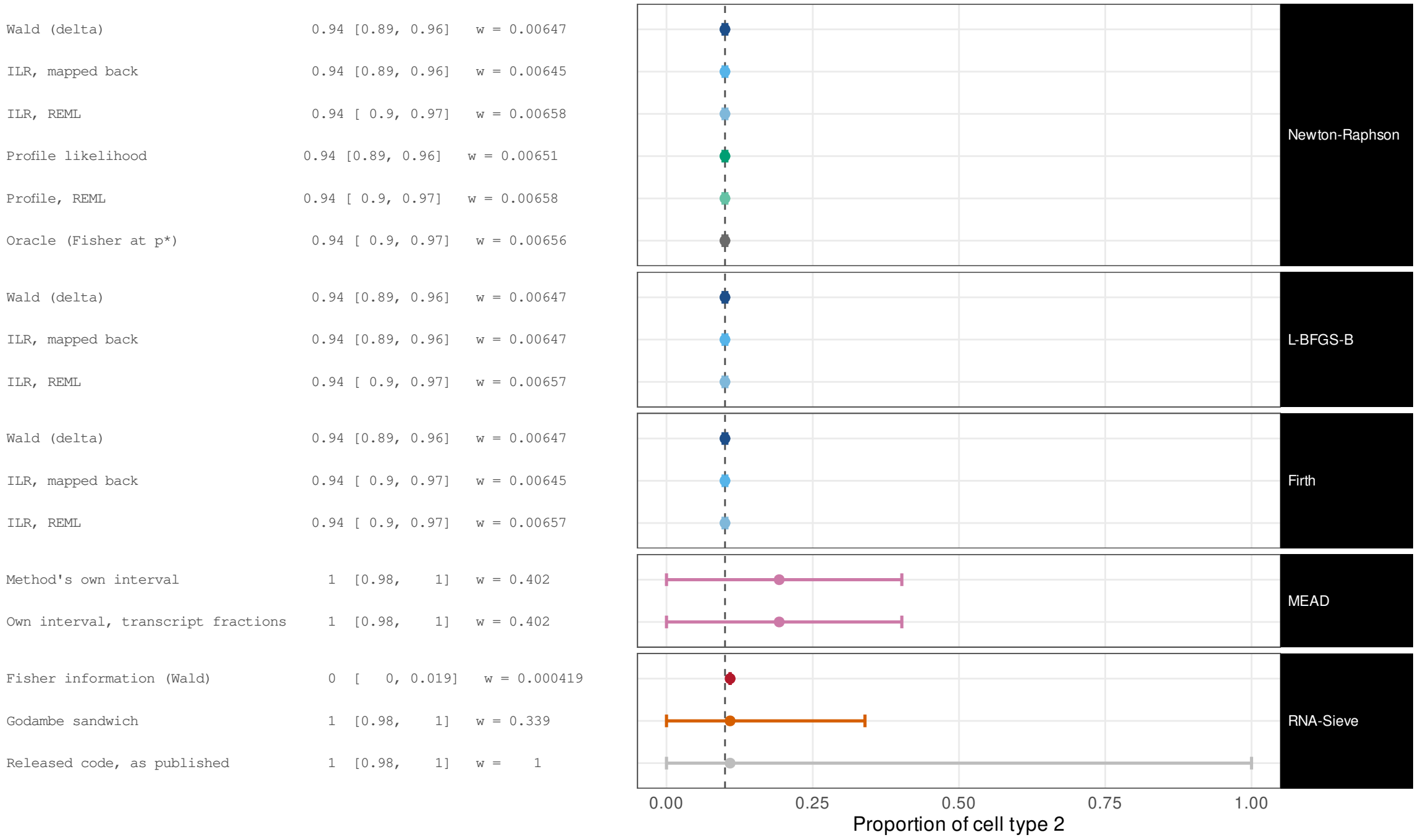
Wald (delta)	0.94 [0.9, 0.97]	w = 0.00652
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.00649
ILR, REML	0.95 [0.91, 0.97]	w = 0.00663
Profile likelihood	0.94 [0.9, 0.97]	w = 0.00655
Profile, REML	0.94 [0.9, 0.97]	w = 0.00663
Oracle (Fisher at p*)	0.94 [0.9, 0.97]	w = 0.00661
Wald (delta)	0.94 [0.9, 0.97]	w = 0.00652
ILR, mapped back	0.93 [0.89, 0.96]	w = 0.00651
ILR, REML	0.94 [0.9, 0.97]	w = 0.00663
Wald (delta)	0.94 [0.9, 0.97]	w = 0.00652
ILR, mapped back	0.94 [0.9, 0.97]	w = 0.00649
ILR, REML	0.94 [0.9, 0.97]	w = 0.00661
Method's own interval	1 [0.98, 1]	w = 0.493
Own interval, transcript fractions	1 [0.98, 1]	w = 0.493
Fisher information (Wald)	0.015 [0.0051, 0.043]	w = 0.000408
Godambe sandwich	1 [0.98, 1]	w = 0.401
Released code, as published	1 [0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / unbalanced / high sibling overlap / a = 5 / equal sizes



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / unbalanced / high sibling overlap / a = 5 / equal sizes

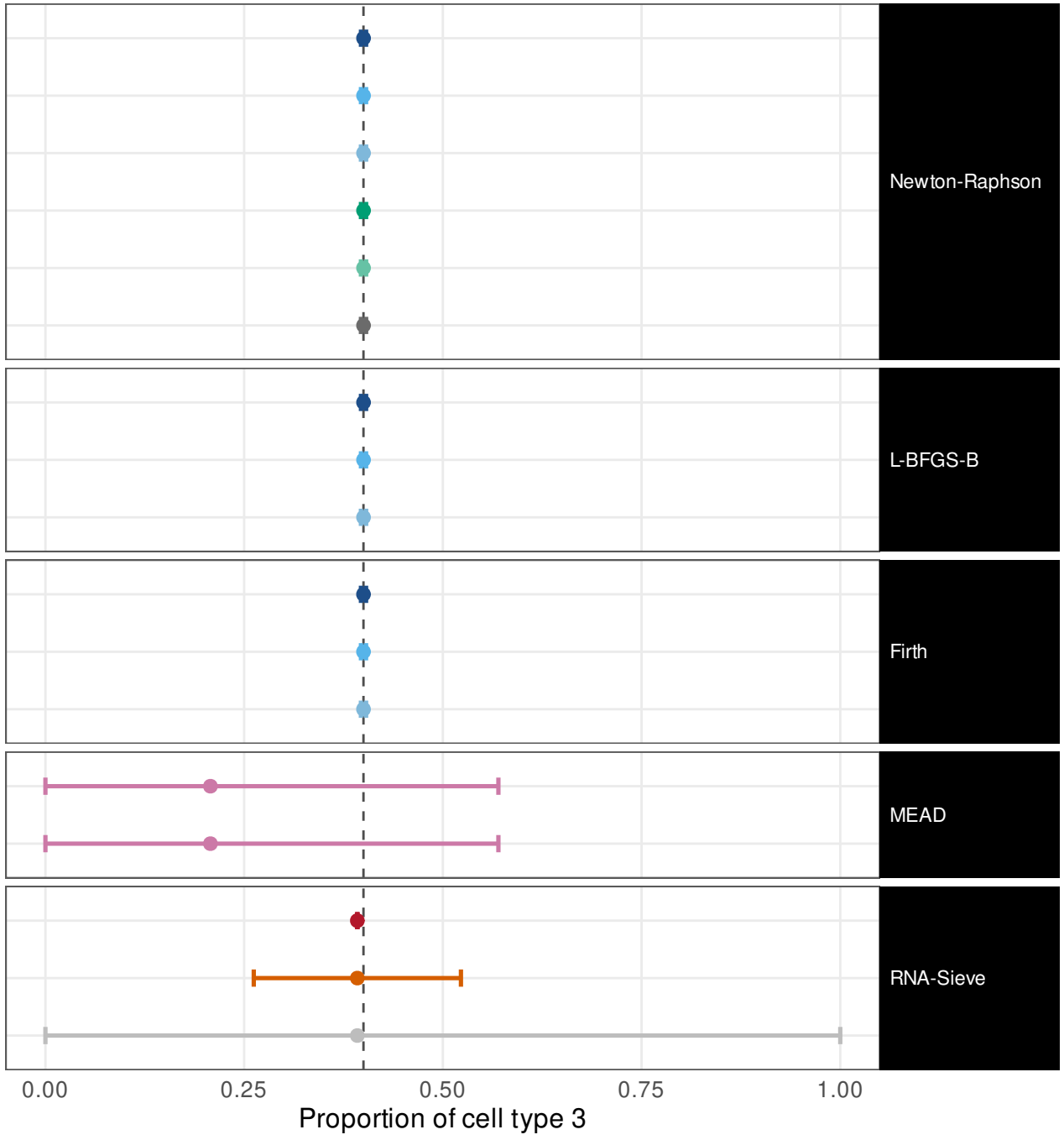
```
Wald (delta)                        0.96 [0.93, 0.98]   w = 0.00644
ILR, mapped back                    0.96 [0.92, 0.98]   w = 0.00644
ILR, REML                           0.96 [0.93, 0.98]   w = 0.00652
Profile likelihood                  0.96 [0.93, 0.98]   w = 0.00647
Profile, REML                       0.96 [0.93, 0.98]   w = 0.00655
Oracle (Fisher at p*)               0.96 [0.93, 0.98]   w = 0.00651

Wald (delta)                        0.96 [0.93, 0.98]   w = 0.00644
ILR, mapped back                    0.96 [0.92, 0.98]   w = 0.00643
ILR, REML                           0.96 [0.93, 0.98]   w = 0.00654

Wald (delta)                        0.96 [0.92, 0.98]   w = 0.00644
ILR, mapped back                    0.96 [0.92, 0.98]   w = 0.0064
ILR, REML                           0.96 [0.93, 0.98]   w = 0.00651

Method's own interval                  1 [0.98,    1]   w = 0.57
Own interval, transcript fractions     1 [0.98,    1]   w = 0.57

Fisher information (Wald)              0 [   0, 0.019]   w = 0.000798
Godambe sandwich                       1 [0.98,    1]   w = 0.261
Released code, as published            1 [0.98,    1]   w =    1
```

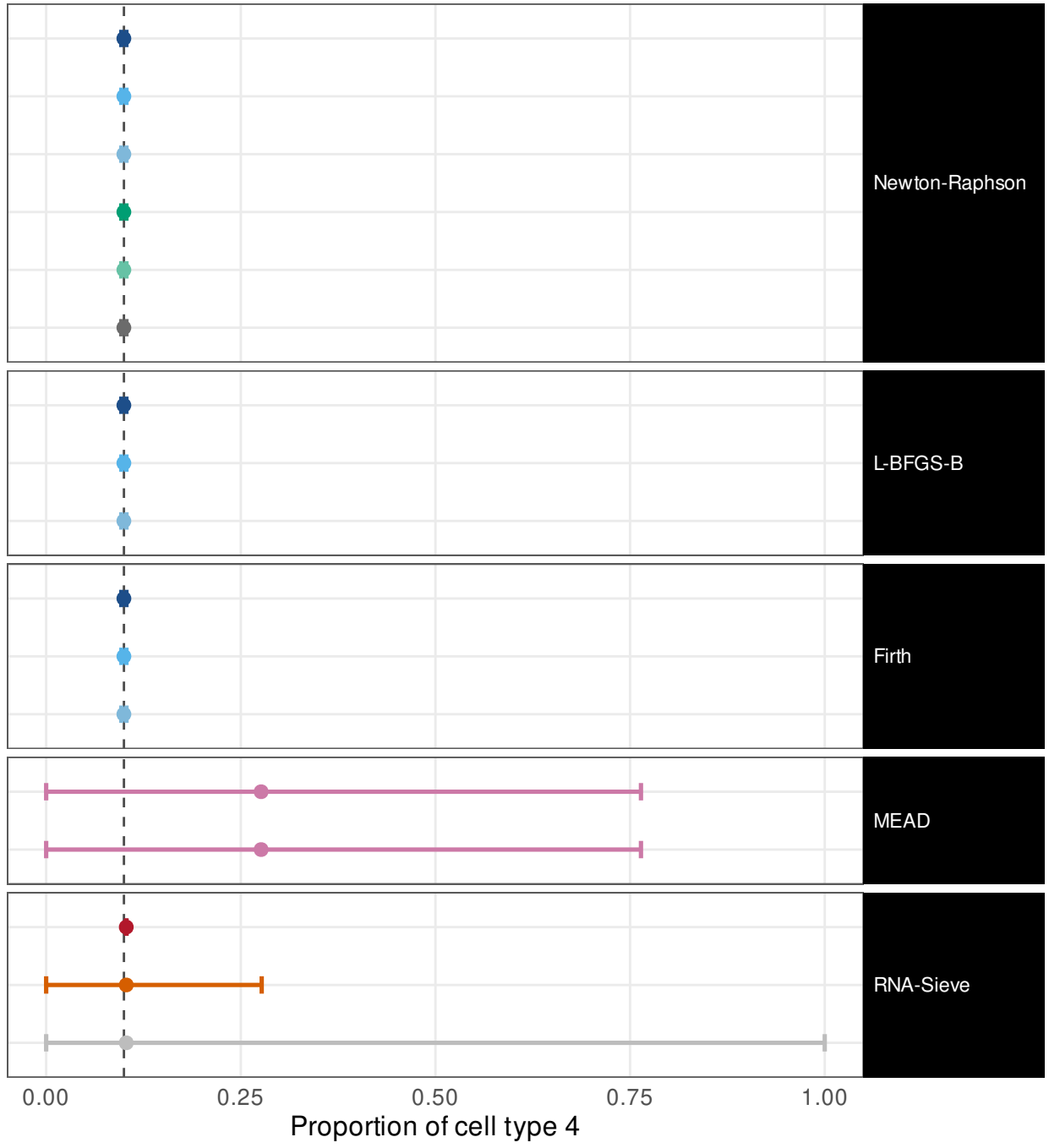


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / unbalanced / high sibling overlap / a = 5 / equal sizes

Wald (delta)	0.94 [0.9, 0.97]	w = 0.00647
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.00648
ILR, REML	0.95 [0.91, 0.97]	w = 0.00654
Profile likelihood	0.95 [0.91, 0.97]	w = 0.0065
Profile, REML	0.96 [0.92, 0.98]	w = 0.00658
Oracle (Fisher at p*)	0.96 [0.92, 0.98]	w = 0.00655
Wald (delta)	0.94 [0.9, 0.97]	w = 0.00647
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.00646
ILR, REML	0.96 [0.92, 0.98]	w = 0.00657
Wald (delta)	0.95 [0.91, 0.97]	w = 0.00647
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.00643
ILR, REML	0.96 [0.92, 0.98]	w = 0.00656
Method's own interval	1 [0.98, 1]	w = 0.764
Own interval, transcript fractions	1 [0.98, 1]	w = 0.764
Fisher information (Wald)	0.045 [0.024, 0.083]	w = 0.000763
Godambe sandwich	1 [0.98, 1]	w = 0.277
Released code, as published	1 [0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / unbalanced / high sibling overlap / a = 5 / sizes 1:1:5:5

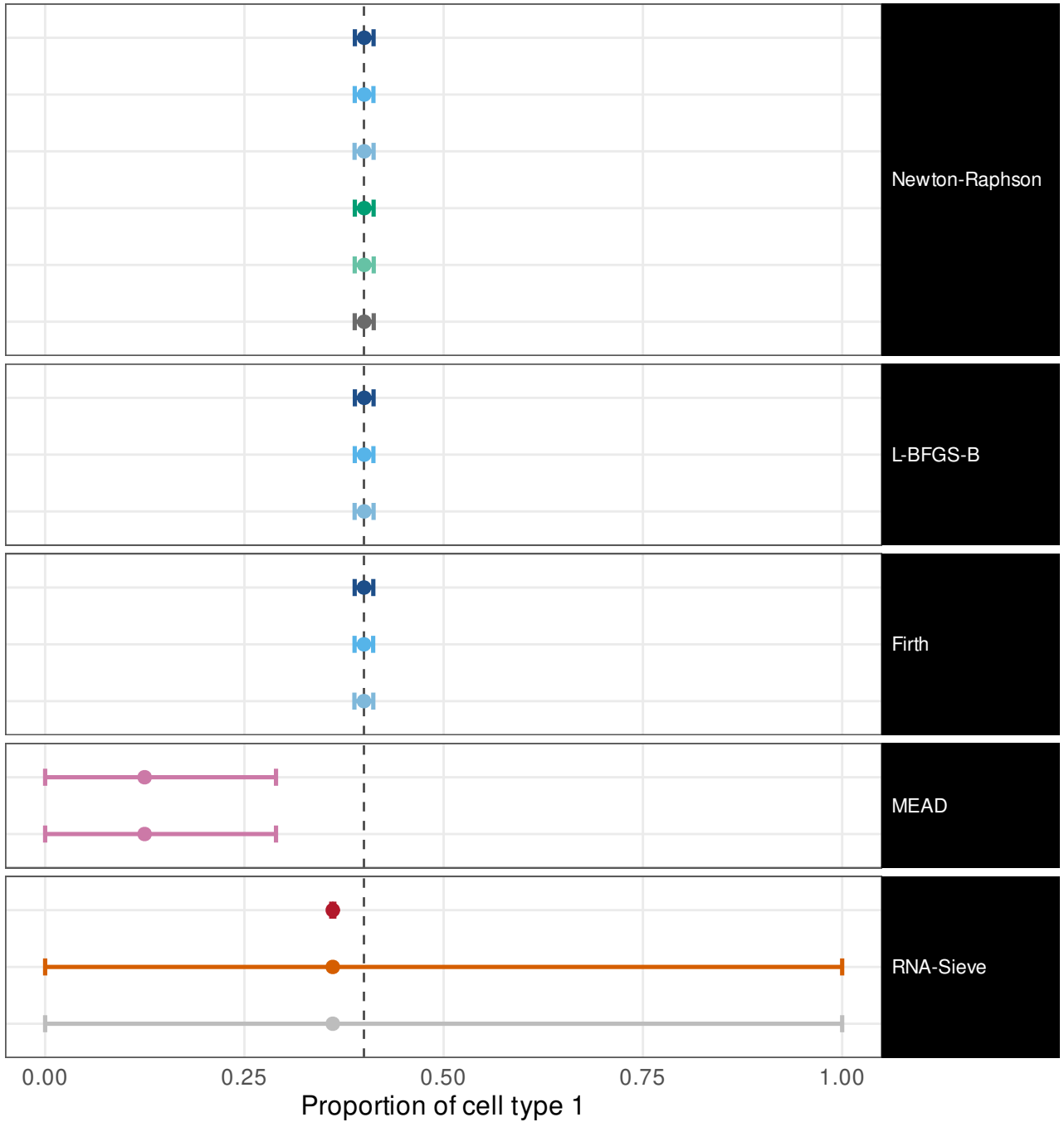
```
Wald (delta)                        0.95 [0.91, 0.97]   w = 0.0235
ILR, mapped back                    0.95 [0.91, 0.97]   w = 0.0234
ILR, REML                           0.95 [0.91, 0.97]   w = 0.0239
Profile likelihood                  0.95 [0.91, 0.97]   w = 0.0236
Profile, REML                       0.95 [0.91, 0.97]   w = 0.0238
Oracle (Fisher at p*)               0.96 [0.92, 0.98]   w = 0.0238

Wald (delta)                        0.95 [0.91, 0.97]   w = 0.0235
ILR, mapped back                    0.95 [0.91, 0.97]   w = 0.0234
ILR, REML                           0.96 [0.92, 0.98]   w = 0.0238

Wald (delta)                        0.95 [0.91, 0.97]   w = 0.0234
ILR, mapped back                    0.96 [0.92, 0.98]   w = 0.0234
ILR, REML                           0.95 [0.91, 0.97]   w = 0.0237

Method's own interval                  0 [   0, 0.019]   w = 0.29
Own interval, transcript fractions     1 [0.98,    1]   w = 0.29

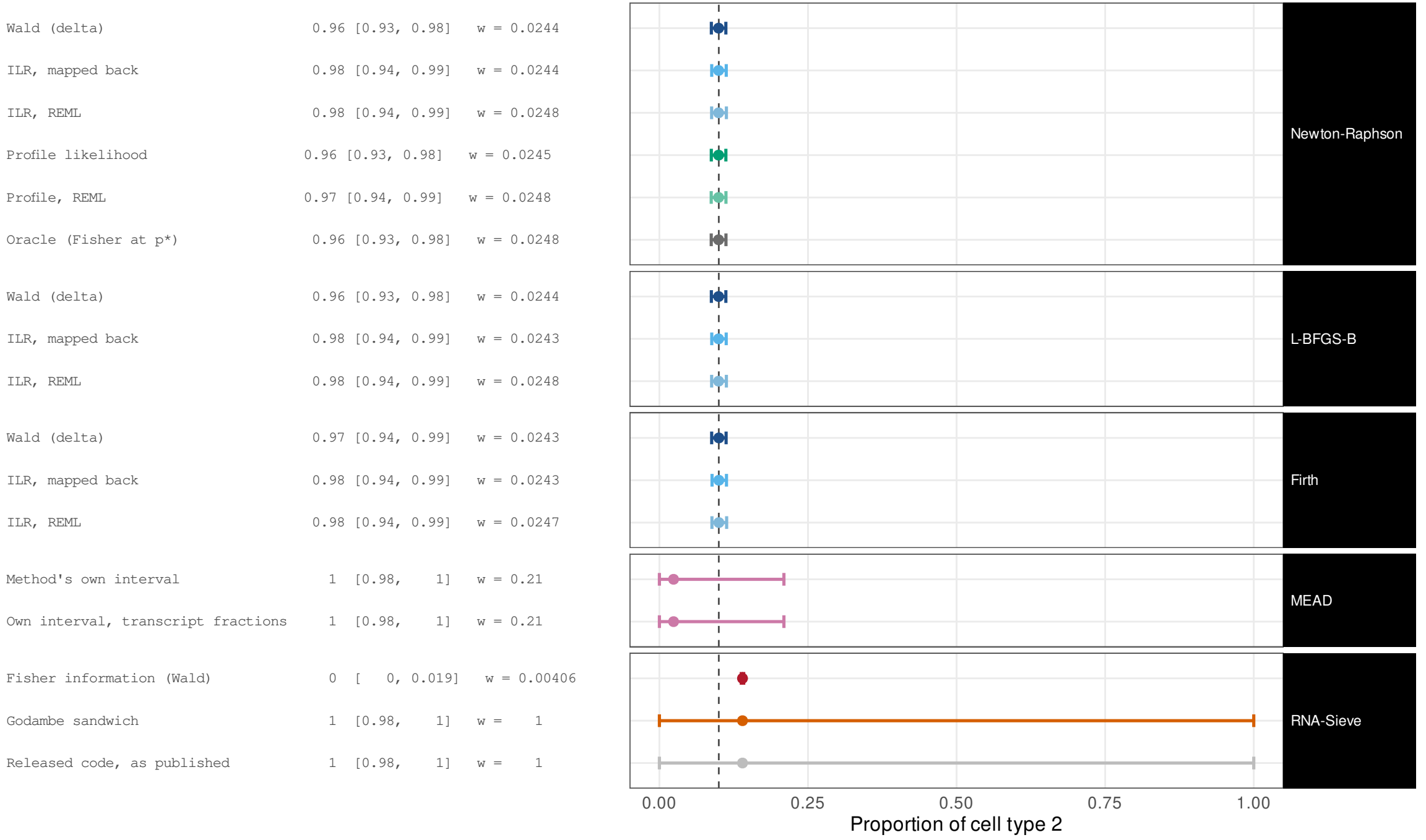
Fisher information (Wald)              0 [   0, 0.019]   w = 0.00382
Godambe sandwich                       1 [0.98,    1]   w =    1
Released code, as published            1 [0.98,    1]   w =    1
```



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / unbalanced / high sibling overlap / a = 5 / sizes 1:1:5:5



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / unbalanced / high sibling overlap / a = 5 / sizes 1:1:5:5

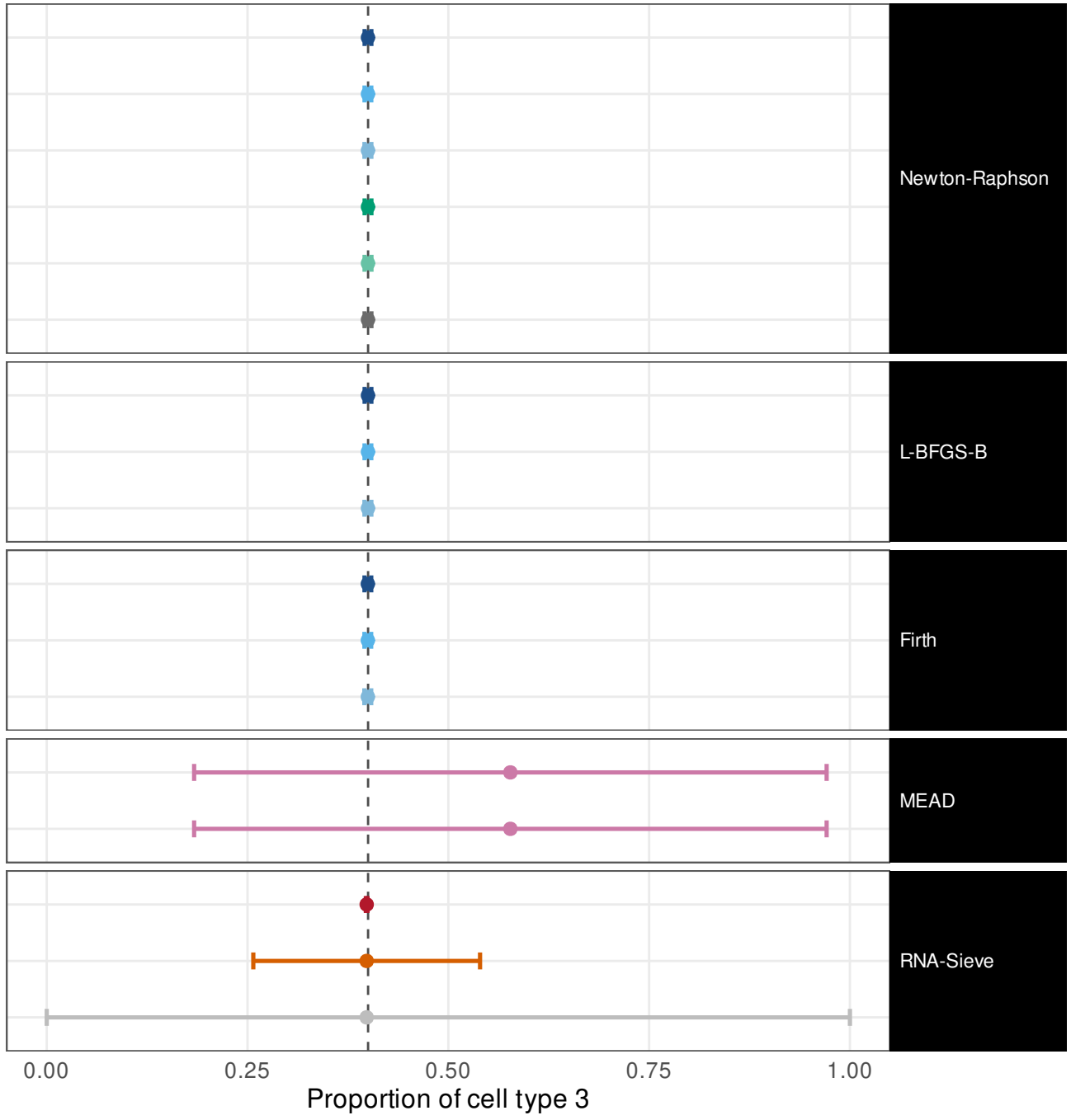
```
Wald (delta)                        0.95 [0.91, 0.97]   w = 0.00842
ILR, mapped back                    0.95 [0.91, 0.97]   w = 0.00843
ILR, REML                           0.96 [0.92, 0.98]   w = 0.00858
Profile likelihood                  0.95 [0.91, 0.97]   w = 0.00846
Profile, REML                       0.95 [0.91, 0.97]   w = 0.00856
Oracle (Fisher at p*)               0.95 [0.91, 0.97]   w = 0.00852

Wald (delta)                        0.95 [0.91, 0.97]   w = 0.00842
ILR, mapped back                    0.94 [ 0.9, 0.97]   w = 0.00842
ILR, REML                           0.94 [ 0.9, 0.97]   w = 0.00857

Wald (delta)                        0.94 [ 0.9, 0.97]   w = 0.00841
ILR, mapped back                    0.95 [0.91, 0.97]   w = 0.00841
ILR, REML                           0.94 [0.89, 0.96]   w = 0.00855

Method's own interval                  1 [0.98,    1]   w = 0.788
Own interval, transcript fractions     1 [0.98,    1]   w = 0.788

Fisher information (Wald)           0.24 [0.19,  0.3]   w = 0.00203
Godambe sandwich                       1 [0.98,    1]   w = 0.282
Released code, as published            1 [0.98,    1]   w =    1
```

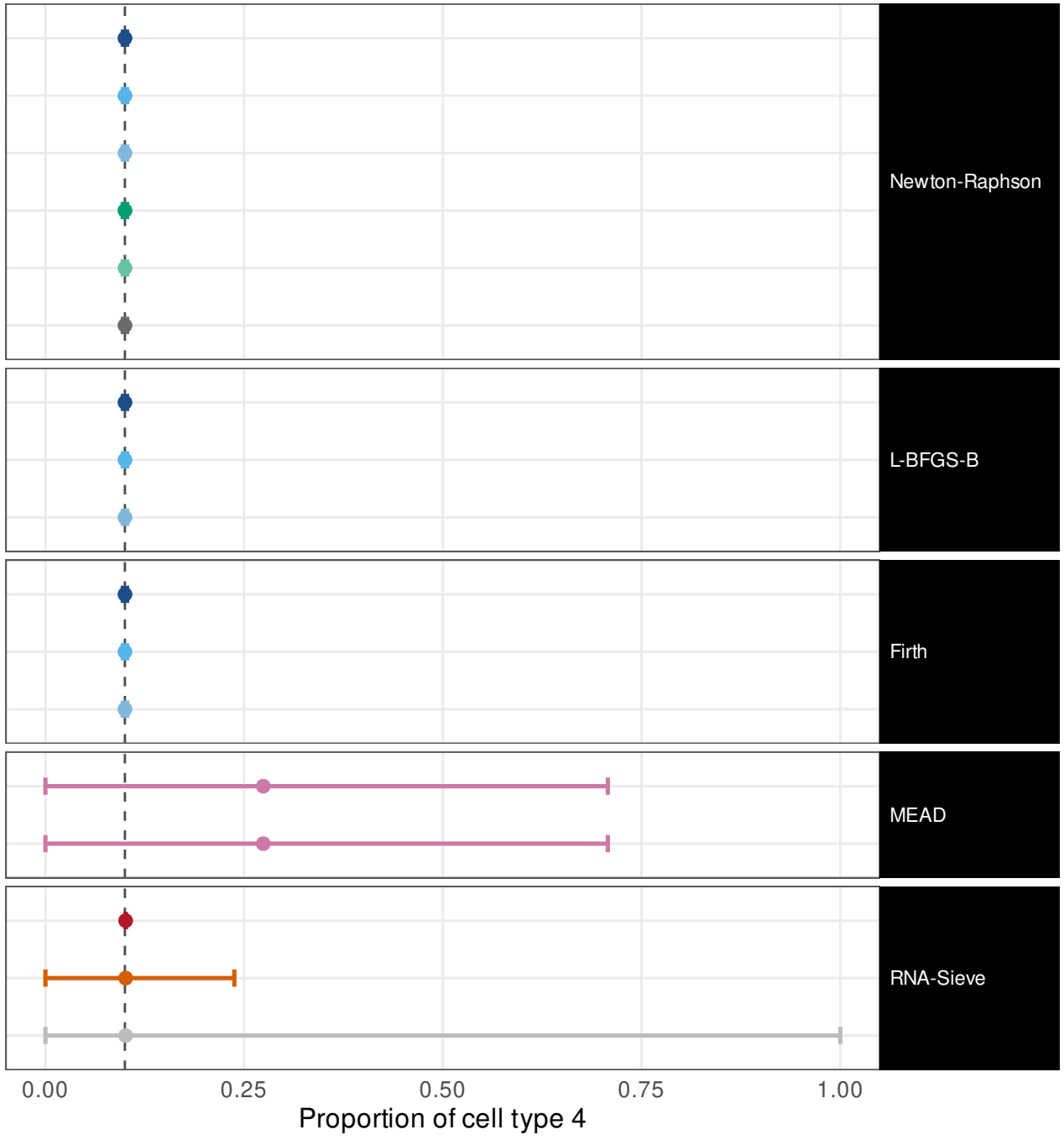


Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.

Confidence intervals: coverage [Wilson 95% interval] and mean width w

G = 200 / 1e+05 cells per bulk / unbalanced / high sibling overlap / a = 5 / sizes 1:1:5:5

Wald (delta)	0.96 [0.93, 0.98]	w = 0.00536
ILR, mapped back	0.97 [0.94, 0.99]	w = 0.00535
ILR, REML	0.98 [0.94, 0.99]	w = 0.00543
Profile likelihood	0.97 [0.94, 0.99]	w = 0.00538
Profile, REML	0.97 [0.94, 0.99]	w = 0.00544
Oracle (Fisher at p*)	0.98 [0.94, 0.99]	w = 0.00542
Wald (delta)	0.96 [0.93, 0.98]	w = 0.00536
ILR, mapped back	0.97 [0.94, 0.99]	w = 0.00535
ILR, REML	0.98 [0.94, 0.99]	w = 0.00546
Wald (delta)	0.96 [0.92, 0.98]	w = 0.00535
ILR, mapped back	0.96 [0.92, 0.98]	w = 0.00534
ILR, REML	0.98 [0.94, 0.99]	w = 0.00543
Method's own interval	1 [0.98, 1]	w = 0.708
Own interval, transcript fractions	1 [0.98, 1]	w = 0.708
Fisher information (Wald)	0.14 [0.094, 0.19]	w = 0.000532
Godambe sandwich	1 [0.98, 1]	w = 0.238
Released code, as published	1 [0.98, 1]	w = 1



Bar: mean lower to mean upper bound over the simulated bulks; point: mean estimate; dashed: mean target.
Methods without intervals are not shown.